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# On Two-Dimensional Chiral Conformal Field Theories with Sporadic Finite Simple Group Symmetries

## （散在型有限単純群対称性をもつ 二次元カイラル共形場理論について）

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**Abstract**

This thesis consists of two studies on two-dimensional chiral conformal field theories with sporadic finite simple group symmetries. The first study develops one approach to investigate the structure of the automorphism group of a lattice vertex operator algebra, and we particularly discuss the odd Leech lattice conformal field theory. The second study focuses on the computation of an orbifold partition function of the Conway moonshine module.

In the first study, we show that the Mathieu groups $M_{24}$ and $M_{23}$ in the isometry group $\mathrm{Aut}(O_{24})$ of the odd Leech lattice $O_{24}$ do not lift to subgroups of the automorphism group $\mathrm{Aut}(V_{O_{24}})$ of its lattice vertex operator (super)algebra $V_{O_{24}}$. In other words, the subgroups $2^{24}.M_{24}$ and $2^{23}.M_{23}$ of the automorphism group $\mathrm{Aut}(V_{O_{24}})$ are non-split extensions. Our method can also confirm a similar result for the Conway group $\mathrm{Co}_0$ and the Leech lattice, which was already shown in [Gri73]. This study is motivated by the moonshine-type observation on the $\mathcal{N} = 2$ extremal elliptic genus of central charge 24 by [BDFK15]. We also investigate weight-1 and weight-$\frac{3}{2}$ currents invariant under the subgroup $2^{24}.M_{24}$ or $2^{23}.M_{23}$ of the automorphism group $\mathrm{Aut}(V_{O_{24}})$, and revisit an $\mathcal{N} = 2$ superconformal algebra in the odd Leech lattice vertex operator algebra $V_{O_{24}}$.

In the second study, we aim to compute the Witten index of the orbifold theory $(V^{f\natural})^{\otimes 24}/\mathrm{Co}_1$ of the Conway moonshine module, or Duncan's module $V^{f\natural}$, by its Conway group symmetry. This study is motivated by a candidate proposed by [AKL22] for a superconformal field theory corresponding to the periodicity element of the spectrum of topological modular forms under the Stolz–Teichner conjecture. In particular, we calculate the twisted partition functions of Duncan's module, for each conjugacy class of commuting pairs of the Conway group. To obtain the Witten index, we further need to determine the phases to trivialize the anomaly. We attempt to do so from certain consistency conditions, and find a conjectural set of phases, although its validity remains an open question.

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# Introduction

## 1 Introduction

### 1.1 General Introduction

Unexpected coincidences between different fields which have been studied independently often occur in the history of science, and they promote the development and understanding of both fields. One notable example is moonshine phenomena.

In finite group theory in mathematics, the classification of finite simple groups was a long-standing problem. Nowadays, it is already settled, and finite simple groups are classified into three infinite series, and 26 sporadic groups. In 1978, when the existence of the largest sporadic group called the monster group $\mathbb{M}$ was still a conjecture, McKay noticed that its smallest nontrivial irreducible representation dimension $196883$ appears in the coefficient of the modular $j$-function

$$j(\tau) = q^{-1} + 774 + 196884q + \cdots \quad (q = e^{2\pi\sqrt{-1}\tau}), \tag{1.1}$$

as $196884 = 196883 + 1$ [McK01]. Furthermore, Thompson observed that the first few coefficients of the modular $j$-function can also be written as simple sums of irreducible representation dimensions of the monster group $\mathbb{M}$ [Tho79b]. It came as a surprise to mathematicians at that time, because the modular $j$-function is a concept important in arithmetic geometry of elliptic curves, a field studied independently of finite group theory. Conway and Norton deepened the relation between the monster group $\mathbb{M}$ and the modular $j$-function, and proposed a conjecture which they call the monstrous moonshine [CN79]. Here, the English word "moonshine" means "insubstantial or unreal" [Gan06a].

The appearance of the irreducible representations of the monster group in the modular $j$-function was theoretically explained by Frenkel, Lepowsky, and Meurman. They constructed an algebraic object called a vertex operator algebra (VOA), whose automorphism group is the monster group, and whose graded character is the modular $j$-function (up to the constant term) [FLM88]. However, this moonshine phenomenon was not confined to mathematics. Surprisingly, the concept of vertex operator algebras provided a mathematical formulation of two-dimensional conformal field theories in physics. Conformal field theory (CFT) is an important framework in physics, which describes the string theory in high-energy physics and the critical phenomena in condensed matter physics.

This triggered off a lot of interaction between mathematics and physics. For example, many other moonshine phenomena were observed between the sporadic groups and modular functions or weak Jacobi forms, and one of the prominent examples, the K3 Mathieu moonshine, was found by physicists Eguchi, Ooguri, and Tachikawa [EH10]. It has been studied by both mathematicians and physicists to this day, but its mysterious nature has not yet been fully understood.

Moreover, conformal field theories with sporadic group symmetries have been extending beyond moonshine phenomena. For example, they are studied in [Wit07, GGK+08] in relation to

three-dimensional gravity. In addition, a specific VOA called the Conway moonshine module is expected to be useful in providing supporting evidence for the Stolz–Teichner conjecture, which also proposes a new correspondence between mathematics and physics [ST11].

This thesis comprises two original studies on two-dimensional conformal field theories with sporadic group symmetries, together with review parts. The first study, presented in Part III, is primarily moonshine-oriented, and develops an approach to investigate the structure of the automorphism group of a lattice VOA. The second study, presented in Part IV, focuses on the Conway moonshine module (which we also call Duncan's module), and calculates its twisted partition functions as an indispensable step toward validating a candidate for a superconformal field theory (SCFT) corresponding to the periodicity element under the Stolz–Teichner conjecture.

## 1.2 Motivation for Part III

**A short introduction**

Consider the CFT $\mathcal{T}$ of chiral bosons whose target spacetime is an $n$-dimensional torus $T^n = \mathbb{R}^n/L$, where $L$ is a lattice of rank $n$. The geometric symmetry of the spacetime, the isometry group $\mathrm{Aut}(L)$ of the lattice, seems to be the symmetry of this theory. However, this is not necessarily the case, because of the effect of cocycle factors.

In fact, we will show that the symmetry $\mathrm{Aut}(L)$ of the lattice does not lift to a subgroup of the symmetry $\mathrm{Aut}(\mathcal{T})$ of the theory, for the cases of the odd Leech lattice and the Leech lattice, which are of particular interest from the perspective of moonshine phenomena. Our computational approach only uses basic data of the isometry group and its action on the lattice, and is in principle applicable to other lattices.

**Motivation from moonshine**

Symmetries of lattices can be key ingredients in understanding moonshine phenomena. The most prominent example is the monster vertex operator algebra (or the moonshine module) $V^\natural$, which was constructed in [FLM88] as a $\mathbb{Z}_2$-orbifold of the Leech lattice VOA, and provided an explanation for the appearance of representation dimensions of the monster group $\mathbb{M}$ in the coefficients of the modular $j$-function. The symmetry, or the automorphism (isometry) group $\mathrm{Aut}(\Lambda_{24})$ of the Leech lattice $\Lambda_{24}$ is the Conway group $\mathrm{Co}_0$, and its quotient $\mathrm{Co}_0/\mathbb{Z}_2$ by its center $\mathbb{Z}_2$ is a sporadic finite simple group denoted by $\mathrm{Co}_1$. Its information is inherited by the automorphism group $\mathrm{Aut}(V^\natural)$ of the monster VOA $V^\natural$, which is the monster group $\mathbb{M}$, in the form of the maximal subgroup[1] $2^{1+24}.\mathrm{Co}_1 \subset \mathbb{M}$. (See Section 5 for more details.)

Another case where we can expect the symmetry of a lattice accounts for a moonshine phenomenon was observed in [BDFK15]. They pointed out that if we decompose a certain weak Jacobi form $Z^{m=4}_{\mathrm{ext}}(\tau, z)$, called the $\mathcal{N} = 2$ extremal elliptic genus of central charge $c = 6m = 24$

[1] Here, $N.G$ denotes any (split or non-split) extension of a group $G$ by a group $N$. In addition, we write $N : G$ if it splits, or equivalently, it is a semidirect product of $N$ and $G$. We also let 2 denote $\mathbb{Z}_2$. See the beginning of Section 2 or Appendix A.1 for more on the notation of groups.

[GGK+08], into the characters $\mathrm{ch}^{\mathcal{N}=2}_{l=\frac{c}{6}-\frac{1}{2};h,Q}(\tau,z)$ of the Ramond representations of the $\mathcal{N}=2$ superconformal algebra (SCA) with $(-1)^{F=J_0}$ inserted,

$$Z^{m=4}_{\mathrm{ext}} = \mathrm{ch}^{\mathcal{N}=2}_{\frac{7}{2};1,4} + 47\mathrm{ch}^{\mathcal{N}=2}_{\frac{7}{2};1,0} + 23\,\mathrm{ch}^{\mathcal{N}=2}_{\frac{7}{2};2,4} + 2024(\mathrm{ch}^{\mathcal{N}=2}_{\frac{7}{2};2,3} + \mathrm{ch}^{\mathcal{N}=2}_{\frac{7}{2};2,-3}) + 14168(\mathrm{ch}^{\mathcal{N}=2}_{\frac{7}{2};2,2} + \mathrm{ch}^{\mathcal{N}=2}_{\frac{7}{2};2,-2}) + \cdots, \tag{1.2}$$

then representation dimensions of the largest Mathieu group $M_{24}$ appear in a nontrivial way. (See for example [CDD+14] for explicit formulae of $\mathrm{ch}^{\mathcal{N}=2}_{l;h,Q}(\tau,z)$.) For example, the coefficients $23$ and $2024$ are dimensions of irreducible representations of $M_{24}$, and the coefficient $14168$ is the sum $10395+3520+253$ of irreducible representation dimensions of $M_{24}$.

In [BDFK15], they further constructed a chiral $\mathcal{N}=2$ SCFT whose elliptic genus[2] coincides with the extremal elliptic genus $Z^{m=4}_{\mathrm{ext}}(\tau,z)$. Their construction was the fermionization of the lattice CFT of the $(A_1)^{24}$ Niemeier lattice, together with a special choice of $\mathcal{N}=2$ SCA generators. The automorphism group of the $(A_1)^{24}$ Niemeier lattice is $2^{24}:M_{24}$ [CS99, Table 16.1, Ch. 18 §4], although the automorphism group of the resulting lattice CFT preserving their $\mathcal{N}=2$ SCA becomes smaller than $M_{24}$. A similar analysis was also done for $\mathcal{N}=4$ extremal SCFT of central charge 24 in [Har16].

This situation of [BDFK15] seems analogous to the celebrated K3 Mathieu moonshine. The K3 Mathieu moonshine was first observed in [EOT10] and studied extensively, but is still missing some nice algebraic structure to explain all the properties such as the twining elliptic genera, their twisted versions, and their modular properties. (See Section 7 for more details.)

Although the lattice symmetry seemingly offers some explanation for the appearance of representation dimensions of $M_{24}$ observed in [BDFK15], if we take a closer look, the situation turns out to be more complicated than expected as follows.

We first note that, the construction of the SCFT in [BDFK15] was the fermionization with respect to the reflection $\mathbb{Z}_2$ symmetry of the $(A_1)^{24}$ Niemeier lattice CFT, but we can obtain an equivalent theory by the fermionization with respect to the shift $\mathbb{Z}_2$ symmetry of the same theory.[3] The Neveu-Schwarz (NS) sector of the resulting CFT is a lattice CFT $V_{O_{24}}$ of the odd Leech lattice $O_{24}$, and the Ramond (R) sector also admits a description in terms of lattices. In fact, the construction of the extremal elliptic genus from an odd Leech lattice was also done in [KY23b] from the perspective of code CFT (see also [GJF18, Example 4.5]).

One might expect that we could take an $\mathcal{N}=2$ SCA in the VOA of the NS sector $V_{O_{24}}$ such that the automorphism group $\mathrm{Aut}_{\mathcal{N}=2}(V_{O_{24}})$ preserving the $\mathcal{N}=2$ SCA is the Mathieu group $M_{24}$. If it were the case, it would justify the appearance of the representation dimensions of $M_{24}$ in the elliptic genus (the partition function of the R sector), under the spectral flow of the $\mathcal{N}=2$

[2]Here, the elliptic genus is in the sense of (3.49).

[3] The equivalence between the reflection $\mathbb{Z}_2$ orbifold and the shift $\mathbb{Z}_2$ orbifold was proved in [DGM94] in the cases of the lattice CFTs constructed from doubly-even self-dual binary codes. In the case of the main text, the $(A_1)^{24}$ Niemeier lattice CFT is the one constructed from the binary Golay code, and fermionization and $\mathbb{Z}_2$-orbifold can be uniformly treated in the modern understanding. See Section 5.3.4 for more details.

SCA. In fact, the automorphism group of the odd Leech lattice is $\mathrm{Aut}(O_{24}) = 2^{12} : M_{24}$ and contains $M_{24}$ as a subgroup.

However, here we encounter a problem. The symmetry $\mathrm{Aut}(V_L)$ of a lattice VOA inherits the information of the lattice symmetry $\mathrm{Aut}(L)$ in the form of the *group extension* of $\mathrm{Aut}(L)$. More precisely, it is the group extension $2^{\mathrm{rank}\, L}.\mathrm{Aut}(L)$ that is the subgroup of $\mathrm{Aut}(V_L)$ [FLM88, DN99]. Therefore, $\mathrm{Aut}(L)$ itself is not a subgroup of $\mathrm{Aut}(V_L)$ unless this group extension splits. In the language of physics, this is because of the cocycle factors appearing in the operator product expansion (OPE) of vertex operators.

**What we will actually do**

The main goal of Part III of this thesis is to answer the following question, in particular in the case of the odd Leech lattice.

**Question.** Does the automorphism group $\mathrm{Aut}(L)$ (or its subgroup) of a lattice $L$ lift to a subgroup of the automorphism group $\mathrm{Aut}(V_L)$ of the lattice VOA $V_L$?

We will analyze the group extension quite explicitly, and finally prove by contradiction that the answer is no in the cases of the subgroups $M_{24}$ and $M_{23}$ of the automorphism group $\mathrm{Aut}(O_{24})$ of the odd Leech lattice (Theorem 9.4). We will also confirm that the answer is no in the case of the Conway group $\mathrm{Co}_0$ of the Leech lattice by a similar method; this result was already shown in [Gri73] (see also [Gri87], [Iva09, Lemma 1.8.8 (iii)]), but we hope that our method is more elementary and easy to follow for non-experts of group theory. Of course, there is a possibility that the techniques of [Gri73, Gri87, Iva09] can also be used to show the result for the odd Leech lattice, but such a result is not explicitly available in the literature to the author's knowledge.

As a result, the odd Leech lattice CFT does not have the $M_{24}$ symmetry in the first place, unless we forget the OPE structure and regard it as just a momentum lattice. Instead, it has a $2^{24}.M_{24}$ symmetry which does not split. We further investigate the weight-1 and weight-$\frac{3}{2}$ currents of the odd Leech lattice CFT invariant under the symmetry $2^{24}.M_{24}$ or its subgroup $2^{23}.M_{23}$, and conclude that there is no such weight-$\frac{3}{2}$ invariant current. Therefore, we cannot find an $\mathcal{N} = 2$ SCA invariant under the $2^{24}.M_{24}$ or $2^{23}.M_{23}$ symmetry in the odd Leech lattice CFT, because an $\mathcal{N} = 2$ SCA needs weight-$\frac{3}{2}$ supercurrents.

This requires a slight modification on the $\mathcal{N} = 2$ SCA proposed by [BDFK15]. Their discussion on how their supercurrents satisfy the OPE of $\mathcal{N} = 2$ SCA is based on the assumption that the CFT would have the $M_{24}$ symmetry, so we have to revisit that point. Still, we can realize an $\mathcal{N} = 2$ SCA which reproduces the $\mathcal{N} = 2$ extremal elliptic genus by modifying the supercurrents as in Appendix B. However, since such supercurrents are no longer invariant under the $2^{24}.M_{24}$ or $2^{23}.M_{23}$ symmetry (and also not invariant under the $M_{24}$ or $M_{23}$ symmetry even if we forget the OPE structure), the situation surrounding this moonshine phenomenon seems to be more mysterious than we thought.

## 1.3 Motivation for Part IV

### A short introduction

For a CFT $\mathcal{T}$ on torus $T^2$ with finite group symmetry $G$, we can construct a $G$-invariant theory $\mathcal{T}/G$ called an orbifold, if there is no obstruction called an anomaly. To construct the orbifold, we take the sum of all the sectors with $G$-twisted boundary conditions in both spatial and temporal directions of $T^2$. However, if an anomaly exists, then the partition functions of the twisted sectors are only defined up to $\mathrm{U}(1)$ phases, and in particular, the modular invariance of the orbifold theory is broken.

For example, the $\mathbb{Z}_2$ symmetry $(-1)^F$ of a single complex chiral fermion is anomalous, but if we consider four complex chiral fermions, then we can make this anomaly vanish. The partition function of the orbifold theory is then[4]

$$Z_{\mathrm{orb}}(\tau) = \frac{1}{2\eta(\tau)^4}\left(\theta_3(\tau)^4 - \theta_4(\tau)^4 - \theta_2(\tau)^4 - \theta_1(\tau)^4\right), \tag{1.3}$$

as is well-known in superstring theory (see e.g. [Pol07b, §10.7]). This partition function $Z_{\mathrm{orb}}(\tau)$ is actually zero because of the supersymmetry, and the equation $Z_{\mathrm{orb}}(\tau) = 0$ is also known as the Jacobi identity (E.36) in mathematics.

The four terms of (1.3) correspond to the $2 \times 2$ choices of the periodic and anti-periodic boundary conditions in the two directions of $T^2$. The signs, or more generally the $\mathrm{U}(1)$ phases, of these four terms must be determined so that the effect of the anomaly vanishes. In the case of the four terms in (1.3), we can fortunately determine their phases by only imposing the modular invariance (up to phases (1.4, 1.5)) on $Z_{\mathrm{orb}}(\tau)$ (except for the term $\theta_1(\tau)^4$ which is in fact zero).

However, in more general orbifolds, we cannot determine these phases only from the modular invariance, and this phase problem poses particular difficulties despite being an essential step in the calculation of the orbifold partition function. We will deal with one such example in Part IV, that is, the orbifold of 576 real chiral fermions (more precisely, the 24-fold tensor product of the Conway moonshine module) by its Conway group symmetry. The partition function of this theory is of particular interest from the perspective of realization of an SCFT corresponding to the periodicity element of topological modular forms under the Stolz–Teichner conjecture.

### Motivation from the Stolz–Teichner conjecture

In moonshine phenomena, the modular functions associated with CFTs (the partition functions or elliptic genera) play a central role. In [ST11], a mathematical conjecture which suggests a stronger relation between them was proposed, and moonshine has also begun to be studied in relation to it [GJF18, JF20, Lin22, AKL22]. This conjecture is called the Stolz–Teichner conjecture.

The Stolz–Teichner conjecture says that the ring $\mathrm{TMF}_\bullet$ of the classes of "topological modular forms" is isomorphic to the ring $\mathrm{SQFT}_\bullet$ of the equivalence classes of two-dimensional $\mathcal{N} = (0,1)$ supersymmetric quantum field theories (SQFT). Its precise statement is beyond the scope of this thesis, but we add more information as follows.

[4] See Appendix E for the notation of the elliptic theta functions $\theta_i(\tau)$ and the Dedekind eta function $\eta(\tau)$.

$\mathrm{TMF}_\bullet$ is a generalized cohomology ring graded by integers $\nu \in \mathbb{Z}$. If we neglect the torsion part of $\mathrm{TMF}_\nu$, that is, if we consider $\mathrm{TMF}_\nu \otimes \mathbb{C}$, then it is isomorphic to the space $(\mathrm{MF}_\mathbb{C})^{\text{w-h}}_{\frac{\nu}{2}}$ of the weakly-holomorphic modular forms (Section 3.2) of weight $\frac{\nu}{2}$. Their addition and multiplication correspond to the direct sum and the tensor product of SQFTs. The equivalence relations of SQFTs are defined properly, and for example they include the continuous deformations.

The degree $\nu$ of the space $\mathrm{SQFT}_\nu$ specifies the gravitational anomaly of the theories belonging to it, and if the theory is a two-dimensional $\mathcal{N} = (0,1)$ SCFT of central charge $(c, \tilde{c})$, then $\nu = 2(\tilde{c} - c)$. The map $\mathrm{SQFT}_\bullet \to \mathrm{TMF}_\bullet \to (\mathrm{MF}_\mathbb{Z})^{\text{w-h}}_{\frac{\bullet}{2}}$ associates to a theory $\mathcal{T}$ its elliptic genus[5] normalized by $\eta(\tau)$ as $\eta(\tau)^\nu Z^{\mathcal{T}}_{\text{ell}}(\tau)$. Here, $(\mathrm{MF}_\mathbb{Z})^{\text{w-h}}_{\frac{\bullet}{2}}$ is the ring of weakly holomorphic modular forms with integral $q$-expansion coefficients. See for example [Tac21, TY23] for more information accessible from the physics side.

If we believe this conjecture, then we can extract nontrivial statements on the space $\mathrm{SQFT}_\bullet$ of SQFTs, by translating the properties of $\mathrm{TMF}_\bullet$. Conversely, verifying such statements serves as a test for the Stolz–Teichner conjecture.

For example, the map $\mathrm{TMF}_\bullet \to (\mathrm{MF}_\mathbb{Z})^{\text{w-h}}_{\frac{\bullet}{2}}$ is not surjective, and an element of its image generated only by $\eta(\tau)$ over $\mathbb{Z}$ is in the form of $\frac{24}{\gcd(24,k)}\eta(\tau)^{24k}$ with $k$ an integer [Hop02, Prop. 4.6]. This suggests that if an $\mathcal{N} = (0,1)$ SCFT with degree $2(\tilde{c} - c) = 24k$ has a constant elliptic genus $Z^{\mathcal{T}}_{\text{ell}}$, then $Z^{\mathcal{T}}_{\text{ell}}$ is divisible by $\frac{24}{\gcd(24,k)}$.

In fact, the Conway moonshine module is such an SCFT with minimal non-zero $k = \pm 1$ and elliptic genus 24. The Conway moonshine module or Duncan's module $V^{f\natural}$ is (the NS sector of) a chiral $\mathcal{N} = 1$ SCFT, whose automorphism group $\mathrm{Aut}_{\mathcal{N}=1}(V^{f\natural})$ preserving its $\mathcal{N} = 1$ superconformal algebra is isomorphic to the sporadic Conway group $\mathrm{Co}_1$. It was first constructed by Duncan in [Dun07], from the theory of 24 real free chiral fermions, through the procedure similar to $\mathbb{Z}_2$-orbifold. So its central charge is $c = \frac{1}{2} \times 24 = 12$. Therefore, by placing it to the right-moving part, and setting the left-moving part trivial,[6] we obtain an $\mathcal{N} = (0,1)$ SCFT belonging to $\mathrm{SQFT}_{24}$. Moreover, the elliptic genus of Duncan's module is a constant (the Witten

[5] Here, the elliptic genus is in the sense of (3.46). We only deal with CFTs in this thesis, but a similar discussion of elliptic genera can be repeated for modular-invariant (up to gravitational anomaly phases) QFTs. The modular transformations of the elliptic genus are [TYY23, §2.1]

$$Z_{\text{ell}}(\tau + 1) = e^{-\pi\sqrt{-1}\frac{\nu}{12}} Z_{\text{ell}}(\tau), \tag{1.4}$$

$$Z_{\text{ell}}(-\frac{1}{\tau}) = e^{-\pi\sqrt{-1}\frac{\nu}{4}} Z_{\text{ell}}(\tau). \tag{1.5}$$

(cf. In the theory of one real free chiral fermion with central charge $c = \frac{1}{2}$, the lowest conformal weight in the R sector is $\frac{1}{16} = \frac{c}{8}$, so $q^{L_0 - \frac{c}{24}} \mapsto e^{\pi\sqrt{-1}\frac{2c}{12}} q^{L_0 - \frac{c}{24}}$ under $\tau \mapsto \tau + 1$.) Here, $Z_{\text{ell}}(\tau) = 0$ unless $\nu$ is a multiple of 4, because we obtain $Z_{\text{ell}}(\tau) = e^{-2\pi\sqrt{-1}\frac{\nu}{4}} Z_{\text{ell}}(\tau)$ by applying the modular S transformation twice. So, either $Z_{\text{ell}}(\tau) = 0$, or the phase of the modular S transformation is just a sign. As a result, by combining the elliptic genus with the Dedekind eta function $\eta(\tau)$ (see Appendix E), we can see that $\eta^\nu(\tau) Z(\tau)$ is in fact a modular function of weight $\frac{\nu}{2}$.

[6] Similarly, if we place Duncan's module to the left-moving part, and set the right-moving part trivial, then we obtain an SCFT belonging to $\mathrm{SQFT}_{-24}$. Of course, its elliptic genus is also 24.

index; see Section 3.4) thanks to the $\mathcal{N} = 1$ supersymmetry, and its value is 24. As a result, Duncan's module $V^{f\natural}$ is a minimal nontrivial SCFT satisfying the divisibility property suggested by the Stolz–Teichner conjecture.

Another example of implications from the Stolz–Teichner conjecture is our main interest. There exists a class called the periodicity element in $\mathrm{TMF}_{-24^2=-576}$, such that multiplying any class of $\mathrm{TMF}_\bullet$ by it gives rise to a bijection $\mathrm{TMF}_\bullet \to \mathrm{TMF}_{\bullet-576}$. This suggests the existence of an $\mathcal{N} = (0,1)$ SQFT of degree $-576$, whose elliptic genus is 1.

In [AKL22], a conjectural construction of such an SCFT was proposed based on Duncan's module $V^{f\natural}$. More precisely, they proposed a chiral $\mathcal{N} = 1$ SCFT of central charge $c = \frac{576}{2} = 288$ with a unique vacuum state in the NS sector, with the expectation that its elliptic genus is 1. If we have such a theory, then by regarding it as the left-moving part coupling to the trivial right-moving part, we obtain an SCFT of degree $\nu = 2(0 - 288) = -576$, corresponding to the periodicity element. The explicit construction of such an SCFT is not yet known,[7] so if we can verify that the proposed theory indeed has the elliptic genus 1, then it provides a piece of evidence supporting the Stolz–Teichner conjecture. See [JFY24] for another study of the periodicity element in $\mathrm{SQFT}_\bullet$.

The SCFT proposed in [AKL22] is expressed as $(V^{f\natural})^{\otimes 24}/A_{24} \times \mathrm{Co}_1$. Let us see how they arrived at this theory.

Since Duncan's module $V^{f\natural}$ has central charge 12, its 24-fold tensor product $(V^{f\natural})^{\otimes 24}$ indeed has the central charge $c = 12 \times 24 = 288$. The elliptic genus of Duncan's module $V^{f\natural}$ is 24, and thus the elliptic genus of $(V^{f\natural})^{\otimes 24}$ is $24^{24}$, which is much larger than 1. Even so, since $V^{f\natural}$ have a quite large symmetry, the $\mathrm{Co}_1$ symmetry, we can try taking the orbifold by this symmetry. Orbifolding is a procedure to make a $G$-invariant theory $\mathcal{T}/G$ from a theory $\mathcal{T}$ with a finite group symmetry $G$. It first adds some states to the theory, but then takes the projection onto the $G$-invariant states, so we can expect the number of states to decrease.

[7] There is also the periodicity element in $\mathrm{TMF}_{576}$. It is known [Dev19] that for any class of $\mathrm{TMF}_\nu$ with positive degree $\nu > 0$, there exists a corresponding SQFT as an $\mathcal{N} = (0,1)$ sigma model, but the explicit construction of its target manifold is highly nontrivial.

We also note that, if we allow multiple vacua in the NS sector, then a construction of a chiral $\mathcal{N} = 1$ SCFT of central charge $c = 288$ and elliptic genus 1 is known as [AKL22, Eq. (1.5)]

$$24697376(V^{f\natural})^{\otimes 24} \oplus 1291795102224619090515486568295959(V^{f\natural})^{\otimes 24}/S_{24}. \tag{1.6}$$

Since the elliptic genera of $(V^{f\natural})^{\otimes 24}$ and $(V^{f\natural})^{\otimes 24}/S_{24}$ are constants (the Witten indices) and coprime, these coefficients are found as a solution to the Bézout equation. However, this theory has massively degenerate vacua.

We expect the existence of an SQFT with a unique vacuum corresponding to the periodicity element because of the following facts. First, according to the classification of the chiral fermionic CFTs of central charge $c \leq 24$ [BSLTZ23, Ray23, HM23], every class of $\mathrm{TMF}_\nu$ with $-48 \leq \nu \leq 0$ is realized by a CFT with a unique vacuum. Second, the above sigma model description guarantees the existence of an SQFT with a unique vacuum for each positive-degree class, so if we had an SQFT corresponding to the periodicity element of degree $-576$ with a unique vacuum, then by using $\mathrm{TMF}_\bullet \to \mathrm{TMF}_{\bullet-576}$, we can conclude the existence of an SQFT with a unique vacuum for any class of $\mathrm{TMF}_\bullet$.

However, we cannot always construct the orbifold theory. There can be an obstruction called the anomaly of the symmetry. One miraculous thing is that the anomaly of the $\mathrm{Co}_1$ symmetry of Duncan's module $V^{f\natural}$ is described as the generator of a group $SH(\mathrm{Co}_1) \cong \mathbb{Z}_{24}$ [JF17, Example 2.4.1], and therefore the anomaly of the diagonal $\mathrm{Co}_1$ symmetry of $(V^{f\natural})^{\otimes 24}$ vanishes. Hence, we can take its orbifold as $(V^{f\natural})^{\otimes 24}/\mathrm{Co}_1$.

Compared with $24^{24} \sim 1.3 \times 10^{33}$, the order $|\mathrm{Co}_1| \sim 4.2 \times 10^{18}$ of $\mathrm{Co}_1$ is still small, so it is conjectured that we have to take the orbifold by a bigger group. In [AKL22], they calculated that the orbifolds of $(V^{f\natural})^{\otimes 24}$ by the symmetry group $S_{24}$ and its alternating subgroup $A_{24}$, which act as permutation of the factors of the tensor product, have the elliptic genera $Z_{\mathrm{ell}}^{(V^{f\natural})^{\otimes 24}/S_{24}} = -25499225$ and $Z_{\mathrm{ell}}^{(V^{f\natural})^{\otimes 24}/A_{24}} = 381058359637574 \sim 3.8 \times 10^{14}$. Since these values are smaller than $|\mathrm{Co}_1|$, it is promising to take the orbifold of these theories by the $\mathrm{Co}_1$ symmetry.

In [AKL22], they further claim that the combined symmetry $S_{24} \times \mathrm{Co}_1$ is anomalous[8] (although only $S_{24}$ is non-anomalous). As a result, they finally conjecture that the desired SCFT with elliptic genus 1 is $(V^{f\natural})^{\otimes 24}/A_{24} \times \mathrm{Co}_1$.

Lastly, we remark that the way of constructing an orbifold $\mathcal{T}/G$ from a given theory $\mathcal{T}$ and its symmetry $G$ is not unique, and there exists a degree of freedom called the discrete torsion. However, once we construct one of the orbifolds in a consistent way, then all the other orbifolds can be constructed from it, in principle (see Section 12.1.3). Therefore, the first step is to calculate the elliptic genus of $(V^{f\natural})^{\otimes 24}/A_{24} \times \mathrm{Co}_1$ in any way. This is one of our ultimate goals, and the main purpose of Part IV of this thesis is to present partial results we have obtained so far.

**What we will actually do**

Before getting into the elliptic genus of $(V^{f\natural})^{\otimes 24}/A_{24} \times \mathrm{Co}_1$, we have to understand how to calculate the elliptic genus of $(V^{f\natural})^{\otimes 24}/\mathrm{Co}_1$, which will also be referred to as the Conway orbifold partition function of Duncan's module. There are three steps as follows. See Section 12.2.4 for their details.

Step 1. Determine the conjugacy classes of commuting pairs of $\mathrm{Co}_1$, and their $\mathrm{SL}(2,\mathbb{Z})$- and $\mathrm{GL}(2,\mathbb{Z})$-orbits.

Step 2. Compute the twisted partition functions of $V^{f\natural}$, twisted both in spatial and temporal directions by $\mathrm{Co}_1$.

Step 3. Determine the phases to trivialize the anomalous phases for all $\mathrm{GL}(2,\mathbb{Z})$-orbits.

[8] Let us review their discussion. In the theory of $n$ real free chiral fermions, the group $\mathrm{SO}(n)$ acts on the NS sector as a genuine representation, but it acts on the R sector only as a projective representation; what really acts on the R sector as a genuine representation is its double cover $\mathrm{Spin}(n) \cong \mathbb{Z}_2.\mathrm{SO}(n)$. This appearance of the projective phase of the $\mathrm{SO}(n)$-action on the R sector is regarded as a part of the fermionic anomaly. Similarly, $\mathrm{Co}_1$ genuinely acts on the NS sector $V^{f\natural}$, but it acts on the R sector $V_{\mathrm{tw}}^{f\natural}$ only projectively, and what genuinely acts on $V_{\mathrm{tw}}^{f\natural}$ is $\mathrm{Co}_0 \cong \mathbb{Z}_2.\mathrm{Co}_1$. If this anomaly is vanishing, then $\mathrm{Co}_1$ also acts on $V_{\mathrm{tw}}^{f\natural}$ genuinely, and the action of $\mathrm{Co}_0$ becomes unfaithful because the action of $\mathbb{Z}_2 \subset \mathrm{Co}_0$ becomes trivial. According to [AKL22], the action of $\mathrm{Co}_0$ on the R sector of the orbifold $(V^{f\natural})^{\otimes 24}/S_{24}$ is faithful, so $\mathrm{Co}_1$ can act on this R sector only projectively, which means this $S_{24} \times \mathrm{Co}_1$ symmetry is anomalous. On the other hand, the action of $\mathrm{Co}_0$ on the R sector of $(V^{f\natural})^{\otimes 24}/A_{24}$ is unfaithful.

We have succeeded in Steps 1 and 2 (Theorem 14.5), and the obtained values of the twisted partition functions are recorded in Table G.3. As for Step 3, there seems no well-established method of achieving it, to the author's knowledge. Therefore, we attempt to determine the phases from certain consistency conditions. We have found one solution (Conjecture 14.6), but its validity as the phases to trivialize the anomalous phases is still an open question.

Suppose we succeeded in calculating the elliptic genus of $(V^{f\natural})^{\otimes 24}/\mathrm{Co}_1$. A machinery for computing the elliptic genus of $(V^{f\natural})^{\otimes 24}/A_{24} \times \mathrm{Co}_1$ is almost already developed in [AKL22], but the phase problem similar to Step 3 also exists in the computation of the elliptic genus of $(V^{f\natural})^{\otimes 24}/A_{24} \times \mathrm{Co}_1$. So our future study might begin with directly determining the phases for $(V^{f\natural})^{\otimes 24}/A_{24} \times \mathrm{Co}_1$.

## 1.4 Reader's Guide

**Relation between the sections**

Part I is a review of basic concepts, and its primary role is to fix the definitions and notations, and collect data for later analysis. Readers already familiar with the topics in Part I can skip it and refer back when necessary.

Part II and Appendix A are also reviews, but some sections introduce the structures of VOAs and groups heavily used in later sections. To access the original study in Part III, readers are advised to read Section 5.2 first. To access the original study in Part IV, readers are advised to read Section 6 first. Optionally, the other parts can be skipped.

Appendices B, C, and D contain the detailed analyses omitted in Part III. Appendices E and F summarize some facts mainly used in Part IV. The last Appendix G presents the data obtained as the main result of Part IV.

**Organization of this thesis**

The organization of the rest of this thesis is as follows.

Part I is a review of basic concepts. In Section 2, we review three important finite simple groups, along with definitions and examples of codes and lattices. In Section 3, we summarize definitions, examples, and fundamental facts about modular functions and weak Jacobi forms. We also introduce the concept of elliptic genus in CFT. In Section 4, we briefly review definitions of concepts related to VOAs.

Part II is a review of moonshine phenomena, but its primary focus is to introduce important examples of VOAs for the main study of this thesis. In Section 5, after a review of the monstrous moonshine, we introduce lattice VOAs. In Section 6, we introduce the Conway moonshine module, also known as Duncan's module. In Section 7, we briefly review the K3 Mathieu moonshine.

In Part III, we develop a technique to answer our Question raised in Section 1.2, that is, whether the lattice isometry group is a subgroup of the automorphism group of the lattice VOA or not. We also discuss the moonshine-type observation of the odd Leech lattice. In Section 8, we look into the details of the group extension structure of the automorphism group of a lattice VOA. We then paraphrase Question into more concrete forms Question 2 and 3, and slightly generalize

it to Question 3′. In Section 9, we negatively settle Question 3′ in the cases of the odd Leech lattice and the Leech lattice. In Section 10, we investigate weight-1 and weight-$\frac{3}{2}$ currents of the odd Leech lattice CFT invariant under the subgroups of its automorphism group. In Section 11, we revisit the moonshine-type observation by [BDFK15] in light of our results.

In Part IV, we describe how to compute the Conway orbifold partition function of Duncan's module. In Section 12, we review the general theory of orbifolds, including the 't Hooft anomaly and concrete steps to compute the orbifold partition function. In Section 13, we compute the twisted partition functions of the theory of free chiral fermions, which are the ingredients of the twisted partition functions of Duncan's module. In Section 14, we aim to carry out the computation of the Conway orbifold partition function. In particular, we calculate the twisted partition functions of Duncan's module, for each $\mathrm{SL}(2,\mathbb{Z})$-orbit of the conjugacy classes of commuting pairs of the Conway group $\mathrm{Co}_1$. We also present one conjectural set of the phases to trivialize the anomaly.

Before getting into appendices, we present a concluding summary in Part V.

The last Part VI is appendices. Appendix A contains an elementary introduction to group extensions. Appendix B explains how we can find supercurrents of the $\mathcal{N} = 2$ SCA in the odd Leech lattice CFT. Appendices C and D provide some details we omitted in Part III. Appendix E summarizes the definitions and modular properties of several theta functions and the Dedekind eta function. Appendix F explains the details of the lift of elements of $\mathrm{SO}(n)$ to $\mathrm{Spin}(n)$, and its implementation on computer for the computation of the twisted partition functions. Appendix G records the data of the twisted partition functions of Duncan's module, which are the main results of Part IV.

**Code repository**

In this thesis, we employ multiple programming languages to find solutions or compute values, rather than to perform numerical simulations. They include Python, C++, Mathematica, and GAP. GAP [GAP22] is a language for computational discrete algebra, in particular for computational group theory, developed by mathematicians in the field of algebra. We will explicitly show the GAP script in this thesis for readers unfamiliar with the language. Our code written in the other languages is available at the following GitHub repositories.

The Python code used in the analysis[9] of Part III is uploaded to

https://github.com/MasakiOkada0101/latticeVOAsymmetry

in the form of Jupyter Notebook (.ipynb file). These files are used in the following sections:

- Section 9.2: `M24_of_odd_Leech.ipynb, M23_of_odd_Leech.ipynb`.
- Section 9.3: `Co0_of_Leech.ipynb`.
- Appendix B: `check_supercurrent_OPE.ipynb, find_supercurrent.ipynb`.

[9]The notation of equations in [Oka24], on which Part III of this thesis is based, uses 0-based indexing, because the calculation was mainly done by Python. However, throughout [Oka24] and [AKL$^+$], we also used C++ (0-based), Mathematica (1-based), and GAP (1-based). Considering that 1-based indexing seems more common in math and physics literature, we unified the notation to 1-based indexing in this thesis, with only minor exceptions.

The C++ and Mathematica code used in the analysis of Part IV is uploaded to

https://github.com/MasakiOkada0101/ConwayTwPF.

Together with the data generated as described in Section 14.1, we uploaded the source code to implement the algorithms explained in Section 14.2.2.

**Notes**

This thesis is based on the following papers.


[Oka24] M. Okada, *On the symmetry of odd Leech lattice CFT*, JHEP **06** (2025) 208, arXiv:2412.19430 [hep-th].

[AKL+] J. Albert, J. Kaidi, Y.-H. Li, M. Okada, and Y. Tachikawa, *Conway Orbifold of Supermoonshine*, to be submitted .


The author's research during his master's and Ph.D. studies also include the following published works.


[LOST22] Y.-H. Lin, M. Okada, S. Seifnashri, and Y. Tachikawa, *Asymptotic density of states in 2d CFTs with non-invertible symmetries*, JHEP **03** (2023) 094, arXiv:2208.05495 [hep-th].

[OW23] M. Okada and T. Watari, *A note on varieties of weak CM-type*, J. Geom. Phys. **197** (2024) 105084, arXiv:2306.10282 [math.AG].

[OT24] M. Okada and Y. Tachikawa, *Noninvertible Symmetries Act Locally by Quantum Operations*, Phys. Rev. Lett. **133** (2024) 191602, arXiv:2403.20062 [hep-th].

[KOW24] A. Kidambi, M. Okada, and T. Watari, *Notes on Characterizations of 2d Rational SCFTs: Algebraicity, Mirror Symmetry, and Complex Multiplication*, Fortsch. Phys. **73** (2025) 2400161, arXiv:2408.00861 [hep-th].

[OSTZ25] M. Okada, S. Shimamura, Y. Tachikawa, and Y. Zhang, *On anomalies and fermionic unitary operators*, JHEP **11** (2025) 122, arXiv:2509.02989 [hep-th].

# Part I
# Review of Basic Concepts

This Part I is a review of basic concepts related to moonshine phenomena. A moonshine phenomenon is, simply put, an observed relationship between a finite group, in particular a simple one, and a modular function. So we review some general facts about finite simple groups in Section 2, and modular functions in Section 3. The most classical example of moonshine phenomena, the monstrous moonshine, was theoretically explained by an underlying object called a vertex operator algebra (VOA). VOAs are also expected to be key ingredients in understanding other moonshine phenomena, and provide connections with two-dimensional conformal field theories (CFT) in physics. We will review the axiomatic definition of a VOA in Section 4 to get the idea of how VOAs mathematically formulate CFTs in physics, but the definition itself is somewhat technical. In practice, in this thesis, it suffices to consider the examples of VOAs reviewed in the next Part II, whose constructions would be more familiar to physicists working with CFTs.

Readers already familiar with these topics can skip this Part I, and come back just to refer to definitions, notations, and data collected here for later analysis if necessary.

## 2 Finite Simple Groups

In this Section 2, we review basic facts about finite simple groups, mainly focusing on important examples of sporadic finite simple groups. After mentioning the classification of finite simple groups in Section 2.1, we will describe the Mathieu groups, the Conway groups, and the monster group in Sections 2.2, 2.3, and 2.4, respectively. Along the way, we also review fundamental concepts such as codes and lattices, partly in order to fix definitions and notations; we mainly follow the definitions in [CS99].

Before proceeding, let us clarify the convention of permutation groups and its actions, and some notations regarding groups and lattice vectors here.

**Convention of permutation groups and its actions**

The symmetric group $S_n$ consists of permutations $\sigma : \Omega_n \to \Omega_n$ where $\Omega_n = \{1, \dots, n\}$, and there are two conventions for the definition of the multiplication:

$$(1,2) \cdot (1,3) = (1,3,2);\ 1 \mapsto 3, 3 \mapsto 2, 2 \mapsto 1, \tag{2.1}$$

$$(1,2) \mathbin{\tilde{\cdot}} (1,3) = (1,2,3);\ 1 \mapsto 2, 2 \mapsto 3, 3 \mapsto 1. \tag{2.2}$$

They are related as $\tau \cdot \sigma = \sigma \mathbin{\tilde{\cdot}} \tau$.

In this thesis, we will adopt the multiplication $\cdot$ (2.1) and the left action of $\sigma \in S_n$ on a vector $k = (k_1, \dots, k_n) \in \mathbb{R}^n$ as

$$\sigma(k) = (k_{\sigma^{-1}(1)}, \dots, k_{\sigma^{-1}(n)}). \tag{2.3}$$

This can also be written as $(\sigma(k))_{\sigma(i)} = k_i$.

In GAP [GAP22] and the webpage of ATLAS of Finite Group Representations [WWT+], on the other hand, they adopt the multiplication $\tilde{\cdot}$ (2.2). So when we cite equations from them, we will convert them into representations in terms of the multiplication $\cdot$ (2.1) in this thesis.

**Notations of groups**

For groups $G$ and $N$, following ATLAS of Finite Groups [CCN+85], we write $N.G$ for any extension of $G$ by $N$.

$$1 \to N \to N.G \to G \to 1. \tag{2.4}$$

If it is a split extension, or equivalently a semidirect product, then we write $N : G$ or $N \rtimes G$. Furthermore, if it is a direct product, then we write $N \times G$. We sometimes use the notation $N \dot{} G$ for a non-split extension. For readers not familiar with these topics, Appendix A contains an elementary introduction to group extensions.

For a group $G$, we let $H^n(G; A)$ denote the $n$-th group cohomology[10] of $G$ with coefficients in $A$. Here, $A$ is a $G$-module, although in most cases in this thesis, the $G$-action on $A$ is trivial.

$R^\times$ denotes the multiplicative group of a ring $R$.

$\mathbb{K}[G]$ denotes the group algebra of $G$ over $\mathbb{K}$.

The cyclic group $\mathbb{Z}_n$ is sometimes denoted by just $n$.

The *general linear group* $\mathrm{GL}(n, \mathbb{K})$ is the group of all the invertible $n \times n$ matrices over the field $\mathbb{K}$. The *special linear group* $\mathrm{SL}(n, \mathbb{K})$ is the subgroup of $\mathrm{GL}(n, \mathbb{K})$ of all the matrices with $\det = 1$. The *projective general linear group* $\mathrm{PGL}(n, \mathbb{K})$ is the quotient group of $\mathrm{GL}(n, \mathbb{K})$ by its center $\mathbb{K}^\times \cdot 1$, which consists of the non-zero scalar matrices. The *projective special linear group* $\mathrm{PSL}(n, \mathbb{K})$ is the quotient group of $\mathrm{SL}(n, \mathbb{K})$ by its center, which consists of the non-zero scalar matrices with $\det = 1$. In some literature, $\mathrm{PGL}(n, \mathbb{Z}_q)$ is also denoted by $\mathrm{PGL}(n, q)$, and $\mathrm{PSL}(n, \mathbb{Z}_q)$ is by $\mathrm{PSL}(n, q)$ or $L_n(q)$.

$1_G$ or simply $1$ denotes the identity element of $G$. $I$ denotes the identity matrix. $\mathrm{id}_X$ denotes the identity map $X \to X$.

**Notations of lattice vectors**

For a lattice vector $k \in L \subset \mathbb{R}^n$ of a lattice $L$, the components of $k$ with respect to the standard basis of $\mathbb{R}^n$ are denoted by subscript as $k = (k_1, \ldots, k_n)$, and the components of $k$ with respect to another basis $e_1, \ldots, e_n$ of $\mathbb{R}^n$ (for example a $\mathbb{Z}$-basis of $L$, but not limited to such a case) are denoted by superscript as $k = \sum_i k^i e_i$.

We let $\mathbb{e}_1, \ldots, \mathbb{e}_n$ denote an orthonormal basis of $\mathbb{R}^n$ where the lattice $L$ is embedded, with respect to the symmetric bilinear form of $L$ (extended by $\mathbb{R}$-linearity).

When the basis is obvious from the context, $\vec{1}$ denotes $(1, \ldots, 1)$.

[10] There is another concept, the cohomology $H^n_{\mathrm{top}}(X; A)$ of a topological space $X$. In this notation, the group cohomology $H^n(G; A)$ in the main text is the cohomology $H^n_{\mathrm{top}}(BG; A)$ of the classifying space $BG$ of $G$.

## 2.1 Classification of Finite Simple Groups

A group $G$ is said to be *simple* if its normal subgroups are the trivial ones $\{1\}$ and $G$ only. A *composition series* of a group $G$ is a finite series $\{1\} = H_0 \subset H_1 \subset \cdots \subset H_n = G$, where $H_{i-1}$ is a proper normal subgroup of $H_i$, and $H_i/H_{i-1}$ is simple, for all $i = 1, \ldots, n$. The groups $H_i/H_{i-1}$ are called the *composition factors*, and $n$ is called the *length* of the composition series. If a group has a composition series, it is unique in the following sense.

**Theorem 2.1** (the Jordan–Hölder theorem)**.** *If a group has a composition series, then any two composition series of it have the same length, and the composition factors of them are the same up to permutation and isomorphism.*

It is known that any finite group has a composition series. In this sense, finite simple groups are fundamental building blocks of finite groups.

The classification of finite simple groups was achieved through the tremendous efforts of many mathematicians.

**Theorem 2.2** (classification of finite simple groups)**.** *Every finite simple group is isomorphic to one of the following groups:*

- *cyclic groups $\mathbb{Z}_p$ of prime order,*
- *alternating groups $A_n$ of degree $n \geq 5$,*
- *finite simple groups of Lie type,*
- *26 sporadic groups, listed in Table 2.1.*

Among the 26 sporadic groups, the largest one is the monster group $\mathbb{M}$, and 20 sporadic groups (including $\mathbb{M}$) are involved in $\mathbb{M}$ as subgroups or quotients of subgroups. They are called the *happy family*, and the other 6 sporadic groups are called *pariahs* in [Gri82]. The happy family is further classified into three generations [Gri82]: the first generation is involved in the Mathieu group $M_{24}$, the second generation the Conway group $\mathrm{Co}_1$, and the third generation the monster group $\mathbb{M}$.

Table 2.1: The list of sporadic groups. For each sporadic group $G$, we also show its name, order $|G|$, number $c(G)$ of conjugacy classes, Schur multiplier $H_2(G;\mathbb{Z})$, which is isomorphic to $H^2(G;\mathrm{U}(1))$ if $G$ is finite [Gri98, (2.12.2)], and generation as the happy family, where "P" denotes a pariah. The notation of sporadic groups and data are taken from [CCN+85].[11]

| $G$ | name | $\lvert G\rvert$ | | $c(G)$ | $H^2(G;\mathrm{U}(1))$ | generation |
|---|---|---|---|---|---|---|
| $M_{11}$ | Mathieu | $2^4 3^2 5.11$ | $= 7920$ | 10 | 1 | 1 |
| $M_{12}$ | Mathieu | $2^6 3^3 5.11$ | $= 95040$ | 15 | $\mathbb{Z}_2$ | 1 |
| $J_1$ | Janko | $2^3 3.5.7.11.19$ | $= 175560$ | 15 | 1 | P |
| $M_{22}$ | Mathieu | $2^7 3^2 5.7.11$ | $= 443520$ | 12 | $\mathbb{Z}_{12}$ | 1 |
| $HJ$, $J_2$ | Hall–Janko | $2^7 3^3 5^2 7$ | $= 604800$ | 21 | $\mathbb{Z}_2$ | 2 |
| $M_{23}$ | Mathieu | $2^7 3^2 5.7.11.23$ | $\sim 1.02 \times 10^7$ | 17 | 1 | 1 |
| $HS$ | Higman–Sims | $2^9 3^2 5^3 7.11$ | $\sim 4.44 \times 10^7$ | 24 | $\mathbb{Z}_2$ | 2 |
| $J_3$ | Janko | $2^7 3^5 5.17.19$ | $\sim 5.02 \times 10^7$ | 21 | $\mathbb{Z}_3$ | P |
| $M_{24}$ | Mathieu | $2^{10} 3^3 5.7.11.23$ | $\sim 2.45 \times 10^8$ | 26 | 1 | 1 |
| $McL$ | McLaughlin | $2^7 3^6 5^3 7.11$ | $\sim 8.98 \times 10^8$ | 24 | $\mathbb{Z}_3$ | 2 |
| $He$ | Held | $2^{10} 3^3 5^2 7^3 17$ | $\sim 4.03 \times 10^9$ | 33 | 1 | 3 |
| $Ru$ | Rudvalis | $2^{14} 3^3 5^3 7.13.29$ | $\sim 1.46 \times 10^{11}$ | 36 | $\mathbb{Z}_2$ | P |
| $Suz$ | Suzuki | $2^{13} 3^7 5^2 7.11.13$ | $\sim 4.48 \times 10^{11}$ | 43 | $\mathbb{Z}_6$ | 2 |
| $O'N$ | O'Nan | $2^9 3^4 5.7^3 11.19.31$ | $\sim 4.61 \times 10^{11}$ | 30 | $\mathbb{Z}_3$ | P |
| $\mathrm{Co}_3$ | Conway | $2^{10} 3^7 5^3 7.11.23$ | $\sim 4.96 \times 10^{11}$ | 42 | 1 | 2 |
| $\mathrm{Co}_2$ | Conway | $2^{18} 3^6 5^3 7.11.23$ | $\sim 4.23 \times 10^{13}$ | 60 | 1 | 2 |
| $Fi_{22}$ | Fischer | $2^{17} 3^9 5^2 7.11.13$ | $\sim 6.46 \times 10^{13}$ | 65 | $\mathbb{Z}_6$ | 3 |
| $HN$ | Harada–Norton | $2^{14} 3^6 5^6 7.11.19$ | $\sim 2.73 \times 10^{14}$ | 54 | 1 | 3 |
| $Ly$ | Lyons | $2^8 3^7 5^6 7.11.31.37.67$ | $\sim 5.18 \times 10^{16}$ | 53 | 1 | P |
| $Th$ | Thompson | $2^{15} 3^{10} 5^3 7^2 13.19.31$ | $\sim 9.07 \times 10^{16}$ | 48 | 1 | 3 |
| $Fi_{23}$ | Fischer | $2^{18} 3^{13} 5^2 7.11.13.17.23$ | $\sim 4.09 \times 10^{18}$ | 98 | 1 | 3 |
| $\mathrm{Co}_1$ | Conway | $2^{21} 3^9 5^4 7^2 11.13.23$ | $\sim 4.16 \times 10^{18}$ | 101 | $\mathbb{Z}_2$ | 2 |
| $J_4$ | Janko | $2^{21} 3^3 5.7.11^3 23.29.31.37.43$ | $\sim 8.68 \times 10^{19}$ | 62 | 1 | P |
| $Fi'_{24}$ | Fischer | $2^{21} 3^{16} 5^2 7^3 11.13.17.23.29$ | $\sim 1.26 \times 10^{24}$ | 108 | $\mathbb{Z}_3$ | 3 |
| $B$ | baby monster (Fisher) | $2^{41} 3^{13} 5^6 7^2 11.13.17.19.23.31.47$ | $\sim 4.15 \times 10^{33}$ | 184 | $\mathbb{Z}_2$ | 3 |
| $\mathbb{M}$ | monster (Fisher–Griess) | $2^{46} 3^{20} 5^9 7^6 11^2 13^3 17.19.23.29.31.41.47.59.71$ | $\sim 8.08 \times 10^{53}$ | 194 | 1 | 3 |

[11] There are also lists of sporadic groups in [Gri98,Har99,Yos24]. Note that some orders $|G|$ on the lists of [Gri98,Yos24] are different from those on [CCN+85], and seem to be typos.

Historically, the sporadic groups first constructed were the Mathieu groups $M_{11}, M_{12}$ in 1861 [Mat61], and $M_{22}, M_{23}, M_{24}$ in 1873 [Mat73]. The problem of classifying all the finite simple groups was questioned in an article [Hö92] by Hölder in 1892. Chevalley and others constructed simple groups called the groups of Lie type around 1955 [Che55], but the advent of a new sporadic group had to await the discovery of the Janko group $J_1$, announced in a one-page paper [Jan65] in 1965, and further clarified in [Jan66]. This was the kick-off of the successive discoveries of all the sporadic groups, which coincidentally ended again with the Janko group $J_4$ in 1976 [Jan76]. Then the finite group theory was oriented toward the completion of the classification theorem. It was declared that the proof was completed in 1981 [Gor82], but it was established as an accumulation of numerous results scattered throughout the literature, so Gorenstein, Lyons, and Solomon launched a project (the GLS project) to simplify the proof and to present it as a streamlined series of books. However, a serious gap was found in the classification of groups called quasithin groups. It was in 2004 that the gap was finally closed by Aschbacher and Smith in their two-volume books [AS04b, AS04a], as many as 1221 pages in total. Nowadays, the classification of finite simple groups is regarded as an established theorem. The books of the GLS project have reached volume 10 [CGLS23] in 2023, and the project is still ongoing.

As for the history of classification of finite simple groups, see [Asc94, §15], [Sol01, Gur18, Sol18a], [Asc04], and in particular for the GLS project, see [Sol18b]. References on sporadic groups include [Asc94], [Gri98], and [CS99]. ATLAS of Finite Groups [CCN+85] is a standard database of finite simple groups, and there is also the webpage ATLAS of Finite Group Representations [WWT+]. Regarding the literature on sporadic groups written in Japanese, the unpublished lecture notes[12] [Kon96] by Kondo and several expository articles were the only available references for a long time, although there was a book [Har99] by Harada focusing on the monster group and the monstrous moonshine. Fortunately, the situation has drastically improved with the publication of Yoshiara's book [Yos24] in 2024.

In the following sections, we will review particularly important sporadic groups, the Mathieu groups, the Conway groups, and the monster group. They can be described as the automorphism groups (or their subgroups or quotients) of some algebraic objects, the binary Golay code, the Leech lattice, and the Griess algebra or the monster VOA, respectively.

## 2.2 The Mathieu Groups and the Golay Codes

There are several ways to define or construct the Mathieu groups, and one of the most approachable ways is to define the largest Mathieu group $M_{24}$ as the automorphism group of the binary Golay code $G_{24}$. So we first review generalities of linear codes and the binary Golay code in Section 2.2.1, and then introduce the Mathieu groups in Section 2.2.2. As for more details of the Mathieu groups and their different constructions, see for example [Gri98, CS99, Iva18, Yos24, Cur25].

[12] The author thanks Masahiko Miyamoto for sharing this notes.

### 2.2.1 The Golay Codes

**Basic definitions for codes**

A *$q$-ary linear code* $\mathcal{C}$ of *dimension* $m$ and *length* $n$ is an $m$-dimensional subspace of the $n$-dimensional $\mathbb{F}_q$-linear space $(\mathbb{F}_q)^n$, where $q$ is a prime or a prime power and $\mathbb{F}_q$ is the finite field of order $q$. In this thesis, codes always refer to linear ones. An element of a code is called a *codeword*, and the *(Hamming) weight* of a codeword $w = (w_1, \ldots, w_n)$ is $\mathrm{wt}(w) := |\{i \mid w_i \neq 0\}|$. When the minimal nonzero weight $\min_{w \in \mathcal{C} \setminus \{0\}} \mathrm{wt}(w)$ is $d$, this linear code is denoted by $[n, m, d]_q$.

Two linear codes are said to be *equivalent* when one is mapped to the other by a *monomial* matrix, which is a matrix containing exactly one nonzero element of $\mathbb{F}_q$ in each row and column. An equivalent map from a code to itself is called an *automorphism*, and the set of all the automorphisms of $\mathcal{C}$ forms the *automorphism group* $\mathrm{Aut}(\mathcal{C})$ of $\mathcal{C}$. Be careful that some literature calls the quotient of $\mathrm{Aut}(\mathcal{C})$ by its center $\mathbb{F}_q^\times \cdot 1$, which consists of the non-zero scalar matrices, the automorphism group of $\mathcal{C}$ (see e.g. [AM66]).

When $q = p^a$ with $p$ prime, the *dual code* $\mathcal{C}^*$ of $\mathcal{C}$ is defined by

$$\mathcal{C}^* := \{v \in (\mathbb{F}_q)^n \mid v \cdot \bar{w} = 0 \text{ for any } w \in \mathcal{C}\}, \tag{2.5}$$

where $\bar{w} := ((w_1)^p, \ldots, (w_n)^p)$ is the *conjugate* of $w$, and $v \cdot w := \sum_i v_i w_i$. A code $\mathcal{C}$ is said to be *self-dual* if $\mathcal{C}^* = \mathcal{C}$. Since $\dim \mathcal{C}^* = n - m$, the length of a self-dual code must be even.

A binary code is said to be *even* if the weight of any codeword is even, and *doubly-even* if a multiple of 4. An even but not doubly-even code is said to be *singly-even*. A singly-even self-dual code is sometimes called *Type I*, and a doubly-even self-dual code *Type II*. A Type II code exists if and only if the length is a multiple of 8 [CS99, Ch. 7 §6 Cor. 18].

**The binary and ternary Golay codes**

The *(extended) binary Golay code* $G_{24}$ is the unique $[24, 12, 8]_2$ linear code up to equivalence. This code is doubly-even and self-dual. A basis of $G_{24}$ can be read off from the proof of [CS99, Ch. 10 §2.1 Thm. 7] as

$$\begin{array}{l}
(1,1,1,1,1,0,0,1,0,0,1,0,1,0,0,0,0,0,0,0,0,0,0,0),\\
(0,1,1,1,1,1,0,0,1,0,0,1,0,1,0,0,0,0,0,0,0,0,0,0),\\
(0,0,1,1,1,1,1,0,0,1,0,0,1,0,1,0,0,0,0,0,0,0,0,0),\\
(0,0,0,1,1,1,1,1,0,0,1,0,0,1,0,1,0,0,0,0,0,0,0,0),\\
(0,0,0,0,1,1,1,1,1,0,0,1,0,0,1,0,1,0,0,0,0,0,0,0),\\
(0,0,0,0,0,1,1,1,1,1,0,0,1,0,0,1,0,1,0,0,0,0,0,0),\\
(0,0,0,0,0,0,1,1,1,1,1,0,0,1,0,0,1,0,1,0,0,0,0,0),\\
(0,0,0,0,0,0,0,1,1,1,1,1,0,0,1,0,0,1,0,1,0,0,0,0),\\
(0,0,0,0,0,0,0,0,1,1,1,1,1,0,0,1,0,0,1,0,1,0,0,0),\\
(0,0,0,0,0,0,0,0,0,1,1,1,1,1,0,0,1,0,0,1,0,1,0,0),\\
(0,0,0,0,0,0,0,0,0,0,1,1,1,1,1,0,0,1,0,0,1,0,1,0),\\
(1,1,1,1,1,1,1,1,1,1,1,1,1,1,1,1,1,1,1,1,1,1,1,1).
\end{array} \tag{2.6}$$

The 24 columns are labeled by $0, 1, \ldots, 22, \infty$ in [CS99], which we will write $1, 2, \ldots, 24$ below.

The binary Golay code has 1, 759, 2576, 759, and 1 codewords with weight 0, 8, 12, 16, and 24, respectively. Codewords with weight 8 and 12 are called an *octad* and a *dodecad*, respectively.

The code obtained by dropping one coordinate of $G_{24}$ is called the *(perfect) binary Golay code* $G_{23}$. We omit "extended" or "perfect" when it is obvious from the context. The perfect Golay code $G_{23}$ is the unique $[23, 12, 7]_2$ linear code up to equivalence. The extended Golay code $G_{24}$ can be recovered from $G_{23}$ by adding a parity bit (a bit which makes the weight of the codeword even).

The *perfect ternary Golay code* $G_{11}$ and the *extended ternary Golay code* $G_{12}$ are respectively the unique $[11, 6, 5]_3$ and $[12, 6, 6]_3$ linear codes up to equivalence. $G_{11}$ can be obtained by dropping one coordinate of $G_{12}$, and $G_{12}$ can be recovered from $G_{11}$ by adding a zero-sum check digit [CS99, (2.8.5)]. $G_{12}$ is self-dual.

The proof of uniqueness of these four Golay codes $G_{24}$, $G_{23}$, $G_{12}$, and $G_{11}$ are comprehensively explained in [MS77, Ch. 20], based on the original papers [Ple68, DG75]. [vL99, §4.2] and [Ple98, §10] also contain some proofs.

### 2.2.2 The Mathieu Groups

**The largest Mathieu group $M_{24}$**

The automorphism group $\mathrm{Aut}(G_{24})$ of a binary Golay code is the largest *Mathieu group* $M_{24}$. It is a sporadic simple group, and 5-transitive as a subgroup of $S_{24}$ acting on 24 points. Here,

**Definition 2.3** ($k$-transitive)**.** An action of a group $G$ on a set $X$ is said to be *$k$-transitive* (*$k$-fold transitive*) if for any $k$ elements $\{i_1, \ldots, i_k\} \subset X$ and $k$ elements $\{j_1, \ldots, j_k\} \subset X$, there is $g \in G$ such that $g \cdot i_t = j_t$. In addition, if such $g$ is unique for any $\{i_1, \ldots, i_k\} \subset X$ and $\{j_1, \ldots, j_k\} \subset X$, then the action is said to be *sharply $k$-transitive*. A 1-transitive action is just called a *transitive* action.

A group $G$ is said to be *$k$-transitive*, *sharply $k$-transitive*, if $G$ is isomorphic to a subgroup of $S_n$ such that its action on $\Omega_n = \{1, \ldots, n\}$ is $k$-transitive, sharply $k$-transitive, respectively. ■

Obviously, a $k$-transitive group is also $k'$-transitive for $k' \leq k$.

In the above basis (2.6), $M_{24}$ is generated by the following four permutations [CS99, Ch. 10 §2.1]:

$$\begin{aligned}
&(1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23),\\
&(16,8,15,6,11,21,18,12,23,22,20)(4,7,13,2,3,5,9,17,10,19,14),\\
&(24,1)(16,4)(8,14)(15,19)(6,10)(11,17)(21,9)(18,5)(12,3)(23,2)(22,13)(20,7),\\
&(15,18,12,20,23)(21,11,8,6,22)(19,5,3,7,2)(9,17,14,10,13).
\end{aligned} \tag{2.7}$$

A presentation of $M_{24}$ can be found in [WWT+] as

$$M_{24} = \langle a, b \mid a^2 = b^3 = (ba)^{23} = [b^{-1}, a]^{12} = [(bab)^{-1}, a]^5 = (b^{-1}ab^{-1}aba)^3(b^{-1}ababa)^3 = ((b^{-1}aba)^3ba)^4 = 1\rangle, \tag{2.8}$$

where $[y^{-1}, x^{-1}] = y^{-1}x^{-1}yx$ and note that the multiplication (2.2) in [WWT+] is converted to (2.1) here. In $M_{24}$ generated by (2.7), these generators $a$ and $b$ can be taken as[13]

$$a = (1,8)(2,7)(3,19)(4,23)(5,20)(6,15)(9,16)(10,11)(12,22)(13,18)(14,24)(17,21), \tag{2.9}$$

$$b = (2,9,12)(3,13,24)(4,17,21)(5,16,15)(8,10,23)(11,14,22). \tag{2.10}$$

**The second largest Mathieu group $M_{23}$**

The *Mathieu Group* $M_{23}$ is the stabilizer group of one point of the action of $M_{24}$ on 24 points. Since $M_{24}$ is transitive, $M_{23}$ is unique up to isomorphism, regardless of the choice of the stabilized point. Equivalently, $M_{23}$ is the automorphism group $\mathrm{Aut}(G_{23})$ of the perfect binary Golay code $G_{23}$. It is also a sporadic simple group.

A presentation of $M_{23}$ can be found in [WWT+] as

$$\begin{aligned} M_{23} = \langle a, b \mid\ & a^2 = b^4 = (ba)^{23} = (b^2a)^6 = [b^{-1}, a]^6 = (b^2ab^{-1}aba)^4 = 1, \\ & (b^{-1}a)^3(ba)^3(b^{-1}aba)^2b^2ab^{-1}a(ba)^3 = b^2ab^{-1}abab^2aba(b^{-1}ab^2a)^2(b^2aba)^3 = 1\rangle. \end{aligned} \tag{2.11}$$

If we take $M_{23}$ as the subgroup of $M_{24}$ generated by (2.7) stabilizing the point 23, these generators $a$ and $b$ can be taken as

$$a = (1,15)(2,20)(3,12)(4,19)(5,17)(7,13)(8,18)(11,14), \tag{2.12}$$

$$b = (1,12,17,21)(2,6,3,8)(4,9,16,20)(5,23)(10,15,22,13)(11,19). \tag{2.13}$$

**More on Mathieu groups**

There are five sporadic Mathieu groups: $M_{24}$, $M_{23}$, $M_{22}$, $M_{12}$, and $M_{11}$.

We already introduced $M_{24}$ and $M_{23}$. $M_{22}$ is the stabilizer group of two points (not as a set; each point must be stabilized) of the action of $M_{24}$ on 24 points. Since $M_{24}$ is 2-transitive, $M_{22}$ is unique up to isomorphism, regardless of the choice of the stabilized points. In a similar manner, since $M_{24}$ is at most 5-transitive, we can also define $M_{21}$, $M_{20}$, and $M_{19}$. $M_{21}$ is simple, but not sporadic because it is isomorphic to $\mathrm{PSL}(3, \mathbb{F}_4)$. $M_{20} \cong 2^4 : A_5$ and $M_{19} \cong 2^4 : 3$ are not simple [Yos24, §4.1].

$M_{12}$ is the quotient $\mathrm{Aut}(G_{12})/\langle -1\rangle$ of the automorphism group of the extended ternary Golay code by its center $\langle -1\rangle = \mathbb{Z}_2$. $\mathrm{Aut}(G_{12}) \cong \mathbb{Z}^2.M_{12}$ is a non-split extension. $M_{12}$ is sharply 5-transitive as a subgroup of $S_{12}$. $M_{11}$ is the point stabilizer of $M_{12}$. The automorphism group $\mathrm{Aut}(G_{11})$ of the perfect ternary Golay code is $\mathbb{Z}_2 \times M_{11}$ [AM66, Theorem 1].

Thanks to the classification of finite simple groups, multiply transitive finite groups are also classified [DM96, §7]. Obviously, $S_n$ is sharply $n$-transitive, and $A_n$ is sharply $(n-2)$-transitive.

[13] These generators (2.9) and (2.10) were found in the result of the GAP [GAP22] command `IsomorphismGroups(MathieuGroup(24), G)`, where `G` is declared as a `Group` generated by the permutations (2.7). This command returns one explicit isomorphism in the form of a map between generators, so it suffices to check the relations in (2.8) for the displayed generators.

Except for them, there are no $k$-transitive finite groups for $k \geq 6$. $M_{12}$ is the only sharply 5-transitive finite group, $M_{24}$ is the only non-sharply 5-transitive one, $M_{11}$ is the only sharply 4-transitive (and not 5-transitive) one, and $M_{23}$ is the only non-sharply 4-transitive (and not 5-transitive) one, except for $S_n$ and $A_n$. $M_{22}$ is 3-transitive, but there are other 3-transitive finite groups, for example $\mathrm{PSL}(2, \mathbb{F}_q)$.

We have introduced the Mathieu groups as the automorphism groups of the Golay codes (or their subgroups or quotients), but there is another well-established construction as the automorphism groups of the Steiner system.

**Definition 2.4** (the Steiner system)**.** The *Steiner system* $S(t, k, n)$ $(1 \leq t \leq k \leq n)$ is a set $\{B_1, \ldots, B_b\}$ of $k$-element subsets $B_i \subset \Omega_n$ of $\Omega_n = \{1, \ldots, n\}$, such that for any $t$-element subset $T \subset \Omega_n$, there is a unique $1 \leq i \leq b$ such that $T \subset B_i$. ■

The Steiner system does not necessarily exist for a given $(t, k, n)$, and if exists, $b = \binom{n}{t} / \binom{k}{t}$, because the number $\binom{n}{t}$ of $t$-element subsets must be equal to $b \cdot \binom{k}{t}$. It is known that the automorphism groups of $S(5, 8, 24)$, $S(4, 7, 23)$, and $S(3, 6, 22)$ are $M_{24}$, $M_{23}$, and $M_{22}.2$, respectively. It is also known that the automorphism groups of $S(5, 6, 12)$ and $S(4, 5, 11)$ are $M_{12}$ and $M_{11}$, respectively.

The Schur multipliers $H^2(G; \mathrm{U}(1))$ of the Mathieu groups $M_{24}, M_{23}, M_{11}$ were correctly calculated to be trivial in [BF66]. Those of $M_{22}, M_{12}$ were also calculated there, but turned out to be wrong, and corrected in [BF68, Maz82] to be the cyclic groups of order 12, 2, respectively.

## 2.3 The Conway Groups and the Leech Lattice

The Conway groups are defined as the automorphism group (or its subgroups or quotients) of the Leech lattice $\Lambda_{24}$. So we first review generalities of lattices and the Leech lattice in Section 2.3.1, and then we introduce the Conway groups in Section 2.3.2. As for the facts cited here and for more on the Leech lattice and the Conway groups, see for example [Gri98, Ch. 9], [Yos24, Sec. 7]. [CS99] contains comprehensive explanations and data on lattices.

### 2.3.1 The Leech Lattice and the Odd Leech Lattice

**Basic definitions for lattices**

A *lattice* of rank $n$ is a free abelian group $L$ of rank $n$ whose basis is an $\mathbb{R}$-basis of a vector space $\mathbb{R}^n$ with a symmetric bilinear form $\langle -, - \rangle : L \times L \to \mathbb{R}$. Such a lattice is denoted by the pair $(L, \langle -, - \rangle)$, or just $L$, and naturally regarded as a subset of $\mathbb{R}^n$. We will only consider the cases where the symmetric bilinear forms are non-degenerate. If the symmetric bilinear form is positive-definite, then the lattice is said to be *positive-definite* or *Euclidean*. If the symmetric bilinear form is indefinite, and of signature $(r, s)$, then the lattice is said to be *indefinite* or *Lorentzian*, and of *signature* $(r, s)$.

A vector $k$ in $L$ is sometimes called a *lattice point* of $L$, and we also write the product $\langle k, k' \rangle$ of two vectors $k, k' \in L$ given by the symmetric bilinear form as $k \cdot k'$. If any vectors $k, k' \in L$ satisfy $k \cdot k' \in \mathbb{Z}$, then the lattice is said to be *integral*. The *squared length* of a vector $k \in L$ is defined as $|k|^2 := k \cdot k$. A vector $k \in L$ with $|k|^2 \in 2\mathbb{Z}$ is called an *even vector*, and $k \in L$ with $|k|^2 \in 2\mathbb{Z} + 1$ is called an *odd vector*. An integral lattice is said to be *even* if its vectors are all even, and *odd* otherwise. If we take a $\mathbb{Z}$-basis $e_1, \ldots, e_n$ of $L$, then the matrix $G = [e_i \cdot e_j]_{i,j}$ is called the *Gram matrix* of $L$.

An *isometry* or *isomorphism* $g : L \to L'$ of lattices $L$ and $L'$ is an isomorphism of free abelian groups compatible with their symmetric bilinear forms. The group of all the isometries $L \to L$, or the *automorphisms* of $L$, is denoted by $\mathrm{Aut}(L)$ or $O(L)$, but note that $\mathrm{Aut}(L)$ sometimes denotes the automorphism group of just a free abelian group $L$ (e.g. Section 5.2.3).

The *dual lattice* $L^*$ of $L$ is defined by

$$L^* := \{l \in \mathbb{R}^n \mid \langle l, k \rangle \in \mathbb{Z} \text{ for any } k \in L\}, \tag{2.14}$$

with the same symmetric bilinear form as that of $L$. $L$ is integral if and only if $L \subset L^*$. $L$ is said to be *self-dual* or *unimodular* if $L^* = L$. An odd self-dual lattice is called *Type I*, and an even self-dual lattice is called *Type II*.

Similarly to Type II codes, a Type II positive-definite lattice exists if and only if the rank is a multiple of 8 [CS99, Ch. 7 §6 Cor. 18]. Up to isometry, the $E_8$ lattice is the only one of rank 8. $E_8 \oplus E_8$ and $D_{16}^+$ are the only ones of rank 16. There are 24 Type II positive-definite lattices of rank 24, and they are called the *Niemeier lattices* [Nie73]. 23 lattices of them have vectors with squared length 2, whereas the last one, the Leech lattice $\Lambda_{24}$, does not. Moreover, a Type II indefinite lattice of signature $(r, s)$ $(r, s > 0)$ exists if and only if $r - s \equiv 0 \bmod 8$, and it is unique up to isometry [Ser73, Ch. V].

### Root lattices

A positive-definite even lattice generated by vectors with squared length 2 is called a *root lattice*.[14] Any root lattice is known to be a direct sum of the *irreducible* root lattices, which are

[14] Note that there is also the concept of a *root* of a semisimple Lie algebra $\mathfrak{g}$, whose squared length is not necessarily 2, and the *root lattice* $L_\mathfrak{g}$ of $\mathfrak{g}$, which is the lattice generated by the roots of $\mathfrak{g}$. If all the roots have squared length 2, then $\mathfrak{g}$ is said to be *simply-laced*.

The subgroup $W_\mathfrak{g}$ of the isometry group $\mathrm{Aut}(L_\mathfrak{g})$ generated by the reflections with respect to the hyperplanes perpendicular to roots is called the *Weyl group* of the Lie algebra $\mathfrak{g}$. The entire isometry group $\mathrm{Aut}(L_\mathfrak{g})$ is known to be the semidirect product $W_\mathfrak{g} : \mathrm{Aut}(\text{Dynkin diagram})$.

The Weyl group $W_G$ is also defined for a connected compact Lie group $G$ as follows. Let $T_G \cong \mathrm{U}(1)^r$ be the maximal torus in $G$, and $N(T_G) := \{g \in G \mid gtg^{-1} \in T_G \text{ for any } t \in T_G\}$ be its normalizer. Since the conjugate action of $T_G$ on $T_G$ is trivial, $T_G$ is a normal subgroup of $N(T_G)$. The *Weyl group* of the Lie group $G$ is then $W_G := N(T_G)/T_G$. It is known that $W_G$ coincides with the Weyl group $W_\mathfrak{g}$ of the Lie algebra $\mathfrak{g}$ of the Lie group $G$. The rank of $L_\mathfrak{g}$ is $r$.

Whether $N(T_G) = T_G.W_G \cong \mathrm{U}(1)^r.W_\mathfrak{g}$ splits or not, which is an analogous problem to our main Question 1 in Part III, is answered in [CWW74, Theorem 2] for simple Lie groups. It depends on the Lie group $G$, and for example, it does not split when $G = E_6, E_7, E_8$.

classified as the $A_n$ lattices ($n \in \mathbb{Z}_{\geq 1}$), the $D_n$ lattices ($n \in \mathbb{Z}_{\geq 4}$), and the $E_6$, $E_7$, $E_8$ lattices, defined as follows [CS99, Ch. 4].

- The $A_n$ *lattice* is a root lattice of rank $n$ defined as

$$A_n := \{(x_1, \dots, x_{n+1}) \in \mathbb{Z}^{n+1} \mid \sum_{i=1}^{n+1} x_i = 0\}, \tag{2.15}$$

on the hyperplane $\mathbb{R}^n \cong \{x_1 + \cdots + x_{n+1} = 0\} \subset \mathbb{R}^{n+1}$, with the standard Euclidean metric on $\mathbb{R}^{n+1}$.

We can take a $\mathbb{Z}$-basis $e_1, \dots, e_n$ of $A_n$ as

$$e_i = (0, \dots, \overset{i}{\check{1}}, -1, \dots, 0) \in \mathbb{R}^{n+1}, \tag{2.16}$$

and then the Gram matrix is

$$\begin{pmatrix} 2 & -1 & 0 & \cdots & 0 & 0 \\ -1 & 2 & -1 & \cdots & 0 & 0 \\ 0 & -1 & 2 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 2 & -1 \\ 0 & 0 & 0 & \cdots & -1 & 2 \end{pmatrix}. \tag{2.17}$$

- The $D_n$ *lattice* is a root lattice of rank $n$ defined as

$$D_n := \{k = (k_1, \dots, k_n) \in \mathbb{Z}^n \mid \sum_{i=1}^{n} k_i \equiv 0 \bmod 2\}, \tag{2.18}$$

with the standard Euclidean metric on $\mathbb{R}^n$.

We can take a $\mathbb{Z}$-basis $e_1, \dots, e_n$ of $D_n$ as

$$e_1 = (-1, -1, 0, \dots, 0), \tag{2.19}$$

$$e_i = (0, \dots, \overset{i-1}{\check{1}}, -1, \dots, 0) \quad (i = 2, \dots, n), \tag{2.20}$$

and then the Gram matrix is

$$\begin{pmatrix} 2 & 0 & -1 & 0 & \cdots & 0 & 0 \\ 0 & 2 & -1 & 0 & \cdots & 0 & 0 \\ -1 & -1 & 2 & -1 & \cdots & 0 & 0 \\ 0 & 0 & -1 & 2 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & 2 & -1 \\ 0 & 0 & 0 & 0 & \cdots & -1 & 2 \end{pmatrix}. \tag{2.21}$$

In addition, the $D_n^+$ *lattice* is defined as

$$D_n^+ := D_n \sqcup (\frac{1}{2}\vec{1} + D_n). \tag{2.22}$$

- The $E_8$ *lattice* is a root lattice of rank $8$ defined as

$$E_8 := \{k = (k_1, \ldots, k_8) \in \mathbb{Z}^8 \sqcup (\mathbb{Z} + \frac{1}{2})^8 \mid \sum_{i=1}^{n} k_i \equiv 0 \bmod 2\}, \tag{2.23}$$

with the standard Euclidean metric on $\mathbb{R}^n$. We have $D_8^+ = E_8$.

We can take a $\mathbb{Z}$-basis $e_1, \ldots, e_8$ of $E_8$ as

$$e_1 = (-1, -1, 0, \ldots, 0), \tag{2.24}$$
$$e_2 = (\frac{1}{2}, -\frac{1}{2}, -\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}), \tag{2.25}$$
$$e_i = (0, \ldots, \overset{i-1}{\check{1}}, -1, \ldots, 0) \quad (i = 3, \ldots, 8), \tag{2.26}$$

and then the Gram matrix is

$$\begin{pmatrix} 2 & 0 & -1 & 0 & \cdots & 0 & 0 \\ 0 & 2 & 0 & -1 & \cdots & 0 & 0 \\ -1 & 0 & 2 & -1 & \cdots & 0 & 0 \\ 0 & -1 & -1 & 2 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & 2 & -1 \\ 0 & 0 & 0 & 0 & \cdots & -1 & 2 \end{pmatrix}. \tag{2.27}$$

The $E_7$ *lattice* and the $E_6$ *lattice* are defined as

$$E_7 := \{(k_1, \ldots, k_8) \in E_8 \mid k_7 = k_8\}, \tag{2.28}$$
$$E_6 := \{(k_1, \ldots, k_8) \in E_8 \mid k_6 = k_7 = k_8\}. \tag{2.29}$$

The Niemeier lattices can be distinguished by their root sublattices, that is, the sublattice generated by its vectors with squared length 2. A Niemeier lattice with its root sublattice being $X$ is often referred to as the $X$ Niemeier lattice, say the $(A_1)^{24}$ Niemeier lattice, except for the Leech lattice $\Lambda_{24}$ whose root sublattice is $0$.

**Lattices from codes**

There are several ways to construct a lattice from a given code. We focus on the case of binary codes here. See [CS99] for more on constructions.

Let $\mathcal{C} \subset (\mathbb{F}_2)^n$ be a binary code of length $n$, and regard each codeword of $\mathcal{C}$ as a vector with entries 0 or 1. We can construct a lattice $\Lambda(\mathcal{C})$ of rank $n$ as

$$\Lambda(\mathcal{C}) := \frac{1}{\sqrt{2}}\mathcal{C} + \sqrt{2}\mathbb{Z}^n \subset \mathbb{R}^n, \tag{2.30}$$

with the standard Euclidean metric $k \cdot k' = \sum_i k_i k'_i$ for $k = (k_i)_i, k' = (k'_i)_i \in \mathbb{R}^n$ as the symmetric bilinear form. This construction is called Construction A [CS99, Ch. 7 §2]. This lattice $\Lambda(\mathcal{C})$ satisfies $\Lambda(\mathcal{C}^*) = \Lambda(\mathcal{C})^*$. Hence, $\Lambda(\mathcal{C})$ is integer if and only if $\mathcal{C}$ satisfies $\mathcal{C} \subset \mathcal{C}^*$, and self-dual if and only if $\mathcal{C}$ is self-dual. In addition, $\Lambda(\mathcal{C})$ is of Type I (odd self-dual) if and only if $\mathcal{C}$ is of Type I (singly-even self-dual), and Type II (even self-dual) if and only if $\mathcal{C}$ is of Type II (doubly-even self-dual). Another construction called Construction B [CS99, Ch. 7 §5] associates the sublattice

$$\Lambda_{\mathrm{B}}(\mathcal{C}) := \{k = (k_1, \dots, k_n) \in \Lambda(\mathcal{C}) \mid \sqrt{2}\sum_{i=1}^{n} k_i \in 4\mathbb{Z}\} \tag{2.31}$$

of $\Lambda(\mathcal{C})$ to a binary code $\mathcal{C}$.

For a doubly-even self-dual binary code $\mathcal{C}$, we further consider the following constructions [DGM94, §5.1]. We first define

$$\mathbb{Z}^n_+ := \{x \in \mathbb{Z}^n \mid |x|^2 \in 2\mathbb{Z}\}, \tag{2.32}$$

$$\mathbb{Z}^n_- := \{x \in \mathbb{Z}^n \mid |x|^2 \in 2\mathbb{Z} + 1\}. \tag{2.33}$$

Recall that the length $n$ of a doubly-even self-dual code is always a multiple of 8, and also define

$$\Lambda_0(\mathcal{C}) := \frac{1}{\sqrt{2}}\mathcal{C} + \sqrt{2}\mathbb{Z}^n_+, \tag{2.34}$$

$$\Lambda_1(\mathcal{C}) := \frac{1}{\sqrt{2}}\mathcal{C} + \sqrt{2}\mathbb{Z}^n_-, \tag{2.35}$$

$$\Lambda_2(\mathcal{C}) := \frac{1}{2\sqrt{2}}\vec{1} + \frac{1}{\sqrt{2}}\mathcal{C} + \sqrt{2}\mathbb{Z}^n_{(-)^{\frac{n}{8}+1}}, \tag{2.36}$$

$$\Lambda_3(\mathcal{C}) := \frac{1}{2\sqrt{2}}\vec{1} + \frac{1}{\sqrt{2}}\mathcal{C} + \sqrt{2}\mathbb{Z}^n_{(-)^{\frac{n}{8}}}, \tag{2.37}$$

where $\vec{1} := (1, \dots, 1)$. All the vectors in $\Lambda_i(\mathcal{C})$ for $i = 0, 1, 3$ are even, whereas those in $\Lambda_2(\mathcal{C})$ are odd. Then we can construct some more lattices as in Table 2.2.

Table 2.2: Lattices constructed from a doubly-even self-dual binary code $\mathcal{C}$. On the right side of the vertical line, the names of lattices constructed from the binary Golay code $G_{24}$ and their isometry groups are shown.

| name of construction | lattice | property | for $\mathcal{C} = G_{24}$ | Aut(lattice) |
|---|---|---|---|---|
| Construction A | $\Lambda(\mathcal{C}) = \Lambda_0(\mathcal{C}) \cup \Lambda_1(\mathcal{C})$ | even self-dual | $(A_1)^{24}$ Niemeier lattice | $2^{24} : M_{24}$ |
| Construction B | $\Lambda_{\mathrm{B}}(\mathcal{C}) = \Lambda_0(\mathcal{C})$ | even | | |
| twisted construction | $\tilde{\Lambda}(\mathcal{C}) = \Lambda_0(\mathcal{C}) \cup \Lambda_3(\mathcal{C})$ | even self-dual | Leech lattice $\Lambda_{24}$ | $\mathrm{Co}_0$ |
| – | $\tilde{\Lambda}'(\mathcal{C}) = \Lambda_0(\mathcal{C}) \cup \Lambda_2(\mathcal{C})$ | odd self-dual | odd Leech lattice $O_{24}$ | $2^{12} : M_{24}$ |

**The odd Leech lattice $O_{24}$**

The *odd Leech lattice* $O_{24}$ is the unique positive-definite odd self-dual lattice of rank 24 without roots (vectors of squared length 2) up to isometry. It can be constructed from the binary Golay code $G_{24}$ as

$$O_{24} = (\frac{1}{\sqrt{2}} G_{24} + \sqrt{2}\mathbb{Z}_+^{24}) \cup (\frac{1}{2\sqrt{2}}\vec{1} + \frac{1}{\sqrt{2}} G_{24} + \sqrt{2}\mathbb{Z}_+^{24}). \tag{2.38}$$

Historically, the odd Leech lattice was first found in [OP44].

If we use the binary Golay code $G_{24}$ with the basis (2.6), then we can take a $\mathbb{Z}$-basis $e_1, \dots, e_{24}$ of (2.38) as

$$\begin{aligned}
e_1 &= \tfrac{1}{2\sqrt{2}}\vec{1}, \\
e_2 &= \tfrac{1}{\sqrt{2}}(\text{the second line from the bottom of (2.6)}), \\
&\vdots \\
e_{12} &= \tfrac{1}{\sqrt{2}}(\text{the first line of (2.6)}), \\
e_{13} &= \sqrt{2}(1,0,0,0,0,0,0,0,0,0,0,0,1,0,\dots,0), \\
e_{14} &= \sqrt{2}(1,0,0,0,0,0,0,0,0,0,0,1,0,0,\dots,0), \\
&\vdots \\
e_{23} &= \sqrt{2}(1,1,0,\dots,0), \\
e_{24} &= \sqrt{2}(2,0,\dots,0).
\end{aligned} \tag{2.39}$$

In fact, $O_{24}$ in (2.38) obviously contains $\mathrm{Span}_{\mathbb{Z}}\{e_i\}_i$, and the opposite direction of the inclusion can be checked as follows. Since it is obvious that $\mathrm{Span}_{\mathbb{Z}}\{e_i\}_i$ contains $\frac{1}{2\sqrt{2}}\vec{1}$ and $\frac{1}{\sqrt{2}}G_{24}$, it suffices to check that it also contains $\sqrt{2}\mathbb{Z}_+^{24}$. Here, $\mathbb{Z}_+^{24}$ is generated by the 24 vectors

$$\begin{aligned}
&(1,0,\dots,0,0,1), \\
&(1,0,\dots,0,1,0), \\
&\vdots \\
&(1,1,0,\dots,0), \\
&(2,0,\dots,0).
\end{aligned}$$

Since the last 12 vectors of $\sqrt{2}\times$(above 24 vectors) are exactly $e_{13}, \dots, e_{24}$, it suffices to check that the first 12 vectors of $\sqrt{2}\times$(above 24 vectors) can be written as $\mathbb{Z}$-linear combinations of $e_1, \dots, e_{24}$, which can be checked by computer.

The isometry group $\mathrm{Aut}(O_{24})$ of the odd Leech lattice is known to be $2^{12} : M_{24}$ [CS99, Ch. 17]. In the construction (2.38), these automorphisms are apparent because $M_{24} = \mathrm{Aut}(G_{24})$ and $2^{12}$ are the maps $k = (k_i)_i \mapsto ((-1)^{w_i} k_i)_i$ where $w = (w_i)_i \in G_{24}$.

If we use Construction A for ternary codes, an odd Leech lattice can also be constructed from any self-dual ternary code of length 24 with the minimal nonzero weight 9. It is known that there are only two such ternary codes up to equivalence [LPS81]: the extended quadratic residue

code $Q_{24}$ (see e.g. [MS77, Ch. 16]), and the symmetric code $P_{24}$ defined by Pless [Ple69, Ple72] [MS77, Ch. 16 §8]. However, the automorphism groups of these ternary codes are $\mathrm{Aut}(Q_{24}) = (\mathbb{F}_3)^\times \times \mathrm{PSL}(23, \mathbb{F}_2)$ [MS77, Apply Theorem 13 in Ch. 16 §5 to $p = 23$] and $\mathrm{Aut}(P_{24}) = \mathbb{Z}_4 \times \mathrm{PGL}(11, \mathbb{F}_2)$ [MPS76, §5.2], so the structure of $\mathrm{Aut}(O_{24}) = 2^{12} : M_{24}$ is not apparent in these constructions of odd Leech lattices. See [GJF18, Example 4.5] for application of these constructions to $\mathcal{N} = 1$ supersymmetries of the lattice VOAs.

**The Leech lattice $\Lambda_{24}$**

The *Leech lattice* $\Lambda_{24}$ is the unique positive-definite even self-dual lattice of rank 24 without roots (vectors of squared length 2) up to isometry. It can be constructed from the binary Golay code $G_{24}$ as

$$\Lambda_{24} = (\frac{1}{\sqrt{2}} G_{24} + \sqrt{2}\mathbb{Z}^{24}_+) \cup (\frac{1}{2\sqrt{2}}\vec{1} + \frac{1}{\sqrt{2}} G_{24} + \sqrt{2}\mathbb{Z}^{24}_-). \tag{2.40}$$

Historically, the Leech lattice was discovered by Leech in [Lee67, §2.31], but it is also said that one of the more than 10 Niemeier lattices, which was reported to be found by Witt in [Wit41, p. 324] without further details, is the Leech lattice [Wit98, p. 328-329]. See an informative video [Bor20] by Borcherds for more details.

For an even lattice $L$, we define

$$L_d := \{k \in L \mid |k|^2 = 2d\}. \tag{2.41}$$

An even lattice $L$ of rank 24 is isomorphic to the Leech lattice $\Lambda_{24}$, if and only if it satisfies

$$|L_1| = 0, \tag{2.42}$$
$$|L_2| = 196560, \tag{2.43}$$
$$|L_3| = 16773120, \tag{2.44}$$
$$|L_4| = 398034000. \tag{2.45}$$

We consider the quotient $\Lambda_{24}/2\Lambda_{24}$ of $\Lambda_{24}$ by the equivalence relation $k \sim k' \Leftrightarrow k - k' \in 2\Lambda_{24}$. We can take a complete set of representatives of $\Lambda_{24}/2\Lambda_{24}$ as follows.

- Take $0 \in \Lambda_{24}$.
- From $(\Lambda_{24})_2$ and $(\Lambda_{24})_3$, if we take $k$, then do not take $-k$.
- For $k \in (\Lambda_{24})_4$, its equivalence class $k + 2\Lambda_{24}$ contains 48 vectors of squared length 8 as $(k + 2\Lambda_{24}) \cap (\Lambda_{24})_4 = \{\pm k, \pm k_2, \ldots, \pm k_{24}\}$, so take one of them. It is known that $\{k, k_2, \ldots, k_{24}\}$ constitutes an orthogonal basis of $\mathbb{R}^{24}$.

### 2.3.2 The Conway Groups

**The Conway groups $\mathrm{Co}_0$ and $\mathrm{Co}_1$**

The isometry group $\mathrm{Aut}(\Lambda_{24})$ of the Leech lattice is the largest *Conway group* $\mathrm{Co}_0$. It is not a simple group, but its quotient by the center $\{\pm 1\}$ is the largest sporadic Conway group $\mathrm{Co}_1$.

From the construction (2.40), it is apparent that $\mathrm{Co}_0$ contains $2^{12} : M_{24}$ as a subgroup, where $M_{24} = \mathrm{Aut}(G_{24})$ and $2^{12}$ are the maps $k = (k_i)_i \mapsto ((-1)^{w_i} k_i)_i$ where $w = (w_i)_i \in G_{24}$. This subgroup $2^{12} : M_{24}$ is called the *monomial subgroup* of $\mathrm{Co}_0$. $\mathrm{Co}_0$ is generated by this monomial subgroup and a specific order-2 element. As a subgroup of $\mathrm{O}(\mathbb{R}^{24})$, $\mathrm{Co}_0$ does not contain matrices of determinant $-1$, and hence $\mathrm{Co}_0$ is a subgroup of $\mathrm{SO}(\mathbb{R}^{24})$.

Generators of $\mathrm{Co}_0$ as a subgroup of $\mathrm{SL}(24, \mathbb{Z})$ can be found on the $\mathrm{Co}_1$ page of [WWT+] as

$$A = \begin{pmatrix}
2 & 0 & 0 & -3 & -2 & -1 & 0 & -2 & 1 & 1 & 0 & 1 & 1 & 0 & 0 & 1 & -1 & -1 & 0 & 0 & -1 & 0 & 0 & 0 \\
-4 & 1 & 0 & 6 & 4 & 2 & 2 & 3 & -1 & -3 & -1 & 0 & -2 & -1 & -1 & -1 & 1 & 1 & 0 & 1 & 0 & 0 & 0 & 0 \\
4 & -1 & 0 & -6 & -4 & -2 & -3 & -4 & 2 & 4 & 3 & -3 & 2 & 1 & 2 & 0 & 1 & -1 & -1 & -1 & 1 & 0 & -1 & 1 \\
4 & 1 & 1 & -6 & -3 & -2 & -1 & -4 & 2 & 2 & 1 & 0 & 1 & 0 & 1 & 2 & 0 & -1 & -1 & 0 & -1 & 0 & 0 & 1 \\
5 & -2 & 0 & -8 & -5 & -3 & -2 & -4 & 2 & 4 & 2 & 0 & 2 & 2 & 1 & 1 & -1 & -1 & 0 & 0 & 0 & 1 & -1 & 0 \\
-3 & 0 & -1 & 4 & 1 & 2 & 1 & 1 & 0 & -2 & 0 & 0 & 0 & -1 & -1 & -1 & 0 & 0 & 0 & 0 & -1 & -1 & 0 & 0 \\
1 & 0 & -1 & -2 & -2 & -1 & 0 & -2 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & -1 & 0 & 0 & 0 & 0 \\
-3 & 1 & -1 & 5 & 3 & 1 & 2 & 2 & -1 & -2 & -1 & 0 & -1 & -1 & -1 & -1 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
-2 & 2 & 0 & 3 & 3 & 0 & 2 & 2 & 0 & -1 & -2 & 1 & 0 & -1 & 0 & 0 & 1 & -1 & 1 & 1 & 0 & 0 & 0 & 1 \\
9 & -1 & -1 & -13 & -9 & -5 & -4 & -9 & 4 & 6 & 4 & -1 & 3 & 1 & 2 & 2 & -1 & -3 & -1 & -1 & -2 & 1 & -1 & 1 \\
1 & -1 & -1 & -3 & -3 & -1 & -1 & -4 & 2 & 2 & 3 & -2 & 1 & 0 & 0 & 0 & 0 & 0 & -1 & -1 & 0 & 0 & -1 & 1 \\
4 & -1 & 1 & -6 & -4 & -2 & -2 & -3 & 2 & 3 & 2 & -1 & 2 & 1 & 1 & 1 & 0 & -1 & -1 & 0 & 0 & 0 & -1 & 1 \\
-4 & 0 & 0 & 5 & 3 & 2 & 2 & 3 & -2 & -2 & -1 & 0 & -2 & 0 & -1 & -1 & 0 & 2 & 0 & 0 & 1 & 0 & 0 & -1 \\
8 & 2 & 2 & -9 & -5 & -4 & 0 & -5 & 2 & 2 & 2 & 1 & 0 & 1 & 1 & 2 & -1 & 0 & -1 & 1 & -1 & 1 & 0 & 1 \\
-3 & 0 & 0 & 3 & 3 & 1 & 0 & 3 & -1 & -1 & -1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 1 & -1 & 0 & 0 \\
-5 & 2 & 0 & 6 & 4 & 3 & 3 & 2 & -1 & -4 & -2 & 1 & -2 & -2 & -2 & 0 & 0 & 1 & 0 & 0 & -1 & -1 & 1 & 0 \\
3 & 0 & 0 & -3 & -3 & -1 & 0 & -2 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & -1 & -1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 1 & 1 & 0 & 0 & 2 & -1 & 0 & -2 & 1 & 0 & 0 & 0 & 0 & 0 & -1 & 1 & 0 & 0 & 0 & 0 & -1 \\
-1 & 0 & -1 & 2 & 0 & 1 & -1 & 0 & 0 & 0 & 0 & -1 & 1 & -1 & 0 & -1 & 1 & -1 & 0 & -1 & 0 & -1 & 0 & 0 \\
-4 & -1 & -1 & 4 & 2 & 2 & 0 & 2 & -1 & -2 & -1 & 1 & -1 & -1 & -1 & 0 & 0 & 1 & 0 & 0 & -1 & 0 & 1 & -1 \\
-9 & 1 & 0 & 11 & 7 & 5 & 3 & 6 & -2 & -5 & -3 & 1 & -2 & -2 & -2 & -1 & 1 & 2 & 0 & 1 & 0 & -1 & 1 & 0 \\
-6 & 1 & 0 & 9 & 5 & 4 & 3 & 4 & -2 & -4 & -1 & -1 & -2 & -1 & -1 & -2 & 1 & 2 & 0 & 0 & 1 & -1 & 0 & 0 \\
5 & 1 & -1 & -7 & -5 & -3 & -1 & -6 & 2 & 3 & 3 & 0 & 1 & 0 & 1 & 1 & -1 & 0 & -1 & -1 & -1 & 1 & 0 & 1 \\
6 & 0 & 1 & -6 & -4 & -2 & -2 & -3 & 1 & 2 & 1 & 0 & 2 & 1 & 1 & 1 & -1 & -1 & 0 & 0 & -1 & 0 & 0 & 0
\end{pmatrix}^T, \tag{2.46}$$

$$B = \begin{pmatrix}
0 & 0 & 1 & 0 & 0 & 0 & -1 & 1 & 1 & 0 & -1 & 0 & 1 & 0 & 1 & 0 & 1 & 0 & 0 & 1 & 0 & 1 & 0 & 1 \\
3 & 1 & -1 & -3 & -2 & -2 & 0 & -3 & 1 & 2 & 2 & -1 & 1 & 0 & 1 & 0 & 0 & -1 & 0 & -1 & 0 & -1 & -1 & 0 \\
-4 & 0 & -1 & 7 & 4 & 3 & 1 & 4 & -3 & -3 & -2 & 0 & -1 & -1 & -1 & -1 & 0 & 0 & 1 & -1 & 0 & -1 & 1 & -1 \\
0 & -1 & 0 & 1 & 0 & 1 & -1 & 1 & 0 & 0 & 0 & -1 & 1 & 0 & 1 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
6 & 0 & 2 & -6 & -4 & -2 & -3 & -2 & 1 & 2 & 1 & -1 & 1 & 1 & 2 & 1 & 1 & -1 & -1 & 0 & 0 & 1 & 0 & 1 \\
2 & 0 & -1 & -5 & -3 & -3 & -1 & -4 & 3 & 3 & 1 & 0 & 2 & 0 & 1 & 1 & 0 & -1 & 0 & 0 & 0 & 1 & -1 & 1 \\
-10 & 0 & 0 & 12 & 8 & 5 & 4 & 7 & -3 & -6 & -3 & 1 & -3 & -1 & -3 & -1 & 0 & 3 & 1 & 1 & 1 & -1 & 1 & -1 \\
-1 & 4 & 1 & 5 & 4 & 1 & 3 & 3 & -2 & -3 & -2 & 1 & -1 & -1 & 0 & 0 & 1 & 1 & 1 & 0 & 0 & -1 & 1 & 0 \\
-1 & 0 & 1 & 3 & 2 & 1 & -1 & 3 & -1 & -1 & -2 & 0 & 1 & 0 & 1 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 1 & 0 \\
2 & 2 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 1 & 0 & 2 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\
-5 & 3 & 0 & 11 & 7 & 4 & 4 & 6 & -4 & -5 & -3 & 0 & -2 & -2 & -1 & -2 & 1 & 1 & 1 & -1 & 1 & -2 & 1 & -1 \\
3 & 0 & 1 & -2 & -1 & -1 & -1 & 0 & 0 & 1 & -1 & 0 & 1 & 0 & 1 & 0 & 0 & -1 & 0 & 0 & 0 & 1 & 0 & 0 \\
-4 & 0 & -1 & 5 & 3 & 2 & 2 & 2 & -1 & -3 & 0 & 0 & -2 & -1 & -2 & -1 & 0 & 1 & 0 & 0 & 0 & -1 & 0 & -1 \\
2 & -1 & 1 & -2 & -1 & -1 & -1 & 0 & 1 & 1 & 1 & -1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & -1 & 0 \\
2 & -1 & 1 & -3 & -1 & -1 & -1 & 0 & -1 & 1 & -1 & 1 & -1 & 1 & 0 & 1 & -1 & 0 & 0 & 0 & 0 & 1 & 1 & -1 \\
-2 & 0 & -1 & 3 & 1 & 1 & 1 & 1 & 0 & -1 & 0 & 0 & 0 & -1 & 0 & -1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
-11 & -2 & -1 & 12 & 7 & 5 & 3 & 7 & -2 & -5 & -3 & 1 & -2 & -1 & -3 & -2 & 0 & 2 & 1 & 1 & 1 & 0 & 0 & -1 \\
-2 & 0 & 1 & 2 & 2 & 1 & 0 & 3 & -1 & -1 & -2 & 1 & -1 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\
-6 & 1 & -1 & 7 & 5 & 2 & 2 & 4 & -2 & -2 & -3 & 1 & -1 & -1 & -1 & -1 & 1 & 0 & 1 & 0 & 1 & 0 & 0 & 0 \\
-3 & 1 & -1 & 3 & 2 & 1 & 2 & 0 & 0 & -1 & 0 & 0 & -1 & -1 & -1 & -1 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 \\
2 & -2 & -2 & -6 & -5 & -2 & -2 & -6 & 3 & 4 & 3 & -1 & 2 & 0 & 0 & 0 & -1 & -1 & -1 & -1 & -1 & 0 & -1 & 0 \\
2 & -1 & -2 & -4 & -4 & -1 & -1 & -5 & 2 & 2 & 4 & -2 & 1 & 0 & 0 & 0 & -1 & 0 & -1 & -2 & 0 & -1 & -1 & 0 \\
-6 & 1 & 0 & 8 & 5 & 4 & 3 & 3 & -2 & -4 & -1 & 0 & -2 & -1 & -2 & -1 & 0 & 2 & 0 & 0 & 0 & -2 & 1 & -1 \\
14 & 1 & 2 & -18 & -11 & -7 & -4 & -10 & 4 & 7 & 4 & 0 & 3 & 2 & 3 & 3 & -1 & -2 & -2 & 0 & -1 & 2 & -1 & 2
\end{pmatrix}^T, \tag{2.47}$$

where the transpose on the right-hand side is taken just for the convenience of notation. These generators satisfy $A^2 = -1$ and $B^3 = 1$.

The Conway group $\mathrm{Co}_0$ generated by these generators $A$ and $B$ is the isometry group of a lattice $(\mathbb{Z}^{24}, q)$, where $q$ is some symmetric bilinear form which makes $(\mathbb{Z}^{24}, q)$ isomorphic to the Leech lattice. The matrix form of $q$ up to scalar multiplication can be found by solving the conditions

$$A^T q A = q, \quad B^T q B = q, \tag{2.48}$$

by Mathematica [Wol23]. Note that the elements of the Conway group $\mathrm{Co}_0$ in this notation act on the column vectors in the lattice $(\mathbb{Z}^{24}, q)$ from the left. The result is

$$q = \begin{pmatrix}
4 & -2 & -2 & 2 & 2 & 2 & -1 & -1 & 2 & 2 & -2 & 2 & -2 & 1 & -1 & 1 & 1 & 2 & 0 & -1 & 0 & -2 & -2 & 2 \\
-2 & 4 & 0 & 0 & -2 & 0 & -1 & 2 & 0 & 0 & 2 & -1 & 1 & 1 & -1 & 1 & -2 & -2 & -1 & 1 & 1 & 2 & 1 & -2 \\
-2 & 0 & 4 & 0 & 0 & -2 & 0 & 0 & 0 & 0 & 2 & 0 & 0 & -2 & 1 & -1 & -1 & -1 & 1 & -1 & -1 & 1 & 0 & -1 \\
2 & 0 & 0 & 4 & 0 & 0 & -2 & -1 & 2 & 2 & 0 & 2 & -2 & 1 & 0 & 2 & -1 & 0 & -1 & 0 & 1 & -1 & -1 & 0 \\
2 & -2 & 0 & 0 & 4 & 0 & -1 & -1 & 1 & 1 & -1 & 1 & -1 & 0 & 0 & -1 & 0 & 2 & 0 & -2 & -2 & -1 & -2 & 2 \\
2 & 0 & -2 & 0 & 0 & 4 & 0 & 0 & 1 & 1 & -1 & 1 & -1 & 1 & -1 & 1 & 1 & 0 & 1 & 0 & 1 & 0 & -1 & 1 \\
-1 & -1 & 0 & -2 & -1 & 0 & 4 & 1 & -1 & -1 & 0 & -2 & 2 & 0 & 0 & 0 & 2 & -1 & 0 & 0 & -1 & 1 & 2 & -1 \\
-1 & 2 & 0 & -1 & -1 & 0 & 1 & 4 & 1 & 1 & 2 & -1 & 1 & 0 & -1 & 1 & -1 & -1 & -1 & -1 & -1 & 1 & 1 & -1 \\
2 & 0 & 0 & 2 & 1 & 1 & -1 & 1 & 4 & 2 & 0 & 2 & -2 & 1 & 0 & 1 & -1 & 0 & -1 & -2 & 0 & -1 & -1 & 0 \\
2 & 0 & 0 & 2 & 1 & 1 & -1 & 1 & 2 & 4 & 0 & 1 & -2 & 0 & -1 & 1 & -1 & 1 & 0 & -1 & -1 & -1 & -1 & 1 \\
-2 & 2 & 2 & 0 & -1 & -1 & 0 & 2 & 0 & 0 & 4 & 0 & 1 & -1 & 0 & 1 & -1 & -2 & 0 & 0 & 0 & 2 & 1 & -2 \\
2 & -1 & 0 & 2 & 1 & 1 & -2 & -1 & 2 & 1 & 0 & 4 & -2 & 0 & 1 & 0 & 0 & 1 & 0 & -1 & 1 & -2 & -2 & 1 \\
-2 & 1 & 0 & -2 & -1 & -1 & 2 & 1 & -2 & -2 & 1 & -2 & 4 & 0 & -1 & 0 & 1 & -1 & -1 & 1 & 0 & 2 & 2 & -2 \\
1 & 1 & -2 & 1 & 0 & 1 & 0 & 0 & 1 & 0 & -1 & 0 & 0 & 4 & -1 & 1 & 0 & -1 & -2 & 0 & 0 & 0 & 0 & 0 \\
-1 & -1 & 1 & 0 & 0 & -1 & 0 & -1 & 0 & -1 & 0 & 1 & -1 & -1 & 4 & -1 & -1 & 0 & 0 & 0 & 0 & -1 & 0 & 0 \\
1 & 1 & -1 & 2 & -1 & 1 & 0 & 1 & 1 & 1 & 0 & 0 & 1 & -1 & 4 & 0 & -1 & -1 & 0 & 1 & 1 & 0 & -1 & \\
1 & -2 & -1 & -1 & 0 & 1 & 2 & -1 & -1 & -1 & -1 & 0 & 1 & 0 & -1 & 0 & 4 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\
2 & -2 & -1 & 0 & 2 & 0 & -1 & -1 & 0 & 1 & -2 & 1 & -1 & -1 & 0 & -1 & 1 & 4 & 1 & -1 & -1 & -2 & -2 & 2 \\
0 & -1 & 1 & -1 & 0 & 1 & 0 & -1 & -1 & 0 & 0 & 0 & -1 & -2 & 0 & -1 & 1 & 1 & 4 & 0 & 0 & 0 & -1 & 1 \\
-1 & 1 & -1 & 0 & -2 & 0 & 0 & -1 & -2 & -1 & 0 & -1 & 1 & 0 & 0 & 0 & 0 & -1 & 0 & 4 & 2 & 0 & 2 & -1 \\
0 & 1 & -1 & 1 & -2 & 1 & -1 & -1 & 0 & -1 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & -1 & 0 & 2 & 4 & 0 & 1 & -1 \\
-2 & 2 & 1 & -1 & -1 & 0 & 1 & 1 & -1 & -1 & 2 & -2 & 2 & 0 & -1 & 1 & 0 & -2 & 0 & 0 & 0 & 4 & 1 & -2 \\
-2 & 1 & 0 & -1 & -2 & -1 & 2 & 1 & -1 & -1 & 1 & -2 & 2 & 0 & 0 & 0 & 0 & -2 & -1 & 2 & 1 & 1 & 4 & -2 \\
2 & -2 & -1 & 0 & 2 & 1 & -1 & -1 & 0 & 1 & -2 & 1 & -2 & 0 & 0 & -1 & 0 & 2 & 1 & -1 & -1 & -2 & -2 & 4
\end{pmatrix}. \tag{2.49}$$

**The Conway groups $\mathrm{Co}_2$ and $\mathrm{Co}_3$**

There are two more sporadic simple Conway groups $\mathrm{Co}_2$ and $\mathrm{Co}_3$. They are the stabilizer groups of one vector in $(\Lambda_{24})_2$ and $(\Lambda_{24})_3$, respectively, under the action of $\mathrm{Co}_0$ on $\Lambda_{24}$. It is known that the actions of $\mathrm{Co}_0$ on $(\Lambda_{24})_2$ and $(\Lambda_{24})_3$ are transitive, so these groups are unique up to isomorphism. Since $-1 \in \mathrm{Co}_0$ does not preserve any non-zero vector, $\mathrm{Co}_2$ and $\mathrm{Co}_3$ are also isomorphic to subgroups of $\mathrm{Co}_1$.

## 2.4 The Monster Group

The monster group $\mathbb{M}$, whose existence was predicted independently by Fischer and Griess in 1973, was first constructed by Griess in 1981 [Gri81, Gri82], as the automorphism group of a certain algebra $B^\natural$ with a bilinear form $\tau$. The algebra $B^\natural$ is called the Griess algebra. More precisely, Griess constructed $\mathbb{M}$ as a subgroup of $\mathrm{Aut}(B^\natural, \tau)$, and $\mathrm{Aut}(B^\natural, \tau) = \mathbb{M}$ was later shown in [Tit84].

As a vector space, $B^\natural$ is a 196884-dimensional representation space of a group $C(\Lambda_{24}) = 2^{1+24}.\mathrm{Co}_1$ over $\mathbb{Q}$. In [Gri82], a specific commutative non-associative product and an associative[15] symmetric bilinear form $\tau$ are introduced to this $C(\Lambda_{24})$-module, in a way compatible with the

[15] A bilinear form $\tau$ is called *associative* if it satisfies $\tau(u, v \cdot w) = \tau(u \cdot v, w)$.

$C(\Lambda_{24})$-action, which defines the $C(\Lambda_{24})$-algebra $B^\natural$ called the *Griess algebra*. Now, $\mathrm{Aut}(B^\natural, \tau)$ contains $C(\Lambda_{24})$, and a specific order-2 element $\sigma$ can be found in $\mathrm{Aut}(B^\natural, \tau) \setminus C(\Lambda_{24})$. Finally, the group generated by $C(\Lambda_{24})$ and $\sigma$ is shown to be simple, and have the right order $|\mathbb{M}|$.

When $B^\natural$ was introduced in [Gri82], the monster VOA $V^\natural$ [FLM88] (explained later in Section 5) was not yet constructed. However, in the modern understanding, it is natural to regard $B^\natural$ as the weight-2 subspace $(V^\natural)_2$ of $V^\natural$. In fact, one of the important results on the monstrous moonshine is that $\mathrm{Aut}(V^\natural) = \mathbb{M}$. As we will see in Section 5, $V^\natural$ is the $\mathbb{Z}_2$-orbifold of the Leech lattice VOA, and hence its weight-2 subspace $(V^\natural)_2 = B^\natural$ consists of

- $\mathrm{Span}\{\mathfrak{a}^i_{-1}\mathfrak{a}^j_{-1}|0\rangle\}_{i,j=1,\dots,24}$, which is the symmetric tensor product of $\mathbf{24}$, so the dimension is $\frac{25\times 24}{2} = 1 + 299$, containing the trivial representation $\mathrm{Span}\{\sum_i \mathfrak{a}^i_{-1}\mathfrak{a}^i_{-1}|0\rangle\}$ of $\mathrm{SO}(24)$.
- $\mathrm{Span}\{|k\rangle + |-k\rangle\}_{k\in(\Lambda_{24})_2}$, whose dimension is $|(\Lambda_{24})_2|/2 = 98280$.
- $\mathrm{Span}\{\mathfrak{c}^i_{-\frac{1}{2}}|s\rangle\}_{i=1,\dots,24,|s\rangle\in\mathcal{X}(\Lambda_{24})}$, where $\mathcal{X}(\Lambda_{24})$ is the $2^{\frac{24}{2}}$-dimensional irreducible representation of a certain gamma matrix algebra $\Gamma(\Lambda_{24})$, so the dimension is $24 \times 2^{\frac{24}{2}} = 98304$.

These three subspaces are denoted by $U$, $V$, and $W$, respectively, in [Gri82]. In the language of VOA (see Section 4), the commutative non-associative product on $B^\natural$ is introduced as

$$v \cdot w = v_{(1)}w, \tag{2.50}$$

and the associative symmetric bilinear form $\tau$ is introduced as

$$\tau(v, w)\mathbf{1} = v_{(3)}w, \tag{2.51}$$

for any $v, w \in (V^\natural)_2$. The one-dimensional subspace in the first $(1 + 299)$-dimensional subspace will turn out to be the trivial representation of $\mathbb{M} = \mathrm{Aut}(B^\natural, \tau)$, and the irreducible decomposition of $B^\natural$ as a representation of $\mathbb{M}$ is $196884 = 1 + 196883$, where $196883$ is the smallest nontrivial irreducible representation of $\mathbb{M}$.

See for example [Gri98, Ch. 11] and [Har99] for more on the Griess algebra and the monster group. A simpler construction of the monster group was provided in [Con85], and reviewed in [CS99, Ch. 29]. See [Iva09] for another construction using the amalgam method.

# 3 Modular Functions and Weak Jacobi Forms

This Section 3 collects the basic definitions and important examples of modular functions and weak Jacobi forms. After mentioning the action of the modular group on the upper half-plane in Section 3.1, we will review modular functions and weak Jacobi forms in Sections 3.2 and 3.3, respectively. In the last Section 3.4, we will review the elliptic genes of CFT, through which weak Jacobi forms appear in physics.

More details on the facts cited here about the modular functions can be found, for example, in [Ser73, Ch. VII] and [DS05, Ch. 1]. The foundational literature on weak Jacobi forms is [EZ85], and we will follow a summary in [DMZ12, §4].

## 3.1 Modular Group

The special linear group

$$\mathrm{SL}(2,\mathbb{Z}) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \;\middle|\; a,b,c,d \in \mathbb{Z}, ad-bc=1 \right\} \tag{3.1}$$

is generated by

$$T := \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad S := \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}. \tag{3.2}$$

It acts on the upper half-plane of the complex plane

$$\mathbb{H} := \{\tau \in \mathbb{C} \mid \mathrm{Im}(\tau) > 0\} \tag{3.3}$$

as

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} : \tau \mapsto \frac{a\tau + b}{c\tau + d}. \tag{3.4}$$

If we define a complex torus as $E_\tau := \mathbb{C}/(\mathbb{Z} \oplus \tau\mathbb{Z})$, then it is known that[16] $E_\tau$ and $E_{\tau'}$ are isomorphic as complex manifolds if and only if there is $g \in \mathrm{SL}(2,\mathbb{Z})$ such that $g \cdot \tau = \tau'$.

Since the center $\mathbb{Z}_2 = \langle -I \rangle$ of $\mathrm{SL}(2,\mathbb{Z})$ acts trivially on $\mathbb{H}$, we can say that $\mathrm{PSL}(2,\mathbb{Z}) = \mathrm{SL}(2,\mathbb{Z})/\langle -I \rangle$ acts on $\mathbb{H}$. This group $\mathrm{PSL}(2,\mathbb{Z})$ is called the *modular group*. However, in the following discussions, it suffices to use $\mathrm{SL}(2,\mathbb{Z})$. In fact, some literature also calls $\mathrm{SL}(2,\mathbb{Z})$ the modular group.

Lastly, we remark that $\mathrm{GL}(2,\mathbb{Z})$ is generated by $T$, $S$, and

$$P := \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}. \tag{3.5}$$

[16] This fact can be understood as follows [ES15, §5.1]. The lattice in the complex plane spanned by $\omega_1, \omega_2 \in \mathbb{C}$ is the same as the one spanned by $\omega_1', \omega_2' \in \mathbb{C}$ such that $\begin{pmatrix} \omega_2' \\ \omega_1' \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} \omega_2 \\ \omega_1 \end{pmatrix}$ where $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}(2,\mathbb{Z})$. (A general $\mathrm{GL}(2,\mathbb{Z})$ transformation may reverse the orientation of the lattice.) In addition, a torus $\mathbb{C}/(\omega_1\mathbb{Z} \oplus \omega_2\mathbb{Z})$ is isomorphic to $\mathbb{C}/(\lambda\omega_1\mathbb{Z} \oplus \lambda\omega_2\mathbb{Z})$ where $\lambda \in \mathbb{C}$, so we can regard the $\mathrm{SL}(2,\mathbb{Z})$ transformations on $\omega_1, \omega_2$ as those on $\tau := \frac{\omega_2}{\omega_1}$.

## 3.2 Modular Functions

**Definition 3.1** (modular function and modular form)**.**

(1) A meromorphic function $f : \mathbb{H} \to \mathbb{C}$ is called a *weakly modular function* of *weight* $k \in \mathbb{Z}$ if it satisfies

$$f(\frac{a\tau + b}{c\tau + d}) = (c\tau + d)^k f(\tau) \quad \text{for any } \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}(2, \mathbb{Z}), \tag{3.6}$$

or equivalently,

$$f(\tau + 1) = f(\tau), \quad f(-\frac{1}{\tau}) = \tau^k f(\tau). \tag{3.7}$$

From this periodicity under $\tau \mapsto \tau + 1$, we can regard $f$ as a function of $q := e^{2\pi\sqrt{-1}\tau}$.

(2) A weakly modular function $f : \mathbb{H} \to \mathbb{C}$ is called a *modular function* if, as a function of $q$, it extends to a meromorphic function at $q = 0$. In such a case, we will simply say that $f$ is meromorphic at $q = 0$ or[17] at $\tau = \infty$.

By this definition, a modular function $f$ admits a Laurent expansion at $q = 0$,

$$f(\tau) = \sum_{n=n_0}^{\infty} c_n q^n. \tag{3.8}$$

(3) A modular function $f : \mathbb{H} \to \mathbb{C}$ is called a *weakly holomorphic modular form* if it is holomorphic on $\mathbb{H}$ but not necessarily at $q = 0$. That is, a pole is allowed to exist only at $q = 0$.

(4) A weakly holomorphic modular form $f : \mathbb{H} \to \mathbb{C}$ is called a *modular form* if it is holomorphic at $q = 0$. In this case, $f(\infty)$ denotes the value of (the extended) $f(\tau)$ at $q = 0$.

(5) A modular form $f : \mathbb{H} \to \mathbb{C}$ is called a *cusp form* if $f(\infty) = 0$.

■

There is no non-zero weakly modular function of odd weight. This is because, if $k$ is odd, we have $f(\tau) = -f(\tau)$ by applying $a = d = -1$ and $b = c = 0$ to (3.6).

Here are some examples of modular functions and modular forms.

[17] Here, $\infty$ denotes the point at infinity. Of course, as a limit, $q \to 0$ as $\mathrm{Im}(\tau) \to \infty$.

- Let $k \geq 4$ be an even integer. The *Eisenstein series* $G_k(\tau)$ of weight $k$ defined as follows is a modular form of weight $k$.

$$G_k(\tau) := \sum_{(m,n)\in\mathbb{Z}^2\setminus\{(0,0)\}} \frac{1}{(m+n\tau)^k} \tag{3.9}$$

$$= 2\zeta(k) + 2\frac{(2\pi\sqrt{-1})^k}{(k-1)!}\sum_{c=1}^{\infty}\sum_{d=1}^{\infty} d^{k-1}q^{cd} \tag{3.10}$$

$$= 2\zeta(k) + 2\frac{(2\pi\sqrt{-1})^k}{(k-1)!}\sum_{n=1}^{\infty}\left(\sum_{d|n} d^{k-1}\right)q^n. \tag{3.11}$$

Here, $\zeta(k) := \sum_{n=1}^{\infty}\frac{1}{n^k}$ is the Riemann zeta function, and its value for even $k$ is known to be $\zeta(k) = -\frac{(2\pi\sqrt{-1})^k}{2k!}B_k$, where $B_k$ is the Bernoulli number defined as $\frac{x}{e^x-1} = \sum_{k=0}^{\infty}\frac{B_k}{k!}x^k$. To calculate it, the recurrence relation $B_k = -\frac{1}{k+1}\sum_{i=0}^{k-1}\binom{k+1}{i}B_i$ and $B_0 = 1$ are useful.

The *normalized Eisenstein series*

$$E_k(\tau) := \frac{1}{2\zeta(k)}G_k(\tau) = 1 - \frac{2k}{B_k}\sum_{n=1}^{\infty}\left(\sum_{d|n} d^{k-1}\right)q^n \tag{3.12}$$

is also frequently used. For example, since $B_4 = -\frac{1}{30}$ and $B_6 = \frac{1}{42}$,

$$E_4(\tau) = \frac{720}{(2\pi)^4}G_4(\tau) = 1 + 240\sum_{n=1}^{\infty}\left(\sum_{d|n} d^3\right)q^n \tag{3.13}$$

$$= 1 + 240q + 2160q^2 + \cdots, \tag{3.14}$$

$$E_6(\tau) = \frac{30240}{(2\pi)^6}G_6(\tau) = 1 - 504\sum_{n=1}^{\infty}\left(\sum_{d|n} d^5\right)q^n \tag{3.15}$$

$$= 1 - 504q - 16632q^2 - \cdots. \tag{3.16}$$

- The *modular discriminant* $\Delta(\tau)$ defined as follows is a cusp form of weight 12.

$$\Delta(\tau) := (60G_4(\tau))^3 - 27(140G_6(\tau))^2 \tag{3.17}$$

$$= \frac{(2\pi)^{12}}{1728}(E_4(\tau)^3 - E_6(\tau)^2) \tag{3.18}$$

$$= (2\pi)^{12}\eta(\tau)^{24} \tag{3.19}$$

$$= q\prod_{n=1}^{\infty}(1-q^n)^{24}, \tag{3.20}$$

where $\eta(\tau)$ is the Dedekind eta function defined in Appendix E as (E.38). The normalized one

$$\tilde{\Delta}(\tau) := \frac{1}{(2\pi)^{12}}\Delta(\tau) = \eta(\tau)^{24} \tag{3.21}$$

is also frequently used.

- The *modular $j$-function* or the *$j$-invariant* $j(\tau)$ defined as follows is a weakly holomorphic modular form of weight 0.

$$j(\tau) := 1728\frac{(60G_4(\tau))^3}{\Delta(\tau)} = \frac{E_4(\tau)^3}{\tilde{\Delta}(\tau)} \tag{3.22}$$

$$= q^{-1} + 744 + 196884q + 21493760q^2 + \cdots . \tag{3.23}$$

  The weight of $j(\tau)$ being 0 means that $j : \mathbb{H} \to \mathbb{C}$ is a function which is also well-defined on $\mathbb{H}/\mathrm{SL}(2,\mathbb{Z})$. In addition, $j : \mathbb{H}/\mathrm{SL}(2,\mathbb{Z}) \to \mathbb{C}$ is known to be a bijection. As remarked in Section 3.1, elements of $\mathbb{H}/\mathrm{SL}(2,\mathbb{Z})$ are in one-to-one correspondence with the isomorphism classes of complex tori as $[\tau] \leftrightarrow [E_\tau]$. Therefore, $j(\tau)$ gives a complete invariant of the isomorphism classes of complex tori.

- Let $L$ be a positive-definite lattice of rank $n$. The *theta function of the lattice $L$* is defined as

$$\Theta_L(\tau) := \sum_{k\in L} q^{\frac{1}{2}|k|^2}. \tag{3.24}$$

  If $L$ is even self-dual, then $\Theta_L(\tau)$ is a modular form of weight $\frac{n}{2}$.

  For example, the theta function of the $E_8$ lattice is $\Theta_{E_8}(\tau) = E_4(\tau)$. We can see it as follows. The constant term of $\Theta_{E_8}(\tau)$ is of course $\Theta_{E_8}(\tau) = 1 + O(q)$. As in the following Theorem 3.2, the ring of modular forms is $\mathbb{C}[E_4, E_6]$, so this constant term determines the modular form $\Theta_{E_8}(\tau)$ of weight 4 as $\Theta_{E_8}(\tau) = E_4(\tau)$. In particular, we can say that

$$j(\tau) = \frac{\Theta_{E_8^{\oplus 3}}(\tau)}{\eta(\tau)^{24}}. \tag{3.25}$$

  More generally, we sometimes define the theta function $\Theta_L(\tau, \overline{\tau})$ of a lattice $L$ of signature $(r, s)$ (see (5.27)).

Obviously, all modular functions of the same weight constitute a $\mathbb{C}$-vector space. In addition, the multiplication of two modular functions of weight $m$ and $m'$ is a modular function of weight $m + m'$. Therefore, if we allow the sum of modular functions of inhomogeneous weights, then we obtain the graded ring of all (the inhomogeneous sums of) modular functions. Furthermore, its subrings are known to have the following structures. Let $R[X_1, \ldots, X_r]$ denote the polynomial ring over a commutative ring $R$.

**Theorem 3.2.**

- *The ring of modular functions of weight 0 constitutes the rational function field $\mathbb{C}(j)$.*
- *The ring of weakly holomorphic modular forms is $\mathbb{C}[E_4, E_6, \tilde{\Delta}^{-1}]$.*

  *In particular, the ring of weakly holomorphic modular forms of weight 0 is $\mathbb{C}[j]$.*

- *The ring of modular forms is* $\mathbb{C}[E_4, E_6]$.

  *In particular, there is no non-zero modular form of weight* $k < 4$.

- *The ring of weakly holomorphic modular forms with integral* $q$*-expansion coefficients (*$c_n \in \mathbb{Z}$ *in (3.8)) is*[18] $(\mathbb{Z}[E_4, E_6, \tilde{\Delta}]/\sim)[\tilde{\Delta}^{-1}]$*, where the relation* $\sim$ *is defined by* $1728\tilde{\Delta} \sim (E_4)^3 - (E_6)^2$.

- *The ring of modular forms with integral* $q$*-expansion coefficients (*$c_n \in \mathbb{Z}$ *in (3.8)) is* $\mathbb{Z}[E_4, E_6, \tilde{\Delta}]/\sim$*, with the above relation* $\sim$.

## 3.3 Weak Jacobi Forms

**Definition 3.3.** Let $\varphi : \mathbb{H} \times \mathbb{C} \to \mathbb{C}$ be a holomorphic function satisfying

$$\varphi(\frac{a\tau+b}{c\tau+d}, \frac{z}{c\tau+d}) = (c\tau+d)^k e^{2\pi\sqrt{-1}m\frac{cz^2}{c\tau+d}}\varphi(\tau, z) \quad \text{for any } \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}(2,\mathbb{Z}), \tag{3.26}$$

$$\varphi(\tau, z+\lambda\tau+\mu) = e^{-2\pi\sqrt{-1}m(\lambda^2\tau+2\lambda z)}\varphi(\tau, z) \qquad \text{for any } \lambda, \mu \in \mathbb{Z}, \tag{3.27}$$

where $k \in \mathbb{Z}$ is called the *weight* and $m \in \mathbb{Z}_{>0}$ is called the *index*. These transformations are equivalent to

$$\varphi(\tau+1, z) = \varphi(\tau, z), \qquad \varphi(-\frac{1}{\tau}, \frac{z}{\tau}) = \tau^k e^{2\pi\sqrt{-1}m\frac{z^2}{\tau}}\varphi(\tau, z), \tag{3.28}$$

$$\varphi(\tau, z+1) = \varphi(\tau, z), \qquad \varphi(\tau, z+\tau) = e^{-2\pi\sqrt{-1}m(\tau+2z)}\varphi(\tau, z). \tag{3.29}$$

(The last transformation $\varphi(\tau, z+\tau)$ also follows from those of $\varphi(-\frac{1}{\tau}, \frac{z}{\tau})$ and $\varphi(\tau, z+1)$.) From the periodicity under $\tau \mapsto \tau+1$ and $z \mapsto z+1$, we can regard $\varphi$ as a function of $q := e^{2\pi\sqrt{-1}\tau}$ and $y := e^{2\pi\sqrt{-1}z}$. We further assume that $\varphi$ admits a Fourier expansion

$$\varphi(\tau, z) = \sum_{n,r\in\mathbb{Z}} c(n,r)q^n y^r. \tag{3.30}$$

(1) $\varphi$ is called a *weakly holomorphic Jacobi form* if there exists $n_0 \in \mathbb{Z}$ such that $c(n,r) = 0$ for $n < n_0$.

(2) $\varphi$ is called a *weak Jacobi form* if $c(n,r) = 0$ for $n < 0$.

(3) $\varphi$ is called a *(holomorphic) Jacobi form* if $c(n,r) = 0$ for $4mn < r^2$.

(4) $\varphi$ is called a *Jacobi cusp form* if $c(n,r) = 0$ for $4mn \leq r^2$.

■

[18] See https://mathoverflow.net/questions/386020/modular-forms-over-mathbbz-vs-modular-forms-with-integral-fourier-coefficie.

If we set $m = 0$, then $\varphi(\tau, z)$ as a function of $z$ is a holomorphic doubly-periodic function, and such a function is known to be constant with respect to $z$. Therefore, a weakly holomorphic Jacobi form of weight $k$ and index 0 is just a modular function of weight $k$. We can also say that if $\varphi(\tau, z)$ is a weakly holomorphic Jacobi form of weight $k$ and index $m$, then $\varphi(\tau, 0)$ is a modular function of weight $k$. Similarly to the case of modular functions, there is no non-zero weakly holomorphic Jacobi form of odd weight.

Important examples of weak Jacobi forms are[19]

$$\varphi_{-2,1}(\tau, z) := -\frac{\theta_1(\tau, z)^2}{\eta(\tau)^6}, \tag{3.31}$$

$$\varphi_{0,1}(\tau, z) := 4\left(\frac{\theta_2(\tau, z)^2}{\theta_2(\tau, 0)^2} + \frac{\theta_3(\tau, z)^2}{\theta_3(\tau, 0)^2} + \frac{\theta_4(\tau, z)^2}{\theta_4(\tau, 0)^2}\right), \tag{3.32}$$

where $\theta_i(\tau, z)$ $(i = 1, 2, 3, 4)$ and $\eta(\tau)$ are the elliptic theta functions and the Dedekind eta function, respectively, introduced in Appendix E. These $\varphi_{k,m}(\tau, z)$ are weak Jacobi forms of weight $k$ and index $m$.

Similarly to the case of modular functions, we have the graded ring of all (the inhomogeneous sums of) weak Jacobi forms. It has the following structure. If we write the $\mathbb{C}$-vector space of modular forms of weight $k$ as $M_k$, then recall that the graded ring of modular forms is $\bigoplus_{k\in\mathbb{Z}} M_k = \mathbb{C}[E_4, E_6]$.

**Theorem 3.4.** *Let $J^{\mathrm{weak}}_{k,m}$ denote the ring of weak modular forms of weight $k$ and index $m$. The ring of weak Jacobi forms has the structure*

$$\bigoplus_{k\in\mathbb{Z}, m\in\mathbb{Z}_{>0}} J^{\mathrm{weak}}_{k,m} = \left(\bigoplus_{k\in\mathbb{Z}} M_k\right)[\varphi_{-2,1}, \varphi_{0,1}] \tag{3.33}$$

$$= \mathbb{C}[E_4, E_6, \varphi_{-2,1}, \varphi_{0,1}]. \tag{3.34}$$

*In particular,*

$$J^{\mathrm{weak}}_{k,m} = \bigoplus_{j=0}^{m} M_{k+2j} \cdot \varphi^j_{-2,1}\varphi^{m-j}_{0,1}. \tag{3.35}$$

Table 3.1: Generators of the ring of weak Jacobi forms.

| | weight $k$ | index $m$ | Fourier expansion |
|---|---|---|---|
| $E_4$ | 4 | 0 | $1 + 240q + 2160q^2 + \cdots$ |
| $E_6$ | 6 | 0 | $1 - 504q - 16632q^2 - \cdots$ |
| $\varphi_{-2,1}$ | $-2$ | 1 | $y - 2 + y^{-1} - 2(y - 2 + y^{-1})^2 q + \cdots$ |
| $\varphi_{0,1}$ | 0 | 1 | $y + 10 + y^{-1} + (10y^2 - 64y + 108 - 64y^{-1} + 10y^{-2})q + \cdots$ |

[19] The minus sign in (3.31) is needed to obtain $\varphi_{-2,1}(\tau, z) = (y^{\frac{1}{2}} - y^{-\frac{1}{2}})^2 + \cdots$, because we have $\sqrt{-1}$ in the definition (E.21) of $\theta_1$.

**Extremal elliptic genus**

As an example, the weak Jacobi form $Z^{m=4}_{\mathrm{ext}}(\tau,z)$ of weight 0 and index 4, starting with the terms

$$Z^{m=4}_{\mathrm{ext}}(\tau,z) = y^4 + 0(y^3+y^2+y) + \cdots \tag{3.36}$$

can be determined from (3.35) and Table 3.1 as

$$Z^{m=4}_{\mathrm{ext}}(\tau,z) = \frac{1}{432}\varphi^4_{0,1} + \frac{1}{8}E_4\varphi^2_{-2,1}\varphi^2_{0,1} + \frac{11}{27}E_6\varphi^3_{-2,1}\varphi_{0,1} + \frac{67}{144}E_4^2\varphi^4_{-2,1}. \tag{3.37}$$

This is an example of the $\mathcal{N}=2$ extremal elliptic genus defined[20] by [GGK+08].

**Definition 3.5** ($\mathcal{N}=2$ extremal elliptic genus)**.** We define the *polar region* of index $m \in \mathbb{Z}_{>0}$ as

$$\mathcal{P}^{(m)} := \{(n,r) \in \mathbb{Z}^2 \mid 0 \le n, 4mn < r^2, 1 \le r \le m\}, \tag{3.38}$$

and the terms $q^n y^r$ of the Fourier expansion of a function $f(\tau,z)$ such that $(n,r) \in \mathcal{P}^{(m)}$ are called the *polar terms* of index $m$ of $f$.

An $\mathcal{N}=2$ *extremal elliptic genus* $Z^m_{\mathrm{ext}}(\tau,z)$ of index $m$ or of central charge $6m$ is a weak Jacobi form of weight 0 and index $m$ such that its polar terms of index $m$ coincide with the polar terms of index $m$ of the character of the Ramond vacuum representation[21] of the $\mathcal{N}=2$ superconformal algebra of central charge $c=6m$. Such polar terms can be explicitly calculated as the polar terms of

$$(-1)^m(1-q)y^m\prod_{l=1}^{\infty}\frac{(1-yq^{l+1})(1-y^{-1}q^l)}{(1-q^l)^2}. \tag{3.39}$$

■

According to [GGK+08], the number of polar terms of index $m$ exceeds the number of $\mathbb{C}$-linearly independent terms of $J^{\mathrm{weak}}_{0,m}$ in (3.35). Therefore, an $\mathcal{N}=2$ extremal elliptic genus $Z^m_{\mathrm{ext}}(\tau,z)$ is unique if it exists. In addition, they showed that it exists only when $m=1,2,3,4,5,7,8,11,13$ for $m \le 36$, and it does not exist for sufficiently large $m$. So they conjecture that it exists only for the listed values of $m$. The $\mathcal{N}=2$ extremal elliptic genera for $m=1,2,3,4$

[20] An *extremal VOA* is defined in [Hoe07] as a self-dual VOA of central charge $c$ such that its minimal non-zero conformal weight is greater than $[\frac{c}{24}]$ (see also [HÖ8]). The extremal VOAs (CFTs) are studied in [Wit07] in relation to three-dimensional gravity. Inspired by [Wit07], the $\mathcal{N}=2$ and $\mathcal{N}=4$ extremal extremal elliptic genera are defined in [GGK+08].

[21] More precisely, the Ramond vacuum representation means the Ramond massless representation of conformal weight $\frac{c}{24}$ and $\mathrm{U}(1)$ charge $\frac{c}{6}$ [CDD+14, §7] (which is also called the Ramond graviton representation of the same weight and charge [Egu04]). The quantity (3.39) is obtained as the $\frac{1}{2}$-spectral flow of the character of the NS vacuum representation (the NS graviton representation of conformal weight 0 and $\mathrm{U}(1)$ charge 0 [Egu04]). This (3.39) is the terms of the character of the Ramond vacuum representation, containing all its polar terms.

are [GGK+08, BDFK15]

$$Z_{\rm ext}^{m=1}(\tau,z)=\varphi_{0,1}, \tag{3.40}$$

$$Z_{\rm ext}^{m=2}(\tau,z)=\frac{1}{6}\varphi_{0,1}^2+\frac{5}{6}E_4\varphi_{-2,1}^2, \tag{3.41}$$

$$Z_{\rm ext}^{m=3}(\tau,z)=\frac{1}{48}\varphi_{0,1}^3+\frac{7}{16}E_4\varphi_{-2,1}^2\varphi_{0,1}+\frac{13}{24}E_6\varphi_{-2,1}^3, \tag{3.42}$$

$$Z_{\rm ext}^{m=4}(\tau,z)=\frac{1}{432}\varphi_{0,1}^4+\frac{1}{8}E_4\varphi_{-2,1}^2\varphi_{0,1}^2+\frac{11}{27}E_6\varphi_{-2,1}^3\varphi_{0,1}+\frac{67}{144}E_4^2\varphi_{-2,1}^4. \tag{3.43}$$

## 3.4 Elliptic Genus

Weak Jacobi forms appear in physics as the elliptic genus of $\mathcal{N}=2$ SCFT. (Be careful not to confuse it with the extremal $\mathcal{N}=2$ elliptic genus, which was purely mathematically defined as above.) To explain what it is, let us begin with the Witten index of a chiral $\mathcal{N}=1$ SCFT.

In the R sector of a chiral $\mathcal{N}=1$ SCFT of central charge $c$, there is a supercharge $G_0$ such that $(G_0)^2=L_0-\frac{c}{24}$ (see (4.23) below). We then have, for a general state $|h\rangle$ with $L_0$-eigenvalue $h$ in the R sector,

$$|G_0|h\rangle|^2=\langle h|(L_0-\frac{c}{24})|h\rangle=h-\frac{c}{24}, \tag{3.44}$$

where $G_0$ is a Hermitian operator. As a result, unless $h=\frac{c}{24}$, $G_0|h\rangle$ is a non-zero state. Since $[G_0,L_0]=0$ and $\{G_0,(-1)^F\}=0$, where $(-1)^F$ is the fermion parity operator, we can say that $G_0|h\rangle$ is a state with the same $L_0$-eigenvalue and opposite parity compared with $|h\rangle$, when $h\neq\frac{c}{24}$. Therefore, $|h\rangle$ and $G_0|h\rangle$ cancel in the trace over the R sector $\mathcal{H}_{\rm R}$

$$\mathrm{Tr}_{\mathcal{H}_{\rm R}}(-1)^F q^{L_0-\frac{c}{24}}, \tag{3.45}$$

and only the states with $h=\frac{c}{24}$ contribute. So the quantity (3.45) is a constant. A state with $h=\frac{c}{24}$ in the R sector is called a *Ramond vacuum*,[22] and the constant (3.45) is counting the number of Ramond vacua with sign $(-1)^F$. This constant is called the *Witten index*. (We can also define the Witten index for a general (not necessarily conformal) supersymmetric theory in a similar way.)

The elliptic genus is a concept similar to the Witten index. We define the *elliptic genus* of an $\mathcal{N}=(0,1)$ SCFT which is non-chiral (consisting of left- and right-moving parts) as

$$Z_{\rm ell}(\tau):=\mathrm{Tr}_{\mathcal{H}_{\rm R\widetilde{R}}}(-1)^{F+\tilde{F}}q^{L_0-\frac{c}{24}}\bar{q}^{\widetilde{L}_0-\frac{\tilde{c}}{24}}, \tag{3.46}$$

where $\mathcal{H}_{\rm R\widetilde{R}}$ is the Hilbert space of the entire R sector, and the objects without tilde and with tilde $\widetilde{\ }$ denote the objects in left- and right-moving parts respectively. The entire R sector has the structure $\mathcal{H}_{\rm R\widetilde{R}}=\bigoplus_i\mathcal{H}_{{\rm R},i}\otimes\mathcal{H}_{\widetilde{\rm R},i}$, and by the discussion similar to above, we can say that the

[22] Be careful not to confuse $|0\rangle_{\rm R}$ introduced in (6.20) with a Ramond vacuum. In fact, the conformal weight of $|0\rangle_{\rm R}$ of the $n$ real free chiral fermions is $\frac{n}{16}=\frac{c}{8}$, and the theory is not supersymmetric.

sectors containing the right-moving Ramond vacua $\bigoplus_{i \text{ such that } \mathcal{H}_{\widetilde{\mathrm{R}},i}=\mathrm{vac}} \mathcal{H}_{\mathrm{R},i} \otimes \mathcal{H}_{\widetilde{\mathrm{R}},i}$ only contribute to the trace (3.46). Therefore, the elliptic genus $Z_{\mathrm{ell}}(\tau)$ is independent of $\bar{q}$.

We regard a non-chiral $\mathcal{N} = (\mathcal{N}_l, \mathcal{N}_r)$ SCFT with $\mathcal{N}_l \geq 0$ and $\mathcal{N}_r \geq 1$ as a special case of non-chiral $\mathcal{N} = (0,1)$ SCFTs. If $\mathcal{N}_l \geq 1$, then it further follows that the elliptic genus $Z_{\mathrm{ell}}(\tau)$ is independent of $q$, and hence a constant. We may then call it the Witten index.

If $\mathcal{N}_l \geq 2$, then the quantity

$$Z_{\mathrm{ell}}(\tau, z) := \mathrm{Tr}_{\mathcal{H}_{\mathrm{R}\widetilde{\mathrm{R}}}} (-1)^{F+\tilde{F}} y^{J_0} q^{L_0 - \frac{c}{24}} \bar{q}^{\widetilde{L}_0 - \frac{\tilde{c}}{24}}, \tag{3.47}$$

where $J_0$ is the $\mathrm{U}(1)$ charge (the zero mode of $J(z)$) of the $\mathcal{N} = 2$ superconformal algebra (B.1)–(B.7), is also called the *elliptic genus*. In fact, this (3.47) is the traditional definition of the elliptic genus, but recently, the scope of the term "elliptic genus" seems to slightly broadened so that it includes the quantity $Z_{\mathrm{ell}}(\tau) = Z_{\mathrm{ell}}(\tau, 0)$ in (3.46).

Moreover, for a fermionic chiral CFT, by regarding it as coupling to a trivial right-moving $\mathcal{N}_r = 1$ theory, its elliptic genus is

$$Z_{\mathrm{ell}}(\tau) = \mathrm{Tr}_{\mathcal{H}_{\mathrm{R}}} (-1)^F q^{L_0 - \frac{c}{24}}. \tag{3.48}$$

If the chiral CFT has $\mathcal{N} = 1$ supersymmetry, then this is just the Witten index. If $\mathcal{N} \geq 2$, then the quantity

$$Z_{\mathrm{ell}}(\tau, z) = \mathrm{Tr}_{\mathcal{H}_{\mathrm{R}}} (-1)^F y^{J_0} q^{L_0 - \frac{c}{24}}, \tag{3.49}$$

is also the elliptic genus.

Let us come back to the weak Jacobi forms. We consider a non-chiral $\mathcal{N} = (2,1)$ SCFT of central charge $(c, \tilde{c})$ or a chiral $\mathcal{N} = 2$ SCFT of central charge $c$, such that the phases caused by the gravitational anomaly in the modular transformations (1.4, 1.5) are trivial, that is, $2(\tilde{c} - c) \equiv 0 \bmod 24$. If the $\mathrm{U}(1)$ charges (the $J_0$-eigenvalues) of the states in the NS sector of the SCFT are all integers, and $\frac{c}{3}$ is a positive integer, then the elliptic genus $Z_{\mathrm{ell}}(\tau, z)$ in (3.47) or (3.49) is known [KYY93] to be a weak Jacobi form of weight 0 and index $\frac{c}{6}$. (There is also the concept of a weak Jacobi form of half-integer index.) The transformation (3.26) follows from the modular invariance, and the transformation (3.27) follows from the spectral flow.

We have defined the extremal $\mathcal{N} = 2$ elliptic genera in Section 3.3. Constructing an SCFT whose elliptic genus coincides with an extremal $\mathcal{N} = 2$ elliptic genus is a nontrivial problem. For example, $Z_{\mathrm{ext}}^{m=1}(\tau, z)$ is the elliptic genus of the K3 CFT (Section 7) divided by 2. Duncan's module will be introduced as an $\mathcal{N} = 1$ chiral SCFT in Section 6.2, but it also admits $\mathcal{N} = 2$ and $\mathcal{N} = 4$ superconformal algebras, and its elliptic genus as an $\mathcal{N} = 2$ theory is $Z_{\mathrm{ext}}^{m=2}(\tau, z)$ [CDD+14]. An $\mathcal{N} = 2$ chiral SCFT with elliptic genus $Z_{\mathrm{ext}}^{m=4}(\tau, z)$ is constructed in [BDFK15] from the odd Leech lattice. There is also the concept of $\mathcal{N} = 4$ extremal elliptic genera $Z_{\mathrm{ext}}^{m,\mathcal{N}=4}(\tau, z)$ [GGK+08]. Duncan's module as $\mathcal{N} = 4$ theory has the elliptic genus $Z_{\mathrm{ext}}^{m=2,\mathcal{N}=4}(\tau, z)$. An $\mathcal{N} = 4$ chiral SCFT with $Z_{\mathrm{ext}}^{m=4,\mathcal{N}=4}(\tau, z)$ is constructed in [Har16]. See [FH17, KY23b] for more on SCFTs with extremal elliptic genera.

# 4 Vertex Operator Algebras

The axiomatic definition of a VOA is somewhat technical. In practice, in this thesis, it suffices to consider the examples of VOAs introduced in the next Part II, whose constructions would be more familiar from the viewpoint of CFTs in physics. We will not go into the proof that these examples actually satisfy the axioms of a VOA. Even so, reviewing the definition of a VOA is more or less helpful for understanding how VOAs mathematically formulate CFTs in physics, so we will summarize it in this Section 4.

The historical development of the concept of VOA is summarized in [FLM88, Introduction §III]. It developed through three stages: vertex operator, vertex algebra, and vertex operator algebra.

**Vertex operator (a historical note)**

Lepowsky and Wilson constructed a certain representation of the affine Lie algebra $\widehat{\mathfrak{sl}}(2,\mathbb{C})$ in [LW78], using the operators which are now called *twisted vertex operators*, which was generalized to other affine Lie algebras in [KKLW81]. Garland noticed the similarity between those operators and the vertex operators in a physics theory called the dual resonance theory, and it was confirmed that this resemblance can actually be made into a complete coincidence in the works by Frenkel and Kac [FK80], and by Segal [Seg81] independently, where representations of the affine Lie algebras were constructed using the (untwisted) *vertex operators*.

In [FLM84], Frenkel, Lepowsky, and Meurman constructed a representation, which we now call the moonshine module $V^\natural$, of a certain algebra $\hat{B}^\natural$, an "affinization" of the Griess algebra $B^\natural$, using vertex operators. They also showed that the monster group $\mathbb{M}$ acts on $V^\natural$, and the graded character of $V^\natural$ is the modular $j$-function (without the constant term).

## 4.1 Vertex Algebras

Motivated by [FLM84], Borcherds [Bor86] introduced the axioms of a *vertex algebra*.[23] Following [Har99, §5.2], they can be summarized as follows.

**Definition 4.1** (vertex algebra)**.** A vector space $V$ over a field $\mathbb{K}$ of characteristic 0 is called a *vertex algebra* if it satisfies the following axioms.

(1) For any $v \in V$ and any $n \in \mathbb{Z}$, a $\mathbb{K}$-linear endomorphism $v_{(n)} \in \mathrm{End}(V)$ is given.

We define a formal series $Y(v,z) \in \mathrm{End}(V)[[z,z^{-1}]]$ of $z$, called the *vertex operator* corresponding to $v$, as

$$Y(v,z) := \sum_{n\in\mathbb{Z}} v_{(n)} z^{-n-1}. \tag{4.1}$$

[23] Borcherds posted on MathOverflow that his definition of a vertex algebra was purely motivated by an attempt to understand the works by Frenkel, Lepowsky, and Meurman, and he did not use any insights from field theories in physics, simply because he was barely familiar with such topics. https://mathoverflow.net/questions/53988/what-is-the-motivation-for-a-vertex-algebra

The map $V \to \mathrm{End}(V)[[z, z^{-1}]]; v \mapsto Y(v, z)$ is often called the *state-field correspondence*, and we require it to be $\mathbb{K}$-linear.

(2) For any $v, w \in V$, there exists $n \in \mathbb{Z}$ such that $v_{(m)}w = 0$ for any $m \geq n$.

(3) For any $u, v, w \in V$ and $m, n, q \in \mathbb{Z}$,

$$\sum_{i \geq 0} \binom{m}{i} (u_{(q+i)}v)_{(m+n-i)}w = \sum_{i \geq 0} (-1)^i \binom{q}{i} [u_{(m+q-i)}(v_{(n+i)}w) - (-1)^q v_{(n+q-i)}(u_{(m+i)}w)]. \tag{4.2}$$

This equation (4.2) is called the *Borcherds identity* (or the *Jacobi identity*).

(4) There is a specific element $\mathbf{1} \in V$ called the *vacuum vector*, which satisfies for any $v \in V$, $v_{(-1)}\mathbf{1} = v$, and $v_{(n)}\mathbf{1} = 0$ for any $n \geq 0$.

■

Following physics convention, any element $v \in V$ of a vertex algebra $V$ is called a *state*, and a vertex operator $Y(v, z)$ a *field*, a *current*, or simply an *operator*. Sometimes $v$ is also called a current. The operator $v_{(n)}$ is called the *$n$-th mode* of the mode expansion of $Y(v, z)$, or simply of $v$.

The Borcherds identity (4.2) is equivalent to

$$z_0^{-1}\delta\left(\frac{z_1 - z_2}{z_0}\right) Y(u, z_1)Y(v, z_2)w - z_0^{-1}\delta\left(\frac{z_2 - z_1}{-z_0}\right) Y(v, z_2)Y(u, z_1)w = z_2^{-1}\delta\left(\frac{z_1 - z_0}{z_2}\right) Y(Y(u, z_0)v, z_2)w, \tag{4.3}$$

where

$$\delta(z) = \sum_{n \in \mathbb{Z}} z^n, \tag{4.4}$$

and we break down

$$\delta\left(\frac{z_1 - z_2}{z_0}\right) = \sum_{n \in \mathbb{Z}} \left(\frac{z_1 - z_2}{z_0}\right)^n \tag{4.5}$$

into an infinite sum of the terms in the form of $z_0^i z_1^j z_2^k$, under the definition

$$(a - b)^n := a^n(1 - \frac{b}{a})^n := \sum_{i=0}^{\infty} \binom{n}{i} a^{n-i}(-b)^i \quad \text{for } n < 0. \tag{4.6}$$

In particular, $(a - b)^n$ is not equal to $(-b + a)^n$ if $n < 0$, under this definition.

There are some variants of the definitions of vertex algebras. For example, [Kac98, CKLW15] adopt the following definition. We will use[24] the notation $[A, B] := AB - BA$.

[24] [Kac98] deal with vertex superalgebras, rather than vertex algebras, throughout the entire book. The definition of the bracket there is $[A, B] := AB - (-1)^{|A||B|}BA$. See also the description of vertex superalgebras in the main text below.

**Definition 4.2** (vertex algebra (another definition))**.** A vector space $V$ over $\mathbb{C}$ is called a *vertex algebra* if it satisfies the following axioms.

(1) For any $v \in V$ and any $n \in \mathbb{Z}$, a $\mathbb{C}$-linear endomorphism $v_{(n)} \in \mathrm{End}(V)$ is given.

The formal series $Y(v,z) := \sum_{n\in\mathbb{Z}} v_{(n)} z^{-n-1}$ is called the *vertex operator* corresponding to $v$, and the *state-field correspondence* $V \to \mathrm{End}(V)[[z,z^{-1}]]; v \mapsto Y(v,z)$ is required to be a linear map.

(2) For any $u, v \in V$, there exists $n \in \mathbb{Z}$ such that $u_{(m)}v = 0$ for any $m \geq n$.

A formal series $\sum_{n\in\mathbb{Z}} a_n z^{-n-1} \in \mathrm{End}(V)[[z,z^{-1}]]$ such that for any $w \in V$, $a_n w = 0$ for sufficient large $k$ is called a *field*. The vertex operators $Y(v,z)$ are fields, and the state-field correspondence is a linear map from $V$ to the space of fields.

(3) For any $v, w \in V$, there exists $N \in \mathbb{Z}_{>0}$ such that

$$(z_0 - z_1)^N [Y(v,z_0), Y(w,z_1)] = 0. \tag{4.7}$$

This condition is called the *locality*.

(4) There is a specific operator $T \in \mathrm{End}(V)$ called the *infinitesimal translation operator*, which satisfies

$$[T, Y(v,z)] = \frac{d}{dz} Y(v,z) \tag{4.8}$$

for any $v \in V$. This condition is called the *translation covariance*.

(5) There is a specific element $\mathbf{1} \in V$ called the *vacuum vector*, which satisfies $T\mathbf{1} = 0$, $Y(\mathbf{1},z) = \mathrm{id}_V$, and $v_{(-1)}\mathbf{1} = v$ for any $v \in V$.

■

Even if we replace the condition $v_{(-1)}\mathbf{1} = v$ with

$$Y(v,z)\mathbf{1}|_{z=0} = v, \tag{4.9}$$

we obtain an equivalent definition [CKLW15, Remark 4.1]. We can also show [Kac98, Cor. 4.4 (c)]

$$[T, Y(v,z)] = Y(Tv,z) = \frac{d}{dz} Y(v,z). \tag{4.10}$$

A vertex algebra of this Definition 4.2 satisfies the Borcherds identity (4.2) of Definition 4.1 [Kac98, §4.8].

## 4.2 Vertex Operator Algebras

Finally, Frenkel, Lepowsky, and Meurman [FLM88] introduced the axioms of a *vertex operator algebra*. Following [Har99, §5.2], they can be summarized as follows.

**Definition 4.3** (vertex operator algebra)**.** A vector space $V$ over a field $\mathbb{K}$ of characteristic 0 is called a *vertex operator algebra (VOA)* if it satisfies the following axioms.

(1) $V$ is a vertex algebra in the sense of Definition 4.1, with the state-field correspondence $v \mapsto Y(v,z)$ and the vacuum vector $\mathbf{1}$.

(2) $Y(\mathbf{1},z) = \mathrm{id}_V$.

(3) There is a specific element $\omega \in V$ called the *Virasoro element* such that if we define $L_n := \omega_{(n+1)}$, that is,

$$Y(\omega,z) = \sum_{n\in\mathbb{Z}} L_n z^{-n-2}, \tag{4.11}$$

then $\{L_n\}_{n\in\mathbb{Z}}$ satisfy the commutation relations of the *Virasoro algebra*

$$[L_m, L_n] = (m-n)L_{m+n} + \frac{c}{12}(m^3-m)\delta_{m,-n}, \tag{4.12}$$

where $c \in \mathbb{K}$ is called the *central charge*.

(4) $L_0$ is diagonalizable on $V$, and its eigenvalues lie in $\mathbb{Z}$. So the eigenspace decomposition of $V$ with respect to $L_0$ defines a $\mathbb{Z}$-grading on $V$ as $V = \bigoplus_{h\in\mathbb{Z}} V_h$.

If $v \in V_h$, then $v$ is called a *homogeneous* element of *(conformal) weight* or *(conformal) dimension* $h =: \mathrm{wt}(v)$.

(5) Every subspace $V_h$ is finite-dimensional, and there is $h_{\min} \in \mathbb{Z}$ such that $V_h = 0$ for any $h < h_{\min}$.

(6) For any $v \in V$,

$$Y(L_{-1}v,z) = \frac{d}{dz}Y(v,z). \tag{4.13}$$

■

In addition, a VOA is said to be *of CFT type* [DLMM98], if $h_{\min} = 0$ and $V_0 = \mathbb{K}\mathbf{1}$.

Following physics convention, the operator $T(z) := Y(\omega,z)$ is called the *energy-momentum tensor*. We can show $\mathrm{wt}(\omega) = 2$, because $L_0\omega = L_0L_{-2}\mathbf{1} = 2L_{-2}\mathbf{1} = 2\omega$, using the Definition 4.1 of the vacuum vector $\mathbf{1}$ and (4.12).

There are some variants of the definitions of VOAs. For example, the corresponding concept in [Kac98, CKLW15] is a *conformal vertex algebra*, which is defined as a vertex algebra $V$ in

the sense of Definition 4.2, with a specific element $\omega \in V$ such that $\{L_n\}_{n\in\mathbb{Z}}$ defined as in (4.11) satisfy (4.12), $L_0$ is diagonalizable on $V$ (which defines a $\mathbb{C}$-grading on $V$ in general), and $L_{-1} = T$ (compare (4.10) and (4.13)).

Here is one caveat on the convention of the weight. From the axioms of VOAs, we can show [Kac98, Eq. (4.9.3) and Theorem 4.10 (e)] that for a homogeneous $v \in V$,

$$[L_0, v_{(n)}] = -(n + 1 - \mathrm{wt}(v))v_{(n)}. \tag{4.14}$$

Therefore, $v_{(n)}$ maps $V_h$ to $V_{h-(n+1-\mathrm{wt}(v))}$. In physics literature, however, the usual convention is

$$Y(v, z) = \sum_{n\in\mathbb{Z}} v_{(n;\mathrm{phys})} z^{-n-\mathrm{wt}(v)}, \tag{4.15}$$

and hence

$$v_{(n)} = v_{(n+1-\mathrm{wt}(v);\mathrm{phys})}. \tag{4.16}$$

Some mathematics literature e.g. [Kac98, CKLW15] also introduces this convention. Then

$$[L_0, v_{(n;\mathrm{phys})}] = -nv_{(n;\mathrm{phys})}, \tag{4.17}$$

and therefore $v_{(n;\mathrm{phys})}$ maps $V_h$ to $V_{h-n}$.

**More on vertex algebras and vertex operator algebras**

If a subspace $W$ of a vertex algebra or a VOA $V$ forms a vertex algebra or a VOA under the same structure as $V$, then $W$ is called a *vertex subalgebra* or a *sub-VOA* of $V$, respectively.

A *module* or a *representation* of a vertex algebra $V$ is a vector space $M$ such that for any $v \in V$, a field

$$Y^M(v, z) = \sum_{n\in\mathbb{Z}} v^M_{(n)} z^{-n-1}, \quad v^M_{(n)} \in \mathrm{End}(M), \tag{4.18}$$

is given, the state-field correspondence $v \mapsto Y^M(v, z)$ is linear, $Y^M(\mathbf{1}, z) = \mathrm{id}_M$, and the Borcherds identity (4.2) for them holds. A vertex algebra $V$ itself is a module of $V$, and sometimes called the *adjoint module*. A *($\mathbb{Z}_N$-)twisted module* of a vertex operator uses $\mathrm{End}(M)[[z^{\frac{1}{N}}, z^{-\frac{1}{N}}]]$ ($N \in \mathbb{Z}_{>0}$) instead of $\mathrm{End}(M)[[z, z^{-1}]]$. Some literature [DL93] also introduces a concept we may call "twisted vertex algebras", which use $\mathrm{End}(V)[[z^{\frac{1}{N}}, z^{-\frac{1}{N}}]]$ instead of $\mathrm{End}(V)[[z, z^{-1}]]$.

A *vertex superalgebra* introduces an additional $\mathbb{Z}_2$-grading $V = V^{\bar{0}} \oplus V^{\bar{1}}$ called the *parity* to a vertex algebra, and the Borcherds identity (4.3) is generalized to the super version

$$z_0^{-1}\delta\left(\frac{z_1 - z_2}{z_0}\right) Y(u, z_1)Y(v, z_2)w - (-1)^{|u||v|} z_0^{-1}\delta\left(\frac{z_2 - z_1}{-z_0}\right) Y(v, z_2)Y(u, z_1)w = z_2^{-1}\delta\left(\frac{z_1 - z_0}{z_2}\right) Y(Y(u, z_0)v, z_2)w, \tag{4.19}$$

where $|v| := 0, 1$ for $v \in V^{\bar{0}}, V^{\bar{1}}$, respectively. Correspondingly, the locality (4.7) is generalized to

$$(z_0 - z_1)^N (Y(v, z_0)Y(w, z_1) - (-1)^{|v||w|} Y(w, z_1)Y(v, z_0)) = 0. \tag{4.20}$$

[DL93] further introduces *generalized vertex algebras*, which subsume both twisted vertex algebras and vertex superalgebras as special cases.

A *vertex operator superalgebra (VOSA)* is a VOA with the parity $\mathbb{Z}_2$-grading $V = V^{\bar{0}} \oplus V^{\bar{1}}$ making it a vertex superalgebra, and the weight $\frac{1}{2}\mathbb{Z}$-grading $V = \bigoplus_{h \in \frac{1}{2}\mathbb{Z}} V_h$ instead of the $\mathbb{Z}$-grading. Sometimes $V^{\bar{0}} = \bigoplus_{h \in \mathbb{Z}} V_h$ and $V^{\bar{1}} = \bigoplus_{h \in \mathbb{Z}+\frac{1}{2}} V_h$ are further required [DMC14, §2.1]. Despite of its name "super," a VOSA just describes a fermionic CFT in physics, not necessarily with a supersymmetry. For the description of SCFT in physics, we introduce the following structures.

An $\mathcal{N} = 1$ *vertex operator superalgebra* requires the existence of a specific element $\tau \in V^{\bar{1}}_{\frac{3}{2}}$ such that if we define $G_r := \tau_{(r+\frac{1}{2})}$, that is,

$$Y(\tau, z) = \sum_{r \in \mathbb{Z}+\frac{1}{2}} G_r z^{-r-\frac{3}{2}}, \tag{4.21}$$

then $\{G_r\}_{r \in \mathbb{Z}+\frac{1}{2}}$ satisfy the $\mathcal{N} = 1$ superconformal algebra

$$[L_m, G_r] = \Big(\frac{1}{2}m - r\Big) G_{m+r}, \tag{4.22}$$

$$\{G_r, G_s\} = 2L_{r+s} + \frac{c}{3}\Big(r^2 - \frac{1}{4}\Big)\delta_{r,-s}, \tag{4.23}$$

together with (4.12). This element $\tau$ or its operator $G(z) := Y(\tau, z)$ is called the *supercurrent*. $\mathcal{N} = 2$ and $\mathcal{N} = 4$ *vertex operator superalgebras* are also defined in a similar way, by requiring the existence of elements whose corresponding fields satisfy the $\mathcal{N} = 2$ and $\mathcal{N} = 4$ superconformal algebras, respectively.

As for more details on these definitions, see for example [DL93, Kac98, FBZ04] and references in [DMC14, §2.1].

VOAs formulate CFTs from the viewpoint of state-field correspondence, while there is another mathematical formulation of CFTs, called *local conformal nets*, which focuses on algebras of local operators. These two formulations are useful in different situations, and the translation between them is one of the research topics in the field of operator algebras. See for example [CKLW15, Kaw15, Kaw17].

# Part II
# Review of Moonshine Phenomena

This Part II is a review of moonshine phenomena and an introduction to important examples of VOAs. The most classical example of moonshine phenomena is the monstrous moonshine, and its underlying object, the monster VOA, is constructed as the $\mathbb{Z}_2$-orbifold of the Leech lattice VOA. So, in Section 5, after the review of the monstrous moonshine, we will introduce the general construction of lattice VOAs. Another well-established moonshine phenomenon is the Conway moonshine. In Section 6, we will review that the Conway moonshine module, also known as Duncan's module, is constructed from a Clifford module VOSA, which describes the CFT of free fermions. In the last Section 7, we will review a relatively new and notable example of moonshine phenomena, the K3 Mathieu moonshine, whose mysterious nature is not fully understood yet.

## 5 Monstrous Moonshine and Lattice VOA

The monstrous moonshine is the first discovered example of a moonshine phenomenon, and the origin of this research field. As we will review in Section 5.1, it was observed as an empirical relationship between the monster group $\mathbb{M}$ and the modular $j$-function $j(\tau)$, and theoretically explained by the existence of an underlying VOA called the monster VOA $V^\natural$. The monster VOA $V^\natural$ is the $\mathbb{Z}_2$-orbifold of the Leech lattice VOA. So we will review the general construction of lattice VOAs in Section 5.2, and their $\mathbb{Z}_2$-orbifolds in Section 5.3. In particular, the structure of the automorphism group of a lattice VOA explained in Section 5.2 will be our main subject of study in Part III.

### 5.1 Monstrous Moonshine

In 1978 and 1979, McKay and Thompson [Tho79b, McK01] observed that the first several coefficients (except for the constant term) of the modular $j$-function

$$j(\tau) = q^{-1} + 744 + 196884q + \cdots \tag{5.1}$$

$$= \sum_{i=-1}^{\infty} c_i q^i, \tag{5.2}$$

can be written as simple sums of irreducible representation dimensions of the monster group $\mathbb{M}$ as follows. The irreducible representation dimensions of $\mathbb{M}$ are, from the smallest one,

| $i$ | 1 | 2 | 3 | 4 | 5 | 6 | 7 | $\cdots$ |
|---|---|---|---|---|---|---|---|---|
| $\chi_i(1_{\mathbb{M}})$ | 1 | 196883 | 21296876 | 842609326 | 18538750076 | 19360062527 | 293553734298 | $\cdots$ |

(5.3)

where $\chi_i$ denotes the $i$-th irreducible character of $\mathbb{M}$. The coefficients of the modular $j$-function are then

$$\begin{aligned}
c_1 &= 196884 = \chi_1(1) + \chi_2(1), && (5.4)\\
c_2 &= 21493760 = \chi_1(1) + \chi_2(1) + \chi_3(1), && (5.5)\\
c_3 &= 864299970 = 2\chi_1(1) + 2\chi_2(1) + \chi_3(1) + \chi_4(1), && (5.6)\\
c_4 &= 20245856256 = 3\chi_1(1) + 3\chi_2(1) + \chi_3(1) + 2\chi_4(1) + \chi_5(1) && (5.7)\\
&= 2\chi_1(1) + 3\chi_2(1) + 2\chi_3(1) + \chi_4(1) + \chi_6(1), && (5.8)\\
c_5 &= 333202640600 = 4\chi_1(1) + 5\chi_2(1) + 3\chi_3(1) + 2\chi_4(1) + \chi_5(1) + \chi_6(1) + \chi_7(1), && (5.9)\\
&\vdots
\end{aligned}$$

The constant term of the modular $j$-function is not important in the sense that, even if we change it, the key property of the modular $j$-function that it is the generator of the rational function field $\mathbb{C}(j)$ of the modular functions of weight 0 (see Thm. 3.2) does not change. So they conjectured that there exists a graded vector space $V = \bigoplus_{i=-1}^{\infty} V_i$ such that it is a representation of the monster group $\mathbb{M}$, and its graded character of the identity element $1_{\mathbb{M}}$ is the modular $j$-function without the constant term

$$J(\tau) := j(\tau) - 744. \tag{5.10}$$

In addition, this graded vector space $V$ must have some nice property; otherwise, we can easily construct it by just stacking the trivial representation $\chi_1$. So they focused on the graded characters

$$J_g(\tau) := \sum_{i=-1}^{\infty} \mathrm{Tr}_{V_i}(g) q^i \tag{5.11}$$

of the elements $g \in \mathbb{M}$ other than the identity $1_{\mathbb{M}}$. This character $J_g(\tau)$ is called the *McKay–Thompson series* associated with $g \in \mathbb{M}$.

Recall (Section 3.2) that $J_{1_{\mathbb{M}}}(\tau) = J(\tau)$ is the generator of the rational function field $\mathbb{C}(j)$ of the modular functions of weight $0$. Such a modular function $f : \mathbb{H} \to \mathbb{C}$ is invariant under the action of the modular group $\mathrm{SL}(2,\mathbb{Z})$, so it is well-defined as a function $f : \mathbb{H}/\mathrm{SL}(2,\mathbb{Z}) \to \mathbb{C}$ on the quotient space $\mathbb{H}/\mathrm{SL}(2,\mathbb{Z})$. It is known that the one-point compactification (the compact space made by adding the point at infinity) of $\mathbb{H}/\mathrm{SL}(2,\mathbb{Z})$, denoted by $(\mathbb{H}/\mathrm{SL}(2,\mathbb{Z}))^*$, is topologically equivalent to a Riemann sphere, which is a surface of genus $0$. Pushing forward this observation (and an observation[25] by Ogg [Ogg75]), Thompson proposed the following conjecture in [Tho79a]: for each element $g \in \mathbb{M}$, there exists a subgroup $\Gamma_g$ of the modular group $\mathrm{SL}(2,\mathbb{Z})$ (more precisely, a congruence subgroup) such that $(\mathbb{H}/\Gamma_g)^*$ is a genus-zero surface, and

[25] In 1975, Ogg already observed that a prime number $p$ divides the order $|\mathbb{M}|$ of the monster group if and only if $(\mathbb{H}/\Gamma_0(p)^+)^*$ has genus 0, where $\Gamma_0(p)^+ = \langle \Gamma_0(p), \frac{1}{\sqrt{p}}\begin{pmatrix} 0 & -1 \\ p & 0 \end{pmatrix} \rangle$. This is an unofficial beginning of the monstrous moonshine [Gan06a].

$J_g(\tau)$ is the generator of the rational function field of the functions $(\mathbb{H}/\Gamma_g)^* \to \mathbb{C}$ over $\mathbb{C}$. This is the *McKay–Thompson conjecture*.

The McKay–Thompson conjecture was proven through the following progression. In [CN79], Conway and Norton proposed the conjectural forms of $\Gamma_g$ and $J_g(\tau)$ for each element $g \in \mathbb{M}$. This is called the *Conway–Norton conjecture*, or the *monstrous moonshine conjecture*. In addition, they also showed that these conjectural $\Gamma_g$ and $J_g(\tau)$ satisfy that $(\mathbb{H}/\Gamma_g)^*$ has genus 0 and $J_g(\tau)$ generates the rational function field of the functions $(\mathbb{H}/\Gamma_g)^* \to \mathbb{C}$. Frenkel, Lepowsky, and Meurman constructed a representation $V^\natural$ of the monster group $\mathbb{M}$ whose graded character is $J(\tau)$ in [FLM84], and clarified its VOA structure and showed that its automorphism group $\mathrm{Aut}(V^\natural)$ is precisely the monster group $\mathbb{M}$ in [FLM88]. Finally, Borcherds showed in [Bor92] that the graded character of $g \in \mathbb{M}$ on $V^\natural$ coincides with the conjectured $J_g(\tau)$ by the Conway–Norton conjecture, for which he was awarded the Fields Medal.

The $\mathbb{M}$-module $V^\natural$ constructed in [FLM88] is called the monster VOA or the moonshine module. The monster VOA $V^\natural$ is constructed as the $\mathbb{Z}_2$-orbifold of the Leech lattice VOA $V_{\Lambda_{24}}$, the details of which are reviewed in the following Sections 5.2 and 5.3. In the language of physics, $V^\natural$ is a chiral bosonic CFT of central charge 24, having the partition function $J(\tau)$ and the symmetry $\mathbb{M}$. Let us briefly review why this is the case.

In general, we can construct a chiral bosonic modular-invariant lattice CFT $V_L$ from a Euclidean even self-dual lattice $L$ of rank $n \equiv 0 \bmod 24$. Such a lattice CFT $V_L$ has central charge $n$, and its partition function is

$$Z^{(V_L)}(\tau) := \mathrm{Tr}_{V_L}[q^{L_0 - \frac{c}{24}}] \tag{5.12}$$

$$= \frac{\Theta_L(\tau)}{\eta(\tau)^n}. \tag{5.13}$$

For this partition function $Z^{(V_L)}(\tau)$ to start with the term $q^{-1}$, we should choose $n = 24$. So the lattice must be an even self-dual lattice of rank 24, that is, a Niemeier lattice.

Now that $Z^{(V_L)}(\tau)$ is modular invariant (a modular function of weight 0) starting with the term $q^{-1}$, it must be in the form of $J(\tau)$ + constant, and in fact, we have

$$Z^{(V_L)}(\tau) = J(\tau) + 24 + |L_1|. \tag{5.14}$$

Here, $|L_1|$ is the number of lattice vectors with squared length 2 (as defined in (2.41)), and hence $|L_1| = 0$ for the Leech lattice $L = \Lambda_{24}$. In addition, 24 is coming from the states $\alpha^i_{-1}|0\rangle$ ($i = 1, \ldots, 24$) of conformal weight ($L_0$-eigenvalue) 1, and these states have odd parity under the $\mathbb{Z}_2$ symmetry $k \mapsto -k$ of the lattice $L$. Therefore, these 24 states are eliminated by taking the $\mathbb{Z}_2$-orbifold, which retains the $\mathbb{Z}_2$-invariant states. As a result, the partition function of the $\mathbb{Z}_2$-orbifold of the Leech lattice CFT $V_{\Lambda_{24}}$ coincides with the modular $j$-function without the constant term $J(\tau)$.

The proof of $\mathrm{Aut}(V^\natural) = \mathbb{M}$ by [FLM88] can be outlined as follows. As mentioned in Section 2.4, the monster group $\mathbb{M}$ is generated by a subgroup $C(\Lambda_{24}) = 2^{1+24}.\mathrm{Co}_1$ and a specific order-2

element $\sigma$. Recall that the isometry group of the Leech lattice $\Lambda_{24}$ is the Conway group $\mathrm{Co}_0$. Before taking the $\mathbb{Z}_2$-orbifold, the automorphism group $\mathrm{Aut}(V_{\Lambda_{24}})$ of the Leech lattice VOA is a group extension $2^{24}.\mathrm{Co}_0$ of $\mathrm{Co}_0$ (Section 5.2.3). When we construct the $\mathbb{Z}_2$-orbifold $V^\natural$, we add the twisted sector $(V_{\Lambda_{24}})_{\mathrm{tw}}$ as $V_{\Lambda_{24}} \oplus (V_{\Lambda_{24}})_{\mathrm{tw}}$, and project them onto the $\mathbb{Z}_2$-invariant states $(V_{\Lambda_{24}})^0 \oplus (V_{\Lambda_{24}})^0_{\mathrm{tw}}$, which is $V^\natural$ (Section 5.3.1). The addition of the twisted sector makes the group $2^{24}.\mathrm{Co}_0$ acting on $V_{\Lambda_{24}}$ enlarge to $2^{1+24}.\mathrm{Co}_0$ acting on $V_{\Lambda_{24}} \oplus (V_{\Lambda_{24}})_{\mathrm{tw}}$, and the projection makes $2^{1+24}.\mathrm{Co}_0$ into its $\mathbb{Z}_2$-quotient $C(\Lambda_{24}) = 2^{1+24}.\mathrm{Co}_1$ (Section 5.3.2).

In this way, the subgroup $C(\Lambda_{24})$ acts on the monster VOA $V^\natural$ naturally, but the construction of the order-2 element $\sigma$ is quite nontrivial. While $C(\Lambda_{24})$ acts on the untwisted sector $(V_{\Lambda_{24}})^0$ and the twisted sector $(V_{\Lambda_{24}})^0_{\mathrm{tw}}$ separately, $\sigma \in \mathrm{Aut}(V^\natural)$ was constructed as an automorphism mixing these two sectors, and called the triality operator in [FLM88]. Its construction was revisited by [DGM90a, DGM94], and explained in relation to an isomorphism between two $\mathbb{Z}_2$-orbifolds by different kinds of $\mathbb{Z}_2$ symmetries (Section 5.3.3). Finally, by showing that $\mathbb{M}$ generated by $C(\Lambda_{24})$ and $\sigma$ is the whole automorphism group $\mathrm{Aut}(V^\natural)$, we have $\mathrm{Aut}(V^\natural) \cong \mathbb{M}$. We also note that another simple construction of $V^\natural$ and proof of $\mathrm{Aut}(V^\natural) \cong \mathbb{M}$ were provided by Miyamoto in [Miy04].

Before proceeding, we remark that the automorphism group $\mathrm{Aut}(V)$ of a VOA $V$ preserves the Virasoro algebra (4.12) of $V$ by its definition, so we should be able to decompose the partition function $J(\tau)$ of $V^\natural$ simultaneously into the characters of the monster group $\mathbb{M}$ and the characters of the Virasoro algebra. The irreducible character[26] $\mathrm{ch}_h(\tau)$ of the Virasoro algebra of highest weight $h$ and central charge $c = 24$ is

$$\mathrm{ch}_0(\tau) = \frac{q^{1/24}}{\eta(q)}(q^{-1} - 1), \tag{5.15}$$

$$\mathrm{ch}_h(\tau) = \frac{q^{1/24}}{\eta(q)} q^{h-1} \quad (h \in \mathbb{Z}_{>0}). \tag{5.16}$$

Here, we have

$$\frac{q^{1/24}}{\eta(q)} = \frac{1}{\prod_{m=1}^\infty (1-q^m)} = \sum_{N=0}^\infty p(N) q^N \tag{5.17}$$

$$= 1 + q + 2q^2 + 3q^3 + 5q^4 + 7q^5 + 11q^6 \cdots, \tag{5.18}$$

[26] We can generate a module $V_h$ of the Virasoro algebra from the highest weight state $|h\rangle$ (a state defined as $L_0|h\rangle = h|h\rangle$ and $L_m|h\rangle = 0$ for any $m > 0$) as $V_h := \mathrm{Span}_{\mathbb{C}}\{L_{-m_1} \cdots L_{-m_l}|h\rangle \mid m_1, \ldots, m_l \in \mathbb{Z}_{>0}\}$. Since the state $L_{-m_1} \cdots L_{-m_l}|h\rangle$ has $L_0$-eigenvalue $m_1 + \cdots + m_l + h$, the character of this module $V_h$ is $\mathrm{Tr}_{V_h}[q^{L_0 - \frac{c}{24}}] = q^{h-\frac{c}{24}} \sum_{N=0}^\infty p(N) q^N = \frac{q^{1/24}}{\eta(q)} q^{h-\frac{c}{24}}$, using (5.17). However, this module $V_h$ is not necessarily irreducible. When $h = 0$, the state $L_{-1}|0\rangle$ satisfies the defining properties of the highest weight state with $h = 1$ (for example, $L_1(L_{-1}|0\rangle) = (2L_0 - L_{-1}L_1)|0\rangle = 0$), so the module $V_{h=0}$ contains the submodule $V_{h=1}$. Therefore, the irreducible module of highest weight 0 has the character $\frac{q^{1/24}}{\eta(q)} q^{-\frac{c}{24}}(1-q)$. Whether the module $V_h$ with $h > 0$ is irreducible or not depends on the values of weight $h$ and central charge $c$, but for $c > 1$, it is known to be irreducible. See for example [DFMS97, Ch. 7], [ES15, Ch. 2] for more details.

where $p(N)$ denotes the number of the partitions of the integer $N$ (the *partition function* in number theory). Therefore, $J(\tau)$ is decomposed as (note that $\mathrm{ch}_h(\tau)$ starts with the term $q^{h-1}$)

$$J(\tau) = \sum_{i=-1}^{\infty} c'_i \mathrm{ch}_{i+1}(\tau), \tag{5.19}$$

where $c'_0 = 0$ and

$$c'_{-1} = \chi_1(1), \tag{5.20}$$
$$c'_1 = \chi_2(1), \tag{5.21}$$
$$c'_2 = \chi_3(1), \tag{5.22}$$
$$c'_3 = \chi_4(1), \tag{5.23}$$
$$c'_4 = \chi_6(1), \tag{5.24}$$
$$c'_5 = \chi_5(1) + \chi_7(1), \tag{5.25}$$
$$\vdots$$

We can see that the same representation $\chi_i$ of $\mathbb{M}$ appearing in the different coefficients $c_i$ (5.4)–(5.9) are bundled into the representation of the Virasoro algebra, and the appearance of the irreducible representation dimensions of $\mathbb{M}$ becomes simple.

Lastly, we briefly mention a generalization of the monstrous moonshine. In the language of physics, the McKay–Thompson series (5.11) is an $\mathbb{M}$-twisted partition function of $V^\natural$ with a twisted boundary condition in the temporal direction. From the perspective of the general theory of orbifolds (Section 12.2), it is also natural to consider the twisted partition functions with twisted boundary conditions in the spatial direction, or both spatial and temporal directions. (We remark that the temporal and spatial twists are referred to as *twining* and *twisted*, respectively, in the terminology of [GHV10a, GPRV12]. In physics literature, both of them are usually called *twisted*.) Moonshine conjectures related to these twisted partition functions were proposed by Norton in [Nor87], as a generalization of the work by Queen [Que81], and called the *generalized moonshine conjecture*. After many steps by mathematicians (see for example [Car17]), the final step of the proof of this generalized moonshine conjecture was announced in [Car12]. For more on monstrous moonshine and its recent developments, see for example [Har99, Gan06a, Gan06b, DGO14, HHP22] and references therein.

## 5.2 Lattice VOA

One typical construction of a VOA uses a lattice as an ingredient, and the resulting VOA is called a lattice VOA. Before getting into its mathematical description, we first present a brief explanation in the language of physics in Section 5.2.1. Then we see the construction of a lattice VOA in Section 5.2.2. We also describe the automorphism group of the lattice VOA in Section 5.2.3, because it plays an important role in moonshine phenomena, and also because it will be our main subject in Part III.

### 5.2.1 Physical Description

For a given lattice $L$, we can consider a CFT having $L$ as a momentum lattice. If the symmetric bilinear form of the lattice is of signature $(r, s)$, and if it is diagonalized with respect to a certain basis $e_1, \dots, e_r, e_{r+1}, \dots, e_{r+s}$ as

$$|k|^2 = |k_+|_+^2 - |k_-|_-^2 \quad \text{for } k = k_+ + k_- = \sum_{i=1}^{r} k^i e_i + \sum_{i=r+1}^{r+s} k^i e_i \in L, \tag{5.26}$$

where $|\bullet|_+^2$ and $|\bullet|_-^2$ are some positive norms, then the partition function of the lattice CFT is

$$Z^{(V_L)}(\tau, \bar{\tau}) = \frac{1}{\eta(\tau)^r \overline{\eta(\tau)^s}} \Theta_L(\tau, \bar{\tau}) \tag{5.27}$$

$$= \frac{1}{\eta(\tau)^r \overline{\eta(\tau)^s}} \sum_{k \in L} q^{\frac{1}{2}|k_+|_+^2} \bar{q}^{\frac{1}{2}|k_-|_-^2}, \tag{5.28}$$

and the central charge of the lattice CFT is $(c, \tilde{c}) = (r, s)$. If the lattice $L$ is even self-dual, then the partition function $Z^{(V_L)}(\tau, \bar{\tau})$ is bosonic and modular invariant[27] up to the phases (1.4, 1.5) from the gravitational anomaly. An even self-dual lattice of signature $(r, s)$ exists if and only if $r - s \equiv 0 \bmod 8$ [Ser73, Ch. V], so the phases from the gravitational anomaly vanish if the lattice $L$ further satisfies $r - s \equiv 0 \bmod 24$, and then the partition function $Z^{(V_L)}(\tau, \bar{\tau})$ is completely modular invariant. If the lattice is Euclidean, then the resulting CFT is chiral. See for example [ES15, §5.5] for more details.

We will mainly deal with the chiral cases in this thesis. The states of a chiral lattice CFT are $\mathbb{C}$-linear combinations of the states in the form of

$$\alpha_{-m_1}^{i_1} \cdots \alpha_{-m_l}^{i_l} |k\rangle, \tag{5.29}$$

where $|k\rangle$ is the state with momentum vector $k \in L$, $m_1, \dots, m_l \in \mathbb{Z}_{>0}$, and $\alpha_m^1, \dots, \alpha_m^n$ ($m \in \mathbb{Z}_{>0}$) are the creation operators corresponding to an $\mathbb{Z}$-basis $e_1, \dots, e_n$ of $L$. Together with the annihilation operators $\alpha_m^i$ ($m \in \mathbb{Z}_{<0}$) and the momentum operators $\alpha_0^i$ such that the state (5.29) is an eigenstate of $\alpha_0^i$ with eigenvalue $\langle e_i, k\rangle$, the operators $\{\alpha_m^i\}_{m \in \mathbb{Z}}^{i=1,\dots,n}$ satisfy the commutation relations $[\alpha_m^i, \alpha_{m'}^{i'}] = \langle e_i, e_{i'}\rangle m \delta_{m+m',0}$. The conformal weight of the state (5.29) is $\frac{1}{2}|k|^2 + m_1 + \cdots + m_l$. We also often use the creation-annihilation operators $\mathfrak{a}_m^1, \dots, \mathfrak{a}_m^n$ with respect to an orthonormal basis $\mathfrak{e}_1, \dots, \mathfrak{e}_n$ of $\mathbb{R}^n$ where the lattice $L$ is embedded. They are related to $\alpha_m^1, \dots, \alpha_m^n$ under a proper $\mathbb{R}$-linear transformation, and their commutation relation is of course $[\mathfrak{a}_m^i, \mathfrak{a}_{m'}^{i'}] = \delta_{i,i'} m \delta_{m+m',0}$.

If we introduce the chiral bosons $X(z) = (X^i(z))_{i=1,\dots,n}$ which satisfy the OPE

$$X^i(z_1) \cdot X^j(z_2) \sim -\delta_{i,j} \log(z_1 - z_2), \tag{5.30}$$

[27] This follows from the formula (see e.g. [ES15, Eq. (5.166)]) of the modular S transformation $\Theta_L(-\frac{1}{\tau}, -\frac{1}{\bar{\tau}}) = \frac{1}{\det G}(-\sqrt{-1}\tau)^{r/2}(\sqrt{-1}\bar{\tau})^{s/2}\Theta_{L^*}(\tau, \bar{\tau})$, where $G$ is the Gram matrix of $L$, and $L^*$ is the dual lattice of $L$. The modular T transformation of $\Theta_L(\tau, \tau)$ is easy.

and whose mode expansions are given as (see e.g. [ES15, §1.8.1])

$$\partial X^i(z) = -\sqrt{-1} \sum_{m=-\infty}^{\infty} \frac{\mathfrak{a}^i_m}{z^{m+1}}, \tag{5.31}$$

then we can describe the state-operator correspondence as (see e.g. [Pol07a, §2.8])

$$\mathfrak{a}^i_{-m}|0\rangle \quad \longleftrightarrow \quad \frac{\sqrt{-1}}{(m-1)!}\partial^m X^i(z) \quad (m \geq 1), \tag{5.32}$$

$$|k\rangle \quad \longleftrightarrow \quad V_k(z) \propto :e^{\sqrt{-1}k\cdot X(z)}:, \tag{5.33}$$

where $\propto$ denotes that we ignore the cocycle factor.

We will look into the details of cocycle factors in Section 8.1, but here we give a brief exposition. To realize the appropriate commutation relations of the vertex operators $V_k(z) \propto :e^{\sqrt{-1}k\cdot X(z)}:$, in accordance with whether $V_k(z)$ is bosonic (the weight $\frac{1}{2}|k|^2$ is an integer) or fermionic (a half-integer), we have to introduce a correction factor $c_k(p)$ satisfying

$$c_k(p+k')c_{k'}(p) = (-1)^{k\cdot k'+|k|^2|k'|^2}c_{k'}(p+k)c_k(p) = \varepsilon(k,k')c_{k+k'}(p) \tag{5.34}$$

to modify the commutation relations of $:e^{\sqrt{-1}k\cdot X(z)}:$'s. As a result, an additional factor called a *cocycle factor* $\varepsilon : L \times L \to \{\pm 1\}$ which satisfies the 2-cocycle condition appears in the OPE of the vertex operators $V_k(z) = :e^{\sqrt{-1}k\cdot X(z)}: c_k(p)$ as

$$V_k(z_1)\cdot V_{k'}(z_2) = (-1)^{|k|^2|k'|^2}V_{k'}(z_2)\cdot V_k(z_1) \sim \varepsilon(k,k')(z_1-z_2)^{k\cdot k'}V_{k+k'}(z_2), \tag{5.35}$$

where we dropped $O((z_1-z_2)^{k\cdot k'+1})$ terms in OPE.

### 5.2.2 Construction of Lattice VOA

The foundational literature on lattice VOAs is [FLM88], and [Kac98, §§5.4-5.5] also provides a detailed description. [DN99, Lam20] contain readable summaries, and we will follow them. These references except for [Kac98] deal with ordinary lattice VOAs constructed from even lattices, but we also consider odd lattices in this thesis, so the resulting VOA can be a VOSA to be precise (see e.g. [DL93, Remark 12.38]). However, this distinction is not important for our purposes, so we will be unconcerned about that point. In this thesis, we always assume that lattices are integral. We also always assume that lattices are Euclidean (positive-definite), and hence the resulting lattice VOA formulates a chiral CFT in the language of physics. As for Lorentzian lattice VOAs or more general VOAs to describe non-chiral CFTs (full CFTs in other words), see [HK05, Mor20, SS23].

The first step to constructing the lattice VOA $V_L$ from a given lattice $L$ is to furnish the lattice $L$ with a cocycle factor $\hat{\varepsilon}$. We will eventually use the specific cocycle factor $\varepsilon$ required by physics as in (5.34), but for a moment, let $\hat{\varepsilon} : L \times L \to \mathbb{Z}_2$ denote a general 2-cocycle to keep generality. We will use some basic concepts of group extensions throughout the rest of this Section 5. See

Appendix A for some basic facts on group extensions, although the description in this Section 5 is intended to be as self-contained as possible.

Let $\hat{L}$ be the central extension[28] of the lattice (as a free abelian group) $L$ by $\mathbb{Z}_2 = \langle \kappa \mid \kappa^2 = 1 \rangle$

$$1 \to \mathbb{Z}_2 \to \hat{L} \to L \to 0, \tag{5.36}$$

specified by a 2-cocycle $\hat{\varepsilon} : L \times L \to \mathbb{Z}_2$. In other words, $\hat{L}$ is $\mathbb{Z}_2 \times L$ as a set, and if we write its element as $\kappa^m e^k$ ($\kappa^m \in \mathbb{Z}_2, k \in L$), then $\hat{L}$ is a group specified by the multiplication

$$\kappa^m e^k \cdot \kappa^{m'} e^{k'} = \hat{\varepsilon}(k, k') \kappa^{m+m'} e^{k+k'}. \tag{5.37}$$

So, $e^k \in \hat{L}$ and $\hat{\varepsilon}$ correspond to $c_k(p)$ and $\varepsilon$ in (5.34) respectively, under $\kappa = e^{\sqrt{-1}\pi}$.

Here is one comment on the normalization of the 2-cocycle. It follows from the 2-cocycle condition

$$\hat{\varepsilon}(k, k') \hat{\varepsilon}(k + k', k'') = \hat{\varepsilon}(k, k' + k'') \hat{\varepsilon}(k', k''), \tag{5.38}$$

that any 2-cocycle $\hat{\varepsilon} : L \times L \to \mathbb{Z}_2$ satisfies

$$\hat{\varepsilon}(k, 0) = \hat{\varepsilon}(0, k) = \hat{\varepsilon}(0, 0) \quad \text{for any } k \in L. \tag{5.39}$$

Furthermore, it is known that there exists a 2-cocycle $\hat{\varepsilon}$ satisfying the normalization condition

$$\hat{\varepsilon}(0, 0) = \kappa^0, \tag{5.40}$$

in any cohomology class in $H^2(L, \mathbb{Z}_2)$, and it defines an equivalent extension $\hat{L}$ to any 2-cocycle in the same cohomology class (see the last paragraph of Section A.1). Therefore, we will always assume that $\hat{\varepsilon}$ is a normalized one as in (5.40), without loss of generality. Then, we can observe that the multiplication of $\hat{L}$ is well-behaved in the sense that

$$\kappa^m e^0 \cdot \kappa^{m'} e^k = \kappa^{m+m'} e^k, \tag{5.41}$$

$$\kappa^m e^k \cdot \kappa^{m'} e^0 = \kappa^{m+m'} e^k, \tag{5.42}$$

and hence there is no confusion if we just write $\kappa^m$ instead of $\kappa^m e^0$. We will also just write $e^k$ instead of $\kappa^0 e^k$.

Now $\hat{L}$ has the multiplication structure reflecting the OPE (5.34), but not yet the structure of a $\mathbb{C}$-vector space of states. So we would like to consider the group algebra $\mathbb{C}[\hat{L}]$ of $\hat{L}$ over $\mathbb{C}$, but we have to identify $\kappa$ with $e^{\sqrt{-1}\pi} \in \mathbb{C}$, in order to properly make $\kappa^m e^k$ and $\kappa^{m'} e^k$ be $\mathbb{C}$-linearly dependent. This is mathematically done as follows. We consider a representation $\mathbb{C}$ of the subgroup $\mathbb{Z}_2$ of $\hat{L}$, where the generator $\kappa$ acts on $\mathbb{C}$ as multiplication by $e^{\sqrt{-1}\pi}$. We define

[28] More generally, [FLM88] deals with the central extension $\hat{L}$ of $L$ by $\mathbb{Z}_s = \langle \kappa \mid \kappa^s = 1 \rangle$. Furthermore, we would like to consider the central extension by $\mathrm{U}(1) \cong \mathbb{R}/2\mathbb{Z} = \langle \kappa \rangle_{\mathbb{R}} / \langle \kappa \rangle_{2\mathbb{Z}}$ later. Therefore, the discussions below avoid using special properties for $\mathbb{Z}_2$, such as $\hat{\varepsilon}(k, k')^{-1} = \hat{\varepsilon}(k, k')$.

the $\hat{L}$-module $\mathbb{C}\{L\}$ as the induced representation of $\hat{L}$ from it, or equivalently, the extension of scalars of the module $\mathbb{C}$ from $\mathbb{C}[\mathbb{Z}_2]$ to $\mathbb{C}[\hat{L}]$:

$$\mathbb{C}\{L\} := \mathrm{Ind}^{\hat{L}}_{\mathbb{Z}_2 \text{ or } \mathrm{U}(1)}\mathbb{C} = \mathbb{C}[\hat{L}] \otimes_{\mathbb{C}[\mathbb{Z}_2]} \mathbb{C}. \tag{5.43}$$

Roughly speaking, $\mathbb{C}\{L\}$ is just a group algebra $\mathbb{C}[\hat{L}]$ with $\kappa^m \cdot e^k = e^{\sqrt{-1}\pi m} e^k$.

The second step is to introduce the creation-annihilation operators. Let $\mathfrak{h} = L \otimes_{\mathbb{Z}} \mathbb{C}$ be the abelian Lie algebra, and $\hat{\mathfrak{h}} = \mathfrak{h} \otimes \mathbb{C}[t, t^{-1}] \oplus \mathbb{C}K$ be its affine Lie algebra with the Lie bracket

$$[\alpha(m), \alpha'(m')] = \langle \alpha, \alpha' \rangle m \delta_{m+m',0} K, \tag{5.44}$$

where $\alpha(m) := \alpha \otimes t^m$ with $\alpha \in \mathfrak{h}$, $m \in \mathbb{Z}$, and the symmetric bilinear form $\langle -, - \rangle$ on $L$ is extended to $\mathfrak{h}$ by $\mathbb{C}$-linearity.

Let $U(\hat{\mathfrak{h}})$ be the universal enveloping algebra of $\hat{\mathfrak{h}}$ (the algebra where $\alpha(m)\alpha'(m') - \alpha'(m')\alpha(m) = [\alpha(m), \alpha'(m')]$ holds), and define the $\hat{\mathfrak{h}}$-module $M(1)$ as

$$M(1) := U(\hat{\mathfrak{h}}) \otimes_{\mathfrak{h} \otimes \mathbb{C}[t] \oplus \mathbb{C}K} \mathbb{C}, \tag{5.45}$$

where $\mathfrak{h} \otimes \mathbb{C}[t]$ acts on $\mathbb{C}$ as $\alpha(m) \cdot \mathbb{C} = 0$ ($m \geq 0$),[29] and $K$ acts on $\mathbb{C}$ as multiplication by 1. As a vector space, $M(1)$ is isomorphic to $U(\hat{\mathfrak{h}}^-)$, which is the universal enveloping algebra of the abelian subalgebra

$$\hat{\mathfrak{h}}^- := \mathfrak{h} \otimes t^{-1}\mathbb{C}[t^{-1}], \tag{5.46}$$

of $\hat{\mathfrak{h}}$.

Finally, the *lattice VOA* $V_L$ is defined as the $\mathbb{C}$ vector space

$$V_L := M(1) \otimes_{\mathbb{C}} \mathbb{C}\{L\}, \tag{5.47}$$

with the VOA structure such as the vacuum vector and the state-field correspondence, which we do not write down here. Any element of $V_L$ therefore can be written as a $\mathbb{C}$-linear combination of elements in the form of

$$\alpha_1(-m_1) \cdots \alpha_l(-m_l) e^k, \tag{5.48}$$

where $\alpha_1, \ldots, \alpha_l \in \mathfrak{h} = L \otimes_{\mathbb{Z}} \mathbb{C}$, $m_1, \ldots, m_l \in \mathbb{Z}_{>0}$, and $k \in L$.

Let $e_1, \ldots, e_n$ be an integral basis of $L$. The element $e_{i_1}(-m_1) \cdots e_{i_l}(-m_l) e^k$ of $V_L$ is usually denoted by $\alpha^{i_1}_{-m_1} \cdots \alpha^{i_l}_{-m_l} |k\rangle$ in physics literature, where the commutation relation of the creation-annihilation operators $\alpha^i_m$ is $[\alpha^i_m, \alpha^{i'}_{m'}] = \langle e_i, e_{i'} \rangle m \delta_{m+m',0}$. We also often use the creation-annihilation operators $\mathfrak{a}^i_m := \mathfrak{e}_i(m)$ ($m \neq 0$; see footnote 29) with respect to an orthonormal

[29] Here, $\alpha(0)$ also annihilates the states, unlike the momentum operators usually denoted by $\alpha^i_0$ in physics as in the previous Section 5.2.1. This is just a problem of conventions.

basis $\mathbb{e}_1, \ldots, \mathbb{e}_n$ of $\mathbb{R}^n$ where the lattice $L$ is embedded. Their commutation relation is of course $[\mathbb{\alpha}^i_m, \mathbb{\alpha}^{i'}_{m'}] = \delta_{i,i'} m \delta_{m+m',0}$.

The vacuum vector $\mathbf{1}$ of the lattice VOA $V_L$ is $\mathbf{1} = e^0$. The Virasoro element $\omega$ is given by $\omega = \frac{1}{2}\sum_i \mathbb{e}_i(-1)^2\mathbf{1}$, and the corresponding operator is $T(z) = -\frac{1}{2}\sum_i :(\partial X^i(z))^2:$. Then the weight of the element (5.48), namely the eigenvalue with respect to $L_0$, can be calculated as

$$\mathrm{wt}(\alpha_1(-m_1)\cdots\alpha_l(-m_l)e^k) = m_1 + \cdots + m_l + \frac{1}{2}|k|^2, \tag{5.49}$$

and this weight introduces the grading of the VOA $V_L$.

### 5.2.3 Automorphism Group of Lattice VOA

**The group $O(\hat{L})$**

Since the lattice VOA $V_L$ is built on the central extension $\hat{L}$ of the lattice $L$, before directly getting into the automorphism group of $V_L$, it is useful to first investigate the automorphism group of $\hat{L}$.

We define the *commutator map* $\hat{c} : L \times L \to \mathbb{Z}_2$ by

$$\hat{c}(k, k') := \hat{\varepsilon}(k, k')\hat{\varepsilon}(k', k)^{-1}. \tag{5.50}$$

If $\hat{\varepsilon}$ here is the specific one $\varepsilon$ in (5.34), then it immediately follows from (5.34) that $\varepsilon$ must satisfy

$$\varepsilon(k, k') = (-1)^{k\cdot k' + |k|^2|k'|^2}\varepsilon(k', k), \tag{5.51}$$

so we have $\hat{c}(k, k') = \kappa^{k\cdot k'+|k|^2|k'|^2}\,|_{\kappa=-1}$.

Forgetting the symmetric bilinear form on $L$, for a moment, we focus on the automorphism group $\mathrm{Aut}(L)$ of just a free abelian group $L$, instead of the isometry group $O(L)$ of the lattice $L$. The proposition [FLM88, Prop. 5.4.1] states that

$$1 \to \mathrm{Hom}(L, \mathbb{Z}_2) \xrightarrow{\sim} \mathrm{Aut}(\hat{L}, \kappa) \xrightarrow{-} \mathrm{Aut}(L, \hat{c}) \to 1 \tag{5.52}$$

is exact. (The proof is also reviewed in Appendix A.4.) The details of (5.52) are as follows. $\mathrm{Hom}(L, \mathbb{Z}_2) \xrightarrow{\sim} \mathrm{Aut}(\hat{L}, \kappa)$ maps $\eta \in \mathrm{Hom}(L, \mathbb{Z}_2)$ to

$$\tilde{\eta} : \quad \hat{L} \to \hat{L} \tag{5.53}$$

$$\kappa^m e^k \mapsto \eta(k)\kappa^m e^k. \tag{5.54}$$

Note that $\eta \in \mathrm{Hom}(L, \mathbb{Z}_2)$ is determined only from the values of $\eta(e_1), \ldots, \eta(e_n)$, where $e_1, \ldots, e_n$ is a basis of $L$, and hence

$$\mathrm{Hom}(L, \mathbb{Z}_2) \cong (\mathbb{Z}_2)^n. \tag{5.55}$$

$\mathrm{Aut}(\hat{L}, \kappa)$ is defined as

$$\mathrm{Aut}(\hat{L}, \kappa) := \{f \in \mathrm{Aut}(\hat{L}) \mid f(\kappa) = \kappa\}, \tag{5.56}$$

and $\mathrm{Aut}(L,\hat{c})$ is defined as

$$\mathrm{Aut}(L,\hat{c}) := \{g \in \mathrm{Aut}(L) \mid \hat{c}(g(k),g(k')) = \hat{c}(k,k')\}. \tag{5.57}$$

$\mathrm{Aut}(\hat{L},\kappa)\bar{\to}\mathrm{Aut}(L,\hat{c})$ maps $f \in \mathrm{Aut}(\hat{L})$ to

$$\bar{f} : L \to L \tag{5.58}$$
$$k \mapsto \overline{f(e^k)}, \tag{5.59}$$

where the natural projection $\hat{L}\bar{\to}L$ of (5.36) is used.

Since 1 and $\kappa$ are the only elements of finite order in $\hat{L}$, it follows that any $f \in \mathrm{Aut}(\hat{L})$ satisfies $f(\kappa) = \kappa$, and hence $\mathrm{Aut}(\hat{L},\kappa)$ reduces to $\mathrm{Aut}(\hat{L})$ in the case at hand. However, this is the special property for the extension by $\mathbb{Z}_{s=2}$. If we consider a more general extension, say by $\mathbb{Z}_{s>2}$, then we cannot reduce $\mathrm{Aut}(\hat{L},\kappa)$ to $\mathrm{Aut}(\hat{L})$ (see footnote 28).

Let us recall that $L$ is a lattice, more than just a free abelian group, and move on to the isometry group $O(L)$ of the lattice from the automorphism group $\mathrm{Aut}(L)$ of the free abelian group. If the commutator map $\hat{c}$ depends only on the bilinear form of the lattice $L$, say $\hat{c}(k,k') = \kappa^{k\cdot k'+|k|^2|k'|^2}$, then the isometry group $O(L)$ is a subgroup of $\mathrm{Aut}(L,\hat{c})$ defined in (5.57). In addition, if we define

$$O(\hat{L}) := \{f \in \mathrm{Aut}(\hat{L},\kappa) \mid \bar{f} \in O(L)\}, \tag{5.60}$$

then we obtain the exact sequence [FLM88, Prop. 6.4.1]

$$1 \to \mathrm{Hom}(L,\mathbb{Z}_2) \tilde{\to} O(\hat{L}) \bar{\to} O(L) \to 1 \tag{5.61}$$

from (5.52).

*Remark* 5.1. For any $g \in O(L)$, it is known that there exists a lift $\hat{g} \in O(\hat{L})$ of $g$ such that $\hat{g}(e^k) = e^k$ for any $k \in L$ fixed by $g$ as $g(k) = k$ [Lep85, §5]. Such lift $\hat{g}$ is called the *standard lift* of $g$, and is sometimes of use in research. See for example [Bor92, Lemma 12.1], [Mö16, §5.3], [vEMS20, §7] for some properties of the standard lift. *(Remark ends.)*

**The automorphism group** $\mathrm{Aut}(V_L)$

Following [FLM88, §8.10] or [DN99, §2.3], the definition of an automorphism of a VOA is as follows.

**Definition 5.2** (automorphism of a VOA)**.** An *automorphism* of a VOA $V$ over $\mathbb{C}$ is a map $F : V \to V$ such that

(1) it is a $\mathbb{C}$-linear automorphism on the $\mathbb{C}$-vector space $V$.

(2) it preserves all the products.[30] More precisely, $F(v_{(m)}(v')) = (F(v))_{(m)}(F(v'))$ for any $v, v' \in V$ and $m \in \mathbb{Z}$. Or equivalently, $F \circ Y(v,z) \circ F^{-1} = Y(F(v),z)$ for any $v \in V$.

(3) it preserves the Virasoro element $F(\omega) = \omega$, and hence preserves the grading of $V$.

■

Any element $f$ of $O(\hat{L})$ induces an automorphism $F$ of a lattice VOA $V_L$, which acts on the state (5.29) as

$$F(\alpha_1(-m_1)\cdots\alpha_l(-m_l)e^k) = \bar{f}(\alpha_1)(-m_1)\cdots\bar{f}(\alpha_l)(-m_l)f(e^k), \tag{5.63}$$

where $\bar{f} \in O(L)$ was defined in (5.59) and extended by $\mathbb{C}$-linearity here. This preserves the group structure of $O(\hat{L})$, and hence $O(\hat{L}) \cong (\mathbb{Z}_2)^n.O(L)$ can be regarded as a subgroup of the automorphism group $\mathrm{Aut}(V_L)$ of the lattice VOA $V_L$. The whole $\mathrm{Aut}(V_L)$ is determined in [DN99] as follows.

**Theorem 5.3** ([DN99, Theorem 2.1])**.** *For a positive-definite even lattice $L$, the automorphism group $\mathrm{Aut}(V_L)$ is generated by $O(\hat{L})$ and $N$ defined as*

$$N := \{\exp(v_{(0)}) \mid v \in V_L, \mathrm{wt}(v) = 1\}. \tag{5.64}$$

*In short,*

$$\mathrm{Aut}(V_L) = N \cdot O(\hat{L}). \tag{5.65}$$

*$N$ is a normal subgroup of $\mathrm{Aut}(V_L)$, and $N \cap O(\hat{L})$ contains $\mathrm{Hom}(L, \mathbb{Z}_2) \subset O(\hat{L})$.*

*Remark* 5.4. We will not need the details of the subgroup $N \subset \mathrm{Aut}(V_L)$ in this thesis, but here we provide some explanation in this Remark.

An *(even) derivation*[31] of a VOA $V$ is a linear endomorphism $D : V \to V$ such that

- $D(v_{(m)}(v')) = (D(v))_{(m)}(v') + v_{(m)}(D(v'))$ for any $v, v' \in V$ and $m \in \mathbb{Z}$,
- $D(\omega) = 0$.

[30] A vertex algebra $V$ in the sense of Definition 4.2 satisfies the *Borcherds OPE formula* [Kac98, Thm. 4.6]

$$Y(v,z_1)Y(w,z_2) = \sum_{n=0}^{\infty} \frac{Y(v_{(n)}w, z_2)}{(z_1 - z_2)^{n+1}} + :Y(v,z_1)Y(w,z_2):, \tag{5.62}$$

in the domain $|z_1| > |z_2|$ for any $v, w \in V$. In view of this, the condition (2) of Definition 5.2 is phrased as "preserve OPE" in Section 11.

[31] An *odd derivation* [Kac98, §4.3] requires $D(v_{(m)}(v')) = (D(v))_{(m)}(v') + (-1)^{|v|}v_{(m)}(D(v'))$ instead of the first equation, but we do not use it in this thesis.

Combining these equations and $L_0 = \omega_{(1)}$, we can see $D(L_0 v) = L_0(Dv)$, and hence a derivation preserves the grading of $V$. We can easily see that $\exp(D)$ is an automorphism of $V$.

For any element $u \in V$ of weight 1, $u_{(0)}$ is a derivation as follows. The first equation follows from $[u_{(0)}, v_{(m)}] = (u_{(0)}v)_{(m)}$ [Kac98, Eq. (4.6.3)]. To see the second equation, we have $[u_{(0)}, L_{m'}] = 0$ from $[u_{(m)}, L_{m'}] = (m - (\mathrm{wt}(u) - 1)m')u_{(m+m')}$ [Kac98, Cor. 4.10 (iii)], and hence $u_{(0)}(\omega) = u_{(0)}L_{-2}\mathbf{1} = L_{-2}u_{(0)}\mathbf{1} = 0$, using the Definition 4.1 of the vacuum vector $\mathbf{1}$.

Therefore, $\exp(u_{(0)})$ with $\mathrm{wt}(u) = 1$ is an automorphism of $V$, called an *inner automorphism* in [Kac98, Remark 4.9c]. So

$$N := \{\exp(u_{(0)}) \mid u \in V, \mathrm{wt}(u) = 1\} \tag{5.66}$$

is indeed a subgroup of $\mathrm{Aut}(V_L)$. Furthermore, $N$ is a normal subgroup, because any $F \in \mathrm{Aut}(V)$ satisfies $F \circ \exp(u_{(0)}) \circ F^{-1} = \exp((F(u))_{(0)})$ and $\mathrm{wt}(F(u)) = \mathrm{wt}(u) = 1$.

For example, in the case of a lattice VOA $V_L$, if we take the weight-1 element $u$ as $e_i(-1)e^0$, or $\alpha^i_{-1}|0\rangle$ in physics notation, then $u_{(0)}$ is the momentum operator $\alpha^i_0$ described in Section 5.2.1 (see also footnote 29), which corresponds to $\langle e_i, - \rangle \in \mathrm{Hom}(L, \mathbb{R})$. Recalling $\omega = \frac{1}{2}\sum_i \mathfrak{e}_i(-1)^2 e^0$, we can explicitly see that $\alpha^i_0(\omega) = 0$. We can also see that $\exp(t\alpha^i_0) \in N$ is in $\mathrm{Hom}(L, \mathbb{Z}_2)$ for appropriate $t \in \mathbb{C}$, and this is the basic argument to show that $N \cap O(\hat{L})$ contains $\mathrm{Hom}(L, \mathbb{Z}_2) \subset O(\hat{L})$.

If the lattice $L$ contains a vector $k$ of squared length 2, then $e^k \in V_L$ is also a weight-1 element. On the other hand, if $L$ does not have a vector of squared length 2, then $N \cap O(\hat{L})$ coincides with $\mathrm{Hom}(L, \mathbb{Z}_2) \subset O(\hat{L})$, and $\mathrm{Aut}(V_L) = N.O(L)$ [Lam20, Remark 2.3]. The Leech lattice $\Lambda_{24}$ is such an example. *(Remark ends.)*

As a study of the structure of $\mathrm{Aut}(V_L)$, we can consider the following question: does the isometry group $O(L)$ of the lattice $L$ lift to a subgroup of the automorphism group $\mathrm{Aut}(V_L)$ of the lattice VOA $V_L$? This is the main subject of Part III. Now that we have seen that $O(\hat{L})$ is a subgroup of $\mathrm{Aut}(V_L)$, this question is equivalent to whether the group extension (5.61) splits or not.

## 5.3 $\mathbb{Z}_2$-Orbifold of Lattice VOA

When a CFT $\mathcal{T}$ has a finite group symmetry $G \subset \mathrm{Aut}(\mathcal{T})$ and it is non-anomalous, then we can construct a new CFT $\mathcal{T}/G$ consisting of $G$-invariant states, which is called the orbifold of $\mathcal{T}$ by $G$. The monster VOA $V^\natural$ is the orbifold of the Leech lattice VOA $V_{\Lambda_{24}}$ by the reflection $\mathbb{Z}_2$ symmetry $X(z) \mapsto -X(z)$. So we review the orbifold $\tilde{V}_L$ of a lattice VOA $V_L$ by the reflection $\mathbb{Z}_2$ symmetry in Section 5.3.1, and the automorphisms of the resulting theory $\tilde{V}_L$ in Section 5.3.2. A lattice VOA has another $\mathbb{Z}_2$ symmetry, the shift $\mathbb{Z}_2$ symmetry $X(z) \mapsto X(z) + \pi\chi$, and we will review it and the relations between two orbifolds by the reflection $\mathbb{Z}_2$ and by the shift $\mathbb{Z}_2$ in Section 5.3.3. This perspective of another symmetry was not explicitly used in the original proof of $\mathrm{Aut}(V^\natural) \cong \mathbb{M}$ by [FLM88], but [DGM90a] revealed that it provides a clear way of

understanding the proof. Lastly, in Section 5.3.4, we will briefly review a uniform treatment of $\mathbb{Z}_2$-orbifold and fermionization, which is not directly related to the monstrous moonshine, but will be used in a part of this thesis.

Readers who wish to proceed directly to Part III can skip this Section 5.3 and refer back to the necessary parts only when necessary.

### 5.3.1 Orbifold by Reflection $\mathbb{Z}_2$ Symmetry

**The reflection automorphism $\theta_0$**

We first explain what the reflection $\mathbb{Z}_2$ symmetry of a lattice VOA $V_L$ is.

**Proposition 5.5.** *Any lift $\theta \in O(\hat{L})$ of $-\mathrm{id}_L \in O(L)$ in the exact sequence (5.61) satisfies $\theta^2 = \mathrm{id}_{\hat{L}}$.*

*Proof.* Since $e^k \cdot e^{-k} = \hat{\varepsilon}(k,-k)$, we have $e^{-k} = \hat{\varepsilon}(k,-k)(e^k)^{-1}$. Hence, for a given $k \in L$, there is $m \in \mathbb{Z}$ such that $\theta(e^k) = \kappa^m (e^k)^{-1}$. Therefore, by using $\theta(\kappa) = \kappa$ from the definition (5.60) of $O(\hat{L})$,

$$\theta^2(e^k) = \kappa^m \theta(e^k)^{-1} = \kappa^m \kappa^{-m} e^k = e^k. \tag{5.67}$$

□

Suppose the 2-cocycle $\hat{\varepsilon} : L \times L \to \mathbb{Z}_2$ is bihomomorphic, or we customarily say it is bilinear. In fact, we will eventually use the bilinear one (8.13) for the 2-cocycle $\varepsilon$ appearing in (5.34). Then, the map $\theta_0 : \hat{L} \to \hat{L}$ defined as

$$\theta_0(\kappa^m e^k) = \kappa^m e^{-k} \tag{5.68}$$

is a homomorphism, because

$$\theta_0(e^k \cdot e^l) = \hat{\varepsilon}(k,l) e^{-k-l}, \quad \theta_0(e^k) \cdot \theta_0(e^l) = \hat{\varepsilon}(-k,-l) e^{-k-l}, \tag{5.69}$$

and $\hat{\varepsilon}(k,l) = \hat{\varepsilon}(-k,-l)$ by the assumption of bilinearity. $\theta_0$ is obviously an automorphism in $O(\hat{L})$, and hence it extends to the automorphism of $V_L$ as in (5.63):

$$\theta_0(\alpha_1(-m_1)\cdots\alpha_l(-m_l)e^k) = (-\alpha_1)(-m_1)\cdots(-\alpha_l)(-m_l)e^{-k}. \tag{5.70}$$

By Proposition 5.5, this $\theta_0$ generates a subgroup $\mathbb{Z}_2$ of $\mathrm{Aut}(V_L)$, and this is the *reflection* $\mathbb{Z}_2$ *symmetry* of the lattice VOA $V_L$.

In physics notation, the action of $\theta_0$ on a state is

$$\theta_0 \alpha^{i_1}_{-m_1} \cdots \alpha^{i_l}_{-m_l} |k\rangle = (-\alpha^{i_1}_{-m_1}) \cdots (-\alpha^{i_l}_{-m_l}) |-k\rangle, \tag{5.71}$$

and this can be described as the reflection symmetry of the chiral bosons (5.31)

$$X^i(z) \mapsto -X^i(z) \quad \text{for all } i = 1, \ldots, n. \tag{5.72}$$

**The $\theta_0$-twisted sector**

The orbifold of a lattice VOA $V_L$ by the reflection $\mathbb{Z}_2$ symmetry $\mathbb{Z}_2 = \langle\theta_0\rangle \subset \mathrm{Aut}(V_L)$ is done by the following two steps.

1. Introduce the $\theta_0$-twisted sector $(V_L)_{\mathrm{tw}}$.
2. Project the whole $V_L \oplus (V_L)_{\mathrm{tw}}$ to the $\theta_0$-invariant states $(V_L)^0 \oplus (V_L)^0_{\mathrm{tw}} =: \tilde{V}_L$.

The sectors[32] can be summarized as

| $\mathbb{Z}_2 = \langle\theta_0\rangle$ | untwisted | twisted |
|---|---|---|
| even | $(V_L)^0$ | $(V_L)^0_{\mathrm{tw}}$ |
| odd | $(V_L)^1$ | $(V_L)^1_{\mathrm{tw}}$ |

. (5.73)

The construction of an orbifold by more general finite group symmetry is also accomplished by similar two steps, as reviewed in Section 12.2.1.

We review the construction of the twisted sector $(V_L)_{\mathrm{tw}}$. Let $L$ be an even self-dual lattice of rank $n$, and the 2-cocycle $\hat{\varepsilon}$ be the specific one $\varepsilon$ in (8.13) associated with the commutator $c(k,k') = \kappa^{k\cdot k'}$ and the quadratic form $q(k) = \kappa^{\frac{1}{2}|k|^2}$, in the rest of this Section 5.3. As a result, the action of $\theta_0$ can be written as

$$\theta_0(e^k) = e^{-k} = \varepsilon(k,-k)(e^k)^{-1} = \kappa^{\frac{1}{2}|k|^2}(e^k)^{-1}, \tag{5.74}$$

where in the last equation, we used the bilinearity of $\varepsilon$ and Lemma A.4 (2) $\varepsilon(k,k) = q(k) = \kappa^{\frac{1}{2}|k|^2}$.

If we follow [FLM88], we first define

$$\begin{aligned} K &:= \{\theta_0(a)^{-1}a \mid a \in \hat{L}\} && (5.75)\\ &= \{\kappa^{\frac{1}{2}|k|^2}(e^k)^2 \mid k \in L\} && (5.76)\\ &= \{e^{2k} \mid k \in L\}. && (5.77) \end{aligned}$$

According to Theorem A.7, $K$ is a subgroup of the center of $\hat{L}$, and we have a central extension

$$1 \to \mathbb{Z}_2 \to \hat{L}/K \to L/2L \to 0. \tag{5.78}$$

Similarly to the notation $\kappa^m e^k$ of an element of $\hat{L}$, we may write an element of $\hat{L}/K$ as $\kappa^m\gamma_{k+2L}$. We will built the twisted sector on an irreducible representation of $Q = \hat{L}/K$.

A more physicist-friendly description of the group $Q$ is introduced in [DGM94, §5.3], so let us follow it below. We define the *gamma matrix algebra* $\Gamma(L)$ *associated with* $L$ as the unital (that is, $\Gamma(L)$ contains 1 and hence $\mathbb{C}$) $\mathbb{C}$-algebra generated by $\{\gamma_k\}_{k\in L}$ satisfying

$$\gamma_k\gamma_{k'} = (-1)^{k\cdot k'}\gamma_{k'}\gamma_k = \varepsilon(k,k')\gamma_{k+k'}, \quad \gamma_k^2 = (-1)^{\frac{1}{2}|k|^2}. \tag{5.79}$$

[32] $(V_L)^0$ and $(V_L)^0_{\mathrm{tw}}$ here in the main text are denoted by $(V_L)^{\theta_0}$ and $(V_L^T)^{\theta_0}$ respectively in [FLM88]. $(V_L)^0, (V_L)^1, (V_L)^0_{\mathrm{tw}}, (V_L)^1_{\mathrm{tw}}$, and $\tilde{V}_L$ here are denoted by $\mathcal{H}^+(L), \mathcal{H}^-(L), \mathcal{H}^+_T(L), \mathcal{H}^-_T(L)$, and $\tilde{\mathcal{H}}(L)$, respectively in [DGM94].

If we identify $\gamma_k$ with $e^k$, then $\Gamma(L)$ is roughly the algebra $\mathbb{C}\{L\}$ with the relation $(e^k)^2 = (-1)^{\frac{1}{2}|k|^2}$ introduced, which corresponds to the quotient by $K$.

We define a group $Q := \{\pm\gamma_k \mid k \in L\}$. From the relations (5.79), it is obvious that $|Q| = 2^{1+n}$. In particular, if we take a $\mathbb{Z}$-basis $e_1, \dots, e_n$ of $L$, then

$$Q = \{\pm\gamma_{e_1}^{t_1} \cdots \gamma_{e_n}^{t_n}\}_{t_i=0,1}\,. \tag{5.80}$$

Moreover, $Q$ is an extraspecial 2-group, which is defined as follows.

**Definition 5.6** (extraspecial $p$-group)**.** A group $G$ is called a *$p$-group*, if its order $|G|$ is a power of a prime $p$. A group $G$ is called an *elementary abelian $p$-group*, if it is isomorphic to $(\mathbb{Z}_p)^n$ for some $n \in \mathbb{Z}_{>0}$. A group $G$ is called an *extraspecial $p$-group* and denoted by $p^{1+n}$, if the center $Z(G)$ of $G$ satisfies $Z(G) \cong \mathbb{Z}_p$ and $G/Z(G) \cong (\mathbb{Z}_p)^n$. ■

It is known that for any extraspecial $p$-group $p^{1+n}$, $n$ is even.

**Proposition 5.7.** *$Q$ is an extraspecial 2-group $2^{1+n}$.*

*Proof.* We can show that the center of $Q$ is $\{\pm 1\}$ as follows. If $\gamma_k$ is in the center of $Q$, then $k \cdot k' \in 2\mathbb{Z}$ for any $k' \in L$. This means $\frac{k}{2} \in L^*$, where $L^*$ is the dual lattice defined as (2.14). Since we assumed $L$ is self-dual, $k \in 2L$, and therefore $\gamma_k = \pm 1$ by using the relations (5.79).

Finally, $Q/\{\pm 1\} \cong L/2L \cong (\mathbb{Z}_2)^n$. □

The irreducible representations of $Q$ are, the $2^n$ one-dimensional ones of $Q/\{\pm 1\} \cong (\mathbb{Z}_2)^n$ extended so that they map the center $\{\pm 1\}$ to 1, and the unique $2^{\frac{n}{2}}$-dimensional faithful one. We can understand this $2^{\frac{n}{2}}$-dimensional irreducible representation as follows. We can show [DGM90b, Appendix C] that $Q$ can be written as

$$Q = \{\pm\tilde{\gamma}_1^{t_1} \cdots \tilde{\gamma}_n^{t_n}\}_{t_i=0,1}\,, \tag{5.81}$$

using the usual gamma matrices (the Clifford algebra)

$$\tilde{\gamma}_i\tilde{\gamma}_j = -\tilde{\gamma}_j\tilde{\gamma}_i\ (i \neq j), \quad \tilde{\gamma}_i^2 = \tilde{\varepsilon}_i, \tag{5.82}$$

or equivalently $\{\tilde{\gamma}_i, \tilde{\gamma}_j\} = 2\tilde{\varepsilon}_i\delta_{ij}$, for some $\tilde{\varepsilon}_i = \pm 1$. Then, the construction of the $2^{\frac{n}{2}}$-dimensional irreducible representation can be done by the usual argument of representing $\tilde{\gamma}_i$ as the tensor product of $\frac{n}{2}$ Pauli matrices.

Among the irreducible representations of $Q$, this $2^{\frac{n}{2}}$-dimensional one is the only one which can extend to the representation of the whole algebra $\Gamma(L)$. (The other one-dimensional ones cannot satisfy for example $\{\tilde{\gamma}_i, \tilde{\gamma}_j\} = 0$.) Therefore, the gamma matrix algebra $\Gamma(L)$ have the unique irreducible representation, which we will write $\mathcal{X}(L)$, of dimension $2^{\frac{n}{2}}$.

*Remark* 5.8. For more general even lattices $L$, the center of the group $\{\pm\gamma_k \mid k \in L\}$ can be bigger than $\{\pm 1\}$. We still can construct its irreducible representations by a similar discussion using the usual gamma matrices, but the resulting representations are not unique in general, because there are degrees of freedom of assigning scalers $\pm 1, \pm\sqrt{-1}$ to the central elements. Such irreducible representations have a common dimension smaller than $2^{\frac{n}{2}}$ in general. See [DGM90b, Appendix C] for more details. *(Remark ends.)*

We introduce the creation-annihilation operators $c(r)$ ($c \in L \otimes_{\mathbb{Z}} \mathbb{C}$, $r \in \mathbb{Z} + \frac{1}{2}$) satisfy

$$[c(r), c'(r')] = \langle c, c' \rangle r \delta_{r+r',0}. \tag{5.83}$$

Then the $\theta_0$-twisted sector $(V_L)_{\mathrm{tw}}$ is constructed as the $\mathbb{C}$-vector space spanned by the elements in the form of

$$c_1(-r_1) \cdots c_l(-r_l)|s\rangle, \tag{5.84}$$

where $c_1, \ldots, c_l \in L \otimes_{\mathbb{Z}} \mathbb{C}$, $r_1, \ldots, r_l \in (\mathbb{Z} + \frac{1}{2})_{>0}$, and $|s\rangle \in \mathcal{X}(L)$. We also often use the creation-annihilation operators $\mathfrak{c}_r^i := \mathfrak{e}_i(r)$ with respect to an orthonormal basis $\mathfrak{e}_1, \ldots, \mathfrak{e}_n$ of $\mathbb{R}^n$ where the lattice $L$ is embedded.

The operators

$$L_m = \frac{1}{2} \sum_{r \in \mathbb{Z} + \frac{1}{2}} : \mathfrak{c}_r \mathfrak{c}_{m-r} : + \frac{n}{16} \delta_{m,0} \tag{5.85}$$

satisfy the Virasoro algebra (4.12) of central charge $n$. Then the state (5.84) is an eigenstate of $L_0$ with eigenvalue

$$r_1 + \cdots + r_l + \frac{n}{16}. \tag{5.86}$$

**Projection to $\theta_0$-invariant states**

To complete the construction of the orbifold, we project the whole $V_L \oplus (V_L)_{\mathrm{tw}}$ to the $\theta_0$-invariant states $(V_L)^0 \oplus (V_L)^0_{\mathrm{tw}} =: \tilde{V}_L$.

Recall that $\theta_0$ acts on $V_L$ as in (5.71)

$$\theta_0 \alpha_m^i \theta_0^{-1} = -\alpha_m^i, \quad \theta_0 |k\rangle = |-k\rangle. \tag{5.87}$$

Therefore, the $\theta_0$-invariant ($\theta_0$-even) sector $(V_L)^0$ and the $\theta_0$-odd sector $(V_L)^1$ are

$$\begin{aligned}(V_L)^0 = \mathrm{Span}_{\mathbb{C}}(\{&(\text{even number of } \alpha_m^i\text{'s})(|k\rangle + |-k\rangle)\} \\ &\oplus \{(\text{odd number of } \alpha_m^i\text{'s})(|k\rangle - |-k\rangle)\}),\end{aligned} \tag{5.88}$$

$$\begin{aligned}(V_L)^1 = \mathrm{Span}_{\mathbb{C}}(\{&(\text{odd number of } \alpha_m^i\text{'s})(|k\rangle + |-k\rangle)\} \\ &\oplus \{(\text{even number of } \alpha_m^i\text{'s})(|k\rangle - |-k\rangle)\}).\end{aligned} \tag{5.89}$$

We extend the action of $\theta_0$ on $V_L$ to the action of $\hat{\theta}_0$ on $V_L \oplus (V_L)_{\mathrm{tw}}$ by defining

$$\hat{\theta}_0 \mathfrak{c}_r^i \hat{\theta}_0^{-1} = -\mathfrak{c}_r^i, \quad \hat{\theta}_0 |s\rangle = (-1)^{\frac{n}{8}} |s\rangle. \tag{5.90}$$

Recall that the rank $n$ of an even self-dual lattice $L$ is a multiple of 8. Then, the $\hat{\theta}_0$-even sector $(V_L)^0_{\mathrm{tw}}$ and the $\hat{\theta}_0$-odd sector $(V_L)^1_{\mathrm{tw}}$ are

$$(V_L)^0_{\mathrm{tw}} = \mathrm{Span}_{\mathbb{C}}\{\text{states in } (V_L)_{\mathrm{tw}} \text{ with an integer } L_0\text{-eigenvalue (5.86)}\}, \tag{5.91}$$

$$(V_L)^1_{\mathrm{tw}} = \mathrm{Span}_{\mathbb{C}}\{\text{states in } (V_L)_{\mathrm{tw}} \text{ with a half-integer } L_0\text{-eigenvalue (5.86)}\}. \tag{5.92}$$

We finally define the orbifold $\tilde{V}_L$ of the lattice VOA $V_L$ by the reflection $\mathbb{Z}_2$ symmetry as

$$\tilde{V}_L := (V_L)^0 \oplus (V_L)^0_{\mathrm{tw}}. \tag{5.93}$$

The results of [FLM88], [DGM94, Lemma 5.3, Theorem 5.4] show that

- $(V_L)^0$ is a sub-VOA of $V_L$. (This holds for a general Euclidean even lattice $L$.)
- $(V_L)^0_{\mathrm{tw}}$ is a module of $(V_L)^0$.
- $\tilde{V}_L = (V_L)^0 \oplus (V_L)^0_{\mathrm{tw}}$ has a structure of VOA, extended from that on $(V_L)^0$. (This holds for a general Euclidean even lattice such that $\sqrt{2}L^*$ is also even.)

From the last point, we can indeed call the $L_0$-eigenvalue (5.86) the weight.

The reflection $\mathbb{Z}_2$ orbifold $\tilde{V}_{\Lambda_{24}}$ of the Leech lattice VOA is called the *monster VOA* or the *moonshine module*, and denoted by $V^\natural$.

### 5.3.2 Automorphisms of Reflection $\mathbb{Z}_2$ Orbifold

As mentioned in Section 2.4, the monster group $\mathbb{M}$ is generated by a subgroup $C(\Lambda_{24}) = 2^{1+24}.\mathrm{Co}_1$ and a specific order-2 element $\sigma$. In the proof of $\mathrm{Aut}(V^\natural) \cong \mathbb{M}$, the subgroup $C(\Lambda_{24})$ is coming from automorphisms acting on $(V_{\Lambda_{24}})^0$ and $(V_{\Lambda_{24}})^0_{\mathrm{tw}}$ separately. In this Section 5.3.2, we will review these automorphisms, following [FLM88, §10.4] and [DGM94, §6.1].

**The group $O(\hat{L})/\langle\theta_0\rangle$ acting on the untwisted $\theta_0$-invariant sector $(V_L)^0$**

Let us begin with the automorphisms of the untwisted $\theta_0$-invariant sector $(V_L)^0$. Recall that $V_L$ was built on $\hat{L}$, and $O(\hat{L})$ is a subgroup of $\mathrm{Aut}(V_L)$. Since $\theta_0 \in O(\hat{L})$ acts on $(V_L)^0$ trivially, $\mathrm{Aut}((V_L)^0)$ has a subgroup $O(\hat{L})/\langle\theta_0\rangle$. In the exact sequence (5.61), $\theta_0$ is a lift of $-\mathrm{id}_L \in O(L)$, so we have

$$1 \to \mathrm{Hom}(L, \mathbb{Z}_2) \to O(\hat{L})/\langle\theta_0\rangle \to O(L)/\{\pm\mathrm{id}_L\} \to 1. \tag{5.94}$$

Since $K$ defined in (5.75) is preserved by $\theta_0$ as a subgroup of $\hat{L}$, the group $O(\hat{L})/\langle\theta_0\rangle$ is a subgroup of $\mathrm{Aut}(\hat{L}/K)$. Recall $Q = \hat{L}/K$. We define the natural group homomorphism

$$\varphi : O(\hat{L}) \to O(\hat{L})/\langle\theta_0\rangle \subset \mathrm{Aut}(Q). \tag{5.95}$$

The entire automorphism group $\mathrm{Aut}(Q)$ of the group $Q$ can be described as follows. Recall that the group $Q = \hat{L}/K$ satisfies the exact sequence (5.78), similar to (5.36). As a result, similarly to (5.52), we have[33] the following exact sequence [FLM88, Prop. 5.4.5]

$$1 \to \mathrm{Hom}(L/2L, \mathbb{Z}_2) \xrightarrow{\tilde{\ }} \mathrm{Aut}(Q) \xrightarrow{\bar{\ }} \mathrm{Aut}(L/2L, q) \to 1, \tag{5.96}$$

[33] Precisely speaking, we first have to consider $\mathrm{Aut}(Q, \kappa) := \{f \in \mathrm{Aut}(Q) \mid f(\kappa) = \kappa\}$ similarly to (5.52), but in the case at hand, since the center of $Q$ is $\{1, \kappa\}$, any automorphism of $Q$ maps $\kappa$ to $\kappa$, so we just have $\mathrm{Aut}(Q, \kappa) = \mathrm{Aut}(Q)$.

where $q(k) = \kappa^{\frac{1}{2}|k|^2}|_{\kappa=-1}$ is the quadratic form and $\mathrm{Aut}(L/2L, q) := \{g \in \mathrm{Aut}(L/2L) \mid q(g(k)) = q(k)\}$. $\eta \in \mathrm{Hom}(L/2L, \mathbb{Z}_2)$ acts on $\gamma_k \in Q$ as $\tilde{\eta}(\gamma_k) = \eta(k)\gamma_k$, and $f \in \mathrm{Aut}(Q)$ maps $\gamma_k \in Q$ to $\pm\gamma_{\bar{f}(k)}$. Here, from the exact sequence (5.78), we regarded $k$ of $\gamma_k \in Q$ as an element of $L/2L$.

Since the quadratic form $q$ only depends on the bilinear form of $L$, $\mathrm{Aut}(L/2L, q)$ contains $O(L)/\{\pm\mathrm{id}_L\}$ as a subgroup. By restricting the exact sequence (5.96) from $\mathrm{Aut}(L/2L, q)$ to $O(L)/\{\pm\mathrm{id}_L\}$, we obtain a subgroup of $\mathrm{Aut}(Q)$. The restricted exact sequence coincides with (5.94), and therefore this subgroup of $\mathrm{Aut}(Q)$ is exactly $O(\hat{L})/\langle\theta_0\rangle$.

**The group $C_{\mathcal{X}(L)}$ acting on the $Q$-module $\mathcal{X}(L)$**

Next, we consider a certain subgroup of the linear automorphism group $\mathrm{Aut}(\mathcal{X}(L))$ of $\mathcal{X}(L)$. Since $\mathcal{X}(L)$ is a $Q$-module, we have $Q \subset \mathrm{Aut}(\mathcal{X}(L))$. The normalizer[34] $N_{\mathrm{Aut}(\mathcal{X}(L))}(Q)$ of $Q$ in $\mathrm{Aut}(\mathcal{X}(L))$ is defined as

$$N_{\mathrm{Aut}(\mathcal{X}(L))}(Q) := \{f \in \mathrm{Aut}(\mathcal{X}(L)) \mid fQ = Qf\} \tag{5.97}$$
$$= \{f \in \mathrm{Aut}(\mathcal{X}(L)) \mid f \bullet f^{-1} \in \mathrm{Aut}(Q)\}. \tag{5.98}$$

In other words, for any $\gamma \in Q$, if we write the action of $f \in \mathrm{Aut}(\mathcal{X}(L))$ as

$$f(\gamma|s\rangle) = f(\gamma)f|s\rangle \quad \text{for } |s\rangle \in \mathcal{X}(L), \tag{5.99}$$

then $f(\gamma) = f \circ \gamma \circ f^{-1}$ is again in $Q$ if $f \in N_{\mathrm{Aut}(\mathcal{X}(L))}(Q)$. It is known [FLM88, Prop. 5.5.3] that we have the following exact sequence

$$1 \to \{\pm\mathrm{id}_{\mathcal{X}(L)}\} \to N_{\mathrm{Aut}(\mathcal{X}(L))}(Q) \xrightarrow{f \mapsto f\bullet f^{-1}} \mathrm{Aut}(Q) \to 1. \tag{5.100}$$

When we extend an automorphism on $\mathcal{X}(L)$ to that on $(V_L)_{\mathrm{tw}}$, we will need $O(L)$ which acts on the creation-annihilation operators $c(r)$. In (5.100), the source of it is the subgroup $O(\hat{L})/\langle\theta_0\rangle$ of $\mathrm{Aut}(Q)$. We define a subgroup $C_{\mathcal{X}(L)}$ of $N_{\mathrm{Aut}(\mathcal{X}(L))}(Q)$ by restricting the exact sequence (5.100) as

$$1 \to \{\pm\mathrm{id}_{\mathcal{X}(L)}\} \to C_{\mathcal{X}(L)} \xrightarrow{f \mapsto f\bullet f^{-1}} O(\hat{L})/\langle\theta_0\rangle \to 1. \tag{5.101}$$

Note that, in addition to (5.100), $N_{\mathrm{Aut}(\mathcal{X}(L))}(Q)$ also satisfies the exact sequence

$$1 \to Q \to N_{\mathrm{Aut}(\mathcal{X}(L))}(Q) \xrightarrow{f \mapsto \overline{f\bullet f^{-1}}} \mathrm{Aut}(L/2L, q) \to 1, \tag{5.102}$$

and correspondingly,

$$1 \to Q \to C_{\mathcal{X}(L)} \xrightarrow{f \mapsto \overline{f\bullet f^{-1}}} O(L)/\{\pm\mathrm{id}_L\} \to 1. \tag{5.103}$$

[34] The *normalizer* of a subset $H$ of a group $G$ is defined as $N_G(H) := \{g \in G \mid gH = Hg\}$. If $H$ is a subgroup of $G$, then $N_G(H)$ is the largest subgroup of $G$ which contains $H$ as a normal subgroup, hence its name.

(5.102) can be shown as follows. Combining (5.96) and (5.100), we have $N_{\mathrm{Aut}(\mathcal{X}(L))}(Q) \cong 2.\mathrm{Aut}(Q) \cong 2.(2^n.\mathrm{Aut}(L/2L,q))$, and the homomorphism $N_{\mathrm{Aut}(\mathcal{X}(L))}(Q) \to \mathrm{Aut}(L/2L,q); f \mapsto \overline{f \bullet f^{-1}}$. The kernel of this homomorphism obviously contains $Q$ because $\gamma_k\gamma_{k'}\gamma_k^{-1} = \pm\gamma_{k'}$, and this group $Q$ has order $2^{1+n}$, which exhausts the order of the whole kernel. So we obtained the exact sequence (5.102).

**The group $\hat{C}(L)$ acting on the whole $V_L \oplus (V_L)_{\mathrm{tw}}$**

In order to extend the action of $C_{\mathcal{X}(L)}$ on $\mathcal{X}(L)$ to the action on $(V_L)_{\mathrm{tw}}$, we want the action of $g \in O(L)$ on the creation-annihilation operators $c(r) \mapsto g(c)(r)$, so we would like to revive $O(\hat{L})$ instead of $O(\hat{L})/\langle\theta_0\rangle$. This also incorporates the action of $O(\hat{L})$ on the untwisted sector $V_L$ at the same time.

We define

$$\hat{C}(L) := \{(f, f_{\mathcal{X}(L)}) \in O(\hat{L}) \times C_{\mathcal{X}(L)} \mid \varphi(f) = f_{\mathcal{X}(L)} \bullet f_{\mathcal{X}(L)}^{-1} \in O(\hat{L})/\langle\theta_0\rangle\}. \tag{5.104}$$

It can be summarized into the following commutative diagram.

$$\begin{array}{ccccc} & & \hat{C}(L) = 2^{1+n}.O(L) & & \\ & \swarrow^{/\langle(\theta_0,\mathrm{id}_{\mathcal{X}(L)})\rangle} & & \searrow^{/\langle(\mathrm{id}_{\hat{L}},-\mathrm{id}_{\mathcal{X}(L)})\rangle} & \\ C_{\mathcal{X}(L)} = 2^{1+n}.(O(L)/\{\pm\mathrm{id}_L\}) & & & & O(\hat{L}) = 2^n.O(L) \\ & \searrow_{f\mapsto f\bullet f^{-1}} & & \swarrow_{\varphi} & \\ & & O(\hat{L})/\langle\theta_0\rangle = 2^n.(O(L)/\{\pm\mathrm{id}_L\}) & & \end{array} \tag{5.105}$$

Here, we have the exact sequence

$$1 \to Q \xrightarrow{\gamma\mapsto(\gamma\bullet\gamma^{-1},\gamma)} \hat{C}(L) \xrightarrow{(f,f_{\mathcal{X}(L)})\mapsto\bar{f}} O(L) \to 1, \tag{5.106}$$

where $\gamma \bullet \gamma^{-1} \in \mathrm{Aut}(Q)$ in (5.100) is regarded as an element of $\mathrm{Hom}(L/2L,\mathbb{Z}_2) \subset \mathrm{Aut}(Q)$ as in (5.96), and extended to an element of $\mathrm{Hom}(L,\mathbb{Z}_2) \subset O(\hat{L})$. In fact, $\gamma_k\gamma_{k'}\gamma_k^{-1} = \eta_k(k')\gamma_{k'}$ for some $\eta_k \in \mathrm{Hom}(L,\mathbb{Z}_2)$.

The action of $F = (f, f_{\mathcal{X}(L)}) \in \hat{C}(L)$ on $V_L$ is the one already defined in (5.63),

$$F(\alpha_1(-m_1)\cdots\alpha_l(-m_l)e^k) = \bar{f}(\alpha_1)(-m_1)\cdots\bar{f}(\alpha_l)(-r_l)f(e^k), \tag{5.107}$$

and the action on the state (5.84) in $(V_L)_{\mathrm{tw}}$ is defined as

$$F(c_1(-r_1)\cdots c_l(-r_l)|s\rangle) = \bar{f}(c_1)(-r_1)\cdots\bar{f}(c_l)(-r_l)f_{\mathcal{X}(L)}(|s\rangle). \tag{5.108}$$

As a result, $\hat{C}(L)$ acts on the whole $V_L \oplus (V_L)_{\mathrm{tw}}$.

**The group $C(L)$ acting on the $\mathbb{Z}_2$-orbifold $\tilde{V}_L = (V_L)^0 \oplus (V_L)^0_{\mathrm{tw}}$**

Now, we can see that $\hat{\theta}_0$ defined in (5.90) is $(\theta_0, (-1)^{\frac{n}{8}}\mathrm{id}_{\mathcal{X}(L)})$ in $\hat{C}(L)$. Therefore, the group $C(L) := \hat{C}(L)/\langle(\theta_0, (-1)^{\frac{n}{8}}\mathrm{id}_{\mathcal{X}(L)})\rangle$ acts on the $\hat{\theta}_0$-invariant sector $\tilde{V}_L = (V_L)^0 \oplus (V_L)^0_{\mathrm{tw}}$. From (5.106), we have

$$1 \to Q \to C(L) \to O(L)/\{\pm\mathrm{id}_L\} \to 1. \tag{5.109}$$

When $n$ is an even multiple of 8, then $C(L)$ coincides with $C_{\mathcal{X}(L)}$. When $n$ is an odd multiple of 8, for example when $L$ is the Leech lattice $\Lambda_{24}$, then $C(L)$ fits into the commutative diagram (5.105) as follows.

$$\begin{array}{ccccc}
 & & \hat{C}(L) = 2^{1+n}.O(L) & & \\
 & \swarrow{\scriptstyle /\langle(\theta_0,\mathrm{id}_{\mathcal{X}(L)})\rangle} & \downarrow{\scriptstyle /\langle(\theta_0,-\mathrm{id}_{\mathcal{X}(L)})\rangle} & \searrow{\scriptstyle /\langle(\mathrm{id}_{\hat{L}},-\mathrm{id}_{\mathcal{X}(L)})\rangle} & \\
C_{\mathcal{X}(L)} = 2^{1+n}.(O(L)/\{\pm\mathrm{id}_L\}) & & C(L) = 2^{1+n}.(O(L)/\{\pm\mathrm{id}_L\}) & & O(\hat{L}) = 2^n.O(L) \\
 & \searrow{\scriptstyle f\mapsto f\bullet f^{-1}} & \downarrow & \swarrow{\scriptstyle \varphi} & \\
 & & O(\hat{L})/\langle\theta_0\rangle = 2^n.(O(L)/\{\pm\mathrm{id}_L\}) & &
\end{array} \tag{5.110}$$

In particular, when $L$ is the Leech lattice $\Lambda_{24}$, then $C(\Lambda_{24}) = 2^{1+24}.\mathrm{Co}_1$, which we mentioned in Section 2.4.

### 5.3.3 Triality and Shift $\mathbb{Z}_2$ Symmetry

**The triality operator $\sigma$**

From a given even self-dual lattice $L$, we have seen that we can construct two VOAs; one is the lattice VOA $V_L$, and the other is the orbifold $\tilde{V}_L$ of it by the reflection $\mathbb{Z}_2$ symmetry. We also explained in Section 2.3.1 that from a double-even self-dual binary code $\mathcal{C}$, we can construct two even self-dual lattices; one is $\Lambda(\mathcal{C})$ by Construction A, and the other is $\tilde{\Lambda}(\mathcal{C})$ by the twisted construction. As a result, from a double-even self-dual binary code $\mathcal{C}$, we can consider four VOAs, that is, $V_{\Lambda(\mathcal{C})}$, $V_{\tilde{\Lambda}(\mathcal{C})}$, $\tilde{V}_{\Lambda(\mathcal{C})}$, and $\tilde{V}_{\tilde{\Lambda}(\mathcal{C})}$. It was pointed out in [DGM90a, DGM94] that there are isomorphisms $\sigma$ among these VOAs, and this will provide one description of the proof of $\mathrm{Aut}(V^\natural) \cong \mathbb{M}$ as $\mathbb{M} = \langle C(\Lambda_{24}), \sigma\rangle$.

Recall that $\Lambda(\mathcal{C}) = \Lambda_0(\mathcal{C}) \sqcup \Lambda_1(\mathcal{C})$ and $\tilde{\Lambda}(\mathcal{C}) = \Lambda_0(\mathcal{C}) \sqcup \Lambda_3(\mathcal{C})$ in the notation[35] of (2.34)–

[35]The correspondence of the notations between [DGM94] and [FLM88, §12] is, in a situation where $n$ is an odd multiple of 8,

| [DGM94] | $\Lambda(\mathcal{C})$ | $\Lambda_2(\mathcal{C}) \sqcup \Lambda_3(\mathcal{C})$ | $\Lambda_0(\mathcal{C})$ | $\Lambda_1(\mathcal{C})$ | $\Lambda_2(\mathcal{C})$ | $\Lambda_3(\mathcal{C})$ | $\sigma, \sigma_3$ | $\sigma_1$ | Nothing |
|---|---|---|---|---|---|---|---|---|---|
| [FLM88] | $L_0$ | $L_1$ | $\Lambda_0$ | $\Lambda_2$ | $\Lambda_3$ | $\Lambda_1$ | $\sigma$ | $\tau$ | $\sigma_1$ |

.

Our main text basically follows [DGM94], but we will use $\tau$ instead of $\sigma_1$ in Remark 5.9. We also use

| this thesis | $V_{\Lambda_i(\mathcal{C})}$ | $V_{\mathcal{X}_i(\Lambda_0(\mathcal{C}))}$ | $(i = 0, 1, 2, 3)$ | $V^0_\bullet$ | $V^1_\bullet$ |
|---|---|---|---|---|---|
| [FLM88] | $V_{\Lambda_j}$ | $V'_{\Lambda_j}$ | $(j = 0, 2, 3, 1)$ | $V^+_\bullet$ | $V^-_\bullet$ |

.

(2.37). If we write the sector consisting of the states in the form of (5.29) with $k \in \Lambda_i(\mathcal{C})$ as $V_{\Lambda_i(\mathcal{C})}$, then

$$V_{\Lambda(\mathcal{C})} = V_{\Lambda_0(\mathcal{C})} \oplus V_{\Lambda_1(\mathcal{C})}, \tag{5.111}$$
$$V_{\tilde{\Lambda}(\mathcal{C})} = V_{\Lambda_0(\mathcal{C})} \oplus V_{\Lambda_3(\mathcal{C})}. \tag{5.112}$$

In particular, since $\Lambda_0(\mathcal{C})$ is also a lattice, $V_{\Lambda_0(\mathcal{C})}$ forms a VOA.

We can also introduce similar decompositions to the twisted sectors $(V_{\Lambda(\mathcal{C})})_{\mathrm{tw}}$ and $(V_{\tilde{\Lambda}(\mathcal{C})})_{\mathrm{tw}}$ as follows. If we decompose the irreducible representation $\mathcal{X}(\Lambda(\mathcal{C}))$ of $\Gamma(\Lambda(\mathcal{C}))$ and that $\mathcal{X}(\tilde{\Lambda}(\mathcal{C}))$ of $\Gamma(\tilde{\Lambda}(\mathcal{C}))$ into irreducible representations of $\Gamma(\Lambda_0(\mathcal{C}))$, then we obtain $\mathcal{X}(\Lambda(\mathcal{C})) = \mathcal{X}_0(\Lambda_0(\mathcal{C})) \oplus \mathcal{X}_1(\Lambda_0(\mathcal{C}))$ and $\mathcal{X}(\tilde{\Lambda}(\mathcal{C})) = \mathcal{X}_0(\Lambda_0(\mathcal{C})) \oplus \mathcal{X}_3(\Lambda_0(\mathcal{C}))$, respectively. The dimensions of $\mathcal{X}(\Lambda(\mathcal{C}))$ and $\mathcal{X}(\tilde{\Lambda}(\mathcal{C}))$ are $2^{\frac{n}{2}}$, and those of $\mathcal{X}_i(\Lambda_0(\mathcal{C}))$ are $2^{\frac{n}{2}-1}$ (see Remark 5.8). If we write the sector consisting of the states in the form of (5.84) with $|s\rangle \in \mathcal{X}_i(\Lambda_0(\mathcal{C}))$ as $V_{\mathcal{X}_i(\Lambda_0(\mathcal{C}))}$, then

$$(V_{\Lambda(\mathcal{C})})_{\mathrm{tw}} = V_{\mathcal{X}_0(\Lambda_0(\mathcal{C}))} \oplus V_{\mathcal{X}_1(\Lambda_0(\mathcal{C}))}, \tag{5.113}$$
$$(V_{\tilde{\Lambda}(\mathcal{C})})_{\mathrm{tw}} = V_{\mathcal{X}_0(\Lambda_0(\mathcal{C}))} \oplus V_{\mathcal{X}_3(\Lambda_0(\mathcal{C}))}. \tag{5.114}$$

We further decompose these sectors into $\hat{\theta}_0$-even sectors, denoted by the superscript $^0$, and $\hat{\theta}_0$-odd sectors, denoted by the superscript $^1$. As a result, the four VOAs can be summarized into the following diagram.

$$\begin{array}{ccccccccccc}
 & & & & V_{\Lambda(\mathcal{C})} & & \tilde{V}_{\tilde{\Lambda}(\mathcal{C})} & & & & \\
 & & & & \| & & \| & & & & \\
 & & & & (V_{\Lambda_0(\mathcal{C})})^0 & & (V_{\Lambda_0(\mathcal{C})})^0 & & & & \\
 & & & & \oplus & & \oplus & & & & \\
V_{\tilde{\Lambda}(\mathcal{C})} & = & (V_{\Lambda_0(\mathcal{C})})^0 & \oplus & (V_{\Lambda_0(\mathcal{C})})^1 & \oplus & (V_{\Lambda_3(\mathcal{C})})^0 & \oplus & (V_{\Lambda_3(\mathcal{C})})^1 & & \\
 & & & & \oplus & & \oplus & & & & \\
\tilde{V}_{\Lambda(\mathcal{C})} & = & (V_{\Lambda_0(\mathcal{C})})^0 & \oplus & (V_{\Lambda_1(\mathcal{C})})^0 & \oplus & (V_{\mathcal{X}_0(\Lambda_0(\mathcal{C}))})^0 & \oplus & (V_{\mathcal{X}_1(\Lambda_0(\mathcal{C}))})^0 & & \\
 & & & & \oplus & & \oplus & & & & \\
 & & & & (V_{\Lambda_1(\mathcal{C})})^1 & & (V_{\mathcal{X}_3(\Lambda_0(\mathcal{C}))})^0 & & & &
\end{array} \tag{5.115}$$

At the end of Section 5.3.1, we mentioned that $(V_L)^0$ is a sub-VOA of $V_L$, for an even lattice $L$. Since $\Lambda_0(\mathcal{C})$ is an even lattice, $(V_{\Lambda_0(\mathcal{C})})^0$ is a sub-VOA of $V_{\Lambda_0(\mathcal{C})}$. Moreover, every sector $(V_{\Lambda_i(\mathcal{C})})^p$ and $(V_{\mathcal{X}_i(\Lambda_0(\mathcal{C}))})^p$ ($i = 0, 1, 2, 3$ and $p = 0, 1$)[36] has the structure of irreducible representation of $(V_{\Lambda_0(\mathcal{C})})^0$ [DGM94, Prop. 7.1].

[36] See [DGM94] for the details of the sectors with $i = 2$.

[DGM90a, DGM94] constructed a map $\sigma$ acting on these sectors as

$$\begin{array}{cccc} & \circlearrowleft & \circlearrowleft & \\ & (V_{\Lambda_0(\mathcal{C})})^0 & (V_{\Lambda_0(\mathcal{C})})^0 & \\ \circlearrowleft & & & \\ (V_{\Lambda_0(\mathcal{C})})^0 & (V_{\Lambda_0(\mathcal{C})})^1 & (V_{\Lambda_3(\mathcal{C})})^0 & (V_{\Lambda_3(\mathcal{C})})^1 \\ & \downarrow\uparrow & \uparrow\downarrow & \downarrow\uparrow \\ (V_{\Lambda_0(\mathcal{C})})^0 & (V_{\Lambda_1(\mathcal{C})})^0 & (V_{\mathcal{X}_0(\Lambda_0(\mathcal{C}))})^0 & (V_{\mathcal{X}_1(\Lambda_0(\mathcal{C}))})^0 \\ \circlearrowleft & & & \\ & (V_{\Lambda_1(\mathcal{C})})^1 & (V_{\mathcal{X}_3(\Lambda_0(\mathcal{C}))})^0 & \\ & \circlearrowleft & \circlearrowleft & \end{array} , \tag{5.116}$$

and

$$\sigma : V_{\Lambda(\mathcal{C})} \to V_{\Lambda(\mathcal{C})}, \tag{5.117}$$

$$\sigma : V_{\tilde{\Lambda}(\mathcal{C})} \to \tilde{V}_{\Lambda(\mathcal{C})}, \quad \sigma : \tilde{V}_{\Lambda(\mathcal{C})} \to V_{\tilde{\Lambda}(\mathcal{C})}, \tag{5.118}$$

$$\sigma : \tilde{V}_{\tilde{\Lambda}(\mathcal{C})} \to \tilde{V}_{\tilde{\Lambda}(\mathcal{C})}, \tag{5.119}$$

are all isomorphisms of VOAs. This map $\sigma$ is called the *triality operator*. This map mixes the untwisted sector and the twisted sector of $\tilde{V}_{\tilde{\Lambda}(\mathcal{C})}$.

*Remark* 5.9. While $\sigma$ preserves the first and fourth rows of (5.116), there also exists a map $\tau$ (see footnote 35) which preserves the first and second rows of (5.116). These operators $\sigma$ and $\tau$ generate a group $\langle \sigma, \tau \rangle$ isomorphic to the symmetric group $S_3$, hence the term triality operator. *(Remark ends.)*

For example, in the case where $\mathcal{C}$ is the binary Golay code $G_{24}$, we can see from Table 2.2 that $V_{\Lambda(\mathcal{C})}$ is the lattice VOA $V_{(A_1)^{24}}$ of the $(A_1)^{24}$ Niemeier lattice, $\tilde{V}_{\Lambda(\mathcal{C})} \cong V_{\tilde{\Lambda}(\mathcal{C})}$ is the Leech lattice VOA $V_{\Lambda_{24}}$, and $\tilde{V}_{\tilde{\Lambda}(\mathcal{C})}$ is the reflection $\mathbb{Z}_2$ orbifold $\tilde{V}_{\Lambda_{24}}$ of the Leech lattice VOA, that is, the monster VOA $V^\natural$.

**The shift $\mathbb{Z}_2$ symmetry**

We briefly review that, from the viewpoint of physics, the CFT $V_{\tilde{\Lambda}(\mathcal{C})}$ introduced above can be regarded as the orbifold of the CFT $V_{\Lambda(\mathcal{C})}$ by its shift $\mathbb{Z}_2$ symmetry.

If we consider an even self-dual lattice $\Lambda(\mathcal{C})$ constructed by Construction A from a doubly-even self-dual code $\mathcal{C}$, the resulting CFT has the *shift* $\mathbb{Z}_2$ *symmetry* with respect to the shift vector $\chi := \frac{1}{\sqrt{2}}\vec{1} \in \Lambda(\mathcal{C})$ as

$$X(z) \mapsto X(z) + \pi\chi, \tag{5.120}$$

$$V_k(z) \propto : e^{\sqrt{-1}k\cdot X(z)} : \ \mapsto e^{\sqrt{-1}\pi k\cdot\chi} V_k(z) = \begin{cases} V_k(z) & (k \in \Lambda_0(\mathcal{C})) \\ -V_k(z) & (k \in \Lambda_1(\mathcal{C})) \end{cases} . \tag{5.121}$$

In terms of $O(\hat{L}) \subset \mathrm{Aut}(V_L)$ in (5.61), if we define $\chi^* := \langle \chi, - \rangle \in \mathrm{Hom}(L, \mathbb{Z})$, the shift $\mathbb{Z}_2$ symmetry is the subgroup $\mathbb{Z}_2 \subset O(\hat{L})$ generated by $(\chi^* \bmod 2) \in \mathrm{Hom}(L, \mathbb{Z}_2) \subset O(\hat{L})$.

Then we can also consider the states under the twisted boundary condition by this shift $\mathbb{Z}_2$ symmetry. The even and odd sectors of the untwisted and twisted sectors under the action of the shift $\mathbb{Z}_2$ symmetry can be described as (see for example [LS19, Appendix A], [KNO23b, §2.2])

| shift $\mathbb{Z}_2$ | untwisted | twisted |
|---|---|---|
| even | $V_{\Lambda_0(\mathcal{C})}$ | $V_{\Lambda_3(\mathcal{C})}$ |
| odd | $V_{\Lambda_1(\mathcal{C})}$ | $V_{\Lambda_2(\mathcal{C})}$ |

. (5.122)

Recall that $V_{\Lambda_i(\mathcal{C})}$ denotes the sector consisting of the states in the form of (5.29) with $k \in \Lambda_i(\mathcal{C})$.

As a result, the orbifold of the CFT $V_{\Lambda(\mathcal{C})}$ by its shift $\mathbb{Z}_2$ symmetry consists of the even sectors $V_{\Lambda_0(\mathcal{C})} \oplus V_{\Lambda_3(\mathcal{C})}$, which is precisely the CFT $V_{\tilde{\Lambda}(\mathcal{C})}$. So we can say that the isomorphism $\sigma : \tilde{V}_{\Lambda(\mathcal{C})} \cong V_{\tilde{\Lambda}(\mathcal{C})}$ in (5.118) is the isomorphism between the reflection $\mathbb{Z}_2$ orbifold and the shift $\mathbb{Z}_2$ orbifold of the lattice CFT $V_{\Lambda(\mathcal{C})}$ constructed from a doubly-even self-dual code $\mathcal{C}$.

In fact, the triality operator $\sigma$ is constructed in [DGM94, §7] as follows. In the CFT $V_{\Lambda(\mathcal{C})}$, for each axis of the lattice $\Lambda(\mathcal{C})$, there exists an $\widehat{\mathfrak{su}}(2)_1$ current

$$J_a(z_1)J_b(z_2) \sim \frac{1}{2}\delta_{ab}\frac{1}{(z_1-z_2)^2} + \sqrt{-1}\epsilon_{abc}\frac{J_c(z_2)}{z_1-z_2}, \tag{5.123}$$

as

$$J_1(z) = \frac{\sqrt{-1}}{\sqrt{2}}\partial X^i(z) \qquad \in (V_{\Lambda_0(\mathcal{C})})^1, \tag{5.124}$$

$$J_2(z) = -\frac{1}{2}(V_{\sqrt{2}\mathbb{e}_i}(z) + V_{-\sqrt{2}\mathbb{e}_i}(z)) \quad \in (V_{\Lambda_1(\mathcal{C})})^0, \tag{5.125}$$

$$J_3(z) = \frac{\sqrt{-1}}{2}(V_{\sqrt{2}\mathbb{e}_i}(z) - V_{-\sqrt{2}\mathbb{e}_i}(z)) \in (V_{\Lambda_1(\mathcal{C})})^1. \tag{5.126}$$

The operator $\sigma$ is constructed so that it implements the $\mathrm{SU}(2)$ rotation $J_1 \leftrightarrow J_2$ and $J_3 \mapsto -J_3$. As a result, the reflection $\mathbb{Z}_2$ symmetry $J_1 \mapsto -J_1$, $J_2 \mapsto J_2$, $J_3 \mapsto -J_3$ is essentially equivalent to the shift $\mathbb{Z}_2$ symmetry $J_1 \mapsto J_1$, $J_2 \mapsto -J_2$, $J_3 \mapsto -J_3$.

We refer the reader to [KNO23b, KY24] for the shift $\mathbb{Z}_2$ symmetries of more general lattice CFTs, and their orbifolds (and fermionizations).

### 5.3.4 $\mathbb{Z}_2$-orbifold and Fermionization

In the modern understanding of fermionization [Tac18, KTT19] (see also [HNT20, BSZ24]), we can uniformly treat $\mathbb{Z}_2$-orbifold and fermionization. Let us briefly review it.

In general, suppose a theory $\mathcal{T}$ has a non-anomalous $\mathbb{Z}_2$ symmetry, and the even and odd sectors of the untwisted and twisted sectors are

| $\mathbb{Z}_2$ | untwisted | twisted |
|---|---|---|
| even | $S$ | $U$ |
| odd | $T$ | $V$ |

. (5.127)

By orbifolding $\mathcal{T}$ by the $\mathbb{Z}_2$ symmetry, a new $\mathbb{Z}_2$ symmetry emerges,[37] and we obtain the following orbifold theory:

| (orbifold) | untwisted | twisted |
|---|---|---|
| even | $S$ | $T$ |
| odd | $U$ | $V$ |

. (5.128)

By fermionizing $\mathcal{T}$ by the $\mathbb{Z}_2$ symmetry, on the other hand, we obtain a fermionic CFT. There are two ways of fermionization:

| (fermionic) | NS | R |
|---|---|---|
| $(-1)^F=+1$ | $S$ | $U$ |
| $(-1)^F=-1$ | $V$ | $T$ |

, 

| (fermionic′) | NS | R |
|---|---|---|
| $(-1)^F=+1$ | $S$ | $T$ |
| $(-1)^F=-1$ | $V$ | $U$ |

. (5.129)

Here, NS and R denote the Neveu-Schwarz sector and the Ramond sector, respectively. In terms of a fermionic field $\psi(z)$, the NS sector satisfies the periodic boundary condition $\psi(e^{2\pi\sqrt{-1}}z)=\psi(z)$, and the R sector satisfies the anti-periodic boundary condition[38] $\psi(e^{2\pi\sqrt{-1}}z)=-\psi(z)$.

*Remark* 5.10. It is tempting to regard the NS/R sectors as the untwisted/twisted sectors by some $\mathbb{Z}_2$ symmetry. In fact, in mathematics, the NS sector and the R sector are usually formulated as a VOSA and its canonically-twisted module respectively (see Section 6.1.1). However, from the perspective of physics, it is natural to distinguish the NS/R sectors and the untwisted/twisted sectors, as the former is specified by the spin structure, but the latter is specified by the $\mathbb{Z}_2$ gauge field of the theory. See also footnote 59. *(Remark ends.)*

*Remark* 5.11. As remarked in Section 5.2.1, if we used an even self-dual lattice of signature $(r,s)$, then we obtain a bosonic lattice CFT of central charge $(c,\tilde{c})=(r,s)$, which is modular invariant up to the phases (1.4, 1.5) from the gravitational anomaly $\nu=2(\tilde{c}-c)$. Recall that the signature $(r,s)$ of an even self-dual lattice satisfies $r-s\equiv 0 \bmod 8$ [Ser73, Ch. V]. More generally, it is known that the gravitational anomaly of a bosonic CFT with central charge $(c,\tilde{c})$ satisfies $2(\tilde{c}-c)\equiv 0 \bmod 16$. Therefore, if we are given a fermionic CFT with central charge satisfying $2(\tilde{c}-c)\equiv 0 \bmod 16$, then we can bosonize it as a reverse operation of the fermionization [BSZ24]. *(Remark ends.)*

If we fermionize the lattice CFT $V_{\Lambda(\mathcal{C})}$ constructed from a doubly-even self dual code $\mathcal{C}$ by the reflection $\mathbb{Z}_2$ symmetry (5.73), then the NS sector of the resulting fermionic CFT is

$$(V_{\Lambda(\mathcal{C})})^0\oplus(V_{\Lambda(\mathcal{C})})^1_{\mathrm{tw}}=(V_{\Lambda_0(\mathcal{C})})^0\oplus(V_{\Lambda_1(\mathcal{C})})^0\oplus(V_{\mathcal{X}_0(\Lambda_0(\mathcal{C}))})^1\oplus(V_{\mathcal{X}_1(\Lambda_0(\mathcal{C}))})^1. \tag{5.130}$$

[37] If we write the partition function of the theory $\mathcal{T}$ on a torus twisted by the original $\mathbb{Z}_2$ symmetry in spatial and temporal directions as $Z^{(\mathcal{T})\,a_t}_{\ a_s}$ ($a_s,a_t\in\{0,1\}$; see also Section 12.2.1), then the new $\mathbb{Z}_2$ symmetry of the orbifold theory $\tilde{\mathcal{T}}$ is $Z^{(\tilde{\mathcal{T}})\,b_t}_{\ b_s}=\frac{1}{2}\sum_{a_s,a_t\in\{0,1\}}(-1)^{a_sb_t-a_tb_s}Z^{(\mathcal{T})\,a_t}_{\ a_s}$. More generally, if the original theory $\mathcal{T}[A]$ is on a surface $\Sigma$ and coupled to the $\mathbb{Z}_2$ gauge field $A\in H^1(M;\mathbb{Z}_2)$, then the orbifold theory is $\tilde{\mathcal{T}}[B]:=\frac{1}{\sqrt{|H^1(\Sigma;\mathbb{Z}_2)|}}\sum_{A\in H^1(\Sigma;\mathbb{Z}_2)}(-1)^{\int_\Sigma A\smile B}\mathcal{T}[A]$. See for example [BT17, §2], [LOZ23, §2.5], [BSZ24, §2] for more details.

[38] If we move to the cylinder coordinate $\sigma=\sigma^1+\sqrt{-1}\sigma^2$ defined as $z=e^{-\sqrt{-1}\sigma}$, since $\psi_{\mathrm{cyl}}(\sigma)=\psi(z)(\frac{\partial z}{\partial\sigma})^{\frac{1}{2}}$, the NS sector is anti-periodic $\psi_{\mathrm{cyl}}(\sigma+2\pi)=-\psi_{\mathrm{cyl}}(\sigma)$, and the R sector is periodic $\psi_{\mathrm{cyl}}(\sigma+2\pi)=\psi_{\mathrm{cyl}}(\sigma)$.

If we fermionize $V_{\Lambda(\mathcal{C})}$ by the shift $\mathbb{Z}_2$ symmetry (5.122), then the NS sector is

$$V_{\Lambda_0(\mathcal{C})} \oplus V_{\Lambda_2(\mathcal{C})} = (V_{\Lambda_0(\mathcal{C})})^0 \oplus (V_{\Lambda_0(\mathcal{C})})^1 \oplus (V_{\Lambda_2(\mathcal{C})})^0 \oplus (V_{\Lambda_2(\mathcal{C})})^1. \tag{5.131}$$

We have seen that the reflection $\mathbb{Z}_2$ symmetry and the shift $\mathbb{Z}_2$ symmetry are essentially equivalent in the previous Section 5.3.3. Therefore, these NS sectors of the fermionizations of $V_{\Lambda(\mathcal{C})}$ by the reflection $\mathbb{Z}_2$ and by the shift $\mathbb{Z}_2$ are isomorphic.[39]

For example, in the case where $\mathcal{C}$ is the binary Golay code $G_{24}$, the NS sector $V_{\Lambda_0(\mathcal{C})} \oplus V_{\Lambda_2(\mathcal{C})}$ of the fermionization of the $(A_1)^{24}$ Niemeier lattice CFT $V_{(A_1)^{24}}$ by the shift $\mathbb{Z}_2$ symmetry is the odd Leech lattice CFT $V_{O_{24}}$ according to Table 2.2, and it is isomorphic to the NS sector $(V_{\Lambda(\mathcal{C})})^0 \oplus (V_{\Lambda(\mathcal{C})})^1_{\mathrm{tw}}$ of the fermionization by the reflection $\mathbb{Z}_2$ symmetry.

Lattice CFTs constructed from more general codes over finite fields, and their orbifolds and fermionizations, are investigated in [GJF18, Yah22, KY23a, KNO23b, KY24]. Recently, the construction of CFTs from quantum codes through Lorentzian lattices was established in [DS20a, DS20b], and has been developed in many directions; see for example [DS20c, DS21, BDR21, ACD22,DK22], [HKM22], [Fur22,Fur23], [BR23], [KNO22,AKN$^+$23,KNO23a,KNO23b,AKN24]. We also note that [HM20] established a relation between a certain quantum code and a particular K3 CFT studied in [GTVW13] in the context of Mathieu moonshine.

[39] We can also see it by persistently following the description of [FLM88] as follows. We have already seen in (5.116) that $\sigma$ maps $(V_{\Lambda_0(\mathcal{C})})^0$ to itself, and interchanges $(V_{\Lambda_1(\mathcal{C})})^0 \leftrightarrow (V_{\Lambda_0(\mathcal{C})})^1$. So it suffices to observe $(V_{\mathcal{X}_0(\Lambda_0(\mathcal{C}))})^1 \leftrightarrow (V_{\Lambda_2(\mathcal{C})})^0$ and $(V_{\mathcal{X}_1(\Lambda_0(\mathcal{C}))})^1 \leftrightarrow (V_{\Lambda_2(\mathcal{C})})^1$. In the notation of [FLM88] (see footnote 35), we would like to see $V'^-_{\Lambda_0} \leftrightarrow V^+_{\Lambda_3}$ and $V'^-_{\Lambda_2} \leftrightarrow V^-_{\Lambda_3}$. The triality operator $\sigma$ is defined in [FLM88, Eq. (12.2.50)] as $\sigma = h \circ \nu(a_{k_0}) \circ \sigma_1$. $h$ is just a multiplication by $\mathrm{U}(1)$ phases. According to [FLM88, Prop. 12.2.7], $\sigma_1$ maps $V'^-_{\Lambda_0} \leftrightarrow V^-_{\Lambda_3}$ and $V'^-_{\Lambda_2} \leftrightarrow V^+_{\Lambda_3}$. Next, $a_{k_0}$ is an object similar to $\gamma_{\sqrt{2}\mathbf{e}_{i_0}}$ for some $i_0 \in \{1, \dots, n\}$ in (5.79) here, but an extension of a certain lattice by $\mathbb{Z}_4$ instead of $\mathbb{Z}_2$ is used in [FLM88, Eq. (12.1.24)], and correspondingly, the relations of $\gamma_k$'s are modified so that they include not only $\pm 1$ but also $\pm\sqrt{-1}$ like $(\gamma_{\sqrt{2}\mathbf{e}_i})^4 = 1$. $\nu$ is a map similar to $Q \to \hat{C}(L)$ in (5.106) here. This explains $\nu(a_{k_0}) : V'^{\pm}_{\Lambda_0} \leftrightarrow V'^{\pm}_{\Lambda_2}$ on the twisted sectors. (In addition, $\nu(a_{k_0}) : V'^{\pm}_{\Lambda_1} \leftrightarrow V'^{\mp}_{\Lambda_3}$. The parity flip is caused by the fact that $\hat{\theta}_0$ multiplies $\mathcal{X}_i(\Lambda_0(\mathcal{C}))$ $(i = 0, 1, 3)$ by $(-1)^{\frac{n}{8}}$ as in (5.90), whereas multiplies $\mathcal{X}_2(\Lambda_0(\mathcal{C}))$ by $-(-1)^{\frac{n}{8}}$ [DGM94, the first paragraph of §7].) On the untwisted sectors, $\gamma \bullet \gamma^{-1}$ acts as $\eta \in \mathrm{Hom}(L, \mathbb{Z}_4) \subset O(\hat{L})$ such that $\gamma\gamma_k\gamma^{-1} = \eta(k)\gamma_k$. If $(\gamma_k)^2 \propto 1$, then $\gamma_{-k} \propto \gamma_k$, and hence $\eta(k) = \eta(-k)$. However, now that $(\gamma_k)^4 \propto 1$, we have $\gamma_{-k} \propto (\gamma_k)^3$, and commutation of $\gamma$'s may produce $\pm\sqrt{-1}$, so $\eta(k)$ may have opposite sign to $\eta(-k)$. Then $|k\rangle + |-k\rangle$ and $|k\rangle - |-k\rangle$ may be interchanged. As a result, $\nu(a_{k_0})$ maps $V^{\pm}_{\Lambda_i}$ to itself for $i = 0, 2$, whereas interchanges $V^+_{\Lambda_i} \leftrightarrow V^-_{\Lambda_i}$ for $i = 1, 3$. These actions of $\nu(a_{k_0})$ are consistent with [FLM88, Eq. (12.2.48)] and the fact that $\sigma$ is an involution $\sigma^2 = \mathrm{id}$.

# 6 Conway Moonshine and Clifford Module VOSA

The Conway moonshine was first established in [Dun07] as an $\mathcal{N}=1$ VOSA $V^{f\natural}$ such that its automorphism group preserving the $\mathcal{N}=1$ structure is isomorphic to the sporadic Conway group $\mathrm{Co}_1$. Compared to the monstrous moonshine, which was led by the existence of the modular function, the Conway moonshine usually emphasizes the existence of such a VO(S)A with the Conway group symmetry itself. The Conway moonshine module $V^{f\natural}$ is a kind of $\mathbb{Z}_2$-orbifold of the VOSA describing 24 real free fermions. So we first review the construction of the VOSA of free fermions as a Clifford algebra module in Section 6.1, and then the construction of $V^{\natural}$ and the Conway group action on it in Section 6.2. We finally review the statement of the Conway moonshine in Section 6.3.

## 6.1 Clifford Module VOSA

In this Section 6.1, we review the construction of the VOSA of free fermions as a Clifford algebra module. In Section 6.1.1, we will introduce the Clifford algebras $\mathrm{Cliff}(\hat{\mathfrak{a}})$, $\mathrm{Cliff}(\hat{\mathfrak{a}}_{\mathrm{tw}})$ and their modules $A(\mathfrak{a})$, $A(\mathfrak{a})_{\mathrm{tw}}$, which describe the NS and R sectors of the fermions, respectively. In Section 6.1.2, we will review the spin group action on the modules.

### 6.1.1 Clifford Module Construction

The *tensor algebra* of a $\mathbb{K}$-vector space $V$ is

$$T(V) := \bigoplus_{d=0}^{\infty} V^{\otimes d}, \tag{6.1}$$

where $V^{\otimes 0} = \mathbb{K}$.

Let $V$ be a $\mathbb{K}$-vector space with a quadratic form $q$ (see Remark A.3 for the definition of a quadratic form). Its *Clifford algebra* $\mathrm{Cliff}(V,q)$ is the quotient of its tensor algebra by the relation[40] $u \otimes u \sim -q(u)$ for $u \in V$.

$$\mathrm{Cliff}(V,q) := T(V)/u \otimes u \sim -q(u). \tag{6.2}$$

If the characteristic of the field $\mathbb{K}$ is not 2, then by using the the associated form $\langle u,v\rangle := \frac{1}{2}(q(u+v)-q(u)-q(v))$ in the sense of (A.46), we have $q(u) = \langle u,u\rangle$, and the relation $u \otimes u = -q(u)$ is equivalent to

$$u \otimes v + v \otimes u = -2\langle u,v\rangle \tag{6.3}$$

for $u,v \in V$.

If the quadratic form $q$ is 0, then the Clifford algebra is the *exterior algebra* $\bigwedge V$ of $V$, and the multiplication $\otimes$ is often denoted by $\wedge$. For example, $u \wedge v = -v \wedge u$ for $u,v \in V$.

[40] Another convention $u \otimes u \sim q(u)$ is also commonly used for the definition of $\mathrm{Cliff}(V,q)$.

**Construction of the VOSA**

Let $\mathfrak{a}$ be an $n$-dimensional $\mathbb{C}$-vector space with a non-degenerate symmetric bilinear form $\langle -, - \rangle$. Define

$$\hat{\mathfrak{a}} := \bigoplus_{r \in \mathbb{Z}+\frac{1}{2}} \mathfrak{a}(r) \tag{6.4}$$

as the direct sum of copies of $\mathfrak{a}$ labeled by $r \in \mathbb{Z} + \frac{1}{2}$, and the symmetric bilinear form on $\hat{\mathfrak{a}}$ as the $\mathbb{C}$-bilinear extension of

$$\langle u(r), v(s) \rangle = \langle u, v \rangle \delta_{r+s,0} \quad (u, v \in \mathfrak{a},\ r, s \in \mathbb{Z} + \frac{1}{2}). \tag{6.5}$$

Here, $u(r)$ ($u \in \mathfrak{a}$) denotes an element in $\mathfrak{a}(r)$.

We define the Clifford algebra $\mathrm{Cliff}(\hat{\mathfrak{a}})$ of $\hat{\mathfrak{a}}$ as the one with respect to the quadratic form $q(\hat{u}) = \langle \hat{u}, \hat{u} \rangle$ ($\hat{u} \in \hat{\mathfrak{a}}$). That is,

$$\mathrm{Cliff}(\hat{\mathfrak{a}}) := T(\hat{\mathfrak{a}}) / \hat{u} \otimes \hat{u} \sim -\langle \hat{u}, \hat{u} \rangle. \tag{6.6}$$

We will omit $\otimes$ below. It is easy to see that the relation $\hat{u}\hat{u} = -\langle \hat{u}, \hat{u} \rangle$ is equivalent to

$$\{u(r), v(s)\} := u(r)v(s) + v(s)u(r) = -2\langle u(r), v(s) \rangle. \tag{6.7}$$

We introduce a polarization[41] $\hat{\mathfrak{a}} = \hat{\mathfrak{a}}^- \oplus \hat{\mathfrak{a}}^+$ as

$$\hat{\mathfrak{a}}^- := \bigoplus_{r \in (\mathbb{Z}+\frac{1}{2})_{<0}} \mathfrak{a}(r), \quad \hat{\mathfrak{a}}^+ := \bigoplus_{r \in (\mathbb{Z}+\frac{1}{2})_{>0}} \mathfrak{a}(r). \tag{6.8}$$

$\mathrm{Cliff}(\hat{\mathfrak{a}}^-) = \bigwedge \hat{\mathfrak{a}}^-$ and $\mathrm{Cliff}(\hat{\mathfrak{a}}^+) = \bigwedge \hat{\mathfrak{a}}^+$ are also defined in the same way as $\mathrm{Cliff}(\hat{\mathfrak{a}})$, and they are subalgebras of $\mathrm{Cliff}(\hat{\mathfrak{a}})$.

Take the $\mathrm{Cliff}(\hat{\mathfrak{a}}^+)$-module $\mathbb{C}$, whose basis is denoted by $|0\rangle$, on which $u(r) \in \mathrm{Cliff}(\hat{\mathfrak{a}}^+)$ acts as $u(r)|0\rangle = 0$ ($u \in V, r > 0$), and $\mathbb{C} \subset \mathrm{Cliff}(\hat{\mathfrak{a}}^+)$ acts as the usual multiplication. We define the $\mathrm{Cliff}(\hat{\mathfrak{a}})$-module $A(\mathfrak{a})$ as the induced module from it:

$$A(\mathfrak{a}) := \mathrm{Cliff}(\hat{\mathfrak{a}}) \otimes_{\mathrm{Cliff}(\hat{\mathfrak{a}}^+)} \mathbb{C}|0\rangle. \tag{6.9}$$

That is, $\hat{\mathfrak{a}}^+$ annihilates $|0\rangle$, and $A(\mathfrak{a})$ is isomorphic to $\mathrm{Cliff}(\hat{\mathfrak{a}}^-)|0\rangle$ as a $\mathrm{Cliff}(\hat{\mathfrak{a}}^-)$-module.

We can equip $A(\mathfrak{a})$ with a VOSA structure of central charge $\frac{n}{2}$. Let $\psi_1, \ldots, \psi_n$ be an orthonormal[42] basis of $\mathfrak{a}$ with $\langle \psi_i, \psi_i \rangle = +1$. The Virasoro element is given as

$$\omega = -\frac{1}{4} \sum_{i=1}^{n} \psi_i(-\frac{3}{2})\psi_i(-\frac{1}{2})|0\rangle. \tag{6.10}$$

[41] An *isotropic* vector of a vector space $V$ with a quadratic form $q$ is a vector $v \in V$ (sometimes required to be non-zero) such that $q(v) = 0$. A *polarization* of $V$ is a decomposition $V = V^+ \oplus V^-$ into maximal isotropic subspaces with respect to $q$. Note that the associated form of the quadratic form (A.45) is also called polarization, although we do not use this terminology in this thesis to avoid confusion.

[42] Since we are considering the symmetric bilinear form $\langle -, - \rangle$, not a Hermitian form, on the $\mathbb{C}$-vector space, there is no concept of the signature of the form, and we can always take an orthonormal basis with squared norm $+1$.

The weight of the state $u_1(-r_1)\cdots u_l(-r_l)|0\rangle$ ($u_i \in \mathfrak{a}$, $r_i \in (\mathbb{Z}+\frac{1}{2})_{>0}$) is

$$\mathrm{wt}(u_1(-r_1)\cdots u_l(-r_l)|0\rangle) = r_1 + \cdots + r_l. \tag{6.11}$$

The map $-\mathrm{id}_{\hat{\mathfrak{a}}} : \hat{\mathfrak{a}} \to \hat{\mathfrak{a}}$ induces the map $\theta : \mathrm{Cliff}(\hat{\mathfrak{a}}) \to \mathrm{Cliff}(\hat{\mathfrak{a}})$ called the *parity involution*. We write its eigenspace decomposition as $\mathrm{Cliff}(\hat{\mathfrak{a}}) = \mathrm{Cliff}(\hat{\mathfrak{a}})^0 \oplus \mathrm{Cliff}(\hat{\mathfrak{a}})^1$ with eigenvalues 1 and $-1$ respectively. We introduce the parity decomposition $A(\mathfrak{a}) = A(\mathfrak{a})^0 \oplus A(\mathfrak{a})^1$ by

$$A(\mathfrak{a})^0 := \mathrm{Cliff}(\hat{\mathfrak{a}})^0|0\rangle = (\overset{\text{even}}{\bigwedge} \hat{\mathfrak{a}}^-)|0\rangle, \tag{6.12}$$

$$A(\mathfrak{a})^1 := \mathrm{Cliff}(\hat{\mathfrak{a}})^1|0\rangle = (\overset{\text{odd}}{\bigwedge} \hat{\mathfrak{a}}^-)|0\rangle. \tag{6.13}$$

This introduces the $\mathbb{Z}_2$-grading of the VOSA on $A(\mathfrak{a})$. Any states in $A(\mathfrak{a})^0$ have integer weights, and states in $A(\mathfrak{a})^1$ have half-integer weights. In particular, the subalgebra $A(\mathfrak{a})^0$ of $A(\mathfrak{a})$ is a VOA.

In physics notation, $\psi_i(r)$ is often denoted by $\psi^i_r$, and the elements of $\hat{\mathfrak{a}}^-$ and $\hat{\mathfrak{a}}^+$ in the form of $u(r)$ are called the creation operators and the annihilation operators, respectively. $A(\mathfrak{a})$ is the NS sector of the $n$ real free chiral fermion theory.

**Construction of the R sector**

In general, for a vertex superalgebra $V = V^{\bar{0}} \oplus V^{\bar{1}}$, its *canonically-twisted module* $M$ is a $\mathbb{Z}_2$-twisted module $M$ of $V$ with $\mathbb{Z}_2$-grading $M^{\bar{0}} \oplus M^{\bar{1}}$ such that the twisted vertex operators (4.18)

$$Y^M(v,z) = \sum_{n \in \frac{1}{2}\mathbb{Z}} v^M_{(n)} z^{-n-1} \tag{6.14}$$

satisfy, for any $v \in V^{\bar{p}}$, $v^M_{(n)}$ maps $M^{\bar{q}}$ to $M^{\overline{p+q}}$, and $v^M_{(n)} = 0$ if $n \notin \mathbb{Z} + \frac{p}{2}$. Any vertex superalgebra has the parity involution $\theta := \mathrm{id}_{V^{\bar{0}}} \oplus (-\mathrm{id}_{V^{\bar{1}}})$, and in light of this, a canonically-twisted module is a $\theta$-twisted supermodule.

We can construct a canonically-twisted module $A(\mathfrak{a})_{\mathrm{tw}}$ of $A(\mathfrak{a})$ as follows. It is the R sector of the $n$ real free chiral fermion theory in the language of physics.

We assume $\underline{n = \dim \mathfrak{a} \text{ is even}}$ below. Define

$$\hat{\mathfrak{a}}_{\mathrm{tw}} := \bigoplus_{r \in \mathbb{Z}} \mathfrak{a}(r) \tag{6.15}$$

and the bilinear form on $\hat{\mathfrak{a}}_{\mathrm{tw}}$ as

$$\langle u(r), v(s)\rangle = \langle u, v\rangle \delta_{r+s,0} \quad (u, v \in \mathfrak{a},\ r, s \in \mathbb{Z}). \tag{6.16}$$

We choose a polarization of $\mathfrak{a}$ as $\mathfrak{a} = \mathfrak{a}^- \oplus \mathfrak{a}^+$. They are usually chosen so that

$$\mathfrak{a}^- = \mathrm{Span}_{\mathbb{C}}\{\Psi_1, \ldots, \Psi_{\frac{n}{2}}\}, \quad \mathfrak{a}^+ = \mathrm{Span}_{\mathbb{C}}\{\overline{\Psi}_1, \ldots, \overline{\Psi}_{\frac{n}{2}}\}, \tag{6.17}$$

where

$$\Psi_i := \frac{1}{\sqrt{2}}(\psi_{2i-1} + \sqrt{-1}\psi_{2i}), \quad \overline{\Psi}_i := \frac{1}{\sqrt{2}}(\psi_{2i-1} - \sqrt{-1}\psi_{2i}), \tag{6.18}$$

for an orthonormal basis $\psi_1, \dots, \psi_n$ of $\mathfrak{a}$. Then we introduce a polarization $\hat{\mathfrak{a}}_{\mathrm{tw}} = \hat{\mathfrak{a}}^-_{\mathrm{tw}} \oplus \hat{\mathfrak{a}}^+_{\mathrm{tw}}$ as

$$\hat{\mathfrak{a}}^-_{\mathrm{tw}} := \mathfrak{a}^-(0) \oplus \bigoplus_{r \in \mathbb{Z}_{<0}} \mathfrak{a}(r), \quad \hat{\mathfrak{a}}^+_{\mathrm{tw}} := \mathfrak{a}^+(0) \oplus \bigoplus_{r \in \mathbb{Z}_{>0}} \mathfrak{a}(r). \tag{6.19}$$

Again, $\mathrm{Cliff}(\hat{\mathfrak{a}}^\pm_{\mathrm{tw}}) = \bigwedge \hat{\mathfrak{a}}^\pm_{\mathrm{tw}}$ are subalgebras of $\mathrm{Cliff}(\hat{\mathfrak{a}})$.

Take the $\mathrm{Cliff}(\hat{\mathfrak{a}}^+_{\mathrm{tw}})$-module $\mathbb{C}$, whose basis is denoted by $|0\rangle_{\mathrm{R}}$, on which $u(r) \in \mathrm{Cliff}(\hat{\mathfrak{a}}^+_{\mathrm{tw}})$ acts as $u(r)|0\rangle_{\mathrm{R}} = 0$, and $\mathbb{C} \subset \mathrm{Cliff}(\hat{\mathfrak{a}}^+)$ acts as the usual multiplication. We define the $\mathrm{Cliff}(\hat{\mathfrak{a}}_{\mathrm{tw}})$-module $A(\mathfrak{a})_{\mathrm{tw}}$ as

$$A(\mathfrak{a})_{\mathrm{tw}} := \mathrm{Cliff}(\hat{\mathfrak{a}}_{\mathrm{tw}}) \otimes_{\mathrm{Cliff}(\hat{\mathfrak{a}}^+_{\mathrm{tw}})} \mathbb{C}|0\rangle_{\mathrm{R}}. \tag{6.20}$$

That is, $\hat{\mathfrak{a}}^+_{\mathrm{tw}}$ annihilates $|0\rangle_{\mathrm{R}}$.

We can equip $A(\mathfrak{a})_{\mathrm{tw}}$ with a canonically-twisted $A(\mathfrak{a})$-module structure. The weight of the state $u_1(-r_1)\cdots u_l(-r_l)u^-_1(0)\cdots u^-_{l'}(0)|0\rangle_{\mathrm{R}}$ ($u_i \in \mathfrak{a}$, $r_i \in \mathbb{Z}_{>0}$, $u^-_i \in \mathfrak{a}^-$) is

$$\mathrm{wt}(u_1(-r_1)\cdots u_l(-r_l)u^-_1(0)\cdots u^-_{l'}(0)|0\rangle_{\mathrm{R}}) = r_1 + \cdots + r_l + \frac{n}{16}. \tag{6.21}$$

The $\mathbb{Z}_2$-grading on $A(\mathfrak{a})_{\mathrm{tw}}$ is introduced as follows: we assign 0 or 1 to the parity of $|0\rangle_{\mathrm{R}}$, and then $(\bigwedge^{\mathrm{even}} \hat{\mathfrak{a}}^-_{\mathrm{tw}})|0\rangle_{\mathrm{R}}$ has the same parity as $|0\rangle_{\mathrm{R}}$, whereas $(\bigwedge^{\mathrm{odd}} \hat{\mathfrak{a}}^-_{\mathrm{tw}})|0\rangle_{\mathrm{R}}$ has the opposite parity. The parity of $|0\rangle_{\mathrm{R}}$ is basically arbitrary. For example, [FFR91] always defines the parity of $|0\rangle_{\mathrm{R}}$ as 0. In this thesis, following [DMC14], we specify it as $\frac{n}{4} \bmod 2$, when $n$ is a multiple of 4, which will be justified in Section 6.1.2 as the eigenvalue $(-1)^{\frac{n}{4}}$ of $|0\rangle_{\mathrm{tw}}$ with respect to the action by the lift $\psi_1 \cdots \psi_n \in \mathrm{Spin}(\mathfrak{a})$ of $-\mathrm{id}_{\mathfrak{a}} \in \mathrm{SO}(\mathfrak{a})$. We write the subspaces of $A(\mathfrak{a})_{\mathrm{tw}}$ with parity 0 and 1 as $A(\mathfrak{a})^0_{\mathrm{tw}}$ and $A(\mathfrak{a})^1_{\mathrm{tw}}$, respectively.

### 6.1.2 Spin Group Action

**Spin group** $\mathrm{Spin}(\mathfrak{a})$

The details of the definition and properties of the spin group are reviewed in Appendix F. Here, we only present the main points.

Recall that $\mathfrak{a}$ is an $n$-dimensional $\mathbb{C}$-vector space with symmetric bilinear form $\langle -, - \rangle$. Similarly to $\mathrm{Cliff}(\hat{\mathfrak{a}})$, we introduce the parity decomposition $\mathrm{Cliff}(\mathfrak{a}) = \mathrm{Cliff}(\mathfrak{a})^0 \oplus \mathrm{Cliff}(\mathfrak{a})^1$ induced by $-\mathrm{id}_{\mathfrak{a}}$. The Clifford algebra $\mathrm{Cliff}(\mathfrak{a})$ contains the *spin group* $\mathrm{Spin}(\mathfrak{a})$ defined by

$$\mathrm{Spin}(\mathfrak{a}) := \{u_1 \cdots u_{2l} \mid u_i \in \mathfrak{a}, \langle u_i, u_i \rangle = 1\} \tag{6.22}$$

$$= \{x \in (\mathrm{Cliff}(\mathfrak{a})^0)^\times \mid x\mathfrak{a}x^{-1} \subset \mathfrak{a}, x^T x = 1\}. \tag{6.23}$$

Here, $(\mathrm{Cliff}(\mathfrak{a})^0)^\times$ is the set of invertible elements of $\mathrm{Cliff}(\mathfrak{a})^0$, and $x^T$ is defined as $x^T = u_l \cdots u_1$ if $x = u_1 \cdots u_l$ ($u_i \in \mathfrak{a}$). If $x \in \mathbb{C}$, then $x^T = x$. (This is the definition of the spin group in the sense of (F.25, F.27).)

An important property of $\mathrm{Spin}(\mathfrak{a})$ is the following fact.

**Proposition 6.1.** $\mathrm{Spin}(\mathfrak{a})$ *is a double cover of* $\mathrm{SO}(\mathfrak{a})$ *defined as*

$$\mathrm{SO}(\mathfrak{a}) := \{O \in \mathrm{SL}(\mathfrak{a}) \mid \langle Ou, Ov\rangle = \langle u, v\rangle \text{ for any } u, v \in \mathfrak{a}\}. \tag{6.24}$$

*More precisely,* $\mathrm{Spin}(\mathfrak{a})$ *is a central extension of* $\mathrm{SO}(\mathfrak{a})$ *by* $\mathbb{Z}_2$

$$1 \to \{\pm 1_{\mathrm{Spin}(\mathfrak{a})}\} \hookrightarrow \mathrm{Spin}(\mathfrak{a}) \xrightarrow{\varphi} \mathrm{SO}(\mathfrak{a}) \to 1, \tag{6.25}$$

*where* $\varphi : \mathrm{Spin}(\mathfrak{a}) \to \mathrm{SO}(\mathfrak{a})$ *is the group homomorphism defined as*

$$\varphi : x \mapsto x \bullet x^{-1}. \tag{6.26}$$

*In particular, if we take an orthonormal basis* $\psi_1, \dots, \psi_n$ *of* $\mathfrak{a}$*, then the lifts* $\pm\hat{O} \in \mathrm{Spin}(\mathfrak{a})$ *of* $O \in \mathrm{SO}(\mathfrak{a})$ *act on* $\sum_i c^i\psi_i \in \mathfrak{a}$ *as (F.34)*

$$(\pm\hat{O}) \left[ \begin{pmatrix}\psi_1 \cdots \psi_n\end{pmatrix} \begin{pmatrix} c^1 \\ \vdots \\ c^n \end{pmatrix} \right] (\pm\hat{O})^{-1} = \begin{pmatrix}\psi_1 \cdots \psi_n\end{pmatrix} O \begin{pmatrix} c^1 \\ \vdots \\ c^n \end{pmatrix}, \tag{6.27}$$

*where* $O$ *is represented as a matrix with respect to the basis* $\{\psi_i\}_i$*.*

**The** $\mathrm{Spin}(\mathfrak{a})$**-action on** $A(\mathfrak{a})$ **and** $A(\mathfrak{a})_{\mathrm{tw}}$

The Clifford module VOSA $A(\mathfrak{a})$ and its twisted module $A(\mathfrak{a})_{\mathrm{tw}}$ naturally admit a $\mathrm{Spin}(\mathfrak{a})$-action as follows.

The action of $x \in \mathrm{Spin}(\mathfrak{a})$ on $u_1(r_1)\cdots u_l(r_l)|0\rangle \in A(\mathfrak{a})$ ($r_i \in \mathbb{Z} + \frac{1}{2}$) is defined as

$$u_1(r_1)\cdots u_k(r_k)|0\rangle \;\mapsto\; (xu_1x^{-1})(r_1)\cdots(xu_kx^{-1})(r_k)|0\rangle. \tag{6.28}$$

This action is well-defined because $\{(xu_ix^{-1})(r_i), (xu_jx^{-1})(r_j)\} = -2\langle xu_ix^{-1}, xu_jx^{-1}\rangle\delta_{r_i+r_j,0} = \{u_i(r_i), u_j(r_j)\}$. Since $-1_{\mathrm{Spin}(\mathfrak{a})} \in \mathrm{Spin}(\mathfrak{a})$ acts on $A(\mathfrak{a})$ trivially, this $\mathrm{Spin}(\mathfrak{a})$-action reduces to $\mathrm{Spin}(\mathfrak{a})/\langle -1_{\mathrm{Spin}(\mathfrak{a})}\rangle \cong \mathrm{SO}(\mathfrak{a})$-action.

To describe the action of $\mathrm{Spin}(\mathfrak{a})$ on $A(\mathfrak{a})_{\mathrm{tw}}$, we identify $\mathrm{Cliff}(\mathfrak{a})$ with $\mathrm{Cliff}(\mathfrak{a}(0)) \subset \mathrm{Cliff}(\hat{\mathfrak{a}}_{\mathrm{tw}})$. Then $x \in \mathrm{Spin}(\mathfrak{a}) \subset \mathrm{Cliff}(\mathfrak{a})$ acts on $|0\rangle_{\mathrm{R}}$. So the action of $x \in \mathrm{Spin}(\mathfrak{a})$ on $u_1(r_1)\cdots u_l(r_l)|0\rangle_{\mathrm{R}} \in A(\mathfrak{a})_{\mathrm{tw}}$ ($r_i \in \mathbb{Z}$) is defined as

$$u_1(r_1)\cdots u_k(r_k)|0\rangle_{\mathrm{R}} \;\mapsto\; (xu_1x^{-1})(r_1)\cdots(xu_kx^{-1})(r_k)x|0\rangle_{\mathrm{R}}. \tag{6.29}$$

The action of $-1_{\mathrm{Spin}(\mathfrak{a})} \in \mathrm{Spin}(\mathfrak{a})$ on $A(\mathfrak{a})_{\mathrm{tw}}$ is nontrivial, and $A(\mathfrak{a})_{\mathrm{tw}}$ is only a projective representation of $\mathrm{SO}(\mathfrak{a})$.

There are two lifts of $-1_{\mathrm{SO}(\mathfrak{a})} \in \mathrm{SO}(\mathfrak{a})$. If we take an orthonormal basis $\psi_1, \dots, \psi_n$ of $\mathfrak{a}$, then they can be explicitly written as $\pm\psi_1\cdots\psi_n \in \mathrm{Spin}(\mathfrak{a})$. Depending on the choice of the polarization $\mathfrak{a} = \mathfrak{a}^- \oplus \mathfrak{a}^+$, one acts as $|0\rangle_{\mathrm{R}} \mapsto \sqrt{-1}^{\frac{n}{2}}|0\rangle_{\mathrm{R}}$, and the other acts as $|0\rangle_{\mathrm{R}} \mapsto -\sqrt{-1}^{\frac{n}{2}}|0\rangle_{\mathrm{R}}$. The former one is denoted by $\mathfrak{z}$ and called the *lift of* $-1_{\mathrm{SO}(\mathfrak{a})}$ *associated with the polarization*. When $n = \dim\mathfrak{a}$ is a multiple of 4, the $\mathbb{Z}_2$-grading of the canonically-twisted module

$A(\mathfrak{a})_{\mathrm{tw}} = A(\mathfrak{a})^0_{\mathrm{tw}} \oplus A(\mathfrak{a})^1_{\mathrm{tw}}$ introduced at the end of the previous Section 6.1.1 coincides with the eigenspace decomposition with respective to $\mathfrak{z}$ with eigenvalue $+1$ and $-1$. The $\mathbb{Z}_2$-grading of the VOSA $A(\mathfrak{a}) = A(\mathfrak{a})^0 \oplus A(\mathfrak{a})^1$ also coincides with the eigenspace decomposition with respect to $\mathfrak{z}$. The eigenvalues of each sector with respect to $\mathfrak{z}$, $-\mathfrak{z}$, and $-1_{\mathrm{Spin}(\mathfrak{a})}$ can be summarized as follows.

| | $A(\mathfrak{a})^0$ | $A(\mathfrak{a})^1$ | $A(\mathfrak{a})^0_{\mathrm{tw}}$ | $A(\mathfrak{a})^1_{\mathrm{tw}}$ |
|---|---|---|---|---|
| eigenvalue w.r.t. $\mathfrak{z}$ | $1$ | $-1$ | $1$ | $-1$ |
| eigenvalue w.r.t. $-\mathfrak{z}$ | $1$ | $-1$ | $-1$ | $1$ |
| eigenvalue w.r.t. $-1_{\mathrm{Spin}(\mathfrak{a})}$ | $1$ | $1$ | $-1$ | $-1$ |

(6.30)

If we choose the polarization as in (6.17), then we can explicitly check

$$\mathfrak{z} = +\psi_1 \cdots \psi_n = \sqrt{-1}^{\frac{n}{2}} (\Psi_1 \overline{\Psi}_1 + 1) \cdots (\Psi_{\frac{n}{2}} \overline{\Psi}_{\frac{n}{2}} + 1). \tag{6.31}$$

Finally, we remark that these $\mathrm{Spin}(\mathfrak{a})$-actions on $A(\mathfrak{a})$ and $A(\mathfrak{a})_{\mathrm{tw}}$ are weight-preserving and parity-preserving.

## 6.2 Duncan's Supermoonshine Module

In this Section 6.2, we review the construction of the Conway moonshine module $V^{f\natural}$. It is constructed from the VOSA of free fermions in a way similar to the $\mathbb{Z}_2$-orbifold in Section 6.2.1, and we will see how the Conway group action is realized on it in Section 6.2.2.

### 6.2.1 Construction of Duncan's Module

We set $\mathfrak{a} = \Lambda_{24} \otimes_{\mathbb{Z}} \mathbb{C}$ where $\Lambda_{24}$ is the Leech lattice with symmetric bilinear form $\langle -, - \rangle$. In particular, $n = \dim \mathfrak{a} = 24$. The $A(\mathfrak{a})^0$-module structure on

$$V^{f\natural} := A(\mathfrak{a})^0 \oplus A(\mathfrak{a})^0_{\mathrm{tw}} \tag{6.32}$$

extends uniquely to an VOSA structure, and the $A(\mathfrak{a})^0$-module structure on

$$V^{f\natural}_{\mathrm{tw}} := A(\mathfrak{a})^1 \oplus A(\mathfrak{a})^1_{\mathrm{tw}} \tag{6.33}$$

extends uniquely to a canonically-twisted $V^{f\natural}$-module structure. There seems no consensus on the name of this VOSA $V^{f\natural}$, or the whole theory $V^{f\natural} \oplus V^{f\natural}_{\mathrm{tw}}$. We call it the *Conway (super)moonshine module* following [TW17], or *Duncan's (supermoonshine) module* following [AKL22], because it was first constructed by Duncan in [Dun07], although it should be noted that it was also revisited with Mack-Crane in [DMC14].

The original theory $A(\mathfrak{a})$ and Duncan's module $V^{f\natural}$ are in the relation similar to the $\mathbb{Z}_2$-orbifold[43] of each other.

| $\mathbb{Z}_2 = \langle \mathfrak{z} \rangle$ | NS $A(\mathfrak{a})$ | R $A(\mathfrak{a})_{\mathrm{tw}}$ |
|---|---|---|
| even | $A^0(\mathfrak{a})$ | $A^0(\mathfrak{a})_{\mathrm{tw}}$ |
| odd | $A^1(\mathfrak{a})$ | $A^1(\mathfrak{a})_{\mathrm{tw}}$ |

'$\langle\mathfrak{z}\rangle$-orbifold' $\rightleftarrows$ '$\langle -1_{\mathrm{Spin}(\mathfrak{a})}\rangle$-orbifold'

| $\mathbb{Z}_2 = \langle -1_{\mathrm{Spin}(\mathfrak{a})} \rangle$ | NS $V^{f\natural}$ | R $V^{f\natural}_{\mathrm{tw}}$ |
|---|---|---|
| even | $A^0(\mathfrak{a})$ | $A^1(\mathfrak{a})$ |
| odd | $A^0(\mathfrak{a})_{\mathrm{tw}}$ | $A^1(\mathfrak{a})_{\mathrm{tw}}$ |

(6.34)

We may consider the similar construction with respect to $-\mathfrak{z}$ (6.30), and the resulting theory is denoted by $V^{s\natural}$.

| $\mathbb{Z}_2 = \langle -\mathfrak{z} \rangle$ | NS $A(\mathfrak{a})$ | R $A(\mathfrak{a})_{\mathrm{tw}}$ |
|---|---|---|
| even | $A^0(\mathfrak{a})$ | $A^1(\mathfrak{a})_{\mathrm{tw}}$ |
| odd | $A^1(\mathfrak{a})$ | $A^0(\mathfrak{a})_{\mathrm{tw}}$ |

'$\langle-\mathfrak{z}\rangle$-orbifold' $\rightleftarrows$ '$\langle -1_{\mathrm{Spin}(\mathfrak{a})}\rangle$-orbifold'

| $\mathbb{Z}_2 = \langle -1_{\mathrm{Spin}(\mathfrak{a})} \rangle$ | NS $V^{s\natural}$ | R $V^{s\natural}_{\mathrm{tw}}$ |
|---|---|---|
| even | $A^0(\mathfrak{a})$ | $A^1(\mathfrak{a})$ |
| odd | $A^1(\mathfrak{a})_{\mathrm{tw}}$ | $A^0(\mathfrak{a})_{\mathrm{tw}}$ |

(6.35)

The $A(\mathfrak{a})^0$-module structure on

$$V^{s\natural} := A(\mathfrak{a})^0 \oplus A(\mathfrak{a})^1_{\mathrm{tw}} \tag{6.36}$$

extends uniquely to an VOSA structure, and the $A(\mathfrak{a})^0$-module structure on

$$V^{s\natural}_{\mathrm{tw}} := A(\mathfrak{a})^1 \oplus A(\mathfrak{a})^0_{\mathrm{tw}} \tag{6.37}$$

extends uniquely to a canonically-twisted $V^{s\natural}$-module structure. This $V^{s\natural}$ is isomorphic to $V^{f\natural}$ as a VOSA, and also called the Conway moonshine module.

*Remark* 6.2 (lattice description). Through the boson-fermion correspondence [Fre81], the VOA $A(\mathfrak{a})^0$ is isomorphic to the lattice VOA $V_{D_{12}}$ of the $D_{12}$ lattice. Recall that

$$D_n := \{(k_1, \dots, k_n) \in \mathbb{Z}^n \mid \sum_{i=1}^{n} k_i \equiv 0 \bmod 2\}. \tag{6.38}$$

It is known that the irreducible representations of a lattice VOA $V_L$ for an even lattice $L$ are in one-to-one correspondence with the elements of the quotient $L^*/L$ of the dual lattice $L^*$, called the *discriminant group*[44] of the lattice $L$, and labeled as $\{V_{\lambda+L}\}_{\lambda+L\in L^*/L}$ [Don93].

The $D_n$ lattice is $\sqrt{2}\times$ the lattice constructed by Construction A from the *even weight code* $\mathcal{E}_n := \{w \in \mathbb{F}_2^n \mid \mathrm{wt}(w) \equiv 0 \bmod 2\}$ [CS99, Ch. 3 §2.3], or $\sqrt{2}\times$ the lattice constructed by

[43] In the viewpoint of footnote 59, we assume that the spin structures on the tori of both the original and orbifold theories are the NS sectors (in the spatial direction), and take the $\mathbb{Z}_2$-orbifold by the $\mathbb{Z}_2$ gauge field. In this sense, $A(\mathfrak{a})_{\mathrm{tw}}$ is in fact the $\mathbb{Z}_2$-twisted sector of the NS sector $A(\mathfrak{a})$. The reason why we can also regard $A(\mathfrak{a})_{\mathrm{tw}}$ as the R sector is that the fermions are subject to the sum $\sigma + A$ of the spin structure $\sigma$ and the $\mathbb{Z}_2$ gauge field $A$.

[44] If a lattice $L$ is integral, we have $L \subset L^*$, and hence the discriminant group $L^*/L$ is well-defined. Letting $G$ denote the Gram matrix $L$, we have $|L^*/L| = \det G$.

Construction B from the trivial code $0_n$ of length $n$. Mimicking (2.34)–(2.37) for doubly-even self dual code $\mathcal{C}$, let us define for $0_n$ with $n \equiv 0 \bmod 4$,

$$\Lambda_0(0_n) := \mathbb{Z}_+^n = D_n, \tag{6.39}$$
$$\Lambda_1(0_n) := \mathbb{Z}_-^n = (1,0,\dots,0) + D_n, \tag{6.40}$$
$$\Lambda_2(0_n) := \frac{1}{2}\vec{1} + \mathbb{Z}^n_{(-)^{\frac{n}{4}+1}} = \frac{1}{2}((-1)^{\frac{n}{4}+1},1,\dots,1) + D_n, \tag{6.41}$$
$$\Lambda_3(0_n) := \frac{1}{2}\vec{1} + \mathbb{Z}^n_{(-)^{\frac{n}{4}}} = \frac{1}{2}((-1)^{\frac{n}{4}},1,\dots,1) + D_n. \tag{6.42}$$

Now, we have $D_{12} = \Lambda_0(0_{12})$, and $D_{12}^* = \bigsqcup_{i=0}^3 \Lambda_i(0_{12})$. $|D_{12}^*/D_{12}| = 4$ and the four elements of $D_{12}^*/D_{12}$ are $\Lambda_i(0_{12})$ $(i = 0,1,2,3)$. As $A(\mathfrak{a})^0$-modules,

$$A(\mathfrak{a})^0 \cong V_{\Lambda_0(0_{12})}, \quad A(\mathfrak{a})^1 \cong V_{\Lambda_2(0_{12})}, \quad A(\mathfrak{a})^0_{\mathrm{tw}} \cong V_{\Lambda_2(0_{12})}, \quad A(\mathfrak{a})^1_{\mathrm{tw}} \cong V_{\Lambda_3(0_{12})}. \tag{6.43}$$

As a result, $V^{f\natural} \cong V_{D_{12}^+}$ as VOSAs, where $D_{12}^+ := D_{12} \sqcup (\frac{1}{2}\vec{1} + D_{12})$ is the unique self-dual positive-definite lattice of rank 12 without vectors of squared length 1. Since $D_{12}^+ \cong D_{12} \sqcup (\frac{1}{2}(-1,1,\dots,1) + D_{12})$, we can see $V^{s\natural} \cong V_{D_{12}^+} \cong V^{f\natural}$. *(Remark ends.)*

### 6.2.2 Conway Group Action

**The** $\mathrm{SemiSpin}(24)$**-action on** $V^{f\natural}$

We focus on $\mathrm{SO}(24) = \mathrm{SO}(\Lambda_{24}\otimes_{\mathbb{Z}}\mathbb{R}) \subset \mathrm{SO}(\mathfrak{a})$, and its double cover $\mathrm{Spin}(24) = \mathrm{Spin}(\Lambda_{24}\otimes_{\mathbb{Z}}\mathbb{R}) \subset \mathrm{Spin}(\mathfrak{a})$.

In general, if $n$ is a multiple of 4, then the center[45] of $\mathrm{Spin}(n)$ is $\mathbb{Z}_2 \times \mathbb{Z}_2 \cong \langle -1_{\mathrm{Spin}(n)}\rangle \times \langle \mathfrak{z}\rangle$. The quotient $\mathrm{Spin}(n)/\langle -1_{\mathrm{Spin}(n)}\rangle$ is isomorphic to $\mathrm{SO}(n)$, whereas the quotient $\mathrm{Spin}(n)/\langle \mathfrak{z}\rangle$ is another group called the *semi-spin group* $\mathrm{SemiSpin}(n)$, not isomorphic to $\mathrm{SO}(n)$ except for $n = 8$.

Recall that $\mathrm{Spin}(24)$ acts on the whole $A(\mathfrak{a}) \oplus A(\mathfrak{a})_{\mathrm{tw}}$. Since the action of $\mathfrak{z}$ on $V^{f\natural}$ is trivial, the $\mathrm{Spin}(24)$-action on $V^{f\natural}$ reduces to the action by the quotient $\mathrm{Spin}(24)/\langle -\mathfrak{z}\rangle \cong \mathrm{SemiSpin}(24)$.

$$\begin{array}{ccc} & \mathrm{Spin}(24) \curvearrowright A(\mathfrak{a}) \oplus A(\mathfrak{a})_{\mathrm{tw}} & \\ {\scriptstyle /\langle \mathfrak{z}\rangle}\swarrow & & \searrow{\scriptstyle /\langle -1_{\mathrm{Spin}(24)}\rangle} \\ \mathrm{SemiSpin}(24) \curvearrowright V^{f\natural} & & \mathrm{SO}(24) \curvearrowright A(\mathfrak{a}) \end{array} \tag{6.45}$$

[45] The center of $\mathrm{Spin}(n)$ is [Bin13, Ch. 8]

$$Z(\mathrm{Spin}(n)) = \begin{cases} \mathbb{Z}_2 \cong \langle -1_{\mathrm{Spin}(n)}\rangle & (n \text{ is odd}), \\ \mathbb{Z}_4 \cong \langle \mathfrak{z}\rangle & (n \equiv 2 \bmod 4), \\ \mathbb{Z}_2 \times \mathbb{Z}_2 \cong \langle -1_{\mathrm{Spin}(n)}\rangle \times \langle \mathfrak{z}\rangle & (n \equiv 0 \bmod 4). \end{cases} \tag{6.44}$$

**The $\mathrm{Co}_1$-action on $V^{f\natural}$**

Take the Conway group $\mathrm{Co}_0 = \mathrm{Aut}(\Lambda_{24}) \subset \mathrm{SO}(24)$.

**Proposition 6.3** ([DMC14, Prop. 3.1])**.** *There is a lift $\hat{G} \subset \mathrm{Spin}(24)$ of $\mathrm{Co}_0 \subset \mathrm{SO}(24)$ such that $\hat{G}$ is isomorphic to $\mathrm{Co}_0$, and such a lift is unique.*

*Proof.* The existence of $\hat{G} \cong \mathrm{Co}_0$ is due to the fact that the Schur multiplier $H_2(\mathrm{Co}_0; \mathbb{Z})$ is trivial [Gri74, §9], and $\mathrm{Co}_0$ is a perfect group[46] $H_1(\mathrm{Co}_0; \mathbb{Z}) = 0$.

We can see the uniqueness as follows. According to the $\mathrm{Co}_1$ page of [WWT$^+$], $\mathrm{Co}_0$ is generated by two elements $A, B$ such that[47] $(ABABABABBABABBABB)^{33} = 1$ and $B^3 = 1$. Since these relations contain odd numbers of $A$ and $B$ respectively, these relations determine which of the two lifts $\pm\hat{A} \in \mathrm{Spin}(24)$ of $A$ is the generator $\hat{A}$ of $\hat{G} \subset \mathrm{Spin}(24)$, and which of $\pm\hat{B}$ is the generator as well. □

We take an orthonormal basis $\psi_1, \dots, \psi_{24}$ of $\Lambda_{24} \otimes_{\mathbb{Z}} \mathbb{R}$, and introduce the polarization $\mathfrak{a} = \mathfrak{a}^- \oplus \mathfrak{a}^+$ as in (6.17). This determines which of the two lifts of $-1_{\mathrm{SO}(24)} \in \mathrm{SO}(24)$ is $\mathfrak{z} \in \mathrm{Spin}(24)$; the other one is $-\mathfrak{z}$. Since $-1_{\mathrm{SO}(24)} \in \mathrm{Co}_0$, one of $\pm\mathfrak{z}$ is contained in $\hat{G}$, and the other is not. Recalling (6.31) $\mathfrak{z} = \psi_1 \cdots \psi_{24}$, we can rename $\psi_1$ and $\psi_2$ so that $\mathfrak{z} \in \hat{G}$. In this way, we always assume that we take the orthonormal basis of $\Lambda_{24} \otimes_{\mathbb{Z}} \mathbb{R}$ and the polarization such that $\mathfrak{z} \in \hat{G}$.

Since the action of $\mathfrak{z}$ on $V^{f\natural}$ is trivial, the action of $\hat{G} \subset \mathrm{Spin}(24)$ on $V^{f\natural}$ reduces to the action by the quotient $\hat{G}/\langle\mathfrak{z}\rangle \cong \mathrm{Co}_1 \subset \mathrm{SemiSpin}(24)$.

We will also write $\hat{G}$ as $\mathrm{Co}_0$ when there is no risk of confusion.

$$\begin{array}{ccc} & \mathrm{Co}_0 \subset \mathrm{Spin}(24) \curvearrowright A(\mathfrak{a}) \oplus A(\mathfrak{a})_{\mathrm{tw}} & \\ {\scriptstyle /\langle\mathfrak{z}\rangle}\swarrow & & \searrow{\scriptstyle \cong} \\ \mathrm{Co}_1 \subset \mathrm{SemiSpin}(24) \curvearrowright V^{f\natural} & & \mathrm{Co}_0 \subset \mathrm{SO}(24) \curvearrowright A(\mathfrak{a}) \end{array} \tag{6.46}$$

*Remark* 6.4. On $V^{s\natural}$, the action of $-\mathfrak{z}$ is trivial, but the action of $\mathfrak{z}$ is not. So we can say that the action of $\mathrm{Spin}(24)$ reduces to the action of $\mathrm{Spin}(24)/\langle-\mathfrak{z}\rangle \cong \mathrm{SemiSpin}(24)$, but since $-\mathfrak{z} \notin \mathrm{Co}_0 \subset \mathrm{Spin}(24)$, the action of $\mathrm{Co}_0$ does not reduce to the action of $\mathrm{Co}_1$. In particular, $V^{f\natural}$ and $V^{s\natural}$ are isomorphic as VOSAs, but they are not as $\mathrm{Co}_0$-modules. *(Remark ends.)*

[46] A group $G$ is said to be *perfect* if its abelianization $G^{\mathrm{ab}} := G/[G, G]$ is trivial. Since the only normal subgroups of $\mathrm{Co}_0$ are the trivial ones 1, $\mathrm{Co}_0$, and its center $\mathbb{Z}_2$ satisfying $\mathrm{Co}_0/\mathbb{Z}_2 \cong \mathrm{Co}_1$, it is obvious that $\mathrm{Co}_0$ is perfect. Using $G^{\mathrm{ab}} \cong H_1(G; \mathbb{Z})$, we have $H_1(\mathrm{Co}_0; \mathbb{Z}) = 0$. By applying $H_1(\mathrm{Co}_0; \mathbb{Z}) = 0$ and the trivial Schur multiplier $H_2(\mathrm{Co}_0; \mathbb{Z}) = 0$ to the universal coefficient theorem, which states $0 \to \mathrm{Ext}_{\mathbb{Z}}(H_1(G; \mathbb{Z}); A) \to H^2(G; A) \to \mathrm{Hom}_{\mathbb{Z}}(H_2(G; \mathbb{Z}); A) \to 0$ is exact, we have $H^2(\mathrm{Co}_0; \mathbb{Z}_2) = 0$. This means any extension of $\mathrm{Co}_0$ by $\mathbb{Z}_2$ splits, and hence there is a group-homomorphic lift of $\mathrm{Co}_0 \subset \mathrm{SO}(24)$ to $\mathrm{Spin}(24)$.

[47] The convention of [WWT$^+$] is the right action, so if we use $A, B$ introduced in (2.46, 2.47) which act on column vectors in $\mathbb{Z}^{24}$ from the left, we have to reverse the order of the multiplications in these relations. Note that the Version 3 webpage https://brauer.maths.qmul.ac.uk/Atlas/v3/spor/Co1/ is not maintained, and there is a typo which lacks the last $B$ as "$(ABABABABBABABBAB)^{33} = 1$," when accessed November 9th, 2025. Instead, the Version 2 webpage https://brauer.maths.qmul.ac.uk/Atlas/spor/Co1/ is maintained when accessed, and presents the correct relation.

## 6.3 Conway Moonshine

Recall that $V^{f\natural} = A(\mathfrak{a})^0 \oplus A(\mathfrak{a})^0_{\mathrm{tw}}$, and states in $A(\mathfrak{a})^0$ have integer weights, whereas those in $A(\mathfrak{a})^0_{\mathrm{tw}}$ have half-integer weights. In particular, the weight-$\frac{3}{2}$ subspace of $A(\mathfrak{a})^0_{\mathrm{tw}}$ is

$$\mathbf{2}^{\mathbf{11}}_{+} := \mathrm{Span}_{\mathbb{C}}\{(\text{even number of } \Psi_i(0)\text{'s})|0\rangle_{\mathrm{R}}\} \subset A(\mathfrak{a})^0_{\mathrm{tw}}. \tag{6.47}$$

This $\mathbf{2}^{\mathbf{11}}_{+}$ and

$$\mathbf{2}^{\mathbf{11}}_{-} := \mathrm{Span}_{\mathbb{C}}\{(\text{odd number of } \Psi_i(0)\text{'s})|0\rangle_{\mathrm{R}}\} \subset A(\mathfrak{a})^1_{\mathrm{tw}}, \tag{6.48}$$

are called the *chiral representations* or the *Weyl spinor representations* of $\mathrm{Spin}(24)$.

We have seen that $\mathrm{Co}_1 \subset \mathrm{SemiSpin}(24)$ acts on $V^{f\natural}$. As a $\mathrm{Co}_1$-module, this weight-$\frac{3}{2}$ subspace $\mathbf{2}^{\mathbf{11}}_{+}$ decomposes into irreducible modules as [DMC14, proof of Prop. 4.4]

$$\mathbf{2}^{\mathbf{11}}_{+} = \mathbf{1} \oplus \mathbf{276} \oplus \mathbf{1771}, \tag{6.49}$$

where $\mathbf{1}$ is the trivial representation, whose states are invariant under the $\mathrm{Co}_1$-action. Moreover, a suitably scaled basis $\tau$ of this $\mathrm{Co}_1$-invariant subspace $\mathbf{1}$ constitutes the supercurrent of an $\mathcal{N} = 1$ superconformal algebra (4.22, 4.23) in $V^{f\natural}$ [DMC14, Prop. 4.4].

Conversely, it is proved [Dun07, Thm. 4.11] that the group $\mathrm{Aut}_{\mathcal{N}=1}(V^{f\natural})$ of all the automorphisms of the VOSA $V^{f\natural}$ preserving this $\mathcal{N} = 1$ structure is precisely isomorphic to the Conway group $\mathrm{Co}_1$. To summarize,

**Theorem 6.5** ([Dun07, Thm. 4.11])**.** *The VOSA $V^{f\natural}$ admits an $\mathcal{N} = 1$ structure such that $\mathrm{Aut}_{\mathcal{N}=1}(V^{f\natural}) \cong \mathrm{Co}_1$.*

Furthermore, its uniqueness is also known as follows, although we do not go into the details.

**Theorem 6.6** ([DMC14, Thm. 4.5])**.** *Let $V$ be a self-dual $C_2$-cofinite rational VOSA of CFT type, with central charge 12 and without weight-$\frac{1}{2}$ states $V_{\frac{1}{2}} = 0$. Then $V$ is isomorphic to $V^{f\natural}$ as a VOSA.*

In particular, $V$ in this Theorem 6.6 admits an $\mathcal{N} = 1$ structure such that $\mathrm{Aut}_{\mathcal{N}=1}(V) \cong \mathrm{Co}_1$.

Of course, we can see the appearance of representation dimensions of $\mathrm{Co}_1$ in the coefficients of the partition function of $V^{f\natural}$ [Dun07, Eq. (1.0.2)]

$$\mathrm{Tr}_{V^{f\natural}} q^{L_0 - \frac{12}{24}} = \frac{\Theta_{E_8}(\tau)\eta(\tau)^8}{\eta(\tau/2)^8\eta(2\tau)^8} - 8 \tag{6.50}$$

$$= q^{-\frac{1}{2}}(1 + 276q + 2^{11}q^{\frac{3}{2}} + 11202q^4 + 49152q^{\frac{5}{2}} + \cdots), \tag{6.51}$$

where 276 is the smallest nontrivial irreducible representation dimension of $\mathrm{Co}_1$, $2^{11} = 1+276+1771$ as in (6.49), $11202 = 1+276+299+1771+8855$, and so on. Since the $\mathrm{Co}_1$-action preserves the $\mathcal{N} = 1$ superconformal algebra, we will still see the representation dimensions of $\mathrm{Co}_1$ even if we decompose the partition function into $\mathcal{N} = 1$ superconformal algebra characters.

Now, we can see that the three important sporadic groups $\mathbb{M}$, $\mathrm{Co}_1$, and $M_{24}$ can be realized as automorphism groups of the following objects.

| Aut | algebraic object | VOA |
|---|---|---|
| $\mathbb{M}$ | the Griess algebra $B^\natural$ | the (bosonic) VOA $V^\natural$ |
| $\mathrm{Co}_1$ | the (quotient of) Leech lattice $\Lambda_{24}/\{\pm 1\}$ | the $\mathcal{N}=1$ VOSA $V^{f\natural}$ |
| $M_{24}$ | the Golay code $G_{24}$ | ? |

(6.52)

*Remark* 6.7. Among the algebraic objects listed above (6.52), we have already mentioned the uniqueness of $G_{24}$, $\Lambda_{24}$, and $V^{f\natural}$. The Golay code $G_{24}$ is the unique binary code of length 24, dimension 12, and minimal nonzero weight 8 (Section 2.2.1). The Leech lattice $\Lambda_{24}$ is the unique even self-dual lattice of rank 24 and without vectors of squared length 2 (Section 2.3.1). Duncan's module $V^{f\natural}$ is the unique VOSA of central charge 12 and without weight-$\frac{1}{2}$ states (Theorem 6.6).

As mentioned in Section 2.4, the Griess algebra $B^\natural$ is the weight-2 subspace $(V^\natural)_2$ of the monster VOA $V^\natural$. The uniqueness of the monster VOA $V^\natural$ as a VOA of central charge 24 and without weight-1 states is still an open problem, called the *Frenkel–Lepowsky–Meurman uniqueness conjecture*. See [DGL07] for a partial result. One interest thing is that VOAs of central charge 24 with weight-1 states $V_1 \neq 0$ are known to be uniquely determined from the structure of the weight-1 subspace $V_1$, and already classified as 70 VOAs; see for example [vELMS21, Lam23]. Therefore, if the uniqueness of the monster VOA is proved, then the proof of the list of 71 VOAs of central charge 24 conjectured by Schellekens [Sch93] will be completed. *(Remark ends.)*

# 7 Mathieu Moonshine

A K3 surface is a complex-two-dimensional Calabi–Yau manifold. More precisely,[48] a complex-two-dimensional compact Kähler manifold $X$ is said to be a *K3 surface* if its canonical bundle is trivial, and it has the Hodge number $h^{0,1}(X) = 0$. It is known that any K3 surfaces are diffeomorphic, but their isomorphism classes as complex manifolds are not unique. The second integral cohomology[49] $H^2(X;\mathbb{Z})$ of a K3 surface together with the symmetric bilinear form defined by the cup product is a lattice of signature $(3,19)$ and isometric to[50] $\mathrm{II}_{1,1}^{\oplus 3} \oplus (-E_8)^{\oplus 2}$. The isomorphism classes of K3 surfaces are parametrized by the real-two-dimensional plane in $H^2(X;\mathbb{R})$, specified by the nowhere-vanishing holomorphic 2-form $\omega$ on $X$. The space of all the isomorphism classes of K3 surfaces is called the *moduli space* of K3 surfaces. See for example [Kon18, Kon20, Asp96] for more details.

We can consider the supersymmetric non-linear sigma model (full or non-chiral two-dimensional SCFT) with its target space being a K3 surface $X$. We call it a *K3 sigma model* for short. It has $\dim_{\mathbb{R}} X$ bosons and fermions, so the central charge is $c = \tilde{c} = 4 \times (1 + \frac{1}{2}) = 6$. Thanks to the hyperKähler structure of the K3 surface, it has $\mathcal{N} = (4,4)$ supersymmetries. Hence, the elliptic genus $Z_{\mathrm{ell}}^{(\mathrm{K3})}(\tau, z)$ (see Section 3.4) of a K3 sigma model is a weak Jacobi form of weight 0 and index 1. From Theorem 3.4, such a weak Jacobi form is $\varphi_{0,1}(\tau, z)$ up to scalar multiplication, and it was calculated in [EOTY89] as

$$Z_{\mathrm{ell}}^{(\mathrm{K3})}(\tau, z) = 2\varphi_{0,1}(\tau, z) = 8\left(\frac{\theta_2(\tau,z)^2}{\theta_2(\tau,0)^2} + \frac{\theta_3(\tau,z)^2}{\theta_3(\tau,0)^2} + \frac{\theta_4(\tau,z)^2}{\theta_4(\tau,0)^2}\right). \tag{7.1}$$

This elliptic genus is independent of the choice of the target K3 surface, or equivalently, constant over the moduli space.

Recall that the elliptic genus is a $(-1)^F$-inserted partition function of the left-moving R sectors coupling to the right-moving Ramond vacua. Since the left-moving part of a K3 sigma model has the $\mathcal{N} = 4$ supersymmetry, we can decompose the elliptic genus $Z_{\mathrm{ell}}^{(\mathrm{K3})}(\tau, z)$ into the $(-1)^F$-inserted characters of irreducible Ramond representations of the $\mathcal{N} = 4$ superconformal algebra of central charge $c = 6$. Such irreducible characters are labeled by the conformal weight $h$ and the spin $l$ (with respect to the $\widehat{\mathfrak{su}}(2)_{\frac{c}{6}}$ algebra consisting of the three supercurrents) of the highest weight state, and classified into

- the massless characters $\mathrm{ch}_{(0)h,l}^{\mathcal{N}=4}(\tau, z)$ with $h = \frac{1}{4}$ and $l = 0, \frac{1}{2}$,

[48] There are several different definitions of a Calabi–Yau manifold. If we define it as a compact Kähler manifold $X$ with trivial canonical bundle, then two-dimensional Calabi–Yau manifolds are only complex-2-tori and K3 surfaces. Between them, K3 surfaces can be specified by the condition $h^{0,1}(X) = 0$ as in the main text, or by the condition that the holonomy group is $\mathrm{SU}(2)$. (The holonomy of a torus is trivial.) So some literature uses the definition of a Calabi–Yau manifold containing the condition that the holonomy is $\mathrm{SU}(n)$ where $n$ is the complex dimension.

[49] In the notation of footnote 10, $H^2(X;\mathbb{Z})$ in the main text here should be written as $H^2_{\mathrm{top}}(X;\mathbb{Z})$.

[50] $\mathrm{II}_{1,1}$ is the even self-dual lattice of signature $(1,1)$. It can be written as $\mathrm{II}_{1,1} = (\mathbb{Z}^2, q)$ where the symmetric bilinear form $q$ has the matrix form $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. $-E_8$ denotes the $E_8$ lattice whose symmetric bilinear form is multiplied by $-1$, and hence of signature $(0,8)$.

- the massive characters $\mathrm{ch}^{\mathcal{N}=4}_{h,l}(\tau,z)$ with $h=\frac{1}{4}+\mathbb{Z}_{>0}$ and $l=\frac{1}{2}$.

Their explicit formulae are derived in [ET88] (also reviewed in [EOT10]) as

$$\mathrm{ch}^{\mathcal{N}=4}_{(0)\frac{1}{4},0}(\tau,z)=\frac{\theta_1(\tau,z)^2}{\eta(\tau)^3}\mu(\tau,z),\quad \mu(\tau,z):=\frac{-\sqrt{-1}y^{1/2}}{\theta_1(\tau,z)}\sum_{l\in\mathbb{Z}}\frac{(-1)^l y^l q^{l(l+1)/2}}{1-yq^l}, \tag{7.2}$$

$$\mathrm{ch}^{\mathcal{N}=4}_{h,\frac{1}{2}}(\tau,z)=\frac{\theta_1(\tau,z)^2}{\eta(\tau)^3}q^{h-\frac{3}{8}}. \tag{7.3}$$

$\mathrm{ch}^{\mathcal{N}=4}_{(0)\frac{1}{4},\frac{1}{2}}(\tau,z)$ can be derived from

$$\mathrm{ch}^{\mathcal{N}=4}_{\frac{1}{4},\frac{1}{2}}(\tau,z)=2\,\mathrm{ch}^{\mathcal{N}=4}_{(0)\frac{1}{4},0}(\tau,z)+\mathrm{ch}^{\mathcal{N}=4}_{(0)\frac{1}{4},\frac{1}{2}}(\tau,z), \tag{7.4}$$

where we allowed $h=\frac{1}{4}$ for the massive character (7.3) as $\mathrm{ch}^{\mathcal{N}=4}_{\frac{1}{4},\frac{1}{2}}(\tau,z)=\frac{\theta_1(\tau,z)^2}{\eta(\tau)^3}q^{-\frac{1}{8}}$.

As a result, the K3 elliptic genus $Z^{(\mathrm{K3})}_{\mathrm{ell}}(\tau,z)$ is decomposed as [Oog89, EOT10]

$$Z^{(\mathrm{K3})}_{\mathrm{ell}}(\tau,z)=20\,\mathrm{ch}^{\mathcal{N}=4}_{(0)\frac{1}{4},0}(\tau,z)-2\,\mathrm{ch}^{\mathcal{N}=4}_{(0)\frac{1}{4},\frac{1}{2}}(\tau,z)+2\sum_{n=1}^{\infty}A_n\mathrm{ch}^{\mathcal{N}=4}_{n+\frac{1}{4},\frac{1}{2}}(\tau,z) \tag{7.5}$$

$$=24\,\mathrm{ch}^{\mathcal{N}=4}_{(0)\frac{1}{4},0}(\tau,z)-2\,\mathrm{ch}^{\mathcal{N}=4}_{\frac{1}{4},\frac{1}{2}}(\tau,z)+2\sum_{n=1}^{\infty}A_n\mathrm{ch}^{\mathcal{N}=4}_{n+\frac{1}{4},\frac{1}{2}}(\tau,z), \tag{7.6}$$

where

| $n$ | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | $\cdots$ |
|---|---|---|---|---|---|---|---|---|---|---|
| $A_n$ | 45 | 231 | 770 | 2277 | 5796 | 13915 | 30843 | 65550 | 132825 | $\cdots$ |

(7.7)

In 2010, Eguchi, Ooguri, and Tachikawa [EOT10] observed that these coefficients can be written as simple sums of irreducible representation dimensions of the largest Mathieu group $M_{24}$. The 26 irreducible representation dimensions of $M_{24}$ are, from the smallest one,

$$1,23,45,45,231,231,252,253,483,770,770,$$
$$990,990,1035,1035,1035,1265,1771,2024,$$
$$2277,3312,3520,5313,5796,5544,10395.$$

So $A_1,\dots,A_5$ are directly the irreducible representation dimensions of $M_{24}$, and

$$A_6=3520+10395, \tag{7.8}$$

$$A_7=1771+2024+5313+5544+5796+10395. \tag{7.9}$$

This observation by [EOT10] triggered the effort to construct a coherent theory analogous to the monstrous moonshine, so-called the *(K3) Mathieu moonshine*. Recall from Section 5.1 that the McKay–Thompson series $J_g(\tau)$ is the character of $g\in\mathbb{M}$ represented on the moonshine module $V^\natural$, and in the language of physics, it is a twisted partition function twisted in the temporal

direction. The analogue of the McKay–Thompson series, that is, the character $\phi_g(\tau, z)$ of $g \in M_{24}$ instead of $1_{M_{24}}$ in the elliptic genus $Z_{\mathrm{ell}}^{(\mathrm{K3})}(\tau, z)$ was calculated in [Che10, GHV10a, GHV10b, EH10], and called the *twining elliptic genus* in [GHV10a, GHV10b]. In addition, the existence of an $M_{24}$-module which reproduces these twining elliptic genera as its characters was shown in [Gan12]. The twisted partition functions twisted in both spatial and temporal directions, called the *twisted twining elliptic genera*, were also calculated in [GPRV12]. Remarkably, their modular transformation properties fit into the theory of anomaly of orbifold (Section 12.1.2), and controlled by a 3-cocycle in $H^3(M_{24}; \mathrm{U}(1))$. This suggests that there exists an underlying VOA with $M_{24}$ symmetry and the elliptic genus $Z_{\mathrm{ell}}^{(\mathrm{K3})}(\tau, z)$.

In spite of these pieces of evidence, such an underlying VOA has not yet been found, and many mysterious aspects still remain. First, the geometric symmetry of any K3 surface is smaller than the subgroup $M_{23}$ of $M_{24}$. More precisely, the following theorem is known.

**Theorem 7.1** (Mukai [Muk88])**.** *For a finite group $G$, the followings are equivalent.*

*(1) There exists a K3 surface $X$ such that its automorphism group $\mathrm{Aut}_\omega(X)$ preserving the nowhere-vanishing holomorphic 2-form $\omega$ on $X$ contains $G$ as a subgroup.*

*(2) $G$ is a subgroup of $M_{23}$, and as a subgroup of $M_{24}$ acting on 24 points $\Omega_{24}$, the action of $G$ on $\Omega_{24}$ has at least 5 orbits in $\Omega_{24}$.*

In the same spirit, the automorphism groups of K3 sigma models preserving the $\mathcal{N} = (4, 4)$ superconformal algebra were studied in [GHV11], through a strategy similar to Kondō's proof [Kon98] of Mukai's Theorem 7.1. They all turned out to be subgroups of the Conway group $\mathrm{Co}_1$, but none of them are $M_{24}$, or contain $M_{24}$ as a subgroup. Instead, some of them are proper subgroups of $M_{24}$, and the others are not even subgroups of $M_{24}$.

One proposal to resolve this problem was provided in [TW11, TW13a, TW13b]. Since the elliptic genus $Z_{\mathrm{ell}}^{(\mathrm{K3})}(\tau, z)$ is constant over the moduli space of K3 surfaces, they glued the automorphism groups of the K3 surfaces at three different points in the moduli space together, and succeeded in constructing the action of a maximal subgroup $2^4 : A_8$ called the octad subgroup of $M_{24}$ on the first massive part $\mathbf{45} \oplus \overline{\mathbf{45}}$ corresponding to the term $2A_1 \mathrm{ch}^{\mathcal{N}=4}_{\frac{5}{4},\frac{1}{2}}(\tau, z)$ in $Z_{\mathrm{ell}}^{(\mathrm{K3})}(\tau, z)$ (7.5). This idea is referred to as *symmetry-surfing*. Their action of $2^4 : A_8$ in fact coincides with the $M_{24}$-action restricted to $2^4 : A_8$ on the irreducible representation $\mathbf{45}$ of $M_{24}$. Its application to the higher-weight parts is discussed in [GKP16].

Another possible way to circumvent the situation is to loosen the condition on the automorphism group that it should preserve the $\mathcal{N} = (4, 4)$ superconformal algebra. In fact, we only need the $\mathcal{N} = (4, 1)$ supersymmetry to consider the elliptic genus and its decomposition into the irreducible characters of the left-moving $\mathcal{N} = 4$ superconformal algebra, so the automorphism group preserving only the $\mathcal{N} = (4, 1)$ supersymmetry is still useful. For example, a K3 sigma model with $2^8 : M_{20}$ symmetry, which is one of the maximal symmetries given in [GHV11] (and not a subgroup of $M_{24}$), was constructed in [GTVW13]. This K3 sigma model was further studied in relation to quantum error-correcting codes in [HM20], and they also showed that the automorphism

group preserving the $\mathcal{N}=(4,1)$ superconformal algebra is indeed larger than $2^8:M_{20}$, although it does not contain $M_{24}$. Of course, we may seek an $\mathcal{N}=(4,1)$ SCFT with $M_{24}$ symmetry and elliptic genus $Z_{\mathrm{ell}}^{(\mathrm{K3})}(\tau,z)$ (so the left-moving central charge should be 6), other than K3 sigma models.

As another approach, the relation to Duncan's module (Section 6) is also being explored. As a naive observation, since a K3 sigma model has central charge $(c,\tilde{c})=(6,6)$, if we "reflect" the right-moving part to the left-moving part, then we obtain a chiral SCFT with central charge $c=12$, which is the same central charge as Duncan's module. In fact, under an appropriate procedure called *reflection*, it was shown [DMC15, TW17] that the reflection of the specific K3 sigma model constructed in [GTVW13] is isomorphic to Duncan's module as a VOSA. As reviewed in Section 6.3, Duncan's module admits an $\mathcal{N}=1$ superconformal algebra, and the automorphism group preserving it is the Conway group $\mathrm{Co}_1$, which contains $M_{24}$ as a subgroup. In addition, it is known that Duncan's module also admits $\mathcal{N}=2$ and $\mathcal{N}=4$ superconformal algebras [CDD+14] (see also Section 3.4). However, the automorphism group of Duncan's module preserving its $\mathcal{N}=2$ and $\mathcal{N}=4$ superconformal algebras are only $M_{23}$ and $M_{22}$, respectively [CDD+14]. It was also pointed out in [Gan12] that if the character of $M_{24}$ appearing in the elliptic genus $Z_{\mathrm{ell}}^{(\mathrm{K3})}(\tau,z)$ is a restriction of a virtual character (a signed sum of irreducible characters) of some representation of $\mathrm{Co}_1$ or $\mathrm{Co}_0$, then such a representation of $\mathrm{Co}_1$ or $\mathrm{Co}_0$ must have an excessively large dimension. (For example, the smallest virtual representation of $\mathrm{Co}_0$ which restricts to the representation $\mathbf{45}\oplus\overline{\mathbf{45}}$ of $M_{24}$ has dimension over 100 billion.)

Lastly, we mention that, as a generalization of the Mathieu moonshine, the *umbral moonshine* was proposed in [CDH12, CDH13]. It relates a certain quotient $\mathrm{Aut}(L)/\mathrm{Weyl}(L)$ of the isometry group to a mock modular form, for each Niemeier lattice $L$ except for the Leech lattice $\Lambda_{24}$, and contains the K3 Mathieu moonshine as a case of $L=(A_1)^{24}$. The result of [Gan12] for the K3 Mathieu moonshine was generalized to the other cases of the umbral moonshine by [DGO15]. For more on Mathieu moonshine and its recent developments, see for example [Tac12, DGO14, Car17, HHP22, JF20] and references therein.

# Part III
# Symmetry of Lattice CFT

In this Part III, we develop a technique to investigate the following problem.

**Question 1.** Does the automorphism group $O(L)$ (or its subgroup) of a lattice $L$ lift to a subgroup of the automorphism group $\mathrm{Aut}(V_L)$ of the lattice VOA $V_L$?

More precisely, our approach provides a technique to show that the answer to this Question 1 is negative for a given lattice $L$ (when the answer is indeed negative for $L$).

In the previous Section 5.2, we reviewed that $\mathrm{Aut}(V_L)$ inherits the information of $O(L)$ in the form of its group extension, because the cocycle factor disturbs the group structure of $O(L)$ directly lifting up to that of $\mathrm{Aut}(V_L)$. In Section 8, by looking into the details of the extension, we will reformulate this Question 1 into a more tractable form. In Section 9, we will negatively settle Question 1 in the cases of the odd Leech lattice and the Leech lattice.

Subsequently, motivated by the moonshine-type observation explained in Introduction 1.2, we discuss the odd Leech lattice CFT $V_{O_{24}}$, which carries the structure of an extremal $\mathcal{N}=2$ SCFT. Since we showed that the extension of $O(L)$, but not $O(L)$ itself, is a genuine symmetry of the lattice VOA in this case, we will investigate if there exists some structure, say a superconformal algebra, preserved by this genuine symmetry in Section 10. In Section 11, we will explore some possibilities of explaining the moonshine-type observation of the odd Leech lattice CFT.

## 8 Cocycle Factor and Automorphism of Lattice VOA

In order to answer Question 1, we have to understand how an isometry of a lattice $L$ is lifted to automorphisms of the lattice VOA $V_L$. In Section 5.2, we have seen that $\mathrm{Aut}(V_L)$ contains a subgroup $O(\hat{L})$, where $\hat{L}$ is the central extension of $L$ by $\mathbb{Z}_2$ specified by the 2-cocycle called the cocycle factor, and $O(\hat{L})$ is a certain extension of $O(L)$ by $\mathrm{Hom}(L,\mathbb{Z}_2)$. We will look into the details of these extensions in this Section 8.

We will describe the concept of cocycle factor in physics in Section 8.1, and keeping the physics notation, we will outline the discussions of the rest of this section in Section 8.2. They are refined in a mathematical language in the following Section 8.3, where the main Question 1 will be paraphrased into more concrete forms Question 2 and 3, and slightly generalized to Question 3′.

### 8.1 Cocycle Factor

To realize the appropriate commutation relations of the vertex operators $V_k(z) \propto :e^{\sqrt{-1}k\cdot X(z)}:$, in accordance with whether $V_k(z)$ is bosonic or fermionic, we have to introduce a correction factor $c_k(p)$ to modify the commutation relations of $:e^{\sqrt{-1}k\cdot X(z)}:$'s. As a result, an additional

factor $\varepsilon(k,k')$ called a cocycle factor appears in the OPEs of the vertex operators. In this Section 8.1, we will review this story in the language of physics. A good reference for cocycle factors is [GSW12, §§6.4.4-6.4.5], but it deals with only even lattices. The cases including odd lattices are discussed in for example [GO85, Appendix], [GTVW13, Appendix A].

As already mentioned, let $L$ be an integral Euclidean lattice of rank $n$ in this thesis. We would like to construct the vertex operators $V_k(z)$ satisfying the commutation relation

$$V_k(z_1)\cdot V_{k'}(z_2) = (-1)^{|k|^2|k'|^2} V_{k'}(z_2)\cdot V_k(z_1), \tag{8.1}$$

for any $k,k' \in L$. The conformal weight of $V_k(z)$ is $\frac{1}{2}|k|^2$, so it is bosonic when $k$ is even and fermionic when $k$ is odd, which accounts for the factor $(-1)^{|k|^2|k'|^2}$ of (8.1).

Recalling that the OPE of operators $:e^{\sqrt{-1}k\cdot X(z)}:$ is

$$:e^{\sqrt{-1}k\cdot X(z_1)}:\cdot:e^{\sqrt{-1}k'\cdot X(z_2)}: = (z_1-z_2)^{k\cdot k'}:e^{\sqrt{-1}k\cdot X(z_1)}e^{\sqrt{-1}k'\cdot X(z_2)}:, \tag{8.2}$$

we can observe

$$:e^{\sqrt{-1}k\cdot X(z_1)}:\cdot:e^{\sqrt{-1}k'\cdot X(z_2)}: = (-1)^{k\cdot k'}:e^{\sqrt{-1}k'\cdot X(z_2)}:\cdot:e^{\sqrt{-1}k\cdot X(z_1)}: \sim (z_1-z_2)^{k\cdot k'}:e^{\sqrt{-1}(k+k')\cdot X(z_2)}:, \tag{8.3}$$

where $O((z_1-z_2)^{k\cdot k'+1})$ terms are dropped. This (8.3) differs from the desired commutation relation (8.1) only by the sign. To modify it, let us introduce the correction factors $c_k(p)$ as

$$V_k(z) = :e^{\sqrt{-1}k\cdot X(z)}: c_k(p), \tag{8.4}$$

where $c_k(p)$ is an operator in the form of a function of the momentum operators $p^i = \alpha_0^i$ $(i = 1,\dots,n)$. Then (8.1) translates to the condition on this correction factor $c_k(p)$ as

$$c_k(p+k')c_{k'}(p) = (-1)^{k\cdot k'+|k|^2|k'|^2} c_{k'}(p+k)c_k(p), \tag{8.5}$$

where we used $c_k(p)\cdot:e^{\sqrt{-1}k'\cdot X(z)}: = :e^{\sqrt{-1}k'\cdot X(z)}:\cdot c_k(p+k')$, because $[p^i, :e^{\sqrt{-1}k'\cdot X(z)}:] = (k')^i:e^{\sqrt{-1}k'\cdot X(z)}:$.

In addition, if we assume that the OPE of $V_k(z_1)\cdot V_{k'}(z_2)$ contains $(z_1-z_2)^{k\cdot k'}V_{k+k'}(z_2)$ as expected from (8.3), then we need $c_k(p+k')c_{k'}(p) = c_{k+k'}(p)$. However, this immediately turns out to be impossible in general, because (8.5) states that $c_k(p+k')c_{k'}(p)$ and $c_{k'}(p+k)c_k(p)$, both of which are supposed to become $c_{k+k'}(p)$, can differ by sign. So, we introduce a factor $\varepsilon : L\times L \to \{\pm 1\}$ so that

$$c_k(p+k')c_{k'}(p) = (-1)^{k\cdot k'+|k|^2|k'|^2} c_{k'}(p+k)c_k(p) = \varepsilon(k,k')c_{k+k'}(p). \tag{8.6}$$

This allows the vertex operators $V_k(z)$ to satisfy the OPE

$$V_k(z_1)\cdot V_{k'}(z_2) = (-1)^{|k|^2|k'|^2}V_{k'}(z_2)\cdot V_k(z_1) \sim \varepsilon(k,k')(z_1-z_2)^{k\cdot k'}V_{k+k'}(z_2), \tag{8.7}$$

where $O((z_1-z_2)^{k\cdot k'+1})$ terms are dropped. It easily follows from (8.6) that $\varepsilon$ must satisfy

$$\varepsilon(k,k') = (-1)^{k\cdot k'+|k|^2|k'|^2}\varepsilon(k',k). \tag{8.8}$$

Furthermore, we impose the associativity on $c_k(p)$ as

$$(c_k(p+k'+k'')c_{k'}(p+k''))c_{k''}(p) = c_k(p+k'+k'')(c_{k'}(p+k'')c_{k''}(p)). \tag{8.9}$$

Then the factor $\varepsilon$ should satisfy the condition

$$\varepsilon(k,k')\varepsilon(k+k',k'') = \varepsilon(k,k'+k'')\varepsilon(k',k''). \tag{8.10}$$

This means that the factor $\varepsilon : L \times L \to \{\pm 1\}$ is a 2-cocycle in the language of group cohomology (where $\{\pm 1\}$ is regarded as an $L$-module by the trivial $L$-action), and hence $\varepsilon$ is called a *cocycle factor*. A 2-cocycle satisfying (8.8) is unique up to coboundary; see Theorem A.2. In the language of Section A.2, $\varepsilon$ is a 2-cocycle for the commutator map $c(k,k') = (-1)^{k\cdot k'+|k|^2|k'|^2}$.

To construct $c_k(p)$ satisfying (8.6) and (8.9), we choose a $\mathbb{Z}$-basis $\{e_i\}_{i=1,\dots,n}$ of $L$ and introduce a bilinear non-commutative product $* : L \times L \to \mathbb{Z}$ on $L$ by

$$k * k' := \sum_{i>j} k^i k'^j (e_i \cdot e_j + |e_i|^2|e_j|^2) + \sum_{i=1}^n k^i k'^i \frac{1}{2}(|e_i|^2 + |e_i|^4), \tag{8.11}$$

where $|e_i|^4 := (|e_i|^2)^2$, for $k = \sum_i k^i e_i$ and $k' = \sum_j k'^j e_j$. This product $*$ depends on the choice of the basis $\{e_i\}_i$. In the language of Section A.3, this definition (8.11) follows the construction (A.49) of a cocycle from the commutator map $c(k,k') = (-1)^{k\cdot k'+|k|^2|k'|^2}$ and the quadratic form $q(k) = (-1)^{\frac{1}{2}(|k|^2+|k|^4)}$. This construction also appears in [Kac98, Remark 5.5a].

We can now construct $c_k(p)$ satisfying (8.6) and (8.9) as

$$c_k(p) = (-1)^{k*p}, \tag{8.12}$$

and then $\varepsilon$ is given by

$$\varepsilon(k,k') = (-1)^{k*k'}, \tag{8.13}$$

which of course satisfies (8.8) and (8.10).

*Proof that $c_k(p)$ in (8.12) satisfies (8.6) and (8.9).* For the first equation of (8.6), it suffices to check

$$(-1)^{k*k'} = (-1)^{k\cdot k'+|k|^2|k'|^2}(-1)^{k'*k}. \tag{8.14}$$

This is already discussed around (A.49), but we can check it explicitly as follows:

$$(-1)^{k*k'+k'*k} = (-1)^{\sum_{i\neq j}(k^i k'^j e_i\cdot e_j + |k^i e_i|^2|k'^j e_j|^2) + \sum_i (k^i k'^i e_i\cdot e_i + |k^i e_i|^2|k'^i e_i|^2)} \tag{8.15}$$

$$= (-1)^{k\cdot k'+|k|^2|k'|^2}, \tag{8.16}$$

where the first equation follows from $k^i \equiv (k^i)^2 \bmod 2$, and the second equation follows from $\sum_i |k^i e_i|^2 \equiv |k|^2 \bmod 2$.

The second equation of (8.6), and (8.9), are obvious. □

*Remark* 8.1. The 2-cocycle satisfying (8.8) is unique only up to coboundary. In particular, the bilinear non-commutative product we can use to construct $c_k(p)$ is not unique. For example, even if we define

$$k *_c k' := \sum_{i>j} k^i k'^j (e_i \cdot e_j + |e_i|^2 |e_j|^2), \tag{8.17}$$

and construct $c_k(p)$ as $(-1)^{k *_c p}$, it satisfies (8.6) and (8.9). In the language of Section A.2, this definition (8.17) follows the construction (A.32) of a cocycle from the commutator map $c(k, k') = (-1)^{k\cdot k' + |k|^2 |k'|^2}$, without considering to specify a quadratic form.

If the lattice $L$ is even, or if the lattice $L$ is odd and we choose the basis so that $e_1$ is odd and $e_2, \dots, e_n$ are even,[51] then this product reduces modulo 2 to

$$k *_c k' \equiv \sum_{i>j} k^i k'^j e_i \cdot e_j \mod 2. \tag{8.18}$$

This is the form adopted in [GSW12, §6.4.5].

This product $*_c$ (8.17) suffices for our discussions in this Part III, but does not work well when we consider the reflection $\mathbb{Z}_2$ symmetry discussed in Section 5.3. Suppose we used this product $*_c$ to construct the cocycle factor $\varepsilon$, which will play the role of the 2-cocycle of the section $k \mapsto e^k$ of the extension $\hat{L}$ (5.36). Then $\varepsilon(k,k) = (-1)^{\frac{1}{2}|k|^2}$ in (5.74) no longer holds, because the information of the quadratic form $q(k) = (-1)^{\frac{1}{2}|k|^2}$ (which coincides with $q(k) = (-1)^{\frac{1}{2}(|k|^2+|k|^4)}$ for an even lattice) is not taken into account (See Lemma A.4 (2)). As a result, the equation (5.74) would be modified by sign, so if we follow the definition of $\theta_0$ in [FLM88, Eq. (6.4.13)] as $\theta_0(e^k) = (-1)^{\frac{1}{2}|k|^2}(e^k)^{-1}$, then $\theta_0(e^k)$ would be $e^{-k}$ only up to sign, and the reflection $\mathbb{Z}_2$ symmetry $|k\rangle \mapsto |-k\rangle$ would need to be modified by some extra signs $|k\rangle \mapsto \pm|-k\rangle$. To avoid this complication, we will adopt the product $*$ (8.11) in this thesis. *(Remark ends.)*

*Remark* 8.2. The product $*$ (8.11) is a little bit cumbersome. If we can take a $\mathbb{Z}$-basis $e_1, \dots, e_n$ of $L$ such that

- if $L$ is even,[52] $|e_i|^2 \in 4\mathbb{Z}$ for any $i = 1, \dots, n$,
- if $L$ is odd, $e_1 \in 4\mathbb{Z} + 3$ and $|e_i|^2 \in 4\mathbb{Z}$ for any $i = 2, \dots, n$,

then (8.11) modulo 2 coincides with the product (8.18)

$$k * k' \equiv \sum_{i>j} k^i k'^j e_i \cdot e_j \mod 2. \tag{8.19}$$

In the analysis in Sections 9.2–9.3 , we will use the bases satisfying these conditions, and this product (8.19) is used in the Python code. *(Remark ends.)*

[51] This is always possible because (odd vector) − (odd vector) = (even vector).

[52] The existence of a $\mathbb{Z}$-basis $e_1, \dots, e_n$ of a lattice $L$ satisfying $|e_i|^2 \in 4\mathbb{Z}$ for any $i = 1, \dots, n$ does not imply that $|k|^2 \in 4\mathbb{Z}$ for any $k \in L$. In fact, we can take such a basis of the Leech lattice, as we can see from the symmetric bilinear form (2.49), but the Leech lattice has vectors whose squared lengths are not multiples of 4.

## 8.2 Brief Description in Physics Notation

In this Section 8.2, we outline what we will do in the following Section 8.3 with a simplified argument. The discussions here will be refined in a mathematical language in the following Section 8.3.

Let us assume that an isometry $g$ of the lattice $L$ can be lifted to an automorphism of the lattice VOA $V_L$, which consists of $V_k(z)$'s in (8.7), as

$$g(V_k(z)) = \zeta_g(k) V_{g(k)}(z), \tag{8.20}$$

where $\zeta_g(k)$ is some factor taking values in $\mathbb{Z}_2 = \{\pm 1\}$, or more generally, in $\mathrm{U}(1)$. Here, we abused notation and let $g$ also denote the automorphism of $V_L$ lifted from the isometry $g$ of $L$. Since $g$ is an automorphism, it must preserve the multiplication (8.7), and hence $\zeta_g$ must satisfy

$$\varepsilon(k,k')\zeta_g(k+k') = \zeta_g(k)\zeta_g(k')\varepsilon(g(k),g(k')). \tag{8.21}$$

If we specify the values of $\zeta_g(e_1), \dots, \zeta_g(e_n)$ for a basis $\{e_i\}_{i=1,\dots,n}$ of $L$, then (8.21) recursively determines all the values $\zeta_g(k)$ for $k \in L$. The resulting $\zeta_g$ is well-defined, because

$$\zeta_g((k+k')+k'') = \zeta_g(k+k')\zeta_g(k'')\frac{\varepsilon(g(k+k'),g(k''))}{\varepsilon(k+k',k'')} \tag{8.22}$$

$$= \zeta_g(k)\zeta_g(k')\zeta_g(k'')\frac{\varepsilon(g(k),g(k'))}{\varepsilon(k,k')}\frac{\varepsilon(g(k+k'),g(k''))}{\varepsilon(k+k',k'')} \tag{8.23}$$

and

$$\zeta_g(k+(k'+k'')) = \zeta_g(k)\zeta_g(k'+k'')\frac{\varepsilon(g(k),g(k'+k''))}{\varepsilon(k,k'+k'')} \tag{8.24}$$

$$= \zeta_g(k)\zeta_g(k')\zeta_g(k'')\frac{\varepsilon(g(k),g(k'+k''))}{\varepsilon(k,k'+k'')}\frac{\varepsilon(g(k'),g(k''))}{\varepsilon(k',k'')} \tag{8.25}$$

are equal since $\varepsilon$ is a 2-cocycle (8.10). Conversely, any $\zeta_g$ determined in such a way by (8.21) can define an automorphism of $V_L$. Therefore, there are $\mathbb{Z}_2^n$, or more generally $\mathrm{U}(1)^n$, degrees of freedom in the way of lifting a lattice isometry $g$ to an automorphism of the lattice VOA.

*Remark* 8.3. Here is another derivation of $\mathbb{Z}_2^n$. If both $\zeta_g$ and $\zeta'_g$ satisfy (8.21), then their difference $\eta(k) = \zeta'_g(k)\zeta_g(k)^{-1}$ turns out to be just a linear map in $\mathrm{Hom}(L,\mathbb{Z}_2)$:

$$\eta(k+k') = \eta(k)\eta(k'). \tag{8.26}$$

Therefore, the degree of freedom in choosing $\zeta_g$ is $\mathrm{Hom}(L,\mathbb{Z}_2) \cong \mathbb{Z}_2^n$. *(Remark ends.)*

Our main Question 1 was whether we can lift each isometry of $L$ to an automorphism of $V_L$, while preserving the whole group structure of the isometry group $O(L)$. If this is the case, we should have

$$g_2(g_1(V_k(z))) = (g_2 g_1)(V_k(z)) \quad \text{for any } g_1, g_2 \in O(L), \tag{8.27}$$

and hence

$$\zeta_{g_2}(g_1(k))\zeta_{g_1}(k) = \zeta_{g_2 g_1}(k) \quad \text{for any } g_1, g_2 \in O(L). \tag{8.28}$$

Therefore, our Question 1 can be restated as follows. "Can we find a collection $\{\zeta_g : L \to \mathbb{Z}_2 \text{ (or } \mathrm{U}(1)) \mid \zeta_g \text{ satisfies (8.21)}\}_{g\in O(L)}$ satisfying the condition (8.28)? Here, we can adjust the values of $\zeta_g(e_1), \dots, \zeta_g(e_n) \in \mathbb{Z}_2$ (or $\mathrm{U}(1)$) for each $g \in O(L)$, from which $\zeta_g$ is completely determined by (8.21)." This is the paraphrased Question 3 or 3′ in the following Section 8.3.

## 8.3 Lift of Lattice Isometries to Automorphisms of Lattice VOA

Recall from Section 5.2 that the lattice VOA $V_L$ is constructed on the central extension of the lattice $L$ by $\mathbb{Z}_2 = \langle \kappa \mid \kappa^2 = 1 \rangle$

$$1 \to \mathbb{Z}_2 \to \hat{L} \xrightarrow{-} L \to 0, \tag{8.29}$$

specified by a 2-cocycle $\hat{\varepsilon} : L \times L \to \mathbb{Z}_2$. Any element of $\hat{L}$ is written in the form of $\kappa^m e^k$ ($k \in L$). We will eventually use the cocycle factor $\varepsilon$ in (8.13) required by physics in the next Section 9. Note that this cocycle $\varepsilon$ (8.13) satisfies the normalization condition (5.40). The automorphism group $\mathrm{Aut}(V_L)$ of the lattice VOA $V_L$ contains a subgroup $O(\hat{L})$ described by the group extension

$$1 \to \mathrm{Hom}(L, \mathbb{Z}_2) \to O(\hat{L}) \to O(L) \to 1, \tag{8.30}$$

where $O(L)$ is the isometry group of the lattice $L$.

Now, our main Question 1 is reduced to how $O(L)$ is lifted into $O(\hat{L})$ in (8.30). More precisely,

**Question 2.** Does the exact sequence (8.30) have a section $O(L) \to O(\hat{L})$ such that it is a group homomorphism?

If the answer is yes, we say that the exact sequence (8.30) splits, and $O(\hat{L})$ is a semidirect product of $\mathrm{Hom}(L, \mathbb{Z}_2)$ and $O(L)$. Otherwise, $O(\hat{L})$ is some non-split extension of $O(L)$, and $O(L)$ does not lift to a subgroup of $O(\hat{L})$ or $\mathrm{Aut}(V_L)$.

We will paraphrase this Question 2 into a more concrete form. Let $S : O(L) \to O(\hat{L})\,; g \mapsto S_g$ be a section of (8.30). Then for each $g \in O(L)$, we can define $\zeta_g : L \to \mathbb{Z}_2$ by

$$S_g(e^k) = \zeta_g(k) e^{g(k)}, \tag{8.31}$$

because $\overline{S_g(e^k)} = g(k)$ (Recall the definition (5.59) of $O(\hat{L}) \to O(L)$). Since $S_g \in O(\hat{L})$ is an automorphism of $\hat{L}$, it must satisfy

$$S_g(e^k \cdot e^{k'}) = S_g(e^k) \cdot S_g(e^{k'}), \tag{8.32}$$

which translates to the condition on $\zeta_g$ as

$$\hat{\varepsilon}(k, k')\zeta_g(k + k') = \zeta_g(k)\zeta_g(k')\hat{\varepsilon}(g(k), g(k')), \tag{8.33}$$

where we also used $S_g \in O(\hat{L})$ satisfies $S_g(\kappa) = \kappa$.

*Remark* 8.4. The equation (8.33) is equivalent to $\hat{\varepsilon}(k,k') = \hat{\varepsilon}(g(k),g(k'))d\zeta_g(k,k')$, which means that the cocycles $\hat{\varepsilon}(-,-)$ and $\hat{\varepsilon}(g(-),g(-))$ are in the same cohomology class. *(Remark ends.)*

Take another section $S' : O(L) \to O(\hat{L})$ of (8.30) and define $\zeta'_g : L \to \mathbb{Z}_2$ in the same way. Since $S'_g(S_g)^{-1}$ is in the kernel of $O(\hat{L}) \to O(L)$ in (8.30), there exists $\eta_g \in \mathrm{Hom}(L,\mathbb{Z}_2)$ satisfying

$$\tilde{\eta}_g = S'_g(S_g)^{-1}, \tag{8.34}$$

which shows that the degree of freedom in taking the value $S_g$ of the section $S : O(L) \to O(\hat{L})$ is the linear map $\eta_g \in \mathrm{Hom}(L,\mathbb{Z}_2)$ for each $g \in O(L)$. In terms of the factors $\zeta_g$ and $\zeta'_g : L \to \mathbb{Z}_2$, (8.34) reduces to

$$\eta_g = \zeta'_g(\zeta_g)^{-1}, \tag{8.35}$$

where the degree of freedom in taking the value $S_g$ is just parametrized by $\zeta_g$.

We can rephrase this degree of freedom $\mathrm{Hom}(L,\mathbb{Z}_2) \cong (\mathbb{Z}_2)^n$ in terms of the factor $\zeta_g : L \to \mathbb{Z}_2$ in the following way. If we define the values of $\zeta_g(e_1),\dots,\zeta_g(e_n) \in \mathbb{Z}_2$, where $e_1,\dots,e_n$ is a basis of $L$, then the condition (8.33) recursively determines all the values of $\zeta_g(k)$ for $k \in L$. This recursive way to determine $\zeta_g$ is well-defined thanks to the fact that $\hat{\varepsilon}$ is a 2-cocycle, as we saw in (8.22)–(8.25). Therefore, the value $S_g \in O(\hat{L})$ is parametrized by only $\zeta_g(e_1),\dots,\zeta_g(e_n)$, and hence the parameter space is $(\mathbb{Z}_2)^n$.

If there exists a section $S : O(L) \to O(\hat{L})$ such that it is a group homomorphism, as in Question 2, then it should satisfy

$$S_{g_2} \circ S_{g_1} = S_{g_2 g_1} \quad \text{for any } g_1, g_2 \in O(L). \tag{8.36}$$

This translates to the condition on $\zeta_g$ as

$$\zeta_{g_2}(g_1(k))\zeta_{g_1}(k) = \zeta_{g_2 g_1}(k) \quad \text{for any } g_1, g_2 \in O(L). \tag{8.37}$$

Therefore, Question 2 can be paraphrased as

**Question 3.** Can we find a collection $\{\zeta_g : L \to \mathbb{Z}_2 \mid \zeta_g \text{ satisfies (8.33)}\}_{g\in O(L)}$ satisfying the condition (8.37), by adjusting each $\zeta_g$ within the degree of freedom $\mathrm{Hom}(L,\mathbb{Z}_2)$? More specifically, we are allowed to adjust the values of $\zeta_g(e_1),\dots,\zeta_g(e_n) \in \mathbb{Z}_2$ for each $g \in O(L)$, from which $\zeta_g$ is completely determined by (8.33).

*Remark* 8.5. Note that even if we take another cocycle $\hat{\varepsilon}'$ cohomologous to $\hat{\varepsilon}$, in the equation (8.33) from which $\zeta_g$ is determined, the answer to Question 3 does not change. This is just because cocycles in the same cohomology class define equivalent extensions $\hat{L}$, but we can also check it explicitly as follows. If $\zeta_g$ satisfies (8.33), then $\zeta'_g$ satisfying (8.33) for $\hat{\varepsilon}' = \hat{\varepsilon}d\delta$, where $d\delta(k,k') = \delta(k')\delta(k+k')^{-1}\delta(k)$ is a coboundary, can be constructed as

$$\zeta'_g(k) = \zeta_g(k)\delta(k)\delta(g(k))^{-1}. \tag{8.38}$$

In addition, if $\{\zeta_g \mid (8.33)\}_g$ satisfies (8.37), then it is easy to check that $\{\zeta'_g\}_g$ constructed by (8.38) also satisfies (8.37). *(Remark ends.)*

Finally, we mention one slight generalization of the question. We can generalize the discussions so far from $\mathbb{Z}_2 = \langle \kappa \mid \kappa^2 = 1 \rangle$ to $\mathrm{U}(1) \cong \mathbb{R}/2\mathbb{Z} = \{\kappa^r \mid r \in \mathbb{R}\}/\langle \kappa^2 \rangle_{\mathbb{Z}}$ (see footnote 28). As a result, we may consider another question

**Question 3′.** Can we find a collection $\{\zeta_g : L \to \mathrm{U}(1) \mid \zeta_g \text{ satisfies } (8.33)\}_{g \in O(L)}$ satisfying the condition (8.37), by adjusting each $\zeta_g$ within the degree of freedom $\mathrm{Hom}(L, \mathrm{U}(1))$? More specifically, We are allowed to adjust the values of $\zeta_g(e_1), \ldots, \zeta_g(e_n) \in \mathrm{U}(1)$ for each $g \in O(L)$, from which $\zeta_g$ is completely determined by (8.33).

Of course, if the answer to Question 3 is yes, then so is the answer to Question 3′, and if the answer to Question 3′ is no, then so is that to Question 3.

# 9 Determining Whether the Lift Preserves Group Structures or Not

In this Section 9, we negatively settle Question 3′ in the cases of the odd Leech lattice and the Leech lattice.

This section presents original work based on [Oka24].

## 9.1 Closed Form of the Factor $\zeta_g$

Although we have pointed out that the relation (8.33) determines the value of $\zeta_g(k)$ for any $k \in L$ recursively from the values of $\zeta_g(e_1), \dots, \zeta_g(e_n)$, it will be convenient if we have a closed form of $\zeta_g(k)$. It can be done at least in the case where the cocycle $\hat{\varepsilon}$ is the specific one $\varepsilon$ in (8.13). Recall its formula:

$$\varepsilon(k, k') = (-1)^{k*k'}, \tag{9.1}$$

where the bilinear non-commutative product $*$ is defined for a given $\mathbb{Z}$-basis $e_1, \dots, e_n$ of the lattice $L$ as

$$k * k' = \sum_{i>j} k^i k'^j (e_i \cdot e_j + |e_i|^2 |e_j|^2) + \sum_{i=1}^{n} k^i k'^i \frac{1}{2}(|e_i|^2 + |e_i|^4). \tag{9.2}$$

**Proposition 9.1.** *If the 2-cocycle $\hat{\varepsilon} : L \times L \to \mathrm{U}(1)$ in (8.33) is the one $\varepsilon$ in (9.1) where the product $*$ is specified by a $\mathbb{Z}$-basis $e_1, \dots, e_n$ of the integral lattice $L$, then the equation (8.33) determines the value of $\zeta_g(k)$ for $g \in O(L)$ and $k = \sum_i k^i e_i \in L$ as*

$$\zeta_g(k) = \prod_{i=0}^{n-1} \zeta_g(e_i)^{k^i} \cdot \prod_{\substack{i'<j' \\ i',j' \in \{i \mid k^i \text{ is odd}\}}} \varepsilon(g(e_{i'}), g(e_{j'})). \tag{9.3}$$

Before getting into the proof of this Proposition 9.1, we prepare the following lemma.

**Lemma 9.2.** *If the 2-cocycle $\hat{\varepsilon} : L \times L \to \mathrm{U}(1)$ in (8.33) is the one $\varepsilon$ in (9.1) where the product $*$ is specified by a $\mathbb{Z}$-basis $e_1, \dots, e_n$ of the integral lattice $L$, then we have*

*(1)* $\varepsilon(2k, k') = 1$ *for any* $k, k' \in L$.

*(2)* $\zeta_g(2k + k') = \zeta_g(2k)\zeta_g(k')$ *for any* $k, k' \in L$ *and* $g \in O(L)$.

*(3)* $\varepsilon(k, k) = (-1)^{\frac{1}{2}(|k|^2 + |k|^4)}$ *for any* $k \in L$.
*In particular,* $\varepsilon(k, k) = \varepsilon(g(k), g(k))$ *for any* $k \in L$ *and* $g \in O(L)$.

*(4)* $\zeta_g((t_1 + t_2)k) = \zeta_g(t_1 k)\zeta_g(t_2 k)$ *for any* $t_1, t_2 \in \mathbb{Z}$, $k \in L$, *and* $g \in O(L)$.

*Proof.* First, note that the values of the specific cocycle $\varepsilon$ (9.1) are still in $\mathbb{Z}_2$, even though we have extended our main question to the $\mathrm{U}(1)$ case as in Question 3′.

(1) This is because $2k * k' \equiv 0 \bmod 2$.

(2) Use (1) to (8.33).

(3) This is the result of Lemma A.4 (2), but we can check it explicitly as follows. We first observe that

$$2k * k = |k|^2 + \Big(\sum_i k_i |e_i|^2\Big)^2. \tag{9.4}$$

Since $|k|^2 \equiv \sum_i |k^i e_i|^2 \equiv \sum_i k^i |e_i|^2 \bmod 2$, we have $|k|^4 \equiv (\sum_i k^i |e_i|^2)^2 \bmod 4$ (because the square of an even number is 0 mod 4, and the square of an odd number is 1 mod 4). Therefore,

$$2k * k \equiv |k|^2 + |k|^4 \mod 4, \tag{9.5}$$

and hence

$$k * k \equiv \frac{1}{2}(|k|^2 + |k|^4) \mod 2. \tag{9.6}$$

This shows $\varepsilon(k,k) = (-1)^{\frac{1}{2}(|k|^2+|k|^4)}$.

(4) Use (3) to (8.33). Note that $\varepsilon(t_1 k, t_2 k) = \varepsilon(k,k)^{t_1 t_2}$ follows from the bilinearity of the product $*$. □

Now, we prove Proposition 9.1.

*Proof of Proposition 9.1.* By using Lemma 9.2 (2) and (4), we have

$$\zeta_g(k) = \left(\prod_{i=0}^{n-1} \zeta_g(e_i)^{2\lfloor \frac{k^i}{2} \rfloor}\right) \cdot \zeta_g\Big(\sum_{i' \in \{i \mid k^i \text{ is odd}\}} e_{i'}\Big). \tag{9.7}$$

If we write the elements of $\{i \mid k^i \text{ is odd}\}$ as $i'_1, \dots, i'_l$ where $i'_1 < \cdots < i'_l$, then by using (8.33) and $\varepsilon(e_i, e_j) = 1$ for $i < j$ repeatedly, the last factor of (9.7) becomes

$$\begin{aligned}
\zeta_g(e_{i'_1} + \cdots + e_{i'_l}) &= \zeta_g(e_{i'_1})\zeta_g(e_{i'_2} + \cdots + e_{i'_l})\varepsilon(g(e_{i'_1}), g(e_{i'_2} + \cdots + e_{i'_l})) && (9.8)\\
&= \zeta_g(e_{i'_1})\zeta_g(e_{i'_2} + \cdots + e_{i'_l}) \prod_{1<m'\le l} \varepsilon(g(e_{i'_1}), g(e_{i'_{m'}})) && (9.9)\\
&\vdots && (9.10)\\
&= \prod_{m=1}^{l} \zeta_g(e_{i'_m}) \prod_{1\le m<m'\le l} \varepsilon(g(e_{i'_m}), g(e_{i'_{m'}})). && (9.11)
\end{aligned}$$

Therefore, we obtain (9.3). □

*Remark* 9.3. [Oka24, Prop. 8.1] showed that if we can take a $\mathbb{Z}$-basis $e_1, \dots, e_n$ of $L$ such that

- if $L$ is even, $|e_i|^2 \in 4\mathbb{Z}$ for any $i = 1, \ldots, n$,
- if $L$ is odd, $e_1$ is odd and $|e_i|^2 \in 4\mathbb{Z}$ for any $i = 2, \ldots, n$,

then the 2-cocycle $\varepsilon$ constructed from the product $*_c$ (8.17) (or the reduced one (8.18)) with such a basis also leads to the same formula (9.3) of $\zeta_g(k)$. *(Remark ends.)*

## 9.2 The Mathieu Group of Odd Leech Lattice

Let us see that the answer to Question 3′ is no in the case of the odd Leech lattice $O_{24}$.

We take $e_1, \ldots, e_{24}$ in (2.39) as a basis of the odd Leech lattice $O_{24}$. This basis satisfies the condition mentioned in Remark 8.2. The isometry group $O(O_{24})$ contains the automorphism group $M_{24}$ of the binary Golay code $G_{24}$ as a subgroup, and its element $g \in M_{24} \subset S_{24}$ acts on $k = (k_1, \ldots, k_{24}) \in O_{24} \subset \mathbb{R}^{24}$ from left as

$$g(k) = (k_{g^{-1}(1)}, \ldots, k_{g^{-1}(24)}), \quad \text{or equivalently,} \quad (g(k))_{g(i)} = k_i. \tag{9.12}$$

Recall from Section 2.2.2 that the Mathieu group $M_{24}$ is generated by the two generators $a, b$ satisfying the relations in (2.8):

$$a^2 = 1, \tag{9.13}$$
$$b^3 = 1, \tag{9.14}$$
$$(ba)^{23} = 1, \tag{9.15}$$
$$(b^2aba)^{12} = 1, \tag{9.16}$$
$$(b^2ab^2ababa)^5 = 1, \tag{9.17}$$
$$(b^2ab^2aba)^3(b^2abababa)^3 = 1, \tag{9.18}$$
$$((b^2aba)^3ba)^4 = 1. \tag{9.19}$$

In our choice of a basis of $O_{24}$, we can take explicit elements of $a, b$ as in (2.9), (2.10).

For the answer to Question 3′ to be yes, we must at least be able to find the values of $\zeta_a(e_i)$ and $\zeta_b(e_i) \in \mathrm{U}(1)$ for $i = 1, \ldots, 24$, which are compatible with the relations (9.13)–(9.19). For example, by applying the relation (9.13) to the condition (8.37) that $\{\zeta_g\}_g$ preserve the group structure, we have

$$\zeta_a(a(e_i))\zeta_a(e_i) = \zeta_1(e_i), \tag{9.20}$$

for any $i = 1, \ldots, 24$. By using the explicit form (9.3) of $\zeta_g(k)$, the left-hand side of (9.20) can be factorized in the form of

$$(-1)^{\mu^i}\zeta_a(e_1)^{m_1^i} \cdots \zeta_a(e_{24})^{m_{24}^i}, \tag{9.21}$$

where $\mu^i \in \mathbb{Z}_2, m_1^i, \ldots, m_{24}^i \in \mathbb{Z}$. We further write $\zeta_a(e_j) \in \mathrm{U}(1)$ as $(e^{\sqrt{-1}\pi})^{\rho_j}$ with $\rho_j \in \mathbb{R}/2\mathbb{Z}$. Then (9.21) is rewritten as

$$(e^{\sqrt{-1}\pi})^{m_1^i\rho_1 + \cdots + m_{24}^i\rho_{24} + \mu^i}. \tag{9.22}$$

On the other hand, the right-hand side of (9.20) is 1, because we can show that $\zeta_1(k) = 1$ for any $k \in L$ from the condition (8.37). Therefore, the equation (9.20) is equivalent to

$$m_1^i \rho_1 + \cdots + m_{24}^i \rho_{24} = \mu^i \quad \text{in } \mathbb{R}/2\mathbb{Z}. \tag{9.23}$$

In this way, from the relation (9.13), we have obtained 24 necessary conditions (9.23; $i = 1, \ldots, 24$) for $\zeta_a(e_j) = (e^{\sqrt{-1}\pi})^{\rho_j}$ to preserve the group structure of $M_{24}$.

In the same way, from the seven relations (9.13)–(9.19), we obtain $7 \times 24$ equations, which we label by $\iota = 1, \cdots, 7 \times 24$,

$$m_1^\iota \rho_1 + \cdots + m_{24}^\iota \rho_{24} + m_{25}^\iota \rho_{25} + \cdots + m_{48}^\iota \rho_{48} = \mu^\iota \text{ in } \mathbb{R}/2\mathbb{Z}, \tag{9.24}$$

with respect to variables $\rho_i \in \mathbb{R}/2\mathbb{Z}$ defined by $\zeta_a(e_j) = (e^{\sqrt{-1}\pi})^{\rho_j}$ and $\zeta_b(e_j) = (e^{\sqrt{-1}\pi})^{\rho_{24+j}}$, with coefficients $m_j^\iota \in \mathbb{Z}$ and $\mu_j^\iota \in \mathbb{Z}/2\mathbb{Z} \subset \mathbb{R}/2\mathbb{Z}$. We write these equations in a matrix form

$$M\vec{\rho} = \vec{\mu}, \tag{9.25}$$

where the coefficient matrix $M$ is $\mathbb{Z}$-valued and has $7 \times 24$ rows and 48 columns, the variable $\vec{\rho}$ is the $\mathbb{R}/2\mathbb{Z}$-valued column vector with 48 entries, and $\vec{\mu}$ is the $\mathbb{Z}/2\mathbb{Z}$-valued and hence $\mathbb{R}/2\mathbb{Z}$-valued column vector with $7 \times 24$ entries.

It is known that for any $t \times u$ integer matrix $M \in \mathbb{Z}^{t \times u}$, there exist invertible square matrices $P \in \mathrm{GL}(t, \mathbb{Z})$ and $Q \in \mathrm{GL}(u, \mathbb{Z})$, such that $D = PMQ$ is in the form of

$$D = \left(\begin{array}{ccc|c} d_1 & & & \\ & \ddots & & O \\ & & d_r & \\ \hline & O & & O \end{array}\right), \quad d_1, \ldots, d_r \in \mathbb{Z}_{\neq 0}, \quad d_i \mid d_{i+1} \text{ for any } i. \tag{9.26}$$

The entries $d_1, \ldots, d_r$ are unique for $M$ up to multiplication by elements in $\mathbb{Z}^\times = \{\pm 1\}$, and called the *elementary divisors* of $M$. $D$ is called the *Smith normal form* of $M$, and $M = P^{-1}DQ^{-1}$ is called the *Smith decomposition* of $M$. The matrices $P, D, Q$ for a given $M$ can be calculated by Mathematica command `SmithDecomposition[`$M$`]`.

Let $D = PMQ$ be the Smith normal form of the coefficient matrix $M$ in (9.25). Then the equation (9.25) is equivalent to

$$DQ^{-1}\vec{\rho} = P\vec{\mu}. \tag{9.27}$$

If the rank of $D$, or equivalently that of $M$, is $r$, and if $P\vec{\mu}$ contains any non-zero element of $\mathbb{R}/2\mathbb{Z}$ somewhere in its $(r+1), \ldots, (7 \times 24)$-th entries, then the equation (9.27) does not have any solution $\vec{\rho}$, and hence we cannot find $\zeta_a(e_i)$ and $\zeta_b(e_i)$ $(i = 1, \ldots, 24)$ preserving the group structure of $M_{24}$ and compatible with the relations (9.13)–(9.19). This means that the answer to Question 3′ is no, and hence the Mathieu group $M_{24}$, a subgroup of the isometry group of the odd Leech lattice $O_{24}$, does not lift to a subgroup of the automorphism group of the lattice VOA $V_{O_{24}}$.

In fact, we only need the $2 \times 24$ equations from the relations (9.13) and (9.18) to bring this matter to a conclusion. The coefficient matrix $M$ and $\vec{\mu}$ for these $2\times 24$ equations are calculated by computer (see `M24_of_odd_Leech.ipynb` for the source code) and recorded in (C.1, C.2). By the Smith decomposition $M = P^{-1}DQ^{-1}$, we obtain the elementary divisors of $M$ as

$$1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 2, 54, \tag{9.28}$$

and hence the rank of $D$ is 25, whereas

$$P\vec{\mu} = \begin{pmatrix} 16018 & 16019 & 22266 & 16022 & 22269 & 22266 & 16020 & 22268 \\ 16020 & 16019 & 16020 & 16020 & 31230 & 77430 & 48739 & 61107 \\ 51039 & 77168 & 54837 & 70020 & 78108 & 53946 & 76206 & 62456 \\ 31928 & 0 & 0 & -2 & 0 & 0 & 22171 & 13702 \\ 23183 & 2 & 5974 & 17416 & 17762 & 0 & 0 & 7460 \\ 28878 & -2 & 2 & 0 & 5156 & 23924 & 0 & 33476 \end{pmatrix}^T, \tag{9.29}$$

where the 31st and 33rd entries are nonzero in $\mathbb{R}/2\mathbb{Z}$. Therefore, we cannot find the values of $\zeta_a(e_i)$ and $\zeta_b(e_i)$ $(i = 1, \ldots, 24)$ compatible with the relations (9.13) and (9.18), and hence the subgroup $M_{24}$ of the isometry group of $O_{24}$ does not lift to a subgroup of the automorphism group of the lattice VOA $V_{O_{24}}$.

In a similar way, we can see that the subgroup $M_{23}$ of $M_{24}$ also does not lift to a subgroup of the automorphism group of $V_{O_{24}}$. Recall that the Mathieu group $M_{23}$ is generated by two generators $a, b$ satisfying the relations in (2.11):

$$a^2 = 1, \tag{9.30}$$
$$b^4 = 1, \tag{9.31}$$
$$(ba)^{23} = 1, \tag{9.32}$$
$$(b^2a)^6 = 1, \tag{9.33}$$
$$(b^3aba)^6 = 1, \tag{9.34}$$
$$(b^2ab^3aba)^4 = 1, \tag{9.35}$$
$$(b^3a)^3(ba)^3(b^3aba)^2b^2ab^3a(ba)^3 = 1, \tag{9.36}$$
$$b^2ab^3abab^2aba(b^3ab^2a)^2(b^2aba)^3 = 1. \tag{9.37}$$

From the relations (9.36) and (9.37), we obtain the equation $M\vec{\rho} = \vec{\mu}$ which $\zeta_a(e_i) = (e^{\sqrt{-1}\pi})^{\rho_i}$ and $\zeta_b(e_i) = (e^{\sqrt{-1}\pi})^{\rho_{24+i}}$ should satisfy, where the coefficient matrix $M$ and $\vec{\mu}$ are calculated by computer (see `M23_of_odd_Leech.ipynb` for the source code) and recorded in (C.3, C.4). By the Smith decomposition $M = P^{-1}DQ^{-1}$, we obtain the elementary divisors of $M$ as

$$1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, \tag{9.38}$$

and hence the rank of $D$ is 24, whereas

$$P\vec{\mu} = (\begin{array}{cccccccc} 0 & -184045 & -57967 & -121500 & -94368 & -45774 & -62078 & -153328 \\ -2276 & -572 & -94178 & -81760 & -132488 & -197704 & -197578 & -182388 \\ -59707 & -185810 & -9243 & -5733 & -86582 & -170898 & -161384 & -99556 \\ 0 & -2422 & -198226 & -188029 & -118824 & -38728 & -109273 & -199800 \\ -172362 & -200168 & -11403 & -113189 & -102530 & -173862 & -50724 & -20268 \\ -165812 & -169514 & -123120 & -137854 & -136654 & -81506 & -25294 & -201016 \end{array})^T, \tag{9.39}$$

where the 28th, 31st, 35th, and 36th entries are nonzero in $\mathbb{R}/2\mathbb{Z}$. Therefore, we cannot find the values of $\zeta_a(e_i)$ and $\zeta_b(e_i)$ $(i = 1, \dots, 24)$ compatible with the relations (9.36) and (9.37), and hence the subgroup $M_{23}$ of the isometry group of $O_{24}$ does not lift to a subgroup of the automorphism group of the lattice VOA $V_{O_{24}}$.

Let us restate our conclusion in the form of a theorem.

**Theorem 9.4.** *Let $L$ be an odd Leech lattice $O_{24}$. The subgroups $(\mathbb{Z}_2)^{24}.M_{24}$ and $(\mathbb{Z}_2)^{24}.M_{23}$ of the automorphism group $\mathrm{Aut}(V_L)$ of the lattice VOA are non-split extensions. In particular, the subgroups $M_{24}$ and $M_{23}$ of the isometry group $O(L)$ do not lift to a subgroup of $\mathrm{Aut}(V_L)$.*

If we use $\mathrm{U}(1)$ instead of $\mathbb{Z}_2$ in the construction of $\hat{L}$ in (8.29) and hence in the construction of $V_L$, then the subgroups $\mathrm{U}(1)^{24}.M_{24}$ and $\mathrm{U}(1)^{24}.M_{23}$ of $\mathrm{Aut}(V_L)$ are non-split. We will see that $(\mathbb{Z}_2$ or $\mathrm{U}(1))^{24}.M_{23}$ contains $(\mathbb{Z}_2$ or $\mathrm{U}(1))^{23}.M_{23}$ as a subgroup around (10.5) and in Appendix D, and of course it is also non-split.

## 9.3 The Conway Group of Leech Lattice

Let us see that the answer to Question 3′ is no in the case of the Leech lattice $\Lambda_{24}$. In fact, it was already shown in [Gri73]; we will check it by the similar discussion to that of the previous Section 9.2.

Recall that the Conway group $\mathrm{Co}_0$ is generated by the two generators $A, B$ in (2.46, 2.47) as a subgroup of $\mathrm{SL}(24, \mathbb{Z})$, and it is the isometry group of the Leech lattice, which is now given as $\mathbb{Z}^{24}$ with the symmetric bilinear form $q$ in (2.49). We take the standard basis $\{\mathbb{e}_i\}_{i=1,\dots,24}$ as a basis of this Leech lattice. From the matrix form (2.49) of the symmetric bilinear form $q$, we can see that this basis satisfies the condition mentioned in Remark 8.2.

We need some relations of the generators $A$ and $B$ to use the discussion similar to that of Section 9.2. Unfortunately, unlike the case of the Mathieu groups, we could not find information on the set of relations constituting the presentation of $\mathrm{Co}_0$ in terms of $A$ and $B$, but the way of finding relations is not important. After a lot of trials, we found some sets of relations which lead to the system of equations showing that there are no values $\zeta_A(\mathbb{e}_1), \dots, \zeta_A(\mathbb{e}_{24})$ and $\zeta_B(\mathbb{e}_1), \dots, \zeta_B(\mathbb{e}_{24})$ for the isometry group $\mathrm{Co}_0$ of the Leech lattice to lift to a subgroup of the automorphism group of its lattice VOA. (It would be valuable if there exists some systematic way to find a set of such relations.)

For example, a set of relations[53]

$$(Z^{21}BBAB)^{11} = 1, \tag{9.40}$$
$$((YX^2YX)^2XYX)^{22} = 1, \tag{9.41}$$
$$Y^3(YX)^5Y^2XB = 1, \tag{9.42}$$

where $X = BA$, $Y = BBA$, and $Z = (BBABA)^2BA$, leads to the equation $M\vec{\rho} = \vec{\mu}$ with the coefficient matrix $M$ and $\vec{\mu}$ recorded in (C.5, C.6); these $M$ and $\vec{\mu}$ are calculated by computer (see `Co0_of_Leech.ipynb` for the source code). By the Smith decomposition $M = P^{-1}DQ^{-1}$, we obtain the elementary divisors of $M$ as

$$1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 2, \tag{9.43}$$

and hence the rank of $D$ is 24, whereas

$$P\vec{\mu} = \begin{pmatrix} 27709164 & 2529357 & 7489045 & 30268789 & 4526426 & 20722962 & 15136149 & 30380229 \\ 5661159 & 9704077 & 3615025 & 17793711 & 1073374 & 8631464 & 30885927 & 27407271 \\ 7011428 & 2117131 & 25166374 & 9655442 & 645641 & 12408278 & 26657215 & -5342424 \\ 0 & 2 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 2 & 2 & 2 & 0 & 0 & 2 & 2 \\ 0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 \\ 10378735 & 0 & 9006865 & 16286994 & 0 & 25479941 & 27049788 & 31316233 \end{pmatrix}^T, \tag{9.44}$$

where for example the last entry is nonzero in $\mathbb{R}/2\mathbb{Z}$. Therefore, we cannot find the values of $\zeta_A(\mathbb{e}_i)$ and $\zeta_B(\mathbb{e}_i)$ $(i = 1, \ldots, 24)$ compatible with the relations (9.40)–(9.42), and hence the isometry group $\mathrm{Co}_0$ of the Leech lattice $\Lambda_{24}$ does not lift to a subgroup of the automorphism group of the lattice VOA $V_{\Lambda_{24}}$.

[53] The relation (9.40) is taken from the semi-presentation of $\mathrm{Co}_1$ on the Version 3 webpage of ATLAS of Finite Group Representations https://brauer.maths.qmul.ac.uk/Atlas/v3/spor/Co1/, the relation (9.41) is taken from the representative of the conjugacy class 22A of $\mathrm{Co}_1$ on the same webpage, and the relation (9.42) was found through a brute-force search by computer.

# 10 Currents Invariant under Specified Symmetry

Now that we understand the symmetry of the odd Leech lattice CFT contains $\mathrm{U}(1)^{24}.M_{24}$, instead of $M_{24}$, our next interest is in investigating what kind of structure can be preserved under its action or the action of its subgroup. Specifically, in this Section 10, we determine weight-1 and weight-$\frac{3}{2}$ elements of the odd Leech lattice VOA, which could be ingredients of a superconformal algebra, invariant under the action of $\mathrm{U}(1)^{24}.M_{24}$ or its subgroup $\mathrm{U}(1)^{23}.M_{23}$.

This section presents original work based on [Oka24].

**The currents invariant under $\mathrm{U}(1)^{24}.M_{24}$**

In this Section 10, let $L$ be the odd Leech lattice $O_{24}$ constructed as in (2.38). Recall from Section 5.2.3 that an element of $O(\hat{L})$ induces an automorphism of the lattice VOA $V_L$ as in (5.63). Then the automorphism group $\mathrm{Aut}(V_L)$ contains $O(\hat{L}) = \mathrm{U}(1)^{24}.O(L)$, and hence $\mathrm{U}(1)^{24}.M_{24}$, as a subgroup. Let us investigate the weight-1 and weight-$\frac{3}{2}$ currents of $V_L$, invariant under the action of $\mathrm{U}(1)^{24}.M_{24}$.

To begin with, any weight-1 current of $V_L$ is in the form of

$$\alpha(-1)e^0, \tag{10.1}$$

where $\alpha \in L \otimes_{\mathbb{Z}} \mathbb{C}$. Since $e^0$ is the identity element of $\hat{L}$, we have $f(e^0) = e^0$ for any $f \in O(\hat{L})$. Therefore, an element (10.1) is invariant under the action (5.63) of $\mathrm{U}(1)^{24}.M_{24} \subset \mathrm{Aut}(V_L)$, if and only if $\alpha \in L \otimes_{\mathbb{Z}} \mathbb{C}$ is invariant under the action of $M_{24} \subset O(L)$ extended by $\mathbb{C}$-linearity. Since the action of $M_{24}$ on $L = O_{24}$ constructed as in (2.38) is the permutation of the coordinates with respect to the standard basis $\mathbb{e}_1, \dots, \mathbb{e}_{24}$ of $\mathbb{R}^{24}$ where $L$ is embedded, and it is transitive, such $\alpha$ is only $\vec{1} = (1, \dots, 1) = \sum_i \mathbb{e}_i$, up to multiplication by $\mathbb{C}$. In conclusion, the weight-1 current of $V_L$ invariant under the action of $\mathrm{U}(1)^{24}.M_{24}$ is

$$j^{M_{24}} = \sum_{i=1}^{24} \mathbb{e}_i(-1)e^0, \tag{10.2}$$

up to multiplication by $\mathbb{C}$. In physics notation,

$$j^{M_{24}} = \sum_{i=1}^{24} \mathbb{a}^i_{-1}|0\rangle. \tag{10.3}$$

Next, any weight-$\frac{3}{2}$ current of $V_L$ is a linear combination of elements in the form of

$$e^k \quad \text{with} \quad k = \frac{1}{2\sqrt{2}}(-1)^w, \quad w \in G_{24}, \tag{10.4}$$

where the codeword $w$ of the binary Golay code $G_{24}$ is regarded as a vector $(w_i)_i$ with entries 0 or 1, and $(-1)^w$ denotes the vector $((-1)^{w_i})_i$. Such elements are not invariant under the action of $\mathrm{Hom}(L, \mathrm{U}(1)) \cong \mathrm{U}(1)^{24} \subset \mathrm{Aut}(V_L)$, recalling its action (5.54). Therefore, there is no weight-$\frac{3}{2}$ current of $V_L$ invariant under the action of $\mathrm{U}(1)^{24}.M_{24}$.

Even if we imposed the invariance under the action of $\mathbb{Z}_2^{24}.M_{24}$ only, there is still no invariant weight-$\frac{3}{2}$ current of $V_L$. The reason is as follows. $k = \frac{1}{2\sqrt{2}}(-1)^w$ is an odd vector, and the $\mathbb{Z}$-basis $e_1, \dots, e_{24}$ of $L$ in (2.39) satisfies the condition that $e_1$ is odd and the other $e_i$'s are even. Then, it follows that the component $k^1$ of the odd vector $k = \sum_i k^i e_i$ with respect to the basis $e_1$ is odd, and therefore its dual $(e_1)^\vee \in \mathrm{Hom}(L, \mathbb{Z}_2)$ always gives a minus sign to (10.4).

**The currents invariant under** $\mathrm{U}(1)^{23}.M_{23}$

We have learned that there is no weight-$\frac{3}{2}$ current fixed by $\mathrm{U}(1)^{24}.M_{24} \subset \mathrm{Aut}(V_L)$. How about a smaller subgroup? For example, the subgroup of $M_{24}$ stabilizing one coordinate, say $k_1$ of $k = \sum_i k_i \mathfrak{e}_i \in L$, is $M_{23}$. In addition, if we consider the subgroup

$$\mathrm{Stab}_{\mathrm{Hom}(L,\mathrm{U}(1))}(\mathfrak{e}_1) := \{\eta \in \mathrm{Hom}(L, \mathrm{U}(1)) \mid \eta(L \cap \mathbb{R}\mathfrak{e}_1) = 1\} \cong \mathrm{U}(1)^{23}, \tag{10.5}$$

of $\mathrm{Hom}(L, \mathrm{U}(1))$, then $\mathrm{Stab}_{\mathrm{Hom}(L,\mathrm{U}(1))}(\mathfrak{e}_1).M_{23}$ is a subgroup of $\mathrm{Hom}(L, \mathrm{U}(1)).M_{24} \subset O(\hat{L})$. To see it,[54] we have to check that the 2-cocycle $E : M_{24} \times M_{24} \to \mathrm{Hom}(L, \mathrm{U}(1))$ associated with the extension $O(\hat{L})$ in (5.61) takes values in $\mathrm{Stab}_{\mathrm{Hom}(L,\mathrm{U}(1))}(\mathfrak{e}_1)$ when restricted to $E|_{M_{23}\times M_{23}}$. See Appendix D for the detailed proof.

Let us investigate the weight-1 and weight-$\frac{3}{2}$ currents of $V_L$, invariant under the action of $\mathrm{Stab}_{\mathrm{Hom}(L,\mathrm{U}(1))}(\mathfrak{e}_1).M_{23} \cong \mathrm{U}(1)^{23}.M_{23}$. By an analysis similar to the case of $\mathrm{U}(1)^{24}.M_{24}$, the weight-1 invariant currents are

$$j_1^{M_{23}} = \mathfrak{e}_1(-1)e^0, \quad j_2^{M_{23}} = \sum_{i=2}^{24} \mathfrak{e}_i(-1)e^0, \tag{10.6}$$

and their $\mathbb{C}$-linear combinations. In physics notation,

$$j_1^{M_{23}} = \mathfrak{a}_{-1}^1|0\rangle, \quad j_2^{M_{23}} = \sum_{i=2}^{24} \mathfrak{a}_{-1}^i|0\rangle. \tag{10.7}$$

There is no weight-$\frac{3}{2}$ invariant current again. Even if we imposed the invariance under the action of $\mathbb{Z}_2^{23}.M_{23}$ only, there are still no invariant weight-$\frac{3}{2}$ currents. This is because the basis vector $e_{24}$ in (2.39) is in fact a primitive vector of $L \cap \mathbb{R}\mathfrak{e}_1$, and hence $\mathrm{Stab}_{\mathrm{Hom}(L,\mathbb{Z}_2)}(\mathfrak{e}_1) = \mathrm{Span}_{\mathbb{Z}_2}\{(e_1)^\vee, \dots, (e_{23})^\vee\}$, where $(e_i)^\vee$ are the dual basis vectors of (2.39). Therefore, $(e_1)^\vee$ is in $\mathrm{Stab}_{\mathrm{Hom}(L,\mathbb{Z}_2)}(\mathfrak{e}_1)$, and it always gives a minus sign to (10.4) by the same discussion as in the case of $\mathbb{Z}_2^{24}.M_{24}$.

[54] The fact that $\mathrm{Stab}_{\mathrm{Hom}(L,\mathrm{U}(1))}(\mathfrak{e}_1).M_{23}$ is a subgroup of $\mathrm{Hom}(L, \mathrm{U}(1)).M_{24}$ is nontrivial in the following sense. Let $\hat{G}$ be an group extension $N.G$ of $G$ by $N$. For any subgroup $G'$ of $G$, $\hat{G}$ contains an extension $N.G'$ as a subgroup. However, for a subgroup $N'$ of $N$, $N'.G$ is not a subgroup of $\hat{G}$ in general. This is because the multiplication law of $\hat{G}$ is like $(n_1, g_1) \cdot (n_2, g_2) = (n_1 g_1(n_2)\epsilon(g_1, g_2), g_1 g_2)$; see (A.4) for the precise expression.

# 11 Discussion on Odd Leech Lattice CFT

In this Section 11, we revisit the moonshine-type observation on the $\mathcal{N} = 2$ extremal elliptic genus by [BDFK15], in light of what we have studied about the odd Leech lattice CFT.

This section presents original work based on [Oka24].

**Recapitulation of Part III**

In this Part III, we have seen that the symmetry $\mathrm{Aut}(V_{O_{24}})$ of the odd Leech lattice VOA inherits the $M_{24}$ or $M_{23}$ symmetry of the odd Leech lattice $O_{24}$ in the form of the non-split extension $\mathrm{U}(1)^{24}.M_{24}$ or $\mathrm{U}(1)^{23}.M_{23}$, and therefore $M_{24}$ and $M_{23}$ are not genuine symmetries of the lattice VOA. We also determined weight-1 and weight-$\frac{3}{2}$ currents in the odd Leech lattice VOA invariant under the symmetry $\mathrm{U}(1)^{24}.M_{24}$ or its subgroup $\mathrm{U}(1)^{23}.M_{23}$. In particular, we concluded that there is no such weight-$\frac{3}{2}$ current.

**The moonshine-type observation on the odd Leech lattice CFT to discuss**

Let us revisit the observation by [BDFK15]. They pointed out that representation dimensions of $M_{24}$ appear in the decomposition (1.2) of the $\mathcal{N} = 2$ extremal elliptic genus $Z_{\mathrm{ext}}^{m=4}(\tau, z)$ of central charge $c = 6m = 24$ into the $(-1)^F$-inserted Ramond characters of an $\mathcal{N} = 2$ SCA. They further constructed a chiral $\mathcal{N} = 2$ SCFT whose partition function of the R sector coincides with the extremal elliptic genus $Z_{\mathrm{ext}}^{m=4}(\tau, z)$, although a slight modification is needed for their supercurrents as in Appendix B. Their theory is the fermionization of the $(A_1)^{24}$ Niemeier lattice CFT with respect to the shift $\mathbb{Z}_2$ symmetry, and as mentioned in Section 5.3.4, its NS sector is equivalent to the odd Leech lattice VOA $V_{O_{24}}$, denoted by $V_L$ for short below.

Now, we notice that there are more obstacles than we naively expected, in trying to explain the moonshine-type observation by this fermionized $(A_1)^{24}$ theory. The apparent problem is why it is the representation dimensions of $M_{24}$ that appear in the decomposition of the partition function of the R sector into the $\mathcal{N} = 2$ SCA characters, even though the genuine symmetry of the VOA of the NS sector is the non-split extension $(\mathbb{Z}_2 \text{ or } \mathrm{U}(1))^{24}.M_{24}$. Let us still try to explore a perspective in which $M_{24}$ or its subgroup $M_{23}$ takes on a meaningful role. As we will see, this is not straightforward at all.

Here, we note that the difference between the NS sector and the R sector is not important. One reason is that, using the spectral flow of the $\mathcal{N} = 2$ SCA, we see that the partition function of the NS sector should also exhibit the representation dimensions of $M_{24}$ under the decomposition into $\mathcal{N} = 2$ SCA characters. Another reason is that, since the R sector is not closed under the OPE, we cannot consider a new CFT consisting of the R sector only, but the NS sector always comes into the story. So there is no hope that the symmetry of CFT contains $M_{24}$ as a subgroup.

*Remark* 11.1. In light of VOA, the R sector can be formulated as a twisted module $M_{\mathrm{R}}$ of the NS sector $V_L$. We may consider the automorphism group of this module $\mathrm{Aut}_{V_L}(M_{\mathrm{R}}) = (\mathrm{End}_{V_L}(M_{\mathrm{R}}))^\times$, but it would be too small to provide any useful implications. For example, if $M_{\mathrm{R}}$ is an irreducible module of $V_L$, then $\mathrm{Aut}_{V_L}(M_{\mathrm{R}}) = \mathbb{C}^\times$ by Schur's lemma. We can try to find a

subalgebra $\mathcal{A}$ of $V_L$, say some algebra containing an $\mathcal{N}=2$ SCA, such that $\mathrm{End}_{\mathcal{A}}(M_{\mathrm{R}})$ contains $\mathbb{C}[M_{24}]$ or $\mathbb{C}[M_{23}]$, but this attempt is not much different from the following approach focusing on the NS sector. *(Remark ends.)*

As such, to explain the appearance of the representation dimensions of $M_{24}$, we have to exclude the effect of the group extension.

**Explore possible explanations**

One possible way to do so is to construct some VOA-like object, for example by using $\hat{L}/\mathbb{Z}_2 \cong L$, on which $(2^{24}.M_{24})/2^{24} \cong M_{24}$ can act. The discussion on whether this idea works is beyond the scope of this thesis, and it might require further study.

Another way[55] is to consider a subVOA of $V_L$ fixed by $2^{24} \subset 2^{24}.M_{24}$. Such a sub-VOA admits the action of $(2^{24}.M_{24})/2^{24} \cong M_{24}$. Recalling how this $2^{24} \cong \mathrm{Hom}(L,\mathbb{Z}_2)$ acts on $V_L$ (5.54), we can see that the fixed subVOA is the lattice VOA $V_{2L}$, where $2L$ is the sublattice of $L$ spanned by twice the vectors of $L$. We can also check that $M_{24}$ in fact acts on this $V_{2L}$ by observing that the cocycle $\varepsilon : L \times L \to \mathbb{Z}_2$ restricted to $2L$ is trivial.

This lattice sub-VOA $V_{2L}$ has $4^{24}$ irreducible modules $V_{\lambda+2L}$ labeled by $[\lambda] \in (2L)^*/2L$ [Don93]. If we write a basis of the odd Leech lattice $L$ as $\{e_i\}_{i=1,\dots,24}$ and its dual basis of $L^*$ as $\{(e_i)^\vee\}_{i=1,\dots,24}$, since the odd Leech lattice is self-dual, we can take a representative of $[\lambda] \in (2L)^*/2L$ in the form of $\lambda = \sum_{i=1}^{24} \lambda^i (e_i)^\vee$ with $\lambda^i \in \{0, \frac{1}{2}, 1, \frac{3}{2}\}$. Then, the original lattice VOA $V_L$ as a module of $V_{2L}$ is the sum $\oplus_{\lambda^i \in \{0,1\}} V_{\lambda+2L}$ of the $2^{24}$ irreducible modules.

*Remark* 11.2. More generally, for each $(e_{j_0})^\vee + \cdots + (e_{j_k})^\vee \in \mathrm{Hom}(L,\mathbb{Z}_2) \cong 2^{24}$, the sum $V_{\frac{1}{2}(e_{j_0})^\vee + \cdots + \frac{1}{2}(e_{j_k})^\vee + L} := \oplus_{\lambda^i \in \{0,1\}} V_{\lambda + \frac{1}{2}(e_{j_0})^\vee + \cdots + \frac{1}{2}(e_{j_k})^\vee + 2L}$ of the $2^{24}$ irreducible modules is the $((e_{j_0})^\vee + \cdots + (e_{j_k})^\vee)$-twisted module of the original lattice VOA $V_L$. (See also Remark 6.2.) *(Remark ends.)*

Since this lattice VOA $V_{2L}$ is rational and $C_2$-cofinite, the characters (bosonic partition functions) of these irreducible modules of $V_{2L}$ are closed under the modular transformations [Zhu96]. Note that $V_{2L}$ itself is also a module of $V_{2L}$. Since $V_{2L}$ has a genuine $M_{24}$ symmetry, the character of $V_{2L}$ itself exhibits representation dimensions of $M_{24}$ under the decomposition into Virasoro characters. We may even define the analogues of the McKay–Thompson series, and their twisted versions, for this module $V_{2L}$.

However, since the modules other than $V_{2L}$ do not admit the action of $M_{24}$ in general (otherwise, $V_L$ would admit the $M_{24}$-action, which is a contradiction), it is difficult to directly explain the appearance of representation dimensions of $M_{24}$ in the partition function of $V_L$, or to define such $M_{24}$-inserted characters for every module of $V_{2L}$ and expect their modular properties. In fact, $M_{24}$ acts on the set $(2L)^*/2L$ of the labels of the irreducible modules as $[\lambda] \mapsto [g(\lambda)]$ ($g \in M_{24}$), and the module $V_{\lambda+2L}$ is just a projective representation of the subgroup $G_{[\lambda]} \subset M_{24}$ stabilizing

[55] The author thanks the anonymous referee of [Oka24] for providing this viewpoint of focusing on a subVOA having an $M_{24}$ symmetry.

$[\lambda] \in (2L)^*/2L$ because of the nontrivial cocycle factor. By putting all these data, particularly on the $2^{24}$ irreducible modules $V_{\lambda+2L}$ ($\lambda^i \in \{0,1\}$) constituting $V_L$, we just come back to the $2^{24}.M_{24}$-action on $V_L$.

Another rather haphazard way is to forget the information of cocycle factors, even though it destroys the OPE structure. Let us investigate this possibility step by step.

**1. Review of the definition of an automorphism of VOA**

As reviewed in Definition 5.2, the definition of an automorphism of the VOA $V_L$ is a map $F : V_L \to V_L$ such that

(1) it is a $\mathbb{C}$-linear automorphism on the $\mathbb{C}$-vector space $V_L$,
(2) it preserves OPE; more precisely, $F(v_{(m)}(v')) = (F(v))_{(m)}(F(v'))$ for any $v, v' \in V_L$,
(3) it preserves the Virasoro element; $F(\omega) = \omega$.

Such maps constitute the automorphism group $\mathrm{Aut}(V_L)$, and it contains $\mathrm{Hom}(L, \mathbb{Z}_2).O(L)$ as a subgroup.

**2. Modifying the conditions to discard the effect of cocycle factors**

To discard the effect of cocycle factors, let us modify these conditions (1)–(3) and consider a map $F : V_L \to V_L$ such that

(1) it is a $\mathbb{C}$-linear automorphism on the $\mathbb{C}$-vector space $V_L$,
(2) it can be written in the form of (5.63),

$$F(\alpha_1(-m_1)\cdots\alpha_l(-m_l)e^k) = \bar{f}(\alpha_1)(-m_1)\cdots\bar{f}(\alpha_l)(-m_l)f(e^k),$$

by using some map $f : \hat{L} \to \hat{L}$ such that $\bar{f} : L \to L; k \mapsto \overline{f(e^k)}$ (or equivalently, $\bar{f} : L \to L$ defined as $F(e^k) = \pm e^{\bar{f}(k)}$) is an isometry on $L$,
(3) it preserves the Virasoro element; $F(\omega) = \omega$,
(4) it preserves the OPE action by $\omega$; $F(\omega_{(m)}(v')) = \omega_{(m)}(F(v'))$ for any $v' \in V_L$.

The condition (2) uses a map $f : \hat{L} \to \hat{L}$ with $\bar{f} \in O(L)$, and such $f$ is homomorphic up to cocycle factors

$$f(e^k e^{k'}) = \kappa^{0 \text{ or } 1} f(e^k) f(e^{k'}). \tag{11.1}$$

Note that the condition (1) requires $f(\kappa^m e^k) = \kappa^m f(e^k)$. The condition (3) also requires $f(e^0) = e^0$ because $\omega = \frac{1}{2}\sum_i \mathfrak{e}_i(-1)^2 e^0$. The last condition (4) automatically follows from the other conditions (1)–(3), in particular because the action of $\omega_{(m)}$ is just a multiplication by creation-annihilation operators.

The set of all maps $F : V_L \to V_L$ satisfying these conditions (1)–(4) constitute a group under the composition, which we write as $\mathrm{LinAut}(V_L; \mathrm{Vir})_{O(L)}$. As a result of above discussions, this

group $\mathrm{LinAut}(V_L;\mathrm{Vir})_{O(L)}$ turns out to be $\mathrm{Map}(L,\mathbb{Z}_2)_0 : O(L)$, where $\mathrm{Map}(L,\mathbb{Z}_2)_0$ is the set of all the maps $L \to \mathbb{Z}_2$ with $0 \mapsto \kappa^0$, which constitutes a group under the multiplication. It is greatly enlarged from $\mathrm{Hom}(L,\mathbb{Z}_2).O(L)$, and in particular contains $M_{24} \subset O(L)$ preserving the momentum lattice $L = O_{24}$.

By the condition (2), $F$ preserves OPE of monomial elements up to cocycle factors; more precisely, for any $v, v' \in V_L$ in the form of $\alpha_1(-m_1)\cdots\alpha_l(-m_l)e^k$,

$$F(v_{(m)}(v')) = \pm(F(v))_{(m)}(F(v')). \tag{11.2}$$

This is because, $F$ preserves OPE of $e^k$ and $e^{k'}$ up to cocycle factors, which can be seen from the explicit formula [DN99, §3.4] of $(e^k)_{(m)}(e^{k'})$, and the commutation relation $[\alpha(m),\alpha'(m')] = \langle\alpha,\alpha'\rangle m\delta_{m+m',0}$ is also preserved as $[\bar{f}(\alpha)(m),\bar{f}(\alpha')(m')] = \langle\alpha,\alpha'\rangle m\delta_{m+m',0}$. In this sense, we have discarded the effect of cocycle factors.

The last condition (4) guarantees that the action of this group $\mathrm{LinAut}(V_L;\mathrm{Vir})_{O(L)}$ does not change the $L_0$-eigenvalues of states of $V_L$, so that we can expect that $\mathrm{LinAut}(V_L;\mathrm{Vir})_{O(L)}$ accounts for some properties of the decomposition of a partition function $\mathrm{Tr}_{V_L} q^{L_0-\frac{c}{24}}$ into the Virasoro characters. Here we considered a simple partition function rather than $\mathrm{Tr}_{V_L}(-1)^F y^{J_0} q^{L_0-\frac{c}{24}}$ respecting the $\mathcal{N}=2$ SCA, just for simplicity.

### 3. Taking the $\mathcal{N}=2$ SCA into account

We only focused on the Virasoro algebra so far. Can we apply a similar discussion to $\mathcal{N}=2$ SCA? The generators of $\mathcal{N}=2$ SCA are the energy-momentum tensor $T(z)$, a $\mathrm{U}(1)$ current $J(z)$ of weight 1, and supercurrents $G^{\pm}(z)$ of weight $\frac{3}{2}$. As discussed in [BDFK15] and reviewed around (B.14), we can choose the $\mathrm{U}(1)$ current $J(z)$ as $2\sqrt{-2}\partial X^0(z)$, to reproduce the extremal elliptic genus as the partition function $\mathrm{Tr}_{\mathrm{R}}(-1)^F y^{J_0} q^{L_0-\frac{c}{24}}$ of the R sector. This $\mathrm{U}(1)$ current is exactly $j_1^{M_{23}} = \mathfrak{a}^0_{-1}|0\rangle$ in (10.7) up to scalar multiplication, and invariant under the action of $2^{23}.M_{23} \subset \mathrm{Aut}(V_L)$. However, there is no weight-$\frac{3}{2}$ current invariant under $2^{23}.M_{23}$ as we have seen in Section 10. In addition, any weight-$\frac{3}{2}$ currents invariant under the subgroup $M_{23}$ of $\mathrm{LinAut}(V_L;\mathrm{Vir})_{O(L)}$ does not satisfy the OPEs of supercurrents of $\mathcal{N}=2$ SCA, as we will show in Remark B.1 of Appendix B. The subgroup of $2^{24}.M_{24} \subset \mathrm{Aut}(V_L)$ preserving the supercurrents we will construct in Appendix B would be too small to provide any useful implications (see the last paragraph of Appendix B).

Therefore, what we can do is to deal with the subalgebra $\mathcal{A}$ of the $\mathcal{N}=2$ SCA generated by $T(z)$ and $J(z)$ only, and consider a group $\mathrm{LinAut}(V_L;\mathcal{A})_{O(L)}$ consisting of maps $F : V_L \to V_L$ such that

(1) it is a $\mathbb{C}$-linear automorphism on the $\mathbb{C}$-vector space $V_L$,

(2) it can be written in the form of (5.63),

$$F(\alpha_1(-m_1)\cdots\alpha_l(-m_l)e^k) = \bar{f}(\alpha_1)(-m_1)\cdots\bar{f}(\alpha_l)(-m_l)f(e^k),$$

by using some map $f : \hat{L} \to \hat{L}$ such that $\bar{f} : L \to L; k \mapsto \overline{f(e^k)}$ is an isometry on $L$,

(3) it preserves the Virasoro element $F(\omega) = \omega$ and the $\mathrm{U}(1)$ current $F(j_1^{M_{23}}) = j_1^{M_{23}}$,

(4) it preserves the OPE action by $\omega$ and $j_1^{M_{23}}$; $F(\omega_{(m)}(v')) = \omega_{(m)}(F(v'))$ and $F((j_1^{M_{23}})_{(m)}(v')) = (j_1^{M_{23}})_{(m)}(F(v'))$ for any $v' \in V_L$.

Then the group $\mathrm{LinAut}(V_L; \mathcal{A})_{O(L)}$ contains $M_{23}$ as a subgroup, through a similar discussion to that on $M_{24} \subset \mathrm{LinAut}(V_L; \mathrm{Vir})_{O(L)}$ above, and it might account for the appearance of representation dimensions of $M_{23}$ in the decomposition of the partition function $\mathrm{Tr}_{V_L}(-1)^F y^{J_0} q^{L_0 - \frac{c}{24}}$ into the characters of the algebra $\mathcal{A}$.

However, in order to deduce some conclusion on the decomposition into the $\mathcal{N} = 2$ SCA characters, we need precise information on how the characters of the $\mathcal{N} = 2$ SCA decompose into the characters of the algebra $\mathcal{A}$. We might be able to say that the coefficients in the decomposition into the $\mathcal{N} = 2$ SCA characters are at least virtual representation dimensions (where not only addition but also subtraction of irreducible representation dimensions are allowed) of $M_{23}$, but it is difficult to say more than that at this stage.

Moreover, in the first place, the whole group $\mathrm{LinAut}(V_L; \mathcal{A})_{O(L)}$ is much larger than $M_{23}$; it is in fact $\mathrm{Map}(L, \mathbb{Z}_2)_0 : (2^{11} : M_{23})$, where $2^{11} : M_{23}$ is the subgroup of $O(L) \cong 2^{12} : M_{24}$ stabilizing the first entry of the lattice vectors. So it is unclear why the subgroup $M_{23}$ is only in effect, compared with other moonshine phenomena such as the monstrous moonshine and the Conway moonshine, where the automorphism groups of their VOAs are precisely the monster group $\mathbb{M}$ and the Conway group $\mathrm{Co}_1$. Anyway, since we have destroyed the OPE structure, and hence the VOA structure, it is difficult to imagine that the discussions here would yield so fruitful outcomes as, for example, the monstrous moonshine.

In conclusion, the situation surrounding the moonshine-type observation on the extremal elliptic genus seems more mysterious than we think. We have only dealt with the VOA of odd Leech lattice CFT, or the fermionized $(A_1)^{24}$ Niemeier lattice CFT, but of course there is a possibility that another new theory explains the observation in a nice way. It would be interesting if that is the case, and if it also provides some implications for other moonshine phenomena including the K3 Mathieu moonshine.

# Part IV

# The Conway Orbifold of Duncan's Supermoonshine Module

In this Part IV, we describe how to compute the Conway orbifold partition function of Duncan's module $V^{f\natural}$. As reviewed in Section 6, Duncan's module $V^{f\natural}$ has a $\mathrm{Co}_1$ symmetry preserving its $\mathcal{N}=1$ superconformal algebra.

In Section 12, we will review the general theory of orbifolds. In general, a symmetry of a theory has an obstruction to the orbifold, called an anomaly, and the anomaly of the $\mathrm{Co}_1$ symmetry of Duncan's module vanishes if we take the 24-fold tensor product $(V^{f\natural})^{\otimes 24}$. We will also write down three concrete steps to compute the partition function of the orbifold $(V^{f\natural})^{\otimes 24}/\mathrm{Co}_1$.

Since Duncan's module $V^{f\natural}$ is constructed from the theory of 24 real free chiral fermions, its twisted partition functions can be calculated from those of the free chiral fermions. So we will compute the twisted partition functions of free chiral fermions, twisted by $\mathrm{SO}(n)$ or $\mathrm{Spin}(n)$, and transformation properties of them in Section 13.

Finally, we will carry out the three steps to compute the Conway orbifold partition function in Section 14. We will accomplish the first two steps: calculate the conjugacy classes of commuting pairs of $\mathrm{Co}_1$, and the twisted partition functions of Duncan's module $V^{f\natural}$. As for the final step, determining the phases to trivialize the anomaly, we will attempt to determine them from consistency conditions. We will find one solution, although its validity requires further investigation.

## 12 Review of Orbifolds

This Section 12 reviews the theoretical foundations of orbifolds, with the goal of understanding how to compute the Conway orbifold partition function of Duncan's module. Orbifold is a procedure to make a new theory from a theory with a finite group symmetry, but in general, such a symmetry has an obstruction to the orbifold, called anomaly. In Section 12.1, we will review the concept of anomaly, and when and how we can make it vanish. In Section 12.2, we will write down the partition function of the orbifold theory, and by exploiting several properties of the theory, we will reduce the expression of the partition function so that we can calculate it in minimal steps.

### 12.1 't Hooft Anomaly

An 't Hooft anomaly [tH80] is a quantum anomaly of a global symmetry $G$ of a theory $\mathcal{T}$. When $\mathcal{T}$ is a bosonic theory of $d+1$ spacetime dimensions with $d=0,1,2$, its 't Hooft anomaly is classified by $H^{d+2}(G;\mathrm{U}(1))$. When it is not trivial, it is an obstruction to orbifolding $\mathcal{T}$ by $G$. In

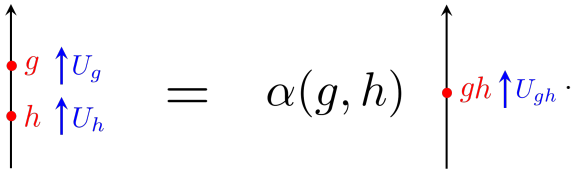
this Section 12.1, we will review it, starting with the case of $0+1$ dimension. At the end, we will describe the anomaly of the Conway symmetry of Duncan's module.

### 12.1.1 Anomaly in $0+1$ Dimension

Let $\mathcal{T}$ be a $0+1$ dimensional quantum field theory (QFT), namely, a quantum mechanical system, with a finite group symmetry $G$. The action of $g \in G$ on the Hilbert space $\mathcal{H}$ is implemented by a unitary operator $U_g : \mathcal{H} \to \mathcal{H}$. The group structure requires that the successive implementation of $U_g$ and $U_h$ should be equivalent to $U_{gh}$, but since the physical states are defined only up to $\mathrm{U}(1)$ phases, an additional phase factor $\alpha(g,h) \in \mathrm{U}(1)$ can appear as

$$U_g U_h = \alpha(g,h) U_{gh}. \tag{12.1}$$

If we draw the one-dimensional timeline, the Hilbert space $\mathcal{H}$ is living on each point, and the action of $U_g$ is depicted as a point operator, as in the following pictorial equation:




(12.2)

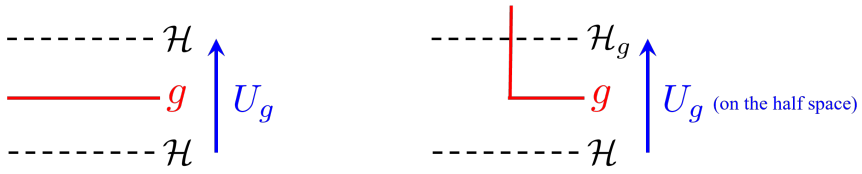
The phase $\alpha$ compensates the change of the intermediate picture, while keeping the action of the symmetries from the starting Hilbert space to the ending Hilbert space.

Therefore, $G$ acts on $\mathcal{H}$ as a projective representation, or a representation of an extension of $G$ by $\mathrm{U}(1)$, and its equivalence classes can be classified by the cohomology classes $[\alpha] \in H^2(G;\mathrm{U}(1))$. If we state it more concretely, the associativity $(U_g U_h)U_k = U_g(U_h U_k)$ (or the fact that the operator obtained as the fusion of ordered three operators is independent of the history of changing pictures) imposes the 2-cocycle condition on $\alpha$, and redefining $U_g$ as $\beta(g)U_g$ ($\beta(g) \in \mathrm{U}(1)$) changes $\alpha$ by the 2-coboundary $d\beta$. In this way, the cohomology class $[\alpha] \in H^2(G;\mathrm{U}(1))$ is associated with the theory $\mathcal{T}$. This is the *'t Hooft anomaly* of the $G$ symmetry of $\mathcal{T}$, and $\alpha$ is referred to as the *anomalous phase*.

When the cohomology class $[\alpha] \in H^2(G;\mathrm{U}(1))$ is trivial, we say that the anomaly is trivial or vanishing, the symmetry is non-anomalous, or the theory is anomaly-free. In such a situation, by the redefinition of $U_g$, we can make $\alpha(g,h) = 1$ for any $g, h \in G$, so that $G$ acts on $\mathcal{H}$ as a genuine representation. Otherwise, the 'projection operator' $P := \frac{1}{|G|}\sum_{g\in G} U_g$ onto $G$-invariant states is not truly the projection, for example because $U_g P \neq P$. Analogously, a nontrivial anomaly in $1+1$ dimensions is an obstruction to orbifolding or gauging the theory, which is a procedure to make a $G$-invariant theory.

### 12.1.2 Anomaly in $1+1$ Dimensions

Next, let $\mathcal{T}$ be a 1+1 dimensional QFT with a finite group symmetry[56] $G$. On the two-dimensional spacetime, the group action $U_g : \mathcal{H} \to \mathcal{H}$ on the time-sliced Hilbert space $\mathcal{H}$ is depicted as a horizontal line. Moreover, if the symmetry action is local, then we may perform it only on a segment or a half-line of the entire one-dimensional space. As a result, the defect, or the twisted boundary condition, appears at the boundary, and it is depicted as a vertical line (see Figure 12.1). We can also define more curved lines by moving the boundary (see for example [Sei23]). We will write the $g$-twisted Hilbert space, realized by the action of $g$ on the right-half space, as $\mathcal{H}_g$. If there are multiple twisted boundary conditions $g_1, \ldots, g_n$ inserted, then the twisted Hilbert space is denoted by $\mathcal{H}_{g_1,\ldots,g_n}$.




(a) (b)

Figure 12.1: (a) The action $U_g : \mathcal{H} \to \mathcal{H}$ of $g \in G$ depicted as a horizontal line. (b) The twisted Hilbert space implemented by the half-space action.

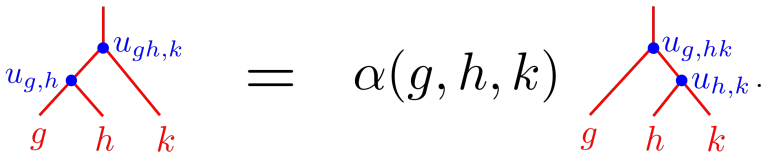
The fusion of such lines is implemented by the fusion operator

$$u_{g,h} : \mathcal{H}_{g,h} \to \mathcal{H}_{gh}. \tag{12.3}$$

The order of the fusion of three lines $\mathcal{H}_{g,h,k} \to \mathcal{H}_{ghk}$ should not affect the physics, which implies the existence of a phase factor $\alpha(g,h,k) \in \mathrm{U}(1)$ such that

$$u_{gh,k}u_{g,h} = \alpha(g,h,k)u_{g,hk}(U_g u_{h,k} U_g^{-1}), \tag{12.4}$$

where $U_g \bullet U_g^{-1}$ appears because we implemented the $g$-twist on the time-sliced Hilbert space by letting $U_g$ act on the right-half space. As a pictorial equation,




(12.5)

[56] As seen in Section 12.1.1, a symmetry $G$ of a $0+1$ dimensional theory acts on the Hilbert space as a projective representation. In $1+1$ dimension, on the other hand, when we say that the theory has a symmetry $G$, we assume that $G$ acts on the untwisted Hilbert space as a genuine representation. Even so, it can acts on the twisted Hilbert spaces as projective representations.

For example, the automorphism group $\mathrm{Aut}(V)$ of a VOA $V$ acts on the untwisted Hilbert space $V$ as a genuine representation, by its mathematical definition. For a monster VOA $V^\natural$, the anomaly of $\mathrm{Aut}(V^\natural) \cong \mathbb{M}$ is known to have order 24 in $H^3(\mathbb{M}; \mathrm{U}(1))$ [JF17, Thm. 1]. So it is conjectured that $H^3(\mathbb{M}; \mathrm{U}(1)) \cong \mathbb{Z}_{24}$ in [JF17].

Again, the phase $\alpha$ compensates the change of the intermediate picture, while keeping the action of the symmetries from the starting Hilbert space to the ending Hilbert space.

We omit the details, but the pentagon identity (the fact that the final picture obtained by changing the order of fusions from the given picture is independent of the history of changing pictures; see Figure 12.2) imposes the 3-cocycle condition on $\alpha$, and redefining $u_{g,h}$ as $\beta(g,h)u_{g,h}$ ($\beta(g,h) \in \mathrm{U}(1)$) changes $\alpha$ by the 3-coboundary $d\beta$.

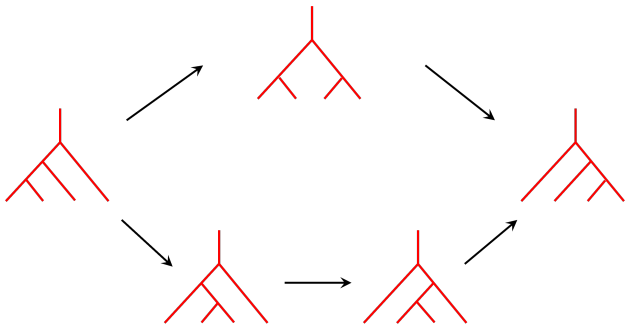
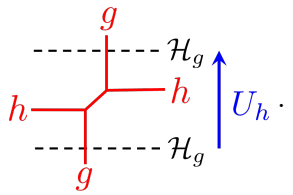


Figure 12.2: The change of pictures to obtain the pentagon identity.

In this way, the cohomology class $[\alpha] \in H^3(G;\mathrm{U}(1))$ is associated with the theory $\mathcal{T}$. This fact was known in the context of algebraic QFT [Mug04], and also established in the context of condensed matter physics in [EN14]; see also [Sei23]. This is the 't Hooft anomaly of the $G$ symmetry of $\mathcal{T}$.

### 12.1.3 Canceling Anomaly and Discrete Torsion

As we will see in Section 12.2, the orbifold of a $(1+1)$-dimensional CFT $\mathcal{T}$ by $G$ is constructed by adding the twisted sectors and projecting them onto the $G$-invariant states. Assume that we define the action of $U_h$ on the twisted sector $\mathcal{H}_g$, where $g$ and $h$ commute, as

$$U_h \ :=$$  (12.6)



(If $g$ and $h$ do not commute, $U_h$ maps $\mathcal{H}_g$ to $\mathcal{H}_{hgh^{-1}}$.) If the anomalous phases $\alpha$ in (12.5) are nontrivial, then the composition $U_hU_k$ of actions on $\mathcal{H}_g$ differs from $U_{hk}$ by a $\mathrm{U}(1)$ phase, so we cannot define the appropriate projection operator as $\frac{1}{|G|}\sum_{g\in G} U_g$, for a similar reason to the case of $0+1$ dimension in Section 12.1.1. This is an obstruction to orbifolding.

Moreover, a nontrivial anomaly is also an obstruction to the modular invariance of the orbifold.

Assume that we define the twisted partition function $Z_g^h(\tau)$ on a torus for commuting $g$ and $h$ as

$$Z_g^h(\tau) := Z\left(\begin{array}{ccc} & g & \\ h & & h \\ & g & \end{array}\right). \tag{12.7}$$

Then, the anomalous phases appear in the modular transformations, say the modular S transformation, as

$$Z_g^h(\tau) = Z\left(\begin{array}{ccc} & g & \\ h & \rightarrow & h \\ & g & \end{array}\right) \overset{S}{\longmapsto} Z\left(\begin{array}{ccc} & h & \\ g & \leftarrow & g \\ & h & \end{array}\right)$$

$$\| \tag{12.8}$$

$$\text{(phase from } \alpha) \; Z\left(\begin{array}{ccc} & h & \\ g & \leftarrow & g \\ & h & \end{array}\right) = \text{(phase from } \alpha) \; Z_h^{g^{-1}}(\tau).$$

Such phases break the modular invariance property of the orbifold theory, unless the anomalous phases $\alpha$ are all trivial.

If the cohomology class $[\alpha] \in H^3(G; \mathrm{U}(1))$ is trivial, then by taking a 2-cochain $\beta_0$ such that $\alpha(g,h,k) = d\beta_0(g,h,k) = \beta_0(h,k)\beta_0(gh,k)^{-1}\beta_0(g,hk)\beta_0(g,h)^{-1}$ and rewriting (12.4) using a new fusion operator $(u_0)_{g,h} := \beta_0(g,h)u_{g,h}$, we can remove the anomalous phase $\alpha$ from (12.4), so that the obstacle to a modular-invariant orbifold theory does not exist. In particular, if we write the twisted partition function defined as in (12.7) using fusion operators $u_{g,h}$ as ${Z^{(u)}}_g^h(\tau)$, then the new twisted partition function is

$${Z^{(u_0)}}_g^h(\tau) = \frac{\beta_0(h,g)}{\beta_0(g,h)} {Z^{(u)}}_g^h(\tau), \tag{12.9}$$

and this ${Z^{(u_0)}}_g^h(\tau)$ is the twisted partition function we can actually use for the computation of the twisted partition function of the orbifold theory.

However, there is a degree of freedom in the choice of the 2-cochain $\beta_0$ which is a solution to the equation $\alpha = d\beta_0$. In fact, even if we multiply one solution $\beta_0$ by a 2-cocycle $\beta$, the resulting $\beta_0 \cdot \beta$ is also a solution.[57] If we define another new fusion operator $(u_0')_{g,h} := \beta_0(g,h)\beta(g,h)u_{g,h}$, then the resulting twisted partition function ${Z^{(u_0')}}_g^h(\tau)$ is different from the originally modified one ${Z^{(u_0)}}_g^h(\tau)$ by $\frac{\beta(h,g)}{\beta(g,h)}$ in general. When the 2-cocycle $\beta$ is just a 2-coboundary $\beta(g,h) = d\gamma(g,h) = \gamma(h)\gamma(gh)^{-1}\gamma(g)$, then this difference $\frac{\beta(h,g)}{\beta(g,h)}$ is just 1 because $g$ and $h$ commute, and hence ${Z^{(u_0')}}_g^h(\tau)$ is the same as ${Z^{(u_0)}}_g^h(\tau)$.

[57] Since $\mathrm{U}(1)$ is a multiplicative group, the linearity of the differential $d$ is $d(\beta_0 \cdot \beta) = d\beta_0 \cdot d\beta$, and the cocycle condition is $d\beta = 1$.

As a result, we can say that if there is one way of canceling the anomaly (using $\beta_0$), then there are $\#H^2(G;\mathrm{U}(1))$ different ways of canceling the anomaly (using $\beta_0\beta$ for $[\beta] \in H^2(G;\mathrm{U}(1))$) leading to different twisted partition functions, and hence different orbifold theories. This degree of freedom is called the *discrete torsion*.

From the perspective of group extension, $H^2(G;\mathrm{U}(1))$ classifies the equivalence classes of the central extensions $1 \to \mathrm{U}(1) \to \mathrm{U}(1).G \to G \to 1$ (Appendix A.1). In addition, if $G$ is abelian, then $\frac{\beta(h,g)}{\beta(g,h)}$ is the commutator $c(h,g) \in \mathrm{U}(1)$ (see Appendix A.2). Even if $G$ is non-abelian, as we will see in Section 12.2.2, the twisted partition function $Z^{(u_0)}{}^h_g(\tau)$ can be non-zero only when $g$ and $h$ commute, so we may regard $\frac{\beta(h,g)}{\beta(g,h)}$ as the commutator.

For example, in the case of $G = \mathrm{Co}_1$, we have $H^2(\mathrm{Co}_1;\mathrm{U}(1)) = \mathbb{Z}_2$ as its Schur multiplier. Its trivial and non-trivial elements correspond to the split extension $U(1) \times \mathrm{Co}_1$ and the non-split extension $U(1)\cdot\mathrm{Co}_1$, respectively. Since $\mathrm{Co}_0 \cong \mathbb{Z}_2\cdot\mathrm{Co}_1$ is a non-split central extension, by writing the multiplication law of $\mathrm{Co}_0$ as $(\xi,g)\cdot(\xi',h) = (\xi\xi'\beta(g,h),gh)$ ($\xi,\xi' \in \{\pm1\}$, $g,h \in \mathrm{Co}_1$, $\beta : \mathrm{Co}_1 \times \mathrm{Co}_1 \to \{\pm1\}$ is a 2-cocycle), we can construct[58] the non-split extension $U(1)\cdot\mathrm{Co}_1$ by just extending $\xi \in \{\pm1\}$ to $\xi \in \mathrm{U}(1)$. When $g$ and $h$ commute, the value of the commutator $\frac{\beta(h,g)}{\beta(g,h)}$ is 1 if the lifts $(1,g),(1,g') \in \mathrm{Co}_0$ of $g,g' \in \mathrm{Co}_1$ also commute, and $-1$ if the lifts anticommute.

When we calculate the twisted partition functions of Duncan's module in Section 14.2.1, we will take the twists by $g,h \in \mathrm{Co}_1$ through their lifts $\check{g},\check{h} \in \mathrm{Co}_0$. If the lifts do not commute, then the twisted partition function just vanishes. Therefore, the phase $\frac{\beta(h,g)}{\beta(g,h)}$ caused by the discrete torsion is always 1 for non-zero twisted partition functions, and hence the orbifold partition function of $(V^{f\natural})^{\otimes 24}/\mathrm{Co}_1$ is unique.

### 12.1.4 Anomaly in Fermionic Systems

Finally, we briefly mention the anomaly in fermionic systems. Let us consider a fermionic theory with a finite group symmetry $G$. We demand that the $G$-action should preserve the parity, that is, it should commute with the symmetry $\mathbb{Z}_2 = \langle(-1)^F\rangle$ defining the fermionic parity.[59] (As a remark, in mathematics, the definition of an automorphism of a VOSA contains the condition that

[58] More rigorously, we have to check that $H^2(\mathrm{Co}_1;\mathbb{Z}_2) \to H^2(\mathrm{Co}_1;\mathrm{U}(1))$ is injective. This can be done as follows. From the short exact sequence $1 \to \mathbb{Z}_2 \to \mathrm{U}(1) \xrightarrow{(\bullet)^2} \mathrm{U}(1) \to 1$, we have the long exact sequence $\cdots \to H^1(\mathrm{Co}_1;\mathrm{U}(1)) \to H^2(\mathrm{Co}_1;\mathbb{Z}_2) \to H^2(\mathrm{Co}_1;\mathrm{U}(1)) \to \cdots$. Since $\mathrm{Co}_1$ is simple, $H^1(\mathrm{Co}_1;\mathrm{U}(1)) = \mathrm{Hom}(\mathrm{Co}_1,\mathrm{U}(1)) = 0$ (because the kernel of a homomorphism is a normal subgroup), so $H^2(\mathrm{Co}_1;\mathbb{Z}_2) \to H^2(\mathrm{Co}_1;\mathrm{U}(1))$ is injective.

[59] A fermionic theory $\mathcal{T}[\sigma]$ on a space-time manifold $M$ depends on the spin structure $\sigma$ on $M$. The set of spin structures is an affine space modeled on $H^1(M;\mathbb{Z}_2)$, so by introducing a $\mathbb{Z}_2$ gauge field $A \in H^1(M;\mathbb{Z}_2)$ as $\mathcal{T}[\sigma,A] := \mathcal{T}[\sigma+A]$ up to anomalous phases [BSZ24], we obtain the canonical $\mathbb{Z}_2$ symmetry $\langle(-1)^F\rangle$ of the fermionic theory. Then the assumption that $G$ commutes with $(-1)^F$ also means that the $G$-action does not change the NS sector to the R sector, and the R sector to the NS sector. See also footnote 43.

Note that, in [DGG21], the action of a symmetry $G$ is assumed not to contain the change of spin structure $\sigma$ (Be careful that this change seems denoted by the action by $(-1)^F$ there, and $G_f = \langle(-1)^F\rangle.G$), but by absorbing the change of $\sigma$ into the change of the gauge field $A$, we can also apply the discussion there to $G$ containing $(-1)^F$.

it preserves the parity.) We also assume that $G$ acts on the NS sector as a genuine representation, but $G$ may act on the R sector as a projective representation, and it is regarded as a part of the fermionic anomaly. A prototypical example[60] is that, $\mathrm{SO}(n)$ acts on the NS sector $A(\mathfrak{a})$ of $n$ real free fermions as a genuine representation, but it acts on the R sector $A(\mathfrak{a})_{\mathrm{tw}}$ only as a projective representation. This is because what really acts on $A(\mathfrak{a})_{\mathrm{tw}}$ as a genuine representation is its double cover $\mathrm{Spin}(n) = \mathbb{Z}_2.\mathrm{SO}(n)$; the projective $\mathrm{SO}(n)$-action on the R sector is the composition of a section $s : \mathrm{SO}(n) \to \mathrm{Spin}(n)$ and the genuine $\mathrm{Spin}(n)$-action.[61]

Whereas the bosonic anomalies in $1+1$ dimensions are classified by the ordinary cohomology $H^3(G;\mathrm{U}(1))$, the fermionic anomalies in $1+1$ dimensions are classified by the supercohomology $SH^3(G)$ [KT17,WG17]. As a set, $SH^3(G)$ is the same as $H^3(G;\mathrm{U}(1))\oplus H^2(G;\mathbb{Z}_2)\oplus H^1(G;\mathbb{Z}_2)$. The first layer $H^3(G;\mathrm{U}(1))$ is coming from the anomalies similar to the bosonic cases. The second layer $H^2(G;\mathbb{Z}_2)$ is coming from the fact that the fusion operators $u_{g,h}$ can have fermionic parities in the fermionic cases (see e.g. [EN14, Tac18, OSTZ25]), and sometimes called the Gu-Wen layer [GW12]. The third layer $H^1(G;\mathbb{Z}_2)$ describes anomalies which occur when the number of Majorana fermions is odd, for example at the edge of the Kitaev chain [Kit00].

As an example, the anomaly of a $\mathbb{Z}_2$ symmetry of one real fermion in $1+1$ dimensions corresponds to the generator of $SH(\mathbb{Z}_2) = H^3(\mathbb{Z}_2;\mathrm{U}(1)).H^2(\mathbb{Z}_2;\mathbb{Z}_2).H^1(\mathbb{Z}_2;\mathbb{Z}_2) \cong \mathbb{Z}_2.\mathbb{Z}_2.\mathbb{Z}_2 \cong \mathbb{Z}_8$. See [DGG21] for more details.

In higher $d+1$ dimensions, the fermionic anomalies are classified by a certain generalized cohomology theory (the Anderson dual of the spin bordism group $\Omega^{\mathrm{spin}}_{d+2}(BG)$) [KTTW14, FH16, GJF17, Yon18], and the supercohomology $SH^{d+2}(G)$ only captures the first three layers of it.

In the case of the $\mathrm{Co}_1$ symmetry of Duncan's module $V^{f\natural}$, it is known [JF17, Example 2.4.1] that its anomaly is the generator of $SH^3(\mathrm{Co}_1) = H^3(\mathrm{Co}_1;\mathrm{U}(1)).H^2(\mathrm{Co}_1;\mathbb{Z}_2) \cong \mathbb{Z}_{12}.\mathbb{Z}_2 \cong \mathbb{Z}_{24}$. Therefore, if we take the 24-fold tensor product of $V^{f\natural}\oplus V^{f\natural}_{\mathrm{tw}}$, its diagonal $\mathrm{Co}_1$-action is non-anomalous, and we can consider the Conway orbifold theory $(V^{f\natural}\oplus V^{f\natural}_{\mathrm{tw}})^{\otimes 24}/\mathrm{Co}_1$. Recall that the $\mathbb{Z}_2$ symmetry defining the fermion parity of $V^{f\natural}\oplus V^{f\natural}_{\mathrm{tw}}$ is $\langle -1_{\mathrm{SemiSpin}(24)}\rangle$. Since $-1_{\mathrm{SemiSpin}(24)}$ is not contained in $\mathrm{Co}_1 \subset \mathrm{SemiSpin}(24)$, this orbifold does not mix the NS sector and the R sector (see footnote 59).

*Remark* 12.1. The anomaly of the $\mathrm{Co}_0$ symmetry of $V^{s\natural}$ is also known [JF17, Example 2.4.1] to be the generator of $SH^3(\mathrm{Co}_0) = H^3(\mathrm{Co}_0;\mathrm{U}(1)) \cong \mathbb{Z}_{24}$. (In fact, we have seen that $H^2(\mathrm{Co}_0;\mathbb{Z}_2)$ is trivial in footnote 46.) So we can repeat the similar arguments for $V^{s\natural}$. *(Remark ends.)*

*Remark* 12.2. Since $SH^3(\mathrm{Co}_0) \cong \mathbb{Z}_{24}$, we may consider the orbifold of $(V^{f\natural}\oplus V^{f\natural}_{\mathrm{tw}})^{\otimes 24}$ by the preimage $\mathrm{Co}_0 \subset \mathrm{Spin}(24)$ of $\mathrm{Co}_1 \subset \mathrm{SemiSpin}(24)$. The action of $\mathrm{Co}_0$ on $V^{f\natural}\oplus V^{f\natural}_{\mathrm{tw}}$ is the same as that of $\mathrm{Co}_1$ (see footnote 60), but the number of twisted sectors are different ($\mathrm{Co}_1$ has

[60] Another example is that, $V^{f\natural}$ admits the action by $\mathrm{SemiSpin}(24)$ (or $\mathrm{Co}_1 \in \mathrm{SemiSpin}(24)$ if we take the $\mathcal{N}=1$ structure into account), but it acts on $V^{f\natural}_{\mathrm{tw}}$ only as a projective representation, because what really acts on $V^{f\natural}_{\mathrm{tw}}$ as a genuine representation is its double cover $\mathrm{Spin}(24)$ (or $\mathrm{Co}_0 \subset \mathrm{Spin}(24)$).

[61] Since $\mathrm{SO}(n)$ is a continuous group, we cannot apply the discussion of Section 12.1.2. In particular, we cannot naively say that "the anomaly of $\mathrm{SO}(n)$ is classified by $H^3(\mathrm{SO}(n);\mathrm{U}(1))$."

101 conjugacy classes, whereas $\mathrm{Co}_0$ has 167 conjugacy classes), so $(V^{f\natural} \oplus V_{\mathrm{tw}}^{f\natural})^{\otimes 24}/\mathrm{Co}_1$ and $(V^{f\natural} \oplus V_{\mathrm{tw}}^{f\natural})^{\otimes 24}/\mathrm{Co}_0$ are different theories. *(Remark ends.)*

## 12.2 Orbifold Partition Function

If a CFT $\mathcal{T}$ has a non-anomalous finite group symmetry $G$, we can construct a new CFT $\mathcal{T}/G$ called the orbifold of $\mathcal{T}$ by $G$, consisting of $G$-invariant states. In this Section 12.2, we will review it, focusing on its partition function. After reviewing the construction of orbifold in Section 12.2.1, we will reduce the expression of its partition function, using the properties of the action of $G$ on twisted sectors in Section 12.2.2. We will investigate the $\mathrm{SL}(2,\mathbb{Z})$ and $\mathrm{GL}(2,\mathbb{Z})$ transformations of the twisted partition functions in Section 12.2.3, and using them, we further reduce the expression of the orbifold partition function, in the case of constant twisted partition functions, in Section 12.2.4. At the end, we will see concrete three steps to compute the Conway orbifold partition function of Duncan's module.

### 12.2.1 Basic Construction of Orbifold

Let $\mathcal{T}$ be a two-dimensional CFT on a torus with a non-anomalous finite group symmetry $G$. The *orbifold* $\mathcal{T}/G$ of $\mathcal{T}$ by $G$ is constructed by the following two steps (see e.g. [Pol07a, §8.5]).

1. Add all the twisted sectors $\mathcal{H}_g$ ($g \in G$) and define
$$\mathcal{H}_{\mathrm{tot}} := \bigoplus_{g\in G} \mathcal{H}_g, \tag{12.10}$$
where $\mathcal{H}_1$ is the Hilbert space of the original theory $\mathcal{T}$.

2. Define the Hilbert space $\mathcal{H}^{(\mathcal{T}/G)}$ of the orbifold $\mathcal{T}/G$ as the $G$-invariant states of $\mathcal{H}_{\mathrm{tot}}$. That is,
$$\mathcal{H}^{(\mathcal{T}/G)} := \left(\frac{1}{|G|}\sum_{g\in G} U_g\right)\mathcal{H}_{\mathrm{tot}}. \tag{12.11}$$

We define the *twisted partition function* of $\mathcal{T}$, with spatial twist $g$ and temporal twist $h$ as
$$Z^{(\mathcal{T})}{}_g^h(\tau) := \mathrm{Tr}_{\mathcal{H}_g}[U_h q^{L_0-\frac{c}{24}}\bar{q}^{\tilde{L}_0-\frac{\tilde{c}}{24}}] \quad (q := e^{2\pi\sqrt{-1}\tau}), \tag{12.12}$$
where $U_h : \mathcal{H}_{\mathrm{tot}} \to \mathcal{H}_{\mathrm{tot}}$ is the unitary action of $h \in G$, and $\tilde{}$ denotes the right-moving part, although we will discuss chiral theories with only left-moving parts after this Section 12.2. Recall that, in general, we need the phase modification (12.9) to make the anomalous phases trivial. Let $Z_0^{(\mathcal{T})}{}_g^h(\tau)$ denote the twisted partition function after the phase modification. The partition function of the orbifold $\mathcal{T}/G$ is then
$$Z^{(\mathcal{T}/G)}(\tau) = \frac{1}{|G|}\sum_{g,h\in G} Z_0^{(\mathcal{T})}{}_g^h(\tau). \tag{12.13}$$

In the rest of this Section 12.2, we will investigate the properties of the above concepts, and rewrite the orbifold partition function (12.13) so that the number of summands will be reduced.

### 12.2.2 Twisted Partition Functions As Class Functions

We have added the twisted sectors. In the field description, a field $\phi(\sigma^1, \sigma^2)$ belonging to the $g$-twisted sector $\mathcal{H}_g$ satisfies

$$\phi(\sigma^1 + 2\pi, \sigma^2) = g \cdot \phi(\sigma^1, \sigma^2). \tag{12.14}$$

Then we can see that the action $U_h : \mathcal{H}_{\text{tot}} \to \mathcal{H}_{\text{tot}}$ of $h \in G$ restricted to $\mathcal{H}_g$ is

$$U_h : \mathcal{H}_g \to \mathcal{H}_{hgh^{-1}}, \tag{12.15}$$

because if $\phi(\sigma^1, \sigma^2)$ belongs to the $g$-twisted sector, then $\phi'(\sigma^1, \sigma^2) := h \cdot \phi(\sigma^1, \sigma^2)$ belongs to the $hgh^{-1}$-twisted sector as

$$\phi'(\sigma^1 + 2\pi, \sigma^2) = hg\phi(\sigma^1, \sigma^2) = hgh^{-1}\phi'(\sigma^1, \sigma^2). \tag{12.16}$$

This fact leads to the following two consequences on the twisted partition functions.

(1) Since $U_h : \mathcal{H}_g \to \mathcal{H}_{hgh^{-1}}$,

$$Z_0^{(\mathcal{T})}{}_g^h(\tau) = \text{Tr}_{\mathcal{H}_g}[U_h q^{L_0 - \frac{c}{24}} \bar{q}^{\tilde{L}_0 - \frac{\tilde{c}}{24}}] = 0 \quad \text{if } hg \neq gh. \tag{12.17}$$

In other words, $\mathcal{H}_g$ is a representation of the centralizer $C_g := \{h \in G \mid hg = gh\}$ of $g$. (If $G$ is anomalous, then $\mathcal{H}_g$ is just a projective representation of $C_g$.)

(2) The representations of the chiral algebra $\mathcal{A}$ on $\mathcal{H}_g$ and $\mathcal{H}_{hgh^{-1}}$ are equivalent under $U_h$. That is, if we write the representation of $\mathcal{A}$ on $\mathcal{H}_g$ as $\rho_g : \mathcal{A} \to \text{End}(\mathcal{H}_g)$, then

$$U_h \rho_g(\bullet) U_h^{-1} = \rho_{hgh^{-1}}(\bullet). \tag{12.18}$$

Therefore,

$$\begin{aligned}
Z_0^{(\mathcal{T})}{}_{kgk^{-1}}^h(\tau) &= \text{Tr}_{\mathcal{H}_{kgk^{-1}}}[U_h q^{\rho_{kgk^{-1}}(L_0) - \frac{c}{24}} \bar{q}^{\tilde{\rho}_{kgk^{-1}}(\tilde{L}_0) - \frac{\tilde{c}}{24}}] && (12.19)\\
&= \text{Tr}_{\mathcal{H}_{kgk^{-1}}}[U_h U_k q^{\rho_g(L_0) - \frac{c}{24}} \bar{q}^{\tilde{\rho}_g(\tilde{L}_0) - \frac{\tilde{c}}{24}} U_k^{-1}] && (12.20)\\
&= \text{Tr}_{\mathcal{H}_g}[U_{k^{-1}} U_h U_k q^{\rho_g(L_0) - \frac{c}{24}} \bar{q}^{\tilde{\rho}_g(\tilde{L}_0) - \frac{\tilde{c}}{24}}] && (12.21)\\
&= Z_0^{(\mathcal{T})}{}_g^{k^{-1}hk}(\tau), && (12.22)
\end{aligned}$$

where we used the cyclicity of the trace in the third equation. In other words, the twisted partition functions are class functions in the sense that

$$Z_0^{(\mathcal{T})}{}_g^h(\tau) = Z_0^{(\mathcal{T})}{}_{kgk^{-1}}^{khk^{-1}}(\tau). \tag{12.23}$$

Combining (12.17) and (12.23), the twisted partition functions $Z_0^{(\mathcal{T})}{}_g^h(\tau)$ can be regarded as a function over the conjugacy classes of the commuting pairs

$$G_{\text{com}}^2/\sim := \{(g, h) \in G \times G \mid gh = hg\}/(g, h) \sim (kgk^{-1}, khk^{-1}). \tag{12.24}$$

Let $G/\sim := G/(g \sim kgk^{-1})$ denote the set of conjugacy classes, and $[g] \in G/\sim$ be the conjugacy class of $g \in G$. We introduce the notation $\sum_{[g]\in G/\sim}$ to represent the sum $\sum_{g\in\{g_1,\dots,g_r\}}$ over a complete set $\{g_1,\dots,g_r\}$ of representatives of the quotient $G/\sim = \{[g_1],\dots,[g_r]\}$. When this notation is used, the sum is supposed to be independent of the choice of the representatives. Now, using

$$\sum_{h\in G} Z_0^{(\mathcal{T})}{}^{h}_{kgk^{-1}}(\tau) = \sum_{h\in G} Z_0^{(\mathcal{T})}{}^{k^{-1}hk}_{g}(\tau) = \sum_{h\in G} Z_0^{(\mathcal{T})}{}^{h}_{g}(\tau), \tag{12.25}$$

the orbifold partition function is

$$\begin{aligned} Z^{(\mathcal{T}/G)}(\tau) &= \frac{1}{|G|}\sum_{g\in G}\sum_{h\in G} Z_0^{(\mathcal{T})}{}^{h}_{g}(\tau) && (12.26)\\ &= \frac{1}{|G|}\sum_{[g]\in G/\sim} |[g]| \sum_{h\in G} Z_0^{(\mathcal{T})}{}^{h}_{g}(\tau) && (12.27)\\ &= \sum_{[g]\in G/\sim} \frac{1}{|C_g|}\sum_{h\in G} Z_0^{(\mathcal{T})}{}^{h}_{g}(\tau) && (12.28)\\ &= \sum_{[g]\in G/\sim} \frac{1}{|C_g|}\sum_{h\in C_g} Z_0^{(\mathcal{T})}{}^{h}_{g}(\tau), && (12.29) \end{aligned}$$

where in the third equation, we used $|G| = |[g]||C_g|$.

We can further decompose the centralizer $C_g$ into conjugacy classes $[h]_{\subset C_g} \in C_g/\sim$. Then, since $Z_0^{(\mathcal{T})}{}^{khk^{-1}}_{g}(\tau) = Z_0^{(\mathcal{T})}{}^{h}_{k^{-1}gk}(\tau) = Z_0^{(\mathcal{T})}{}^{h}_{g}(\tau)$ for $k \in C_g$,

$$Z^{(\mathcal{T}/G)}(\tau) = \sum_{[g]\in G/\sim}\ \sum_{[h]_{\subset C_g}\in C_g/\sim} \frac{|[h]_{\subset C_g}|}{|C_g|} Z_0^{(\mathcal{T})}{}^{h}_{g}(\tau). \tag{12.30}$$

Finally, recall that $Z_{g,h}(\tau)$ can be regarded as a function over $G^2_{\mathrm{com}}/\sim$ defined in (12.24). The summands of (12.30) are labeled by[62] $[h]_{\subset C_g} \in \bigsqcup_{[g]\in G/\sim} C_g/\sim$. It is straightforward to show that the map $[h]_{\subset C_g} \mapsto [(g,h)]$ defines a well-defined[63] bijection $\bigsqcup_{[g]\in G/\sim} C_g/\sim \to G^2_{\mathrm{com}}/\sim$. Therefore, we have

$$Z^{(\mathcal{T}/G)}(\tau) = \sum_{[(g,h)]\in G^2_{\mathrm{com}}/\sim} \frac{|[h]_{\subset C_g}|}{|C_g|} Z_0^{(\mathcal{T})}{}^{h}_{g}(\tau). \tag{12.31}$$

[62] Although the set $\bigsqcup_{[g]\in G/\sim} C_g/\sim := (C_{g_1}/\sim) \sqcup \cdots \sqcup (C_{g_r}/\sim)$ depends on the choice of the representatives $\{g_1,\dots,g_r\}$ of $G/\sim$, the sum (12.30) is independent of such a choice.

[63] It is well-defined in the sense that if $[h]_{\subset C_g} = [h']_{\subset C_g}$ then $[(g,h)] = [(g,h')]$.

### 12.2.3 Modular Transformations of Twisted Partition Functions

Under the modular transformations, the twisted partition functions transform as (see e.g. [Pol07a, §8.5])

$$Z_0^{(\mathcal{T})}{}_g^h(\tau+1) = Z_0^{(\mathcal{T})}{}_g^{gh}(\tau), \tag{12.32}$$

$$Z_0^{(\mathcal{T})}{}_g^h(-\frac{1}{\tau}) = Z_0^{(\mathcal{T})}{}_h^{g^{-1}}(\tau), \tag{12.33}$$

if the phases caused by the gravitational anomaly in the modular transformations (1.4, 1.5) are trivial, that is, $2(\tilde{c}-c) \equiv 0 \bmod 24$; the gravitational anomaly phases cannot be canceled by the phase modification (12.9) to cancel the 't Hooft anomaly. It then follows that the orbifold partition function $Z^{(\mathcal{T}/G)}(\tau) = \sum_{g,h\in G} Z_0^{(\mathcal{T})}{}_g^h(\tau)$ is modular invariant.

**Modular transformations of fermionic twisted partition functions**

Assume that the theory $\mathcal{T}$ is fermionic and let $\langle(-1)^F\rangle$ denote the $\mathbb{Z}_2$ symmetry defining the fermion parity. We demand that the symmetry $G$ should commute with $(-1)^F$. We then define the twisted partition functions for $g, h \in G$ as

$$Z^{(\mathcal{T})}{}_{\mathrm{NS}g}^{\mathrm{NS}h}(\tau) := \mathrm{Tr}_{\mathcal{H}_{\mathrm{NS}g}}[U_h q^{L_0-\frac{c}{24}} \bar{q}^{\tilde{L}_0-\frac{\tilde{c}}{24}}], \tag{12.34}$$

$$Z^{(\mathcal{T})}{}_{\mathrm{NS}g}^{\mathrm{R}h}(\tau) := \mathrm{Tr}_{\mathcal{H}_{\mathrm{NS}g}}[(-1)^F U_h q^{L_0-\frac{c}{24}} \bar{q}^{\tilde{L}_0-\frac{\tilde{c}}{24}}], \tag{12.35}$$

$$Z^{(\mathcal{T})}{}_{\mathrm{R}g}^{\mathrm{NS}h}(\tau) := \mathrm{Tr}_{\mathcal{H}_{\mathrm{R}g}}[U_h q^{L_0-\frac{c}{24}} \bar{q}^{\tilde{L}_0-\frac{\tilde{c}}{24}}], \tag{12.36}$$

$$Z^{(\mathcal{T})}{}_{\mathrm{R}g}^{\mathrm{R}h}(\tau) := \mathrm{Tr}_{\mathcal{H}_{\mathrm{R}g}}[(-1)^F U_h q^{L_0-\frac{c}{24}} \bar{q}^{\tilde{L}_0-\frac{\tilde{c}}{24}}], \tag{12.37}$$

and again, let $Z_0^{(\mathcal{T})}{}_{\mathrm{X}g}^{\mathrm{Y}h}(\tau)$ ($\mathrm{X},\mathrm{Y} \in \{\mathrm{NS},\mathrm{R}\}$) denote the ones after the phase modifications (12.9), when $G$ is non-anomalous.

The modular transformations of these twisted partition functions can be derived by the path integral formalism on a torus as follows. We have seen in footnote 38 that, in the cylinder coordinates, the NS sector is anti-periodic and the R sector is periodic in the spatial direction. In addition, unlike the bosonic path integral, the fermionic path integral which is periodic in temporal direction corresponds to the trace with $(-1)^F$ inserted, and the anti-periodic one corresponds to the usual trace [Pol07a, Appendix A.2]. As a result, we may regard NS and R in (12.34)–(12.37) as the nontrivial element and the identity element of $\mathbb{Z}_2 = \langle(-1)^F\rangle$, respectively. This discussion leads to the modular transformations of the fermionic twisted partition functions as

$$Z_0^{(\mathcal{T})}{}_{\mathrm{NS}g}^{\mathrm{NS}h}(\tau+1) = Z_0^{(\mathcal{T})}{}_{\mathrm{NS}g}^{\mathrm{R}gh}(\tau), \qquad Z_0^{(\mathcal{T})}{}_{\mathrm{NS}g}^{\mathrm{NS}h}(-\frac{1}{\tau}) = Z_0^{(\mathcal{T})}{}_{\mathrm{NS}h}^{\mathrm{NS}g^{-1}}(\tau), \tag{12.38}$$

$$Z_0^{(\mathcal{T})}{}_{\mathrm{NS}g}^{\mathrm{R}h}(\tau+1) = Z_0^{(\mathcal{T})}{}_{\mathrm{NS}g}^{\mathrm{NS}gh}(\tau), \qquad Z_0^{(\mathcal{T})}{}_{\mathrm{NS}g}^{\mathrm{R}h}(-\frac{1}{\tau}) = Z_0^{(\mathcal{T})}{}_{\mathrm{R}h}^{\mathrm{NS}g^{-1}}(\tau), \tag{12.39}$$

$$Z_0^{(\mathcal{T})}{}_{\mathrm{R}g}^{\mathrm{NS}h}(\tau+1) = Z_0^{(\mathcal{T})}{}_{\mathrm{R}g}^{\mathrm{NS}gh}(\tau), \qquad Z_0^{(\mathcal{T})}{}_{\mathrm{R}g}^{\mathrm{NS}h}(-\frac{1}{\tau}) = Z_0^{(\mathcal{T})}{}_{\mathrm{NS}h}^{\mathrm{R}g^{-1}}(\tau), \tag{12.40}$$

$$Z_0^{(\mathcal{T})}{}_{\mathrm{R}g}^{\mathrm{R}h}(\tau+1) = Z_0^{(\mathcal{T})}{}_{\mathrm{R}g}^{\mathrm{R}gh}(\tau), \qquad Z_0^{(\mathcal{T})}{}_{\mathrm{R}g}^{\mathrm{R}h}(-\frac{1}{\tau}) = Z_0^{(\mathcal{T})}{}_{\mathrm{R}h}^{\mathrm{R}g^{-1}}(\tau), \tag{12.41}$$

if the phases caused by the gravitational anomaly are trivial. More schematically,

$$ {}_S\circlearrowleft Z_0^{(\mathcal{T})\,\mathrm{NS}}{}_{\mathrm{NS}} \overset{T}{\leftrightarrow} Z_0^{(\mathcal{T})\,\mathrm{R}}{}_{\mathrm{NS}} \overset{S}{\leftrightarrow} Z_0^{(\mathcal{T})\,\mathrm{NS}}{}_{\mathrm{R}} \circlearrowleft {}_T \quad , \quad Z_0^{(\mathcal{T})\,\mathrm{R}}{}_{\mathrm{R}} \circlearrowleft {}_{T,S} \quad . \tag{12.42} $$

If $G$ contains $(-1)^F$, then we should have

$$ Z_0^{(\mathcal{T})\,\mathrm{Y}(-1)^F h}{}_{\mathrm{X}g}(\tau) = Z_0^{(\mathcal{T})\,\overline{\mathrm{Y}}h}{}_{\mathrm{X}g}(\tau), \tag{12.43} $$

from the definitions (12.34)–(12.37) of the twisted partition functions, and

$$ Z_0^{(\mathcal{T})\,\mathrm{Y}h}{}_{\mathrm{X}(-1)^F g}(\tau) = Z_0^{(\mathcal{T})\,\mathrm{Y}h}{}_{\overline{\mathrm{X}}g}(\tau), \tag{12.44} $$

from the discussion above. Here, we introduced the notation

$$ \overline{\mathrm{NS}} := \mathrm{R}, \quad \overline{\mathrm{R}} := \mathrm{NS}. \tag{12.45} $$

**$\mathrm{GL}(2,\mathbb{Z})$ transformations of twisted partition functions**

By adding the action of $P : \tau \mapsto -\bar{\tau}$ to the T and S transformations, the $\mathrm{SL}(2,\mathbb{Z})$-action enlarges to the $\mathrm{GL}(2,\mathbb{Z})$-action. Under the P transformation, the twisted partition function transforms as

$$ \begin{aligned} Z_0^{(\mathcal{T})\,h}{}_{g}(-\bar{\tau}) &= \mathrm{Tr}_{\mathcal{H}_g} U_h e^{-2\pi\sqrt{-1}\bar{\tau}((L_0-\frac{c}{24})-(\tilde{L}_0-\frac{\tilde{c}}{24}))} && (12.46)\\ &= \left[\mathrm{Tr}_{\mathcal{H}_g} e^{2\pi\sqrt{-1}\tau((L_0^\dagger-\frac{c}{24})-(\tilde{L}_0^\dagger-\frac{\tilde{c}}{24}))} U_h^\dagger\right]^* && (12.47)\\ &= \left[\mathrm{Tr}_{\mathcal{H}_g} U_{h^{-1}} e^{2\pi\sqrt{-1}\tau((L_0-\frac{c}{24})-(\tilde{L}_0-\frac{\tilde{c}}{24}))}\right]^* && (12.48)\\ &= \left[Z_0^{(\mathcal{T})\,h^{-1}}{}_{g}(\tau)\right]^*, && (12.49) \end{aligned} $$

where in the second equation, we simply used $\mathrm{Tr}M = [\mathrm{Tr}M^\dagger]^*$, and in the third equation, we used the fact that $L_0$ is Hermitian and $U_h$ is unitary. It then follows that the orbifold partition function $Z^{(\mathcal{T}/G)}(\tau) = \sum_{g,h\in G} Z_0^{(\mathcal{T})\,h}{}_{g}(\tau)$ satisfies $Z^{(\mathcal{T}/G)}(-\bar{\tau}) = Z^{(\mathcal{T}/G)}(\tau)^*$. Similarly, for a fermionic twisted partition functions $Z_0^{(\mathcal{T})\,\mathrm{Y}h}{}_{\mathrm{X}g}(\tau)$ ($\mathrm{X},\mathrm{Y} \in \{\mathrm{NS},\mathrm{R}\}$), we have $Z_0^{(\mathcal{T})\,\mathrm{Y}h}{}_{\mathrm{X}g}(-\bar{\tau}) = [Z_0^{(\mathcal{T})\,\mathrm{Y}h^{-1}}{}_{\mathrm{X}g}(\tau)]^*$.

### 12.2.4 The Case of Constant Twisted Partition Functions

Since the supersymmetry of Duncan's module $V^{f\natural}$ is preserved by its $\mathrm{Co}_1$ symmetry, the $(-1)^F$-inserted R twisted partition function $Z_0^{((V^{f\natural})^{\otimes 24})\,\mathrm{R}h}{}_{\mathrm{R}g}(\tau) = \mathrm{Tr}_{(V_{\mathrm{tw},g}^{f\natural})^{\otimes 24}}[(-1)^F U_h q^{L_0-\frac{24\times 12}{24}}]$ for any $g,h \in \mathrm{Co}_1$ is a constant (see the discussion of the Witten index in Section 3.4), which is the character of $(-1)^F U_h$ represented on the Ramond vacuum states (the weight-$\frac{24\times 12}{24}$ subspace) of the sector $(V_{\mathrm{tw},g}^{f\natural})^{\otimes 24}$. Note that the $(-1)^F$-inserted R twisted partition functions are closed under the modular transformations (12.41).

In such a case where the twisted partition functions $Z_0^{(\mathcal{T})\,h}{}_{g}(\tau)$ are constants, the modular properties (12.32, 12.33) of the twisted partition functions show that the values of $Z_0^{(\mathcal{T})\,h}{}_{g}$ are the same

for $(g,h)$ in the same $\mathrm{SL}(2,\mathbb{Z})$-orbit of $G\times G$, if we introduce the $\mathrm{SL}(2,\mathbb{Z})$-action on $G\times G$ as $T:(g,h)\mapsto(g,gh)$ and $S:(g,h)\mapsto(h,g^{-1})$.

In this case, we can further reduce the expression (12.31) of the orbifold partition function into the sum of $\mathrm{SL}(2,\mathbb{Z})$-orbits. First, it is straightforward to see that the $\mathrm{SL}(2,\mathbb{Z})$-action on $G\times G$ induces the well-defined action on $G^2_{\mathrm{com}}/\!\sim$. In addition, the factor $\frac{|[h]_{\subset C_g}|}{|C_g|}$ is also constant over an $\mathrm{SL}(2,\mathbb{Z})$-orbit of $G^2_{\mathrm{com}}/\!\sim$, which can be checked as follows.

- The invariance under the T transformation: There exists a bijection $g:[h]_{\subset C_g}\to[gh]_{\subset C_g}$, so it is obvious that $\frac{|[h]_{\subset C_g}|}{|C_g|}=\frac{|[gh]_{\subset C_g}|}{|C_g|}$.
- The invariance under the S transformation: Recall that the well-known equation $|G|=|[g]||C_g|$ is the consequence of the orbit-stabilizer theorem applied to the conjugation. Similarly, we have

$$|C_g|=|[h]_{\subset C_g}||(C_h)_{\subset C_g}|, \tag{12.50}$$

where

$$\begin{aligned}(C_h)_{\subset C_g}&:=\{x\in C_g\mid xh=hx\} &(12.51)\\ &=\{x\in G\mid xg=gx,xh=hx\} &(12.52)\\ &=(C_{g^{-1}})_{\subset C_h}. &(12.53)\end{aligned}$$

Therefore, under the S transformation, we have

$$\frac{|[h]_{\subset C_g}|}{|C_g|}=\frac{1}{|(C_h)_{\subset C_g}|}=\frac{1}{|(C_{g^{-1}})_{\subset C_h}|}=\frac{|[g^{-1}]_{\subset C_h}|}{|C_h|}. \tag{12.54}$$

As a result, the orbit partition function (12.31) can be rewritten as

$$Z^{(\mathcal{T}/G)}(\tau)=\sum_{[[(g,h)]]\in(G^2_{\mathrm{com}}/\sim)/\mathrm{SL}(2,\mathbb{Z})}|\mathrm{SL}(2,\mathbb{Z})\cdot[(g,h)]|\frac{|[h]_{\subset C_g}|}{|C_g|}Z_0^{(\mathcal{T})^h_g}(\tau), \tag{12.55}$$

where $\mathrm{SL}(2,\mathbb{Z})\cdot[(g,h)]$ is the $\mathrm{SL}(2,\mathbb{Z})$-orbit of $[(g,h)]\in G^2_{\mathrm{com}}/\!\sim$.

We may further consider the action of $P:(g,h)\mapsto(g,h^{-1})$. This action is again well-defined on $G^2_{\mathrm{com}}/\!\sim$, and $\frac{|[h]_{\subset C_g}|}{|C_g|}=\frac{|[h^{-1}]_{\subset C_g}|}{|C_g|}$. In addition, for any $[(g,h)]\in G^2_{\mathrm{com}}/\!\sim$, the action $P:G^2_{\mathrm{com}}/\!\sim\to G^2_{\mathrm{com}}/\!\sim$ can be restricted to $P:\mathrm{SL}(2,\mathbb{Z})\cdot[(g,h)]\to\mathrm{SL}(2,\mathbb{Z})\cdot[(g,h^{-1})]$, because $PT=T^{-1}P$ and $PS=S^{-1}T$. Since $P:\mathrm{SL}(2,\mathbb{Z})\cdot[(g,h^{-1})]\to\mathrm{SL}(2,\mathbb{Z})\cdot[(g,h)]$ is its inverse, we have $|\mathrm{SL}(2,\mathbb{Z})\cdot[(g,h)]|=|\mathrm{SL}(2,\mathbb{Z})\cdot[(g,h^{-1})]|$. Recalling the P transformation of the twisted partition functions (12.49), there are two possibilities as follows.

- If the $\mathrm{SL}(2,\mathbb{Z})$-orbit $\mathrm{SL}(2,\mathbb{Z})\cdot[(g,h)]$ is mapped to itself under $P$, that is, if $[(g,h^{-1})]\in\mathrm{SL}(2,\mathbb{Z})\cdot[(g,h)]$, then $Z_0^{(\mathcal{T})^h_g}(\tau)$ is a real number.

- If the $\mathrm{SL}(2,\mathbb{Z})$-orbit $\mathrm{SL}(2,\mathbb{Z})\cdot[(g,h)]$ is mapped to another orbit $\mathrm{SL}(2,\mathbb{Z})\cdot[(g,h^{-1})]$ under $P$, that is, if $[(g,h^{-1})]\notin \mathrm{SL}(2,\mathbb{Z})\cdot[(g,h)]$, then $Z_0^{(\mathcal{T})^h_g}(\tau)$ and $Z_0^{(\mathcal{T})^{h^{-1}}_g}(\tau)$ are complex conjugate to each other.

As a result, the orbifold partition function (12.55) can be further rewritten as

$$\begin{aligned} Z^{(\mathcal{T}/G)}(\tau) = &\sum_{\substack{[[(g,h)]]\in(G^2_{\mathrm{com}}/\sim)/\mathrm{GL}(2,\mathbb{Z})\\ [(g,h^{-1})]\in\mathrm{SL}(2,\mathbb{Z})\cdot[(g,h)]}} |\mathrm{SL}(2,\mathbb{Z})\cdot[(g,h)]|\frac{|[h]_{\subset C_g}|}{|C_g|}Z_0^{(\mathcal{T})^h_g}(\tau)\\ &+\sum_{\substack{[[(g,h)]]\in(G^2_{\mathrm{com}}/\sim)/\mathrm{GL}(2,\mathbb{Z})\\ [(g,h^{-1})]\notin\mathrm{SL}(2,\mathbb{Z})\cdot[(g,h)]}} |\mathrm{SL}(2,\mathbb{Z})\cdot[(g,h)]|\frac{|[h]_{\subset C_g}|}{|C_g|}2\mathrm{Re}[Z_0^{(\mathcal{T})^h_g}(\tau)] \qquad (12.56)\\ = &\sum_{[[(g,h)]]\in(G^2_{\mathrm{com}}/\sim)/\mathrm{GL}(2,\mathbb{Z})} |\mathrm{GL}(2,\mathbb{Z})\cdot[(g,h)]|\frac{|[h]_{\subset C_g}|}{|C_g|}\mathrm{Re}[Z_0^{(\mathcal{T})^h_g}(\tau)], \qquad (12.57)\end{aligned}$$

where $\mathrm{Re}$ denotes the real part of the complex number.

Consider the situation where we can only compute the twisted partition functions up to $\mathrm{U}(1)$ phases, that is, we should multiply the computed twisted partition functions $Z^{(\mathcal{T})^h_g}(\tau)$ by some phases as in (12.9) to cancel the anomalous phases and obtain $Z_0^{(\mathcal{T})^h_g}(\tau)$. Let $\beta_{\mathrm{anom}}(g,h)\in\mathrm{U}(1)$ denote the phase we multiplied $Z^{(\mathcal{T})^h_g}(\tau)$ by to cancel the anomalous phases, which we call the *anomaly-canceling phase* in this thesis. As an equation, we should have

$$Z_0^{(\mathcal{T})^h_g}(\tau)=\beta_{\mathrm{anom}}(g,h)Z^{(\mathcal{T})^h_g}(\tau). \qquad (12.58)$$

Thus, the orbifold partition function (12.57) is

$$Z^{(\mathcal{T}/G)}(\tau) = \sum_{[[(g,h)]]\in(G^2_{\mathrm{com}}/\sim)/\mathrm{GL}(2,\mathbb{Z})} |\mathrm{GL}(2,\mathbb{Z})\cdot[(g,h)]|\frac{|[h]_{\subset C_g}|}{|C_g|}\mathrm{Re}[\beta_{\mathrm{anom}}(g,h)Z^{(\mathcal{T})^h_g}(\tau)]. \qquad (12.59)$$

In addition, if the $\mathrm{SL}(2,\mathbb{Z})$-orbit $\mathrm{SL}(2,\mathbb{Z})\cdot[(g,h)]$ is mapped to itself under $P$, then $\beta_{\mathrm{anom}}(g,h)\in\mathrm{U}(1)$ can be determined up to a sign by the condition that $\beta_{\mathrm{anom}}(g,h)Z^{(\mathcal{T})^h_g}(\tau)$ is a real number. If there are discrete torsions, once we find a set of anomaly-canceling phases $\{\beta_{\mathrm{anom}}(g,h)\}_{[[(g,h)]]\in(G^2_{\mathrm{com}}/\sim)/\mathrm{GL}(2,\mathbb{Z})}$, then the other sets can be systematically constructed as in Section 12.1.3.

In the case of the Conway orbifold of Duncan's module, we will compute the twisted partition functions of Duncan's module based on the twisted partition functions of free fermions, but there is no guarantee that the phases of the resulting twisted partition functions are adjusted so that the anomalous phases are trivial. So we have to use the expression (12.59) to obtain the correct

orbifold partition function as

$$
\begin{aligned}
&Z^{((V^{f\natural})^{\otimes 24}/\mathrm{Co}_1)}{}^{\mathrm{R}}_{\mathrm{R}}(\tau) \\
&= \sum_{\substack{[[(g,h)]]\in(G_{\mathrm{com}}^2/\sim)/\mathrm{GL}(2,\mathbb{Z}) \\ \text{for } G=\mathrm{Co}_1}} |\mathrm{GL}(2,\mathbb{Z})\cdot[(g,h)]|\frac{|[h]_{\subset C_g}|}{|C_g|}\mathrm{Re}\Big[\beta_{\mathrm{anom}}(g,h)\big(Z^{(V^{f\natural})}{}^{\mathrm{R}h}_{\mathrm{R}g}(\tau)\big)^{24}\Big]. \quad (12.60)
\end{aligned}
$$

Here, we used

$$
Z^{((V^{f\natural})^{\otimes 24})}{}^{\mathrm{R}h}_{\mathrm{R}g}(\tau) = \big(Z^{(V^{f\natural})}{}^{\mathrm{R}h}_{\mathrm{R}g}(\tau)\big)^{24}, \quad (12.61)
$$

because we are considering the diagonal $\mathrm{Co}_1$-action. Note that the $\mathrm{Co}_1$-action on a single Duncan's module $V^{f\natural}$ is anomalous, so there is no concept like '$Z_0^{(V^{f\natural})}{}^{\mathrm{R}h}_{\mathrm{R}g}(\tau)$.' We only have

$$
Z_0^{((V^{f\natural})^{\otimes 24})}{}^{\mathrm{R}h}_{\mathrm{R}g}(\tau) = \beta_{\mathrm{anom}}(g,h)Z^{((V^{f\natural})^{\otimes 24})}{}^{\mathrm{R}h}_{\mathrm{R}g}(\tau). \quad (12.62)
$$

In addition, as explained in Section 12.1.3, the phases of non-zero twisted partition functions $Z_0^{((V^{f\natural})^{\otimes 24})}{}^{\mathrm{R}h}_{\mathrm{R}g}(\tau) \neq 0$ are not affected by the discrete torsion, so the correct set $\{\beta_{\mathrm{anom}}(g,h) \mid Z^{((V^{f\natural})^{\otimes 24})}{}^{\mathrm{R}h}_{\mathrm{R}g}(\tau) \neq 0\}_{[[(g,h)]]\in(G_{\mathrm{com}}^2/\sim)/\mathrm{GL}(2,\mathbb{Z})}$ of such anomaly-canceling phases is unique, and hence this orbifold partition function (12.60) is also unique.

As a result, we see that there are the following three steps to compute the Conway orbifold partition function (12.60).

Step 1. Determine $G_{\mathrm{com}}^2/\sim$ and its $\mathrm{SL}(2,\mathbb{Z})$- and $\mathrm{GL}(2,\mathbb{Z})$-orbit decompositions for $G=\mathrm{Co}_1$.

Step 2. Compute the twisted partition functions $Z^{(V^{f\natural})}{}^{\mathrm{R}h}_{\mathrm{R}g}(\tau)$.

Step 3. Determine the anomaly-canceling phases $\beta_{\mathrm{anom}}(g,h)$ for all $\mathrm{GL}(2,\mathbb{Z})$-orbits of $G_{\mathrm{com}}^2/\sim$.

# 13 Twisted Partition Functions of Free Fermions

Recall that Duncan's module is constructed from the theory of 24 real free chiral fermions through the procedure similar to $\mathbb{Z}_2$-orbifolding, and its $\mathrm{Co}_1$ symmetry originates from $\mathrm{Co}_0 \subset \mathrm{Spin}(24)$. Using this fact, we will calculate the twisted partition functions of Duncan's module based on those of free chiral fermions twisted by $\mathrm{Spin}(n)$, in the later Section 14.2.1. As preliminaries, in this Section 13, we calculate the twisted partition functions of free chiral fermions. We begin with the case of one complex fermion in Section 13.1, and then move to the case of multiple complex fermions in Section 13.2, both with $\mathrm{SO}(n)$-twists. We also calculate some transformation properties of them in Section 13.3, and lastly discuss how to regard them as twisted by $\mathrm{Spin}(n)$ in Section 13.4.

## 13.1 One Complex Fermion

Let us begin with two real fermions $\psi_1(z), \psi_2(z)$, which are equivalent to one complex fermion $\Psi(z)$ as

$$\Psi(z) = \frac{1}{\sqrt{2}}(\psi_1(z) + \sqrt{-1}\psi_2(z)), \quad \overline{\Psi}(z) = \frac{1}{\sqrt{2}}(\psi_1(z) - \sqrt{-1}\psi_2(z)). \tag{13.1}$$

In order to make the $\mathrm{SO}(2)$ symmetry on them consistent with the $\mathrm{Spin}(n)$-action on $A(\mathfrak{a})$ and $A(\mathfrak{a})_{\mathrm{tw}}$ introduced in Section 6.1.2, we should consider the action of $\begin{pmatrix} \cos 2\pi\lambda & -\sin 2\pi\lambda \\ \sin 2\pi\lambda & \cos 2\pi\lambda \end{pmatrix} \in \mathrm{SO}(2)$ as

$$\begin{pmatrix} \psi_1(z) & \psi_2(z) \end{pmatrix} \begin{pmatrix} c^1 \\ c^2 \end{pmatrix} \mapsto \begin{pmatrix} \psi_1(z) & \psi_2(z) \end{pmatrix} \begin{pmatrix} \cos 2\pi\lambda & -\sin 2\pi\lambda \\ \sin 2\pi\lambda & \cos 2\pi\lambda \end{pmatrix} \begin{pmatrix} c^1 \\ c^2 \end{pmatrix}, \tag{13.2}$$

or equivalently,

$$\Psi(z) \mapsto e^{-2\pi\sqrt{-1}\lambda}\Psi(z), \quad \overline{\Psi}(z) \mapsto e^{2\pi\sqrt{-1}\lambda}\overline{\Psi}(z). \tag{13.3}$$

However, in physics, for example in [Pol07b], it is natural to regard $\psi_i(z)$ as the coordinates, and consider the action of $\begin{pmatrix} \cos 2\pi\lambda & -\sin 2\pi\lambda \\ \sin 2\pi\lambda & \cos 2\pi\lambda \end{pmatrix} \in \mathrm{SO}(2)$ as

$$\begin{pmatrix} \psi_1 \\ \psi_2 \end{pmatrix}(z) \mapsto \begin{pmatrix} \cos 2\pi\lambda & -\sin 2\pi\lambda \\ \sin 2\pi\lambda & \cos 2\pi\lambda \end{pmatrix} \begin{pmatrix} \psi_1 \\ \psi_2 \end{pmatrix}(z), \tag{13.4}$$

or equivalently,

$$\Psi(z) \mapsto e^{2\pi\sqrt{-1}\lambda}\Psi(z), \quad \overline{\Psi}(z) \mapsto e^{-2\pi\sqrt{-1}\lambda}\overline{\Psi}(z). \tag{13.5}$$

We adopt the former convention (13.2), and introduce the notation $\lambda' := -\lambda$ for a moment (until (13.29)), which enables us to formally follow the convention of (13.4) and [Pol07b]. We will

compute the twisted partition functions, following [Pol07b, §10.7]. Note that this $\mathrm{SO}(2)$ symmetry is anomalous, so there is no choice of the phases of twisted partition functions such that the anomalous phases are trivial. This $\mathrm{SO}(2)$ can act on each sector as projective representations, and it coincides with the action of $\mathrm{Spin}(2)$. The $\mathrm{Spin}(2)$ acts as a genuine representation at least on the NS and R sectors but still can be projective on more general twisted sectors, as already mentioned in Section 12.1.4.

For the sector twisted by $g = \begin{pmatrix} \cos 2\pi\lambda_g & -\sin 2\pi\lambda_g \\ \sin 2\pi\lambda_g & \cos 2\pi\lambda_g \end{pmatrix} \in \mathrm{SO}(2)$ as

$$\Psi(e^{2\pi\sqrt{-1}}z) = e^{2\pi\sqrt{-1}\lambda'_g}\Psi(z), \quad \overline{\Psi}(e^{2\pi\sqrt{-1}}z) = e^{-2\pi\sqrt{-1}\lambda'_g}\overline{\Psi}(z), \tag{13.6}$$

we introduce the mode expansion as

$$\Psi(z) = \sum_{r\in\mathbb{Z}+\frac{1}{2}-\lambda'_g} \Psi_r z^{-r-1/2}, \quad \overline{\Psi}(z) = \sum_{s\in\mathbb{Z}-\frac{1}{2}+\lambda'_g} \overline{\Psi}_s z^{-s-1/2}. \tag{13.7}$$

These modes satisfy the anticommutation relations

$$\{\Psi_r, \Psi_{r'}\} = \{\overline{\Psi}_s, \overline{\Psi}_{s'}\} = 0, \quad \{\Psi_r, \overline{\Psi}_s\} = \delta_{r+s,0}. \tag{13.8}$$

We assume $\lambda'_g \in [-\frac{1}{2}, \frac{1}{2})$. A basis of the twisted sector $\mathcal{H}_g^{(1)}$, where the superscript $^{(1)}$ denotes the number of complex fermions, consists of the states in the form of

$$\left(\prod_{m=1}^{\infty} (\text{one of } \{1, \Psi_{-m+\frac{1}{2}-\lambda'_g}, \overline{\Psi}_{-m+\frac{1}{2}+\lambda'_g}, \Psi_{-m+\frac{1}{2}-\lambda'_g}\overline{\Psi}_{-m+\frac{1}{2}+\lambda'_g}\})\right)|0\rangle_g, \tag{13.9}$$

where $|0\rangle_g$ is the ground state satisfying

$$\Psi_{-m+\frac{1}{2}-\lambda'_g}|0\rangle_g = \overline{\Psi}_{-m+\frac{1}{2}+\lambda'_g}|0\rangle_g = 0 \quad \text{for } m \le 0. \tag{13.10}$$

We can observe that the NS sector $A(\mathfrak{a})$ and the R sector $A(\mathfrak{a})_{\mathrm{tw}}$ with $\dim \mathfrak{a} = 2$ constructed in Section 6.1.1 coincide with the sectors $\mathcal{H}_g^{(1)}$ for $\lambda'_g = 0$ and $\lambda'_g = -\frac{1}{2}$, respectively. In particular, in the R sector $\lambda'_g = -\frac{1}{2}$, we have $\overline{\Psi}_0|0\rangle_g = 0$ but $\Psi_0|0\rangle_g \neq 0$ as expected.

The energy-momentum tensor is

$$T(z) = -\frac{1}{2} : (\Psi\partial\overline{\Psi}(z) + \overline{\Psi}\partial\Psi(z)) : \tag{13.11}$$

$$= \sum_{m\in\mathbb{Z}} L_m z^{-m-2}, \tag{13.12}$$

and therefore

$$L_m = \sum_{r\in\mathbb{Z}+\frac{1}{2}-\lambda'_g} (r - \frac{m}{2})^{\circ}_{\circ}\overline{\Psi}_{m-r}\Psi_{r^{\circ}_{\circ}} + a\delta_{m,0}, \tag{13.13}$$

where the creation-annihilation normal ordering ${}^{\circ}_{\circ}\overline{\Psi}_{m-r}\Psi_{r}{}^{\circ}_{\circ}$ is defined as

$$ {}^{\circ}_{\circ}\overline{\Psi}_{m-r}\Psi_{r}{}^{\circ}_{\circ} := \begin{cases} -\Psi_r\overline{\Psi}_{m-r} & (r \le \min\{0,m\}), \\ \overline{\Psi}_{m-r}\Psi_r & (\text{otherwise}), \end{cases} \tag{13.14} $$

and $a$ is the so-called normal-ordering constant. It can be determined as follows.

$$ \begin{aligned} a|0\rangle_g &= L_0|0\rangle_g && (13.15)\\ &= \frac{1}{2}(L_1L_{-1} - L_{-1}L_1)|0\rangle_g && (13.16)\\ &= \frac{1}{2}L_1L_{-1}|0\rangle_g && (13.17)\\ &= \frac{1}{2}\left(\sum_{r\in\mathbb{Z}+\frac{1}{2}-\lambda'_g}(r-\frac{1}{2}){}^{\circ}_{\circ}\overline{\Psi}_{1-r}\Psi_{r}{}^{\circ}_{\circ}\right)\left(\sum_{r\in\mathbb{Z}+\frac{1}{2}-\lambda'_g}(r+\frac{1}{2}){}^{\circ}_{\circ}\overline{\Psi}_{-1-r}\Psi_{r}{}^{\circ}_{\circ}\right)|0\rangle_g && (13.18)\\ &= \frac{1}{2}\left(\sum_{r\in\mathbb{Z}+\frac{1}{2}-\lambda'_g}(r-\frac{1}{2}){}^{\circ}_{\circ}\overline{\Psi}_{1-r}\Psi_{r}{}^{\circ}_{\circ}\right)(-\lambda'_g)\overline{\Psi}_{-\frac{1}{2}+\lambda'_g}\Psi_{-\frac{1}{2}-\lambda'_g}|0\rangle_g && (13.19)\\ &= \frac{(\lambda'_g)^2}{2}|0\rangle_g, && (13.20) \end{aligned} $$

where the second equation follows from the Virasoro algebra (4.12), the fourth equation is from (13.13), and the other equations follow because $|0\rangle_g$ is annihilated as in (13.10). As a result,

$$ L_0 = \sum_{r\in\mathbb{Z}+\frac{1}{2}-\lambda'_g} r{}^{\circ}_{\circ}\overline{\Psi}_{-r}\Psi_{r}{}^{\circ}_{\circ} + \frac{(\lambda'_g)^2}{2}. \tag{13.21} $$

As already mentioned, there are no canonical $\mathrm{U}(1)$ phases of the twisted partition functions. We will adjust them later in Section 13.2. For now, we follow [Pol07b, §10.7] and adopt the $\mathrm{U}(1)$ phases naturally arising from the assumption

$$ U_h|0\rangle_g = e^{2\pi\sqrt{-1}\lambda'_g\lambda'_h}|0\rangle_g, \tag{13.22} $$

for $h = \begin{pmatrix} \cos 2\pi\lambda_h & -\sin 2\pi\lambda_h \\ \sin 2\pi\lambda_h & \cos 2\pi\lambda_h \end{pmatrix} \in \mathrm{SO}(2)$. This phase is consistent with the action of $\mathfrak{z} = (-1)^F$ on the NS sector $\mathfrak{z}|0\rangle = |0\rangle$ and on the R sector $\mathfrak{z}|0\rangle_{\mathrm{R}} = \sqrt{-1}|0\rangle_{\mathrm{R}}$ for $\dim\mathfrak{a} = 2$, if we identify $|0\rangle_g|_{\lambda'_g=0}$ with $|0\rangle$, $|0\rangle_g|_{\lambda'_g=-\frac{1}{2}}$ with $|0\rangle_{\mathrm{R}}$, and $U_h|_{\lambda'_h=-\frac{1}{2}}$ with $\mathfrak{z}$.

Then, on the subspace $\mathcal{H}^{(1)}_g|_m$ of $\mathcal{H}^{(1)}_g$ spanned by

$$ |0\rangle_g, \quad \Psi_{-m+\frac{1}{2}-\lambda'_g}|0\rangle_g, \quad \overline{\Psi}_{-m+\frac{1}{2}+\lambda'_g}|0\rangle_g, \quad \Psi_{-m+\frac{1}{2}-\lambda'_g}\overline{\Psi}_{-m+\frac{1}{2}+\lambda'_g}|0\rangle_g, \tag{13.23} $$

the twisted partition function with the temporal twist $h$ is

$$\mathrm{Tr}_{\mathcal{H}_g^{(1)}|_m}[U_h q^{L_0-\frac{1}{24}}] \tag{13.24}$$

$$= q^{\frac{(\lambda'_g)^2}{2}-\frac{1}{24}}\mathrm{Tr}\left[e^{2\pi\sqrt{-1}\lambda'_g\lambda'_h}\begin{pmatrix}1 & & & \\ & e^{2\pi\sqrt{-1}\lambda'_h} & & \\ & & e^{-2\pi\sqrt{-1}\lambda'_h} & \\ & & & 1\end{pmatrix}\begin{pmatrix}1 & & & \\ & q^{m-\frac{1}{2}+\lambda'_g} & & \\ & & q^{m-\frac{1}{2}-\lambda'_g} & \\ & & & q^{m-\frac{1}{2}+\lambda'_g}q^{m-\frac{1}{2}-\lambda'_g}\end{pmatrix}\right] \tag{13.25}$$

$$= q^{\frac{(\lambda'_g)^2}{2}-\frac{1}{24}}(1+e^{2\pi\sqrt{-1}\lambda'_h}q^{m-\frac{1}{2}+\lambda'_g})(1+e^{-2\pi\sqrt{-1}\lambda'_h}q^{m-\frac{1}{2}-\lambda'_g}). \tag{13.26}$$

Recalling the structure (13.9) of the whole twisted sector $\mathcal{H}_g^{(1)}$, we obtain

$$\mathrm{Tr}_{\mathcal{H}_g^{(1)}}[U_h q^{L_0-\frac{1}{24}}] = e^{2\pi\sqrt{-1}\lambda'_g\lambda'_h}q^{\frac{(\lambda'_g)^2}{2}-\frac{1}{24}}\prod_{m=1}^{\infty}(1+e^{2\pi\sqrt{-1}\lambda'_h}q^{m-\frac{1}{2}+\lambda'_g})(1+e^{-2\pi\sqrt{-1}\lambda'_h}q^{m-\frac{1}{2}-\lambda'_g}) \tag{13.27}$$

$$= \frac{1}{\eta(\tau)}\theta\begin{bmatrix}\lambda'_g\\ \lambda'_h\end{bmatrix}(\tau,0), \tag{13.28}$$

$$= \frac{1}{\eta(\tau)}\theta\begin{bmatrix}\lambda_g\\ \lambda_h\end{bmatrix}(\tau,0), \tag{13.29}$$

where $\theta\begin{bmatrix}a\\ b\end{bmatrix}(\tau,z)$ is the theta function with characteristics, and $\eta(\tau)$ is the Dedekind eta function, whose notations are summarized in Appendix E. In the last equation, we used (E.17)

$$\theta\begin{bmatrix}-\lambda_g\\ -\lambda_h\end{bmatrix}(\tau,0) = \theta\begin{bmatrix}\lambda_g\\ \lambda_h\end{bmatrix}(\tau,0), \tag{13.30}$$

and therefore we observe that the two conventions (13.2) and (13.4) lead to the same twisted partition functions.

So far, we assumed $\lambda'_g \in [-\frac{1}{2},\frac{1}{2})$. However, since (E.15)

$$\theta\begin{bmatrix}\lambda_g+l\\ \lambda_h+m\end{bmatrix}(\tau,0) = e^{2\pi\sqrt{-1}\lambda_g m}\theta\begin{bmatrix}\lambda_g\\ \lambda_h\end{bmatrix}(\tau,0)\quad(m,l\in\mathbb{Z}), \tag{13.31}$$

under a fixed $\lambda_h$, we may replace $\lambda_g$ with $\lambda_g+\mathbb{Z}$. So we no longer restrict the range of $\lambda_g$. Although the replacement of $\lambda_h$ with $\lambda_h+\mathbb{Z}$ produces a phase, this is inevitable because of the anomaly. Due to this fact, we will regard the twisted partition function $Z^{(\mathcal{T})\substack{\mathrm{Y}e^{2\pi\sqrt{-1}\lambda_h}\\ \mathrm{X}e^{2\pi\sqrt{-1}\lambda_g}}}(\tau)$ as a function of $\lambda_g,\lambda_h\in\mathbb{R}$, and write it as $Z^{(\mathcal{T})\substack{\mathrm{Y}\lambda_h\\ \mathrm{X}\lambda_g}}(\tau)$ below.

Recall that the NS and R sectors are described by $\lambda_g=0$ and $\lambda_g=\frac{1}{2}$ ($\lambda'_g=-\frac{1}{2}$), respectively. In addition, the action of $\mathfrak{z}=(-1)^F$ on the NS sector $A(\mathfrak{a})$ and on the R sector $A(\mathfrak{a})_{\mathrm{tw}}$ for

$\dim \mathfrak{a} = 2$ coincides with the action of $U_h|_{\lambda_h=\frac{1}{2}}$. So we define

$$Z^{(1)\,\mathrm{NS}\lambda_h}_{\mathrm{Pol}\mathrm{NS}\lambda_g}(\tau) := \frac{1}{\eta(\tau)}\theta\begin{bmatrix}\lambda_g\\ \lambda_h\end{bmatrix}(\tau,0) \tag{13.32}$$

$$= e^{2\pi\sqrt{-1}\lambda_g\lambda_h} q^{\lambda_g^2/2}\frac{\theta_3(\tau,\lambda_g\tau+\lambda_h)}{\eta(\tau)}, \tag{13.33}$$

$$Z^{(1)\,\mathrm{R}\lambda_h}_{\mathrm{Pol}\mathrm{NS}\lambda_g}(\tau) := \frac{1}{\eta(\tau)}\theta\begin{bmatrix}\lambda_g\\ \frac{1}{2}+\lambda_h\end{bmatrix}(\tau,0) \tag{13.34}$$

$$= e^{2\pi\sqrt{-1}\lambda_g(\frac{1}{2}+\lambda_h)} q^{\lambda_g^2/2}\frac{\theta_4(\tau,\lambda_g\tau+\lambda_h)}{\eta(\tau)}, \tag{13.35}$$

$$Z^{(1)\,\mathrm{NS}\lambda_h}_{\mathrm{Pol}\mathrm{R}\lambda_g}(\tau) := \frac{1}{\eta(\tau)}\theta\begin{bmatrix}\frac{1}{2}+\lambda_g\\ \lambda_h\end{bmatrix}(\tau,0) \tag{13.36}$$

$$= e^{2\pi\sqrt{-1}\lambda_g\lambda_h} q^{\lambda_g^2/2}\frac{\theta_2(\tau,\lambda_g\tau+\lambda_h)}{\eta(\tau)}, \tag{13.37}$$

$$Z^{(1)\,\mathrm{R}\lambda_h}_{\mathrm{Pol}\mathrm{R}\lambda_g}(\tau) := \frac{1}{\eta(\tau)}\theta\begin{bmatrix}\frac{1}{2}+\lambda_g\\ \frac{1}{2}+\lambda_h\end{bmatrix}(\tau,0) \tag{13.38}$$

$$= -e^{2\pi\sqrt{-1}\lambda_g(\frac{1}{2}+\lambda_h)} q^{\lambda_g^2/2}\frac{\theta_1(\tau,\lambda_g\tau+\lambda_h)}{\eta(\tau)}, \tag{13.39}$$

where $\theta_i(\tau,z)$ $(i=1,2,3,4)$ are the elliptic theta functions also summarized in Appendix E, and the second equation of each definition follows from (E.16). The superscript $^{(1)}$ denotes the number of complex fermions, and the subscript $_{\mathrm{Pol}}$ indicates that the $\mathrm{U}(1)$ phases of these twisted partition functions are the ones of [Pol07b, Eq. (10.7.7)].

## 13.2 Multiple Complex Fermions

Next, we will consider $n$ real fermions $\psi_1(z),\dots,\psi_n(z)$ with $n$ even, which are equivalent to $\frac{n}{2}$ complex fermions $\Psi_1(z),\dots,\Psi_{\frac{n}{2}}(z)$ as

$$\Psi_i(z) = \frac{1}{\sqrt{2}}(\psi_{2i-1}(z)+\sqrt{-1}\psi_{2i}(z)), \quad \overline{\Psi}_i(z) = \frac{1}{\sqrt{2}}(\psi_{2i-1}(z)-\sqrt{-1}\psi_{2i}(z)), \tag{13.40}$$

and their $\mathrm{SO}(n)$ symmetry, where the action of $O\in\mathrm{SO}(n)$ is

$$\big(\psi_1(z)\cdots\psi_n(z)\big)\begin{pmatrix}c^1\\ \vdots\\ c^n\end{pmatrix} \mapsto \big(\psi_1(z)\cdots\psi_n(z)\big)\,O\begin{pmatrix}c^1\\ \vdots\\ c^n\end{pmatrix}. \tag{13.41}$$

To calculate the twisted partition functions, we use the fact (Lemma F.2) that any $g\in\mathrm{SO}(n)$

can be block-diagonalized as

$$O^{-1}gO = \begin{pmatrix} \cos 2\pi(\lambda_g)_1 & -\sin 2\pi(\lambda_g)_1 & & & \\ \sin 2\pi(\lambda_g)_1 & \cos 2\pi(\lambda_g)_1 & & & \\ & & \ddots & & \\ & & & \cos 2\pi(\lambda_g)_{\frac{n}{2}} & -\sin 2\pi(\lambda_g)_{\frac{n}{2}} \\ & & & \sin 2\pi(\lambda_g)_{\frac{n}{2}} & \cos 2\pi(\lambda_g)_{\frac{n}{2}} \end{pmatrix} \tag{13.42}$$

by some $O \in \mathrm{SO}(n)$. In addition, if two matrices $g, h \in \mathrm{SO}(n)$ commute, then they are simultaneously block-diagonalizable[64] into the form of (13.42) by a matrix in $\mathrm{SO}(n)$. We introduce the notations

$$R_{\vec{\lambda}} := \begin{pmatrix} \cos 2\pi\lambda_1 & -\sin 2\pi\lambda_1 & & & \\ \sin 2\pi\lambda_1 & \cos 2\pi\lambda_1 & & & \\ & & \ddots & & \\ & & & \cos 2\pi\lambda_{\frac{n}{2}} & -\sin 2\pi\lambda_{\frac{n}{2}} \\ & & & \sin 2\pi\lambda_{\frac{n}{2}} & \cos 2\pi\lambda_{\frac{n}{2}} \end{pmatrix} \quad \text{for } \vec{\lambda} = (\lambda_1, \dots, \lambda_{\frac{n}{2}}), \tag{13.43}$$

and

$$R^O_{\vec{\lambda}} := OR_{\vec{\lambda}}O^{-1} \quad \text{for } O \in \mathrm{SO}(n). \tag{13.44}$$

Recall from (12.23) that the twisted partition functions satisfy ${Z^{(\mathcal{T})}}^h_g(\tau) \sim {Z^{(\mathcal{T})}}^{k^{-1}hk}_{k^{-1}gk}(\tau)$, where $\sim$ denotes the equality up to a $\mathrm{U}(1)$ phase because of anomaly. In our case of the twisted

[64] The proof is as follows. First block-diagonalize $g$ as in (13.42). This means there is a basis $\psi^g_1, \dots, \psi^g_n$ such that $g(\psi^g_{2i-1}) = \psi^g_{2i-1}\cos 2\pi(\lambda_g)_i + \psi^g_{2i}\sin 2\pi(\lambda_g)_i$ and $g(\psi^g_{2i}) = -\psi^g_{2i-1}\sin 2\pi(\lambda_g)_i + \psi^g_{2i}\cos 2\pi(\lambda_g)_i$. If we define $\Psi^g_i := \frac{1}{\sqrt{2}}(\psi^g_{2i-1} + \sqrt{-1}\psi^g_{2i})$ and $\overline{\Psi}^g_i := \frac{1}{\sqrt{2}}(\psi^g_{2i-1} - \sqrt{-1}\psi^g_{2i})$, then $g(\Psi^g_i) = e^{-2\pi\sqrt{-1}(\lambda_g)_i}\Psi^g_i$ and $g(\overline{\Psi}^g_i) = e^{2\pi\sqrt{-1}(\lambda_g)_i}\overline{\Psi}^g_i$. Let $\Psi^g_{i_1}, \dots, \Psi^g_{i_m}$ be the eigenvectors of $g$ with eigenvalue $e^{-2\pi\sqrt{-1}\lambda_g}$. (We omit the discussion of the case of $\lambda_g \in \mathbb{Z}$ below.) Since $gh = hg$, $h(\Psi^g_{i_1}), \dots, h(\Psi^g_{i_m})$ are again in the eigenspace of $g$ with eigenvalue $e^{-2\pi\sqrt{-1}\lambda_g}$. So $h$ can be restricted to this eigenspace $V_{-\lambda_g} := \mathrm{Span}\{\Psi^g_{i_1}, \dots, \Psi^g_{i_m}\}$. Similarly, $h$ can also be restricted to the eigenspace $V_{\lambda_g} := \mathrm{Span}\{\overline{\Psi}^g_{i_1}, \dots, \overline{\Psi}^g_{i_m}\}$. This restricted matrix $h|_{V_{-\lambda_g}\oplus V_{\lambda_g}}$ is unitary as a whole, because $h^Th = I$ shows the real matrix $h$ restricted to $\mathrm{Span}\{\psi^g_{2i_1-1}, \psi^g_{2i_1}, \dots, \psi^g_{2i_m-1}, \psi^g_{2i_m}\}$ is orthogonal, and the change-of-basis matrix from $\{\psi^g_i\}_i$ to $\{\Psi^g_i, \overline{\Psi}^g_i\}_i$ is unitary. Then both $h|_{V_{-\lambda_g}}$ and $h|_{V_{\lambda_g}}$ are unitary. Therefore, similarly to Lemma F.2, $h|_{V_{-\lambda_g}}$ can be diagonalized by a unitary matrix $U$. If $\Psi^h_l := \sum_{j=1}^m \Psi^g_{i_j}U^j_l$ is the resulting eigenvector of $h$, whose eigenvalue must be in the form of $e^{-2\pi\lambda_h}$ since $h \in \mathrm{SO}(n)$, then $\overline{\Psi}^h_l := \sum_{j=1}^m \overline{\Psi}^g_{i_j}(U^j_l)^*$ is an eigenvector of $h$ with eigenvalue $e^{2\pi\lambda_h}$, because $h$ is a real matrix. By defining $\psi^h_{2l-1} := \frac{1}{2}(\Psi^h_l + \overline{\Psi}^h_l)$ and $\psi^h_{2l} := \frac{1}{2\sqrt{-1}}(\Psi^h_l - \overline{\Psi}^h_l)$, we have both $g(\psi^h_{2l-1}) = \psi^h_{2l-1}\cos 2\pi\lambda_g + \psi^h_{2l}\sin 2\pi\lambda_g$, $g(\psi^h_{2l}) = -\psi^h_{2l-1}\sin 2\pi\lambda_g + \psi^h_{2l}\cos 2\pi\lambda_g$, and $h(\psi^h_{2l-1}) = \psi^h_{2l-1}\cos 2\pi\lambda_h + \psi^h_{2l}\sin 2\pi\lambda_h$, $h(\psi^h_{2l}) = -\psi^h_{2l-1}\sin 2\pi\lambda_h + \psi^h_{2l}\cos 2\pi\lambda_h$. If we write $U^j_l = u^{2j-1}_l + \sqrt{-1}u^{2j}_l$ ($u^{2j-1}_l, u^{2j}_l \in \mathbb{R}$), then $\psi^h_{2l-1} = \sum_{j=1}^m(\psi^g_{2i_j-1}u^{2j-1}_l - \psi^g_{2i_j}u^{2j}_l)$ and $\psi^h_{2l} = \sum_{j=1}^m(\psi^g_{2i_j-1}u^{2j-1}_l + \psi^g_{2i_j}u^{2j}_l)$ are real vectors. These equations also show the change-of-basis matrix from $\{\psi^g_i\}_i$ to $\{\psi^h_i\}_i$ in terms of $\{u^i_j\}_{i,j}$. Since $U$ is unitary, we can show that this change-of-basis matrix is orthogonal. Finally, if the simultaneously block-diagonalizing matrix for $g$ and $h$ obtained in this way has $\det = -1$, then by multiplying it by $\mathrm{diag}(-1, 1, \dots, 1)$, which only changes $(\lambda_g)_1$ and $(\lambda_h)_1$ to $-(\lambda_g)_1$ and $-(\lambda_h)_1$ respectively, we obtain the simultaneously block-diagonalizing matrix in $\mathrm{SO}(n)$.

partition functions $Z^{(n)}{}^{\mathrm{Y}h}_{\mathrm{X}g}(\tau)$ of $n$ real fermions for commuting $g, h \in \mathrm{SO}(n)$, by simultaneously block-diagonalizing $g$ and $h$ as $g = R^O_{\vec{\lambda}_g}$ and $h = R^O_{\vec{\lambda}_h}$, we have $Z^{(n)}{}^{\mathrm{Y}h}_{\mathrm{X}g}(\tau) \sim Z^{(n)}{}^{\mathrm{Y}R_{\vec{\lambda}_h}}_{\mathrm{X}R_{\vec{\lambda}_g}}(\tau)$. The twisted sector $\mathcal{H}^{(\frac{n}{2})}_{R_{\vec{\lambda}_g}}$ of $\frac{n}{2}$ complex fermions is then the tensor product $\bigotimes_{i=1}^{\frac{n}{2}} \mathcal{H}^{(1)}_{e^{2\pi\sqrt{-1}(\lambda_g)_i}}$ of the twisted sectors of one complex fermion. Therefore, the twisted partition function $Z^{(n)}{}^{\mathrm{Y}R_{\vec{\lambda}_h}}_{\mathrm{X}R_{\vec{\lambda}_g}}(\tau)$ is the product $\prod_{i=1}^{\frac{n}{2}} Z^{(1)}{}^{\mathrm{Y}e^{2\pi\sqrt{-1}(\lambda_h)_i}}_{\mathrm{X}e^{2\pi\sqrt{-1}(\lambda_g)_i}}(\tau)$. One choice of the $\mathrm{U}(1)$ phases of the twisted partition functions of one complex fermion is $Z^{(1)}{}^{\mathrm{Y}\lambda_h}_{\mathrm{Pol}\mathrm{X}\lambda_g}$ in (13.33)–(13.39), which is a function of $\lambda_g, \lambda_h \in \mathbb{R}$ instead of $\lambda_g + \mathbb{Z}, \lambda_h + \mathbb{Z} \in \mathbb{R}/\mathbb{Z}$, as remarked after (13.31). So, one choice of the twisted partition functions of $\frac{n}{2}$ complex fermions is

$$Z^{(\frac{n}{2})}{}^{\mathrm{Y}\vec{\lambda}_h}_{\mathrm{Pol}\ \mathrm{X}\vec{\lambda}_g}(\tau) := \prod_{i=1}^{\frac{n}{2}} Z^{(1)}{}^{\mathrm{Y}(\lambda_h)_i}_{\mathrm{Pol}\mathrm{X}(\lambda_g)_i}(\tau) \ , \tag{13.45}$$

which again should be regarded as a function of $\vec{\lambda}_g, \vec{\lambda}_h \in \mathbb{R}^{\frac{n}{2}}$, instead of $\vec{\lambda}_g + \mathbb{Z}^{\frac{n}{2}}, \vec{\lambda}_h + \mathbb{Z}^{\frac{n}{2}} \in (\mathbb{R}/\mathbb{Z})^{\frac{n}{2}}$.

Now, we would like to adjust the $\mathrm{U}(1)$ phases of these twisted partition functions of fermions so that the twisted partition functions $Z^{(V^{f\natural})}{}^{\mathrm{Y}h}_{\mathrm{X}g}(\tau)$ $(g, h \in \mathrm{Co}_1)$ of Duncan's module satisfy some consistency conditions. Specifically, we consider the following consistency conditions, which will be explained in more detail in Section 14.2.1.

(1) The modular transformations of $Z^{(V^{f\natural})}{}^{\mathrm{Y}\check{h}}_{\mathrm{X}\check{g}}$ satisfy (12.38)–(12.41) up to $\mathrm{U}(1)$ phases.

(2) $Z^{(V^{f\natural})}{}^{\mathrm{Y}\check{h}}_{\mathrm{X}-\check{g}}(\tau) \sim Z^{(V^{f\natural})}{}^{\mathrm{Y}\check{h}}_{\overline{\mathrm{X}}\check{g}}(\tau)$ and $Z^{(V^{f\natural})}{}^{\mathrm{Y}-\check{h}}_{\mathrm{X}\check{g}}(\tau) \sim Z^{(V^{f\natural})}{}^{\overline{\mathrm{Y}}\check{h}}_{\mathrm{X}\check{g}}(\tau)$, in accordance with (12.44) and (12.43).

(3) $Z^{(V^{f\natural})}{}^{\mathrm{Y}\check{h}}_{\mathrm{X}\mathfrak{z}\check{g}}(\tau) \sim Z^{(V^{f\natural})}{}^{\mathrm{Y}\check{h}}_{\mathrm{X}\check{g}}(\tau)$ and $Z^{(V^{f\natural})}{}^{\mathrm{Y}\mathfrak{z}\check{h}}_{\mathrm{X}\check{g}}(\tau) \sim Z^{(V^{f\natural})}{}^{\mathrm{Y}\check{h}}_{\mathrm{X}\check{g}}(\tau)$.

Since Duncan's module is constructed from the theory of 12 complex free fermions through the procedure similar to $\mathbb{Z}_2$-orbifolding, the twisted partition functions of Duncan's module will be calculated as signed sums (14.12)–(14.15) of $Z^{(12)}{}^{\mathrm{Y}\vec{\lambda}_h}_{\mathrm{X}\vec{\lambda}_g}(\tau)$ for $\mathrm{X}, \mathrm{Y} \in \{\mathrm{NS}, \mathrm{R}\}$. Then, we will see that for the first condition (1) on the modular transformations to be satisfied, the phases appearing in the modular transformations of $Z^{(12)}{}^{\mathrm{Y}\vec{\lambda}_h}_{\mathrm{X}\vec{\lambda}_g}(\tau)$ must at least be the same up to sign for $\mathrm{X}, \mathrm{Y} \in$

$\{\mathrm{NS},\mathrm{R}\}$. However, $Z^{(12)\mathrm{Y}\vec{\lambda}_h}_{\mathrm{Pol}\ \ \mathrm{X}\vec{\lambda}_g}(\tau)$ in (13.45) does not satisfy this condition.[65]

As we will see in Section 14.2.1, the above consistency conditions are satisfied if we use the following twisted partition functions:

$$Z^{(1)\mathrm{NS}\lambda_h}_{\mathrm{NS}\lambda_g}(\tau) := q^{\lambda_g^2/2}\frac{\theta_3(\tau,\lambda_g\tau+\lambda_h)}{\eta(\tau)} \tag{13.50}$$

$$= e^{-2\pi\sqrt{-1}\lambda_g\lambda_h}\frac{1}{\eta(\tau)}\theta\begin{bmatrix}\lambda_g\\ \lambda_h\end{bmatrix}(\tau,0) \tag{13.51}$$

$$= \frac{1}{\eta(\tau)}\sum_{n=-\infty}^{\infty} e^{2\pi\sqrt{-1}n\lambda_h}q^{\frac{1}{2}(n+\lambda_g)^2}, \tag{13.52}$$

$$Z^{(1)\mathrm{R}\lambda_h}_{\mathrm{NS}\lambda_g}(\tau) := q^{\lambda_g^2/2}\frac{\theta_4(\tau,\lambda_g\tau+\lambda_h)}{\eta(\tau)} \tag{13.53}$$

$$= e^{-2\pi\sqrt{-1}\lambda_g(\frac{1}{2}+\lambda_h)}\frac{1}{\eta(\tau)}\theta\begin{bmatrix}\lambda_g\\ \frac{1}{2}+\lambda_h\end{bmatrix}(\tau,0) \tag{13.54}$$

$$= \frac{1}{\eta(\tau)}\sum_{n=-\infty}^{\infty} e^{2\pi\sqrt{-1}n(\frac{1}{2}+\lambda_h)}q^{\frac{1}{2}(n+\lambda_g)^2}, \tag{13.55}$$

$$Z^{(1)\mathrm{NS}\lambda_h}_{\mathrm{R}\lambda_g}(\tau) := q^{\lambda_g^2/2}\frac{\theta_2(\tau,\lambda_g\tau+\lambda_h)}{\eta(\tau)} \tag{13.56}$$

$$= e^{-2\pi\sqrt{-1}\lambda_g\lambda_h}\frac{1}{\eta(\tau)}\theta\begin{bmatrix}\frac{1}{2}+\lambda_g\\ \lambda_h\end{bmatrix}(\tau,0) \tag{13.57}$$

$$= \frac{1}{\eta(\tau)}e^{\pi\sqrt{-1}\lambda_h}\sum_{n=-\infty}^{\infty} e^{2\pi\sqrt{-1}n\lambda_h}q^{\frac{1}{2}(n+\frac{1}{2}+\lambda_g)^2}, \tag{13.58}$$

[65] For example, the modular S transformations of $Z^{(12)\mathrm{Y}\vec{\lambda}_h}_{\mathrm{Pol}\ \ \mathrm{X}\vec{\lambda}_g}(\tau)$ are

$$Z^{(12)\mathrm{NS}\vec{\lambda}_h}_{\mathrm{Pol}\ \ \mathrm{NS}\vec{\lambda}_g}(-\frac{1}{\tau}) = e^{2\pi\sqrt{-1}\vec{\lambda_g}\cdot\vec{\lambda_h}}Z^{(12)\mathrm{NS}(-\vec{\lambda}_g)}_{\mathrm{Pol}\ \ \mathrm{NS}\vec{\lambda}_h}(\tau), \tag{13.46}$$

$$Z^{(12)\mathrm{R}\vec{\lambda}_h}_{\mathrm{Pol}\ \ \mathrm{NS}\vec{\lambda}_g}(-\frac{1}{\tau}) = e^{2\pi\sqrt{-1}\vec{\lambda_g}\cdot(\vec{\lambda_h}+\frac{1}{2}\vec{1})}Z^{(12)\mathrm{NS}(-\vec{\lambda}_g)}_{\mathrm{Pol}\ \ \mathrm{R}\vec{\lambda}_h}(\tau), \tag{13.47}$$

$$Z^{(12)\mathrm{NS}\vec{\lambda}_h}_{\mathrm{Pol}\ \ \mathrm{R}\vec{\lambda}_g}(-\frac{1}{\tau}) = e^{2\pi\sqrt{-1}(\vec{\lambda_g}-\frac{1}{2}\vec{1})\cdot\vec{\lambda_h}}Z^{(12)\mathrm{R}(-\vec{\lambda}_g)}_{\mathrm{Pol}\ \ \mathrm{NS}\vec{\lambda}_h}(\tau), \tag{13.48}$$

$$Z^{(12)\mathrm{R}\vec{\lambda}_h}_{\mathrm{Pol}\ \ \mathrm{R}\vec{\lambda}_g}(-\frac{1}{\tau}) = e^{2\pi\sqrt{-1}(\vec{\lambda_g}-\frac{1}{2}\vec{1})\cdot(\vec{\lambda_h}+\frac{1}{2}\vec{1})}Z^{(12)\mathrm{R}(-\vec{\lambda}_g)}_{\mathrm{Pol}\ \ \mathrm{R}\vec{\lambda}_h}(\tau). \tag{13.49}$$

$$Z^{(1)\mathrm{R}\lambda_h}_{\mathrm{R}\lambda_g}(\tau) := q^{\lambda_g^2/2}\frac{\theta_1(\tau,\lambda_g\tau+\lambda_h)}{\eta(\tau)} \tag{13.59}$$

$$= -e^{-2\pi\sqrt{-1}\lambda_g(\frac{1}{2}+\lambda_h)}\frac{1}{\eta(\tau)}\theta\begin{bmatrix}\frac{1}{2}+\lambda_g\\ \frac{1}{2}+\lambda_h\end{bmatrix}(\tau,0) \tag{13.60}$$

$$= \frac{1}{\eta(\tau)}(-\sqrt{-1})e^{\pi\sqrt{-1}\lambda_h}\sum_{n=-\infty}^{\infty}e^{2\pi\sqrt{-1}n(\frac{1}{2}+\lambda_h)}q^{\frac{1}{2}(n+\frac{1}{2}+\lambda_g)^2}, \tag{13.61}$$

and

$$Z^{(\frac{n}{2})\mathrm{Y}\vec{\lambda_h}}_{\mathrm{X}\vec{\lambda_g}}(\tau) := \prod_{i=1}^{\frac{n}{2}} Z^{(1)\mathrm{Y}(\lambda_h)_i}_{\mathrm{X}(\lambda_g)_i}(\tau)\ . \tag{13.62}$$

So we will use these twisted partition functions below. If we use these twisted partition functions with $\vec{\lambda}_h = \vec{0}$ to calculate $Z^{(V^{f\natural})\mathrm{Y}}_{\mathrm{R}g}(\tau)$ as in (14.14) and (14.15), then it will reproduce the ones in [JFF23, Lemma 3.2.1].

## 13.3 Modular Transformations and Angle Shifts

Let us calculate the modular transformations of the twisted partition functions $Z^{(\frac{n}{2})\mathrm{Y}\vec{\lambda_h}}_{\mathrm{X}\vec{\lambda_g}}(\tau)$, and the transformation properties of them under some shifts of angles $\vec{\lambda}_g, \vec{\lambda}_h$. We begin with the case of one complex fermion.

**One complex fermion**

Under the modular T transformation, from (E.28)–(E.31) and (E.39),

$$Z^{(1)\mathrm{NS}\lambda_h}_{\mathrm{NS}\lambda_g}(\tau+1) = (qe^{2\pi\sqrt{-1}})^{\lambda_g^2/2}\frac{\theta_3(\tau+1,\lambda_g(\tau+1)+\lambda_h)}{\eta(\tau+1)} \tag{13.63}$$

$$= e^{\pi\sqrt{-1}\lambda_g^2}q^{\lambda_g^2/2}\frac{\theta_4(\tau,\lambda_g\tau+\lambda_g+\lambda_h)}{e^{\pi\sqrt{-1}/12}\eta(\tau)} \tag{13.64}$$

$$= e^{-\pi\sqrt{-1}/12}e^{\pi\sqrt{-1}\lambda_g^2}Z^{(1)\mathrm{R}(\lambda_g+\lambda_h)}_{\mathrm{NS}\lambda_g}(\tau), \tag{13.65}$$

$$Z^{(1)\mathrm{R}\lambda_h}_{\mathrm{NS}\lambda_g}(\tau+1) = (qe^{2\pi\sqrt{-1}})^{\lambda_g^2/2}\frac{\theta_4(\tau+1,\lambda_g(\tau+1)+\lambda_h)}{\eta(\tau+1)} \tag{13.66}$$

$$= e^{\pi\sqrt{-1}\lambda_g^2}q^{\lambda_g^2/2}\frac{\theta_3(\tau,\lambda_g\tau+\lambda_g+\lambda_h)}{e^{\pi\sqrt{-1}/12}\eta(\tau)} \tag{13.67}$$

$$= e^{-\pi\sqrt{-1}/12}e^{\pi\sqrt{-1}\lambda_g^2}Z^{(1)\mathrm{NS}(\lambda_g+\lambda_h)}_{\mathrm{NS}\lambda_g}(\tau), \tag{13.68}$$

$$Z^{(1)\mathrm{NS}\lambda_h}_{\mathrm{R}\lambda_g}(\tau+1) = (qe^{2\pi\sqrt{-1}})^{\lambda_g^2/2}\frac{\theta_2(\tau+1,\lambda_g(\tau+1)+\lambda_h)}{\eta(\tau+1)} \tag{13.69}$$

$$= e^{\pi\sqrt{-1}\lambda_g^2}q^{\lambda_g^2/2}\frac{e^{\pi\sqrt{-1}/4}\theta_2(\tau,\lambda_g\tau+\lambda_g+\lambda_h)}{e^{\pi\sqrt{-1}/12}\eta(\tau)} \tag{13.70}$$

$$= e^{\pi\sqrt{-1}/6}e^{\pi\sqrt{-1}\lambda_g^2}Z^{(1)\mathrm{NS}(\lambda_g+\lambda_h)}_{\mathrm{R}\lambda_g}(\tau), \tag{13.71}$$

$$Z^{(1)\mathrm{R}\lambda_h}_{\mathrm{R}\lambda_g}(\tau+1) = (qe^{2\pi\sqrt{-1}})^{\lambda_g^2/2}\frac{\theta_1(\tau+1,\lambda_g(\tau+1)+\lambda_h)}{\eta(\tau+1)} \tag{13.72}$$

$$= e^{\pi\sqrt{-1}\lambda_g^2}q^{\lambda_g^2/2}\frac{e^{\pi\sqrt{-1}/4}\theta_1(\tau,\lambda_g\tau+\lambda_g+\lambda_h)}{e^{\pi\sqrt{-1}/12}\eta(\tau)} \tag{13.73}$$

$$= e^{\pi\sqrt{-1}/6}e^{\pi\sqrt{-1}\lambda_g^2}Z^{(1)\mathrm{R}(\lambda_g+\lambda_h)}_{\mathrm{R}\lambda_g}(\tau). \tag{13.74}$$

Under the modular S transformation, from (E.14), (E.15) and (E.40),

$$Z^{(1)\mathrm{NS}\lambda_h}_{\mathrm{NS}\lambda_g}(-\frac{1}{\tau}) = e^{-2\pi\sqrt{-1}\lambda_g\lambda_h}\frac{1}{\eta(-\frac{1}{\tau})}\theta\begin{bmatrix}\lambda_g\\ \lambda_h\end{bmatrix}(-\frac{1}{\tau},0) \tag{13.75}$$

$$= \frac{1}{\eta(\tau)}\theta\begin{bmatrix}\lambda_h\\ -\lambda_g\end{bmatrix}(\tau,0) \tag{13.76}$$

$$= e^{-2\pi\sqrt{-1}\lambda_g\lambda_h}Z^{(1)\mathrm{NS}(-\lambda_g)}_{\mathrm{NS}\lambda_h}(\tau), \tag{13.77}$$

$$Z^{(1)\mathrm{R}\lambda_h}_{\mathrm{NS}\lambda_g}(-\frac{1}{\tau}) = e^{-2\pi\sqrt{-1}\lambda_g(\frac{1}{2}+\lambda_h)}\frac{1}{\eta(-\frac{1}{\tau})}\theta\begin{bmatrix}\lambda_g\\ \frac{1}{2}+\lambda_h\end{bmatrix}(-\frac{1}{\tau},0) \tag{13.78}$$

$$= \frac{1}{\eta(\tau)}\theta\begin{bmatrix}\frac{1}{2}+\lambda_h\\ -\lambda_g\end{bmatrix}(\tau,0) \tag{13.79}$$

$$= e^{-2\pi\sqrt{-1}\lambda_g\lambda_h}Z^{(1)\mathrm{NS}(-\lambda_g)}_{\mathrm{R}\lambda_h}(\tau), \tag{13.80}$$

$$Z^{(1)\mathrm{NS}\lambda_h}_{\mathrm{R}\lambda_g}(-\frac{1}{\tau}) = e^{-2\pi\sqrt{-1}\lambda_g\lambda_h}\frac{1}{\eta(-\frac{1}{\tau})}\theta\begin{bmatrix}\frac{1}{2}+\lambda_g\\ \lambda_h\end{bmatrix}(-\frac{1}{\tau},0) \tag{13.81}$$

$$= e^{2\pi\sqrt{-1}(\frac{1}{2}\lambda_h)}\frac{1}{\eta(\tau)}\theta\begin{bmatrix}\lambda_h\\ -\frac{1}{2}-\lambda_g\end{bmatrix}(\tau,0) \tag{13.82}$$

$$= e^{2\pi\sqrt{-1}(-\frac{1}{2}\lambda_h)}\frac{1}{\eta(\tau)}\theta\begin{bmatrix}\lambda_h\\ \frac{1}{2}-\lambda_g\end{bmatrix}(\tau,0) \tag{13.83}$$

$$= e^{-2\pi\sqrt{-1}\lambda_g\lambda_h}Z^{(1)\mathrm{R}(-\lambda_g)}_{\mathrm{NS}\lambda_h}(\tau), \tag{13.84}$$

$$Z^{(1)\mathrm{R}\lambda_h}_{\mathrm{R}\lambda_g}(-\frac{1}{\tau}) = -e^{-2\pi\sqrt{-1}\lambda_g(\frac{1}{2}+\lambda_h)}\frac{1}{\eta(-\frac{1}{\tau})}\theta\begin{bmatrix}\frac{1}{2}+\lambda_g\\ \frac{1}{2}+\lambda_h\end{bmatrix}(-\frac{1}{\tau},0) \tag{13.85}$$

$$= -e^{2\pi\sqrt{-1}(\frac{1}{2}(\frac{1}{2}+\lambda_h))}\frac{1}{\eta(\tau)}\theta\begin{bmatrix}\frac{1}{2}+\lambda_h\\ -\frac{1}{2}-\lambda_g\end{bmatrix}(\tau,0) \tag{13.86}$$

$$= -e^{2\pi\sqrt{-1}(-\frac{1}{2}(\frac{1}{2}+\lambda_h))}\frac{1}{\eta(\tau)}\theta\begin{bmatrix}\frac{1}{2}+\lambda_h\\ \frac{1}{2}-\lambda_g\end{bmatrix}(\tau,0) \tag{13.87}$$

$$= -\sqrt{-1}e^{-2\pi\sqrt{-1}\lambda_g\lambda_h}Z^{(1)\mathrm{R}(-\lambda_g)}_{\mathrm{R}\lambda_h}(\tau). \tag{13.88}$$

Under the shifts $\lambda_g \mapsto \lambda_g + \frac{1}{2}$ or $\lambda_h \mapsto \lambda_h + \frac{1}{2}$, from the expressions (13.51)–(13.61) of the twisted partition functions,

$$Z^{(1)\,\mathrm{NS}\lambda_h}_{\mathrm{NS}(\lambda_g+\frac{1}{2})}(\tau) = e^{-\pi\sqrt{-1}\lambda_h} Z^{(1)\,\mathrm{NS}\lambda_h}_{\mathrm{R}\lambda_g}(\tau), \qquad Z^{(1)\,\mathrm{NS}(\lambda_h+\frac{1}{2})}_{\mathrm{NS}\lambda_g}(\tau) = Z^{(1)\,\mathrm{R}\lambda_h}_{\mathrm{NS}\lambda_g}(\tau), \tag{13.89}$$

$$Z^{(1)\,\mathrm{R}\lambda_h}_{\mathrm{NS}(\lambda_g+\frac{1}{2})}(\tau) = \sqrt{-1} e^{-\pi\sqrt{-1}\lambda_h} Z^{(1)\,\mathrm{R}\lambda_h}_{\mathrm{R}\lambda_g}(\tau), \qquad Z^{(1)\,\mathrm{R}(\lambda_h+\frac{1}{2})}_{\mathrm{NS}\lambda_g}(\tau) = Z^{(1)\,\mathrm{NS}\lambda_h}_{\mathrm{NS}\lambda_g}(\tau), \tag{13.90}$$

$$Z^{(1)\,\mathrm{NS}\lambda_h}_{\mathrm{R}(\lambda_g+\frac{1}{2})}(\tau) = e^{-\pi\sqrt{-1}\lambda_h} Z^{(1)\,\mathrm{NS}\lambda_h}_{\mathrm{NS}\lambda_g}(\tau), \qquad Z^{(1)\,\mathrm{NS}(\lambda_h+\frac{1}{2})}_{\mathrm{R}\lambda_g}(\tau) = -Z^{(1)\,\mathrm{R}\lambda_h}_{\mathrm{R}\lambda_g}(\tau), \tag{13.91}$$

$$Z^{(1)\,\mathrm{R}\lambda_h}_{\mathrm{R}(\lambda_g+\frac{1}{2})}(\tau) = \sqrt{-1} e^{-\pi\sqrt{-1}\lambda_h} Z^{(1)\,\mathrm{R}\lambda_h}_{\mathrm{NS}\lambda_g}(\tau), \qquad Z^{(1)\,\mathrm{R}(\lambda_h+\frac{1}{2})}_{\mathrm{R}\lambda_g}(\tau) = Z^{(1)\,\mathrm{NS}\lambda_h}_{\mathrm{R}\lambda_g}(\tau). \tag{13.92}$$

Under the shifts $\lambda_g \mapsto \lambda_g + m$ or $\lambda_h \mapsto \lambda_h + m$ ($m \in \mathbb{Z}$), from the expressions (13.51)–(13.61) of the twisted partition functions,

$$Z^{(1)\,\mathrm{NS}\lambda_h}_{\mathrm{NS}(\lambda_g+m)}(\tau) = e^{-2\pi\sqrt{-1}m\lambda_h} Z^{(1)\,\mathrm{NS}\lambda_h}_{\mathrm{NS}\lambda_g}(\tau), \qquad Z^{(1)\,\mathrm{NS}(\lambda_h+m)}_{\mathrm{NS}\lambda_g}(\tau) = Z^{(1)\,\mathrm{NS}\lambda_h}_{\mathrm{NS}\lambda_g}(\tau), \tag{13.93}$$

$$Z^{(1)\,\mathrm{R}\lambda_h}_{\mathrm{NS}(\lambda_g+m)}(\tau) = (-1)^m e^{-2\pi\sqrt{-1}m\lambda_h} Z^{(1)\,\mathrm{R}\lambda_h}_{\mathrm{NS}\lambda_g}(\tau), \qquad Z^{(1)\,\mathrm{R}(\lambda_h+m)}_{\mathrm{NS}\lambda_g}(\tau) = Z^{(1)\,\mathrm{R}\lambda_h}_{\mathrm{NS}\lambda_g}(\tau), \tag{13.94}$$

$$Z^{(1)\,\mathrm{NS}\lambda_h}_{\mathrm{R}(\lambda_g+m)}(\tau) = e^{-2\pi\sqrt{-1}m\lambda_h} Z^{(1)\,\mathrm{NS}\lambda_h}_{\mathrm{R}\lambda_g}(\tau), \qquad Z^{(1)\,\mathrm{NS}(\lambda_h+m)}_{\mathrm{R}\lambda_g}(\tau) = (-1)^m Z^{(1)\,\mathrm{NS}\lambda_h}_{\mathrm{R}\lambda_g}(\tau), \tag{13.95}$$

$$Z^{(1)\,\mathrm{R}\lambda_h}_{\mathrm{R}(\lambda_g+m)}(\tau) = (-1)^m e^{-2\pi\sqrt{-1}m\lambda_h} Z^{(1)\,\mathrm{R}\lambda_h}_{\mathrm{R}\lambda_g}(\tau), \qquad Z^{(1)\,\mathrm{R}(\lambda_h+m)}_{\mathrm{R}\lambda_g}(\tau) = (-1)^m Z^{(1)\,\mathrm{R}\lambda_h}_{\mathrm{R}\lambda_g}(\tau). \tag{13.96}$$

**Multiple complex fermions**

The properties of the twisted partition functions $Z^{(N)\,\mathrm{Y}\vec{\lambda}_h}_{\mathrm{X}\vec{\lambda}_g}(\tau)$ of $N$ complex fermions immediately follow from those of $Z^{(1)\,\mathrm{Y}\vec{\lambda}_h}_{\mathrm{X}\vec{\lambda}_g}(\tau)$.

The modular transformations are

$$Z^{(N)\,\mathrm{NS}\vec{\lambda}_h}_{\mathrm{NS}\vec{\lambda}_g}(\tau+1) = e^{-N\pi\sqrt{-1}/12} e^{\pi\sqrt{-1}\vec{\lambda}_g\cdot\vec{\lambda}_g} Z^{(N)\,\mathrm{R}(\vec{\lambda}_g+\vec{\lambda}_h)}_{\mathrm{NS}\vec{\lambda}_g}(\tau), \tag{13.97}$$

$$Z^{(N)\,\mathrm{R}\vec{\lambda}_h}_{\mathrm{NS}\vec{\lambda}_g}(\tau+1) = e^{-N\pi\sqrt{-1}/12} e^{\pi\sqrt{-1}\vec{\lambda}_g\cdot\vec{\lambda}_g} Z^{(N)\,\mathrm{NS}(\vec{\lambda}_g+\vec{\lambda}_h)}_{\mathrm{NS}\vec{\lambda}_g}(\tau), \tag{13.98}$$

$$Z^{(N)\,\mathrm{NS}\vec{\lambda}_h}_{\mathrm{R}\vec{\lambda}_g}(\tau+1) = e^{N\pi\sqrt{-1}/6} e^{\pi\sqrt{-1}\vec{\lambda}_g\cdot\vec{\lambda}_g} Z^{(N)\,\mathrm{NS}(\vec{\lambda}_g+\vec{\lambda}_h)}_{\mathrm{R}\vec{\lambda}_g}(\tau), \tag{13.99}$$

$$Z^{(N)\,\mathrm{R}\vec{\lambda}_h}_{\mathrm{R}\vec{\lambda}_g}(\tau+1) = e^{N\pi\sqrt{-1}/6} e^{\pi\sqrt{-1}\vec{\lambda}_g\cdot\vec{\lambda}_g} Z^{(N)\,\mathrm{R}(\vec{\lambda}_g+\vec{\lambda}_h)}_{\mathrm{R}\vec{\lambda}_g}(\tau), \tag{13.100}$$

and

$$Z^{(N)\,\mathrm{NS}\vec{\lambda}_h}_{\mathrm{NS}\vec{\lambda}_g}(-\frac{1}{\tau}) = e^{-2\pi\sqrt{-1}\vec{\lambda}_g\cdot\vec{\lambda}_h} Z^{(N)\,\mathrm{NS}(-\vec{\lambda}_g)}_{\mathrm{NS}\vec{\lambda}_h}(\tau), \tag{13.101}$$

$$Z^{(N)\,\mathrm{R}\vec{\lambda}_h}_{\mathrm{NS}\vec{\lambda}_g}(-\frac{1}{\tau}) = e^{-2\pi\sqrt{-1}\vec{\lambda}_g\cdot\vec{\lambda}_h} Z^{(N)\,\mathrm{NS}(-\vec{\lambda}_g)}_{\mathrm{R}\vec{\lambda}_h}(\tau), \tag{13.102}$$

$$Z^{(N)\,\mathrm{NS}\vec{\lambda}_h}_{\mathrm{R}\vec{\lambda}_g}(-\frac{1}{\tau}) = e^{-2\pi\sqrt{-1}\vec{\lambda}_g\cdot\vec{\lambda}_h} Z^{(N)\,\mathrm{R}(-\vec{\lambda}_g)}_{\mathrm{NS}\vec{\lambda}_h}(\tau), \tag{13.103}$$

$$Z^{(N)\,\mathrm{R}\vec{\lambda}_h}_{\mathrm{R}\vec{\lambda}_g}(-\frac{1}{\tau}) = (-\sqrt{-1})^N e^{-2\pi\sqrt{-1}\vec{\lambda}_g\cdot\vec{\lambda}_h} Z^{(N)\,\mathrm{R}(-\vec{\lambda}_g)}_{\mathrm{R}\vec{\lambda}_h}(\tau). \tag{13.104}$$

These are consistent with (12.38)–(12.41) up to anomalous phases.

Under the angle shifts,

$$Z^{(N)\,\mathrm{Y}\vec{\lambda}_h}_{\mathrm{X}(\vec{\lambda}_g+\frac{1}{2}\vec{1})}(\tau) = \begin{cases} e^{-\pi\sqrt{-1}\sum_i(\lambda_h)_i} Z^{(N)\,\mathrm{Y}\vec{\lambda}_h}_{\mathrm{X}\vec{\lambda}_g}(\tau) & (\mathrm{Y}=\mathrm{NS}), \\ \sqrt{-1}^N e^{-\pi\sqrt{-1}\sum_i(\lambda_h)_i} Z^{(N)\,\mathrm{Y}\vec{\lambda}_h}_{\mathrm{X}\vec{\lambda}_g}(\tau) & (\mathrm{Y}=\mathrm{R}), \end{cases} \tag{13.105}$$

$$Z^{(N)\,\mathrm{Y}(\vec{\lambda}_h+\frac{1}{2}\vec{1})}_{\mathrm{X}\vec{\lambda}_g}(\tau) = \begin{cases} (-1)^N Z^{(N)\,\mathrm{R}\vec{\lambda}_h}_{\mathrm{R}\vec{\lambda}_g}(\tau) & (\mathrm{X}=\mathrm{R},\mathrm{Y}=\mathrm{NS}), \\ Z^{(N)\,\overline{\mathrm{Y}}\vec{\lambda}_h}_{\mathrm{X}\vec{\lambda}_g}(\tau) & (\text{otherwise}), \end{cases} \tag{13.106}$$

where $\vec{1} := (1,\dots,1)$. Recall that the symmetry $\mathbb{Z}_2 = \langle(-1)^F\rangle$ defining the fermion parity of $A(\mathfrak{a}) \oplus A(\mathfrak{a})_{\mathrm{tw}}$ is $(-1)^F = \mathfrak{z} \in \mathrm{Spin}(n)$, which is a lift of $-1_{\mathrm{SO}(n)} = R_{\frac{1}{2}\vec{1}} \in \mathrm{SO}(n)$. As we will see in the next Section 13.4, $\mathfrak{z} = \hat{R}_{\frac{1}{2}\vec{1}}$ in the notation (13.109), and hence these transformations (13.105) and (13.106) are consistent with (12.44) and (12.43) up to anomalous phases.

We also have, for $\vec{m} \in \mathbb{Z}^N$,

$$Z^{(N)\,\mathrm{Y}\vec{\lambda}_h}_{\mathrm{X}(\vec{\lambda}_g+\vec{m})}(\tau) = \begin{cases} e^{-2\pi\sqrt{-1}\vec{m}\cdot\vec{\lambda}_h} Z^{(N)\,\mathrm{Y}\vec{\lambda}_h}_{\mathrm{X}\vec{\lambda}_g}(\tau) & (\mathrm{Y}=\mathrm{NS}), \\ (-1)^{\sum_i m_i} e^{-2\pi\sqrt{-1}\vec{m}\cdot\vec{\lambda}_h} Z^{(N)\,\mathrm{Y}\vec{\lambda}_h}_{\mathrm{X}\vec{\lambda}_g}(\tau) & (\mathrm{Y}=\mathrm{R}), \end{cases} \tag{13.107}$$

$$Z^{(N)\,\mathrm{Y}(\vec{\lambda}_h+\vec{m})}_{\mathrm{X}\vec{\lambda}_g}(\tau) = \begin{cases} Z^{(N)\,\mathrm{Y}\vec{\lambda}_h}_{\mathrm{X}\vec{\lambda}_g}(\tau) & (\mathrm{X}=\mathrm{NS}), \\ (-1)^{\sum_i m_i} Z^{(N)\,\mathrm{Y}\vec{\lambda}_h}_{\mathrm{X}\vec{\lambda}_g}(\tau) & (\mathrm{X}=\mathrm{R}). \end{cases} \tag{13.108}$$

These are consistent up to anomalous phases with the fact that $R^O_{\vec{\lambda}} = R^O_{\vec{\lambda}+\vec{m}} \in \mathrm{SO}(n)$. We will see in the next Section 13.4 that these anomalous phases stem from the difference between the two lifts of $R^O_{\vec{\lambda}} \in \mathrm{SO}(n)$ to $\mathrm{Spin}(n)$.

## 13.4 Twists by $\mathrm{Spin}(n)$

Suppose we want to calculate the twisted partition function $Z^{(\frac{n}{2})\,\mathrm{Y}h}_{\mathrm{X}g}(\tau)$ of $\frac{n}{2}$ complex fermions twisted by commuting $g, h \in \mathrm{Spin}(n)$. Since $g$ and $h$ commute, their images $\varphi(g)$ and $\varphi(h)$ under the projection $\varphi : \mathrm{Spin}(n) \to \mathrm{SO}(n)$ also commute. Then, as explained in Section 13.2, we can simultaneously block-diagonalize them as $\varphi(g) = R^O_{\vec{\lambda}_{\varphi(g)}}$ and $\varphi(g) = R^O_{\vec{\lambda}_{\varphi(h)}}$.

As shown in Appendix F.2, the lifts of $R_{\vec{\lambda}} \in \mathrm{SO}(n)$ to $\mathrm{Spin}(n)$ are

$$\pm\hat{R}_{\vec{\lambda}} := \pm\prod_{i=1}^{\frac{n}{2}} (\cos\pi\lambda_i + \psi_{2i-1}\psi_{2i}\sin\pi\lambda_i), \tag{13.109}$$

and the lifts of $R^O_{\vec{\lambda}} \in \mathrm{SO}(n)$ to $\mathrm{Spin}(n)$ are

$$\pm\hat{R}^O_{\vec{\lambda}} := \pm\hat{O}\hat{R}_{\vec{\lambda}}\hat{O}^{-1} = \pm\prod_{i=1}^{\frac{n}{2}} (\cos\pi\lambda_i + (\vec{\psi}O)_{2i-1}(\vec{\psi}O)_{2i}\sin\pi\lambda_i), \tag{13.110}$$

where $\pm\hat{O} \in \mathrm{Spin}(n)$ are the lifts of $O \in \mathrm{SO}(n)$, and $\vec{\psi} := (\psi_1, \dots \psi_n)$ is a row vector. As a result, $g$ is $\hat{R}^O_{\vec{\lambda}_{\varphi(g)}}$ or $-\hat{R}^O_{\vec{\lambda}_{\varphi(g)}}$, and $h$ is $\hat{R}^O_{\vec{\lambda}_{\varphi(h)}}$ or $-\hat{R}^O_{\vec{\lambda}_{\varphi(h)}}$.

From (13.110), we can see that if all the entries of $\vec{m}$ are integers, then for any $O \in \mathrm{SO}(n)$,

$$\hat{R}^O_{\vec{m}} = (-1_{\mathrm{Spin}(n)})^{\sum_i m_i} \quad (m_i \in \mathbb{Z}). \tag{13.111}$$

In addition,

$$\hat{R}^O_{\vec{\lambda}_1} \hat{R}^O_{\vec{\lambda}_2} = \hat{R}^O_{\vec{\lambda}_1 + \vec{\lambda}_2}. \tag{13.112}$$

Therefore, $g$ is $\hat{R}^O_{\vec{\lambda}_{\varphi(g)}}$ or $\hat{R}^O_{\vec{\lambda}_{\varphi(g)}+(0,\dots,0,1)}$. Let $\vec{\lambda}_{g,O}$ denote the correct one of $\vec{\lambda}_{\varphi(g)}$ or $\vec{\lambda}_{\varphi(g)} + (0, \dots, 0, 1)$. Similarly, let $\vec{\lambda}_{h,O}$ denote the correct one of $\vec{\lambda}_{\varphi(h)}$ or $\vec{\lambda}_{\varphi(h)} + (0, \dots, 0, 1)$. One caveat is that the correct angles $\vec{\lambda}_{g,O}$ depend not only on $g \in \mathrm{Spin}(n)$ but also on the block-diagonalizing matrix $O \in \mathrm{SO}(n)$, because different angles, including both $\vec{\lambda}_{\varphi(g)}$ and $\vec{\lambda}_{\varphi(g)} + (0, \dots, 0, 1)$, can be $\mathrm{Spin}(n)$-conjugate in general.[66] Hence we have included the subscript $_O$ in the notation $\vec{\lambda}_{g,O}$.

We finally obtain

$$Z^{(\frac{n}{2})\substack{\mathrm{Y}h\\ \mathrm{X}g}}(\tau) = Z^{(\frac{n}{2})\substack{\mathrm{Y}\hat{R}^O_{\vec{\lambda}_{h,O}}\\ \mathrm{X}\hat{R}^O_{\vec{\lambda}_{g,O}}}}(\tau) \sim Z^{(\frac{n}{2})\substack{\mathrm{Y}\hat{R}_{\vec{\lambda}_{h,O}}\\ \mathrm{X}\hat{R}_{\vec{\lambda}_{g,O}}}}(\tau) \sim Z^{(\frac{n}{2})\substack{\mathrm{Y}\vec{\lambda}_{h,O}\\ \mathrm{X}\vec{\lambda}_{g,O}}}(\tau), \tag{13.117}$$

where $\sim$ denotes the equality up to a phase, because of the anomaly. In the second equation, we used $Z^{(\mathcal{T})\,h}_{\ g}(\tau) \sim Z^{(\mathcal{T})\,k^{-1}hk}_{\ \ k^{-1}gk}(\tau)$. In the last equation, we used the fact that the (projective) $\mathrm{SO}(n)$-action should coincides with the (projective) $\mathrm{Spin}(n)$-action, as mentioned in Section 12.1.4. Then, by using the definition (13.62) of $Z^{(\frac{n}{2})\substack{\mathrm{Y}\vec{\lambda}_h\\ \mathrm{X}\vec{\lambda}_g}}$, we can calculate the twisted partition function $Z^{(\frac{n}{2})\substack{\mathrm{Y}h\\ \mathrm{X}g}}(\tau)$ twisted by $g, h \in \mathrm{Spin}(n)$.

We have seen in (13.107) and (13.108) that the shift $\vec{\lambda} \mapsto \vec{\lambda} + (0, \dots, 0, 1)$ produces additional phases of the twisted partition functions. Now we can see that these phases stem from the

[66] For example, for $\vec{\lambda} = (\lambda_1, \dots, \lambda_{\frac{n}{2}-2}, 0, \frac{1}{2})$, we can see that $\hat{R}_{\vec{\lambda}}$ and $\hat{R}_{\vec{\lambda}+(0,\dots,0,1)}$ are in fact $\mathrm{Spin}(n)$-conjugate as follows. If we take

$$O = \begin{pmatrix} 1 & & & & & & & \\ & \ddots & & & & & & \\ & & 1 & & & & & \\ & & & 0 & 1 & & & \\ & & & 1 & 0 & & & \\ & & & & & 0 & 1 & \\ & & & & & 1 & 0 & \\ & & & & & & & \end{pmatrix} \in \mathrm{SO}(n), \tag{13.113}$$

then

$$\hat{R}^O_{\vec{\lambda}} = \hat{O}(\cos\pi\lambda_1 + \psi_1\psi_2 \sin\pi\lambda_1) \cdots (\cos\pi\lambda_{\frac{n}{2}-2} + \psi_{n-5}\psi_{n-4}\sin\pi\lambda_{\frac{n}{2}-2})\psi_{n-1}\psi_n\hat{O}^{-1} \tag{13.114}$$

$$= -(\cos\pi\lambda_1 + \psi_1\psi_2 \sin\pi\lambda_1) \cdots (\cos\pi\lambda_{\frac{n}{2}-2} + \psi_{n-5}\psi_{n-4}\sin\pi\lambda_{\frac{n}{2}-2})\psi_{n-1}\psi_n \tag{13.115}$$

$$= \hat{R}_{\vec{\lambda}+(0,\dots,0,1)}. \tag{13.116}$$

difference between the two lifts $\hat{R}^O_{\vec{\lambda}}$ and $-\hat{R}^O_{\vec{\lambda}} = \hat{R}^O_{\vec{\lambda}+(0,\dots,0,1)}$ of $R^O_{\vec{\lambda}} \in \mathrm{SO}(n)$ to $\mathrm{Spin}(n)$. In addition, recall that the symmetry $\mathbb{Z}_2 = \langle (-1)^F \rangle$ defining the fermion parity and the NS/R sectors of $A(\mathfrak{a}) \oplus A(\mathfrak{a})_{\mathrm{tw}}$ is $(-1)^F = \mathfrak{z} \in \mathrm{Spin}(n)$. From the two lifts of $-1_{\mathrm{SO}(n)} = R_{\frac{1}{2}\vec{1}} \in \mathrm{SO}(n)$, $\mathfrak{z}$ can be specified as $\mathfrak{z} = +\psi_1 \cdots \psi_n$ as in (6.31), so we have

$$\mathfrak{z} = \hat{R}^O_{\frac{1}{2}\vec{1}} \,, \tag{13.118}$$

for any $O \in \mathrm{SO}(n)$ in the notation here. Then we can see that the transformations (13.105) and (13.106) under the shift $\vec{\lambda} \mapsto \vec{\lambda} + \frac{1}{2}\vec{1}$ are consistent with (12.44) and (12.43) up to anomalous phases.

# 14 The Conway Orbifold Partition Function

In this Section 14, we aim to calculate the Conway orbifold partition function $Z^{((V^{f\natural})^{\otimes 24}/\mathrm{Co}_1)\mathrm{R}}_{\mathrm{R}}(\tau)$. At the end of Section 12.2.4, we saw that the Conway orbifold partition function can be expressed as

$$\begin{aligned} &Z^{((V^{f\natural})^{\otimes 24}/\mathrm{Co}_1)\mathrm{R}}_{\mathrm{R}}(\tau) \\ &= \sum_{\substack{[[(g,h)]]\in(G^2_{\mathrm{com}}/\sim)/\mathrm{GL}(2,\mathbb{Z}) \\ \text{for } G=\mathrm{Co}_1}} |\mathrm{GL}(2,\mathbb{Z})\cdot[(g,h)]|\frac{|[h]_{\subset C_g}|}{|C_g|}\mathrm{Re}\Big[\beta_{\mathrm{anom}}(g,h)\big(Z^{(V^{f\natural})\mathrm{R}h}_{\mathrm{R}g}(\tau)\big)^{24}\Big], \end{aligned} \tag{14.1}$$

and this can be evaluated by the following three steps.

Step 1. Determine the conjugacy classes of the commuting pairs

$$G^2_{\mathrm{com}}/\sim := \{(g,h)\in G\times G \mid gh=hg\}/(g,h)\sim(kgk^{-1},khk^{-1}) \tag{14.2}$$

for $G=\mathrm{Co}_1$, and their $\mathrm{SL}(2,\mathbb{Z})$- and $\mathrm{GL}(2,\mathbb{Z})$-orbits.

Step 2. Compute the twisted partition functions $Z^{(V^{f\natural})\mathrm{R}h}_{\mathrm{R}g}(\tau)$.

Step 3. Determine the anomaly-canceling phases $\beta_{\mathrm{anom}}(g,h)$ for all $\mathrm{GL}(2,\mathbb{Z})$-orbits of $G^2_{\mathrm{com}}/\sim$.

We will carry out Step 1 and 2 in Section 14.1 and 14.2, respectively. As for Step 3, we do not have a decisive way to determine all the anomaly-canceling phases, but we will attempt to determine them from consistency conditions, and find one conjectural set of anomaly-canceling phases in Section 14.3.

This section presents original work based on [AKL+].

## 14.1 Step 1. Conjugacy Classes of Commuting Pairs

### 14.1.1 Conjugacy Classes of Commuting Pairs

We compute the conjugacy classes of commuting pairs of $\mathrm{Co}_1$. Recall from Section 2.3 that $\mathrm{Co}_1$ is the quotient of $\mathrm{Co}_0\cong\mathrm{Aut}(\Lambda_{24})$ by its center $\{\pm 1_{\mathrm{Co}_0}\}$, where $\Lambda_{24}$ is the Leech lattice, and generators of $\mathrm{Co}_0$ as a subgroup of $\mathrm{SL}(24,\mathbb{Z})$ are $A$ and $B$ in (2.46, 2.47). Our basic strategy is as follows. First, we rewrite the matrices $A$ and $B$ to the permutations of the vectors of squared length 4 in $\Lambda_{24}$, which are the shortest non-zero vectors, using Mathematica [Wol23]. Then, we can pass them to GAP [GAP22], which is good at handling permutation groups, to compute all what we need.[67]

[67] The author thanks Gerald Höhn for the information that this computation can be done on a modern laptop with only the standard features of Magma or GAP, and that some of the algorithms used by them are probabilistic and can require multiple runs.

**Represent the generators $A$ and $B$ as permutations**

Recall that the $\mathrm{Co}_0$ generated by $A$ and $B$ in (2.46, 2.47) as a subgroup of $\mathrm{SL}(24,\mathbb{Z})$ is the isometry group of $(\mathbb{Z}^{24}, q)$ with the symmetric bilinear form $q$ in (2.49), which makes $(\mathbb{Z}^{24}, q)$ isomorphic to the Leech lattice. According to the symmetric bilinear form $q$ (2.49), all the standard basis vectors $e_i = (0,\ldots,\overset{i}{\check{1}},\ldots,0)^T$ have squared length 4. By applying[68] $A$ and $B$ repeatedly on $e_i$'s, we can generate all the vectors of squared length 4, the set of which was denoted by $(\Lambda_{24})_2$ as defined in (2.41). We have $|(\Lambda_{24})_2| = 196560$ vectors.

$\mathrm{Co}_1$ acts on $(\Lambda_{24})_2/\{\pm 1\}$, so we choose one decomposition $(\Lambda_{24})_2 = P \sqcup N$ so that if $v \in P$ then $-v \in N$. We also impose $e_i \in P$ for all $i = 1,\ldots,24$. Now, we can represent the generators $\pm A$ and $\pm B$ of $\mathrm{Co}_1 = \mathrm{Co}_0/\{\pm 1_{\mathrm{Co}_0}\}$ as the permutations $\pi_{\pm A}$ and $\pi_{\pm B}$ of the $\frac{196560}{2} = 98280$ elements in $P$. Everything so far can be done by Mathematica.

**Compute the conjugacy classes of commuting pairs**

We pass the permutations $\pi_{\pm A}$ and $\pi_{\pm B}$ to GAP. More precisely, $\pi_{\pm A}$ and $\pi_{\pm B}$ should be represented as the permutations of the positions of elements of the ordered set $P$, to be passed to GAP. We can generate the group `Co1` from $\pi_{\pm A}$ and $\pi_{\pm B}$, and check that its order is in fact $|\mathrm{Co}_1| = 4157776806543360000$ by

```
Co1:=Group(A,B);
Order(Co1);
```

In Section 12.2.2 (see footnote 62), we introduced $\bigsqcup_{[g]\in G/\sim} C_g/\sim = (C_{g_1}/\sim)\sqcup\cdots\sqcup(C_{g_r}/\sim)$, where $\{g_1,\ldots,g_r\}$ is a complete set of representatives of the conjugacy classes in $G/\sim$, and mentioned that there is a bijection $\bigsqcup_{[g]\in G/\sim} C_g/\sim \to G^2_{\mathrm{com}}/\sim$ such that $[h]_{\subset C_g} \mapsto [(g,h)]$. Therefore, it suffices to compute $\bigsqcup_{[g]\in G/\sim} C_g/\sim$.

To do so, we first obtain the conjugacy classes in $G/\sim$ by

```
cc:=ConjugacyClasses(Co1);
```

The number of conjugacy classes is $r = 101$ for $G = \mathrm{Co}_1$. We take a representative $g_i$ from each conjugacy class by

```
conjrep:=List(cc,Representative);
```

Then, we compute the centralizers $C_{g_i}$ of the representatives by

```
centralizers:=[];
for i in [2..101] do
    Display(i);
    centralizers[i]:=Centralizer(Co1,conjrep[i]);
od;
centralizers[1]:=Co1;
```

[68] Recall that in our notation, $A$ and $B$ in (2.46, 2.47) act on the column vectors in the lattice $(\mathbb{Z}^{24}, q)$.

In the last line, we took $g_1 := 1_G$ and hence $C_{g_1} = G$. Finally, we can compute the conjugacy classes $C_{g_i}/\sim$ of the centralizers as

```
centralizerconjugacyclasses:=[];
for i in [2..101] do
    Display(i);
    centralizerconjugacyclasses[i]:=ConjugacyClasses(
       centralizers[i]);
od;
centralizerconjugacyclasses[1]:=cc;
```

In the last line, we stored the conjugacy classes $[g_i] \in G/\sim$ as the conjugacy classes $[h]_{\subset C_{g_1}}$ of the centralizer $C_{g_1} = G$. Note that the internal algorithms of some commands of GAP, for example `ConjugacyClasses`, are probabilistic, and sometimes stalls and/or eats up all the memory. In those cases, we need to rerun the loop from the point it stalled or got stuck.

We can compute the sizes $|[h]_{\subset C_g}|$ of the conjugacy classes and export the data as

```
sizes:=List(centralizerconjugacyclasses,x->List(x,Size));
PrintTo("sizes",sizes);
```

We find the total number of conjugacy classes is $|\bigsqcup_{[g]\in G/\sim} C_g/\sim| = |G^2_{\mathrm{com}}/\sim| = 7578$.

We also extract a representative $h$ from each conjugacy class $[h]_{\subset C_g}$ as

```
centralizerconjugacyclassreps:=List(centralizerconjugacyclasses,
   x->List(x,Representative));
```

In particular, the representative of the conjugacy class $[g_i]_{\subset C_{g_1}}$ of the centralizer $C_{g_1} = G$ is taken as $g_i$.

**Remark: Give the ATLAS names to the conjugacy classes in $\mathrm{Co}_1/\sim$**

*Remark* 14.1. The 101 conjugacy classes $[g]$ of $\mathrm{Co}_1$ are named "the order of $g$ + alphabet" as 1A, 2A, 2B, 2C, 3A, ..., 60A by ATLAS [CCN+85]. If we observe the character table in ATLAS, we find that the conjugacy classes can be distinguished, except for the pairs (23A, 23B) and (39A, 39B), by the triple

- the size $|[g]|$ of the conjugacy class,
- the character in the representation $\mathbf{276} = \wedge^2\mathbf{24}$,
- the character in the representation $\mathbf{299} = \mathrm{Sym}^2\mathbf{24} - \mathbf{1}$.

Here, $\mathbf{24}$ is the representation of $\mathrm{Co}_0 \cong \mathrm{Aut}(\Lambda_{24})$ on $\Lambda_{24} \otimes_{\mathbb{Z}} \mathbb{R}$, where $\Lambda_{24}$ is the Leech lattice. So $\mathrm{Co}_1$ only acts as a projective representation on $\mathbf{24}$, but the projective phases take values in $\pm 1$, and hence they vanish in the calculations (14.3, 14.4) of the characters. These data are also recorded in GAP, and we can retrieve them by

```
ch:=CharacterTable("Co1");
ClassNames(ch);
SizesConjugacyClasses(ch);
Irr(ch)[2];
Irr(ch)[3];
```

We have computed the conjugacy classes $\mathrm{Co}_1/\sim$. We will shortly see how to reconstruct a matrix $M_g \in \mathrm{SL}(24,\mathbb{Z})$ such that $\pm M_g$ is the matrix representation of $g \in \mathrm{Co}_1 = \mathrm{Co}_0/\{\pm 1_{\mathrm{Co}_0}\}$, from the permutation representation $\pi_g$ we have obtained. If we would like to give the ATLAS name to the conjugacy class $[g]$, we can compute its characters by using the formulae

$$\chi_{\wedge^2\rho}(M) = \frac{1}{2}\big(\chi_\rho(M)^2 - \chi_\rho(M^2)\big), \tag{14.3}$$

$$\chi_{\mathrm{Sym}^2\rho}(M) = \frac{1}{2}\big(\chi_\rho(M)^2 + \chi_\rho(M^2)\big). \tag{14.4}$$

The characters of the conjugacy classes 23A and 23B only differ in the complex representations $\chi_{17}$ and $\chi_{18} = \overline{\chi}_{17}$ of dimension 673750, and their values are just complex conjugate to each other $\chi_{17}(23\mathrm{A}) = \overline{\chi_{17}(23\mathrm{B})} = \frac{-1+\sqrt{-23}}{2}$. Similarly, the characters of 39A and 39B only differ in $\chi_{27}$ and $\chi_{28} = \overline{\chi}_{27}$ of dimension 2464749, and $\chi_{27}(39\mathrm{A}) = \overline{\chi_{27}(39\mathrm{B})} = \frac{-1+\sqrt{-39}}{2}$. This makes them extremely hard to distinguish. In practice, we will find no problem if we assign 23A, 23B and 39A, 39B arbitrarily to the conjugacy classes in $\mathrm{Co}_1/\sim$ in this thesis. See Table G.1 for the ATLAS names of the conjugacy classes in our code. *(Remark ends.)*

**Back to matrices in** $\mathrm{SL}(24,\mathbb{Z})$

Now, we have a representative $h \in \mathrm{Co}_1$ of each conjugacy class $[h]_{C_g}$ in the form of a permutation $\pi_h$ of the elements of $P$. We would like to reconstruct a matrix $M_h \in \mathrm{SL}(24,\mathbb{Z})$ such that $\pm M_h$ is the matrix representation of $h \in \mathrm{Co}_1 = \mathrm{Co}_0/\{\pm 1_{\mathrm{Co}_0}\}$.

To do so, recalling that we imposed $e_i \in P$, we first have to know the images $\pi_h(e_i)$ of the basis vectors $e_i$. Let $e_i$ be the $n_i$-th entry of $P$. We find the position of each image $\pi_h(e_i)$ in $P$ by

```
basis:=[n1,n2, ... , n24];
images:=List(centralizerconjugacyclassreps,
    x->List(x,y->List(basis,z->z^y)));
PrintTo("images",images);
```

Then we can look up the images $\pi_h(e_i) \in P$ in Mathematica.

Next, we choose the signs of

$$M_h e_i = \pm \pi_h(e_i) \tag{14.5}$$

for each $i = 1, \ldots, 24$, so that they satisfy

$$(M_h e_i)^T q (M_h e_j) = e_i^T q e_j. \tag{14.6}$$

In practice, we first choose $M_h e_1 = +\pi_h(e_1)$. This choice corresponds to the overall sign $\pm M_h$, so we can be careless about it. Then we recursively determine the signs of $M_h e_i = \pm\pi_h(e_i)$ for $i = 2, \ldots, 24$ using (14.6).

Since the representative of the conjugacy class $[g_i]_{\subset C_{g_1}}$ of the centralizer $C_{g_1} = G$ is taken as $g_i$, we have also constructed a matrix $M_{g_i} \in \mathrm{SL}(24, \mathbb{Z})$ such that $\pm M_{g_i}$ is the matrix representation of $g_i \in \mathrm{Co}_1$ in the above procedure. In conclusion, we have obtained a pair $(M_g, M_h)$ of such matrices for the chosen representative $(g, h)$ of each conjugacy class of commuting pairs $[(g, h)] \in G^2_{\mathrm{com}}/\sim$, or equivalently the chosen $(g, h)$ of each $[h]_{\subset C_g} \in \bigsqcup_{[g]\in G/\sim} C_g/\sim$, for $G = \mathrm{Co}_1$. The obtained data are stored in `data.m`.

### 14.1.2 Their $\mathrm{SL}(2, \mathbb{Z})$- and $\mathrm{GL}(2, \mathbb{Z})$-Orbits

We continue to use GAP to compute the $\mathrm{GL}(2, \mathbb{Z})$-action on the conjugacy classes $[h]_{\subset C_g}$. We introduce an abbreviation

```
cr:=centralizerconjugacyclassreps;;
```

(The double semicolon suppresses the output of the computation.)

**Modular T transformation**

We first compute the action of the modular T transformation $[h]_{\subset C_g} \mapsto [gh]_{\subset C_g}$. We prepare a function **findclass** such that

$$\textbf{findclass}(C_g, \text{representatives } \{h_1, \ldots, h_r\} \text{ of } C_g/\sim, gh) = i \text{ such that } h_i \sim gh$$

as

```
findclass := function(group,cc,elem)
    local i;
    for i in [1..Length(cc)] do
        if IsConjugate(group,cc[i],elem) then
            return i;
        fi;
    od;
end;
```

Then the results of the modular T transformation can be obtained as

```
taction:=[];
for i in [1..101] do
    taction[i]:=List(cr[i],
        x-> findclass(centralizers[i],cr[i],cr[1][i] * x));
od;
```

**Modular S transformation**

Next, we compute the action of the modular S transformation $[h]_{\subset C_g} \mapsto [kg^{-1}k^{-1}]_{\subset C_{khk^{-1}}}$, where $k \in \mathrm{Co}_1$ is chosen so that $khk^{-1}$ is one of the representatives $\{g_1, \ldots, g_r\}$ of $G/\sim$ used in the construction of $\bigsqcup_{[g]\in G/\sim} C_g$. Using **findclass** defined above, we prepare a function **findclassS** such that

$$\textbf{findclassS}(g,h) = (i,j) \text{ such that } [kg^{-1}k^{-1}]_{\subset C_{khk^{-1}}} = [h_j]_{\subset C_{g_i}}$$

as

```
findclassS := function(g,h)
    local i,k,d,j;
    i:=findclass(Co1,conjrep,h);
    k:=RepresentativeAction(Co1,conjrep[i],h);
    d:=k*Inverse(g)*Inverse(k);
    j:=findclass(centralizers[i],cr[i],d);
    return [i,j];
end;
```

Then the results of the modular S transformation can be obtained as

```
saction:=[];
for i in [1..101] do
    saction[i]:=List(cr[i], x-> findclassS(conjrep[i],x));
od;
```

**The P transformation**

The results of the P transformation $[h]_{\subset C_g} \mapsto [h^{-1}]_{\subset C_g}$ can be obtained in a way similar to that for the modular T transformation as

```
paction:=[];
for i in [1..101] do
    taction[i]:=List(cr[i],
        x-> findclass(centralizers[i],cr[i],x^-1));
od;
```

The data obtained so far are stored in `gl2z-action.m`.

**The $\mathrm{SL}(2,\mathbb{Z})$- and $\mathrm{GL}(2,\mathbb{Z})$-orbits**

Finally, for the computation of the $\mathrm{SL}(2,\mathbb{Z})$-orbits in $\bigsqcup_{[g]\in G/\sim} C_g/\sim$, it is useful to pass the data of the modular T and S transformations obtained above to Mathematica, and use the command `Graph[vertices, edges]`. We can find there are 445 $\mathrm{SL}(2,\mathbb{Z})$-orbits. See `SL2ZOrbits.nb` for the source code.

To compute the P transformation of the $\mathrm{SL}(2,\mathbb{Z})$-orbits in Mathematica, we only have to take a representative of each $\mathrm{SL}(2,\mathbb{Z})$-orbit, and find the $\mathrm{SL}(2,\mathbb{Z})$-orbit to which the image of the representative under the P transformation belongs. We find there are 9 pairs of $\mathrm{SL}(2,\mathbb{Z})$-orbits interchanged under the P transformation, and the remaining 427 $\mathrm{SL}(2,\mathbb{Z})$-orbits are mapped to themselves under the P transformation.

## 14.2 Step 2. Twisted Partition Functions of Duncan's Module

### 14.2.1 Theoretical Foundations

Recall from Section 6.2 that the construction of Duncan's module $V^{f\natural}$ is similar to the $\mathbb{Z}_2$-orbifold of the theory $A(\mathfrak{a})$ of 24 real free chiral fermions as in (6.34).

| $\mathbb{Z}_2=\langle \mathfrak{z}\rangle$ | NS | R |
|---|---|---|
| | $A(\mathfrak{a})$ | $A(\mathfrak{a})_{\mathrm{tw}}$ |
| even | $A^0(\mathfrak{a})$ | $A^0(\mathfrak{a})_{\mathrm{tw}}$ |
| odd | $A^1(\mathfrak{a})$ | $A^1(\mathfrak{a})_{\mathrm{tw}}$ |

'$\langle\mathfrak{z}\rangle$-orbifold' $\rightleftarrows$ '$\langle -1_{\mathrm{Spin}(\mathfrak{a})}\rangle$-orbifold'

| $\mathbb{Z}_2=\langle -1_{\mathrm{Spin}(\mathfrak{a})}\rangle$ | NS | R |
|---|---|---|
| | $V^{f\natural}$ | $V^{f\natural}_{\mathrm{tw}}$ |
| even | $A^0(\mathfrak{a})$ | $A^1(\mathfrak{a})$ |
| odd | $A^0(\mathfrak{a})_{\mathrm{tw}}$ | $A^1(\mathfrak{a})_{\mathrm{tw}}$ |

(14.7)

Note that this symmetry $\mathbb{Z}_2=\langle\mathfrak{z}\rangle$ of $A(\mathfrak{a})$ is non-anomalous because $SH(\mathbb{Z}_2)=\mathbb{Z}_8$ and there are 24 real free fermions. We would like to calculate the twisted partition functions of Duncan's module twisted by $\mathrm{Co}_1\subset\mathrm{SemiSpin}(24)$, which fits into the diagram (6.46).

$$\begin{array}{ccc} & \mathrm{Co}_0\subset \mathrm{Spin}(24)\curvearrowright A(\mathfrak{a})\oplus A(\mathfrak{a})_{\mathrm{tw}} & \\ {\scriptstyle /\langle\mathfrak{z}\rangle}\swarrow & & \searrow{\scriptstyle \cong} \\ \mathrm{Co}_1\subset \mathrm{SemiSpin}(24)\curvearrowright V^{f\natural} & & \mathrm{Co}_0\subset \mathrm{SO}(24)\curvearrowright A(\mathfrak{a}) \end{array} \tag{14.8}$$

As mentioned in Section 12.1.4, this $\mathrm{Co}_1$ symmetry is anomalous, so there are no canonical $\mathrm{U}(1)$ phases of the twisted partition functions such that the anomalous phases are trivial.

Before getting into the symmetry $\mathrm{Co}_1\subset\mathrm{SemiSpin}(24)$, we begin with the twisted partition functions $Z^{(V^{f\natural})^{\mathrm{Yh}}_{\mathrm{X}g}}(\tau)$ twisted by $g,h\in\mathrm{Spin}(24)$. From its construction (14.7), it is expected that $Z^{(V^{f\natural})^{\mathrm{Yh}}_{\mathrm{X}g}}(\tau)$ can be represented as a signed sum of the twisted partition functions $Z^{(12)^{\mathrm{Yh}}_{\mathrm{X}g}}(\tau)$ of 12 complex free fermions $A(\mathfrak{a})$. For example, schematically,

$$\begin{aligned} Z^{(V^{f\natural})^{\mathrm{R}}_{\mathrm{R}}} &= \mathrm{Tr}_{V^{\natural}_{\mathrm{tw}}}[(-1_{\mathrm{Spin}(\mathfrak{a})})q^{L_0-\frac{1}{2}}] && (14.9)\\ &= \mathrm{Tr}_{A^1(\mathfrak{a})}[q^{L_0-\frac{1}{2}}]-\mathrm{Tr}_{A^1(\mathfrak{a})_{\mathrm{tw}}}[q^{L_0-\frac{1}{2}}] && (14.10)\\ &= \frac{1}{2}(Z^{(12)^{\mathrm{NS}}_{\mathrm{NS}}}-Z^{(12)^{\mathrm{R}}_{\mathrm{NS}}})-\frac{1}{2}(Z^{(12)^{\mathrm{NS}}_{\mathrm{R}}}-Z^{(12)^{\mathrm{R}}_{\mathrm{R}}}). && (14.11)\end{aligned}$$

The twisted partition functions $Z^{(12)^{\mathrm{Yh}}_{\mathrm{X}g}}(\tau)$ twisted by $\mathrm{Spin}(24)$ were already discussed in Section 13.4. However, as noted in Section 13, since the $\mathrm{Spin}(24)$ symmetry is anomalous, there are no canonical $\mathrm{U}(1)$ phases of $Z^{(12)^{\mathrm{Yh}}_{\mathrm{X}g}}(\tau)$ such that the anomalous phases are trivial. The phases of

$Z^{(12)\mathrm{Y}h}_{\mathrm{X}g}(\tau)$ affect the result of their signed sums, so we have to adjust them so that the resulting $Z^{(V^{f\natural})\mathrm{Y}h}_{\mathrm{X}g}(\tau)$'s satisfy some consistency conditions.

We adopt the phases of $Z^{(12)\mathrm{Y}h}_{\mathrm{X}g}(\tau)$ as defined in (13.50)–(13.62), and represent the twisted partition functions of Duncan's module $V^{f\natural}$ as their signed sums as

$$Z^{(V^{f\natural})\mathrm{NS}\vec{\lambda}_h}_{\mathrm{NS}\vec{\lambda}_g}(\tau) := \frac{1}{2}\Big(Z^{(12)\mathrm{NS}\vec{\lambda}_h}_{\mathrm{NS}\vec{\lambda}_g}(\tau) + Z^{(12)\mathrm{R}\vec{\lambda}_h}_{\mathrm{NS}\vec{\lambda}_g}(\tau) + Z^{(12)\mathrm{NS}\vec{\lambda}_h}_{\mathrm{R}\vec{\lambda}_g}(\tau) + Z^{(12)\mathrm{R}\vec{\lambda}_h}_{\mathrm{R}\vec{\lambda}_g}(\tau)\Big), \tag{14.12}$$

$$Z^{(V^{f\natural})\mathrm{R}\vec{\lambda}_h}_{\mathrm{NS}\vec{\lambda}_g}(\tau) := \frac{1}{2}\Big(Z^{(12)\mathrm{NS}\vec{\lambda}_h}_{\mathrm{NS}\vec{\lambda}_g}(\tau) + Z^{(12)\mathrm{R}\vec{\lambda}_h}_{\mathrm{NS}\vec{\lambda}_g}(\tau) - Z^{(12)\mathrm{NS}\vec{\lambda}_h}_{\mathrm{R}\vec{\lambda}_g}(\tau) - Z^{(12)\mathrm{R}\vec{\lambda}_h}_{\mathrm{R}\vec{\lambda}_g}(\tau)\Big), \tag{14.13}$$

$$Z^{(V^{f\natural})\mathrm{NS}\vec{\lambda}_h}_{\mathrm{R}\vec{\lambda}_g}(\tau) := \frac{1}{2}\Big(Z^{(12)\mathrm{NS}\vec{\lambda}_h}_{\mathrm{NS}\vec{\lambda}_g}(\tau) - Z^{(12)\mathrm{R}\vec{\lambda}_h}_{\mathrm{NS}\vec{\lambda}_g}(\tau) + Z^{(12)\mathrm{NS}\vec{\lambda}_h}_{\mathrm{R}\vec{\lambda}_g}(\tau) - Z^{(12)\mathrm{R}\vec{\lambda}_h}_{\mathrm{R}\vec{\lambda}_g}(\tau)\Big), \tag{14.14}$$

$$Z^{(V^{f\natural})\mathrm{R}\vec{\lambda}_h}_{\mathrm{R}\vec{\lambda}_g}(\tau) := \frac{1}{2}\Big(Z^{(12)\mathrm{NS}\vec{\lambda}_h}_{\mathrm{NS}\vec{\lambda}_g}(\tau) - Z^{(12)\mathrm{R}\vec{\lambda}_h}_{\mathrm{NS}\vec{\lambda}_g}(\tau) - Z^{(12)\mathrm{NS}\vec{\lambda}_h}_{\mathrm{R}\vec{\lambda}_g}(\tau) + Z^{(12)\mathrm{R}\vec{\lambda}_h}_{\mathrm{R}\vec{\lambda}_g}(\tau)\Big). \tag{14.15}$$

Then we can see that these twisted partition functions $Z^{(V^{f\natural})\mathrm{Y}\vec{\lambda}_h}_{\mathrm{X}\vec{\lambda}_g}(\tau)$ satisfy some consistency conditions as follows.

(1) The modular transformations of $Z^{(V^{f\natural})\mathrm{Y}\vec{\lambda}_h}_{\mathrm{X}\vec{\lambda}_g}(\tau)$ follow from the modular transformations (13.97)–(13.104) of $Z^{(12)\mathrm{Y}h}_{\mathrm{X}g}(\tau)$ as

$$Z^{(V^{f\natural})\mathrm{NS}\vec{\lambda}_h}_{\mathrm{NS}\vec{\lambda}_g}(\tau+1) = -e^{\pi\sqrt{-1}\vec{\lambda}_g\cdot\vec{\lambda}_g} Z^{(V^{f\natural})\mathrm{R}(\vec{\lambda}_g+\vec{\lambda}_h)}_{\mathrm{NS}\vec{\lambda}_g}(\tau), \tag{14.16}$$

$$Z^{(V^{f\natural})\mathrm{R}\vec{\lambda}_h}_{\mathrm{NS}\vec{\lambda}_g}(\tau+1) = -e^{\pi\sqrt{-1}\vec{\lambda}_g\cdot\vec{\lambda}_g} Z^{(V^{f\natural})\mathrm{NS}(\vec{\lambda}_g+\vec{\lambda}_h)}_{\mathrm{NS}\vec{\lambda}_g}(\tau), \tag{14.17}$$

$$Z^{(V^{f\natural})\mathrm{NS}\vec{\lambda}_h}_{\mathrm{R}\vec{\lambda}_g}(\tau+1) = \quad e^{\pi\sqrt{-1}\vec{\lambda}_g\cdot\vec{\lambda}_g} Z^{(V^{f\natural})\mathrm{NS}(\vec{\lambda}_g+\vec{\lambda}_h)}_{\mathrm{R}\vec{\lambda}_g}(\tau), \tag{14.18}$$

$$Z^{(V^{f\natural})\mathrm{R}\vec{\lambda}_h}_{\mathrm{R}\vec{\lambda}_g}(\tau+1) = \quad e^{\pi\sqrt{-1}\vec{\lambda}_g\cdot\vec{\lambda}_g} Z^{(V^{f\natural})\mathrm{R}(\vec{\lambda}_g+\vec{\lambda}_h)}_{\mathrm{R}\vec{\lambda}_g}(\tau), \tag{14.19}$$

and

$$Z^{(V^{f\natural})\mathrm{NS}\vec{\lambda}_h}_{\mathrm{NS}\vec{\lambda}_g}(-\frac{1}{\tau}) = e^{-2\pi\sqrt{-1}\vec{\lambda}_g\cdot\vec{\lambda}_h} Z^{(V^{f\natural})\mathrm{NS}(-\vec{\lambda}_g)}_{\mathrm{NS}\vec{\lambda}_h}(\tau), \tag{14.20}$$

$$Z^{(V^{f\natural})\mathrm{R}\vec{\lambda}_h}_{\mathrm{NS}\vec{\lambda}_g}(-\frac{1}{\tau}) = e^{-2\pi\sqrt{-1}\vec{\lambda}_g\cdot\vec{\lambda}_h} Z^{(V^{f\natural})\mathrm{NS}(-\vec{\lambda}_g)}_{\mathrm{R}\vec{\lambda}_h}(\tau), \tag{14.21}$$

$$Z^{(V^{f\natural})\mathrm{NS}\vec{\lambda}_h}_{\mathrm{R}\vec{\lambda}_g}(-\frac{1}{\tau}) = e^{-2\pi\sqrt{-1}\vec{\lambda}_g\cdot\vec{\lambda}_h} Z^{(V^{f\natural})\mathrm{R}(-\vec{\lambda}_g)}_{\mathrm{NS}\vec{\lambda}_h}(\tau), \tag{14.22}$$

$$Z^{(V^{f\natural})\mathrm{R}\vec{\lambda}_h}_{\mathrm{R}\vec{\lambda}_g}(-\frac{1}{\tau}) = e^{-2\pi\sqrt{-1}\vec{\lambda}_g\cdot\vec{\lambda}_h} Z^{(V^{f\natural})\mathrm{R}(-\vec{\lambda}_g)}_{\mathrm{R}\vec{\lambda}_h}(\tau). \tag{14.23}$$

These are consistent with (12.38)–(12.41) up to $\mathrm{U}(1)$ phases.

(2) The transformations of $Z^{(V^{f\natural})\mathrm{Y}\vec{\lambda}_h}_{\mathrm{X}\vec{\lambda}_g}(\tau)$ under the angle shift $\vec{\lambda} \mapsto \vec{\lambda} + \vec{m}$ with $\vec{m} \in \mathbb{Z}^{12}$ follow

from the transformations (13.107, 13.108) of $Z^{(12)\mathrm{Y}h}_{\mathrm{X}g}(\tau)$ as

$$Z^{(V^{f\natural})\mathrm{Y}\vec{\lambda}_h}_{\mathrm{X}(\vec{\lambda}_g+\vec{m})}(\tau) = \begin{cases} e^{-2\pi\sqrt{-1}\vec{m}\cdot\vec{\lambda}_h} Z^{(V^{f\natural})\mathrm{Y}\vec{\lambda}_h}_{\mathrm{X}\vec{\lambda}_g}(\tau) & (\textstyle\sum_i m_i \text{ is even}), \\ e^{-2\pi\sqrt{-1}\vec{m}\cdot\vec{\lambda}_h} Z^{(V^{f\natural})\mathrm{Y}\vec{\lambda}_h}_{\overline{\mathrm{X}}\vec{\lambda}_g}(\tau) & (\textstyle\sum_i m_i \text{ is odd}), \end{cases} \tag{14.24}$$

$$Z^{(V^{f\natural})\mathrm{Y}(\vec{\lambda}_h+\vec{m})}_{\mathrm{X}\vec{\lambda}_g}(\tau) = \begin{cases} Z^{(V^{f\natural})\mathrm{Y}\vec{\lambda}_h}_{\mathrm{X}\vec{\lambda}_g}(\tau) & (\textstyle\sum_i m_i \text{ is even}), \\ Z^{(V^{f\natural})\overline{\mathrm{Y}}\vec{\lambda}_h}_{\mathrm{X}\vec{\lambda}_g}(\tau) & (\textstyle\sum_i m_i \text{ is odd}). \end{cases} \tag{14.25}$$

Recall that $-1_{\mathrm{Spin}(24)} = \hat{R}_{\vec{m}}$ if $\sum_i m_i$ is odd. Since the symmetry $\langle -1_{\mathrm{Spin}(24)} \rangle$ defines the fermion parity and the NS/R sectors of $V^{f\natural} \oplus V^{f\natural}_{\mathrm{tw}}$, these transformations are consistent with (12.44, 12.43) up to $\mathrm{U}(1)$ phases.

(3) The transformations of $Z^{(V^{f\natural})\mathrm{Y}\vec{\lambda}_h}_{\mathrm{X}\vec{\lambda}_g}(\tau)$ under the angle shift $\vec{\lambda} \mapsto \vec{\lambda} + \frac{1}{2}\vec{1}$ follow from the transformations (13.105, 13.106) of $Z^{(12)\mathrm{Y}h}_{\mathrm{X}g}(\tau)$ as

$$Z^{(V^{f\natural})\mathrm{Y}\vec{\lambda}_h}_{\mathrm{X}(\vec{\lambda}_g+\frac{1}{2}\vec{1})}(\tau) = \begin{cases} e^{-\pi\sqrt{-1}\sum_i (\lambda_h)_i} Z^{(V^{f\natural})\mathrm{Y}\vec{\lambda}_h}_{\mathrm{X}\vec{\lambda}_g}(\tau) & (\mathrm{Y} = \mathrm{NS}), \\ -e^{-\pi\sqrt{-1}\sum_i (\lambda_h)_i} Z^{(V^{f\natural})\mathrm{Y}\vec{\lambda}_h}_{\mathrm{X}\vec{\lambda}_g}(\tau) & (\mathrm{Y} = \mathrm{R}), \end{cases} \tag{14.26}$$

$$Z^{(V^{f\natural})\mathrm{Y}(\vec{\lambda}_h+\frac{1}{2}\vec{1})}_{\mathrm{X}\vec{\lambda}_g}(\tau) = \begin{cases} Z^{(V^{f\natural})\mathrm{Y}\vec{\lambda}_h}_{\mathrm{X}\vec{\lambda}_g}(\tau) & (\mathrm{X} = \mathrm{NS}), \\ -Z^{(V^{f\natural})\mathrm{Y}\vec{\lambda}_h}_{\mathrm{X}\vec{\lambda}_g}(\tau) & (\mathrm{X} = \mathrm{R}). \end{cases} \tag{14.27}$$

Under the natural projection $\mathrm{Spin}(24) \to \mathrm{SemiSpin}(24)$, two elements $g$ and $\mathfrak{z}g$ in $\mathrm{Spin}(24)$ are mapped to the same element in $\mathrm{SemiSpin}(24)$. Recalling (13.118) $\mathfrak{z} = \hat{R}_{\frac{1}{2}\vec{1}}$, we can see from these transformations that the twisted partition functions $Z^{(V^{f\natural})\mathrm{Y}\vec{\lambda}_h}_{\mathrm{X}\vec{\lambda}_g}(\tau)$ twisted by $\mathrm{Spin}(24)$ are also well-defined as the twisted partition functions twisted by $\mathrm{SemiSpin}(24)$, up to $\mathrm{U}(1)$ phases. This is consistent with the fact that the projective $\mathrm{SemiSpin}(24)$-action can be regarded as the $\mathrm{Spin}(24)$-action (see footnote 60), which is still projective on twisted sectors in general.

Lastly, we also note that these twisted partition functions $Z^{(V^{f\natural})\mathrm{Y}}_{\mathrm{R}\vec{\lambda}_g}(\tau)$ of the R sector for $\vec{\lambda}_h = \vec{0}$ coincides with the formulae of [JFF23, Lemma 3.2.1].

We finally consider the twisted partition functions $Z^{(V^{f\natural})\mathrm{Y}h}_{\mathrm{X}g}(\tau)$ twisted by commuting $g, h \in \mathrm{Co}_1 \subset \mathrm{SemiSpin}(24)$. There are two lifts of $g$ to $\mathrm{Spin}(24)$, and if we write[69] one of them as $\check{g}$, then the other one is $\mathfrak{z}\check{g}$. We map $\check{g}$ to $\varphi(\check{g}) \in \mathrm{SO}(24)$. In the actual calculation, we would like to use the matrices $M_g$ and $M_h$ obtained in Section 14.1.1 as $\varphi(\check{g})$ and $\varphi(\check{h})$ respectively, but these matrices $M_g$ are represented with respect to the basis such that the symmetric bilinear form of the Leech lattice is represent as $q$ in (2.49), so we have to change the basis to the orthonormal one $\psi_1, \dots, \psi_{24}$ such that $\mathfrak{z} = \psi_1 \cdots \psi_{24}$ under the polarization (6.17) is in $\mathrm{Co}_0 \subset \mathrm{Spin}(24)$ (see

[69] We let $\hat{g}$ denote a lift of $g \in \mathrm{SO}(24)$ to $\mathrm{Spin}(24)$, and $\check{g}$ denote a lift of $g \in \mathrm{SemiSpin}(24)$ to $\mathrm{Spin}(24)$.

Section 6.2.2). We write the matrix after the change of the basis as $M_g^{\mathrm{ort}}$, and then it can be used as $\varphi(\check{g})$. Note that the matrix representation $\pm M_g^{\mathrm{ort}}$ of $g \in \mathrm{Co}_1 = \mathrm{Co}_0/\{\pm 1_{\mathrm{Co}_0}\}$ is only defined up to a sign, but this sign can be absorbed to the degree of freedom in naming which of the two lifts of $g$ to $\mathrm{Spin}(24)$ $\check{g}$ is the other one $\mathfrak{z}\check{g}$. We can summarize some relevant elements of the groups as in the following diagram, in accordance with (14.8).

$$\begin{array}{c|c} \check{g} \in \mathrm{Co}_0 & -\check{g} \notin \mathrm{Co}_0 \\ \hline \mathfrak{z}\check{g} \in \mathrm{Co}_0 & -\mathfrak{z}\check{g} \notin \mathrm{Co}_0 \end{array}$$

$$\swarrow^{/\langle \mathfrak{z} \rangle} \qquad \searrow^{/\langle -1_{\mathrm{Spin}(24)} \rangle}$$

$$g \in \mathrm{Co}_1 \mid -g \notin \mathrm{Co}_1 \qquad \frac{M_g^{\mathrm{ort}} \in \mathrm{Co}_0}{-M_g^{\mathrm{ort}} \in \mathrm{Co}_0} \tag{14.28}$$

Even though $g$ and $h$ commute as $gh = hg$ in $\mathrm{SemiSpin}(24)$, their lifts $\check{g}$ and $\check{h}$ does not necessarily commute. There are two possibilities: $\check{g}$ and $\check{h}$ also commute as $\check{g}\check{h} = \check{h}\check{g}$, or they anticommute as $\check{g}\check{h} = \mathfrak{z}\check{h}\check{g}$ in $\mathrm{Spin}(24)$. We have $M_g M_h = M_h M_g$ in the former case, and $M_g M_h = -M_h M_g$ in the latter case. When $\check{g}$ and $\check{h}$ anticommute, then the twisted partition function vanishes as $Z^{(V^{f\natural})\mathrm{Y}h}_{\mathrm{X}g}(\tau) \sim Z^{(V^{f\natural})\mathrm{Y}\check{h}}_{\mathrm{X}\check{g}}(\tau) = 0$, as in (12.17). So we will focus on the commuting case $M_g M_h = M_h M_g$ below. Among the 445 $\mathrm{SL}(2,\mathbb{Z})$-orbits, there are 97 $\mathrm{SL}(2,\mathbb{Z})$-orbits where $\check{g}$ and $\check{h}$ anticommute, and these $\mathrm{SL}(2,\mathbb{Z})$-orbits are all mapped to themselves under the P transformation.

We simultaneously diagonalize $M_g^{\mathrm{ort}}$ and $M_h^{\mathrm{ort}}$ as $M_g^{\mathrm{ort}} = R^O_{\vec{\lambda}_{M_g^{\mathrm{ort}}}}$ and $M_h^{\mathrm{ort}} = R^O_{\vec{\lambda}_{M_h^{\mathrm{ort}}}}$ by $O \in \mathrm{SO}(24)$. As seen in Section 13.4, $\check{g}$ is then either $\hat{R}^O_{\vec{\lambda}_{M_g^{\mathrm{ort}}}}$ or $\hat{R}^O_{\vec{\lambda}_{M_g^{\mathrm{ort}}}+(0,\dots,0,1)}$. Let $\vec{\lambda}_{\check{g},O}$ denote the correct one of $\vec{\lambda}_{M_g^{\mathrm{ort}}}$ or $\vec{\lambda}_{M_g^{\mathrm{ort}}} + (0,\dots,0,1)$. Similarly, let $\vec{\lambda}_{\check{h},O}$ denote the correct one of $\vec{\lambda}_{M_h^{\mathrm{ort}}}$ or $\vec{\lambda}_{M_h^{\mathrm{ort}}} + (0,\dots,0,1)$. Then, from the discussion (3) above, our desired twisted partition function $Z^{(V^{f\natural})\mathrm{Y}h}_{\mathrm{X}g}(\tau)$ can be calculated as $Z^{(V^{f\natural})\mathrm{Y}\vec{\lambda}_{\check{h},O}}_{\mathrm{X}\vec{\lambda}_{\check{g},O}}(\tau)$ defined in (14.12)–(14.15).

The remaining problem is how to detect which of $\vec{\lambda}_{M_g^{\mathrm{ort}}}$ or $\vec{\lambda}_{M_g^{\mathrm{ort}}} + (0,\dots,0,1)$ is $\vec{\lambda}_{\check{g},O}$. If we mistake it, then the corresponding element $-\check{g} \in \mathrm{Spin}(24)$ is not contained in $\mathrm{Co}_0 \subset \mathrm{Spin}(24)$, or from the perspective of discussion (2) above, we end up obtaining the twisted partition function of the opposite sector of NS/R.

Recall that the $\mathrm{Co}_0$ symmetry preserves the $\mathcal{N} = 1$ superconformal algebra of $V^{f\natural}$. Conversely, the only element of the weight-$\frac{3}{2}$ subspace $\mathbf{2}^{\mathbf{11}}_{+}$ of $A(\mathfrak{a})^0_{\mathrm{tw}}$ preserved by the $\mathrm{Co}_0$-action is the supercurrent of $V^{f\natural}$, up to scaler multiplication. Therefore, by explicitly examining which of $\hat{R}^O_{\vec{\lambda}_{M_g^{\mathrm{ort}}}}$ or $\hat{R}^O_{\vec{\lambda}_{M_g^{\mathrm{ort}}}+(0,\dots,0,1)}$ preserves the supercurrent of $V^{f\natural}$, we can detect which of them is the correct one $\check{g} \in \mathrm{Co}_0$. The wrong one reverses the sign of the supercurrent.

*Remark* 14.2. Suppose we would like to calculate the R-R twisted partition function $Z^{(V^{f\natural})\mathrm{R}h}_{\mathrm{R}g}(\tau)$. Since the $\mathcal{N} = 1$ superconformal algebra of $V^{f\natural}$ is preserved by the $\mathrm{Co}_1$ symmetry, the R-R

twisted partition functions are constant, whereas the other ones (NS-NS, NS-R, and R-NS) are not constant in general. This fact can be used to detect which of $\vec{\lambda}_{M_g^{\mathrm{ort}}}$ or $\vec{\lambda}_{M_g^{\mathrm{ort}}} + (0, \dots, 0, 1)$ is $\vec{\lambda}_{\check{g},O}$. This strategy is adopted in [JFF23], and it indeed enables us to calculate the R-R twisted partition functions for the trivial temporal twist $h = 1_{\mathrm{Co}_1}$.

Unfortunately, however, if we consider the twisted partition functions with both spatial and temporal twists nontrivial, we encounter the cases where all the twisted partition functions calculated with $\vec{\lambda}_{M_g^{\mathrm{ort}}}$ or $\vec{\lambda}_{M_g^{\mathrm{ort}}} + (0, \dots, 0, 1)$, and $\vec{\lambda}_{M_h^{\mathrm{ort}}}$ or $\vec{\lambda}_{M_h^{\mathrm{ort}}} + (0, \dots, 0, 1)$ are constants, and their absolute values are different depending on whether we added $(0, \dots, 0, 1)$ or not. The above strategy does not work in such cases.

One important caveat is that we cannot assign 'the correct angles' to each element $\check{g} \in \mathrm{Co}_0$ using the above strategy, because the correct angles $\vec{\lambda}_{\check{g},O}$ depend not only on $\check{g} \in \mathrm{Spin}(n)$ but also on the block-diagonalizing matrix $O \in \mathrm{SO}(n)$, as we remarked in Section 13.4. So, we have no choice but to calibrate the angles for each given commuting pair $\check{g}, \check{h} \in \mathrm{Co}_0$, individually.[70] *(Remark ends.)*

### 14.2.2 Implementation Methods

Having explained our theoretical approach to compute the twisted partition functions of Duncan's module $V^{f\natural}$, we now describe how to practically carry it out using Mathematica and C++ in this Section 14.2.2. See `PF.nb`, `FindSupercurrent.cpp`, and `CalibrateAngles.cpp` for the source code.

#### Change the basis to an appropriate orthonormal one

Recall that the matrix $M_g$ obtained in Section 14.1.1 is represented with respect to the standard basis $e_1, \dots, e_{24}$ of $\mathbb{Z}^{24}$, where the symmetric bilinear form $q$ in (2.49) makes $(\mathbb{Z}^{24}, q)$ isometric to the Leech lattice $\Lambda_{24}$. That is, $M_g$ acts on the vectors of $\mathbb{Z}^{24}$ as[71]

$$\begin{pmatrix} e_1 & \cdots & e_{24} \end{pmatrix} \begin{pmatrix} c_1 \\ \vdots \\ c_{24} \end{pmatrix} \mapsto \begin{pmatrix} e_1 & \cdots & e_{24} \end{pmatrix} M_g \begin{pmatrix} c_1 \\ \vdots \\ c_{24} \end{pmatrix}, \tag{14.29}$$

and satisfies $M_g^T q M_g = q$.

We diagonalize the symmetric matrix $q$ as $V^{-1} q V = \mathrm{diag}(d_1, \dots, d_{24})$ by $V \in \mathrm{O}(24)$. By defining $W := V \mathrm{diag}(d_1^{-\frac{1}{2}}, \dots, d_{24}^{-\frac{1}{2}})$, we have $W^T q W = I$. Then the basis $(e_1 \cdots e_{24})W$ is an orthonormal basis of $\Lambda_{24} \otimes_{\mathbb{Z}} \mathbb{R}$, and with respect to this basis, $M_g$ is represented as $W^{-1} M_g W$.

[70] If the $\mathrm{Spin}(24)$-conjugacy class $[\check{g}]$ of $\check{g} \in \mathrm{Spin}(24)$ does not contain both $\hat{R}_{\vec{\lambda}_{M_g^{\mathrm{ort}}}}$ and $\hat{R}_{\vec{\lambda}_{M_g^{\mathrm{ort}}} + (0,\dots,0,1)}$, then once we find the correct angle $\vec{\lambda}_{\check{g},O}$ for $\check{g}$, we can calibrate the angles for any other elements in the same conjugacy class $[\check{g}]$. However, this is not the case for all of the elements $\check{g}$ in $\mathrm{Co}_0$. That is, there are elements $\check{g} \in \mathrm{Co}_0$ such that its $\mathrm{Spin}(24)$-conjugacy class $[\check{g}]$ contains both $\hat{R}_{\vec{\lambda}_{M_g^{\mathrm{ort}}}}$ and $\hat{R}_{\vec{\lambda}_{M_g^{\mathrm{ort}}} + (0,\dots,0,1)}$.

[71] In our Mathematica code, the generators $A$ and $B$ are the matrices (2.46) and (2.47) without their transpose, following the convention of [WWT+]. So, instead of the equations in the main text, we used $M_g q M_g^T = q$, $W q W^T = I$, and $W M_g W^{-1}$ in our code.

This is not the end of the story. Recall from Section 6.1.2 that if we write an orthonormal basis of $\Lambda_{24} \otimes_{\mathbb{Z}} \mathbb{R}$ as $\psi_1 \dots, \psi_{24}$ and take the polarization of $\mathfrak{a} = \Lambda_{24} \otimes_{\mathbb{Z}} \mathbb{C}$ as in (6.17), then the specific lift $\mathfrak{z} \in \mathrm{Spin}(24)$ of $-1_{\mathrm{SO}(24)} \in \mathrm{SO}(24)$ is $\mathfrak{z} = \psi_1 \cdots \psi_{24}$. In Section 6.2.2, we rename $\psi_1$ and $\psi_{24}$ so that $\mathfrak{z}$ is in $\mathrm{Co}_0 \subset \mathrm{Spin}(24)$. In the notation here, this requires us to order the first and second columns of $W$ properly so that $(e_1 \cdots e_{24})W$ can be used as the renamed basis $\psi_1, \dots, \psi_{24}$. This determines whether $\det V$ is $+1$ or $-1$.

In our basis $e_1, \dots, e_{24}$ such that the symmetric bilinear form is represented as $q$ in (2.49) (or the generators $A$ and $B$ are represented as the matrices (2.46) and (2.47)), we find that the correct choice is $\det V = -1$, so we define $M_g^{\mathrm{ort}} := W^{-1} M_g W$ using such $V$. This correct choice $\det V = -1$ was found because if we choose $\det V = +1$, then there exist pairs $(M_g, M_h)$ such that all the twisted partition functions calculated with $\vec{\lambda}_{M_g^{\mathrm{ort}}}$ or $\vec{\lambda}_{M_g^{\mathrm{ort}}} + (0, \dots, 0, 1)$, and $\vec{\lambda}_{M_h^{\mathrm{ort}}}$ or $\vec{\lambda}_{M_h^{\mathrm{ort}}} + (0, \dots, 0, 1)$ are not constant. This contradicts to the fact that the R-R twisted partition functions $Z^{(V^{f\natural})}_{\mathrm{R}g}{}^{\mathrm{R}h}(\tau)$ are constant (see Remark 14.2).

**Block-diagonalization within** $\mathrm{SO}(24)$

We would like to simultaneously block-diagonalize $M_g^{\mathrm{ort}}$ and $M_h^{\mathrm{ort}}$ as $M_g^{\mathrm{ort}} = R^O_{\vec{\lambda}_{M_g^{\mathrm{ort}}}}$ and $M_h^{\mathrm{ort}} = R^O_{\vec{\lambda}_{M_h^{\mathrm{ort}}}}$ by $O \in \mathrm{SO}(24)$. We basically follow the idea in the proof of Lemma F.2.

Let us begin with the block-diagonalization of a single matrix $M_g^{\mathrm{ort}}$. The Mathematica command `SchurDecomposition` performs the real Schur decomposition. As explained in the proof of Lemma F.2, we obtain $\tilde{O} \in \mathrm{O}(24)$ such that $\tilde{O}^{-1} M_g^{\mathrm{ort}} \tilde{O} = \mathrm{diag}(B_1, \dots, B_m)$, where $B_i \in \mathrm{O}(2)$ or $B_i = \pm 1$. If there is $B_i \in \mathrm{O}(2)$ with $\det B_i = -1$, then we have to change $\tilde{O} \in \mathrm{O}(24)$ so that $B_i$ is further diagonalized into $\mathrm{diag}(1, -1)$. Fortunately, we have not encountered such a case where `SchurDecomposition` produces $B_i \in \mathrm{O}(2)$ with $\det B_i = -1$, so it probably performs the block-diagonalization while making the blocks as small as possible. If $B_i = \pm 1$, then there must be $j \neq i$ such that $B_j = B_i$, so we reorder the columns of $\tilde{O}$ so that $\tilde{O}^{-1} M_g^{\mathrm{ort}} \tilde{O}$ becomes in the form of $R_{\vec{\lambda}}$. Finally, if $\det \tilde{O} = -1$, then we interchange the first and second columns of $\tilde{O}$, which changes $\lambda_1$ to $-\lambda_1$. This gives us $O = \tilde{O} \in \mathrm{SO}(24)$ and $\vec{\lambda}_{M_g^{\mathrm{ort}}} = \vec{\lambda}$ such that $M_g^{\mathrm{ort}} = R^O_{\vec{\lambda}_{M_g^{\mathrm{ort}}}}$.

In order to simultaneously block-diagonalize $M_g^{\mathrm{ort}}$ and $M_h^{\mathrm{ort}}$, it suffices to perform the above procedure for $M_g^{\mathrm{ort}} + \pi M_h^{\mathrm{ort}}$. This is because, $M_g$ and $M_h$ are $\mathbb{Z}$-valued, $M_g^{\mathrm{ort}}$ and $M_h^{\mathrm{ort}}$ are $\mathbb{R}_{\mathrm{alg}}$-valued, where $\mathbb{R}_{\mathrm{alg}} := \mathbb{R} \cap \overline{\mathbb{Q}}$ is the field of all the real algebraic numbers, and therefore any eigenvalue $\mu_g$ of $M_g^{\mathrm{ort}}$ and any eigenvalue $\mu_h$ of $M_h^{\mathrm{ort}}$ are algebraic numbers. Then $\mu_g + \pi \mu_h = \mu_g' + \pi \mu_h'$ if and only if $\mu_g = \mu_g'$ and $\mu_h = \mu_h'$, because $\pi$ is a transcendental number. This means that the eigenspace decomposition with respect to $M_g^{\mathrm{ort}} + \pi M_h^{\mathrm{ort}}$ is a refinement of both the eigenspace decomposition with respect to $M_g^{\mathrm{ort}}$ and $M_h^{\mathrm{ort}}$. Therefore, the eigenvectors which diagonalize $M_g^{\mathrm{ort}} + \pi M_h^{\mathrm{ort}}$ also diagonalize both $M_g^{\mathrm{ort}}$ and $M_h^{\mathrm{ort}}$.

*Remark* 14.3. If we do not need the block-diagonalizing matrix $O \in \mathrm{SO}(24)$ and only need the angles $\vec{\lambda}_{M_g^{\mathrm{ort}}}$ for $M_g^{\mathrm{ort}}$, then we can also use the following method.

We calculate the eigenvalues of $M_g^{\mathrm{ort}}$. Then they constitute 12 pairs $(e^{2\pi\sqrt{-1}\lambda_i}, e^{-2\pi\sqrt{-1}\lambda_i})$

($i = 1, \dots, 12$), and we can say that $\tilde{O}^{-1} M_g^{\rm ort} \tilde{O} = R_{\vec{\lambda}}$ for some $\tilde{O} \in {\rm O}(24)$. The problem is whether $\det \tilde{O}$ is $+1$ or $-1$.

To judge it, we use the property ${\rm Pf}(\tilde{O}^T X \tilde{O}) = \det(\tilde{O}) {\rm Pf}(X)$ of Pfaffian.[72] Calculate ${\rm Pf}(M_g^{\rm ort} - (M_g^{\rm ort})^T)$ and

$$
{\rm Pf}(R_{\vec{\lambda}} - R_{\vec{\lambda}}^T) = \prod_{i=1}^{12} (-2 \sin 2\pi\lambda_i). \tag{14.31}
$$

- If ${\rm Pf}(R_{\vec{\lambda}} - R_{\vec{\lambda}}^T) \neq 0$, then by comparing the signs of ${\rm Pf}(M_g^{\rm ort} - (M_g^{\rm ort})^T)$ and ${\rm Pf}(R_{\vec{\lambda}} - R_{\vec{\lambda}}^T)$, we can judge whether $\det \tilde{O}$ is $+1$ or $-1$. In the latter case, we redefine $\lambda_i$ as $-\lambda_i$ for one $i$, say $i = 12$. Then $\vec{\lambda}$ is the angle $\vec{\lambda}_{M_g^{\rm ort}}$ such that $M_g^{\rm ort}$ and $R_{\vec{\lambda}_{M_g^{\rm ort}}}$ are ${\rm SO}(24)$-conjugate.

- If ${\rm Pf}(R_{\vec{\lambda}} - R_{\vec{\lambda}}^T) = 0$, then one of the entries of $\vec{\lambda}$, say $\lambda_i$, is in $\frac{1}{2}\mathbb{Z}$. In this case, by interchanging the $(2i-1)$-th and $2i$-th columns of $\tilde{O}$, we can flip the sign of $\det \tilde{O}$, while retaining $\vec{\lambda}$. Therefore, we can always say that $\vec{\lambda}$ is the angle $\vec{\lambda}_{M_g^{\rm ort}}$ such that $M_g^{\rm ort}$ and $R_{\vec{\lambda}_{M_g^{\rm ort}}}$ are ${\rm SO}(24)$-conjugate.

In order to obtain the angles $\vec{\lambda}_{M_g^{\rm ort}}$ and $\vec{\lambda}_{M_h^{\rm ort}}$ of the simultaneously block-diagonalization of $M_g^{\rm ort}$ and $M_g^{\rm ort}$, we again perform the above procedure for $M_g^{\rm ort} + \pi M_h^{\rm ort}$. This time, however, checking the signs of ${\rm Pf}(M_g^{\rm ort} - (M_g^{\rm ort})^T)$ and ${\rm Pf}(M_g^{\rm ort} - (M_g^{\rm ort})^T)$ separately does not suffice, because there is a possibility that both of the Pfaffians are zero but $((\lambda_g)_i = 0, (\lambda_h)_i \neq 0)$ and $((\lambda_g)_j \neq 0, (\lambda_h)_j = 0)$ modulo $\frac{1}{2}\mathbb{Z}$. In such a case, it is not guaranteed that we can flip the sign of $\det \tilde{O}$ while retaining both $\vec{\lambda}_g$ and $\vec{\lambda}_h$. So we check whether ${\rm Pf}((M_g^{\rm ort} + \sqrt{-1} M_h^{\rm ort}) - (M_g^{\rm ort} + \sqrt{-1} M_h^{\rm ort})^T)$ and ${\rm Pf}((R_{\vec{\lambda}_g} + \sqrt{-1} R_{\vec{\lambda}_h}) - (R_{\vec{\lambda}_g} + \sqrt{-1} R_{\vec{\lambda}_h})^T)$ are the same or the negative of each other, and redefine $((\lambda_g)_{12}, (\lambda_h)_{12})$ as $(-(\lambda_g)_{12}, -(\lambda_h)_{12})$ in the latter case. When $((\lambda_g)_i = 0, (\lambda_h)_i = 0) \bmod \frac{1}{2}\mathbb{Z}$ for some $i$, this Pfaffian is zero, but in such a case, we can flip the sign of $\det \tilde{O}$ while retaining both $\vec{\lambda}_g$ and $\vec{\lambda}_h$. *(Remark ends.)*

**Construct the supercurrent**

After the block-diagonalization $M_g^{\rm ort} = R^O_{\vec{\lambda}_{M_g^{\rm ort}}}$, we have to detect which of $\hat{R}^O_{\vec{\lambda}_{M_g^{\rm ort}}}$ or $\hat{R}^O_{\vec{\lambda}_{M_g^{\rm ort}} + (0,\dots,0,1)}$ is $\check{g} \in {\rm Co}_0 \subset {\rm Spin}(24)$. We can achieve it by examining which of them preserves the supercurrent $G$ of $V^{f\natural}$; the wrong one reverses the sign of $G$. (Here, we write $\tau$ in (4.21) as $G$.)

[72] Let $A = [a_{i,j}]$ be an antisymmetric matrix $A^T = -A$ of size $2n \times 2n$. Its *Pfaffian* ${\rm Pf}(A)$ is defined as

$$
{\rm Pf}(A) := \frac{1}{2^n n!} \sum_{\sigma \in S_{2n}} {\rm sgn}(\sigma) a_{\sigma(1),\sigma(2)} \cdots a_{\sigma(2n-1),\sigma(2n)}. \tag{14.30}
$$

It satisfies ${\rm Pf}(A)^2 = \det A$, and ${\rm Pf}(B^T A B) = \det(B) {\rm Pf}(A)$ for any $2n \times 2n$ matrix $B$.

To do so, we first have to implement the action of $\hat{R}^O_{\vec{\lambda}} \in \mathrm{Spin}(24)$ on the chiral representation $\mathbf{2^{11}_+}$, because $G \in \mathbf{2^{11}_+}$. Our implementation method is explained in Appendix F.3. For speed reasons, we use C++ instead of Mathematica below.

Next, we find the supercurrent $G \in \mathbf{2^{11}_+}$ up to scaler multiplication as the invariant subspace of $\mathbf{2^{11}}$ under the $\mathrm{Co}_0$-action, which is one-dimensional. This is done as follows, using the Eigen library in C++.

- First, for the generators $A, B$ in (2.46, 2.47) of $\mathrm{Co}_0 \subset \mathrm{SL}(24, \mathbb{Z})$, by repeating our discussion, $A^{\mathrm{ort}}$ and $B^{\mathrm{ort}}$ are generators of $\mathrm{Co}_0 \subset \mathrm{SO}(24)$, and they are uniquely lifted to generators $\hat{A}$ and $\hat{B}$ of $\mathrm{Co}_0 \subset \mathrm{Spin}(24)$. After block-diagonalizing $A^{\mathrm{ort}}$ as $A^{\mathrm{ort}} = R^O_{\vec{\lambda}_{A^{\mathrm{ort}}}}$ by $O \in \mathrm{SO}(24)$, we somehow determine which of $\hat{R}^O_{\vec{\lambda}_{A^{\mathrm{ort}}}}$ or $\hat{R}^O_{\vec{\lambda}_{A^{\mathrm{ort}}}+(0,\dots,0,1)} = -\hat{R}^O_{\vec{\lambda}_{A^{\mathrm{ort}}}}$ is $\hat{A}$. Similarly, for a block-diagonalization $B^{\mathrm{ort}} = R^{O'}_{\vec{\lambda}_{B^{\mathrm{ort}}}}$ by $O' \in \mathrm{SO}(24)$, we somehow determine which of $\hat{R}^{O'}_{\vec{\lambda}_{B^{\mathrm{ort}}}}$ or $\hat{R}^{O'}_{\vec{\lambda}_{B^{\mathrm{ort}}}+(0,\dots,0,1)} = -\hat{R}^{O'}_{\vec{\lambda}_{B^{\mathrm{ort}}}}$ is $\hat{B}$. In practice, we only have to try all the $2 \times 2$ possibilities of $\pm\hat{R}^O_{\vec{\lambda}_{A^{\mathrm{ort}}}}$ and $\pm\hat{R}^{O'}_{\vec{\lambda}_{B^{\mathrm{ort}}}}$; if we used a wrong combination, then $G$ will not be found by the following procedure.

- Next, using the implementation explained in Appendix F.3, we prepare the $2^{11}$-dimensional representation matrices $\hat{A}^{\mathbf{2^{11}_+}}$ and $\hat{B}^{\mathbf{2^{11}_+}}$ of $\hat{A}$ and $\hat{B}$ on $\mathbf{2^{11}_+}$, respectively.

- The supercurrent $G \in \mathbf{2^{11}_+}$ is a basis of $\ker(\hat{A}^{\mathbf{2^{11}_+}} - I) \cap \ker(\hat{B}^{\mathbf{2^{11}_+}} - I)$, which must be a one-dimensional subspace. To compute this subspace, we would like to obtain bases of $\ker(\hat{A}^{\mathbf{2^{11}_+}} - I)$ and $\ker(\hat{B}^{\mathbf{2^{11}_+}} - I)$. The singular value decomposition (SVD) `Eigen::BDCSVD` of the $2^{11}$-dimensional matrix $\hat{A}^{\mathbf{2^{11}_+}} - I$ halted, probably because of a lack of computational resources. So we use the LU decomposition `Eigen::FullPivLU`. There is a command[73] `Eigen::FullPivLU::kernel()` to obtain a $\dim M \times \dim \ker M$ matrix $Q_M$ whose column vectors constitute a basis of $\ker M$. For numerical stability, we orthonormalize the column vectors of $Q_M$ by the QR decomposition[74] `Eigen::HouseholderQR`.

- Finally, $G$ is the only vector up to scalar multiplication such that $Q_{\hat{A}^{\mathbf{2^{11}_+}} - I} G = Q_{\hat{B}^{\mathbf{2^{11}_+}} - I} G$. So, we perform the SVD[75] of $[Q_{\hat{A}^{\mathbf{2^{11}_+}} - I}, -Q_{\hat{B}^{\mathbf{2^{11}_+}} - I}]$, which is a $2^{11} \times (1056+680)$ matrix and `Eigen::JacobiSVD` can handle it without halting. Then there is only one zero singular value, and the corresponding vector is the supercurrent $G$.

[73] The LU factorization with full pivoting `Eigen::FullPivLU` of $M$ calculates $PMQ = LU$, where $P$ and $Q$ are permutation matrices which only reorder the rows and columns of $M$ respectively, $L$ is lower-unitriangular (lower-triangular and all the diagonal elements are 1), and $U$ is upper-triangular. So, $v \in \ker M$ satisfies $UQ^{-1}v = 0$. Therefore, `Eigen::FullPivLU::kernel()` computes $Q \cdot \ker U$.

[74] The QR decomposition of an $m \times n$ matrix $M$ is $M = QR$, where $Q$ is an $m \times \min(m, n)$ matrix, whose first $\mathrm{rank}\, M$ columns are orthonormal (as complex vectors), and $R$ is a $\min(m, n) \times n$ upper-triangular matrix.

[75] The SVD of $M$ is $M = USV^*$, where $U$ and $V$ are unitary, and $S$ is a diagonal matrix, whose diagonal elements are the singular values of $M$.

*Remark* 14.4. The execution of the above algorithm took about 30 minutes even under the -O3 optimization of GCC. In particular, the SVD of $[Q_{\hat{A}^{\mathbf{2^{11}}}_{+}-I}, -Q_{\hat{B}^{\mathbf{2^{11}}}_{+}-I}]$ was the primary bottleneck. If we use $AB$ and $BA$ as generators of $\mathrm{Co}_0$, then since their orders are both 40, their eigenvalues are scattered over the 40 values $e^{2\pi\sqrt{-1}k/40}(k = 0, \dots, 39)$. So the dimensions of $\ker(\hat{A}^{\mathbf{2^{11}}}_{+}\hat{B}^{\mathbf{2^{11}}}_{+} - I)$ and $\ker(\hat{B}^{\mathbf{2^{11}}}_{+}\hat{A}^{\mathbf{2^{11}}}_{+} - I)$ are expected to be smaller than those of $\ker(\hat{A}^{\mathbf{2^{11}}}_{+} - I)$ and $\ker(\hat{B}^{\mathbf{2^{11}}}_{+} - I)$. In fact, $[Q_{\hat{A}^{\mathbf{2^{11}}}_{+}\hat{B}^{\mathbf{2^{11}}}_{+}-I}, -Q_{\hat{B}^{\mathbf{2^{11}}}_{+}\hat{A}^{\mathbf{2^{11}}}_{+}-I}]$ was found to be a $2^{11} \times (68+68)$ matrix. This reduced the runtime to just one minute. *(Remark ends.)*

Now that we have obtained the supercurrent $G$ up to scaler multiplication, we can detect which of $\hat{R}^{O}_{\vec{\lambda}_{M_g^{\mathrm{ort}}}}$ or $\hat{R}^{O}_{\vec{\lambda}_{M_g^{\mathrm{ort}}+(0,\dots,0,1)}}$ is $\check{g} \in \mathrm{Co}_0 \subset \mathrm{Spin}(24)$ by examining which of them preserves $G$. Finally, back in Mathematica, by plugging the correct angles $\vec{\lambda}_{\check{g},O}$ and $\vec{\lambda}_{\check{h},O}$ into $\vec{\lambda}_g$ and $\vec{\lambda}_h$ in the formula $Z^{(V^{f\natural})^{\mathrm{Y}\vec{\lambda}_h}_{\mathrm{X}\vec{\lambda}_g}}$ (14.12)–(14.15) respectively, we obtain the twisted partition functions of Duncan's module $V^{f\natural}$.

**Theorem 14.5.** *The values of the ($(-1)^F$-inserted and Ramond-sector) twisted partition functions $Z^{(V^{f\natural})^{\mathrm{R}h}_{\mathrm{R}g}}$ of Duncan's module $V^{f\natural}$ twisted by $\mathrm{Co}_1$ are those listed in Table G.3 in Appendix G. Note that they can be determined up to phases for each $\mathrm{SL}(2,\mathbb{Z})$-orbit of the conjugacy classes of the commuting pairs of $\mathrm{Co}_1$.*

## 14.3 Step 3. Anomaly-Canceling Phases

Finally, we have to determine the anomaly-canceling phases $\beta_{\mathrm{anom}}(g,h)$ in

$$Z^{((V^{f\natural})^{\otimes 24}/\mathrm{Co}_1)^{\mathrm{R}}_{\mathrm{R}}}(\tau) = \sum_{\substack{[[(g,h)]] \in (G^2_{\mathrm{com}}/\sim)/\mathrm{GL}(2,\mathbb{Z}) \\ \text{for } G=\mathrm{Co}_1}} |\mathrm{GL}(2,\mathbb{Z}) \cdot [(g,h)]| \frac{|[h]_{\subset C_g}|}{|C_g|} \mathrm{Re}\Big[\beta_{\mathrm{anom}}(g,h)\big(Z^{(V^{f\natural})^{\mathrm{R}h}_{\mathrm{R}g}}(\tau)\big)^{24}\Big]. \quad (14.32)$$

Recall from Section 12.2.4 that $\beta_{\mathrm{anom}}(g,h)$ denotes the phase by which we multiply the twisted partition functions to cancel the anomalous phases:

$$Z_0^{((V^{f\natural})^{\otimes 24})^{\mathrm{R}h}_{\mathrm{R}g}}(\tau) = \beta_{\mathrm{anom}}(g,h) Z^{((V^{f\natural})^{\otimes 24})^{\mathrm{R}h}_{\mathrm{R}g}}(\tau). \quad (14.33)$$

As remarked in (12.61), the partition function $Z^{((V^{f\natural})^{\otimes 24})^{\mathrm{R}h}_{\mathrm{R}g}}(\tau)$ twisted by the diagonal $\mathrm{Co}_1$ symmetry is the 24th power of the partition function $Z^{(V^{f\natural})^{\mathrm{R}h}_{\mathrm{R}g}}(\tau)$ twisted by $\mathrm{Co}_1$. In the previous Section 14.2, we saw that we can calculate $Z^{(V^{f\natural})^{\mathrm{R}h}_{\mathrm{R}g}}(\tau)$ as $Z^{(V^{f\natural})^{\mathrm{R}\vec{\lambda}_{\check{h},O}}_{\mathrm{R}\vec{\lambda}_{\check{g},O}}}(\tau)$ using the formula (14.15) (when $\check{g}$ and $\check{h}$ commute in $\mathrm{Spin}(24)$). Combining them, we can compute the twisted partition function $Z^{(V^{f\natural})^{\mathrm{R}h}_{\mathrm{R}g}}(\tau)$ as

$$Z^{((V^{f\natural})^{\otimes 24})^{\mathrm{R}h}_{\mathrm{R}g}}(\tau) = \left(Z^{(V^{f\natural})^{\mathrm{R}\vec{\lambda}_{\check{h},O}}_{\mathrm{R}\vec{\lambda}_{\check{g},O}}}(\tau)\right)^{24}. \quad (14.34)$$

We have to multiply this quantity (14.34) by some $\mathrm{U}(1)$ phase to cancel the anomalous phases and obtain $Z_0^{((V^{f\natural})^{\otimes 24})\substack{\mathrm{R}h\\ \mathrm{R}g}}(\tau)$. It shortly turns out to be convenient to define the anomaly-canceling phase $\beta^0_{\mathrm{anom}}(g,h)$ as the phase of $Z_0^{((V^{f\natural})^{\otimes 24})\substack{\mathrm{R}h\\ \mathrm{R}g}}(\tau)$ itself. (Recall that $Z_0^{((V^{f\natural})^{\otimes 24})\substack{\mathrm{R}h\\ \mathrm{R}g}}(\tau)$ is a constant thanks to the supersymmetry.) That is,

$$Z_0^{((V^{f\natural})^{\otimes 24})\substack{\mathrm{R}h\\ \mathrm{R}g}}(\tau) = \beta^0_{\mathrm{anom}}(g,h)\left|Z^{(V^{f\natural})\substack{\mathrm{R}\vec{\lambda}_{\tilde{h},O}\\ \mathrm{R}\vec{\lambda}_{\tilde{g},O}}}(\tau)\right|^{24}. \tag{14.35}$$

As explained in Section 12.1.3, the phases of non-zero twisted partition functions $Z_0^{((V^{f\natural})^{\otimes 24})\substack{\mathrm{R}h\\ \mathrm{R}g}}(\tau) \neq 0$ are not affected by the discrete torsion, so the correct set $\{\beta^0_{\mathrm{anom}}(g,h) \mid Z^{(V^{f\natural})\substack{\mathrm{R}\vec{\lambda}_{\tilde{h},O}\\ \mathrm{R}\vec{\lambda}_{\tilde{g},O}}}(\tau) \neq 0, [[(g,h)]] \in (G^2_{\mathrm{com}}/\sim)/\mathrm{GL}(2,\mathbb{Z})\}$ of such phases must be unique.

We computed in Section 14.1.2 that the conjugacy classes $G^2_{\mathrm{com}}/\sim$ of the commuting pairs of $G = \mathrm{Co}_1$ decompose into 445 $\mathrm{SL}(2,\mathbb{Z})$-orbits, and 427 of them are mapped to themselves under the P transformation, whereas the remaining $\frac{18}{2} = 9$ pairs are interchanged under the P transformation. Since $Z_0^{((V^{f\natural})^{\otimes 24})\substack{\mathrm{R}h\\ \mathrm{R}g}}(\tau)$ is a constant thanks to the supersymmetry, by applying the discussion of the $\mathrm{GL}(2,\mathbb{Z})$ transformation in Section 12.2.4, we can say that

- For each 427 $\mathrm{SL}(2,\mathbb{Z})$-orbit which is mapped to itself, the twisted partition function $Z_0^{((V^{f\natural})^{\otimes 24})\substack{\mathrm{R}h\\ \mathrm{R}g}}(\tau)$ is a real number. Therefore, $\beta^0_{\mathrm{anom}}(g,h)$ is either $+1$ or $-1$.

  In particular, 137 out of the 427 orbits have non-zero twisted partition functions.

- For each of the remaining 9 pairs, say a pair $(\mathrm{SL}(2,\mathbb{Z})\cdot[(g,h)], \mathrm{SL}(2,\mathbb{Z})\cdot[(g',h')])$, the twisted partition functions $Z_0^{((V^{f\natural})^{\otimes 24})\substack{\mathrm{R}h\\ \mathrm{R}g}}(\tau)$ and $Z_0^{((V^{f\natural})^{\otimes 24})\substack{\mathrm{R}h'\\ \mathrm{R}g'}}(\tau)$ are the complex conjugate to each other. Therefore, once $\beta^0_{\mathrm{anom}}(g,h) \in \mathrm{U}(1)$ is determined, then $\beta^0_{\mathrm{anom}}(g',h')$ is determined as $\beta^0_{\mathrm{anom}}(g,h)^*$.

  All of these orbits have non-zero twisted partition functions.

It would be ideal if we could determine the anomalous-canceling phases $\beta^0_{\mathrm{anom}}(g,h)$ by some bottom-up approach, for example by determining the values of the anomaly 3-cocycle $\alpha(g,h,k)$ of $V^{f\natural}$ for each $g,h,k \in \mathrm{Co}_1$. Unfortunately, however, there seems no well-established method of doing so, to the author's knowledge, except for the case of the $\mathrm{U}(1)$ symmetry [OSTZ25]. Therefore, we attempt to determine $\beta^0_{\mathrm{anom}}(g,h)$ by using the property of the twisted partition functions as the characters of the centralizers as follows.

Recall from Section 12.2.2 that the twisted sector $\mathcal{H}_g$ of the orbifold theory constitutes a genuine representation of the centralizer $C_g$ if the symmetry $G$ is non-anomalous. In particular, the $(-1)^F$-inserted R twisted partition function $Z_0^{((V^{f\natural})^{\otimes 24})\substack{\mathrm{R}h\\ \mathrm{R}g}}(\tau) = \mathrm{Tr}_{(V^{f\natural}_{\mathrm{tw},g})^{\otimes 24}}[(-1)^F U_h q^{L_0 - \frac{24\times 12}{24}}]$ is the character of $(-1)^F U_h$ ($h \in C_g$) represented on the Ramond vacuum states, or the weight-$\frac{24\times 12}{24}$ subspace $\mathcal{H}_{\mathrm{R},g}|_{\mathrm{vac}}$, of the sector $\mathcal{H}_{\mathrm{R},g} = (V^{f\natural}_{\mathrm{tw},g})^{\otimes 24}$ (see Section 12.2.4). In other words, it is a virtual character (a signed sum of irreducible characters) of $h \in C_g$.

As a special case, for the untwisted R sector $\mathcal{H}_{\mathrm{R},1} = (V_{\mathrm{tw}}^{f\natural})^{\otimes 24}$, the centralizer $C_1$ is $\mathrm{Co}_1$ itself, and we know that its weight-$(24\times\frac{1}{2})$ subspace is $\mathbf{24}^{\otimes 24}$, where $\mathbf{24} = \mathrm{Span}_{\mathbb{C}}\{\psi_i(-\frac{1}{2})|0\rangle\}_{i=1,\ldots,24} \subset A(\mathfrak{a})^1$ has the positive parity under $(-1)^F = -1_{\mathrm{Spin}(\mathfrak{a})}$. We also know how the centralizer $\mathrm{Co}_1$ acts on $V_{\mathrm{tw}}^{f\natural}$ projectively, because it is just the $\mathrm{Co}_0$-action on $V_{\mathrm{tw}}^{f\natural}$ reviewed in Section 6.2.2. In Section 14.1.1, we have already obtained the matrices $M_{g_i}$ on this $\mathbf{24}$ such that $\pm M_{g_i}$ is the (projective) representation of $g_i \in \mathrm{Co}_1$, for each representative $g_i$ of the conjugacy classes $[g_i] \in \mathrm{Co}_1/\sim$. Therefore, we can explicitly calculate

$$Z_0^{((V^{f\natural})^{\otimes 24})\mathrm{R}g_i}_{\mathrm{R}1}(\tau) = \mathrm{Tr}[M_{g_i}]^{24}. \tag{14.36}$$

Since $M_{g_i}$'s are real matrices, the results are always non-negative. Therefore we can conclude that $\beta^0_{\mathrm{anom}}(1, g_i)$ in (14.35) are all $+1$. This determines the anomaly-canceling phases of the 99 $\mathrm{SL}(2,\mathbb{Z})$-orbits containing the conjugacy classes $[(1, g_i)]$, out of the 427 $\mathrm{SL}(2,\mathbb{Z})$-orbits which are mapped to themselves by the P transformation. In particular, 56 out of the 99 orbits have non-zero twisted partition functions. We can summarize the number of $\mathrm{SL}(2,\mathbb{Z})$-orbits as follows.

| the number of $\mathrm{SL}(2,\mathbb{Z})$-orbits | non-zero tw. p.f. | zero tw. p.f. | total |
|---|---|---|---|
| P-invariant, contain $[(1, g_i)]$ | 56 | 43 | 99 |
| P-invariant, not contain $[(1, g_i)]$ | 81 | 247 | 328 |
| not P-invariant | 18 | 0 | 18 |
| total | 155 | 290 | 445 |

For the twisted sectors $\mathcal{H}_{\mathrm{R},g\neq 1}$, such an explicit calculation is difficult, but we can extract some constraints on the anomaly-canceling phases as follows. If we label the irreducible characters of $C_g$ as $\chi_a$ $(a = 1\ldots, |C_g/\sim|)$, then the multiplicity of the irreducible representation $a$ in the representation $\mathcal{H}_{\mathrm{R},g}|_{\mathrm{vac}}$ is, from the orthogonality of the irreducible characters,

$$N(g,a) := \frac{1}{|C_g|}\sum_{h\in C_g} \chi_a(h)^* Z_0^{((V^{f\natural})^{\otimes 24})\mathrm{R}h}_{\mathrm{R}g}(\tau) \tag{14.37}$$

$$= \frac{1}{|C_g|}\sum_{[h]_{\subset C_g}\in C_g/\sim} |[h]_{\subset C_g}|\chi_a(h)^* Z_0^{((V^{f\natural})^{\otimes 24})\mathrm{R}h}_{\mathrm{R}g}(\tau) \tag{14.38}$$

$$= \frac{1}{|C_g|}\sum_{[h]_{\subset C_g}\in C_g/\sim} |[h]_{\subset C_g}|\chi_a(h)^* \beta^0_{\mathrm{anom}}(g,h) \left| Z^{(V^{f\natural})\mathrm{R}\vec{\lambda}_{\tilde{h},O}}_{\mathrm{R}\vec{\lambda}_{\tilde{g},O}}(\tau)\right|^{24}. \tag{14.39}$$

These $N(g,a)$ must be all integers, which can be used as the constraints on $\beta^0_{\mathrm{anom}}(g,h)$.

Now, we have to compute the irreducible characters $\chi_a$. In principle, the character table of the centralizer $C_{g_i}$ for each representative $g_i$ of the conjugacy classes $[g_i] \in \mathrm{Co}_1/\sim$ can be computed by GAP as

```
c:=centralizers[i];
irreps:=Irr(c);
chartbl:=List(irreps, chi->ValuesOfClassFunction(chi));
```

where **centralizers** is already computed in Section 14.1.1. This algorithm is fast enough for the centralizer of reasonable size; see Table G.2 for the sizes of the centralizers $C_{g_i}$. It computes the character table for each centralizer $C_{g_i}$ in a few minutes to a few hours, except for the two largest centralizers $C_{g_1}$ and $C_{g_5}$.

For the first one, we have $g_1 = 1_{\mathrm{Co}_1}$ and $C_{g_1} = \mathrm{Co}_1$, and $\beta^0_{\mathrm{anom}}(1_{\mathrm{Co}_1}, h)$'s are already determined to be 1. If we need the character table of $\mathrm{Co}_1$, then it is given in ATLAS [CCN+85] and the GAP library, and the ATLAS names of the conjugacy classes in our code are already determined as in Remark 14.1. For the second one, the ATLAS name of the conjugacy class $[g_5]$ is 3A, and the centralizer is $C_{g_5} \cong 3.Suz$ according to ATLAS. Here, $Suz$ is the sporadic Suzuki group, and the character table of $3.Suz$ is given in ATLAS and the GAP library. So it suffices to determine the ATLAS name of each conjugacy class $[h] \in C_{g_5}/\sim$ in our code. We omit the details,[76] but this can be done by exploiting the data such as the order of a representative $h_j$, the size $|[h_j]|$ of the conjugacy class, the T transformation $[h_j] \mapsto [g_5 h_j]$, the P transformation $[h_j] \mapsto [h_j^{-1}]$, and some characters similarly to Remark 14.1.

In this way, we have obtained $\sum_{i=1}^{101} |C_{g_i}/\sim| = 7578$ constraints on the anomaly-canceling phases $\beta^0_{\mathrm{anom}}(g, h)$, such that $N(g_i, a)$ in (14.39) must be an integer. However, many of $N(g_i, a)$ are computed to be zero, so the actual number of these integrality conditions containing nontrivial information is much smaller. After we substitute $+1$ for all $\beta^0_{\mathrm{anom}}(1, h)$, the number of the integrality conditions which still contain unknown phases $\beta^0_{\mathrm{anom}}(g, h)$ is 960.

These integrality conditions in fact can determine some of the anomaly-canceling phases. For example, one of the multiplicities $\{N(g_{91}, a)\}_a$ gives a constraint

$$-\frac{\beta^0_{\mathrm{anom}}(176)}{9} - \frac{\beta^0_{\mathrm{anom}}(255)}{27} + \frac{4}{27} \in \mathbb{Z}. \tag{14.40}$$

Here, $\beta^0_{\mathrm{anom}}(i)$ denotes the anomaly-canceling phase $\beta^0_{\mathrm{anom}}(g, h)$ for the $i$-th $\mathrm{SL}(2, \mathbb{Z})$-orbit

[76] Here we add more information. The centralizer $C_{g_5} \cong 3.Suz$ is a non-split central extension, and the central $\mathbb{Z}_3$ is generated by $g_5$. The character table of $3.Suz$ has the table automorphisms (the permutations of conjugacy classes and irreducible characters preserving the character table; 23A↔23B and 39A↔39B for $\mathrm{Co}_1$ in Remark 14.1 are the examples). The table automorphisms constitutes $(\mathbb{Z}_2)^4$; one $\mathbb{Z}_2$ is implemented by $g_3 \leftrightarrow (g_3)^2$, and if we fix it, then each $\mathbb{Z}_2$ of the remaining $(\mathbb{Z}_2)^3$ is the exchange of certain two connected components of the graph such that the conjugacy classes of $3.Suz$ are the vertices, and the T and P transformations are the edges. We can only assign the names of the conjugacy classes arbitrarily for these ambiguities form the table automorphisms.

Together with the information of the order and the size, the above graph can specify the conjugacy classes of $3.Suz$ except for 18 conjugacy classes. Among these 18 conjugacy classes, 18b can be uniquely determined by the properties that the order is 18, the character with respect to $\mathbf{66}_\pm$ is $-1$, and the sixth power is 3g. The other 17 conjugacy classes can be reached from 18b by the T and P transformations and the powers.

In this way, we found that the conjugacy classes of the centralizer $C_{g_5} \cong 3.Suz$ in our code are ordered as follows: 1a, 3a, 3b, 3i, 5a, 15a, 15b, 15e, 15f, 5b, 15c, 15d, 3d, 3e, 3c, 15g, 15h, 15i, 21b, 7a, 21a, 21e, 21c, 21d, 21g, 21h, 21f, 6b, 2a, 6a, 12a, 12b, 4a, 6f, 6g, 6e, 24a, 24b, 8a, 12j, 12k, 12i, 24i, 24g, 24h, 3h, 3f, 3g, 6m, 6k, 6l, 6i, 6j, 6h, 9c, 9a, 9b, 9d, 9e, 9f, 18e, 18f, 18d, 18c, 18a, 18b, 10a, 30a, 30b, 60b, 20a, 60a, 2b, 6c, 6d, 10b, 30c, 30d, 42a, 42b, 14a, 6o, 6p, 6n, 12m, 12n, 12l, 4b, 12c, 12d, 12u, 12s, 12t, 12h, 4d, 12g, 6q, 12r, 33a, 33b, 11a, 24d, 8b, 24c, 13a, 39a, 39b, 39d, 13b, 39c, 12f, 4c, 12e, 24f, 8c, 24e, 12q, 12o, 12p.

$\mathrm{SL}(2,\mathbb{Z})\cdot[(g,h)]$ in our code. According to the P transformation property, $\beta^0_{\mathrm{anom}}(176)$ and $\beta^0_{\mathrm{anom}}(255)$ are $+1$ or $-1$. By multiplying (14.40) by 9, we find $\beta^0_{\mathrm{anom}}(255)=+1$. We then have $-\frac{\beta^0_{\mathrm{anom}}(176)}{9}+\frac{1}{9}\in\mathbb{Z}$, so we can also determine $\beta^0_{\mathrm{anom}}(176)=+1$.

However, the situation is more difficult for the anomaly-canceling phases $\beta^0_{\mathrm{anom}}(g,h)$ and $\beta^0_{\mathrm{anom}}(g,h^{-1})$ for the $\mathrm{SL}(2,\mathbb{Z})$-orbits interchanged under the P transformation. In this case, $\beta^0_{\mathrm{anom}}(g,h)$ and $\beta^0_{\mathrm{anom}}(g,h^{-1})$ are complex conjugate to each other, and the coefficients of them in $N(g,a)$'s are also complex conjugate to each other as $c\beta^0_{\mathrm{anom}}(g,h)+c^*\beta^0_{\mathrm{anom}}(g,h^{-1})$, since $\chi_a(h^{-1})=\chi_a(h)^*$. Such coefficients often have large absolute values $|c|$, and we can then fine-tune the argument of $\beta^0_{\mathrm{anom}}(g,h)$ so that $c\beta^0_{\mathrm{anom}}(g,h)+c^*\beta^0_{\mathrm{anom}}(g,h^{-1})$ produces any integer in the range $[-2|c|,2|c|]$.

Due to this difficulty associated with the complex conjugate pairs, we cannot currently conclude that we have exhausted all the integrality conditions above, but at least we have not yet succeeded in determining all the anomaly-canceling phases from these conditions. Notably, however, all the phases determined from these conditions so far have been found to be $+1$.

Furthermore, in spite of the naive definition (14.35) of the anomaly-canceling phases $\beta^0_{\mathrm{anom}}(g,h)$, where the absolute value is taken and the information of the phases is discarded, we found that simply setting $\beta^0_{\mathrm{anom}}(g,h)=1$ for all $\mathrm{SL}(2,\mathbb{Z})$-orbits satisfies all the integrality conditions above. So we propose the following conjecture.

**Conjecture 14.6.** *The anomaly-canceling phases $\beta^0_{\mathrm{anom}}(g,h)$ defined as in (14.35) are all $+1$.*

We repeat that this choice $\beta^0_{\mathrm{anom}}(g,h)=1$ renders all $N(g_i,a)$ integer values. This is a nontrivial observation, in particular given that we have encountered the phases other than 1 even in a simple example, the $\mathbb{Z}_2$-orbifold of four complex fermions (1.3).

If this choice of anomaly-canceling phases is correct, then the Conway orbifold partition function is

$$Z^{((V^{f\natural})^{\otimes 24}/\mathrm{Co}_1)^{\mathrm{R}}_{\mathrm{R}}}(\tau)=665834752697050. \tag{14.41}$$

Even if not, we can observe that the Conway orbifold partition function $Z^{((V^{f\natural})^{\otimes 24}/\mathrm{Co}_1)^{\mathrm{R}}_{\mathrm{R}}}(\tau)$ must be greater than 1 as follows. The orbifold partition function $Z^{((V^{f\natural})^{\otimes 24}/\mathrm{Co}_1)^{\mathrm{R}}_{\mathrm{R}}}(\tau)$ turns out to be written as the sum of a large positive constant term and a positive linear combination of the phases $\beta^0_{\mathrm{anom}}(g,h)$ not yet determined, and even if we assume all the phases $\beta^0_{\mathrm{anom}}(g,h)$ not yet determined to be $-1$, the whole orbifold partition function is still greater than 1.

# Part V
# Conclusion

In this thesis, we studied two subjects concerning two-dimensional chiral CFTs with sporadic group symmetries: the symmetry of lattice VOAs, and the Conway orbifold partition function of Duncan's module.

In Part III, we investigated whether the isometry group $O(L)$ of a lattice $L$ is a subgroup of the automorphism group $\mathrm{Aut}(V_L)$ of the lattice VOA $V_L$ or not. In general, the automorphism group $\mathrm{Aut}(V_L)$ is known to contain a group extension of $O(L)$ by $(\mathbb{Z}_2)^{\mathrm{rank}L}$, which comes from the cocycle factors, as a subgroup. So we explicitly analyzed the structure of this extension, and provided one approach to show that the isometry group $O(L)$ does not lift to a subgroup of the automorphism group $\mathrm{Aut}(V_L)$, if this is indeed the case. In particular, we demonstrated that the subgroups $M_{24}$ and $M_{23}$ of the isometry group of the odd Leech lattice $O_{24}$ do not lift to subgroups of the automorphism group of the odd Leech lattice VOSA $V_{O_{24}}$. We also confirmed that the isometry group $\mathrm{Co}_0$ of the Leech lattice $\Lambda_{24}$ does not lift to a subgroup of the Leech lattice VOA $V_{\Lambda_{24}}$, as already shown in [Gri73].

We then revisited the moonshine-type observation by [BDFK15], which states that (a theory equivalent to) the odd Leech lattice SCFT contains an $M_{23}$-invariant $\mathcal{N}=2$ superconformal algebra, which reproduces the $\mathcal{N}=2$ extremal elliptic genus of central charge 24. Now that we know that the genuine symmetry of the odd Leech lattice VOSA $V_{O_{24}}$ (a subgroup of $\mathrm{Aut}(V_{O_{24}})$) is the non-split extension $2^{24}.M_{24}$, but not $M_{24}$, we investigated the weight-1 and weight-$\frac{3}{2}$ currents in the odd Leech lattice VOSA invariant under the symmetry $2^{24}.M_{24}$ or its subgroup $2^{23}.M_{23}$. Particularly, we concluded that there is no such weight-$\frac{3}{2}$ current. We still have an $\mathcal{N}=2$ superconformal algebra in the odd Leech lattice VOSA, which reproduces the $\mathcal{N}=2$ extremal elliptic genus, by modifying the supercurrents proposed in [BDFK15] as in Appendix B. However, such an $\mathcal{N}=2$ superconformal algebra is no longer invariant under the symmetry $2^{23}.M_{23}$, so the situation of this moonshine-type observation seems more mysterious than we think.

Exploring an explanation of this moonshine-type observation, by the odd Leech lattice VOSA or another new theory, is a future work. It would be interesting if it also provides some implications for other moonshine phenomena including the K3 Mathieu moonshine.

In Part IV, we aimed to calculate the Witten index of the orbifold $(V^{f\natural})^{\otimes 24}/\mathrm{Co}_1$ of Duncan's module by its $\mathrm{Co}_1$ symmetry. We specified three steps to compute it as follows.

Step 1. Determine the conjugacy classes of the commuting pairs

$$G^2_{\mathrm{com}}/\sim := \{(g,h)\in G\times G \mid gh=hg\}/(g,h)\sim(kgk^{-1},khk^{-1}) \tag{14.42}$$

for $G=\mathrm{Co}_1$, and their $\mathrm{SL}(2,\mathbb{Z})$- and $\mathrm{GL}(2,\mathbb{Z})$-orbits.

Step 2. Compute the twisted partition functions $Z^{(V^{f\natural})\mathrm{R}h}_{\mathrm{R}g}(\tau)$.

Step 3. Determine the anomaly-canceling phases $\beta_{\mathrm{anom}}(g,h)$ for all $\mathrm{GL}(2,\mathbb{Z})$-orbits of $G^2_{\mathrm{com}}/\sim$.

We carried out Step 1 by representing the Conway group $\mathrm{Co}_1$ as a permutation group on the vectors of squared length $4$ in the Leech lattice, and passing it to GAP. We also accomplished Step 2, which required careful treatment of many subtleties, such as the phases of the twisted partition functions and detecting the correct lift of each element of $\mathrm{Co}_1 \subset \mathrm{SemiSpin}(24)$ to $\mathrm{Co}_0 \subset \mathrm{Spin}(24)$. The obtained values of the twisted partition functions are recorded in Table G.3. As for Step 3, we utilized the fact that the $g$-twisted sector is a representation of the centralizer of $g$, as consistency conditions on the anomaly-canceling phases, and we presented a conjectural set of anomaly-canceling phases, although its validity requires further investigation.

An apparent future work is to verify the validity of the anomaly-canceling phases. As explained in Introduction 1.3, our next goal is to calculate the Witten index of the alternating orbifold $(V^{f\natural})^{\otimes 24}/A_{24} \times \mathrm{Co}_1$, which is a candidate proposed by [AKL22] for an SCFT corresponding to the periodicity element under the Stolz–Teichner conjecture. A machinery for computing it is almost already prepared in [AKL22], but the phase problem similar to Step 3 also exists. So our future study might begin with directly determining the anomaly-canceling phases for $(V^{f\natural})^{\otimes 24}/A_{24} \times \mathrm{Co}_1$. If its Witten index, or the Witten index of its discrete torsion, turns out to be $1$, then it is an SCFT realizing the periodicity element. If not, then our next study would include seeking for another SCFT corresponding to the periodicity element.

As a more fundamental problem, in order to calculate the partition functions of orbifold theories in general, we need to develop a systematic way of determining the anomaly-canceling phases. The way of calculating explicit values of the anomalous phases is known in the case of $\mathrm{U}(1)$ symmetry [OSTZ25], but such a bottom-up approach for other cases is still not well-established, and remains a subject for future work.

# Part VI
# Appendices

## A Group Extension

In this Section A, we give an elementary introduction to group extensions and a few related topics. In Section A.1, we review basic notions of group extensions following [Bro82, Ch. IV], and see that the equivalence classes of group extensions of a group $G$ by a $G$-module $N$ are in one-to-one correspondence with the cohomology classes of the second group cohomology $H^2(G, N)$. In the next Section A.2, we focus on central extensions of abelian groups, and establish another one-to-one correspondence with commutator maps in the case of free abelian groups, following [FLM88, §5.2]. When a central extension of a free abelian group $L$ is by $\mathbb{Z}_2$, we can further consider additional information, quadratic forms, and corresponding extensions of $L/2L$ by $\mathbb{Z}_2$. This is of particular importance in the construction of the $\mathbb{Z}_2$-twisted sector of a lattice VOA, so reviewed in Section A.3. The last Section A.4 is a review of a certain theorem [FLM88, Prop. 5.4.1] on the automorphism group of a central extension, which plays a vital role in the discussion of the main text.

### A.1 Group Extensions and Group Cohomology

An *extension* of a group $G$ by a group $N$ is a short exact sequence of groups and homomorphisms

$$1 \to N \xrightarrow{i} \hat{G} \xrightarrow{\pi} G \to 1. \tag{A.1}$$

(Be aware that some literature calls it an extension *of* $N$ *by* $G$.) When there is no risk of confusion in the homomorphisms constituting the short exact sequence, we only say that $\hat{G}$ is an extension of $G$ by $N$. (But note that the equivalence of group extension, defined later, classifies not only $\hat{G}$, but also the whole short exact sequence.) We also use the notation $N.G$ for any extension of $G$ by $N$. Since $i(N)$ is the kernel of $\pi$, $N$ can be regarded as a normal subgroup of $\hat{G}$, and the quotient group $\hat{G}/i(N)$ is isomorphic to $G$.

Let us take a set-theoretical section (not necessarily a group homomorphism) $s : G \to \hat{G}$. Since $s(g)s(g')s(gg')^{-1} \in \ker \pi = i(N)$ for $g, g' \in G$, we can define a function $\epsilon : G \times G \to N$ which measures how $s$ failures to be a homomorphism by

$$s(g)s(g') = i(\epsilon(g, g'))s(gg'). \tag{A.2}$$

To look into the structure of $\hat{G}$, we use the bijection $N \times G \to \hat{G}\,; (n, g) \mapsto i(n)s(g)$. Recalling that $i(N)$ is a normal subgroup of $\hat{G}$, we can define the action of $h \in \hat{G}$ on $N$ by

$$\hat{\varphi}_h(n) = i^{-1}(hi(n)h^{-1}), \tag{A.3}$$

and then the multiplication law of $\hat{G}$ can be calculated as

$$i(n)s(g) \cdot i(n')s(g') = i(n \cdot \hat{\varphi}_{s(g)}(n') \cdot \epsilon(g, g'))s(gg'). \tag{A.4}$$

This multiplication law (i.e. the group structure of $\hat{G}$) becomes simpler in the following cases:

- If we can take a section $s$ which is a group homomorphism (i.e. the short exact sequence (A.1) *splits*), then the cocycle becomes trivial $\epsilon(g, g') = 1_N$. In addition, we can define a $G$-action $\sigma$ on $N$ by $\sigma_g := \hat{\varphi}_{s(g)}$. As a result, $\hat{G}$ is isomorphic to the semidirect product $N \rtimes_\sigma G$, and it is also denoted by $N : G$. In particular, $G$ can be regarded as a subgroup of $\hat{G}$ by $s$.
  - Moreover, if the $G$-action $\sigma$ on $N$ is trivial, then $\hat{G}$ is isomorphic to the direct product $N \times G$.
- If $N$ is abelian, then for a given $g \in G$, any element $h \in \hat{G}$ such that $\pi(h) = g$ defines the same action $\hat{\varphi}_h$ on $N$, so the $\hat{G}$-action $\hat{\varphi}$ reduces to a $G$-action $\varphi$ on $N$.
  - Moreover, if (and only if) $i(N)$ is in the center of $\hat{G}$, the $G$-action $\varphi$ becomes trivial. In this case, $\hat{G}$ is called a *central extension*.

We also use the notation $N \cdot G$ for an extension $\hat{G}$ such that the short exact sequence (A.1) does not split.

In the following, we assume $N$ is abelian. An abelian group on which a group $G$ acts is called a $G$-*module*. Since there is the $G$-action $\varphi$ on $N$, $N$ is a $G$-module.

From the associativity $(s(g)s(g'))s(g'') = s(g)(s(g')s(g''))$, the function $\epsilon$ turns out to be an $N$-valued 2-cocycle:

$$\epsilon(g, g')\epsilon(gg', g'') = \varphi_g(\epsilon(g', g''))\epsilon(g, g'g''). \tag{A.5}$$

If we take another section $s'$, then the cocycle $\epsilon'$ for it differs from $\epsilon$ by only coboundary. To see it, we define $\zeta : G \to N$ as the difference of $s(g)$ and $s'(g)$:

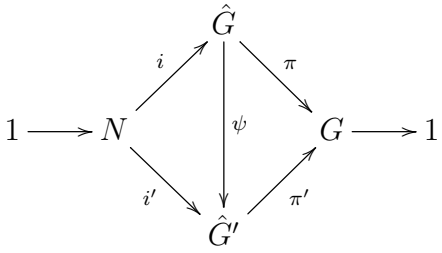
$$s'(g) = i(\zeta(g))s(g). \tag{A.6}$$

Then we can calculate $\epsilon'$ based on the definition (A.2) as

$$\epsilon'(g, g') = \epsilon(g, g')\varphi_g(\zeta(g'))\zeta(gg')^{-1}\zeta(g), \tag{A.7}$$

which shows that $\epsilon'$ and $\epsilon$ differ by the coboundary $d\zeta$. Conversely, if a cocycle $\epsilon'$ is cohomologous to the cocycle $\epsilon$ of the section $s$ as $\epsilon' = \epsilon d\zeta$, then $\epsilon'$ is the cocycle of the section $s'$ such that $s'(g) = i(\zeta(g))s(g)$.

Two extensions $1 \to N \xrightarrow{i} \hat{G} \xrightarrow{\pi} G \to 1$ and $1 \to N \xrightarrow{i'} \hat{G}' \xrightarrow{\pi'} G \to 1$ are said to be *equivalent* if there exists a homomorphism $\psi : \hat{G} \to \hat{G}'$ such that the diagram




(A.8)

commutes. Such $\psi$ is an isomorphism by the short five lemma.[77] The difference of the cocycle $\epsilon$ for a section $s : G \to \hat{G}$ and the cocycle $\epsilon'$ for a section $s' : G \to \hat{G}'$ is again a coboundary, because $\epsilon$ is also the cocycle for the section $\psi \circ s : G \to \hat{G}'$. Note that two extensions can be non-equivalent even if $\hat{G}$ and $\hat{G}'$ are isomorphic as groups.[78] In this sense, the equivalence of group extension classifies not only $\hat{G}$, but also how $N$ and $G$ are incorporated in it.

So far, for a given $G$-module $N$, we have established a map

$$\flat : \left\{ \begin{array}{c} \text{extensions of } G \text{ by } N \\ \text{compatible with the action } G \curvearrowright N \end{array} \right\} / \text{equivalence} \to H^2(G, N). \tag{A.9}$$

We can construct a map $\sharp$ in the inverse direction of (A.9) as follows. For a given $G$-module $N$ and a 2-cocycle $\epsilon : G \times G \to N$, we can construct a group extension $\hat{G}_\epsilon$ as a set $N \times G$ with the multiplication[79]

$$(n, g) \cdot (n', g') = (n \cdot g(n') \cdot \epsilon(g, g'), gg'), \tag{A.10}$$

together with group homomorphisms

$$i : N \to \hat{G}_\epsilon \, ; n \mapsto (n \cdot \epsilon(1_G, 1_G)^{-1}, 1_G), \tag{A.11}$$

$$\pi : \hat{G}_\epsilon \to G \, ; (n, g) \mapsto g, \tag{A.12}$$

constituting the short exact sequence.

[77] For the short five lemma to be applied, $\psi$ should be a homomorphism, not just a map. In fact, for any two extensions $\hat{G}, \hat{G}'$ of $G$ by $N$, there exists a map $\psi$ such that the diagram (A.8) commutes (e.g. just $i(n)s(g) \mapsto i'(n)s'(g)$ for sections $s : G \to \hat{G}$ and $s' : G \to \hat{G}'$ normalized as in (A.21)), but it is of course not an isomorphism in general.

[78] For example, for additive groups $G = \mathbb{Z}_2 \times \mathbb{Z}_2$ and $N = \mathbb{Z}_2$, (i) $\hat{G} = \mathbb{Z}_4 \times \mathbb{Z}_2$ with $i(1) = (2, 0)$ and $\pi((a, b)) = (a, b) \mod 2$, and (ii) $\hat{G} = \mathbb{Z}_4 \times \mathbb{Z}_2$ with $i(1) = (2, 0)$ and $\pi((a, b)) = (b, a) \mod 2$ are inequivalent extensions.

[79] Under this multiplication, the identity element of $\hat{G}_\epsilon$ is $(\epsilon(1_G, 1_G)^{-1}, 1_G)$, and the inverse of $(n, g)$ is $(\epsilon(1_G, 1_G)^{-1} g^{-1}(n^{-1})\epsilon(g^{-1}, g)^{-1}, g^{-1})$. We can check them by using $\epsilon(1_G, g) = \epsilon(1_G, 1_G)$, $\epsilon(g, 1_G) = g(\epsilon(1_G, 1_G))$, and $\epsilon(g, g^{-1})\epsilon(1_G, g) = g(\epsilon(g^{-1}, g))\epsilon(g, 1_G)$, which all follow from the cocycle condition (A.5).

For another 2-cocycle $\epsilon'$, if it differs from $\epsilon$ by the coboundary $d\zeta$ as in (A.7), then

$$\psi : \hat{G}_\epsilon \to \hat{G}_{\epsilon'} \, ; (n, g) \mapsto (n\zeta(g)^{-1}, g) \tag{A.13}$$

defines an homomorphism to show the equivalence of $\hat{G}_\epsilon$ and $\hat{G}_{\epsilon'}$. Therefore, we have established the map $\sharp$ in the inverse direction of (A.9).

In fact, $\flat$ and $\sharp$ are the inverse of each other. To see it, we first note that any 2-cocycle $\epsilon$ satisfies

$$\epsilon(1_G, g) = \epsilon(1_G, 1_G) \quad \text{for any } g \in G, \tag{A.14}$$

which follow from the cocycle condition (A.5). We also have

$$s(1_G) = i(\epsilon(1_G, 1_G)), \tag{A.15}$$

from (A.2), for any section $s : G \to \hat{G}$ of an extension and the cocycle $\epsilon$ for it.

To see $\sharp \circ \flat$ is an identity map, for a given extension $\hat{G}$, take a section $s : G \to \hat{G}$ and construct the extension $\hat{G}_\epsilon$ from the cocycle $\epsilon$ for $s$. Then it is equivalent to the original extension $\hat{G}$ by

$$\psi : \hat{G}_\epsilon \to \hat{G} \, ; (n, g) \mapsto i(n)s(g). \tag{A.16}$$

The most nontrivial part is $\psi \circ (i \text{ for } \hat{G}_\epsilon) = (i \text{ for } \hat{G})$, which follows from

$$\psi((n\epsilon(1_G, 1_G)^{-1}, 1_G)) = i(n\epsilon(1_G, 1_G)^{-1})s(1_G) \overset{(A.15)}{=} i(n). \tag{A.17}$$

To see $\flat \circ \sharp$ is an identity map, starting from a given cocycle $\epsilon$, construct the extension $\hat{G}_\epsilon$ from it. Then the section $s : G \to \hat{G}_\epsilon \, ; g \mapsto (1_N, g)$ gives back the cocycle $\epsilon$ because

$$s(g)s(g') = (\epsilon(g, g'), gg') = (\epsilon(g, g')\epsilon(1_G, gg')^{-1}, 1_G) \cdot (1_N, gg') \tag{A.18}$$

$$\overset{(A.14)}{=} (\epsilon(g, g')\epsilon(1_G, 1_G)^{-1}, 1_G) \cdot (1_N, gg') = i(\epsilon(g, g'))s(gg'). \tag{A.19}$$

To summarize the above discussions, we finally obtained the following theorem.

**Theorem A.1.** *For a group $G$ and a $G$-module $N$, there exists a one-to-one correspondence*

$$\left\{ \begin{array}{c} \textit{extensions of } G \textit{ by } N \\ \textit{compatible with the action } G \curvearrowright N \end{array} \right\} / \textit{equivalence} \underset{\sharp}{\overset{\flat}{\rightleftarrows}} H^2(G, N). \tag{A.20}$$

Lastly, we mention the normalization of sections and cocycles. If we take a section $s : G \to \hat{G}$ satisfying the normalization condition

$$s(1_G) = 1_{\hat{G}}, \tag{A.21}$$

then the cocycle $\epsilon$ for it satisfies the normalization condition

$$\epsilon(1_G, 1_G) = 1_N. \tag{A.22}$$

Therefore, by Theorem A.1, any cohomology class in $H^2(G, N)$ has at least one normalized cocycle. In fact, we can construct it explicitly; for any cocycle $\epsilon'$, if we define $\zeta(g) := \epsilon'(g, g)^{-1}$, then the modified cocycle $\epsilon := \epsilon' \cdot d\zeta$ satisfies the normalization (A.22). Hence, in many cases, we can restrict our attention to the normalized sections and cocycles, without loss of generality.

## A.2 Central Extensions of Free Abelian Group and Commutator Maps

Let $G$ be an abelian group and $1 \to N \xrightarrow{i} \hat{G} \xrightarrow{\pi} G \to 1$ be a central extension of $G$, that is, $N$ is a $G$-module by the trivial $G$-action. Now, any commutator $[h, h'] := hh'h^{-1}h'^{-1}$ of $h, h' \in \hat{G}$ is in $\ker \pi = i(N)$, and hence in the center of $\hat{G}$.

If we take a section $s : G \to \hat{G}$, we can define a function $c : G \times G \to N$ by

$$c(g, g') = i^{-1}([s(g), s(g')]). \tag{A.23}$$

If we take another section $s' : G \to \hat{G}$, then we have $[s'(g), s'(g')] = [s(g), s(g')]$, which follows from the fact that $s'(g)s(g)^{-1}$ is in $\ker \pi = i(N)$ and hence in the center of $\hat{G}$. Therefore, the function $c$ does not depend on the choice of the section. This $c$ is called the *commutator map* associated with the central extension.

The properties of the commutator, $[h, h] = 1$ and $[h, h'] = [h', h]^{-1}$, respectively translates to the *alternating* property

$$c(g, g) = 1_N, \tag{A.24}$$

and the *antisymmetric* property

$$c(g, g') = c(g', g)^{-1}, \tag{A.25}$$

of the commutator map $c$.

From the fact that any commutator of $\hat{G}$ is in the center of $\hat{G}$, we also have the properties

$$[hh', h''] = [h, h''][h', h''], \tag{A.26}$$
$$[h, h'h''] = [h, h'][h, h''], \tag{A.27}$$

for any $h, h', h'' \in \hat{G}$. They translates to the *bilinearity*

$$c(gg', g'') = c(g, g'')c(g', g''), \tag{A.28}$$
$$c(g, g'g'') = c(g, g')c(g, g''), \tag{A.29}$$

of the commutator map $c$. Under the bilinearity (A.28, A.29), the antisymmetric property (A.25) follows from the alternating property (A.24).[80]

In terms of the cocycle $\epsilon$ of the section $s$, defined in (A.2), the commutator map is

$$c(g, g') = \epsilon(g, g')\epsilon(g', g)^{-1}. \tag{A.30}$$

This (A.30) holds even if we replace $\epsilon$ with another cocycle $\epsilon'$ cohomologous to $\epsilon$, because such $\epsilon'$ is, as we have seen around (A.7), just a cocycle for another section. (Or, we can explicitly calculate $\epsilon'(g, g')\epsilon'(g', g)^{-1} = \epsilon(g, g')\epsilon(g', g)^{-1}$.) Therefore, (A.30) establishes a map

$$\flat_c : H^2(G, N) \to \left\{ \begin{array}{c} \text{alternating bilinear maps} \\ c : G \times G \to N \end{array} \right\} \tag{A.31}$$

[80] The converse ("antisymmetric $\Rightarrow$ alternating" under bilinearity) holds if $N$ does not have order-2 elements.

for abelian groups $G$ and $N$, where $G$ acts on $N$ trivially.

To obtain the inverse $\sharp_c$ of this map $\flat_c$, we further assume that $G$ be a free abelian group of finite rank $r$, and take a $\mathbb{Z}$-basis $e_1, \dots, e_r$ of $G$. For a given alternating $\mathbb{Z}$-bilinear map $c : G \times G \to N$, we define a $\mathbb{Z}$-bilinear map $\epsilon_c : G \times G \to N$ by the bilinear extension of

$$\epsilon_c(e_i, e_j) = \begin{cases} c(e_i, e_j) & (i > j), \\ 1_N & (i \leq j). \end{cases} \tag{A.32}$$

Since the action of $G$ on $N$ is trivial in the current situation, a $\mathbb{Z}$-bilinear map $G \times G \to N$ is automatically a 2-cocycle. Now, this cocycle $\epsilon_c$ satisfies (A.30) for the given $c$. Therefore, $c \mapsto \epsilon_c$ establishes a map

$$\sharp_c : \left\{ \begin{array}{c} \text{alternating } \mathbb{Z}\text{-bilinear maps} \\ c : G \times G \to N \end{array} \right\} \to H^2(G, N) \tag{A.33}$$

for a free abelian group $G$ of finite rank and an abelian group $N$, and $\flat_c \circ \sharp_c$ is an identity map.

To see that $\sharp_c \circ \flat_c$ is also an identity map, we start from a given cocycle $\epsilon$, consider the commutator map $c_\epsilon$ for $\epsilon$ as in (A.30), and show that the cocycle $\epsilon_{c_\epsilon}$ constructed from this $c_\epsilon$ as in (A.32) is in the same cohomology class as the original cocycle $\epsilon$. Let $\hat{G}_\epsilon$ be the central extension constructed from $\epsilon$ as in (A.10). Recall that the cocycle for the section $s : G \to \hat{G}_\epsilon \,; g \mapsto (1_N, g)$ is $\epsilon$ itself, and hence the commutator map associated with $\hat{G}_\epsilon$ is of course $c_\epsilon$. Take another section $s' : G \to \hat{G}_\epsilon$ as follows: set $s'(e_1), \dots, s'(e_r)$ to any element satisfying $\pi(s'(e_i)) = e_i$, say $s'(e_i) = s(e_i)$, and for a general element $e_1^{k^1} \cdots e_r^{k^r} \in G$, define

$$s'(e_1^{k^1} \cdots e_r^{k^r}) = s'(e_1)^{k^1} \cdots s'(e_r)^{k^r}. \tag{A.34}$$

This $s' : G \to \hat{G}_\epsilon$ is well-defined because $G$ is a free abelian group, that is, torsion-free.[81] Note that this $s'$ is not necessarily linear, or a homomorphism, because $s'(e_i)$'s are not necessarily commutative. Then the cocycle $\epsilon'$ for this section $s'$ coincides with $\epsilon_{c_\epsilon}$, because

$$s'(e_1^{k^1} \cdots e_r^{k^r}) \cdot s'(e_1^{l^1} \cdots e_r^{l^r}) \tag{A.35}$$

$$= s'(e_1)^{k^1} \cdots s'(e_r)^{k^r} \cdot s'(e_1)^{l^1} \cdots s'(e_r)^{l^r} \tag{A.36}$$

$$= s'(e_1)^{k^1} \cdots s'(e_{r-1})^{k^{r-1}} \cdot s'(e_1)^{l^1} \cdots s'(e_r)^{k^r + l^r} i(\prod_{r>j} c_\epsilon(e_r, e_j)^{k^r l^j}) \tag{A.37}$$

$$= \cdots = s'(e_1)^{k^1 + l^1} \cdots s'(e_r)^{k^r + l^r} i(\prod_{i>j} c_\epsilon(e_i, e_j)^{k^i l^j}) \tag{A.38}$$

$$= s'(e_1^{k^1 + l^1} \cdots e_r^{k^r + l^r}) i(\epsilon_{c_\epsilon}(e_1^{k^1} \cdots e_r^{k^r}, e_1^{l^1} \cdots e_r^{l^r})), \tag{A.39}$$

[81] If $G$ is a finitely generated abelian group, where torsion is allowed, then we cannot take a well-defined section $s'$ in the form of (A.34) in general, and in particular, we cannot transform (A.38) into (A.39). In fact, there are inequivalent extensions of $G$ with the same commutator map in this case; see Remark A.6 for example. If we further assume that $N$ is a divisible group, then we can take a well-defined section $s'$ in the form of (A.34) even if $G$ has a torsion part, and therefore we can still establish the one-to-one correspondence with commutator maps. For example, [Tam00, Prop. 2.6] deals with the case of $N = k^\times$, which is the multiplicative group of a field $k$.

where in the second equation, we used the fact that the commutator map $c_\epsilon$ does not depend on the choice of the section $s$ or $s'$, as we saw above. Therefore, $\epsilon$ and $\epsilon_{c_\epsilon}$ are the cocycles of the different sections $s$ and $s'$ of the same extension $\hat{G}_\epsilon$, and hence they differ only by the coboundary, as we have seen around (A.7).

We finally obtained the following theorem.

**Theorem A.2.** *For a free abelian group $G$ of finite rank and an abelian group $N$, there exist one-to-one correspondences*

$$\{\textit{central extensions of } G \textit{ by } N\} \,/\textit{equivalence} \underset{\sharp}{\overset{\flat}{\rightleftarrows}} H^2(G,N) \tag{A.40}$$

$$\underset{\sharp_c}{\overset{\flat_c}{\rightleftarrows}} \left\{ \begin{array}{c} \textit{alternating } \mathbb{Z}\textit{-bilinear maps} \\ c : G \times G \to N \end{array} \right\}, \tag{A.41}$$

*where in $H^2(G,N)$, $N$ is regarded as a $G$-module by the trivial $G$-action.*

## A.3 Central Extension of Free Abelian Group by $\mathbb{Z}_2$ and Quadratic Forms

A central extension of a free abelian group, say a lattice $L$, by $\mathbb{Z}_2$ plays a major role in discussions of a lattice VOA. In this Section A.3, following discussions around [FLM88, Prop. 5.3.4], we will see that we can induce central extensions of $L/2L$ by $\mathbb{Z}_2$ by specifying quadratic forms, which is particularly important when we construct the $\mathbb{Z}_2$-twisted sector in Section 5.3.1.

Let $L$ be a free abelian group of finite rank $r$, and we regard it as an additive group, that is, the operation is denoted by $+$ and the identity element is $0$. Let $\hat{L}$ be the central extension of $L$ by $\mathbb{Z}_2 = \langle \kappa \mid \kappa^2 = 1 \rangle$

$$1 \to \mathbb{Z}_2 \xrightarrow{i} \hat{L} \xrightarrow{\pi} L \to 0, \tag{A.42}$$

specified by a commutator map $c : L \times L \to \mathbb{Z}_2$. Since $c$ is bilinear, $c$ induces $\tilde{c} : L/2L \times L/2L \to \mathbb{Z}_2$. $\tilde{c}$ is also alternating and bilinear. $L/2L$ is isomorphic to $(\mathbb{F}_2)^r$ as an $\mathbb{F}_2$-linear vector space.

In general, for a given alternating bilinear form $\tilde{c} : (\mathbb{F}_2)^r \times (\mathbb{F}_2)^r \to \mathbb{Z}_2 = \langle \kappa \mid \kappa^2 = 1 \rangle$, there is a map $\tilde{q} : (\mathbb{F}_2)^r \to \mathbb{Z}_2$ such that

$$\tilde{c}(x,y) = \tilde{q}(x+y)\tilde{q}(x)^{-1}\tilde{q}(y)^{-1}, \tag{A.43}$$

and the set of all such maps is in the form of $\{\tilde{q}\eta \mid \eta : (\mathbb{F}_2)^r \to \mathbb{Z}_2 \text{ linear}\}$ [FLM88, Remark 5.3.2].[82] Conversely, a map $\tilde{q} : (\mathbb{F}_2)^r \to \mathbb{Z}_2$ such that the map $\tilde{c} : (\mathbb{F}_2)^r \times (\mathbb{F}_2)^r \to \mathbb{Z}_2$ defined by (A.43) is bilinear is called a *quadratic form*, and $\tilde{c}$ is called the *associated form* of $\tilde{q}$. In such a situation, $\tilde{c}$ is obviously alternating (or equivalently, symmetric in $\mathbb{Z}_2$) by the definition (A.43).

[82] The proof of this statement is as follows. Take a $\mathbb{F}_2$-basis $e_1, \dots, e_r$ of $(\mathbb{F}_2)^r$, and set $\tilde{q}(e_i)$ $(i = 1, \dots, r)$ as arbitrary values of $\mathbb{Z}_2$. Define $\tilde{\epsilon} : (\mathbb{F}_2)^r \times (\mathbb{F}_2)^r \to \mathbb{Z}_2$ as the bilinear extension of (A.49) for $\tilde{c}$ and $\tilde{q}$ instead of $c$ and $q$. Then $\tilde{q}(-) := \tilde{\epsilon}(-,-)$ satisfies (A.43). The degrees of freedom in choosing the values of $\tilde{q}(e_i)$ $(i = 1, \dots, r)$ correspond to the multiplication of linear maps $\eta : (\mathbb{F}_2)^r \to \mathbb{Z}_2$.

*Remark* A.3. More generally, a *quadratic form* on a $\mathbb{K}$-vector space $V$ is a map $\tilde{q} : V \to \mathbb{K}$ such that

$$\tilde{q}(tv) = t^2\tilde{q}(v) \quad (t \in \mathbb{K}, v \in V), \tag{A.44}$$

and its *associated form* or *polarization* $\tilde{b} : V \times V \to \mathbb{K}$ defined by

$$\tilde{b}(v, w) := \tilde{q}(v + w) - \tilde{q}(v) - \tilde{q}(w) \tag{A.45}$$

is bilinear.

If the characteristic of the field $\mathbb{K}$ is not 2, then the quadratic forms and the symmetric bilinear forms are in one-to-one correspondence under $\tilde{q} \mapsto \tilde{b}$; the converse $\tilde{b} \mapsto \tilde{q}$ is $\tilde{q}(v) = \frac{1}{2}\tilde{b}(v, v)$. In this case, the associated form is sometimes defined as

$$\tilde{b}'(v, w) := \frac{1}{2}\left(\tilde{q}(v + w) - \tilde{q}(v) - \tilde{q}(w)\right). \tag{A.46}$$

Note that if we represent a symmetric bilinear form $\tilde{b}'(v, w)$ as $v^T Bw$ by a symmetric matrix $B$, then the corresponding quadratic form $\tilde{q}(v)$ is $v^T Bv$.

However, if the characteristic of $\mathbb{K}$ is 2, then this one-to-one correspondence does not hold. In the special case $\mathbb{K} = \mathbb{F}_2$, the condition (A.44) is always satisfied, and the definition of the quadratic form reduces to the one we already defined before this Remark A.3. As mentioned above, there are multiple quadratic forms with the same associated form, and they are parametrized by linear maps $\eta : V \to \mathbb{F}_2$. One explanation of this special property of $\mathbb{F}_2$ is that any linear map $\eta$ satisfies (A.44). Another explanation is that, over $\mathbb{F}_2$, a linear map $\eta : V \to \mathbb{F}_2$ can be represented as $\eta(v) = v^T \mathrm{diag}(\eta(e_1), \ldots, \eta(e_r))v$ with an $\mathbb{F}_2$-basis $e_1, \ldots, e_r$ of $V$. Therefore, we can add arbitrary linear maps $\eta$ to a quadratic form to get another quadratic form with the same associated form. The resulting quadratic form is $\tilde{q}(v) = v^T[\tilde{\epsilon}(e_i, e_j)]_{i,j}v$ in the language of footnote 82. *(Remark ends.)*

Getting back to the $\tilde{c} : L/2L \times L/2L \to \mathbb{Z}_2$ induced from the commutator map $c$ of the central extension $\hat{L}$ (A.42), let us take a quadratic map $\tilde{q} : L/2L \to \mathbb{Z}_2$ with associated form $\tilde{c}$. Now, we will show that a lift of $2L$

$$K_{\tilde{q}} := \{i(\tilde{q}(k + 2L))\hat{k}^2 \mid \hat{k} \in \hat{L}, k = \pi(\hat{k})\} \tag{A.47}$$

is in fact a subgroup of $\hat{L}$ contained in the center of $\hat{L}$.

Let $q : L \to \mathbb{Z}_2$ be the pullback $(L \to L/2L) \circ \tilde{q}$ of $\tilde{q}$. Obviously we have

$$c(k, k') = q(k + k')q(k)^{-1}q(k')^{-1}. \tag{A.48}$$

By taking a $\mathbb{Z}$-basis $e_1, \ldots, e_r$ of $L$, we can construct a cocycle $\epsilon : L \times L \to \mathbb{Z}_2$ whose commutator $\epsilon(k, k')\epsilon(k', k)^{-1}$ coincides with $c(k, k')$, by the bilinear extension of

$$\epsilon(e_i, e_j) = \begin{cases} c(e_i, e_j) & (i > j), \\ q(e_i) & (i = j), \\ 1 & (i < j). \end{cases} \tag{A.49}$$

(Note that this construction of a cocycle can produce a different one from (A.32) in general.)

**Lemma A.4.**

*(1)* $q((t_1+t_2)k) = q(t_1k)q(t_2k)$ *for any* $t_1, t_2 \in \mathbb{Z}$ *and* $k \in L$.

*(2)* $\epsilon(k,k) = q(k)$ *for any* $k \in L$.

*Proof.* (1) Since $c$ is alternating and bilinear, $c(t_1k, t_2k) = 1$. Use it to (A.48).

(2) Let us write $k = \sum_i k^i e_i$. Using (1) and the bilinearity of $c$ to the definition (A.49) of $\epsilon$, we have

$$\epsilon(k,k) = \prod_{i>j} c(k^i e_i, k^j e_j) \prod_i q(k^i e_i). \tag{A.50}$$

Then, using (A.48) and the bilinearity of $c$ repeatedly,

$$\begin{aligned}
\epsilon(k,k) &= \prod_{i\geq 3, i>j} c(k^i e_i, k^j e_j) \cdot c(k^2 e_2, k^1 e_1) q(k^1 e_1) q(k^2 e_2) \cdot \prod_{i\geq 3} q(k^i e_i) && \text{(A.51)}\\
&= \prod_{i\geq 3, i>j} c(k^i e_i, k^j e_j) \cdot q(k^1 e_1 + k^2 e_2) \cdot \prod_{i\geq 3} q(k^i e_i) && \text{(A.52)}\\
&= \prod_{i\geq 4, i>j} c(k^i e_i, k^j e_j) \cdot c(k^3 e_3, k^1 e_1 + k_2 e^2) q(k^1 e_1 + k^2 e_2) q(k^3 e_3) \cdot \prod_{i\geq 4} q(k^i e_i) && \text{(A.53)}\\
&= \prod_{i\geq 4, i>j} c(k^i e_i, k^j e_j) \cdot q(k^1 e_1 + k^2 e_2 + k^3 e_3) \cdot \prod_{i\geq 4} q(k^i e_i) && \text{(A.54)}\\
&\vdots && \text{(A.55)}\\
&= q(k^1 e_1 + \cdots + k^r e_r) && \text{(A.56)}\\
&= q(k). && \text{(A.57)}
\end{aligned}$$

□

We take a section $s : L \to \hat{L}$ of (A.42) such that its cocycle is $\epsilon$. Then, for any $\kappa^m s(k) \in \hat{L}$ (we omit $i : \mathbb{Z}_2 \to \hat{L}$),

$$\begin{aligned}
q(\pi(\kappa^m s(k)))(\kappa^m s(k))^2 &= q(k) s(k) s(k) && \text{(A.58)}\\
&= q(k)\epsilon(k,k) s(2k) && \text{(A.59)}\\
&= s(2k), && \text{(A.60)}
\end{aligned}$$

where we used Lemma A.4 (2) in the last equation. As a result, $K_{\tilde{q}}$ defined in (A.47) can be written as

$$K_{\tilde{q}} = \{ s(2k) \mid k \in L \}. \tag{A.61}$$

In addition, since $\epsilon$ is bilinear, $\epsilon(2k, k') = \epsilon(k', 2k) = 1$. This shows that $K_{\tilde{q}}$ is a subgroup of $\hat{L}$, and it is in the center of $\hat{L}$.

Now, we have a central extension

$$1 \to \mathbb{Z}_2 \xrightarrow{\iota} \hat{L}/K_{\tilde{q}} \to L/2L \to 0, \tag{A.62}$$

such that for any $\kappa^m s(k) \in \hat{L}$,

$$(\kappa^m s(k) K_{\tilde{q}})^2 = \epsilon(k,k) s(2k) K_{\tilde{q}} = q(k) K_{\tilde{q}} = \iota(\tilde{q}(k+2L)), \tag{A.63}$$

where we used Lemma A.4 (2) in the second equation.

The following Theorem A.5 is known. From this viewpoint of Theorem A.5 (2), the equation (A.63) shows that the central extension (A.62) of $L/2L$ by $\mathbb{Z}_2$ is the one specified by the quadratic form $\tilde{q}$.

**Theorem A.5** ([FLM88, Prop. 5.3.3])**.**

*(1) For a central extension*

$$1 \to \mathbb{Z}_2 \xrightarrow{i} \hat{E} \xrightarrow{\pi} (\mathbb{F}_2)^r \to 0, \tag{A.64}$$

*whose commutator map is $\tilde{c} : (\mathbb{F}_2)^r \times (\mathbb{F}_2)^r \to \mathbb{Z}_2$, if we define a map $\tilde{q} : (\mathbb{F}_2)^r \to \mathbb{Z}_2$ by*

$$a^2 = i(\tilde{q}(\pi(a))) \quad \textit{for} \quad a \in \hat{E}, \tag{A.65}$$

*then $\tilde{q}$ is a quadratic form with its associated form $\tilde{c}$.*

*(2) The association of a quadratic form to a central extension in (1) establishes a one-to-one correspondence*

$$\{\textit{central extensions of } (\mathbb{F}_2)^r \textit{ by } \mathbb{Z}_2\} \,/\textit{equivalence} \leftrightarrow \left\{ \begin{array}{c} \textit{quadratic forms} \\ \tilde{q} : (\mathbb{F}_2)^r \to \mathbb{Z}_2 \end{array} \right\}. \tag{A.66}$$

*Remark* A.6. As we have mentioned below (A.43), there are multiple quadratic forms with the same associated form. Therefore, Theorem A.5 is an example where the one-to-one correspondence between equivalence classes of central extensions and commutator maps like Theorem A.2 does not hold. (See also footnote 81.) *(Remark ends.)*

To summarize the above discussions, we obtained the following theorem.

**Theorem A.7** ([FLM88, Prop. 5.3.4])**.** *Let $\hat{L}$ be the central extension of a free abelian group $L$ by $\mathbb{Z}_2 = \langle \kappa \mid \kappa^2 = 1 \rangle$*

$$1 \to \mathbb{Z}_2 \xrightarrow{i} \hat{L} \xrightarrow{\pi} L \to 0, \tag{A.67}$$

*specified by a commutator map $c : L \times L \to \mathbb{Z}_2$. $c$ induces the alternating bilinear map $\tilde{c} : L/2L \times L/2L \to \mathbb{Z}_2$. For any quadratic form $\tilde{q} : L/2L \to \mathbb{Z}_2$ with associated form $\tilde{c}$,*

$$K_{\tilde{q}} := \{ i(\tilde{q}(k+2L)) \hat{k}^2 \mid \hat{k} \in \hat{L}, k = \pi(\hat{k}) \} \tag{A.68}$$

*is a central subgroup of $\hat{L}$, and*

$$1 \to \mathbb{Z}_2 \to \hat{L}/K \to L/2L \to 0 \tag{A.69}$$

*is a central extension with quadratic form $\tilde{q}$.*

## A.4 A Theorem on Automorphism Group of Central Extension

This Section A.4 is a review of [FLM88, Prop. 5.4.1].

Let $1 \to N \xrightarrow{i} \hat{G} \xrightarrow{\pi} G \to 1$ be a central extension of a free abelian group $G$ of finite rank by an abelian group $N$, and $c : G \times G \to N$ be the associated commutator map.

We first define some necessary objects. We define a subgroup $\mathrm{Aut}(\hat{G}, N)$ of the automorphism group $\mathrm{Aut}(\hat{G})$ as

$$\mathrm{Aut}(\hat{G}, N) := \{\chi \in \mathrm{Aut}(\hat{G}) \mid \chi(i(n)) = i(n) \text{ for any } n \in N\}. \tag{A.70}$$

Since $N$ is abelian, $\mathrm{Hom}(G, N)$ has a group structure under the multiplication of functions: for $\eta, \eta' \in \mathrm{Hom}(G, N)$, $(\eta \cdot \eta')(g) = \eta(g)\eta'(g)$. Now we can define a group homomorphism $I : \mathrm{Hom}(G, N) \to \mathrm{Aut}(\hat{G}, N)$ which maps $\eta \in \mathrm{Hom}(G, N)$ to

$$I(\eta) : \hat{G} \to \hat{G} \tag{A.71}$$

$$h \mapsto i(\eta(\pi(h)))h. \tag{A.72}$$

In fact, $I(\eta)$ is an element of $\mathrm{Aut}(\hat{G}, N)$, and $I$ satisfies $I(\eta \cdot \eta') = I(\eta) \circ I(\eta')$.

We also define a subgroup $\mathrm{Aut}(G, c)$ of $\mathrm{Aut}(G)$ as

$$\mathrm{Aut}(G, c) := \{\phi \in \mathrm{Aut}(G) \mid c(\phi(g), \phi(g')) = c(g, g') \text{ for any } g, g' \in G\}, \tag{A.73}$$

and a group homomorphism $\Pi : \mathrm{Aut}(\hat{G}, N) \to \mathrm{Aut}(G, c)$ which maps $\chi \in \mathrm{Aut}(\hat{G}, N)$ to

$$\Pi(\chi) : G \to G \tag{A.74}$$

$$g \mapsto \pi(\chi(\hat{g})), \tag{A.75}$$

where $\hat{g}$ is an element of $\hat{G}$ such that $\pi(\hat{g}) = g$. Here, $\Pi(\chi)(g)$ does not depend on the choice of $\hat{g}$, because such another $\hat{g}'$ is an element of $\hat{g}i(N)$, and hence

$$\pi(\chi(\hat{g}')) \in \pi(\chi(\hat{g}i(N))) = \pi(\chi(\hat{g})i(N)) = \{\pi(\chi(\hat{g}))\}. \tag{A.76}$$

To see that $\Pi(\chi)$ is in fact an element of $\mathrm{Aut}(G, c)$, it suffices to calculate

$$c(\Pi(\chi)(g), \Pi(\chi)(g')) = i^{-1}([\widehat{\pi(\chi(\hat{g}))}, \widehat{\pi(\chi(\widehat{g'}))}]) \tag{A.77}$$

$$= i^{-1}([\chi(\hat{g}), \chi(\widehat{g'})]) = i^{-1}(\chi([\hat{g}, \widehat{g'}])) = i^{-1}([\hat{g}, \widehat{g'}]) = c(g, g'), \tag{A.78}$$

where we used the definition (A.23) of the commutator map $c$ in the form of $c(-,-) = i^{-1}([\hat{-}, \hat{-}])$; recall that it does not depend on the choice of $\hat{-}$. Lastly, it is easy to see that $\Pi$ is in fact a group homomorphism: $\Pi(\chi \circ \chi') = \Pi(\chi) \circ \Pi(\chi')$.

Now, here is the theorem.

**Theorem A.8** ([FLM88, Prop. 5.4.1]). *The following sequence is exact.*

$$1 \to \mathrm{Hom}(G, N) \xrightarrow{I} \mathrm{Aut}(\hat{G}, N) \xrightarrow{\Pi} \mathrm{Aut}(G, c) \to 1. \tag{A.79}$$

*Proof.* $\ker I = 1_N$ (constant function in $\mathrm{Hom}(G, N)$) and $\mathrm{im}\, I \subset \ker \Pi$ are easy.

$\ker \Pi \subset \mathrm{im}\, I$ can be checked as follows. Let $\chi \in \mathrm{Aut}(\hat{G}, N)$ satisfy $\Pi(\chi) = \mathrm{id}_G$. Since $\pi(\chi(\hat{g})) = g$ for any $g \in G$, we have $\pi(\chi(h)) = \pi(h)$ for any $h \in \hat{G}$, and hence the difference $\chi(h)h^{-1}$ of $\chi(h)$ and $h$ is in $i(N)$. Furthermore, it only depends on $\pi(h)$, because if we take another $h' \in \hat{G}$ such that $\pi(h') = \pi(h)$, then $h' \in hi(N)$ and hence $\chi(h')(h')^{-1} = \chi(h)h^{-1}$. Therefore, there exists a map $\eta : G \to N$ such that $\chi(h)h^{-1} = i(\eta(\pi(h)))$. The linearity of $\eta$ follows from, for any $h, h' \in \hat{G}$,

$$\chi(hh')(hh')^{-1} = \chi(h)\chi(h')(h')^{-1}h^{-1} = \chi(h)h^{-1}\chi(h')(h')^{-1}, \tag{A.80}$$

where the last equation follows from $\chi(h')(h')^{-1} \in i(N) \subset \mathrm{Center}(\hat{G})$. As a result, $\eta \in \mathrm{Hom}(G, N)$ and $I(\eta) = \chi$.

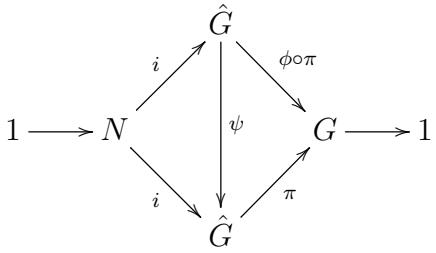
The surjectivity of $\Pi$ can be shown as follows. For $\phi \in \mathrm{Aut}(G, c)$, consider a new central extension

$$1 \to N \xrightarrow{i} \hat{G} \xrightarrow{\phi\circ\pi} G \to 1. \tag{A.81}$$

If $s : G \to \hat{G}$ is a section of the original central extension, i.e. $\pi \circ s = \mathrm{id}_G$, then $s \circ \phi^{-1}$ is a section of (A.81). Then, the commutator map $c_\phi$ associated with the new central extension coincides with the original commutator map $c$, because

$$c_\phi(g, g') = c(\phi^{-1}(g), \phi^{-1}(g)) = c(g, g'). \tag{A.82}$$

Therefore, by Theorem A.2, there exists $\psi \in \mathrm{Aut}(\hat{G})$ such that the diagram




(A.83)

commutes. It is obvious that $\psi \in \mathrm{Aut}(\hat{G}, N)$ from the diagram, and $\Pi(\psi) = \phi$ because

$$\Pi(\psi)(g) = \pi(\psi(s(g))) = \phi(\pi(s(g))) = \phi(g). \tag{A.84}$$

□

# B $\mathcal{N}=2$ SCA in Odd Leech Lattice CFT

In this Section B, we discuss how to realize an $\mathcal{N}=2$ superconformal algebra (SCA) of central charge 24 in the odd Leech lattice CFT, such that the partition function of the corresponding R sector coincides with the $\mathcal{N}=2$ extremal elliptic genus.

The $\mathcal{N}=2$ SCA of central charge $c$ consists of the operators $T(z), J(z), G^+(z), G^-(z)$ satisfying the following OPEs:

$$T(z_1)T(z_2) \sim \frac{c/2}{(z_1-z_2)^4} + \frac{2}{(z_1-z_2)^2}T(z_2) + \frac{1}{z_1-z_2}\partial T(z_2), \tag{B.1}$$

$$T(z_1)J(z_2) \sim \frac{1}{(z_1-z_2)^2}J(z_2) + \frac{1}{z_1-z_2}\partial J(z_2), \tag{B.2}$$

$$T(z_1)G^{\pm}(z_2) \sim \frac{3/2}{(z_1-z_2)^2}G^{\pm}(z_2) + \frac{1}{z_1-z_2}\partial G^{\pm}(z_2), \tag{B.3}$$

$$J(z_1)J(z_2) \sim \frac{c/3}{(z_1-z_2)^2}, \tag{B.4}$$

$$J(z_1)G^{\pm}(z_2) \sim \pm\frac{1}{z_1-z_2}G^{\pm}(z_2), \tag{B.5}$$

$$G^+(z_1)G^-(z_2) \sim \frac{2c/3}{(z_1-z_2)^3} + \frac{2}{(z_1-z_2)^2}J(z_2) + \frac{1}{z_1-z_2}(2T(z_2)+\partial J(z_2)), \tag{B.6}$$

$$G^{\pm}(z_1)G^{\pm}(z_2) \sim 0. \tag{B.7}$$

These operators, $T(z)$, $J(z)$, and $G^{\pm}(z)$ are called the *energy-momentum tensor*, the $\mathrm{U}(1)$ *current*, and the *supercurrents*, respectively.

**The candidate of supercurrents proposed by [BDFK15]**

Let us first review the candidate of $c=24$ $\mathcal{N}=2$ SCA proposed by [BDFK15]. The theory of [BDFK15] uses the reflection $\mathbb{Z}_2$ symmetry $X(z)\to -X(z)$ of the chiral bosonic lattice CFT constructed from the $(A_1)^{24}$ Niemeier lattice. Recalling from Section 5.3.1 the action of the reflection $\mathbb{Z}_2$, the currents of weight $\leq \frac{3}{2}$ in each sector of this $(A_1)^{24}$ theory are

| reflection $\mathbb{Z}_2$ | untwisted | twisted |
|---|---|---|
| even | $V_{\sqrt{2}\mathfrak{e}_i}(z)+V_{-\sqrt{2}\mathfrak{e}_i}(z)$ | |
| odd | $V_{\sqrt{2}\mathfrak{e}_i}(z)-V_{-\sqrt{2}\mathfrak{e}_i}(z)$, $\partial X^i(z)$ | $\lvert s\rangle \in \mathcal{X}((A_1)^{24})$ ($2^{\frac{24}{2}}$-dimensional) |

, (B.8)

where $\{\mathfrak{e}_i\}_{i=1,\dots,24}$ is the standard basis. The authors of [BDFK15] fermionized this $(A_1)^{24}$ theory by the reflection $\mathbb{Z}_2$ symmetry; as reviewed in Section 5.3.4, the NS sector of the resulting theory is the even-untwisted and odd-twisted sectors above, and the R sector is the odd-untwisted and even-twisted sectors. They further proposed a $c=24$ $\mathcal{N}=2$ SCA in the NS sector such that the partition function of the R sector coincides with the extremal $\mathcal{N}=2$ elliptic genus of central charge 24.

As mentioned in Section 5.3.4, the fermionized theories by the reflection $\mathbb{Z}_2$ and by the shift $\mathbb{Z}_2$ are equivalent. With respect to the shift $\mathbb{Z}_2$ symmetry (5.121), the currents of weight $\leq \frac{3}{2}$ in each sector of the $(A_1)^{24}$ theory are

| shift $\mathbb{Z}_2$ | untwisted | twisted |
|---|---|---|
| even | $\partial X^i(z)$ | |
| odd | $V_{\sqrt{2}\mathbb{e}_i}(z), V_{-\sqrt{2}\mathbb{e}_i}(z)$ | $V_{(-1)^w/2\sqrt{2}}(z), w \in G_{24}$ in (10.4) |

, (B.9)

and the NS sector of the fermionized theory by this shift $\mathbb{Z}_2$ is the odd Leech lattice CFT. So we can translate their theory in [BDFK15] into the odd Leech lattice CFT.

By comparing (B.8) and (B.9), we can translate the candidate of $c = 24$ $\mathcal{N} = 2$ SCA proposed by [BDFK15] in terms of the odd Leech lattice CFT as

$$T(z) = -\frac{1}{2}\sum_{i=1}^{24} :(\partial X^i(z))^2:, \tag{B.10}$$

$$J(z) = 2\sqrt{-2}\partial X^1(z), \tag{B.11}$$

$$G^+(z) = \frac{1}{8\sqrt{2}} \sum_{w\in G_{24} \text{ such that } w_1=0} V_{\frac{1}{2\sqrt{2}}(-1)^w}(z), \tag{B.12}$$

$$G^-(z) = \frac{1}{8\sqrt{2}} \sum_{w\in G_{24} \text{ such that } w_1=1} V_{\frac{1}{2\sqrt{2}}(-1)^w}(z), \tag{B.13}$$

where $w = (w_1, \dots, w_{24})$ and $(-1)^w = ((-1)^{w_1}, \dots, (-1)^{w_{24}})$ as in (10.4).

These operators in fact satisfy the $TT$, $TJ$, $JJ$, $TG^\pm$, and $JG^\pm$ OPEs (B.1)–(B.5). However, if we calculate the $G^\pm G^\pm$ OPE and $G^+G^-$ OPE for them, while taking into account the cocycle factor given by (8.13) and the choice of the basis (2.39), then they fail to satisfy the proper OPEs (B.6) and (B.7); their OPEs contain unnecessary weight-2 currents $V_{\frac{1}{\sqrt{2}}w^\pm}(z_2)$, where $w^\pm$ denotes an octad whose non-zero entries are 1 or $-1$, in the $(z_1 - z_2)^{-1}$ term. See `check_supercurrent_OPE.ipynb` for the explicit calculation by computer. In fact, the discussion of [BDFK15, Appendix A.2] on how these unnecessary weight-2 currents decouple from the OPEs is based on the assumption that the subgroup $M_{24}$ of the automorphism group of the lattice directly lifts to a subgroup of the automorphism group of the lattice CFT, but in reality, it is disturbed by the cocycle factors as in Theorem 9.4.

Their proposal is still valuable in the sense that the choice of the $\mathrm{U}(1)$ current $J(z)$ in (B.11) is enough for the elliptic genus (see Section 3.4) to reproduce the $\mathcal{N} = 2$ extremal elliptic genus of central charge 24. The $\mathrm{U}(1)$ current (B.11) determines the first few terms of the elliptic genus as

$$\mathrm{Tr}_{\mathcal{H}_\mathrm{R}}(-1)^F y^{J_0} q^{L_0 - \frac{c}{24}} = y^{-4} + 46 + y^4 + O(q), \tag{B.14}$$

and such a weak Jacobi form of weight 0 and index $\frac{c}{6} = 4$ is uniquely determined to be the $\mathcal{N} = 2$ extremal elliptic genus of central charge 24 (see the discussion around (3.36)). So we only have to

modify the supercurrents $G^{\pm}(z)$ so that these operators satisfy the correct OPEs (B.1)–(B.7) for the $\mathcal{N}=2$ SCA.

*Remark* B.1. *(A long remark; ends at (B.20).)* One of the advantageous points of the candidate (B.12, B.13) of the supercurrents was that they are invariant under the $M_{23}$ symmetry fixing the first entry of $G_{24}$, if we forget the effect of cocycle factors of the odd Leech lattice CFT. Unfortunately, however, we cannot find correct supercurrents while retaining this property.

To state it more precisely, we introduce some notations. Let $G_{24}^{+}$ and $G_{24}^{-}$ denote the set of all codewords of $G_{24}$ with the first entry 0 and 1, respectively. Let $G_{24}^{(x)}$ denote the set of all codewords of $G_{24}$ with Hamming weight $x$, and also decompose it into $G_{24}^{(x),\pm}$ according to the first entry. For example, $G_{24}^{(0)}=G_{24}^{(0),+}=\{\underline{0}\}$ and $G_{24}^{(24)}=G_{24}^{(24),-}=\{\vec{1}\}$. The orbit decomposition of $G_{24}^{+}$ and $G_{24}^{-}$ under the action of $M_{23}$ is $\{\underline{0}\}\sqcup G_{24}^{(8),+}\sqcup G_{24}^{(12),+}\sqcup G_{24}^{(16),+}$ and $G_{24}^{(8),-}\sqcup G_{24}^{(12),-}\sqcup G_{24}^{(16),-}\sqcup\{\vec{1}\}$, respectively.[83] Therefore, if a current $G^{\pm}(z)$ of weight $\frac{3}{2}$ and $\mathrm{U}(1)$ charge $\pm1$ is invariant under the $M_{23}$ symmetry, then it should be in the form of the linear combination

$$G^{+}(z)=c^{(0),+}G^{(0),+}(z)+c^{(8),+}G^{(8),+}(z)+c^{(12),+}G^{(12),+}(z)+c^{(16),+}G^{(16),+}(z), \tag{B.15}$$

$$G^{-}(z)=c^{(8),-}G^{(8),-}(z)+c^{(12),-}G^{(12),-}(z)+c^{(16),-}G^{(16),-}(z)+c^{(24),-}G^{(24),-}(z), \tag{B.16}$$

where we defined

$$G^{(x),\pm}(z):=\sum_{w\in G_{24}^{(x),\pm}}V_{\frac{1}{2\sqrt{2}}(-1)^{w}}(z). \tag{B.17}$$

However, we cannot find the coefficients $c^{(x),\pm}$ such that these $G^{+}(z),G^{-}(z)$ together with $T(z)$ in (B.10) and $J(z)$ in (B.11) satisfy the correct $\mathcal{N}=2$ SCA OPEs (B.1)–(B.7).

To see it, we first use the $G^{+}G^{+}$ OPE. In the calculation of the $G^{+}G^{+}$ OPE for (B.15), the only sources of singular terms are

$$V_{\frac{1}{2\sqrt{2}}(-1)^{w}}(z_1)V_{\frac{1}{2\sqrt{2}}(-1)^{w'}}(z_2)\sim\varepsilon(\tfrac{1}{2\sqrt{2}}(-1)^{w},\tfrac{1}{2\sqrt{2}}(-1)^{w'})\frac{1}{z_1-z_2}V_{\frac{1}{2\sqrt{2}}((-1)^{w}+(-1)^{w'})}(z_2), \tag{B.18}$$

for $w,w'\in G_{24}^{+}$ such that $(-1)^{w}+(-1)^{w'}$ has precisely eight non-zero entries $\pm2$, and hence $V_{\frac{1}{2\sqrt{2}}((-1)^{w}+(-1)^{w'})}(z_2)$ is of weight 2. By focusing on such terms, while taking into account the cocycle factors, we obtain the conditions on $c^{(x),+}$ for $G^{+}(z)$ in (B.15) to satisfy the correct OPE $G^{+}G^{+}\sim0$ as follows.

[83] This can be seen as follows. $M_{23}$ acts on $G_{24}^{(x)}$ as a permutation group, so we can consider a $|G_{24}^{(x)}|$-dimensional representation $R^{(x)}:=\bigoplus_{w\in G_{24}^{(x)}}\mathbb{C}|w\rangle$ of $M_{23}$. If $O\subset G_{24}^{(x)}$ is an orbit of some element under the action of $M_{23}$, then $\mathbb{C}(\sum_{w\in O}|w\rangle)$ is the 1-dimensional irreducible subrepresentation of $R^{(x)}$. Therefore, we have (the number of orbits in $G_{24}^{(x)}$) $\leq$ (the multiplicity of the 1-dimensional irrep. in $R^{(x)}$). By using the character table of $M_{23}$ from the ATLAS book [CCN+85], we can calculate by computer the multiplicity of the 1-dimensional irrep. in $R^{(x)}$ as $1,2,2,2,1$ for $x=0,8,12,16,24$, respectively. This justifies the orbit decompositions in the main text. In particular, $M_{23}$ acts transitively on each $G_{24}^{(x),\pm}$. In a similar way, we can also check that $M_{24}$ acts transitively on each $G_{24}^{(x)}$.

- $c^{(0),+}c^{(16),+} - (c^{(8),+})^2 = 0$ from the terms of $w, w' \in G_{24}^+$ such that
  $(-1)^w + (-1)^{w'} = (2,0,2,0,0,0,0,0,0,0,0,2,0,0,2,0,0,0,0,0,2,2,2,2)$.
- $c^{(0),+}c^{(16),+} + 15(c^{(8),+})^2 = 0$ from the terms of $w, w' \in G_{24}^+$ such that
  $(-1)^w + (-1)^{w'} = (2,0,0,0,2,2,0,0,0,0,0,0,0,0,0,0,0,2,0,2,0,2,2,2)$.
- $(c^{(12),+})^2 - c^{(8),+}c^{(16),+} = 0$ from the terms of $w, w' \in G_{24}^+$ such that
  $(-1)^w + (-1)^{w'} = (2,0,2,0,0,-2,0,-2,0,-2,0,0,-2,0,0,0,0,0,0,0,0,0,2,2)$.

See `check_supercurrent_OPE.ipynb` again for the computation of these conditions. The solution to them is $c^{(8),+} = c^{(12),+} = 0$ and $c^{(0),+}c^{(16),+} = 0$.

As a result, if $G^+(z)$ in (B.15) satisfies the OPE $G^+G^+ \sim 0$, then it must be $G^{(0),+}(z)$ or $G^{(16),+}$ up to scalar multiplication. If we assume $G^+(z) \propto G^{(0),+}(z)$, then the OPE $G^+G^+ \sim 0$ is achieved, and $G^-(z)$ in (B.16) must contain $G^{(24),-}(z)$ for their $G^+G^-$ OPE to have a $(z_1 - z_2)^{-3}$ term. However, its $(z_1 - z_2)^{-2}$ term is proportional to

$$\sum_{w \in G_{24}^{(0),+}} (-1)^w \cdot \partial X(z) = \sum_{i=1}^{24} \partial X^i(z), \tag{B.19}$$

where $(-1)^w \cdot X(z) := \sum_{i=1}^{24} (-1)^{w_i} \partial X^i(z)$, and this is not $J(z)$ in (B.11). So the $G^+G^-$ OPE fails. Let us assume $G^+(z) \propto G^{(16),+}(z)$, then. The OPE $G^+G^+ \sim 0$ is achieved, because it is easy to see that $(-1)^w + (-1)^{w'}$ cannot have precisely eight non-zero entries when $w, w' \in G^{(16),+}$. Now, $G^-(z)$ in (B.16) must contain $G^{(8),-}(z)$, for their $G^+G^-$ OPE to have a $(z_1 - z_2)^{-3}$ term. However, its $(z_1 - z_2)^{-2}$ term is proportional to

$$\sum_{w \in G_{24}^{(16),+}} (-1)^w \cdot \partial X(z) = 253\, \partial X^1(z) - 99 \sum_{i=2}^{24} \partial X^i(z), \tag{B.20}$$

so the $G^+G^-$ OPE fails again. This concludes that we have to give up the $M_{23}$ symmetry to modify the supercurrents so that they satisfy the correct $\mathcal{N} = 2$ SCA OPE. *(Remark ends.)*

**Construct supercurrents**

To modify the supercurrents, we first point out that we can construct a $c = 1$ $\mathcal{N} = 2$ SCA from a codeword $w = (w_1, \ldots, w_{24})$ of the binary Golay code $G_{24}$ as

$$T^w(z) = -\frac{1}{48} : \left((-1)^w \cdot \partial X(z)\right)^2 :, \tag{B.21}$$

$$J^w(z) = \frac{\sqrt{-1}}{6\sqrt{2}} (-1)^w \cdot \partial X(z), \tag{B.22}$$

$$G^{+,w}(z) = \sqrt{\frac{2}{3}} V_{\frac{1}{2\sqrt{2}}(-1)^w}(z), \tag{B.23}$$

$$G^{-,w}(z) = \sqrt{\frac{2}{3}} V_{-\frac{1}{2\sqrt{2}}(-1)^w}(z), \tag{B.24}$$

where $(-1)^w \cdot X(z) := \sum_{i=1}^{24} (-1)^{w_i} \partial X^i(z)$. This is just a well-known construction of a $c = 1$ $\mathcal{N} = 2$ SCA from one free boson [Wat86], if we change the variables as $Y(z) := \frac{1}{2\sqrt{6}} (-1)^w \cdot X(z)$. The cocycle factor does not cause a problem here because

$$\varepsilon(\frac{1}{2\sqrt{2}}(-1)^w, -\frac{1}{2\sqrt{2}}(-1)^w) = 1, \tag{B.25}$$

which follows from the bilinearity of $\varepsilon$ and Lemma A.4 (2) $\varepsilon(k,k) = q(k) = \kappa^{\frac{1}{2}(|k|^2+|k|^4)}$.

In addition, if $w, w' \in G_{24}$ satisfy the orthogonality condition

$$(-1)^w \cdot (-1)^{w'} = \sum_{i=1}^{24} (-1)^{w_i + w'_i} = 0, \tag{B.26}$$

then any OPE between $T^w$, $J^w$, $G^{\pm,w}$ and $T^{w'}$, $J^{w'}$, $G^{\pm,w'}$ vanishes. Therefore, $T^w + T^{w'}$, $J^w + J^{w'}$, and $G^{\pm,w} + G^{\pm,w'}$ constitute an $c = 1 + 1$ $\mathcal{N} = 2$ SCA.

We can find the following 24 codewords $w^{(1)}, \dots, w^{(24)} \in G_{24}$

$$\begin{aligned}
(-1)^{w^{(1)}} &= (\ 1\ \ 1\ \ 1\ \ 1\ \ 1\ \ 1\ \ 1\ \ 1\ \ 1\ \ 1\ \ 1\ \ 1\ \ 1\ \ 1\ \ 1\ \ 1\ \ 1\ \ 1\ \ 1\ \ 1\ \ 1\ \ 1\ \ 1\ \ 1\ \ 1\ \ 1), \\
(-1)^{w^{(2)}} &= (\ 1\ -1\ \ 1\ -1\ -1\ -1\ \ 1\ -1\ -1\ -1\ -1\ -1\ -1\ \ 1\ -1\ -1\ \ 1\ \ 1\ \ 1\ \ 1\ \ 1\ \ 1\ \ 1\ \ 1), \\
(-1)^{w^{(3)}} &= (\ 1\ \ 1\ -1\ -1\ -1\ \ 1\ -1\ \ 1\ \ 1\ -1\ -1\ \ 1\ \ 1\ -1\ -1\ -1\ -1\ -1\ -1\ \ 1\ \ 1\ \ 1\ \ 1\ \ 1), \\
(-1)^{w^{(4)}} &= (\ 1\ -1\ -1\ \ 1\ \ 1\ -1\ -1\ -1\ -1\ \ 1\ \ 1\ -1\ -1\ -1\ \ 1\ \ 1\ -1\ -1\ -1\ \ 1\ \ 1\ \ 1\ \ 1\ \ 1), \\
(-1)^{w^{(5)}} &= (\ 1\ -1\ -1\ -1\ -1\ -1\ \ 1\ \ 1\ \ 1\ \ 1\ \ 1\ -1\ \ 1\ \ 1\ \ 1\ -1\ -1\ \ 1\ -1\ -1\ -1\ -1\ \ 1\ \ 1), \\
(-1)^{w^{(6)}} &= (\ 1\ \ 1\ -1\ \ 1\ \ 1\ \ 1\ \ 1\ -1\ -1\ -1\ -1\ \ 1\ -1\ \ 1\ -1\ \ 1\ -1\ \ 1\ -1\ -1\ -1\ -1\ \ 1\ \ 1), \\
(-1)^{w^{(7)}} &= (\ 1\ \ 1\ -1\ -1\ -1\ -1\ -1\ -1\ -1\ \ 1\ \ 1\ \ 1\ \ 1\ -1\ -1\ \ 1\ \ 1\ \ 1\ \ 1\ -1\ -1\ \ 1\ -1\ \ 1), \\
(-1)^{w^{(8)}} &= (\ 1\ -1\ -1\ \ 1\ \ 1\ \ 1\ -1\ \ 1\ \ 1\ -1\ -1\ -1\ -1\ -1\ \ 1\ -1\ \ 1\ \ 1\ \ 1\ -1\ -1\ \ 1\ -1\ \ 1), \\
(-1)^{w^{(9)}} &= (\ 1\ -1\ \ 1\ \ 1\ \ 1\ -1\ \ 1\ \ 1\ -1\ \ 1\ -1\ \ 1\ \ 1\ -1\ -1\ -1\ -1\ -1\ \ 1\ \ 1\ -1\ -1\ -1\ \ 1), \\
(-1)^{w^{(10)}} &= (\ 1\ \ 1\ \ 1\ -1\ -1\ \ 1\ \ 1\ -1\ \ 1\ -1\ \ 1\ -1\ -1\ -1\ \ 1\ \ 1\ -1\ -1\ \ 1\ \ 1\ -1\ -1\ -1\ \ 1), \\
(-1)^{w^{(11)}} &= (\ 1\ \ 1\ \ 1\ \ 1\ -1\ -1\ -1\ -1\ \ 1\ \ 1\ -1\ \ 1\ -1\ \ 1\ \ 1\ -1\ \ 1\ -1\ -1\ -1\ \ 1\ -1\ -1\ \ 1), \\
(-1)^{w^{(12)}} &= (\ 1\ -1\ \ 1\ -1\ \ 1\ \ 1\ -1\ \ 1\ -1\ -1\ \ 1\ -1\ \ 1\ \ 1\ -1\ \ 1\ \ 1\ -1\ -1\ -1\ \ 1\ -1\ -1\ \ 1), \\
(-1)^{w^{(13)}} &= (\ 1\ -1\ -1\ -1\ -1\ \ 1\ \ 1\ \ 1\ -1\ \ 1\ -1\ \ 1\ -1\ -1\ \ 1\ \ 1\ \ 1\ \ 1\ -1\ \ 1\ \ 1\ -1\ -1\ -1), \\
(-1)^{w^{(14)}} &= (\ 1\ \ 1\ -1\ \ 1\ \ 1\ -1\ \ 1\ -1\ \ 1\ -1\ \ 1\ -1\ \ 1\ -1\ -1\ -1\ \ 1\ \ 1\ -1\ \ 1\ \ 1\ -1\ -1\ -1), \\
(-1)^{w^{(15)}} &= (\ 1\ \ 1\ \ 1\ \ 1\ -1\ -1\ -1\ \ 1\ -1\ -1\ -1\ -1\ \ 1\ \ 1\ \ 1\ \ 1\ -1\ \ 1\ -1\ \ 1\ -1\ \ 1\ -1\ -1), \\
(-1)^{w^{(16)}} &= (\ 1\ -1\ \ 1\ -1\ \ 1\ \ 1\ -1\ -1\ \ 1\ \ 1\ \ 1\ \ 1\ -1\ \ 1\ -1\ -1\ -1\ \ 1\ -1\ \ 1\ -1\ \ 1\ -1\ -1), \\
(-1)^{w^{(17)}} &= (\ 1\ \ 1\ -1\ -1\ \ 1\ \ 1\ \ 1\ -1\ -1\ \ 1\ -1\ -1\ \ 1\ \ 1\ \ 1\ -1\ -1\ -1\ \ 1\ -1\ \ 1\ \ 1\ -1\ -1), \\
(-1)^{w^{(18)}} &= (\ 1\ -1\ -1\ \ 1\ -1\ -1\ \ 1\ \ 1\ \ 1\ -1\ \ 1\ \ 1\ -1\ \ 1\ -1\ \ 1\ -1\ -1\ \ 1\ -1\ \ 1\ \ 1\ -1\ -1), \\
(-1)^{w^{(19)}} &= (\ 1\ \ 1\ \ 1\ \ 1\ -1\ \ 1\ -1\ \ 1\ -1\ \ 1\ \ 1\ -1\ -1\ -1\ -1\ -1\ -1\ \ 1\ \ 1\ -1\ \ 1\ -1\ \ 1\ -1), \\
(-1)^{w^{(20)}} &= (\ 1\ -1\ \ 1\ -1\ \ 1\ -1\ -1\ -1\ \ 1\ -1\ -1\ \ 1\ \ 1\ -1\ \ 1\ \ 1\ -1\ \ 1\ \ 1\ -1\ \ 1\ -1\ \ 1\ -1), \\
(-1)^{w^{(21)}} &= (\ 1\ \ 1\ -1\ -1\ \ 1\ -1\ -1\ \ 1\ \ 1\ \ 1\ -1\ -1\ -1\ \ 1\ -1\ \ 1\ \ 1\ -1\ \ 1\ \ 1\ -1\ -1\ \ 1\ -1), \\
(-1)^{w^{(22)}} &= (\ 1\ -1\ -1\ \ 1\ -1\ \ 1\ -1\ -1\ -1\ -1\ \ 1\ \ 1\ \ 1\ \ 1\ \ 1\ -1\ \ 1\ -1\ \ 1\ \ 1\ -1\ -1\ \ 1\ -1), \\
(-1)^{w^{(23)}} &= (\ 1\ -1\ \ 1\ \ 1\ -1\ \ 1\ \ 1\ -1\ \ 1\ \ 1\ -1\ -1\ \ 1\ -1\ -1\ \ 1\ \ 1\ -1\ -1\ -1\ -1\ \ 1\ \ 1\ -1), \\
(-1)^{w^{(24)}} &= (\ 1\ \ 1\ \ 1\ -1\ \ 1\ -1\ \ 1\ \ 1\ -1\ -1\ \ 1\ \ 1\ -1\ -1\ \ 1\ -1\ \ 1\ -1\ -1\ -1\ -1\ \ 1\ \ 1\ -1),
\end{aligned} \tag{B.27}$$

by searching 23 dodecads $w^{(2)}, \dots, w^{(24)}$ such that any two of the 24 vectors, $(-1)^{w^{(1)}} = \vec{1}$ and $(-1)^{w^{(i)}}$ for those dodecads, satisfy the orthogonality condition (B.26). As a result, the sum

$$T(z) = \sum_{i=1}^{24} T^{w^{(i)}}(z), \quad J(z) = \sum_{i=1}^{24} J^{w^{(i)}}(z), \quad G^{\pm}(z) = \sum_{i=1}^{24} G^{\pm,w^{(i)}}(z), \tag{B.28}$$

constitute a $c = 24$ $\mathcal{N} = 2$ SCA. Moreover, we searched the 23 dodecads under the condition that the first entry of $(-1)^{w^{(i)}}$ is $(-1)^{w_1^{(i)}} = 1$, so the first column vector of $[(-1)^{w_j^{(i)}}]_{i,j}$ is $\vec{1}$, and the other column vectors are all orthogonal to $\vec{1}$ because it is an orthogonal matrix (up to scalar multiplication by $\frac{1}{2\sqrt{6}}$). As a result, the sum $J(z) = \sum_i J^{w^{(i)}}(z)$ turns out to be

$$J(z) = 2\sqrt{-2}\partial X^1(z), \tag{B.29}$$

which coincides with $J(z)$ in (B.11) proposed by [BDFK15]. We can also check that

$$T(z) = -\frac{1}{2}\sum_{i=1}^{24} :(\partial X^i(z))^2:, \tag{B.30}$$

by explicit calculation. See `find_supercurrent.ipynb` for the source code to find $w^{(1)}$, $\ldots, w^{(24)}$ and to check (B.29, B.30).

This construction of $c = 24$ $\mathcal{N} = 2$ SCA can be explained in connection with ternary codes as follows. We have constructed an odd Leech lattice $O_{24}^{\text{bin}}$ from the binary Golay code $G_{24}$, but we can also construct odd Leech lattices $O_{24}^{\text{tern}}$ from certain ternary codes as we reviewed in Section 2.3.1 (see also [GJF18, Example 4.5]). Let $W$ denote the orthogonal matrix $[\frac{1}{2\sqrt{6}}(-1)^{w_j^{(i)}}]_{i,j}$. Then $O_{24}^{\text{tern}} := O_{24}^{\text{bin}} W^T$ is in fact an odd Leech lattice constructed from a ternary code, and $W$ can be regarded as an isometry between $O_{24}^{\text{bin}}$ and $O_{24}^{\text{tern}}$; for example, the row vector $\frac{1}{2\sqrt{2}}(-1)^{w^{(i)}} \in O_{24}^{\text{bin}}$ is mapped to $\frac{1}{2\sqrt{2}}(-1)^{w^{(i)}} W^T = \sqrt{3}\mathbb{e}_i$, which should be contained in any lattice constructed from a ternary code by Construction A. To see that the lattice $O_{24}^{\text{tern}}$ in fact can be constructed from some ternary code, it suffices to observe that the lattice $O_{24}^{\text{tern}}$ is a subset of $\frac{1}{\sqrt{3}}\mathbb{Z}^{24}$, and contains $\{\sqrt{3}\mathbb{e}_i\}_{i=1,\ldots,24}$ (which is a *3-frame* in the terminology of [HMV09, HM09]). In a lattice CFT constructed from ternary codes, [GJF18] pointed out that we can always find a $c = 1$ $\mathcal{N} = 2$ SCA associated with each direction of the standard basis $\mathbb{e}_i$, and hence the sum of them constitute a $c = n = 24$ $\mathcal{N} = 2$ SCA with the $\mathrm{U}(1)$ current in the direction of $\sum_i \mathbb{e}_i = (1, \ldots, 1)$. This direction in $O_{24}^{\text{tern}}$ is mapped by $W$ to the direction of $\mathbb{e}_1 = (1, 0, \ldots, 0)$ in $O_{24}^{\text{bin}}$, and therefore we obtained the $c = 24$ $\mathcal{N} = 2$ SCA with the $\mathrm{U}(1)$ current $J(z)$ in the direction of $\mathbb{e}_1$ as above (B.29).

What is the subgroup of $\mathrm{U}(1)^{24}.M_{24} \subset \mathrm{Aut}(V_{O_{24}})$ preserving the $\mathcal{N} = 2$ SCA (B.28) we have constructed? Since the supercurrents have been constructed from a specific choice of 23 dodecads with the first entry 0, it should be a subgroup of some extension of $M'$, where $M'$ is the subgroup of $M_{24}$ stabilizing the set of these 23 dodecads. The group $M_{23}$ acts transitively on the set $G_{24}^{(12),+}$ of all the dodecads with the first entry 0 (see footnote 83), so such $M'$ is at least smaller than $M_{23}$, and would be too small to provide any useful implications.

# C Data of the Equations to Show the Lift Does Not Preserve the Group Structures

Here, we record the coefficient matrix $M$ and the vector $\vec{\mu}$ of the equation (9.25) for the cases we mentioned in Section 9.

For $M_{24}$ of odd Leech lattice,

$$M = \left(\begin{array}{cccccccccccccccccccccccc}
2&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0\\
2&0&0&0&-1&-1&2&1&-3&0&4&-1&-2&3&0&-3&1&0&-2&-1&0&-2&-2&4\\
2&-1&1&1&-1&-2&2&1&-4&0&6&-3&-3&5&0&-4&3&1&-3&0&0&-2&-2&3\\
0&0&1&1&-1&0&2&0&-1&1&0&-4&0&3&-1&0&2&-1&-1&2&2&2&2&-4\\
0&0&1&0&0&0&1&-1&0&2&-1&-3&1&2&-1&1&2&-1&0&2&1&1&2&-4\\
2&-1&-1&1&0&-1&0&2&-1&-2&4&2&-3&1&2&-3&-1&1&-2&-3&-2&-2&-3&7\\
0&0&0&0&0&0&2&0&0&1&0&-1&0&1&-1&0&0&-1&-1&0&0&0&1&1\\
0&0&1&1&0&-1&0&1&0&1&0&-3&-1&1&-1&0&2&0&0&1&1&1&2&-2\\
2&0&-1&-1&1&0&-2&0&3&-1&-2&4&1&-3&2&2&-3&0&1&-2&-2&-1&-1&2\\
0&1&0&-1&1&2&-1&-2&2&2&-5&0&3&-2&-1&3&-1&-1&3&2&1&2&3&-6\\
2&0&0&-1&0&1&-1&-2&2&2&-3&-1&3&-1&-1&3&0&-1&1&1&0&1&2&-4\\
0&1&0&-1&1&2&-1&-2&2&0&-5&2&3&-3&0&3&-1&0&3&2&1&2&2&-6\\
0&0&2&0&-2&0&2&-2&-2&4&0&-8&2&5&-3&1&7&-1&-1&4&3&2&4&-9\\
0&0&0&0&0&0&0&0&0&0&0&0&0&1&1&0&1&0&0&0&0&0&0&-1\\
0&0&0&0&0&0&0&0&0&0&0&0&0&0&1&1&0&1&0&0&0&0&0&0\\
0&0&0&0&0&0&0&0&2&0&-2&0&1&-1&0&2&0&-1&0&0&0&1&1&-1\\
0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&2&0&0&0&0&0&0&0\\
0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&1&1&0&0&0&0&0&1\\
0&0&0&0&0&0&0&0&0&2&0&-2&0&1&-1&0&2&-1&0&0&0&0&1&0\\
0&0&0&0&2&0&-2&0&2&-2&-2&4&0&-4&1&1&-2&1&2&0&-1&0&-1&1\\
0&2&0&-2&0&2&-2&-2&4&0&-8&2&5&-4&1&6&-1&-1&4&2&2&3&3&-10\\
0&0&0&0&0&2&0&-2&0&2&-2&-2&2&0&-2&1&2&-1&1&2&1&2&2&-4\\
0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&1&1&0&0&0&0&1&-1\\
0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&2&0&0&0&0&0&0&0\\
\hline
18&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0\\
14&2&-2&0&-1&-1&-2&5&-1&-7&2&8&-4&-3&6&-3&-4&1&1&-5&0&-2&-7&8\\
12&3&-1&0&-2&0&-1&8&-1&-12&-1&11&-4&-5&8&-3&-7&3&2&-4&3&1&-5&3\\
10&2&2&1&-7&0&9&1&-11&3&7&-13&-2&15&-2&-6&13&-2&-5&4&8&2&2&-9\\
12&0&1&3&-2&-3&2&1&-6&-1&8&-5&-5&7&1&-5&7&3&-3&1&2&-1&-2&-1\\
12&-1&-1&0&-1&1&5&-1&-6&3&7&-3&-2&6&-3&-6&3&-1&-4&-1&-1&-3&-2&8\\
10&-1&3&2&-2&-4&3&-2&-3&8&5&-15&-1&12&-3&0&13&-1&-4&4&2&0&5&-9\\
10&0&2&1&4&-3&-7&2&8&-4&-7&5&0&-8&4&6&-4&3&5&1&-1&2&2&-6\\
12&-2&0&-1&3&2&-3&-5&5&5&-5&-1&5&-3&-3&5&-1&-1&2&2&-3&1&4&-4\\
14&-3&-2&-1&6&3&-3&-5&5&3&-3&6&3&-9&-5&2&-8&-3&2&-1&-7&-2&-3&13\\
10&-1&2&0&2&2&-2&-5&5&7&-8&-7&6&-2&-6&7&2&-2&3&6&1&5&9&-13\\
14&-1&-2&-3&0&3&3&-1&-1&1&6&2&-1&0&-2&-6&-2&-2&-3&-3&-3&-4&10\\
4&4&2&0&-4&-2&0&8&-2&-10&-2&4&-3&1&9&-1&-1&4&2&0&7&3&0&-11\\
8&0&-2&-4&-2&4&4&2&-4&-4&2&8&1&1&3&-5&-6&-2&-2&-4&-1&-3&-6&10\\
8&0&-2&0&-2&0&4&2&-8&-4&10&4&-5&5&3&-9&0&2&-4&-4&-1&-5&-8&12\\
4&0&4&-2&-6&0&6&-2&-4&8&0&-16&5&15&-2&2&13&-3&-4&6&6&3&9&-20\\
8&0&0&0&-4&-4&4&4&-8&-4&12&0&-6&10&6&-8&4&4&-6&-4&0&-6&-6&6\\
4&0&0&-2&4&0&-4&2&8&-4&-8&10&2&-9&4&5&-10&1&4&-2&-3&0&1&1\\
8&-2&0&-2&2&0&-2&0&2&0&0&4&1&-2&1&0&-4&-1&0&-2&-4&-3&-3&7\\
8&-2&0&-2&0&0&0&0&2&-2&0&4&2&-1&2&1&-4&1&-2&-2&-3&-2&-1&3\\
8&2&-2&-4&-2&4&0&0&0&-4&-4&8&4&-2&5&1&-5&-1&2&-2&1&0&-3&-2\\
8&-2&0&-2&-4&2&10&-2&-10&6&10&-8&0&12&-6&-8&5&-4&-8&0&0&-4&-2&9\\
4&0&0&-2&4&0&-4&2&8&-4&-8&10&2&-9&4&5&-10&1&4&-2&-3&0&1&1\\
8&0&0&-4&-4&0&4&4&-4&-4&4&4&0&6&6&-4&-2&0&-4&-4&0&-4&-4&4
\end{array}\right.$$

$$\left.\begin{array}{cccccccccccccccccccccccc}
0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0\\
0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0\\
0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0\\
0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0\\
0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0\\
0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0\\
0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0\\
0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0\\
0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0\\
0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0\\
0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0\\
0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0\\
0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0\\
0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0\\
0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0\\
0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0\\
0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0\\
0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0\\
0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0\\
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0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0\\
0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0\\
0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0\\
\hline
27&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0\\
-5&-10&8&8&-10&0&8&-8&-2&3&-1&-6&1&-3&-10&12\\
-12&-14&19&5&-19&5&9&-15&3&11&-5&-7&5&-5&-13&15\\
-15&3&11&-16&-2&20&0&-6&17&-2&-6&6&9&1&1&-14\\
-5&-3&4&0&-2&4&1&-2&1&1&0&2&2&-1&-4\\
-2&-5&-1&-7&11&8&-21&-11&15&-1&-2&18&5&-7&8&12&6&12&18&-38\\
-18&-4&15&-9&2&11&2&1&8&9&-16\\
-15&-4&5&-10&-6&-6&9&-3&-3&4&2&-5&1&6&-5\\
-3&-2&-6&4&6&-1&5&3&-7&-9&-2&-9&-5&0&-3&-10&-5&-3&22\\
-7&7&-3&-4&2&3&-3&2&-8&-4\\
-2&-6&18&-7&-13&11&-1&-13&-1&-2&-8&-6&16\\
-10&6&-4&-10&6&10&-2&10&10&0&2&12&4&0&-19\\
-10&18&-1&-14&14&5&-11&-3&0&-9&-11&13\\
-22&12&12&-2&16&20&-1&2&14&10&10&18&-48\\
-2&14&3&0&14&3&-4&5&8&3&3&-20\\
-12&-2&-12&6&4&-12&3&11&-5&6&8&1&-1&4&5&-13\\
-4&-2&-3&7&3&-1&3&3&-1&2&5&-9\\
-6&-4&-4&-2&-3&7&3&-1&3&-1&2&5&-9\\
-10&5&6&-4&6&9&1&-3&4&1&1&7&-14\\
-3&-2&-12&-2&-4&11&3&-3&2&5&-1&-3&-8\\
-16&15&0&-15&0&-1&-9&-7&15\\
-2&-12&0&5&-8&1&11&-2&1&5&3&0&5&8&-14\\
-16&-2&16&2&2&20&6&-4&6&8&2&4&-25
\end{array}\right), \tag{C.1}$$

$$\vec{\mu} = (\,0\;\;1\;\;0\;\;0\;\;0\;\;1\;\;1\;\;1\;\;0\;\;0\;\;0\;\;0\;\;0\;\;0\;\;0\;\;0\;\;0\;\;0\;\;0\;\;0\;\;0\;\;0\;\;0\;\;0\;\;0\;\;0\;\;0\;\;1\;\;1\;\;1\;\;0\;\;1\;\;1\;\;1\;\;0\;\;1\;\;1\;\;0\;\;0\;\;0\;\;0\;\;0\;\;0\;\;0\;\;0\;\;0\;\;0\,)^T. \tag{C.2}$$

For $M_{23}$ of odd Leech lattice,

$$M = \left(\begin{array}{cccccccccccccccccccccccc|cccccccccccccccccccccccc}
15&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&28&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0\\
0&2&6&4&-2&4&4&-8&-4&14&0&-24&1&11&-12&3&17&-2&-1&12&11&7&14&-23&0&5&11&4&1&9&2&-11&6&16&-15&-27&7&2&-16&11&13&-5&4&18&16&16&24&-36\\
0&5&4&0&0&7&0&-6&4&5&-14&-8&8&-3&-5&10&4&-2&8&12&10&11&13&-30&0&10&10&0&1&11&0&-11&11&10&-28&-20&14&-6&-11&19&7&-6&12&24&19&22&27&-53\\
0&1&5&1&2&4&3&-8&7&11&-13&-16&9&0&-11&10&7&-6&4&10&8&12&18&-25&0&3&9&5&6&3&2&-4&8&4&-13&-14&1&-6&-10&8&5&-2&7&11&15&17&19&-27\\
0&3&3&-1&4&10&0&-11&8&8&-15&-12&11&-3&-11&11&4&-7&8&12&10&12&17&-28&0&6&5&0&11&15&-4&-14&20&2&-28&-4&13&-19&-12&18&-7&-8&17&15&13&20&21&-33\\
0&3&2&2&5&6&-2&-4&9&0&-13&-4&4&-8&-5&9&-2&-4&8&8&8&10&12&-18&0&6&5&1&11&14&-8&-12&23&0&-34&1&15&-23&-9&22&-9&-7&21&14&13&22&22&-39\\
0&3&3&-1&4&5&-1&-5&9&4&-15&-4&9&-7&-4&9&-1&-4&9&7&5&11&13&-21&0&6&9&2&2&6&4&-7&5&8&-18&-20&7&-1&-11&11&12&-4&11&19&18&22&26&-48\\
0&9&5&1&0&2&-5&3&10&-6&-23&-2&5&-11&3&13&-2&-1&14&14&12&18&15&-38&0&19&8&-6&1&11&-16&-6&31&-6&-63&9&30&-34&8&44&-10&-4&35&27&19&35&32&-90\\
0&7&3&-3&2&11&-6&-7&13&2&-27&0&15&-13&-3&19&-5&-5&16&13&10&15&16&-38&0&14&6&-6&7&19&-18&-17&37&2&-64&8&34&-36&-2&46&-13&-8&35&24&15&32&33&-80\\
0&8&5&-1&-2&3&1&0&3&2&-17&-12&8&-1&-1&9&7&-3&8&16&13&18&16&-40&0&15&11&-1&-6&5&4&-2&2&4&-31&-22&14&0&-3&18&15&-3&17&32&26&30&32&-80\\
0&9&2&-6&5&12&-10&-5&26&-7&-43&12&22&-27&2&28&-18&-8&23&13&9&22&19&-45&0&17&2&-8&13&19&-24&-12&47&-16&-75&31&36&-55&8&51&-31&-7&46&21&14&36&28&-79\\
0&6&3&-1&3&9&-3&-6&9&1&-20&-4&10&-9&-4&12&-1&-4&11&13&11&14&15&-32&0&10&6&0&6&14&-4&-10&16&0&-36&-6&17&-16&-8&21&-3&-7&22&23&19&26&26&-56\\
0&2&4&0&-2&2&4&-4&-2&8&-4&-16&4&7&-6&2&11&-3&0&10&8&8&11&-21&0&6&6&0&-2&2&4&-2&0&4&-8&-14&3&2&-4&5&8&-3&3&13&11&11&13&-26\\
0&0&4&2&-4&4&6&-8&-6&16&2&-26&3&15&-12&3&18&-5&-5&11&10&5&12&-20&0&0&6&4&2&4&4&-6&0&10&-2&-18&0&5&-12&1&8&-5&-2&8&9&6&11&-9\\
0&0&6&2&-6&-2&12&-4&-14&18&10&-36&-1&25&-12&-6&28&-4&-10&13&11&7&13&-21&0&0&10&8&-10&-8&24&6&-32&14&30&-46&-18&39&-14&-25&36&-1&-18&14&17&4&11&-9\\
0&0&4&2&0&-2&2&-2&6&6&-8&-12&3&1&-4&8&5&-4&0&5&4&8&12&-15&0&0&6&4&4&-4&2&2&4&0&-6&-6&-3&-4&-3&4&2&-1&2&3&9&9&9&-11\\
0&0&4&2&0&2&4&-6&-6&10&4&-18&-1&10&-9&-3&14&-1&-2&8&7&3&9&-12&0&0&6&4&-2&4&10&-6&-10&14&6&-28&-2&17&-15&-6&19&-5&-4&12&11&7&14&-16\\
0&2&4&2&-4&-2&6&0&-8&8&4&-18&-2&12&-3&-3&15&0&-4&8&7&5&7&-15&0&4&10&2&-10&-2&14&-2&-16&18&4&-38&-1&25&-11&-3&30&-2&-6&19&16&11&20&-38\\
0&4&6&-2&-6&2&4&-4&-2&12&-10&-22&9&10&-6&8&15&-5&2&15&11&13&16&-37&0&8&8&0&-8&0&6&0&-6&8&-10&-22&5&10&-3&6&18&-3&4&19&15&15&19&-45\\
0&6&6&2&-6&-2&2&2&-4&4&-6&-18&-1&8&-1&4&14&0&1&14&12&11&12&-31&0&12&8&-2&-8&2&2&0&0&4&-22&-16&11&2&1&14&11&-3&9&23&17&18&20&-55\\
0&2&4&0&-2&2&4&-4&-2&8&-4&-16&4&7&-6&3&11&-3&-1&10&8&8&11&-21&0&4&6&2&0&0&4&0&0&2&-8&-12&1&1&-4&4&6&-2&3&11&12&11&12&-23\\
0&2&4&0&-2&2&4&-4&-2&8&-4&-16&4&7&-6&3&11&-3&-1&10&8&8&11&-21&0&2&6&4&-4&2&14&-2&-16&10&8&-30&-4&21&-13&-10&21&-3&-4&16&15&9&14&-22\\
0&2&4&0&-2&2&4&-4&-2&8&-4&-16&4&7&-6&2&11&-3&0&10&8&8&11&-21&0&2&6&4&-4&0&10&0&-12&8&6&-24&-4&16&-9&-7&19&-2&-2&12&12&9&13&-21\\
0&0&8&4&-8&-4&12&-4&-16&20&12&-40&-4&28&-12&-6&32&-2&-12&14&12&6&14&-23&0&0&12&8&-12&-8&24&4&-32&20&28&-52&-16&42&-16&-22&40&-2&-20&16&18&4&14&-14\\
\hline
15&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&28&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0\\
0&2&4&0&3&7&-1&-8&9&9&-13&-10&9&-3&-9&9&2&-6&3&8&6&10&15&-18&0&5&8&3&5&12&-4&-12&14&10&-21&-14&10&-9&-13&15&4&-6&11&14&12&17&22&-33\\
0&5&8&0&-4&4&3&-8&3&15&-14&-26&9&8&-10&12&16&-6&3&17&13&15&22&-41&0&7&13&4&-4&6&4&-8&1&18&-17&-33&8&8&-14&12&21&-6&5&23&20&20&29&-52\\
0&2&2&2&4&7&-1&-7&7&6&-9&-7&5&-4&-9&6&1&-4&5&6&6&7&11&-12&0&3&6&4&9&15&-4&-14&15&8&-25&-11&12&-11&-16&15&0&-7&15&15&14&19&23&-35\\
0&2&4&2&1&4&2&-7&4&9&-4&-13&3&3&-9&5&7&-3&1&6&5&6&11&-13&0&2&9&5&3&10&4&-12&3&16&-4&-27&2&5&-19&3&14&-4&1&13&12&12&20&-21\\
0&5&9&-3&-6&9&4&-14&4&21&-19&-28&15&9&-13&17&17&-8&5&20&14&14&25&-49&0&7&16&-1&-8&15&6&-25&3&42&-27&-54&22&18&-27&24&35&-13&4&34&22&23&44&-78\\
0&4&5&1&-1&4&1&-6&5&8&-11&-9&6&-1&-6&8&5&-1&4&8&6&8&12&-21&0&7&13&0&-4&12&4&-18&5&28&-23&-39&16&9&-20&19&24&-7&6&27&19&21&35&-63\\
0&5&9&1&-5&3&5&-7&-3&13&-12&-24&7&8&-9&10&18&-1&4&19&15&14&20&-45&0&10&15&0&-8&6&8&-11&-1&24&-26&-42&15&13&-15&19&28&-4&7&33&26&26&38&-80\\
0&2&5&5&-1&-1&4&2&-2&0&5&-9&-6&5&-3&-6&6&2&-3&5&5&5&4&-4&0&9&11&0&-3&13&4&-15&4&20&-23&-33&15&6&-17&16&19&-7&9&28&20&22&32&-61\\
0&4&5&3&0&1&2&0&-2&1&-6&-5&0&-2&-3&2&5&4&4&9&8&8&8&-19&0&10&12&1&-4&3&6&-4&1&10&-22&-27&9&4&-8&14&17&2&9&26&23&24&29&-65\\
0&4&2&4&5&3&-2&2&4&-8&-8&1&-1&-9&-1&1&-3&1&8&8&9&11&7&-15&0&11&4&1&8&9&-4&-1&15&-14&-29&6&9&-22&0&14&-11&-2&22&17&17&24&17&-41\\
0&6&2&0&3&5&-2&0&7&-5&-13&2&4&-9&0&6&-4&-1&9&8&7&11&8&-19&0&12&6&-1&4&11&-4&-6&15&-4&-32&-3&14&-17&-3&19&-3&-4&21&21&17&25&23&-53\\
0&8&4&-6&-2&8&-4&-8&12&8&-32&-10&19&-7&-1&23&3&-8&12&17&12&18&22&-51&0&12&6&-4&-2&6&-6&-4&14&0&-38&-2&19&-14&3&26&-2&-5&18&21&15&21&21&-59\\
0&8&2&-8&-2&10&-6&-8&18&6&-28&-2&18&-8&1&22&-5&-10&9&9&4&12&15&-32&0&12&4&-8&0&20&-12&-20&24&12&-44&-4&30&-18&-6&34&-4&-12&20&22&8&18&24&-56\\
0&12&4&-10&-4&10&-8&-8&18&8&-44&-2&25&-14&3&32&-3&-8&17&17&10&19&23&-58&0&16&10&-8&-10&8&-6&-10&14&14&-48&-16&29&-7&1&37&10&-6&18&31&18&25&32&-88\\
0&8&2&-8&0&12&-8&-10&20&6&-38&0&23&-15&0&28&-7&-10&16&13&8&16&20&-45&0&12&4&-8&2&16&-14&-16&28&6&-52&4&31&-25&-1&39&-9&-10&25&21&10&22&25&-64\\
0&8&4&-6&-4&6&-2&-6&8&8&-26&-8&16&-4&0&19&4&-5&11&14&9&14&17&-44&0&12&10&-6&-10&6&0&-10&6&18&-34&-24&22&3&-4&27&18&-7&12&28&17&22&31&-77\\
0&8&8&-8&-8&8&0&-12&8&16&-32&-16&22&0&-2&26&10&-6&10&20&12&16&24&-61&0&12&14&-8&-14&8&2&-18&6&34&-40&-40&29&11&-11&35&28&-8&8&35&20&23&40&-94\\
0&12&2&-8&-2&8&-8&-4&16&0&-38&4&20&-16&6&26&-6&-5&18&14&9&18&17&-50&0&20&6&-12&-6&12&-8&-10&20&2&-58&-2&34&-19&6&41&-2&-7&28&34&19&30&31&-95\\
0&8&2&-4&0&4&-4&2&8&-6&-22&4&8&-11&6&13&-5&-2&11&11&8&13&10&-31&0&16&4&-8&-2&8&-6&-4&16&-6&-44&4&23&-19&7&29&-6&-4&24&25&16&25&22&-70\\
0&8&2&-6&0&8&-6&-4&16&0&-32&2&17&-14&3&20&-7&-7&13&12&8&16&16&-38&0&14&4&-6&0&8&-10&-4&20&-6&-42&8&21&-23&7&29&-10&-4&22&19&12&21&18&-56\\
0&8&2&-6&0&8&-6&-4&12&0&-28&2&15&-12&3&19&-5&-5&14&12&8&14&14&-38&0&14&4&-8&0&10&-10&-8&22&-2&-46&6&25&-23&5&33&-8&-6&24&21&12&23&22&-64\\
0&12&2&-10&0&12&-10&-8&20&0&-46&6&26&-21&5&32&-9&-7&23&17&11&21&21&-60&0&16&4&-8&-2&10&-10&-6&20&-4&-50&6&27&-22&7&34&-7&-5&27&25&15&26&24&-75\\
0&16&4&-16&-4&16&-12&-12&28&8&-64&0&38&-22&6&46&-8&-14&26&24&14&28&32&-83&0&24&8&-16&-8&16&-16&-16&32&8&-80&0&48&-28&8&60&-4&-12&36&40&20&36&40&-120
\end{array}\right), \tag{C.3}$$

$$\vec{\mu} = (\,0\;\;0\;\;0\;\;0\;\;1\;\;1\;\;0\;\;0\;\;1\;\;1\;\;1\;\;1\;\;0\;\;0\;\;0\;\;0\;\;0\;\;0\;\;0\;\;0\;\;0\;\;0\;\;0\;\;0\;\;0\;\;1\;\;0\;\;1\;\;0\;\;0\;\;1\;\;0\;\;1\;\;0\;\;1\;\;0\;\;0\;\;0\;\;0\;\;0\;\;0\;\;0\;\;0\;\;0\;\;0\;\;0\;\;0\,)^T. \tag{C.4}$$

For $\mathrm{Co}_0$ of Leech lattice,

$$M = \begin{pmatrix}
364 & 84 & 37 & -495 & -288 & -227 & -27 & -367 & 155 & 245 & 186 & -31 & 119 & 45 & 103 & 87 & -12 & -24 & -51 & -9 & 16 & 38 & -73 & 100 & -180 & -155 & -105 & -31 & -53 & 31 & -50 & -10 & 31 & 62 & 8 & 2 & -18 & 3 & -55 & 16 & -20 & -6 & -9 & 0 & -17 & 10 & -52 & -35 \\
-718 & 43 & 13 & 1070 & 690 & 424 & 199 & 774 & -327 & -524 & -418 & 95 & -232 & -122 & -149 & -136 & 109 & 88 & 103 & 71 & 17 & -89 & 125 & -96 & 49 & 332 & 190 & 415 & 363 & 104 & 240 & 290 & -213 & -310 & -175 & 70 & -118 & -81 & -12 & -31 & 32 & 69 & 44 & 31 & 8 & -72 & 130 & 0 \\
-496 & 24 & 10 & 762 & 477 & 322 & 123 & 535 & -237 & -375 & -277 & 44 & -159 & -87 & -108 & -105 & 74 & 62 & 67 & 31 & 5 & -79 & 100 & -72 & 107 & 209 & 123 & 166 & 169 & 23 & 140 & 105 & -96 & -164 & -71 & 41 & -55 & -42 & -1 & -10 & 3 & 35 & 15 & 20 & -4 & -41 & 81 & 7 \\
-850 & 151 & 60 & 1337 & 879 & 519 & 295 & 942 & -409 & -654 & -509 & 108 & -272 & -164 & -154 & -154 & 171 & 126 & 119 & 93 & 38 & -130 & 152 & -68 & -24 & 386 & 208 & 550 & 479 & 158 & 330 & 385 & -278 & -412 & -238 & 113 & -191 & -120 & -68 & -25 & 15 & 98 & 50 & 51 & -13 & -103 & 159 & -28 \\
-80 & 266 & 135 & 257 & 247 & 23 & 198 & 192 & -94 & -120 & -155 & 81 & -40 & -48 & 51 & 34 & 126 & 64 & 24 & 82 & 57 & -11 & 13 & 104 & -199 & 64 & -2 & 251 & 200 & 144 & 138 & 197 & -143 & -168 & -119 & 55 & -157 & -83 & -104 & 26 & -25 & 53 & 2 & 12 & -43 & -75 & 37 & -68 \\
470 & -103 & -54 & -757 & -506 & -286 & -174 & -547 & 243 & 370 & 309 & -78 & 159 & 90 & 80 & 79 & -106 & -75 & -71 & -61 & -23 & 64 & -93 & 34 & -44 & -214 & -113 & -224 & -211 & -68 & -164 & -156 & 135 & 197 & 99 & -47 & 97 & 64 & 25 & 0 & 1 & -47 & -13 & -16 & 15 & 63 & -85 & 13 \\
824 & -230 & -104 & -1332 & -908 & -483 & -346 & -954 & 415 & 649 & 541 & -142 & 272 & 167 & 126 & 128 & -203 & -139 & -123 & -123 & -56 & 115 & -145 & 30 & 87 & -391 & -198 & -608 & -521 & -203 & -354 & -436 & 317 & 445 & 266 & -119 & 233 & 142 & 92 & 15 & -11 & -110 & -48 & -47 & 24 & 125 & -163 & 48 \\
-354 & 127 & 50 & 575 & 402 & 197 & 172 & 407 & -172 & -279 & -232 & 64 & -113 & -77 & -46 & -49 & 97 & 64 & 52 & 62 & 33 & -51 & 52 & 4 & -131 & 177 & 85 & 384 & 310 & 135 & 190 & 280 & -182 & -248 & -167 & 72 & -136 & -78 & -67 & -15 & 12 & 63 & 35 & 31 & -9 & -62 & 78 & -35 \\
364 & 84 & 37 & -495 & -288 & -227 & -27 & -367 & 155 & 245 & 186 & -31 & 119 & 45 & 103 & 87 & -12 & -24 & -51 & -9 & 16 & 38 & -73 & 100 & -180 & -155 & -105 & -31 & -53 & 31 & -50 & -10 & 31 & 62 & 8 & 2 & -18 & 3 & -55 & 16 & -20 & -6 & -9 & 0 & -17 & 10 & -52 & -35 \\
-486 & 235 & 97 & 842 & 591 & 292 & 268 & 575 & -254 & -409 & -323 & 77 & -153 & -119 & -51 & -67 & 159 & 102 & 68 & 84 & 54 & -92 & 79 & 32 & -204 & 231 & 103 & 519 & 426 & 189 & 280 & 375 & -247 & -350 & -230 & 115 & -209 & -117 & -123 & -9 & -5 & 92 & 41 & 51 & -30 & -93 & 107 & -63 \\
-1188 & 146 & 67 & 1827 & 1196 & 710 & 373 & 1321 & -570 & -894 & -727 & 173 & -391 & -212 & -229 & -215 & 215 & 163 & 174 & 132 & 40 & -153 & 218 & -130 & 93 & 546 & 303 & 639 & 574 & 172 & 404 & 446 & -348 & -507 & -274 & 117 & -215 & -145 & -37 & -31 & 31 & 116 & 57 & 47 & -7 & -135 & 215 & -13 \\
-824 & 230 & 104 & 1332 & 908 & 483 & 346 & 954 & -415 & -649 & -541 & 142 & -272 & -167 & -126 & -128 & 203 & 139 & 123 & 123 & 56 & -115 & 145 & -30 & -87 & 391 & 198 & 608 & 521 & 203 & 354 & 436 & -317 & -445 & -266 & 119 & -233 & -142 & -92 & -15 & 11 & 110 & 48 & 47 & -24 & -125 & 163 & -48 \\
-718 & 43 & 13 & 1070 & 690 & 424 & 199 & 774 & -327 & -524 & -418 & 95 & -232 & -122 & -149 & -136 & 109 & 88 & 103 & 71 & 17 & -89 & 125 & -96 & 49 & 332 & 190 & 415 & 363 & 104 & 240 & 290 & -213 & -310 & -175 & 70 & -118 & -81 & -12 & -31 & 32 & 69 & 44 & 31 & 8 & -72 & 130 & 0 \\
132 & -108 & -47 & -267 & -189 & -95 & -96 & -168 & 82 & 130 & 91 & -13 & 40 & 42 & 5 & 18 & -62 & -38 & -16 & -22 & -21 & 41 & -27 & -28 & 73 & -54 & -18 & -135 & -116 & -54 & -90 & -95 & 65 & 102 & 63 & -43 & 73 & 39 & 56 & -6 & 17 & -29 & -6 & -20 & 21 & 31 & -29 & 28 \\
-824 & 230 & 104 & 1332 & 908 & 483 & 346 & 954 & -415 & -649 & -541 & 142 & -272 & -167 & -126 & -128 & 203 & 139 & 123 & 123 & 56 & -115 & 145 & -30 & -87 & 391 & 198 & 608 & 521 & 203 & 354 & 436 & -317 & -445 & -266 & 119 & -233 & -142 & -92 & -15 & 11 & 110 & 48 & 47 & -24 & -125 & 163 & -48 \\
-718 & 43 & 13 & 1070 & 690 & 424 & 199 & 774 & -327 & -524 & -418 & 95 & -232 & -122 & -149 & -136 & 109 & 88 & 103 & 71 & 17 & -89 & 125 & -96 & 49 & 332 & 190 & 415 & 363 & 104 & 240 & 290 & -213 & -310 & -175 & 70 & -118 & -81 & -12 & -31 & 32 & 69 & 44 & 31 & 8 & -72 & 130 & 0 \\
1310 & -465 & -201 & -2174 & -1499 & -775 & -614 & -1529 & 669 & 1058 & 864 & -219 & 425 & 286 & 177 & 195 & -362 & -241 & -191 & -207 & -110 & 207 & -224 & -2 & 291 & -622 & -301 & -1127 & -947 & -392 & -634 & -811 & 564 & 795 & 496 & -234 & 442 & 259 & 215 & 24 & -6 & -202 & -89 & -98 & 54 & 218 & -270 & 111 \\
26 & 79 & 44 & -5 & 29 & -36 & 51 & 12 & -6 & 5 & -32 & 34 & 0 & -3 & 28 & 26 & 32 & 13 & 4 & 30 & 18 & 15 & -7 & 38 & -63 & 5 & -10 & 58 & 42 & 45 & 24 & 51 & -39 & -33 & -28 & 6 & 42 & -25 & -24 & 10 & -4 & 12 & -2 & -4 & -11 & -22 & 4 & -20 \\
824 & -230 & -104 & -1332 & -908 & -483 & -346 & -954 & 415 & 649 & 541 & -142 & 272 & 167 & 126 & 128 & -203 & -139 & -123 & -123 & -56 & 115 & -145 & 30 & 87 & -391 & -198 & -608 & -521 & -203 & -354 & -436 & 317 & 445 & 266 & -119 & 233 & 142 & 92 & 15 & -11 & -110 & -48 & -47 & 24 & 125 & -163 & 48 \\
-26 & -79 & -44 & 5 & -29 & 36 & -51 & -12 & 6 & -5 & 32 & -34 & 0 & 3 & -28 & -26 & -32 & -13 & -4 & -30 & -18 & -15 & 7 & -38 & 63 & -5 & 10 & -58 & -42 & -15 & -24 & -51 & 39 & 33 & 28 & -6 & -42 & 45 & 24 & -10 & 4 & -12 & 2 & 4 & 11 & 22 & -4 & 20 \\
338 & 5 & -7 & -490 & -317 & -191 & -78 & -379 & 161 & 240 & 218 & -65 & 119 & 48 & 75 & 61 & -44 & -37 & -55 & -39 & -2 & 23 & -66 & 62 & -117 & -160 & -95 & -89 & -95 & -14 & -74 & -61 & 70 & 95 & 36 & -4 & 24 & 25 & -31 & 6 & -16 & -18 & -7 & 4 & -6 & 32 & -56 & -15 \\
-364 & -84 & -37 & 495 & 288 & 227 & 27 & 367 & -155 & -245 & -186 & 31 & -119 & -45 & -103 & -87 & 12 & 24 & 51 & 9 & -16 & -38 & 73 & -100 & 180 & 155 & 105 & 31 & 53 & -31 & 50 & 10 & -31 & -62 & -8 & -2 & 18 & -3 & 55 & -16 & 20 & 6 & 9 & 0 & 17 & -10 & 52 & 35 \\
338 & 5 & -7 & -490 & -317 & -191 & -78 & -379 & 161 & 240 & 218 & -65 & 119 & 48 & 75 & 61 & -44 & -37 & -55 & -39 & -2 & 23 & -66 & 62 & -117 & -160 & -95 & -89 & -95 & -14 & -74 & -61 & 70 & 95 & 36 & -4 & 24 & 25 & -31 & 6 & -16 & -18 & -7 & 4 & -6 & 32 & -56 & -15 \\
364 & 84 & 37 & -495 & -288 & -227 & -27 & -367 & 155 & 245 & 186 & -31 & 119 & 45 & 103 & 87 & -12 & -24 & -51 & -9 & 16 & 38 & -73 & 100 & -180 & -155 & -105 & -31 & -53 & 31 & -50 & -10 & 31 & 62 & 8 & 2 & -18 & 3 & -55 & 16 & -20 & -6 & -9 & 0 & -17 & 10 & -52 & -35 \\
\hline
375 & -19 & 3 & -549 & -350 & -226 & -120 & -359 & 161 & 250 & 192 & -35 & 112 & 65 & 80 & 85 & -39 & -40 & -50 & -24 & -18 & 50 & -51 & 56 & -125 & -55 & -58 & 66 & 17 & 33 & -2 & 36 & 11 & -6 & -34 & 10 & -3 & -21 & -32 & -8 & -4 & -6 & 9 & -4 & -5 & 13 & -26 & -10 \\
-216 & -10 & -21 & 301 & 184 & 130 & 59 & 181 & -82 & -133 & -90 & 14 & -58 & -37 & -54 & -56 & 9 & 16 & 31 & 3 & 5 & -31 & 27 & -40 & 138 & -11 & 21 & -165 & -108 & -66 & -52 & -109 & 44 & 75 & 73 & -26 & 43 & 37 & 37 & 12 & -5 & -13 & -19 & -5 & 2 & 8 & -5 & 15 \\
159 & -29 & -18 & -248 & -166 & -96 & -61 & -178 & 79 & 117 & 102 & -21 & 54 & 28 & 26 & 29 & -30 & -24 & -19 & -21 & -13 & 19 & -24 & 16 & 13 & -66 & -37 & -99 & -91 & -33 & -54 & -73 & 55 & 69 & 39 & -16 & 40 & 16 & 5 & 4 & -9 & -19 & -10 & -9 & -3 & 21 & -31 & 5 \\
318 & -58 & -36 & -496 & -332 & -192 & -122 & -356 & 158 & 234 & 204 & -42 & 108 & 56 & 52 & 58 & -60 & -48 & -38 & -42 & -26 & 38 & -48 & 32 & 26 & -132 & -74 & -198 & -182 & -66 & -108 & -146 & 110 & 138 & 78 & -32 & 80 & 32 & 10 & 8 & -18 & -38 & -20 & -18 & -6 & 42 & -62 & 10 \\
318 & -58 & -36 & -496 & -332 & -192 & -122 & -356 & 158 & 234 & 204 & -42 & 108 & 56 & 52 & 58 & -60 & -48 & -38 & -42 & -26 & 38 & -48 & 32 & 26 & -132 & -74 & -198 & -182 & -66 & -108 & -146 & 110 & 138 & 78 & -32 & 80 & 32 & 10 & 8 & -18 & -38 & -20 & -18 & -6 & 42 & -62 & 10 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-204 & 136 & 114 & 390 & 296 & 124 & 126 & 350 & -152 & -202 & -228 & 56 & -100 & -38 & 4 & -4 & 102 & 64 & 14 & 78 & 42 & -14 & 42 & 16 & -328 & 286 & 106 & 726 & 580 & 264 & 320 & 510 & -308 & -426 & -302 & 116 & -246 & -138 & -94 & -40 & 46 & 102 & 78 & 46 & 8 & -100 & 134 & -50 \\
-102 & 68 & 57 & 195 & 148 & 62 & 63 & 175 & -76 & -101 & -114 & 28 & -50 & -19 & 2 & -2 & 51 & 32 & 7 & 39 & 21 & -7 & 21 & 8 & -164 & 143 & 53 & 363 & 290 & 132 & 160 & 255 & -154 & -213 & -151 & 58 & -123 & -69 & -47 & -20 & 23 & 51 & 39 & 23 & 4 & -50 & 67 & -25 \\
-159 & 29 & 18 & 248 & 166 & 96 & 61 & 178 & -79 & -117 & -102 & 21 & -54 & -28 & -26 & -29 & 30 & 24 & 19 & 21 & 13 & -19 & 24 & -16 & -13 & 66 & 37 & 99 & 91 & 33 & 54 & 73 & -55 & -69 & -39 & 16 & -40 & -16 & -5 & -4 & 9 & 19 & 10 & 9 & 3 & -21 & 31 & -5 \\
-216 & -10 & -21 & 301 & 184 & 130 & 59 & 181 & -82 & -133 & -90 & 14 & -58 & -37 & -54 & -56 & 9 & 16 & 31 & 3 & 5 & -31 & 27 & -40 & 138 & -11 & 21 & -165 & -108 & -66 & -52 & -109 & 44 & 75 & 73 & -26 & 43 & 37 & 37 & 12 & -5 & -13 & -19 & -5 & 2 & 8 & -5 & 15 \\
216 & 10 & 21 & -301 & -184 & -130 & -59 & -181 & 82 & 133 & 90 & -14 & 58 & 37 & 54 & 56 & -9 & -16 & -31 & -3 & -5 & 31 & -27 & 40 & -138 & 11 & -21 & 165 & 108 & 66 & 52 & 109 & -44 & -75 & -73 & 26 & -43 & -37 & -37 & -12 & 5 & 13 & 19 & 5 & -2 & -8 & 5 & -15 \\
273 & 49 & 60 & -354 & -202 & -164 & -57 & -184 & 85 & 149 & 78 & -7 & 62 & 46 & 82 & 83 & 12 & -8 & -43 & 15 & 3 & 43 & -30 & 64 & -289 & 88 & -5 & 429 & 307 & 165 & 158 & 291 & -143 & -219 & -185 & 68 & -126 & -90 & -79 & -28 & 19 & 45 & 48 & 19 & -1 & -37 & 41 & -35 \\
375 & -19 & 3 & -549 & -350 & -226 & -120 & -359 & 161 & 250 & 192 & -35 & 112 & 65 & 80 & 85 & -39 & -40 & -50 & -24 & -18 & 50 & -51 & 56 & -125 & -55 & -58 & 66 & 17 & 33 & -2 & 36 & 11 & -6 & -34 & 10 & -3 & -21 & -32 & -8 & -4 & -6 & 9 & -4 & -5 & 13 & -26 & -10 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-375 & 19 & -3 & 549 & 350 & 226 & 120 & 359 & -161 & -250 & -192 & 35 & -112 & -65 & -80 & -85 & 39 & 40 & 50 & 24 & 18 & -50 & 51 & -56 & 125 & 55 & 58 & -66 & -17 & -33 & 2 & -36 & -11 & 6 & 34 & -10 & 3 & 21 & 32 & 8 & 4 & 6 & -9 & 4 & 5 & -13 & 26 & 10 \\
-216 & -10 & -21 & 301 & 184 & 130 & 59 & 181 & -82 & -133 & -90 & 14 & -58 & -37 & -54 & -56 & 9 & 16 & 31 & 3 & 5 & -31 & 27 & -40 & 138 & -11 & 21 & -165 & -108 & -66 & -52 & -109 & 44 & 75 & 73 & -26 & 43 & 37 & 37 & 12 & -5 & -13 & -19 & -5 & 2 & 8 & -5 & 15 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-318 & 58 & 36 & 496 & 332 & 192 & 122 & 356 & -158 & -234 & -204 & 42 & -108 & -56 & -52 & -58 & 60 & 48 & 38 & 42 & 26 & -38 & 48 & -32 & -26 & 132 & 74 & 198 & 182 & 66 & 108 & 146 & -110 & -138 & -78 & 32 & -80 & -32 & -10 & -8 & 18 & 38 & 20 & 18 & 6 & -42 & 62 & -10 \\
-57 & -39 & -39 & 53 & 18 & 34 & -2 & 3 & -3 & -16 & 12 & -7 & -4 & -9 & -28 & -27 & -21 & -8 & 12 & -18 & -8 & -12 & 3 & -24 & 151 & -77 & -16 & -264 & -199 & -99 & -106 & -182 & 99 & 144 & 112 & -42 & 83 & 53 & 42 & 16 & -14 & -32 & -29 & -14 & -1 & 29 & -36 & 20 \\
\hline
35 & -1 & -2 & -57 & -34 & -25 & -8 & -40 & 19 & 26 & 20 & -2 & 10 & 5 & 6 & 9 & -5 & -7 & -5 & -2 & -1 & 5 & -7 & 7 & 16 & -13 & -8 & -44 & -34 & -12 & -14 & -33 & 16 & 21 & 15 & -1 & 9 & 2 & 0 & 6 & -7 & -4 & -7 & -2 & -6 & 6 & -6 & 4 \\
-2 & -1 & -1 & 8 & 1 & 6 & 0 & 1 & -2 & -1 & 3 & -6 & 2 & 0 & 1 & -5 & -1 & 2 & -1 & -3 & 1 & -4 & -1 & -2 & -2 & -2 & 0 & 1 & 4 & -4 & -2 & 6 & -1 & 3 & -3 & 1 & -1 & 4 & 3 & 1 & 3 & -2 & 4 & 2 & 3 & 2 & 0 & -2 \\
-54 & -3 & 1 & 79 & 50 & 32 & 10 & 59 & -25 & -34 & -31 & 5 & -14 & -7 & -9 & -11 & 8 & 7 & 9 & 5 & 3 & -4 & 9 & -9 & -12 & 16 & 9 & 42 & 32 & 14 & 15 & 28 & -16 & -23 & -13 & 1 & -9 & -3 & 0 & -5 & 5 & 6 & 5 & 0 & 4 & -7 & 8 & -4 \\
25 & 2 & 0 & -33 & -20 & -15 & -3 & -23 & 10 & 15 & 12 & -2 & 5 & 2 & 5 & 4 & -2 & -4 & -4 & -1 & 0 & 3 & -4 & 5 & 17 & -9 & -6 & -40 & -28 & -12 & -11 & -29 & 12 & 18 & 12 & 1 & 7 & 2 & 1 & 8 & -8 & -4 & -5 & -2 & -6 & 4 & -3 & 2 \\
10 & -4 & 2 & -16 & -10 & -8 & -7 & -5 & 2 & 7 & 0 & 3 & 1 & 4 & 2 & 3 & -1 & -2 & 0 & 0 & -1 & 5 & 1 & 0 & 27 & 6 & 4 & -25 & -16 & -10 & -9 & -14 & 7 & 10 & 6 & -5 & 7 & 2 & 10 & 3 & 2 & -4 & -2 & -2 & -1 & 2 & -1 & 4 \\
40 & -8 & -5 & -66 & -43 & -26 & -14 & -49 & 22 & 33 & 28 & -9 & 15 & 7 & 8 & 8 & -8 & -8 & -7 & -6 & 0 & 4 & -10 & 5 & -9 & -21 & -12 & -25 & -21 & -4 & -12 & -21 & 14 & 18 & 11 & -1 & 6 & 4 & -3 & 3 & -4 & -1 & -4 & -1 & -1 & 6 & -7 & 1 \\
0 & 3 & -1 & -3 & -1 & -2 & 4 & -4 & 3 & -1 & 2 & 2 & 1 & 1 & -1 & 2 & -2 & 0 & 1 & 1 & -1 & -1 & -1 & 3 & -27 & 6 & 1 & 40 & 26 & 14 & 14 & 24 & -12 & -18 & -12 & 3 & -9 & -7 & -9 & -7 & 4 & 4 & 3 & 4 & 2 & -5 & 1 & -2 \\
-15 & -3 & -2 & 19 & 11 & 9 & 4 & 9 & -5 & -5 & -3 & -2 & 1 & 2 & -2 & -5 & -3 & 3 & 4 & -2 & 3 & -4 & 0 & -3 & 1 & 1 & 1 & 0 & 1 & -4 & -3 & 3 & 1 & 4 & -1 & -1 & 0 & 3 & 4 & -1 & 6 & -3 & 3 & 3 & 2 & 1 & -1 & -1 \\
-16 & -3 & -3 & 14 & 10 & 4 & 4 & 8 & -1 & -3 & -5 & 2 & 0 & 0 & -3 & -2 & -1 & 0 & 4 & 1 & 2 & -2 & -1 & -1 & 26 & 2 & 2 & -32 & -21 & -13 & -10 & -20 & 8 & 13 & 8 & -1 & 5 & 3 & 6 & 5 & 0 & -5 & -3 & 0 & -4 & 4 & 0 & 3 \\
18 & -5 & -1 & -29 & -18 & -12 & -5 & -21 & 8 & 18 & 12 & -4 & 8 & 6 & 5 & 2 & -5 & -4 & -2 & -3 & 3 & 3 & -5 & 3 & 20 & -13 & -7 & -51 & -36 & -16 & -18 & -36 & 18 & 27 & 18 & -4 & 9 & 7 & 6 & 8 & -4 & -5 & -5 & -1 & -4 & 5 & -6 & 2 \\
-24 & -8 & 2 & 39 & 23 & 20 & 1 & 28 & -16 & -14 & -10 & -6 & -5 & 0 & -2 & -9 & 2 & 6 & 3 & -2 & 6 & -4 & 3 & -9 & -25 & -7 & -5 & 23 & 15 & 10 & 2 & 12 & -1 & -3 & -1 & 3 & 0 & -4 & 4 & 4 & 4 & -2 & 2 & 6 & -3 & 1 & -5 \\
-3 & -9 & 1 & 2 & 1 & 1 & -6 & 6 & -3 & 1 & -3 & -1 & -2 & 2 & 0 & -1 & 0 & 1 & 0 & 0 & 2 & 4 & 1 & -4 & 16 & 0 & 1 & -21 & -13 & -6 & -9 & -13 & 8 & 9 & 4 & -1 & 7 & 3 & 8 & 5 & 0 & -2 & -1 & -3 & -2 & 3 & 2 & 2 \\
-6 & -4 & 1 & 10 & 4 & 10 & -2 & 5 & -4 & -5 & 3 & -7 & -1 & 1 & -1 & -4 & 0 & 4 & -2 & -2 & 1 & -4 & 0 & -4 & -46 & -5 & -4 & 58 & 37 & 20 & 13 & 37 & -12 & -20 & -14 & 1 & -8 & -2 & -8 & -11 & 6 & 7 & 8 & 2 & 8 & -4 & 2 & -6 \\
43 & 7 & -2 & -57 & -39 & -27 & -5 & -42 & 20 & 22 & 20 & 1 & 10 & 4 & 7 & 8 & -6 & -6 & -5 & -2 & -6 & 4 & -5 & 10 & 28 & 11 & 6 & -25 & -12 & -16 & -5 & -8 & 1 & 8 & -4 & 4 & 3 & -2 & 5 & 6 & 1 & -9 & 0 & 4 & -4 & 4 & -2 & 5 \\
-20 & 3 & 4 & 33 & 22 & 10 & 6 & 29 & -13 & -17 & -21 & 9 & -10 & -2 & -4 & -3 & 3 & 4 & 5 & 4 & 0 & 2 & 7 & -4 & 21 & 24 & 15 & 8 & 13 & -1 & 5 & 13 & -10 & -9 & -10 & 0 & -3 & -2 & 11 & 1 & 7 & -2 & 3 & 1 & -3 & 7 & 2 \\
46 & 1 & -2 & -63 & -40 & -26 & -8 & -46 & 19 & 28 & 26 & -6 & 12 & 5 & 9 & 8 & -6 & -6 & -7 & -4 & -1 & 4 & -8 & 7 & -22 & -20 & -14 & -7 & -10 & 3 & -4 & -11 & 7 & 9 & 6 & 2 & 2 & -1 & -9 & 2 & -7 & 1 & -3 & -2 & -2 & 2 & -6 & -2 \\
-5 & -5 & -1 & 1 & 1 & 3 & -2 & 2 & 0 & -2 & 2 & -1 & -1 & 0 & -2 & 1 & 0 & 1 & -1 & 1 & 0 & 0 & -1 & -2 & -48 & -6 & -5 & 56 & 32 & 25 & 14 & 30 & -11 & -23 & -11 & 1 & -8 & -6 & -13 & -10 & 2 & 10 & 3 & -1 & 4 & -5 & 2 & -4 \\
-9 & 2 & 2 & 10 & 9 & 3 & 2 & 10 & -4 & -5 & -8 & 5 & -4 & -2 & -2 & 1 & 3 & 1 & 1 & 3 & 0 & 2 & 2 & -1 & -2 & 2 & 3 & 9 & 5 & 6 & 3 & 7 & -4 & -7 & -2 & -1 & -3 & -1 & 0 & -1 & 0 & 4 & -1 & -2 & 1 & -1 & 2 & 0 \\
-37 & -1 & -2 & 47 & 33 & 19 & 7 & 32 & -13 & -17 & -16 & 0 & -6 & -7 & -5 & -6 & 7 & 2 & 5 & 2 & 5 & -4 & 2 & -5 & -28 & -11 & -5 & 24 & 12 & 15 & 7 & 10 & -4 & -8 & 3 & -3 & -5 & 2 & -8 & -6 & -2 & 9 & -1 & -2 & 5 & -3 & -2 & -4 \\
16 & 3 & 1 & -19 & -12 & -4 & -1 & -18 & 6 & 8 & 13 & -8 & 3 & -1 & 3 & 1 & 0 & -2 & -6 & -3 & 1 & -3 & -4 & 2 & -21 & -20 & -12 & -3 & -6 & 4 & -2 & -9 & 6 & 6 & 9 & 0 & 1 & 2 & -9 & -1 & -7 & 3 & -3 & -1 & 0 & 0 & -3 & -2 \\
8 & -4 & 0 & -13 & -9 & 0 & -5 & -13 & 5 & 8 & 12 & -10 & 4 & 0 & 2 & 0 & 0 & 0 & -5 & -3 & 2 & -3 & -4 & -1 & -18 & -24 & -15 & -11 & -13 & 1 & -6 & -15 & 11 & 11 & 12 & -1 & 4 & 4 & -7 & 0 & -7 & 2 & -3 & -4 & 0 & 3 & -4 & -2 \\
23 & -2 & -1 & -29 & -22 & -9 & -8 & -22 & 8 & 13 & 15 & -7 & 8 & 4 & 5 & 2 & -4 & -2 & -4 & -2 & -1 & -1 & -3 & 2 & -25 & -7 & -6 & 21 & 13 & 7 & 4 & 12 & -2 & -4 & -4 & 0 & -3 & -1 & 5 & -4 & 2 & 2 & 3 & 1 & 4 & 0 & -3 & -3 \\
1 & 2 & 2 & 2 & 1 & 5 & 3 & -5 & -1 & -1 & 6 & -6 & 1 & 0 & 0 & -1 & -1 & 2 & -2 & -3 & 2 & -5 & -1 & 0 & -11 & -17 & -10 & -13 & -12 & -1 & -6 & -15 & 10 & 11 & 11 & -2 & 3 & 3 & -5 & 0 & -4 & 0 & -3 & -1 & -1 & 0 & -3 & -2 \\
45 & 2 & -2 & -65 & -40 & -31 & -11 & -44 & 18 & 29 & 18 & 2 & 11 & 5 & 8 & 11 & -5 & -10 & -4 & -4 & -4 & 8 & -4 & 8 & 49 & -6 & -1 & -77 & -53 & -26 & -21 & -50 & 23 & 32 & 22 & -4 & 15 & 5 & 8 & 11 & -8 & -9 & -10 & -2 & -7 & 7 & -9 & 8
\end{pmatrix}, \tag{C.5}$$

$$\vec{\mu} = (\,1\ \ 1\ \ 0\ \ 0\ \ 1\ \ 0\ \ 0\ \ 0\ \ 1\ \ 1\ \ 1\ \ 0\ \ 1\ \ 1\ \ 0\ \ 1\ \ 1\ \ 0\ \ 0\ \ 0\ \ 1\ \ 1\ \ 1\ \ 1$$
$$0\ \ 0\ \ 0\ \ 0\ \ 0\ \ 0\ \ 0\ \ 0\ \ 0\ \ 0\ \ 0\ \ 0\ \ 0\ \ 0\ \ 0\ \ 0\ \ 0\ \ 0\ \ 0\ \ 0\ \ 0\ \ 0$$
$$0\ \ 1\ \ 0\ \ 0\ \ 0\ \ 0\ \ 1\ \ 0\ \ 0\ \ 1\ \ 0\ \ 1\ \ 0\ \ 1\ \ 1\ \ 0\ \ 0\ \ 1\ \ 1\ \ 1\ \ 0\ \ 0\ \ 0\ \ 1\,)^T. \tag{C.6}$$

# D A Subgroup $\mathrm{U}(1)^{23}.M_{23}$ of $\mathrm{U}(1)^{24}.M_{24}$

Here, we write down the detailed proof that the group $\mathrm{Stab}_{\mathrm{Hom}(L,\mathrm{U}(1))}(\mathbb{e}_1).M_{23}$, introduced around (10.5), is a subgroup of $\mathrm{Hom}(L,\mathrm{U}(1)).M_{24} \subset O(\hat{L})$, where $L$ is the odd Leech lattice $O_{24}$ constructed as in (2.38).

Let us take a section $S : O(L) \to O(\hat{L}); g \mapsto S_g$ of (8.30). For each $g \in O(L)$, we have defined $\zeta_g : L \to \mathrm{U}(1)$ in (8.31) as

$$S_g(e^k) = \zeta_g(k) e^{g(k)}. \tag{D.1}$$

Define another section $S' : O(L) \to O(\hat{L})$ by

$$S'_g = \zeta_g(r\mathbb{e}_1)^{-1} S_g, \tag{D.2}$$

where $r\mathbb{e}_1$ is a primitive vector of $L \cap \mathbb{R}\mathbb{e}_1$, say $r\mathbb{e}_1 = \sqrt{2}(2,0,\ldots,0)$. Then $\{\tilde{\eta} \circ S'_g \mid \eta \in \mathrm{Stab}_{\mathrm{Hom}(L,\mathrm{U}(1))}(\mathbb{e}_1), g \in M_{23}\}$, where $\tilde{\ } : \mathrm{Hom}(L,\mathrm{U}(1)) \to O(\hat{L})$ is defined as in (5.53, 5.54), is a subgroup of $\mathrm{Hom}(L,\mathrm{U}(1)).M_{24}$ as follows.

The multiplication of two elements $\tilde{\eta}_1 \circ S'_{g_1}$ and $\tilde{\eta}_2 \circ S'_{g_2}$ in $\mathrm{Hom}(L,\mathrm{U}(1)).M_{24}$ is

$$(\tilde{\eta}_1 \circ S'_{g_1}) \circ (\tilde{\eta}_2 \circ S'_{g_2}) = \tilde{\eta}_1 \circ \widetilde{\eta_2 \circ g_1^{-1}} \circ \widetilde{E'(g_1,g_2)} \circ S'_{g_1 g_2}, \tag{D.3}$$

where we used

$$S'_{g_1} \circ \tilde{\eta}_2 \circ (S'_{g_1})^{-1} = \widetilde{\eta_2 \circ g_1^{-1}}, \tag{D.4}$$

and the 2-cocycle $E' : O(L) \times O(L) \to \mathrm{Hom}(L,\mathrm{U}(1))$ is specified by

$$S'_{g_1} \circ S'_{g_2} = \widetilde{E'(g_1,g_2)} \circ S'_{g_1 g_2}. \tag{D.5}$$

If $g_1 \in M_{23}$ and $\eta_2 \in \mathrm{Stab}_{\mathrm{Hom}(L,\mathrm{U}(1))}(\mathbb{e}_1)$, then $\eta_2 \circ g_1^{-1} \in \mathrm{Stab}_{\mathrm{Hom}(L,\mathrm{U}(1))}(\mathbb{e}_1)$. If $g_1, g_2 \in M_{23}$, the both-hand sides of (D.5) map $e^{r\mathbb{e}_1}$ as

$$e^{r\mathbb{e}_1} = E'(g_1,g_2)(r\mathbb{e}_1) e^{r\mathbb{e}_1}, \tag{D.6}$$

and hence $E'(g_1,g_2) \in \mathrm{Stab}_{\mathrm{Hom}(L,\mathrm{U}(1))}(\mathbb{e}_1)$. Therefore, the set $\{\tilde{\eta} \circ S'_g \mid \eta \in \mathrm{Stab}_{\mathrm{Hom}(L,\mathrm{U}(1))}(\mathbb{e}_1), g \in M_{23}\}$ is closed under the multiplication (D.3), and it turns out to form a subgroup $\mathrm{Stab}_{\mathrm{Hom}(L,\mathrm{U}(1))}(\mathbb{e}_1).M_{23}$ of $\mathrm{Hom}(L,\mathrm{U}(1)).M_{24}$.

# E Notations of Theta Functions

In this Section E, we review the notations of several theta functions and the Dedekind eta function, and their modular transformations. We mainly follow [Pol07a, Eqs. (7.2.31)–(7.2.44)], but $z_{\text{Polchinski}} = e^{2\pi\sqrt{-1}\nu_{\text{Polchinski}}}$ is $y = e^{2\pi\sqrt{-1}z}$ here, and the order of the arguments is $(\nu, \tau)$ in [Pol07a] but $(\tau, z)$ here, following [ES15, Appendix]. As always, $q = e^{2\pi\sqrt{-1}\tau}$.

**The (basic) Jacobi theta function**

We first define the *(basic) Jacobi theta function* as

$$\vartheta(\tau, z) := \sum_{n=-\infty}^{\infty} q^{n^2/2} y^n = \prod_{m=1}^{\infty} (1-q^m)(1+yq^{m-1/2})(1+y^{-1}q^{m-1/2}). \tag{E.1}$$

The second equation is called the Jacobi triple product identity.

Its modular transformations are

$$\vartheta(\tau+1, z) = \vartheta(\tau, z+\frac{1}{2}), \tag{E.2}$$

$$\vartheta(-\frac{1}{\tau}, \frac{z}{\tau}) = (-\sqrt{-1}\tau)^{1/2} e^{\pi\sqrt{-1}z^2/\tau} \vartheta(\tau, z). \tag{E.3}$$

The modular T transformation (E.2) is obvious. The modular S transformation (E.3) can be shown as follows, which is the so-called Poisson summation formula. Using the Fourier expansion of the periodic delta function

$$\sum_{n=-\infty}^{\infty} \delta(x-n) = \sum_{k=-\infty}^{\infty} e^{2\pi\sqrt{-1}kx}, \tag{E.4}$$

we can calculate that

$$\vartheta(-\frac{1}{\tau}, \frac{z}{\tau}) = \sum_{k=-\infty}^{\infty} \int_{-\infty}^{\infty} dx\, e^{-\frac{\pi\sqrt{-1}}{\tau}x^2 + 2\pi\sqrt{-1}(\frac{z}{\tau}+k)x} \tag{E.5}$$

$$= \sum_{k=-\infty}^{\infty} e^{\frac{\pi\sqrt{-1}}{\tau}(z+\tau k)^2} \int_{-\infty}^{\infty} dx\, e^{-\frac{\pi\sqrt{-1}}{\tau}(x-(z+\tau k))^2} \tag{E.6}$$

$$= \sum_{k=-\infty}^{\infty} e^{2\pi\sqrt{-1}(\tau\frac{k^2}{2} + zk + \frac{z^2}{2\tau})} \cdot (-\sqrt{-1}\tau)^{1/2} \tag{E.7}$$

$$= (-\sqrt{-1}\tau)^{1/2} e^{\pi\sqrt{-1}z^2/\tau} \vartheta(\tau, z). \tag{E.8}$$

In the third equation, we used the Gauss integral

$$\int_{-\infty}^{\infty} dx\, e^{-a(x-b)^2} = \sqrt{\frac{\pi}{a}} \quad (\mathrm{Re}(a) > 0,\ b \in \mathbb{C}). \tag{E.9}$$

**The theta function with characteristics**

The *theta function with characteristics* is defined as

$$\theta\left[\begin{array}{c} a \\ b \end{array}\right](\tau, z) := e^{2\pi\sqrt{-1}ab} q^{\frac{a^2}{2}} y^a \vartheta(\tau, z + a\tau + b) \tag{E.10}$$

$$= \sum_{n=-\infty}^{\infty} e^{2\pi\sqrt{-1}(n+a)b} q^{\frac{1}{2}(n+a)^2} y^{n+a} \tag{E.11}$$

$$= e^{2\pi\sqrt{-1}ab} q^{\frac{a^2}{2}} y^a \prod_{m=1}^{\infty} (1-q^m)(1+e^{2\pi\sqrt{-1}b} y q^{m-1/2+a})(1+e^{-2\pi\sqrt{-1}b} y^{-1} q^{m-1/2-a}). \tag{E.12}$$

Its modular transformations follow from (E.2, E.3) as

$$\theta\left[\begin{array}{c} a \\ b \end{array}\right](\tau+1, z) = e^{-\pi\sqrt{-1}(a^2+a)} \theta\left[\begin{array}{c} a \\ a+b+\frac{1}{2} \end{array}\right](\tau, z), \tag{E.13}$$

$$\theta\left[\begin{array}{c} a \\ b \end{array}\right](-\frac{1}{\tau}, \frac{z}{\tau}) = (-\sqrt{-1}\tau)^{1/2} e^{\pi\sqrt{-1}z^2/\tau} e^{2\pi\sqrt{-1}ab} \theta\left[\begin{array}{c} b \\ -a \end{array}\right](\tau, z). \tag{E.14}$$

The following equations, which follow from (E.10) and (E.11), are also useful.

$$\theta\left[\begin{array}{c} a+l \\ b+m \end{array}\right](\tau, z) = e^{2\pi\sqrt{-1}am} \theta\left[\begin{array}{c} a \\ b \end{array}\right](\tau, z) \quad (l, m \in \mathbb{Z}), \tag{E.15}$$

$$\theta\left[\begin{array}{c} a+a' \\ b+b' \end{array}\right](\tau, z) = e^{2\pi\sqrt{-1}a'(b+b')} q^{\frac{(a')^2}{2}} y^{a'} \theta\left[\begin{array}{c} a \\ b \end{array}\right](\tau, z + a'\tau + b'), \tag{E.16}$$

$$\theta\left[\begin{array}{c} -a \\ -b \end{array}\right](\tau, z) = \theta\left[\begin{array}{c} a \\ b \end{array}\right](\tau, -z). \tag{E.17}$$

**The elliptic theta functions**

The *elliptic theta functions*, also often called the *Jacobi theta functions*, are defined as

$$\theta_3(\tau, z) := \theta\left[\begin{array}{c} 0 \\ 0 \end{array}\right](\tau, z) = \vartheta(\tau, z) = \sum_{n=-\infty}^{\infty} q^{n^2/2} y^n, \tag{E.18}$$

$$\theta_4(\tau, z) := \theta\left[\begin{array}{c} 0 \\ 1/2 \end{array}\right](\tau, z) = \sum_{n=-\infty}^{\infty} (-1)^n q^{n^2/2} y^n, \tag{E.19}$$

$$\theta_2(\tau, z) := \theta\left[\begin{array}{c} 1/2 \\ 0 \end{array}\right](\tau, z) = \sum_{n=-\infty}^{\infty} q^{\frac{1}{2}(n+\frac{1}{2})^2} y^{n+\frac{1}{2}}, \tag{E.20}$$

$$\theta_1(\tau, z) := -\theta\left[\begin{array}{c} 1/2 \\ 1/2 \end{array}\right](\tau, z) = \sqrt{-1} \sum_{n=-\infty}^{\infty} (-1)^n q^{\frac{1}{2}(n-\frac{1}{2})^2} y^{n-\frac{1}{2}}. \tag{E.21}$$

We also write $\theta(\tau)$ for $\theta(\tau, 0)$. Their product representations are, from (E.12),

$$\theta_3(\tau, z) = \prod_{m=1}^{\infty}(1-q^m)(1+yq^{m-1/2})(1+y^{-1}q^{m-1/2}), \tag{E.22}$$

$$\theta_4(\tau, z) = \prod_{m=1}^{\infty}(1-q^m)(1-yq^{m-1/2})(1-y^{-1}q^{m-1/2}), \tag{E.23}$$

$$\theta_2(\tau, z) = q^{1/8}y^{1/2}\prod_{m=1}^{\infty}(1-q^m)(1+yq^m)(1+y^{-1}q^{m-1}) \tag{E.24}$$

$$= q^{1/8}(2\cos\pi z)\prod_{m=1}^{\infty}(1-q^m)(1+yq^m)(1+y^{-1}q^m), \tag{E.25}$$

$$\theta_1(\tau, z) = -\sqrt{-1}q^{1/8}y^{1/2}\prod_{m=1}^{\infty}(1-q^m)(1-yq^m)(1-y^{-1}q^{m-1}) \tag{E.26}$$

$$= q^{1/8}(2\sin\pi z)\prod_{m=1}^{\infty}(1-q^m)(1-yq^m)(1-y^{-1}q^m). \tag{E.27}$$

Their modular transformations follow from (E.13, E.14) as

$$\theta_3(\tau+1, z) = \theta_4(\tau, z), \tag{E.28}$$

$$\theta_4(\tau+1, z) = \theta_3(\tau, z), \tag{E.29}$$

$$\theta_2(\tau+1, z) = e^{\frac{1}{4}\pi\sqrt{-1}}\theta_2(\tau, z), \tag{E.30}$$

$$\theta_1(\tau+1, z) = e^{\frac{1}{4}\pi\sqrt{-1}}\theta_1(\tau, z), \tag{E.31}$$

and

$$\theta_3(-\frac{1}{\tau}, \frac{z}{\tau}) = (-\sqrt{-1}\tau)^{1/2}e^{\pi\sqrt{-1}z^2/\tau}\theta_3(\tau, z), \tag{E.32}$$

$$\theta_4(-\frac{1}{\tau}, \frac{z}{\tau}) = (-\sqrt{-1}\tau)^{1/2}e^{\pi\sqrt{-1}z^2/\tau}\theta_2(\tau, z), \tag{E.33}$$

$$\theta_2(-\frac{1}{\tau}, \frac{z}{\tau}) = (-\sqrt{-1}\tau)^{1/2}e^{\pi\sqrt{-1}z^2/\tau}\theta_4(\tau, z), \tag{E.34}$$

$$\theta_1(-\frac{1}{\tau}, \frac{z}{\tau}) = -\sqrt{-1}(-\sqrt{-1}\tau)^{1/2}e^{\pi\sqrt{-1}z^2/\tau}\theta_1(\tau, z). \tag{E.35}$$

Jacobi's identity is

$$\theta_3(\tau, z)^4 - \theta_4(\tau, z)^4 - \theta_2(\tau, z)^4 + \theta_1(\tau, z)^4 = 0. \tag{E.36}$$

Note that

$$\theta_1(\tau, 0) = 0. \tag{E.37}$$

**The Dedekind eta function**

The *Dedekind eta function* is defined as

$$\eta(\tau) := q^{1/24}\prod_{m=1}^{\infty}(1-q^m) = \sum_{n=-\infty}^{\infty}(-1)^n q^{\frac{3}{2}(n-\frac{1}{6})^2}. \tag{E.38}$$

The second equation can be obtained by replacing $q$ and $y$ in $\theta_3(\tau,z)$ with $q^3$ and $-q^{-\frac{1}{2}}$, respectively. The modular transformations are

$$\eta(\tau+1) = e^{\pi\sqrt{-1}/12}\eta(\tau), \tag{E.39}$$

$$\eta(-\frac{1}{\tau}) = (-\sqrt{-1}\tau)^{1/2}\eta(\tau). \tag{E.40}$$

The modular T transformation (E.39) is obvious. To see the modular S transformation (E.40), we first observe that, from (E.27),

$$2\pi\eta(\tau)^3 = \partial_z\theta_1(\tau,z)|_{z=0}. \tag{E.41}$$

Then we have

$$2\pi\eta(-\frac{1}{\tau})^3 = \partial_z\theta_1(-\frac{1}{\tau},z)|_{z=0} \tag{E.42}$$

$$= \tau\partial_z\theta_1(-\frac{1}{\tau},\frac{z}{\tau})|_{z=0} \tag{E.43}$$

$$= \tau(-\sqrt{-1})(-\sqrt{-1}\tau)^{1/2}\partial_z\theta_1(\tau,z)|_{z=0} \tag{E.44}$$

$$= (-\sqrt{-1}\tau)^{3/2}2\pi\eta(\tau)^3. \tag{E.45}$$

# F Lift of $\mathrm{SO}(n)$ To $\mathrm{Spin}(n)$ and Its Action on Spinors

This Section F is a review of the spin group $\mathrm{Spin}(n)$ and the lift of elements of $\mathrm{SO}(n)$ to $\mathrm{Spin}(n)$. The spin group $\mathrm{Spin}(n)$ plays an important role in understanding the structures of the theory $A(\mathfrak{a})$ of free fermions and Duncan's module $V^{f\natural}$, which are constructed as the Clifford algebra modules in Section 6. We will review its definition and fundamental property as a double cover of $\mathrm{SO}(n)$ in Section F.1. In addition, the explicit form of a lift of an element of $\mathrm{SO}(n)$ to $\mathrm{Spin}(n)$ and its action on chiral spinors are heavily used in the computation of twisted partition functions of the above theories in Sections 13 and 14. So we will review them in Sections F.2 and F.3, including their implementation on computer.

## F.1 Definitions of Pin Group and Spin Group

In this Section F.1, we review the definitions of pin group and spin group, and their fundamental properties as (double) covers of $\mathrm{O}(V)$ and $\mathrm{SO}(V)$, respectively. We will follow [LM89, Ch. I §2], so the details of the facts cited here can be found there.

To keep generality, let $V$ be a vector space over a field $\mathbb{K}$ with a symmetric bilinear form $\langle -, - \rangle$. In addition, we assume that the definition of the Clifford algebra of $V$ is

$$\mathrm{Cliff}(V) = T(V)/u \otimes u \sim \epsilon\langle u, u\rangle \quad (u \in V), \tag{F.1}$$

where $\epsilon$ is either $+1$ or $-1$ (cf. footnote 40). We will omit $\otimes$. The relation $u^2 = \epsilon\langle u, u\rangle$ is equivalent to

$$uv + vu = 2\epsilon\langle u, v\rangle \quad (u, v \in V). \tag{F.2}$$

The map $-\mathrm{id}_V : V \to V$ induces the parity involution $\theta : \mathrm{Cliff}(V) \to \mathrm{Cliff}(V)$, which defines the parity decomposition $\mathrm{Cliff}(V) = \mathrm{Cliff}(V)^0 \oplus \mathrm{Cliff}(V)^1$.

We can observe that, within the vector space $V$, the reflection with respect to the hyperplane perpendicular to a vector $v$ can be realized by the *minus* conjugate action by $v$. That is,

**Lemma F.1.** *Any $v \in V$ such that $\langle v, v\rangle \neq 0$ satisfies, for any $c \in V$,*

$$-vcv^{-1} = c - 2\frac{\langle c, v\rangle}{\langle v, v\rangle}v. \tag{F.3}$$

*Proof.* $-vcv^{-1} = -vc(\frac{\epsilon}{\langle v,v\rangle}v) = -v\frac{\epsilon}{\langle v,v\rangle}(-vc + 2\epsilon\langle c, v\rangle) = c - 2\frac{\langle c,v\rangle}{\langle v,v\rangle}v.$ □

It is easy to see that this map $V \to V; c \mapsto c - 2\frac{\langle c,v\rangle}{\langle v,v\rangle}v$ for $\langle v, v\rangle \neq 0$ is an element of

$$\mathrm{O}(V) := \{O \in \mathrm{GL}(V) \mid \langle Ou, Ov\rangle = \langle u, v\rangle \text{ for any } u, v \in V\}. \tag{F.4}$$

Based on the above observation, we define

$$P(V) := (\text{the group generated by } \{v \in V \mid \langle v, v\rangle \neq 0\}) \subset \mathrm{Cliff}(V)^\times, \tag{F.5}$$

$$\tilde{P}(V) := \{x \in \mathrm{Cliff}(V)^\times \mid \theta(x) V x^{-1} = V\}, \tag{F.6}$$

where $\mathrm{Cliff}(V)^\times$ denotes the group of all the invertible elements of $\mathrm{Cliff}(V)$. We have $P(V) \subset \tilde{P}(V)$ by Lemma F.1. We also have[84] $\mathbb{K}^\times \subset P(V)$, where $\mathbb{K}^\times = \mathbb{K} \setminus \{0\}$, because if we take $v \in V$ such that $\langle v, v\rangle \neq 0$, then any $k \in \mathbb{K}^\times$ can be represented as $k = \frac{\epsilon k}{\langle v, v\rangle} v \cdot v$, where both $\frac{\epsilon k}{\langle v, v\rangle} v$ and $v$ are in $P(V)$. If $V$ is finite-dimensional and $\langle -, -\rangle$ is non-degenerate, we can show that the map

$$\varphi : \tilde{P}(V) \to \mathrm{O}(V); x \mapsto \theta(x) \bullet x^{-1} \tag{F.7}$$

is a group homomorphism satisfying

- $\ker \varphi = \mathbb{K}^\times$,
- $\varphi|_{P(V)} : P(V) \to \mathrm{O}(V)$ is surjective.

Taking all of the above into account, we have an exact sequence

$$1 \to \mathbb{K}^\times \to P(V) = \tilde{P}(V) \xrightarrow{\varphi} O(V) \to 1. \tag{F.8}$$

**Pin group** $\mathrm{Pin}(V)$

We define the *pin group* $\mathrm{Pin}(V)$ as

$$\mathrm{Pin}(V) := \{v_1 \cdots v_m \mid v_i \in V, \langle v_i, v_i\rangle = \pm 1\} \subset P(V). \tag{F.9}$$

Let us assume that $V$ is finite-dimensional and $\langle -, -\rangle$ is non-degenerate again. If we can normalize any $v \in V$ with $\langle v, v\rangle \neq 0$ so that $\langle tv, tv\rangle = \pm 1$ ($t \in \mathbb{K}^\times$), then $\varphi|_{\mathrm{Pin}(V)} : \mathrm{Pin}(V) \to \mathrm{O}(V)$ is still surjective because $\varphi(v) = \varphi(tv)$. We say that the field $\mathbb{K}$ is *spin* if $\mathbb{K}^\times = (\mathbb{K}^\times)^2 \cup -(\mathbb{K}^\times)^2$ (hence at least one of $t^2 = \langle v, v\rangle^{-1}$ or $t^2 = -\langle v, v\rangle^{-1}$ has a solution $t \in \mathbb{K}^\times$) and the characteristic of $\mathbb{K}$ is not 2. For example, $\mathbb{R}$ and $\mathbb{C}$ are spin. If $\mathbb{K}$ is spin, then we have an exact sequence

$$1 \to F \to \mathrm{Pin}(V) \xrightarrow{\varphi} O(V) \to 1, \tag{F.10}$$

where

$$F = \begin{cases} \mathbb{Z}_2 \cong \{\pm 1\} & (\sqrt{-1} \notin \mathbb{K}), \\ \mathbb{Z}_4 \cong \{\pm 1, \pm\sqrt{-1}\} & (\sqrt{-1} \in \mathbb{K}). \end{cases} \tag{F.11}$$

[84] [LM89, p. 19] claims that, if we define $\mathbb{K}_0^\times := \mathbb{K}^\times \cap P(V)$, then $\mathbb{K}^\times = \mathbb{K}_0^\times$ or $\mathbb{K}^\times = \mathbb{K}_0^\times \cup (-\mathbb{K}_0^\times)$ holds when $\mathbb{K}$ is spin. However, it seems that $\mathbb{K}^\times = \mathbb{K}_0^\times$ always follows, regardless of whether $\mathbb{K}$ is spin or not, as in the main text.

In addition, under the same assumptions (finite-dimensional $V$, non-degenerate $\langle -,-\rangle$, spin $\mathbb{K}$), we have[85]

$$\mathrm{Pin}(V) = \{x \in \mathrm{Cliff}(V)^\times \mid \theta(x) V x^{-1} = V, \theta(x)^T x = \pm 1\} \subset \tilde{P}(V), \tag{F.12}$$

where $x^T$ is defined as $x^T = u_m \cdots u_1$ if $x = u_1 \cdots u_m$ ($u_i \in V$), and $x^T = x$ if $x \in \mathbb{R}$. We can show $\theta(x)^T x \in \mathbb{K}^\times$ for any $x \in \tilde{P}(V)$, and it is called the *spinor norm* of $x$.

Here are some important remarks on the definition of $\mathrm{Pin}(V)$.

- Suppose $\mathbb{K} = \mathbb{R}$. If $\langle -,-\rangle$ is positive- or negative-definite, then we may retain only one corresponding sign of the condition $\langle v_i, v_i\rangle = \pm 1$ of the definition (F.9).
- Suppose $\mathbb{K} = \mathbb{R}$. If $\langle -,-\rangle$ is positive-definite and $\epsilon = 1$, or $\langle -,-\rangle$ is negative-definite and $\epsilon = -1$, then by taking an orthogonal basis $\psi_1, \ldots, \psi_n$ of $V$ with $\langle \psi_i, \psi_i\rangle = \pm 1$, we have $(\psi_i)^2 = +1$ for all $i$. In this case, $\mathrm{Pin}(V)$ is also written as $\mathrm{Pin}^+(n)$. (F.12) holds even if we replace $\theta(x)^T x = \pm 1$ with $x^T x = +1$.
- Suppose $\mathbb{K} = \mathbb{R}$. If $\langle -,-\rangle$ is positive-definite and $\epsilon = -1$ (e.g. the construction of Duncan's module in this thesis), or $\langle -,-\rangle$ is negative-definite and $\epsilon = 1$, then by taking an orthogonal basis as above, we have $(\psi_i)^2 = -1$ for all $i$. In this case, $\mathrm{Pin}(V)$ is also written as $\mathrm{Pin}^-(n)$. (F.12) holds even if we replace $\theta(x)^T x = \pm 1$ with $\theta(x)^T x = +1$.
- Suppose $\mathbb{K} = \mathbb{C}$. Since we are considering the symmetric bilinear form, not a Hermitian form, there is no concept of the signature of $\langle -,-\rangle$, and we can always take an orthonormal basis $\psi_1, \ldots, \psi_n$ of $V$ with $\langle \psi_i, \psi_i\rangle = +1$. In this case, the pin group defined as in (F.9) is not a double cover of $\mathrm{O}(V)$, as in (F.10, F.11). To avoid it and retain the property that $\mathrm{Pin}(V)$ is a double cover of $\mathrm{O}(V)$, some literature defines (e.g. [Lou01, §17.3])

$$\mathrm{Pin}(V_{\mathbb{C}}) := \{v_1 \cdots v_m \mid v_i \in V_{\mathbb{C}}, \langle v_i, v_i\rangle = +1\} \tag{F.13}$$

$$= \begin{cases} \{x \in \mathrm{Cliff}(V_{\mathbb{C}})^\times \mid \theta(x) V_{\mathbb{C}} x^{-1} = V_{\mathbb{C}}, x^T x = +1\} & (\epsilon = +1) \\ \{x \in \mathrm{Cliff}(V_{\mathbb{C}})^\times \mid \theta(x) V_{\mathbb{C}} x^{-1} = V_{\mathbb{C}}, \theta(x)^T x = +1\} & (\epsilon = -1) \end{cases} \tag{F.14}$$

$$\supset \{\pm \psi_i\}_i, \tag{F.15}$$

or (e.g. [Dun07])

$$\mathrm{Pin}(V_{\mathbb{C}}) := \{v_1 \cdots v_m \mid v_i \in V_{\mathbb{C}}, \langle v_i, v_i\rangle = -1\} \tag{F.16}$$

$$= \begin{cases} \{x \in \mathrm{Cliff}(V_{\mathbb{C}})^\times \mid \theta(x) V_{\mathbb{C}} x^{-1} = V_{\mathbb{C}}, \theta(x)^T x = +1\} & (\epsilon = +1) \\ \{x \in \mathrm{Cliff}(V_{\mathbb{C}})^\times \mid \theta(x) V_{\mathbb{C}} x^{-1} = V_{\mathbb{C}}, x^T x = +1\} & (\epsilon = -1) \end{cases} \tag{F.17}$$

$$\supset \{\pm \sqrt{-1} \psi_i\}_i. \tag{F.18}$$

[85] The details are as follows. It is obvious that $\mathrm{Pin}(V)$ defined in (F.9) is in the set on the right-hand side of (F.12). Conversely, since $P(V) = \tilde{P}(V)$, any element $x$ in the set on the right-hand side of (F.12) can be written in the form of $u_1 \cdots u_l$ ($u_i \in V$) with $q(u_i) \neq 0$. Since $\theta(x)^T x = (-1)^l u_l \cdots u_1 \cdot u_1 \cdots u_l = (-\epsilon)^l \prod_i \langle u_i, u_i\rangle$, the condition $\theta(x)^T x = \pm 1$ translates to $\prod_i \langle u_i, u_i\rangle = \pm 1$. Since $\mathbb{K}$ is spin, we can take $t_i \in \mathbb{K}^\times$ such that $t_i^2 \langle u_i, u_i\rangle = \pm 1$. As a result, $\prod_i t_i^{-2} = \pm 1$, and $\prod_i t_i^{-1} \in F$, where $F \subset \mathrm{Pin}(V)$ is defined in (F.11). Therefore, $x = \prod_i t_i^{-1} \cdot t_1 u_1 \cdots t_l u_l$ is in $\mathrm{Pin}(V)$ defined in (F.9).

These $\mathrm{Pin}(V_{\mathbb{C}})$ are double covers of $\mathrm{O}(V_{\mathbb{C}})$, that is,

$$1 \to \{\pm 1\} \to \mathrm{Pin}(V_{\mathbb{C}}) \xrightarrow{\varphi} \mathrm{O}(V_{\mathbb{C}}) \to 1. \tag{F.19}$$

**Spin group** $\mathrm{Spin}(V)$

We further define the *spin group* $\mathrm{Spin}(V)$ as

$$\begin{aligned} \mathrm{Spin}(V) &:= \mathrm{Pin}(V) \cap \mathrm{Cliff}(V)^0 && \text{(F.20)} \\ &= \{v_1 \cdots v_{2l} \mid v_i \in V, \langle v_i, v_i \rangle = \pm 1\} && \text{(F.21)} \\ &= \{x \in (\mathrm{Cliff}(V)^0)^\times \mid xVx^{-1} = V, x^T x = \pm 1\}. && \text{(F.22)} \end{aligned}$$

If we define $\mathrm{SO}(V)$ as

$$\mathrm{SO}(V) := \mathrm{O}(V) \cap \mathrm{SL}(V), \tag{F.23}$$

then in parallel with (F.10), we have an exact sequence

$$1 \to F \to \mathrm{Spin}(V) \xrightarrow{\varphi} SO(V) \to 1, \tag{F.24}$$

where $F$ is defined as in (F.11).

Again, here are some important remarks on the definition of $\mathrm{Spin}(n)$.

- Suppose $\mathbb{K} = \mathbb{R}$. If $\langle -, - \rangle$ is positive- or negative-definite, then we may retain only one corresponding sign of the condition $\langle v_i, v_i \rangle = \pm 1$ of (F.21). (F.22) holds even if we replace $x^T x = \pm 1$ with $x^T x = +1$. In this case, $\mathrm{Spin}(V)$ with $\dim V = n$ is also written as $\mathrm{Spin}(n)$, which is a subgroup of $\mathrm{Pin}^{\pm}(n)$.

- Suppose $\mathbb{K} = \mathbb{C}$. In this case, the spin group defined as in (F.20) is not a double cover of $\mathrm{SO}(V)$ again, as in (F.24, F.11). To avoid it and retain the property that $\mathrm{Spin}(V)$ is a double cover of $\mathrm{SO}(V)$, some literature defines $\mathrm{Spin}(V_{\mathbb{C}})$ as the intersection of $\mathrm{Pin}(V_{\mathbb{C}})$ defined in (F.13) or (F.16) and $\mathrm{Cliff}(V)^0$. That is,

$$\begin{aligned} \mathrm{Spin}(V_{\mathbb{C}}) &:= \{v_1 \cdots v_{2l} \mid v_i \in V_{\mathbb{C}}, \langle v_i, v_i \rangle = +1\} && \text{(F.25)} \\ &= \{v_1 \cdots v_{2l} \mid v_i \in V_{\mathbb{C}}, \langle v_i, v_i \rangle = -1\} && \text{(F.26)} \\ &= \{x \in (\mathrm{Cliff}(V_{\mathbb{C}})^0)^\times \mid xV_{\mathbb{C}}x^{-1} = V_{\mathbb{C}}, x^T x = +1\}. && \text{(F.27)} \end{aligned}$$

  This $\mathrm{Spin}(V_{\mathbb{C}})$ is a double cover of $\mathrm{SO}(V_{\mathbb{C}})$, that is,

$$1 \to \{\pm 1\} \to \mathrm{Spin}(V_{\mathbb{C}}) \xrightarrow{\varphi} \mathrm{SO}(V_{\mathbb{C}}) \to 1. \tag{F.28}$$

## F.2 Explicit Form of Lift of $\mathrm{SO}(n)$ to $\mathrm{Spin}(n)$

From now on, we assume $V$ is an $n$-dimensional $\mathbb{R}$-vector space, and the symmetric bilinear form $\langle -, - \rangle$ on $V$ is positive-definite. We fix the definition of the Clifford algebra as $\mathrm{Cliff}(V) := T(V)/u \otimes u \sim -\langle u, u \rangle$. In this Section F.2, we explicitly describe the lifts of elements of $\mathrm{SO}(n)$ to $\mathrm{Spin}(n)$.

To begin with, let us see that $\varphi : \mathrm{Spin}(n) \to \mathrm{SO}(n); x \mapsto x \bullet x^{-1}$ is in fact surjective in this case. Recall that the reflection with respect to the hyperplane perpendicular to a vector $v \in V$ can be realized as in Lemma F.1.

- First, any two-dimensional rotation is a composition of two reflections, which can be shown as follows. Suppose we take a basis of $V$ so that the first two basis vectors span the two-dimensional plane to rotate. Then the matrix representation of the reflection with respect to the hyperplane perpendicular to $\vec{v} = (\cos\theta_v, \sin\theta_v, 0, \ldots, 0)$ is
$$\begin{pmatrix} c_1 \\ c_2 \end{pmatrix} \mapsto \begin{pmatrix} -2v_1^2 + 1 & -2v_1 v_2 \\ -2v_1 v_2 & -2v_2^2 + 1 \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} = \begin{pmatrix} -\cos 2\theta_v & -\sin 2\theta_v \\ -\sin 2\theta_v & \cos 2\theta_v \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix}, \tag{F.29}$$
and the other components are just mapped by the identity matrix. Therefore, the composition of two reflections is
$$\begin{pmatrix} -\cos 2\theta_{v'} & -\sin 2\theta_{v'} \\ -\sin 2\theta_{v'} & \cos 2\theta_{v'} \end{pmatrix} \begin{pmatrix} -\cos 2\theta_v & -\sin 2\theta_v \\ -\sin 2\theta_v & \cos 2\theta_v \end{pmatrix} = \begin{pmatrix} \cos 2(\theta_{v'} - \theta_v) & -\sin 2(\theta_{v'} - \theta_v) \\ \sin 2(\theta_{v'} - \theta_v) & \cos 2(\theta_{v'} - \theta_v) \end{pmatrix}. \tag{F.30}$$
- Second, as in the following Lemma F.2, any matrix in $\mathrm{SO}(n)$ can be block-diagonalized into two-dimensional rotation matrices, with a diagonalizing matrix in $\mathrm{SO}(n)$. That is, for any rotation in $\mathrm{SO}(n)$, there is an orthonormal basis such that the whole rotation is a composition of the two-dimensional rotations of the planes spanned by two axes among them.

**Lemma F.2.** *For any $M \in \mathrm{SO}(n)$, there is $O \in \mathrm{SO}(n)$ which block-diagonalizes $M$ as*
$$O^{-1}MO = \begin{pmatrix} \cos 2\pi\lambda_1 & -\sin 2\pi\lambda_1 & & & \\ \sin 2\pi\lambda_1 & \cos 2\pi\lambda_1 & & & \\ & & \ddots & & \\ & & & \cos 2\pi\lambda_{\frac{n}{2}} & -\sin 2\pi\lambda_{\frac{n}{2}} \\ & & & \sin 2\pi\lambda_{\frac{n}{2}} & \cos 2\pi\lambda_{\frac{n}{2}} \end{pmatrix}, \tag{F.31}$$
*if $n$ is even, and*
$$O^{-1}MO = \begin{pmatrix} \cos 2\pi\lambda_1 & -\sin 2\pi\lambda_1 & & & & \\ \sin 2\pi\lambda_1 & \cos 2\pi\lambda_1 & & & & \\ & & \ddots & & & \\ & & & \cos 2\pi\lambda_{\frac{n-1}{2}} & -\sin 2\pi\lambda_{\frac{n-1}{2}} & \\ & & & \sin 2\pi\lambda_{\frac{n-1}{2}} & \cos 2\pi\lambda_{\frac{n-1}{2}} & \\ & & & & & 1 \end{pmatrix}, \tag{F.32}$$
*if $n$ is odd.*

*Proof.* The real Schur decomposition states that there exists $\tilde{O} \in \mathrm{O}(n)$ such that $\tilde{O}^{-1}M\tilde{O}$ is upper quasi-triangular, which means

$$\tilde{O}^{-1}M\tilde{O} = \begin{pmatrix} B_{1,1} & B_{1,2} & \cdots & B_{1,m} \\ & B_{2,2} & \cdots & B_{2,m} \\ & & \ddots & \vdots \\ & & & B_{m,m} \end{pmatrix}, \tag{F.33}$$

where each $B_{i,j}$ is a $2 \times 2$ matrix or a number. Using $(\tilde{O}^{-1}M\tilde{O})^T(\tilde{O}^{-1}M\tilde{O}) = I$, we can further show that $\tilde{O}^{-1}M\tilde{O}$ is quasi-diagonal $\mathrm{diag}(B_{1,1}, \ldots, B_{m,m})$, and $B_{i,i} \in \mathrm{O}(2)$ or $B_{i,i} = \pm 1$. If $B_{i,i} \in \mathrm{O}(2)$ and $\det B_{i,i} = -1$, then such $B_{i,i} \in \mathrm{O}(2)$ can be written as $\begin{pmatrix} \cos\theta & \sin\theta \\ \sin\theta & -\cos\theta \end{pmatrix}$, so we diagonalize it to $\mathrm{diag}(1,-1)$ by $\begin{pmatrix} \cos\frac{\theta}{2} & -\sin\frac{\theta}{2} \\ \sin\frac{\theta}{2} & \cos\frac{\theta}{2} \end{pmatrix} \in \mathrm{SO}(2)$. Then, since $\det(\tilde{O}^{-1}M\tilde{O}) = 1$, there are even number of $B_{i,i} = -1$, and the rest of $B_{i,i}$'s are 1 or elements of $\mathrm{SO}(2)$. Now, by reordering the columns of $\tilde{O}$, we can make $\tilde{O}^{-1}M\tilde{O}$ into the form of (F.31, F.32) within $\tilde{O} \in \mathrm{O}(n)$. If $\det\tilde{O} = -1$, then we just define $O$ as $\tilde{O}$ and obtain the lemma. If $\det\tilde{O} = -1$, then we define $O$ as $\tilde{O}\mathrm{diag}(-1,1,\ldots,1)$, which only changes $B_{1,1}$ from the rotation matrix of angle $\theta$ to that of angle $-\theta$, and we obtain the lemma. □

Let $\psi_1, \ldots, \psi_n$ be an orthonormal basis of $V$. From the above discussions, it is obvious that for any $O \in \mathrm{SO}(n)$ and its lift $\pm\hat{O}$ to $\mathrm{Spin}(n)$,

$$(\pm\hat{O}) \left[ \begin{pmatrix} \psi_1 \cdots \psi_n \end{pmatrix} \begin{pmatrix} c^1 \\ \vdots \\ c^n \end{pmatrix} \right] (\pm\hat{O})^{-1} = \begin{pmatrix} \psi_1 \cdots \psi_n \end{pmatrix} O \begin{pmatrix} c^1 \\ \vdots \\ c^n \end{pmatrix}, \tag{F.34}$$

where $O$ is represented as a matrix with respect to the basis $\{\psi_i\}_i$.

Let $n$ be even below. Recall that the algebra of $\mathrm{Cliff}(V)$ is

$$\{\psi_i, \psi_j\} = -2\delta_{ij}, \quad (\psi_i)^2 = -1. \tag{F.35}$$

We introduce

$$\Psi_i := \frac{1}{\sqrt{2}}(\psi_{2i-1} + \sqrt{-1}\psi_{2i}), \quad \overline{\Psi}_i := \frac{1}{\sqrt{2}}(\psi_{2i-1} - \sqrt{-1}\psi_{2i}). \tag{F.36}$$

They satisfy

$$\{\Psi_i, \Psi_j\} = \{\overline{\Psi}_i, \overline{\Psi}_j\} = 0, \quad \{\Psi_i, \overline{\Psi}_j\} = -2\delta_{ij}. \tag{F.37}$$

Setting $\theta_{v'} = \pi\lambda_i$ and $\theta_v = 0$ in (F.30), we can see that the lifts of $R_{(0,\ldots,\lambda_i,\ldots,0)} := (\Psi_i \mapsto e^{2\pi\sqrt{-1}\lambda_i}\Psi_i, \Psi_{j\neq i} \mapsto \Psi_j) \in \mathrm{SO}(n)$ are

$$\pm((\cos\pi\lambda_i)\psi_{2i-1} + (\sin\pi\lambda_i)\psi_{2i})\psi_{2i-1} = \mp((\cos\pi\lambda_i)1 + (\sin\pi\lambda_i)\psi_{2i-1}\psi_{2i}) \tag{F.38}$$

$$=: \mp\hat{R}_{(0,\ldots,\lambda_i,\ldots,0)}. \tag{F.39}$$

In particular, if we define

$$r_{2i-1,2i} := \psi_{2i-1}\psi_{2i} \tag{F.40}$$

$$= \frac{\sqrt{-1}}{2}(\Psi_i\overline{\Psi}_i - \overline{\Psi}_i\Psi_i) = \sqrt{-1}(-1 - \overline{\Psi}_i\Psi_i) = \sqrt{-1}(\Psi_i\overline{\Psi}_i + 1), \tag{F.41}$$

then we can see that $(r_{2i-1,2i})^2 = -1$, and therefore

$$\hat{R}_{(0,\dots,\lambda_i,\dots,0)} = e^{\pi\lambda_i r_{2i-1,2i}}. \tag{F.42}$$

So, $r_{2i-1,2i}$ corresponds to $2\begin{pmatrix} O & & \\ & \begin{matrix} 0 & -1 \\ 1 & 0 \end{matrix} & \\ & & O \end{pmatrix}$ in $\mathfrak{so}(n)$.

Another way to check that $e^{\pi\lambda_i r_{2i-1,2i}}$ is a lift of $R_{(0,\dots,\lambda_i,\dots,0)}$ is showing $e^{\pi\lambda_i r_{2i-1,2i}}\Psi_j e^{-\pi\lambda_i r_{2i-1,2i}} = e^{2\pi\sqrt{-1}\lambda_i\delta_{i,j}}\Psi_j$, which follows from $\Psi_i r_{2i-1,2i} = -r_{2i-1,2i}\Psi_i = \sqrt{-1}\Psi_i$.

For more general $M \in \mathrm{SO}(n)$, using Lemma F.2, we block-diagonalize it as

$$O^{-1}MO = R_{\vec{\lambda}} := \begin{pmatrix} \cos 2\pi\lambda_1 & -\sin 2\pi\lambda_1 & & & \\ \sin 2\pi\lambda_1 & \cos 2\pi\lambda_1 & & & \\ & & \ddots & & \\ & & & \cos 2\pi\lambda_{\frac{n}{2}} & -\sin 2\pi\lambda_{\frac{n}{2}} \\ & & & \sin 2\pi\lambda_{\frac{n}{2}} & \cos 2\pi\lambda_{\frac{n}{2}} \end{pmatrix}, \tag{F.43}$$

by $O \in \mathrm{SO}(n)$. The lifts of $R_{\vec{\lambda}}$ are

$$\pm\hat{R}_{\vec{\lambda}} := \pm\prod_{i=1}^{\frac{n}{2}}(\cos\pi\lambda_i + \psi_{2i-1}\psi_{2i}\sin\pi\lambda_i) = \pm\prod_{i=1}^{\frac{n}{2}} e^{\pi\lambda_i r_{2i-1,2i}}, \tag{F.44}$$

and therefore the lifts of $M = R^O_{\vec{\lambda}} := OR_{\vec{\lambda}}O^{-1}$ are

$$\pm\hat{R}^O_{\vec{\lambda}} := \pm\hat{O}\hat{R}_{\vec{\lambda}}\hat{O}^{-1} = \pm\prod_{i=1}^{\frac{n}{2}}(\cos\pi\lambda_i + (\hat{O}\psi_{2i-1}\hat{O}^{-1})(\hat{O}\psi_{2i}\hat{O}^{-1})\sin\pi\lambda_i) \tag{F.45}$$

$$= \pm\prod_{i=1}^{\frac{n}{2}}(\cos\pi\lambda_i + (\vec{\psi}O)_{2i-1}(\vec{\psi}O)_{2i}\sin\pi\lambda_i), \tag{F.46}$$

where $\vec{\psi} := (\psi_1, \dots \psi_n)$ is a row vector, and we used (F.34) in the last equation.

## F.3 Action of Lift of $\mathrm{SO}(n)$ on Spinors

In the computation of twisted partition functions of Duncan's module $V^{f\natural}$ in Section 14.2, we need the action of $\hat{R}^O_{\vec{\lambda}} \in \mathrm{Spin}(24)$ on the chiral representation of $\mathrm{Spin}(24)$

$$\mathbf{2}^{\mathbf{11}}_{+} := \mathrm{Span}_{\mathbb{C}}\{(\text{even number of } \Psi_i\text{'s})|0\rangle_{\mathrm{R}}\}, \tag{F.47}$$

where $\overline{\Psi}_i$'s annihilate $|0\rangle_{\rm R}$. For example, the action of $\hat{R}_{\vec{\lambda}}$ on $|0\rangle_{\rm R}$ is

$$\hat{R}_{\vec{\lambda}}|0\rangle_{\rm R} = \prod_{i=1}^{\frac{n}{2}} e^{\pi\lambda_i r_{2i-1,2i}}|0\rangle_{\rm R} = e^{\pi\sqrt{-1}\sum_i \lambda_i}|0\rangle_{\rm R}. \tag{F.48}$$

Any vector in $\mathbf{2}_+^{\mathbf{11}} \oplus \mathbf{2}_-^{\mathbf{11}}$ is called a (full) spinor, and a vector in $\mathbf{2}_+^{\mathbf{11}}$ is called a chiral spinor. If we want to distinguish a vector in $\mathbf{2}_+^{\mathbf{11}}$ and in $\mathbf{2}_-^{\mathbf{11}}$, then we call it a positive-chiral spinor and a negative-chiral spinor, respectively. Note that

$$\mathbf{2}_-^{\mathbf{11}} := {\rm Span}_{\mathbb{C}}\{(\text{odd number of } \Psi_i\text{'s})|0\rangle_{\rm R}\}. \tag{F.49}$$

In this Section F.3, we explain how to implement the action of $\hat{R}^O_{\vec{\lambda}} \in {\rm Spin}(24)$ on a chiral spinor of $\mathbf{2}_+^{\mathbf{11}}$ on computer. In principle, this can be implemented by representing $\psi_i$ as a tensor product of 12 Pauli matrices. Then, any spinor is a tensor product of 12 qubits (two-dimensional vectors), say $|0\rangle_{\rm R} = |\downarrow\rangle \otimes \cdots \otimes |\downarrow\rangle$. The details are as follows.

**Action on full spinors**

First, we consider implementing the full $2^{12}$-dimensional spinors, instead of only the $2^{11}$-dimensional positive-chiral spinors. Let an array $S$ with $2^{12}$ entries $S[i] \in \mathbb{C}$ $(i = 0, \ldots, 2^{12}-1)$ represent a spinor

$$|S\rangle = \sum_{d=0}^{2^{12}-1} S[d] \cdot \Psi_{12}^{d_{11}} \cdots \Psi_1^{d_0}|0\rangle_{\rm R} \tag{F.50}$$

$$=: \sum_{d=0}^{2^{12}-1} S[d] \cdot \Psi^{d_{(2)}}|0\rangle_{\rm R}, \tag{F.51}$$

where $d_{11}\cdots d_0$ is the binary representation of $d$. For example, $\Psi^{d_{(2)}} = \Psi_{i+1}$ for $d = 2^i$.

We rename $\Psi_1, \overline{\Psi}_1, \ldots, \Psi_{12}, \overline{\Psi}_{12}$ to $\mathbf{P}_1, \mathbf{P}_2, \ldots, \mathbf{P}_{23}, \mathbf{P}_{24}$, and represent the action of $\mathbf{P}_i$ as a two-dimensional array $P_i$ such that $P_i[d] = [{\rm BitFlip}(d, \lceil \frac{i}{2} \rceil - 1),\ 0 \text{ or } \pm 1 \text{ or } \pm 2]$, where ${\rm BitFlip}(d, i)$ denotes $d_{11}\cdots(d_i + 1 \bmod 2)\cdots d_0$ as a binary, so that

$$\mathbf{P}_i \cdot \Psi^{P_i[d,1]_{(2)}}|0\rangle_{\rm R} = P_i[d,2] \cdot \Psi^{d_{(2)}}|0\rangle_{\rm R}. \tag{F.52}$$

(That is, the first entry of $P_i[d]$ denotes from which $\Psi^{\bullet_{(2)}}$ $\Psi^{d_{(2)}}$ is coming, and the second entry of $P_i[d]$ denotes the resulting factor of $\Psi^{d_{(2)}}$ in $\mathbf{P}_i \cdot \Psi^{\bullet_{(2)}}$.) Then,

$$\mathbf{P}_i|S\rangle = \sum_{d=0}^{2^{12}-1} P_i[d,2] \cdot S[P_i[d,1]] \cdot \Psi^{d_{(2)}}|0\rangle_{\rm R} \tag{F.53}$$

$$= |P_i[\bullet,2] \cdot S[P_i[\bullet,1]]\rangle. \tag{F.54}$$

In this way, the action of $\mathbf{P}_i$ on a spinor is calculated in $O(N)$ time, where $N = 2^{12}$.

We prepare the $24 \times 24$ matrix

$$C = \begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{\sqrt{-1}}{\sqrt{2}} & & & \\ \frac{1}{\sqrt{2}} & \frac{\sqrt{-1}}{\sqrt{2}} & & & \\ & & \ddots & & \\ & & & \frac{1}{\sqrt{2}} & -\frac{\sqrt{-1}}{\sqrt{2}} \\ & & & \frac{1}{\sqrt{2}} & \frac{\sqrt{-1}}{\sqrt{2}} \end{pmatrix} \tag{F.55}$$

to convert $\vec{\Psi} := (\Psi_1, \overline{\Psi}_1, \cdots, \Psi_{12}, \overline{\Psi}_{12})$ to $\vec{\psi} := (\psi_1, \cdots, \psi_{12})$ as $\vec{\Psi}C = \vec{\psi}$ in advance, and calculate $CO$ in $O(24^3)$ time. Then we can calculate the action of

$$\hat{R}^O_{\vec{\lambda}} = \prod_{i=1}^{12} (\cos \pi\lambda_i + (\vec{\Psi}CO)_{2i-1} (\vec{\Psi}CO)_{2i} \sin \pi\lambda_i) \tag{F.56}$$

on $|0\rangle_{\mathrm{R}}$ in $O(24 \times 24 \times N)$ time (where the addition of spinors and the multiplication by $\cos \pi\lambda$ and $\sin \pi\lambda$ are not taken into account, because they affect the time only by a factor of two or three).

**Retain the positive-chiral spinors only**

Next, we will see how to retain the positive-chiral spinors only. In theory, this seems to reduce the computational cost by half, but in practice, the following implementation is somehow slower than the above implementation of full spinors. Even so, it surely reduces the size of the representation matrix of $\hat{R}^O_{\vec{\lambda}}$ on $\mathbf{2}^{\mathbf{11}}_+$ to a quarter, and it enables us to perform matrix operations such as the LU decomposition in a shorter time and with less memory.

Using $P_i$'s, the action of $\mathbf{P}_i\mathbf{P}_j$ can be represented as

$$\mathbf{P}_i\mathbf{P}_j \cdot \Psi^{P_j[P_i[d,1]]_{(2)}} |0\rangle_{\mathrm{R}} = P_j[P_i[d,1],2]P_i[d,2] \cdot \Psi^{d_{(2)}} |0\rangle_{\mathrm{R}}. \tag{F.57}$$

Therefore, if we define a two-dimensional array $PP_{i,j}$ as $PP_{i,j}[d] := [P_j[P_i[d,1]], P_j[P_i[d,1],2]P_i[d,2]]$, then

$$\mathbf{P}_i\mathbf{P}_j \cdot \Psi^{PP_{i,j}[d,1]_{(2)}} |0\rangle_{\mathrm{R}} = PP_{i,j}[d,2] \cdot \Psi^{d_{(2)}} |0\rangle_{\mathrm{R}}, \tag{F.58}$$

and hence

$$\mathbf{P}_i\mathbf{P}_j|S\rangle = \sum_{d=0}^{2^{12}-1} PP_{i,j}[d,2] \cdot S[PP_{i,j}[d,1]] \cdot \Psi^{d_{(2)}} |0\rangle_{\mathrm{R}} \tag{F.59}$$

$$= |PP_{i,j}[\bullet,2] \cdot S[P_{i,j}[\bullet,1]]\rangle. \tag{F.60}$$

At this point, if $|S\rangle$ is a positive-chiral spinor, that is, a linear combination of $\Psi^{d_{(2)}}|0\rangle_{\mathrm{R}}$ with $\mathrm{PopCount}[d]$ even, then so is $\mathbf{P}_i\mathbf{P}_j|S\rangle$. Here, $\mathrm{Popcount}[d]$ is the so-called population count, which counts the number of ones in the binary representation $d_{(2)}$. Therefore, as long as we deal

with positive-chiral spinors only, we can recover $d$ from $\mathrm{BitShiftRight}[d]$, which is $d_{11}\cdots d_1$ for $d = d_{11}\cdots d_0$ as a binary. This reduces the size of the two-dimensional array $PP_{i,j}$ from $2^{12}\times 2$ to $2^{11}\times 2$.

Finally, we prepare the function to calculate the multiplication of two linear combinations of $\mathbf{P}_i$'s. That is, from two arrays $[a_1,\ldots,a_{24}]$ and $[b_1,\ldots,b_{24}]$, this function calculates the array $[c_1,\ldots,c_{\binom{24}{2}},c_{\binom{24}{2}+1}]$, such that

$$\left(\sum_{i=1}^{24} a_i\mathbf{P}_i\right)\left(\sum_{i=1}^{24} b_i\mathbf{P}_i\right) = c_1\mathbf{P}_1\mathbf{P}_2+\cdots+c_{23}\mathbf{P}_1\mathbf{P}_{24} \tag{F.61}$$

$$+ c_{23+1}\mathbf{P}_2\mathbf{P}_3+\cdots+c_{23+22}\mathbf{P}_2\mathbf{P}_{24} \tag{F.62}$$

$$+\cdots \tag{F.63}$$

$$+ c_{\binom{24}{2}}\mathbf{P}_{23}\mathbf{P}_{24} \tag{F.64}$$

$$+ c_{\binom{24}{2}+1}. \tag{F.65}$$

Now, we can calculate the action of $\hat{R}^O_{\vec{\lambda}} = \prod_{i=1}^{12}(\cos\pi\lambda_i + (\vec{\Psi}CO)_{2i-1}(\vec{\Psi}CO)_{2i}\sin\pi\lambda_i)$ on $|0\rangle_{\mathrm{R}}$, while dealing with positive-chiral spinors only.

# G Values of Twisted Partition Functions of Duncan's Module

In Table G.3, we record the absolute values $|Z^{(V^{f\natural})}_{\mathrm{R}g}{}^{\mathrm{R}h}(\tau)|$ of the $(-1)^F$-inserted Ramond twisted partition functions of Duncan's module $V^{f\natural}$ for each $\mathrm{SL}(2,\mathbb{Z})$-orbit $\mathrm{SL}(2,\mathbb{Z})\cdot[(g,h)]$ of the conjugacy classes of the commuting pairs $g,h\in \mathrm{Co}_1$, which are calculated in Section 14. The columns of Table G.3 are defined as follows.

- The column "No.":

  The index of the $\mathrm{SL}(2,\mathbb{Z})$-orbit $\mathrm{SL}(2,\mathbb{Z})\cdot[(g,h)]$ of

  $$[(g,h)]\in G^2_{\mathrm{com}}/\!\sim = \{(g,h)\in \mathrm{Co}_1\times\mathrm{Co}_1 \mid gh=hg\}/(g,h)\sim(kgk^{-1},khk^{-1}) \quad \text{(G.1)}$$

  in our code. There are 445 orbits.

- The column "P":

  This column is blank if the $\mathrm{SL}(2,\mathbb{Z})$-orbit is mapped to itself under the P transformation $(g,h)\mapsto(g,h^{-1})$, but if the P transformation interchanges it and another orbit, then the index of the other orbit is recorded.

- The column "com.":

  This column is marked "A" if the lifts $\check{g},\check{h}\in\mathrm{Spin}(24)$ of the commuting pair $g,h\in\mathrm{Co}_1\subset\mathrm{SemiSpin}(24)$ anticommute as $\check{g}\check{h}=\mathfrak{z}\check{h}\check{g}$. Otherwise, $\check{g}$ and $\check{h}$ also commute, and the column is left blank.

- The column "size":

  The size $|\mathrm{SL}(2,\mathbb{Z})\cdot[(g,h)]|$ of the $\mathrm{SL}(2,\mathbb{Z})$-orbit. That is, the number of elements of $G^2_{\mathrm{com}}/\!\sim$ in this $\mathrm{SL}(2,\mathbb{Z})$-orbit. The total number is $|G^2_{\mathrm{com}}/\!\sim| = 7578$.

- The column "rep.":

  A representative $(g,h)$ of a representative $[(g,h)]\in G^2_{\mathrm{com}}/\!\sim$ of the $\mathrm{SL}(2,\mathbb{Z})$-orbit $\mathrm{SL}(2,\mathbb{Z})\cdot[(g,h)]$. $1$ denotes the identity element of $\mathrm{Co}_1$. $\{g_i\}_{i=1\ldots,101}$ are taken as representatives of the conjugacy classes of $\mathrm{Co}_1$, and the ATLAS name [CCN+85] of each conjugacy class $[g_i]$ is given in Table G.1. $h^{(i)}_j$ denotes a representative of the $j$-th conjugacy class $[h^{(i)}_j]_{\subset C_{g_i}}$ of the centralizer $C_{g_i}$ of $g_i$ in our code.

  As a remark, $[(1,g_6)]$ and $[(1,g_7)]$ are in the same orbit, and $[(1,g_{39})]$ and $[(1,g_{40})]$ are in the same orbit. These $g_i$ are the representatives of 39A, 39B and 23A, 23B, respectively.

- The column "$[h]_{\subset C_g}/|C_g|$":

  The value of $|[h]_{\subset C_g}|/|C_g|$ for each $\mathrm{SL}(2,\mathbb{Z})$-orbit. Note that the values of $|[h]_{\subset C_g}|$ and $|C_g|$ depend on the choice of the representative $(g,h)$, but their ratio $|[h]_{\subset C_g}|/|C_g|$ is independent of such a choice, as shown in Section 12.2.4.

- The column "$|Z^{\mathrm{R}h}_{\mathrm{R}g}|$":

  The absolute value $|Z^{(V^{f\natural})\mathrm{R}h}_{\mathrm{R}g}(\tau)|$ of the twisted partition function calculated using the formula (14.15) of $Z^{(V^{f\natural})\mathrm{R}\vec{\lambda}_{\check{h},O}}_{\mathrm{R}\vec{\lambda}_{\check{g},O}}(\tau)$. Note that the twisted partition functions are only defined up to phases because the $\mathrm{Co}_1$ symmetry is anomalous.

- The columns "$\vec{\lambda}_{\check{g},O}$" and "$\vec{\lambda}_{\check{h},O}$":

  The vectors $\vec{\lambda}_{\check{g},O}$ and $\vec{\lambda}_{\check{h},O}$ of the representative $(g,h)$ written in the column "rep." They denote the angles obtained by the simultaneous block-diagonalization of the representation matrices of $g$ and $h$ within $\mathrm{SO}(24)$, and by the appropriate choice of $\vec{\lambda}$ or $\vec{\lambda}+(0,\dots,0,1)$. They depend on the choice of the representative $(g,h)$ and the way of block-diagonalizing. They are used to calculate the twisted partition function, but its absolute value does not depend on the above choices. See Section 14.2.1 for more details.

  If $\check{g}$ and $\check{h}$ anticommute, then there are no such angles, so these columns are marked "–".

Lastly, for each conjugacy class $[g_i]$ of $\mathrm{Co}_1$, the size $|C_{g_i}|$ of the centralizer and the number $|C_{g_i}/\!\sim|$ of the conjugacy classes of the centralizer are recorded in Table G.2; these values are independent of the choice of the representative $g_i$ of the conjugacy class $[g_i]$. Since the cyclic group $\langle g_i\rangle$ is a normal subgroup of the centralizer $C_{g_i}$, the size $|C_{g_i}|$ is a multiple of the order of $g_i$. The size of each conjugacy class can be calculated as $|[g_i]| = \frac{|\mathrm{Co}_1|}{|C_{g_i}|}$ where $|\mathrm{Co}_1| = 4157776806543360000$. Of course we have $\sum_{i=1}^{101}|[g_i]| = |\mathrm{Co}_1|$. In addition, since $\bigsqcup_{[g_i]\in\mathrm{Co}_1/\sim} C_{g_i}/\!\sim$ and $G^2_{\mathrm{com}}/\!\sim$ are in one-to-one correspondence (see Section 12.2.2), we have $\sum_{i=1}^{101}|C_{g_i}/\!\sim| = |G^2_{\mathrm{com}}/\!\sim| = 7578$.

Table G.1: The ATLAS names [CCN+85] of the conjugacy classes $[g_i]$ of $\mathrm{Co}_1$ in our code. They are determined as in Remark 14.1. (23A, 23B) and (39A, 39B) are assigned arbitrarily.

| $[g_1]$ | $[g_2]$ | $[g_3]$ | $[g_4]$ | $[g_5]$ | $[g_6]$ | $[g_7]$ | $[g_8]$ | $[g_9]$ | $[g_{10}]$ | $[g_{11}]$ | $[g_{12}]$ | $[g_{13}]$ | $[g_{14}]$ | $[g_{15}]$ | $[g_{16}]$ | $[g_{17}]$ | $[g_{18}]$ | $[g_{19}]$ | $[g_{20}]$ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1A | 2B | 13A | 26A | 3A | 39A | 39B | 2A | 3B | 4A | 6E | 8B | 12E | 24E | 2C | 5B | 10F | 10D | 5A | 10A |
| $[g_{21}]$ | $[g_{22}]$ | $[g_{23}]$ | $[g_{24}]$ | $[g_{25}]$ | $[g_{26}]$ | $[g_{27}]$ | $[g_{28}]$ | $[g_{29}]$ | $[g_{30}]$ | $[g_{31}]$ | $[g_{32}]$ | $[g_{33}]$ | $[g_{34}]$ | $[g_{35}]$ | $[g_{36}]$ | $[g_{37}]$ | $[g_{38}]$ | $[g_{39}]$ | $[g_{40}]$ |
| 5C | 10E | 3D | 6H | 10B | 15C | 30C | 3C | 6D | 6F | 6C | 15D | 4C | 12H | 8E | 9C | 7B | 21C | 23A | 23B |
| $[g_{41}]$ | $[g_{42}]$ | $[g_{43}]$ | $[g_{44}]$ | $[g_{45}]$ | $[g_{46}]$ | $[g_{47}]$ | $[g_{48}]$ | $[g_{49}]$ | $[g_{50}]$ | $[g_{51}]$ | $[g_{52}]$ | $[g_{53}]$ | $[g_{54}]$ | $[g_{55}]$ | $[g_{56}]$ | $[g_{57}]$ | $[g_{58}]$ | $[g_{59}]$ | $[g_{60}]$ |
| 20C | 7A | 35A | 18C | 21A | 14A | 6B | 4D | 8F | 6A | 12C | 24D | 4E | 12L | 14B | 28A | 8A | 12A | 24A | 6G |
| $[g_{61}]$ | $[g_{62}]$ | $[g_{63}]$ | $[g_{64}]$ | $[g_{65}]$ | $[g_{66}]$ | $[g_{67}]$ | $[g_{68}]$ | $[g_{69}]$ | $[g_{70}]$ | $[g_{71}]$ | $[g_{72}]$ | $[g_{73}]$ | $[g_{74}]$ | $[g_{75}]$ | $[g_{76}]$ | $[g_{77}]$ | $[g_{78}]$ | $[g_{79}]$ | $[g_{80}]$ |
| 24C | 4B | 12G | 12K | 9B | 9A | 4F | 6I | 12M | 11A | 33A | 8C | 12B | 24B | 8D | 16A | 20A | 15A | 30A | 60A |
| $[g_{81}]$ | $[g_{82}]$ | $[g_{83}]$ | $[g_{84}]$ | $[g_{85}]$ | $[g_{86}]$ | $[g_{87}]$ | $[g_{88}]$ | $[g_{89}]$ | $[g_{90}]$ | $[g_{91}]$ | $[g_{92}]$ | $[g_{93}]$ | $[g_{94}]$ | $[g_{95}]$ | $[g_{96}]$ | $[g_{97}]$ | $[g_{98}]$ | $[g_{99}]$ | $[g_{100}]$ |
| 15E | 30E | 21B | 10C | 15B | 30B | 20B | 16B | 22A | 12I | 12D | 40A | 24F | 30D | 12J | 18B | 42A | 28B | 18A | 36A |
| $[g_{101}]$ | | | | | | | | | | | | | | | | | | | |
| 12F | | | | | | | | | | | | | | | | | | | |

Table G.2: For each conjugacy class (cc) of $\mathrm{Co}_1$ specified by the ATLAS name [CCN+85], the corresponding conjugacy class $[g_i]$ in our code (the same as Table G.1), the size $|C_{g_i}|$ of the centralizer, and the number $|C_{g_i}/\sim|$ of the conjugacy classes of the centralizer are given.

| cc | $[g_i]$ | $\lvert C_{g_i}\rvert$ | $\lvert C_{g_i}/\sim\rvert$ |
|---|---|---|---|
| 1A | $[g_1]$ | 4157776806543360000 | 101 |
| 2A | $[g_8]$ | 89181388800 | 154 |
| 2B | $[g_2]$ | 2012774400 | 112 |
| 2C | $[g_{15}]$ | 389283840 | 96 |
| 3A | $[g_5]$ | 1345036492800 | 119 |
| 3B | $[g_9]$ | 58786560 | 134 |
| 3C | $[g_{28}]$ | 12597120 | 126 |
| 3D | $[g_{23}]$ | 544320 | 54 |
| 4A | $[g_{10}]$ | 371589120 | 140 |
| 4B | $[g_{62}]$ | 1179648 | 187 |
| 4C | $[g_{33}]$ | 737280 | 109 |
| 4D | $[g_{48}]$ | 294912 | 167 |
| 4E | $[g_{53}]$ | 48384 | 64 |
| 4F | $[g_{67}]$ | 15360 | 66 |
| 5A | $[g_{19}]$ | 3024000 | 105 |
| 5B | $[g_{16}]$ | 36000 | 100 |
| 5C | $[g_{21}]$ | 15000 | 50 |
| 6A | $[g_{50}]$ | 9953280 | 171 |
| 6B | $[g_{47}]$ | 483840 | 104 |
| 6C | $[g_{31}]$ | 155520 | 120 |
| 6D | $[g_{29}]$ | 155520 | 102 |
| 6E | $[g_{11}]$ | 20736 | 189 |
| 6F | $[g_{30}]$ | 15552 | 168 |
| 6G | $[g_{60}]$ | 3456 | 70 |
| 6H | $[g_{24}]$ | 1440 | 57 |
| 6I | $[g_{68}]$ | 576 | 39 |
| 7A | $[g_{42}]$ | 17640 | 63 |
| 7B | $[g_{37}]$ | 1176 | 42 |
| 8A | $[g_{57}]$ | 92160 | 98 |
| 8B | $[g_{12}]$ | 6144 | 92 |
| 8C | $[g_{72}]$ | 6144 | 112 |
| 8D | $[g_{75}]$ | 2048 | 110 |
| 8E | $[g_{35}]$ | 768 | 60 |
| 8F | $[g_{49}]$ | 768 | 68 |
| 9A | $[g_{66}]$ | 1944 | 72 |
| 9B | $[g_{65}]$ | 486 | 66 |
| 9C | $[g_{36}]$ | 486 | 66 |
| 10A | $[g_{20}]$ | 9600 | 80 |
| 10B | $[g_{25}]$ | 1200 | 100 |
| 10C | $[g_{84}]$ | 1200 | 100 |
| 10D | $[g_{18}]$ | 600 | 50 |
| 10E | $[g_{22}]$ | 600 | 45 |
| 10F | $[g_{17}]$ | 160 | 70 |
| 11A | $[g_{70}]$ | 66 | 33 |
| 12A | $[g_{58}]$ | 138240 | 147 |
| 12B | $[g_{73}]$ | 9216 | 138 |
| 12C | $[g_{51}]$ | 4608 | 147 |
| 12D | $[g_{91}]$ | 2592 | 96 |
| 12E | $[g_{13}]$ | 1728 | 126 |
| 12F | $[g_{101}]$ | 864 | 64 |
| 12G | $[g_{63}]$ | 576 | 78 |
| 12H | $[g_{34}]$ | 288 | 48 |
| 12I | $[g_{90}]$ | 288 | 60 |
| 12J | $[g_{95}]$ | 288 | 108 |
| 12K | $[g_{64}]$ | 288 | 84 |
| 12L | $[g_{54}]$ | 72 | 36 |
| 12M | $[g_{69}]$ | 48 | 30 |
| 13A | $[g_3]$ | 156 | 52 |
| 14A | $[g_{46}]$ | 168 | 63 |
| 14B | $[g_{55}]$ | 56 | 35 |
| 15A | $[g_{78}]$ | 5400 | 85 |
| 15B | $[g_{85}]$ | 900 | 75 |
| 15C | $[g_{26}]$ | 180 | 60 |
| 15D | $[g_{32}]$ | 90 | 45 |
| 15E | $[g_{81}]$ | 30 | 30 |
| 16A | $[g_{76}]$ | 64 | 40 |
| 16B | $[g_{88}]$ | 64 | 40 |
| 18A | $[g_{99}]$ | 216 | 63 |
| 18B | $[g_{96}]$ | 54 | 54 |
| 18C | $[g_{44}]$ | 54 | 54 |
| 20A | $[g_{77}]$ | 480 | 80 |
| 20B | $[g_{87}]$ | 40 | 40 |
| 20C | $[g_{41}]$ | 20 | 20 |
| 21A | $[g_{45}]$ | 252 | 84 |
| 21B | $[g_{83}]$ | 63 | 63 |
| 21C | $[g_{38}]$ | 21 | 21 |
| 22A | $[g_{89}]$ | 22 | 22 |
| 23A | $[g_{39}]$ | 23 | 23 |
| 23B | $[g_{40}]$ | 23 | 23 |
| 24A | $[g_{59}]$ | 576 | 96 |
| 24B | $[g_{74}]$ | 192 | 84 |
| 24C | $[g_{61}]$ | 144 | 72 |
| 24D | $[g_{52}]$ | 96 | 60 |
| 24E | $[g_{14}]$ | 48 | 48 |
| 24F | $[g_{93}]$ | 24 | 24 |
| 26A | $[g_4]$ | 52 | 52 |
| 28A | $[g_{56}]$ | 28 | 28 |
| 28B | $[g_{98}]$ | 28 | 28 |
| 30A | $[g_{79}]$ | 120 | 75 |
| 30B | $[g_{86}]$ | 60 | 60 |
| 30C | $[g_{27}]$ | 60 | 60 |
| 30D | $[g_{94}]$ | 30 | 30 |
| 30E | $[g_{82}]$ | 30 | 30 |
| 33A | $[g_{71}]$ | 33 | 33 |
| 35A | $[g_{43}]$ | 35 | 35 |
| 36A | $[g_{100}]$ | 36 | 36 |
| 39A | $[g_6]$ | 39 | 39 |
| 39B | $[g_7]$ | 39 | 39 |
| 40A | $[g_{92}]$ | 40 | 40 |
| 42A | $[g_{97}]$ | 84 | 84 |
| 60A | $[g_{80}]$ | 60 | 60 |

Table G.3: The absolute values of the twisted partition functions of Duncan's module $V^{f\natural}$ for each $\mathrm{SL}(2,\mathbb{Z})$-orbit of conjugacy classes of commuting pairs of $\mathrm{Co}_1$.

| No. | P | com. | size | rep. | $[h]_{\subset C_g}/\lvert C_g\rvert$ | $\lvert Z^{\mathrm{R}h}_{\mathrm{R}g}\rvert$ | $\vec{\lambda}_{\tilde{g},O}$ | $\vec{\lambda}_{\tilde{h},O}$ |
|---|---|---|---|---|---|---|---|---|
| 1 | | | 1 | $(1,g_1)$ | 1/4157776806543360000 | 24 | $(0,0,0,0,0,0,0,0,0,0,0,0)$ | $(0,0,0,0,0,0,0,0,0,0,0,0)$ |
| 2 | | | 3 | $(1,g_2)$ | 1/2012774400 | 0 | $(0,0,0,0,0,0,0,0,0,0,0,0)$ | $(-\frac{1}{4},\frac{1}{4},\frac{1}{4},-\frac{1}{4},\frac{1}{4},\frac{1}{4},\frac{1}{4},\frac{1}{4},-\frac{1}{4},\frac{1}{4},-\frac{1}{4},\frac{1}{4})$ |
| 3 | | | 14 | $(1,g_3)$ | 1/156 | 2 | $(0,0,0,0,0,0,0,0,0,0,0,0)$ | $(-\frac{11}{26},\frac{11}{26},\frac{9}{26},-\frac{9}{26},\frac{7}{26},-\frac{7}{26},\frac{5}{26},\frac{5}{26},-\frac{1}{26},\frac{3}{26},\frac{3}{26})$ |
| 4 | | | 42 | $(1,g_4)$ | 1/52 | 0 | $(0,0,0,0,0,0,0,0,0,0,0,0)$ | $(\frac{25}{52},\frac{23}{52},-\frac{21}{52},\frac{19}{52},\frac{17}{52},-\frac{15}{52},\frac{11}{52},\frac{9}{52},\frac{7}{52},-\frac{5}{52},\frac{3}{52},\frac{51}{52})$ |
| 5 | | | 4 | $(1,g_5)$ | 1/1345036492800 | 12 | $(0,0,0,0,0,0,0,0,0,0,0,0)$ | $(\frac{1}{6},-\frac{1}{6},-\frac{1}{6},\frac{1}{6},\frac{1}{6},-\frac{1}{6},\frac{1}{6},-\frac{1}{6},\frac{1}{6},\frac{1}{6},\frac{1}{6},\frac{1}{6})$ |
| 6 | | | 112 | $(1,g_6)$ | 1/39 | 1 | $(0,0,0,0,0,0,0,0,0,0,0,0)$ | $(-\frac{19}{39},\frac{17}{39},\frac{16}{39},\frac{14}{39},\frac{11}{39},\frac{10}{39},\frac{8}{39},-\frac{7}{39},-\frac{1}{39},\frac{2}{39},-\frac{4}{39},\frac{5}{39})$ |
| 7 | | | 3 | $(1,g_8)$ | 1/89181388800 | 8 | $(0,0,0,0,0,0,0,0,0,0,0,0)$ | $(\frac{1}{2},\frac{1}{2},\frac{1}{2},\frac{1}{2},0,0,0,0,0,0,0,0)$ |
| 8 | | | 4 | $(1,g_9)$ | 1/58786560 | 6 | $(0,0,0,0,0,0,0,0,0,0,0,0)$ | $(\frac{1}{3},\frac{1}{3},\frac{1}{3},-\frac{1}{3},-\frac{1}{3},-\frac{1}{3},0,0,0,0,0,0)$ |
| 9 | | | 6 | $(1,g_{10})$ | 1/371589120 | 8 | $(0,0,0,0,0,0,0,0,0,0,0,0)$ | $(-\frac{1}{4},\frac{1}{4},-\frac{1}{4},\frac{1}{4},-\frac{1}{4},\frac{1}{4},-\frac{1}{4},\frac{1}{2},\frac{1}{2},\frac{1}{2},\frac{3}{2})$ |
| 10 | | | 12 | $(1,g_{11})$ | 1/20736 | 2 | $(0,0,0,0,0,0,0,0,0,0,0,0)$ | $(-\frac{1}{3},-\frac{1}{3},\frac{1}{6},-\frac{1}{6},-\frac{1}{6},-\frac{1}{6},0,0,\frac{1}{2},\frac{1}{2},\frac{1}{2},\frac{3}{2})$ |
| 11 | | | 12 | $(1,g_{12})$ | 1/6144 | 0 | $(0,0,0,0,0,0,0,0,0,0,0,0)$ | $(\frac{3}{8},-\frac{3}{8},-\frac{3}{8},-\frac{3}{8},\frac{1}{2},\frac{1}{2},-\frac{1}{8},-\frac{1}{8},-\frac{1}{8},\frac{1}{8},\frac{1}{8},0,0)$ |
| 12 | | | 24 | $(1,g_{13})$ | 1/1728 | 2 | $(0,0,0,0,0,0,0,0,0,0,0,0)$ | $(\frac{5}{12},-\frac{5}{12},-\frac{1}{3},-\frac{1}{3},-\frac{1}{12},-\frac{1}{12},-\frac{1}{12},-\frac{1}{4},\frac{1}{4},\frac{1}{4},-\frac{1}{4},0,0)$ |
| 13 | | | 48 | $(1,g_{14})$ | 1/48 | 0 | $(0,0,0,0,0,0,0,0,0,0,0,0)$ | $(\frac{5}{24},-\frac{1}{6},-\frac{7}{24},\frac{1}{3},\frac{11}{24},-\frac{1}{24},\frac{1}{8},-\frac{1}{8},0,-\frac{3}{8},\frac{3}{8},\frac{1}{2})$ |
| 14 | | | 3 | $(1,g_{15})$ | 1/389283840 | 0 | $(0,0,0,0,0,0,0,0,0,0,0,0)$ | $(\frac{1}{2},\frac{1}{2},\frac{1}{2},\frac{1}{2},\frac{1}{2},0,0,0,0,0,0,-\frac{1}{2})$ |
| 15 | | | 6 | $(1,g_{16})$ | 1/36000 | 4 | $(0,0,0,0,0,0,0,0,0,0,0,0)$ | $(\frac{2}{5},-\frac{2}{5},-\frac{2}{5},-\frac{2}{5},-\frac{1}{5},\frac{1}{5},-\frac{1}{5},\frac{1}{5},0,0,0,0)$ |
| 16 | | | 18 | $(1,g_{17})$ | 1/160 | 0 | $(0,0,0,0,0,0,0,0,0,0,0,0)$ | $(\frac{1}{10},-\frac{1}{10},\frac{1}{5},\frac{1}{5},\frac{3}{10},\frac{3}{10},-\frac{2}{5},\frac{2}{5},\frac{1}{2},\frac{1}{2},0,0)$ |
| 17 | | | 18 | $(1,g_{18})$ | 1/600 | 2 | $(0,0,0,0,0,0,0,0,0,0,0,0)$ | $(-\frac{3}{10},\frac{1}{10},-\frac{1}{5},\frac{1}{5},-\frac{1}{5},-\frac{2}{5},-\frac{2}{5},-\frac{2}{5},0,0,\frac{1}{2},\frac{1}{2})$ |
| 18 | | | 6 | $(1,g_{19})$ | 1/3024000 | 6 | $(0,0,0,0,0,0,0,0,0,0,0,0)$ | $(\frac{1}{5},-\frac{1}{5},\frac{1}{5},\frac{1}{5},\frac{1}{5},\frac{1}{5},-\frac{2}{5},\frac{2}{5},\frac{2}{5},-\frac{2}{5},-\frac{2}{5},\frac{2}{5})$ |
| 19 | | | 18 | $(1,g_{20})$ | 1/9600 | 2 | $(0,0,0,0,0,0,0,0,0,0,0,0)$ | $(\frac{2}{5},-\frac{2}{5},-\frac{1}{5},-\frac{1}{5},\frac{1}{10},-\frac{1}{10},-\frac{1}{10},\frac{1}{10},\frac{3}{10},\frac{3}{10},\frac{3}{10},-\frac{3}{10})$ |
| 20 | | | 6 | $(1,g_{21})$ | 1/15000 | 1 | $(0,0,0,0,0,0,0,0,0,0,0,0)$ | $(\frac{1}{2},\frac{1}{2},\frac{3}{10},\frac{3}{10},-\frac{3}{10},\frac{3}{10},\frac{3}{10},\frac{1}{10},-\frac{1}{10},\frac{1}{10},-\frac{1}{10},-\frac{1}{10},-\frac{1}{10},-\frac{1}{10})$ |
| 21 | | | 18 | $(1,g_{22})$ | 1/600 | 3 | $(0,0,0,0,0,0,0,0,0,0,0,0)$ | $(-\frac{1}{5},\frac{1}{5},-\frac{2}{5},\frac{2}{5},\frac{1}{10},-\frac{1}{10},\frac{1}{10},-\frac{3}{10},\frac{3}{10},\frac{1}{2},\frac{1}{2})$ |
| 22 | | | 4 | $(1,g_{23})$ | 1/544320 | 0 | $(0,0,0,0,0,0,0,0,0,0,0,0)$ | $(-\frac{1}{3},-\frac{1}{3},\frac{1}{3},-\frac{1}{3},\frac{1}{3},\frac{1}{3},\frac{1}{3},\frac{1}{3},0,0,0,0)$ |
| 23 | | | 12 | $(1,g_{24})$ | 1/1440 | 0 | $(0,0,0,0,0,0,0,0,0,0,0,0)$ | $(-\frac{5}{12},\frac{5}{12},-\frac{5}{12},\frac{5}{12},-\frac{1}{4},-\frac{1}{4},\frac{1}{4},-\frac{1}{4},\frac{1}{12},-\frac{1}{12},\frac{1}{12},-\frac{1}{12})$ |
| 24 | | | 18 | $(1,g_{25})$ | 1/1200 | 0 | $(0,0,0,0,0,0,0,0,0,0,0,0)$ | $(\frac{9}{20},\frac{9}{20},-\frac{9}{20},\frac{7}{20},-\frac{7}{20},\frac{7}{20},\frac{3}{20},-\frac{3}{20},\frac{3}{20},\frac{1}{20},-\frac{1}{20},\frac{19}{20})$ |
| 25 | | | 24 | $(1,g_{26})$ | 1/180 | 0 | $(0,0,0,0,0,0,0,0,0,0,0,0)$ | $(-\frac{1}{15},\frac{1}{15},-\frac{2}{15},\frac{2}{15},-\frac{2}{15},-\frac{4}{15},-\frac{4}{15},\frac{1}{5},\frac{1}{5},\frac{2}{5},\frac{2}{5},-\frac{1}{15})$ |
| 26 | | | 72 | $(1,g_{27})$ | 1/60 | 0 | $(0,0,0,0,0,0,0,0,0,0,0,0)$ | $(\frac{1}{60},\frac{1}{20},\frac{7}{60},\frac{3}{20},-\frac{11}{60},-\frac{13}{60},-\frac{17}{60},\frac{19}{60},-\frac{7}{20},-\frac{23}{60},-\frac{9}{20},\frac{89}{60})$ |
| 27 | | | 4 | $(1,g_{28})$ | 1/12597120 | 3 | $(0,0,0,0,0,0,0,0,0,0,0,0)$ | $(0,0,0,-\frac{1}{3},-\frac{1}{3},-\frac{1}{3},-\frac{1}{3},-\frac{1}{3},\frac{1}{3},\frac{1}{3},\frac{4}{3},\frac{4}{3})$ |
| 28 | | | 12 | $(1,g_{29})$ | 1/155520 | 5 | $(0,0,0,0,0,0,0,0,0,0,0,0)$ | $(-\frac{1}{3},\frac{1}{3},\frac{1}{3},\frac{1}{3},-\frac{1}{6},\frac{1}{6},\frac{1}{6},-\frac{1}{6},-\frac{1}{6},\frac{1}{2},\frac{1}{2},\frac{1}{2})$ |
| 29 | | | 12 | $(1,g_{30})$ | 1/15552 | 1 | $(0,0,0,0,0,0,0,0,0,0,0,0)$ | $(0,\frac{1}{3},\frac{1}{3},\frac{1}{3},\frac{1}{2},\frac{1}{2},\frac{1}{6},\frac{1}{6},\frac{1}{6},-\frac{1}{6},-\frac{1}{6},\frac{5}{6})$ |
| 30 | | | 12 | $(1,g_{31})$ | 1/155520 | 4 | $(0,0,0,0,0,0,0,0,0,0,0,0)$ | $(\frac{1}{3},-\frac{1}{6},\frac{1}{6},-\frac{1}{6},-\frac{1}{6},-\frac{1}{6},\frac{1}{2},\frac{1}{2},\frac{1}{2},\frac{1}{2},0,0,1)$ |
| 31 | | | 24 | $(1,g_{32})$ | 1/90 | 1 | $(0,0,0,0,0,0,0,0,0,0,0,0)$ | $(-\frac{7}{15},\frac{1}{15},-\frac{2}{15},\frac{4}{15},\frac{2}{5},-\frac{2}{5},\frac{1}{3},\frac{1}{3},-\frac{1}{5},\frac{1}{5},0,0)$ |
| 32 | | | 6 | $(1,g_{33})$ | 1/737280 | 4 | $(0,0,0,0,0,0,0,0,0,0,0,0)$ | $(\frac{1}{4},\frac{1}{4},-\frac{1}{4},\frac{1}{4},0,0,0,\frac{1}{2},\frac{1}{2},\frac{1}{2},\frac{1}{2},\frac{3}{2})$ |
| 33 | | | 24 | $(1,g_{34})$ | 1/288 | 1 | $(0,0,0,0,0,0,0,0,0,0,0,0)$ | $(-\frac{5}{12},-\frac{5}{12},\frac{1}{2},\frac{1}{6},-\frac{1}{6},-\frac{1}{12},\frac{1}{12},-\frac{1}{3},\frac{1}{3},\frac{1}{3},0,1)$ |
| 34 | | | 12 | $(1,g_{35})$ | 1/768 | 2 | $(0,0,0,0,0,0,0,0,0,0,0,0)$ | $(\frac{1}{8},\frac{1}{8},\frac{3}{8},\frac{3}{8},-\frac{1}{4},\frac{1}{4},-\frac{1}{4},0,0,\frac{1}{2},\frac{1}{2},-\frac{1}{2})$ |
| 35 | | | 12 | $(1,g_{36})$ | 1/486 | 3 | $(0,0,0,0,0,0,0,0,0,0,0,0)$ | $(\frac{1}{3},\frac{4}{9},-\frac{4}{9},-\frac{4}{9},\frac{2}{9},\frac{2}{9},\frac{2}{9},-\frac{1}{9},-\frac{1}{9},\frac{1}{9},0,0)$ |
| 36 | | | 8 | $(1,g_{37})$ | 1/1176 | 3 | $(0,0,0,0,0,0,0,0,0,0,0,0)$ | $(-\frac{1}{14},\frac{1}{14},-\frac{1}{14},\frac{3}{14},-\frac{3}{14},\frac{3}{14},-\frac{3}{14},\frac{5}{14},\frac{5}{14},\frac{5}{14},\frac{1}{2},\frac{1}{2},\frac{1}{2})$ |
| 37 | | | 32 | $(1,g_{38})$ | 1/21 | 0 | $(0,0,0,0,0,0,0,0,0,0,0,0)$ | $(\frac{10}{21},-\frac{3}{7},\frac{8}{21},\frac{2}{7},-\frac{5}{21},\frac{4}{21},\frac{1}{7},\frac{2}{21},\frac{1}{21},-\frac{1}{3},-\frac{1}{3},0)$ |
| 38 | | | 48 | $(1,g_{39})$ | 1/23 | 1 | $(0,0,0,0,0,0,0,0,0,0,0,0)$ | $(\frac{1}{46},\frac{3}{46},-\frac{5}{46},\frac{7}{46},\frac{9}{46},-\frac{11}{46},\frac{13}{46},\frac{15}{46},\frac{17}{46},\frac{19}{46},-\frac{21}{46},-\frac{1}{2})$ |
| 39 | | | 36 | $(1,g_{41})$ | 1/20 | 1 | $(0,0,0,0,0,0,0,0,0,0,0,0)$ | $(-\frac{1}{20},\frac{1}{10},-\frac{3}{20},\frac{3}{10},-\frac{7}{20},-\frac{9}{20},0,\frac{1}{5},-\frac{1}{5},\frac{2}{5},\frac{2}{5},\frac{1}{2})$ |
| 40 | | | 8 | $(1,g_{42})$ | 1/17640 | 4 | $(0,0,0,0,0,0,0,0,0,0,0,0)$ | $(-\frac{5}{14},\frac{5}{14},\frac{5}{14},\frac{5}{14},\frac{3}{14},\frac{3}{14},\frac{3}{14},-\frac{3}{14},\frac{1}{14},-\frac{1}{14},\frac{1}{14},-\frac{1}{14})$ |
| 41 | | | 48 | $(1,g_{43})$ | 1/35 | 1 | $(0,0,0,0,0,0,0,0,0,0,0,0)$ | $(\frac{3}{70},\frac{1}{70},\frac{9}{70},\frac{11}{70},\frac{13}{70},-\frac{17}{70},-\frac{19}{70},\frac{23}{70},-\frac{27}{70},\frac{29}{70},\frac{31}{70},\frac{103}{70})$ |
| 42 | | | 36 | $(1,g_{44})$ | 1/54 | 1 | $(0,0,0,0,0,0,0,0,0,0,0,0)$ | $(-\frac{1}{18},\frac{5}{18},\frac{1}{3},\frac{7}{18},0,\frac{1}{9},-\frac{1}{9},\frac{2}{9},-\frac{2}{9},-\frac{4}{9},\frac{4}{9},\frac{3}{2})$ |
| 43 | | | 32 | $(1,g_{45})$ | 1/252 | 2 | $(0,0,0,0,0,0,0,0,0,0,0,0)$ | $(\frac{1}{42},\frac{1}{42},-\frac{5}{42},-\frac{5}{42},\frac{11}{42},\frac{11}{42},-\frac{13}{42},-\frac{13}{42},-\frac{19}{42},\frac{19}{42},-\frac{17}{42},\frac{17}{42})$ |
| 44 | | | 24 | $(1,g_{46})$ | 1/168 | 0 | $(0,0,0,0,0,0,0,0,0,0,0,0)$ | $(-\frac{1}{28},\frac{1}{28},\frac{3}{28},\frac{3}{28},\frac{5}{28},\frac{5}{28},\frac{9}{28},-\frac{9}{28},-\frac{11}{28},\frac{11}{28},-\frac{13}{28},\frac{13}{28})$ |
| 45 | | | 12 | $(1,g_{47})$ | 1/483840 | 0 | $(0,0,0,0,0,0,0,0,0,0,0,0)$ | $(\frac{1}{12},-\frac{1}{12},\frac{1}{12},-\frac{1}{12},\frac{1}{12},-\frac{1}{12},\frac{1}{12},\frac{5}{12},\frac{5}{12},-\frac{5}{12},\frac{5}{12},-\frac{5}{12})$ |
| 46 | | | 6 | $(1,g_{48})$ | 1/294912 | 0 | $(0,0,0,0,0,0,0,0,0,0,0,0)$ | $(\frac{1}{4},\frac{1}{4},\frac{1}{4},-\frac{1}{4},\frac{1}{2},\frac{1}{2},\frac{1}{2},0,0,0,0,\frac{3}{2})$ |
| 47 | | | 12 | $(1,g_{49})$ | 1/768 | 0 | $(0,0,0,0,0,0,0,0,0,0,0,0)$ | $(-\frac{1}{8},\frac{1}{8},-\frac{3}{8},\frac{3}{8},0,0,\frac{1}{4},-\frac{1}{4},-\frac{1}{4},\frac{1}{4},\frac{1}{2},\frac{1}{2})$ |
| 48 | | | 12 | $(1,g_{50})$ | 1/9953280 | 4 | $(0,0,0,0,0,0,0,0,0,0,0,0)$ | $(-\frac{1}{3},-\frac{1}{3},\frac{1}{3},\frac{1}{3},\frac{1}{6},\frac{1}{6},-\frac{1}{6},-\frac{1}{6},-\frac{1}{6},\frac{1}{6},\frac{1}{6},\frac{1}{6})$ |
| 49 | | | 24 | $(1,g_{51})$ | 1/4608 | 0 | $(0,0,0,0,0,0,0,0,0,0,0,0)$ | $(\frac{1}{12},\frac{1}{12},-\frac{5}{12},\frac{5}{12},-\frac{1}{6},-\frac{1}{6},\frac{1}{6},\frac{1}{6},\frac{1}{3},-\frac{1}{3},-\frac{1}{3},-\frac{1}{3})$ |
| 50 | | | 48 | $(1,g_{52})$ | 1/96 | 0 | $(0,0,0,0,0,0,0,0,0,0,0,0)$ | $(-\frac{11}{24},-\frac{7}{24},-\frac{5}{24},\frac{1}{24},\frac{5}{12},\frac{5}{12},-\frac{1}{3},\frac{1}{3},-\frac{1}{6},-\frac{1}{6},-\frac{1}{12},-\frac{1}{12})$ |
| 51 | | | 6 | $(1,g_{53})$ | 1/48384 | 0 | $(0,0,0,0,0,0,0,0,0,0,0,0)$ | $(\frac{3}{8},\frac{3}{8},-\frac{3}{8},\frac{3}{8},-\frac{3}{8},-\frac{3}{8},-\frac{1}{8},-\frac{1}{8},\frac{1}{8},-\frac{1}{8},-\frac{1}{8},-\frac{1}{8})$ |

| No. | P | com. | size | rep. | $[h]_{\subset C_g}/\lvert C_g\rvert$ | $\lvert Z_{\mathrm{R}g}^{\mathrm{R}h}\rvert$ | $\vec{\lambda}_{\bar{g},O}$ | $\vec{\lambda}_{\bar{h},O}$ |
|---|---|---|---|---|---|---|---|---|
| 52 | | | 24 | $(1, g_{54})$ | 1/72 | 0 | $(0,0,0,0,0,0,0,0,0,0,0,0)$ | $(\frac{1}{24},\frac{1}{24},\frac{1}{8},-\frac{1}{8},-\frac{5}{24},\frac{5}{24},-\frac{7}{24},\frac{7}{24},\frac{3}{8},\frac{3}{8},\frac{11}{24},-\frac{11}{24})$ |
| 53 | | | 24 | $(1, g_{55})$ | 1/56 | 1 | $(0,0,0,0,0,0,0,0,0,0,0,0)$ | $(-\frac{5}{14},-\frac{1}{14},\frac{3}{14},-\frac{2}{7},-\frac{2}{7},\frac{1}{7},-\frac{1}{7},\frac{3}{7},-\frac{3}{7},\frac{2}{7},0,0)$ |
| 54 | | | 48 | $(1, g_{56})$ | 1/28 | 1 | $(0,0,0,0,0,0,0,0,0,0,0,0)$ | $(-\frac{1}{28},\frac{1}{14},-\frac{3}{28},-\frac{5}{28},-\frac{3}{14},-\frac{9}{28},-\frac{5}{14},\frac{11}{28},\frac{13}{28},-\frac{1}{4},-\frac{1}{4},\frac{1}{2})$ |
| 55 | | | 12 | $(1, g_{57})$ | 1/92160 | 0 | $(0,0,0,0,0,0,0,0,0,0,0,0)$ | $(-\frac{3}{8},-\frac{3}{8},\frac{3}{8},-\frac{3}{8},-\frac{1}{8},\frac{1}{8},\frac{1}{8},\frac{1}{4},-\frac{1}{4},\frac{1}{4},-\frac{1}{4})$ |
| 56 | | | 24 | $(1, g_{58})$ | 1/138240 | 4 | $(0,0,0,0,0,0,0,0,0,0,0,0)$ | $(-\frac{5}{12},-\frac{5}{12},-\frac{5}{12},\frac{5}{12},\frac{1}{12},\frac{1}{12},\frac{1}{12},\frac{1}{6},\frac{1}{6},-\frac{1}{6},\frac{1}{6})$ |
| 57 | | | 48 | $(1, g_{59})$ | 1/576 | 0 | $(0,0,0,0,0,0,0,0,0,0,0,0)$ | $(\frac{11}{24},-\frac{11}{24},-\frac{5}{12},-\frac{5}{12},\frac{7}{24},-\frac{7}{24},\frac{5}{24},-\frac{5}{24},-\frac{1}{12},\frac{1}{12},-\frac{1}{24},-\frac{1}{24})$ |
| 58 | | | 12 | $(1, g_{60})$ | 1/3456 | 0 | $(0,0,0,0,0,0,0,0,0,0,0,0)$ | $(\frac{1}{3},-\frac{1}{3},\frac{1}{3},-\frac{1}{6},-\frac{1}{6},\frac{1}{2},\frac{1}{2},0,0,0,\frac{1}{2})$ |
| 59 | | | 48 | $(1, g_{61})$ | 1/144 | 0 | $(0,0,0,0,0,0,0,0,0,0,0,0)$ | $(\frac{1}{24},-\frac{1}{12},\frac{5}{24},\frac{11}{24},\frac{5}{12},\frac{7}{24},\frac{1}{8},-\frac{1}{8},\frac{3}{8},-\frac{3}{8},\frac{1}{4},-\frac{1}{4})$ |
| 60 | | | 6 | $(1, g_{62})$ | 1/1179648 | 0 | $(0,0,0,0,0,0,0,0,0,0,0,0)$ | $(0,0,\frac{1}{2},\frac{1}{2},\frac{1}{4},\frac{1}{4},-\frac{1}{4},-\frac{1}{4},\frac{1}{4},-\frac{1}{4},-\frac{1}{4},\frac{5}{4})$ |
| 61 | | | 24 | $(1, g_{63})$ | 1/576 | 0 | $(0,0,0,0,0,0,0,0,0,0,0,0)$ | $(-\frac{1}{3},\frac{1}{6},\frac{1}{2},\frac{5}{12},\frac{5}{12},\frac{1}{12},-\frac{1}{12},0,-\frac{1}{4},\frac{1}{4},\frac{1}{4},-\frac{1}{4})$ |
| 62 | | | 24 | $(1, g_{64})$ | 1/288 | 3 | $(0,0,0,0,0,0,0,0,0,0,0,0)$ | $(\frac{1}{3},\frac{1}{4},-\frac{1}{4},-\frac{1}{6},\frac{1}{6},0,-\frac{5}{12},-\frac{5}{12},\frac{5}{12},-\frac{1}{12},\frac{1}{12},\frac{1}{2})$ |
| 63 | | | 12 | $(1, g_{65})$ | 1/486 | 0 | $(0,0,0,0,0,0,0,0,0,0,0,0)$ | $(\frac{1}{6},-\frac{1}{6},\frac{1}{2},-\frac{1}{18},-\frac{1}{18},\frac{1}{18},\frac{5}{18},\frac{5}{18},\frac{5}{18},-\frac{7}{18},-\frac{7}{18},\frac{7}{18})$ |
| 64 | | | 12 | $(1, g_{66})$ | 1/1944 | 3 | $(0,0,0,0,0,0,0,0,0,0,0,0)$ | $(\frac{1}{9},-\frac{1}{9},\frac{1}{9},-\frac{2}{9},\frac{2}{9},\frac{2}{9},\frac{4}{9},-\frac{4}{9},\frac{1}{9},-\frac{1}{3},-\frac{1}{3},\frac{1}{3})$ |
| 65 | | | 6 | $(1, g_{67})$ | 1/15360 | 0 | $(0,0,0,0,0,0,0,0,0,0,0,0)$ | $(0,0,0,-\frac{1}{4},\frac{1}{4},\frac{1}{4},\frac{1}{4},-\frac{1}{4},\frac{1}{4},\frac{1}{2},\frac{1}{2},-\frac{1}{2})$ |
| 66 | | | 12 | $(1, g_{68})$ | 1/576 | 0 | $(0,0,0,0,0,0,0,0,0,0,0,0)$ | $(0,0,-\frac{1}{6},-\frac{1}{6},\frac{1}{6},-\frac{1}{6},-\frac{1}{3},\frac{1}{3},-\frac{1}{3},-\frac{1}{3},\frac{1}{2},\frac{3}{2})$ |
| 67 | | | 24 | $(1, g_{69})$ | 1/48 | 0 | $(0,0,0,0,0,0,0,0,0,0,0,0)$ | $(0,-\frac{1}{12},\frac{1}{12},\frac{1}{6},\frac{1}{6},\frac{1}{4},\frac{1}{4},-\frac{1}{3},\frac{1}{3},\frac{5}{12},\frac{5}{12},\frac{1}{2})$ |
| 68 | | | 12 | $(1, g_{70})$ | 1/66 | 2 | $(0,0,0,0,0,0,0,0,0,0,0,0)$ | $(-\frac{5}{11},-\frac{5}{11},\frac{4}{11},\frac{4}{11},-\frac{4}{11},\frac{2}{11},\frac{2}{11},\frac{3}{11},\frac{3}{11},-\frac{3}{11},\frac{1}{11},\frac{1}{11},0,0)$ |
| 69 | | | 48 | $(1, g_{71})$ | 1/33 | 1 | $(0,0,0,0,0,0,0,0,0,0,0,0)$ | $(\frac{1}{66},\frac{5}{66},-\frac{7}{66},-\frac{31}{66},\frac{29}{66},\frac{25}{66},-\frac{23}{66},-\frac{13}{66},\frac{19}{66},-\frac{17}{66},-\frac{1}{6},\frac{1}{6})$ |
| 70 | | | 12 | $(1, g_{72})$ | 1/6144 | 4 | $(0,0,0,0,0,0,0,0,0,0,0,0)$ | $(-\frac{1}{4},\frac{1}{4},\frac{1}{8},\frac{1}{8},\frac{1}{8},\frac{1}{8},-\frac{3}{8},\frac{3}{8},-\frac{3}{8},\frac{3}{8},\frac{1}{2},\frac{1}{2})$ |
| 71 | | | 24 | $(1, g_{73})$ | 1/9216 | 0 | $(0,0,0,0,0,0,0,0,0,0,0,0)$ | $(\frac{1}{6},\frac{1}{6},-\frac{1}{3},-\frac{1}{3},\frac{1}{12},\frac{1}{12},-\frac{1}{12},\frac{1}{12},-\frac{5}{12},\frac{5}{12},-\frac{5}{12},\frac{7}{12})$ |
| 72 | | | 48 | $(1, g_{74})$ | 1/192 | 2 | $(0,0,0,0,0,0,0,0,0,0,0,0)$ | $(\frac{5}{12},-\frac{1}{12},\frac{11}{24},-\frac{11}{24},\frac{7}{24},\frac{7}{24},-\frac{5}{24},-\frac{5}{24},-\frac{1}{6},\frac{1}{6},\frac{1}{24},\frac{1}{24})$ |
| 73 | | | 12 | $(1, g_{75})$ | 1/2048 | 0 | $(0,0,0,0,0,0,0,0,0,0,0,0)$ | $(\frac{1}{2},\frac{1}{4},-\frac{1}{4},0,\frac{3}{8},\frac{3}{8},-\frac{3}{8},-\frac{3}{8},\frac{1}{8},\frac{1}{8},\frac{1}{8},\frac{1}{8})$ |
| 74 | | | 24 | $(1, g_{76})$ | 1/64 | 0 | $(0,0,0,0,0,0,0,0,0,0,0,0)$ | $(\frac{1}{8},-\frac{3}{8},\frac{1}{2},\frac{7}{16},-\frac{7}{16},-\frac{5}{16},\frac{5}{16},-\frac{3}{16},\frac{3}{16},-\frac{1}{16},-\frac{1}{16},1)$ |
| 75 | | | 36 | $(1, g_{77})$ | 1/480 | 2 | $(0,0,0,0,0,0,0,0,0,0,0,0)$ | $(-\frac{1}{20},-\frac{1}{20},-\frac{3}{20},\frac{3}{20},\frac{1}{5},-\frac{1}{5},-\frac{7}{20},\frac{7}{20},-\frac{2}{5},\frac{2}{5},-\frac{9}{20},\frac{9}{20})$ |
| 76 | | | 24 | $(1, g_{78})$ | 1/5400 | 3 | $(0,0,0,0,0,0,0,0,0,0,0,0)$ | $(-\frac{1}{30},-\frac{1}{30},\frac{1}{30},-\frac{7}{30},-\frac{7}{30},\frac{7}{30},\frac{11}{30},-\frac{11}{30},-\frac{13}{30},\frac{13}{30},\frac{13}{30},\frac{43}{30})$ |
| 77 | | | 72 | $(1, g_{79})$ | 1/120 | 1 | $(0,0,0,0,0,0,0,0,0,0,0,0)$ | $(\frac{2}{15},-\frac{1}{15},-\frac{4}{15},-\frac{7}{15},\frac{1}{30},-\frac{1}{30},\frac{7}{30},\frac{7}{30},\frac{11}{30},-\frac{11}{30},\frac{13}{30},\frac{13}{30})$ |
| 78 | | | 144 | $(1, g_{80})$ | 1/60 | 1 | $(0,0,0,0,0,0,0,0,0,0,0,0)$ | $(-\frac{1}{30},\frac{1}{60},\frac{7}{60},\frac{11}{60},\frac{13}{60},-\frac{7}{30},-\frac{17}{60},\frac{19}{60},\frac{11}{30},-\frac{23}{60},\frac{13}{30},\frac{31}{60})$ |
| 79 | | | 24 | $(1, g_{81})$ | 1/30 | 2 | $(0,0,0,0,0,0,0,0,0,0,0,0)$ | $(-\frac{2}{5},\frac{1}{3},\frac{1}{5},-\frac{7}{15},\frac{7}{15},-\frac{4}{15},\frac{4}{15},\frac{2}{15},\frac{2}{15},\frac{1}{15},\frac{1}{15},0)$ |
| 80 | | | 72 | $(1, g_{82})$ | 1/30 | 0 | $(0,0,0,0,0,0,0,0,0,0,0,0)$ | $(\frac{1}{30},-\frac{1}{15},\frac{1}{10},-\frac{2}{15},\frac{1}{6},\frac{1}{30},\frac{4}{15},\frac{1}{10},\frac{1}{30},-\frac{1}{30},\frac{7}{15},\frac{3}{2})$ |
| 81 | | | 32 | $(1, g_{83})$ | 1/63 | 1 | $(0,0,0,0,0,0,0,0,0,0,0,0)$ | $(\frac{19}{42},-\frac{17}{42},\frac{13}{42},-\frac{11}{42},\frac{5}{42},\frac{1}{42},-\frac{5}{14},\frac{5}{14},-\frac{3}{14},\frac{3}{14},\frac{1}{14},-\frac{1}{14})$ |
| 82 | | | 18 | $(1, g_{84})$ | 1/1200 | 0 | $(0,0,0,0,0,0,0,0,0,0,0,0)$ | $(-\frac{1}{20},\frac{1}{20},\frac{3}{20},-\frac{3}{20},\frac{7}{20},-\frac{7}{20},\frac{9}{20},\frac{9}{20},\frac{1}{4},-\frac{1}{4},\frac{1}{4},\frac{1}{4})$ |
| 83 | | | 24 | $(1, g_{85})$ | 1/900 | 2 | $(0,0,0,0,0,0,0,0,0,0,0,0)$ | $(-\frac{1}{15},\frac{1}{15},-\frac{2}{15},\frac{2}{15},-\frac{7}{15},\frac{7}{15},-\frac{4}{15},\frac{4}{15},\frac{1}{3},\frac{1}{3},-\frac{1}{3},-\frac{1}{3})$ |
| 84 | | | 72 | $(1, g_{86})$ | 1/60 | 0 | $(0,0,0,0,0,0,0,0,0,0,0,0)$ | $(-\frac{1}{60},-\frac{7}{60},\frac{11}{60},-\frac{13}{60},\frac{17}{60},-\frac{19}{60},\frac{29}{60},\frac{23}{60},\frac{1}{12},-\frac{1}{12},-\frac{5}{12},\frac{5}{12})$ |
| 85 | | | 36 | $(1, g_{87})$ | 1/40 | 0 | $(0,0,0,0,0,0,0,0,0,0,0,0)$ | $(\frac{1}{20},\frac{1}{10},\frac{1}{20},-\frac{1}{5},\frac{3}{10},-\frac{7}{20},-\frac{2}{5},-\frac{9}{20},0,\frac{1}{4},-\frac{1}{4},-\frac{1}{2})$ |
| 86 | | | 24 | $(1, g_{88})$ | 1/64 | 2 | $(0,0,0,0,0,0,0,0,0,0,0,0)$ | $(-\frac{3}{8},\frac{1}{4},\frac{1}{8},-\frac{7}{16},-\frac{7}{16},-\frac{5}{16},\frac{5}{16},-\frac{3}{16},\frac{3}{16},\frac{3}{16},0,\frac{1}{16},-\frac{1}{16})$ |
| 87 | | | 36 | $(1, g_{89})$ | 1/22 | 0 | $(0,0,0,0,0,0,0,0,0,0,0,0)$ | $(\frac{1}{22},\frac{1}{11},\frac{3}{22},-\frac{2}{11},\frac{4}{22},-\frac{3}{11},-\frac{7}{22},\frac{4}{11},\frac{5}{22},-\frac{5}{11},0,\frac{1}{2})$ |
| 88 | | | 24 | $(1, g_{90})$ | 1/288 | 2 | $(0,0,0,0,0,0,0,0,0,0,0,0)$ | $(\frac{1}{6},\frac{1}{12},-\frac{5}{12},-\frac{1}{4},\frac{1}{4},-\frac{1}{3},-\frac{1}{3},\frac{1}{3},0,0,\frac{1}{2},\frac{1}{2})$ |
| 89 | | | 24 | $(1, g_{91})$ | 1/2592 | 1 | $(0,0,0,0,0,0,0,0,0,0,0,0)$ | $(-\frac{1}{4},\frac{1}{4},\frac{1}{2},-\frac{5}{12},-\frac{5}{12},\frac{5}{12},\frac{1}{12},\frac{1}{12},\frac{1}{12},\frac{1}{6},\frac{1}{6},\frac{7}{6})$ |
| 90 | | | 72 | $(1, g_{92})$ | 1/40 | 0 | $(0,0,0,0,0,0,0,0,0,0,0,0)$ | $(-\frac{19}{40},-\frac{9}{20},-\frac{17}{40},-\frac{7}{20},-\frac{13}{40},-\frac{11}{40},-\frac{9}{40},\frac{7}{40},-\frac{3}{20},-\frac{3}{40},\frac{1}{40},-\frac{1}{20})$ |
| 91 | | | 48 | $(1, g_{93})$ | 1/24 | 1 | $(0,0,0,0,0,0,0,0,0,0,0,0)$ | $(\frac{5}{12},-\frac{11}{24},\frac{1}{3},\frac{1}{24},-\frac{1}{4},\frac{1}{12},-\frac{1}{24},-\frac{5}{24},\frac{1}{2},0,-\frac{1}{6},\frac{1}{6})$ |
| 92 | | | 72 | $(1, g_{94})$ | 1/30 | 1 | $(0,0,0,0,0,0,0,0,0,0,0,0)$ | $(-\frac{1}{30},\frac{1}{10},-\frac{1}{6},-\frac{1}{5},-\frac{7}{30},\frac{3}{10},\frac{1}{3},\frac{11}{30},\frac{2}{5},\frac{13}{30},\frac{1}{2},0)$ |
| 93 | | | 24 | $(1, g_{95})$ | 1/288 | 0 | $(0,0,0,0,0,0,0,0,0,0,0,0)$ | $(\frac{5}{12},-\frac{1}{12},-\frac{1}{3},\frac{1}{3},\frac{1}{6},-\frac{1}{6},\frac{1}{4},-\frac{1}{4},\frac{1}{2},0,0,-\frac{1}{2})$ |
| 94 | | | 36 | $(1, g_{96})$ | 1/54 | 2 | $(0,0,0,0,0,0,0,0,0,0,0,0)$ | $(-\frac{7}{18},\frac{1}{3},\frac{5}{18},\frac{1}{6},-\frac{1}{18},-\frac{4}{9},-\frac{4}{9},-\frac{2}{9},-\frac{2}{9},\frac{1}{9},-\frac{1}{9},0)$ |
| 95 | | | 96 | $(1, g_{97})$ | 1/84 | 0 | $(0,0,0,0,0,0,0,0,0,0,0,0)$ | $(\frac{41}{84},\frac{37}{84},-\frac{31}{84},-\frac{29}{84},-\frac{1}{84},-\frac{5}{84},\frac{11}{84},\frac{13}{84},\frac{25}{84},-\frac{23}{84},-\frac{17}{84},\frac{19}{84})$ |
| 96 | | | 48 | $(1, g_{98})$ | 1/28 | 0 | $(0,0,0,0,0,0,0,0,0,0,0,0)$ | $(-\frac{1}{56},\frac{3}{56},-\frac{5}{56},\frac{27}{56},\frac{25}{56},\frac{23}{56},-\frac{9}{56},-\frac{1}{56},-\frac{19}{56},\frac{13}{56},-\frac{17}{56},\frac{15}{56})$ |
| 97 | | | 36 | $(1, g_{99})$ | 1/216 | 1 | $(0,0,0,0,0,0,0,0,0,0,0,0)$ | $(-\frac{1}{18},\frac{1}{6},\frac{7}{18},-\frac{5}{18},-\frac{1}{9},-\frac{1}{9},\frac{4}{9},-\frac{4}{9},-\frac{1}{3},-\frac{1}{3},-\frac{2}{9},\frac{2}{9})$ |
| 98 | | | 72 | $(1, g_{100})$ | 1/36 | 1 | $(0,0,0,0,0,0,0,0,0,0,0,0)$ | $(-\frac{1}{36},-\frac{1}{12},-\frac{1}{9},-\frac{5}{36},\frac{7}{36},\frac{2}{9},-\frac{11}{36},\frac{1}{3},-\frac{13}{36},\frac{17}{36},-\frac{4}{9},-\frac{5}{12})$ |
| 99 | | | 24 | $(1, g_{101})$ | 1/864 | 0 | $(0,0,0,0,0,0,0,0,0,0,0,0)$ | $(\frac{1}{24},-\frac{1}{24},-\frac{1}{24},-\frac{5}{24},-\frac{5}{24},-\frac{5}{24},\frac{7}{24},\frac{7}{24},\frac{7}{24},\frac{11}{24},-\frac{11}{24},\frac{35}{24})$ |
| 100 | | A | 1 | $(g_2, h_3^{(2)})$ | 1/1006387200 | 0 | – | – |
| 101 | | A | 4 | $(g_2, h_4^{(2)})$ | 1/720 | 0 | – | – |
| 102 | | A | 12 | $(g_2, h_7^{(2)})$ | 1/1200 | 0 | – | – |
| 103 | | A | 48 | $(g_2, h_{13}^{(2)})$ | 1/60 | 0 | – | – |
| 104 | | A | 1 | $(g_2, h_{16}^{(2)})$ | 1/48384 | 0 | – | – |
| 105 | | A | 4 | $(g_2, h_{18}^{(2)})$ | 1/72 | 0 | – | – |

| No. | P | com. | size | rep. | $[h]_{\subset C_g}/\lvert C_g\rvert$ | $\lvert Z_{\mathrm{R}g}^{\mathrm{R}h}\rvert$ | $\vec{\lambda}_{\tilde{g},O}$ | $\vec{\lambda}_{\tilde{h},O}$ |
|---|---|---|---|---|---|---|---|---|
| 106 | | A | 4 | $(g_2, h_{19}^{(2)})$ | 1/241920 | 0 | – | – |
| 107 | | A | 4 | $(g_2, h_{23}^{(2)})$ | 1/864 | 0 | – | – |
| 108 | | A | 1 | $(g_2, h_{24}^{(2)})$ | 1/245760 | 0 | – | – |
| 109 | | | 3 | $(g_2, h_{25}^{(2)})$ | 1/491520 | 0 | $(-\frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, \frac{1}{4})$ | $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, 0, 0, 0, 0, 0, 0, 0, 0)$ |
| 110 | | A | 8 | $(g_2, h_{29}^{(2)})$ | 1/84 | 0 | – | – |
| 111 | | A | 8 | $(g_2, h_{31}^{(2)})$ | 1/28 | 0 | – | – |
| 112 | | | 6 | $(g_2, h_{32}^{(2)})$ | 1/4096 | 0 | $(-\frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, -\frac{1}{4})$ | $(0, 0, \frac{1}{2}, \frac{1}{2}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, \frac{5}{4})$ |
| 113 | | A | 3 | $(g_2, h_{34}^{(2)})$ | 1/2048 | 0 | – | – |
| 114 | | | 12 | $(g_2, h_{35}^{(2)})$ | 1/256 | 0 | $(\frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, -\frac{1}{4})$ | $(\frac{1}{4}, -\frac{1}{4}, -\frac{3}{8}, -\frac{3}{8}, \frac{1}{8}, \frac{1}{8}, \frac{3}{8}, \frac{3}{8}, 0, 0, \frac{1}{8}, \frac{1}{8})$ |
| 115 | | A | 6 | $(g_2, h_{37}^{(2)})$ | 1/128 | 0 | – | – |
| 116 | | A | 24 | $(g_2, h_{38}^{(2)})$ | 1/32 | 0 | – | – |
| 117 | | | 24 | $(g_2, h_{39}^{(2)})$ | 1/32 | 0 | $(\frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, -\frac{1}{4})$ | $(\frac{3}{16}, -\frac{1}{16}, \frac{5}{16}, -\frac{3}{8}, \frac{7}{16}, \frac{1}{2}, \frac{7}{16}, -\frac{1}{16}, -\frac{1}{8}, -\frac{3}{16}, -\frac{5}{16}, \frac{1}{4})$ |
| 118 | | | 6 | $(g_2, h_{42}^{(2)})$ | 1/6144 | 0 | $(\frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, \frac{1}{4})$ | $(\frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, 0, 0, 0, 0, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2})$ |
| 119 | | A | 3 | $(g_2, h_{44}^{(2)})$ | 1/3072 | 0 | – | – |
| 120 | | A | 4 | $(g_2, h_{45}^{(2)})$ | 1/768 | 0 | – | – |
| 121 | | | 12 | $(g_2, h_{46}^{(2)})$ | 1/1536 | 0 | $(\frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, \frac{1}{4})$ | $(-\frac{1}{12}, -\frac{1}{12}, \frac{5}{12}, -\frac{5}{12}, -\frac{1}{12}, -\frac{1}{12}, \frac{1}{12}, \frac{1}{12}, -\frac{5}{12}, -\frac{5}{12}, -\frac{5}{12}, \frac{5}{12})$ |
| 122 | | A | 6 | $(g_2, h_{48}^{(2)})$ | 1/192 | 0 | – | – |
| 123 | | | 6 | $(g_2, h_{49}^{(2)})$ | 1/192 | 0 | $(\frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, -\frac{1}{4})$ | $(\frac{3}{8}, \frac{3}{8}, \frac{3}{8}, -\frac{3}{8}, -\frac{1}{8}, \frac{1}{8}, \frac{1}{4}, \frac{1}{4}, 0, 0, -\frac{1}{8}, \frac{1}{8})$ |
| 124 | | | 48 | $(g_2, h_{50}^{(2)})$ | 1/384 | 0 | $(-\frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, \frac{1}{4})$ | $(-\frac{1}{12}, \frac{1}{12}, -\frac{5}{12}, -\frac{5}{12}, -\frac{1}{6}, -\frac{1}{6}, -\frac{1}{3}, \frac{1}{3}, -\frac{1}{6}, \frac{1}{6}, \frac{1}{3}, -\frac{1}{3})$ |
| 125 | | A | 24 | $(g_2, h_{52}^{(2)})$ | 1/192 | 0 | – | – |
| 126 | | A | 48 | $(g_2, h_{56}^{(2)})$ | 1/48 | 0 | – | – |
| 127 | | | 48 | $(g_2, h_{57}^{(2)})$ | 1/48 | 0 | $(-\frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, -\frac{1}{4})$ | $(-\frac{5}{24}, -\frac{1}{12}, \frac{1}{24}, \frac{1}{24}, -\frac{7}{24}, \frac{1}{3}, -\frac{5}{12}, -\frac{11}{24}, \frac{11}{24}, \frac{5}{24}, \frac{1}{3}, \frac{31}{24})$ |
| 128 | | | 6 | $(g_2, h_{60}^{(2)})$ | 1/12288 | 0 | $(-\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, -\frac{1}{4})$ | $(-\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2})$ |
| 129 | | A | 3 | $(g_2, h_{62}^{(2)})$ | 1/6144 | 0 | – | – |
| 130 | | | 24 | $(g_2, h_{63}^{(2)})$ | 1/384 | 0 | $(-\frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4})$ | $(-\frac{1}{6}, \frac{1}{6}, -\frac{1}{3}, \frac{1}{3}, \frac{1}{12}, -\frac{1}{12}, \frac{1}{6}, -\frac{1}{6}, -\frac{1}{3}, -\frac{1}{3}, -\frac{5}{12}, -\frac{5}{12})$ |
| 131 | | A | 12 | $(g_2, h_{65}^{(2)})$ | 1/192 | 0 | – | – |
| 132 | | | 3 | $(g_2, h_{66}^{(2)})$ | 1/30720 | 0 | $(\frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, \frac{1}{4})$ | $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, 0, 0, 0, 0, 0, 0)$ |
| 133 | | A | 3 | $(g_2, h_{68}^{(2)})$ | 1/15360 | 0 | – | – |
| 134 | | | 12 | $(g_2, h_{69}^{(2)})$ | 1/96 | 0 | $(\frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, -\frac{1}{4})$ | $(\frac{1}{4}, -\frac{1}{4}, \frac{1}{12}, -\frac{1}{12}, -\frac{5}{12}, -\frac{5}{12}, -\frac{1}{12}, -\frac{1}{12}, \frac{1}{4}, \frac{1}{4}, \frac{5}{12}, \frac{5}{12})$ |
| 135 | | A | 12 | $(g_2, h_{71}^{(2)})$ | 1/48 | 0 | – | – |
| 136 | | | 12 | $(g_2, h_{72}^{(2)})$ | 1/256 | 0 | $(-\frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4})$ | $(-\frac{3}{8}, -\frac{3}{8}, -\frac{1}{4}, \frac{1}{4}, \frac{1}{8}, -\frac{1}{8}, -\frac{1}{8}, -\frac{1}{8}, -\frac{3}{8}, -\frac{3}{8}, -\frac{1}{4}, -\frac{1}{4})$ |
| 137 | | A | 6 | $(g_2, h_{74}^{(2)})$ | 1/128 | 0 | – | – |
| 138 | | A | 3 | $(g_2, h_{75}^{(2)})$ | 1/768 | 0 | – | – |
| 139 | | | 6 | $(g_2, h_{76}^{(2)})$ | 1/768 | 0 | $(-\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, \frac{1}{4})$ | $(-\frac{3}{8}, \frac{3}{8}, -\frac{1}{8}, \frac{1}{8}, -\frac{1}{8}, -\frac{1}{8}, -\frac{1}{8}, \frac{1}{8}, \frac{3}{8}, \frac{3}{8}, \frac{3}{8}, -\frac{3}{8})$ |
| 140 | | | 24 | $(g_2, h_{77}^{(2)})$ | 1/96 | 0 | $(-\frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4})$ | $(-\frac{7}{24}, -\frac{11}{24}, \frac{5}{24}, \frac{1}{24}, -\frac{5}{24}, -\frac{5}{24}, \frac{1}{24}, -\frac{1}{24}, \frac{7}{24}, -\frac{7}{24}, -\frac{11}{24}, \frac{13}{24})$ |
| 141 | | A | 12 | $(g_2, h_{78}^{(2)})$ | 1/96 | 0 | – | – |
| 142 | | | 6 | $(g_2, h_{79}^{(2)})$ | 1/128 | 0 | $(\frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4})$ | $(\frac{1}{4}, \frac{1}{4}, -\frac{1}{8}, \frac{1}{8}, 0, 0, \frac{1}{4}, -\frac{1}{4}, \frac{1}{2}, \frac{1}{2}, -\frac{3}{8}, \frac{5}{8})$ |
| 143 | | A | 6 | $(g_2, h_{80}^{(2)})$ | 1/128 | 0 | – | – |
| 144 | | A | 12 | $(g_2, h_{81}^{(2)})$ | 1/80 | 0 | – | – |
| 145 | | | 18 | $(g_2, h_{83}^{(2)})$ | 1/80 | 0 | $(-\frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, -\frac{1}{4})$ | $(-\frac{9}{20}, \frac{7}{20}, -\frac{1}{20}, -\frac{3}{20}, \frac{7}{20}, -\frac{7}{20}, -\frac{3}{20}, \frac{3}{20}, -\frac{9}{20}, \frac{9}{20}, \frac{1}{20}, \frac{21}{20})$ |
| 146 | | A | 64 | $(g_2, h_{87}^{(2)})$ | 1/84 | 0 | – | – |
| 147 | | A | 12 | $(g_2, h_{89}^{(2)})$ | 1/1200 | 0 | – | – |
| 148 | | A | 48 | $(g_2, h_{95}^{(2)})$ | 1/60 | 0 | – | – |
| 149 | | | 3 | $(g_2, h_{97}^{(2)})$ | 1/192 | 0 | $(\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, -\frac{1}{4})$ | $(\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, 0, 0, 0, \frac{1}{4}, \frac{1}{4}, \frac{1}{4})$ |
| 150 | | A | 6 | $(g_2, h_{98}^{(2)})$ | 1/192 | 0 | – | – |
| 151 | | | 6 | $(g_2, h_{99}^{(2)})$ | 1/384 | 0 | $(-\frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, -\frac{1}{4})$ | $(-\frac{1}{4}, -\frac{1}{4}, \frac{3}{8}, \frac{3}{8}, \frac{1}{2}, \frac{1}{2}, -\frac{1}{4}, -\frac{1}{4}, 0, 0, \frac{1}{8}, \frac{1}{8})$ |
| 152 | | A | 6 | $(g_2, h_{100}^{(2)})$ | 1/384 | 0 | – | – |
| 153 | | A | 24 | $(g_2, h_{101}^{(2)})$ | 1/48 | 0 | – | – |

| No. | P | com. | size | rep. | $[h]_{\subset C_g}/\lvert C_g\rvert$ | $\lvert Z_{\mathrm{R}g}^{\mathrm{R}h}\rvert$ | $\vec{\lambda}_{\tilde{g},O}$ | $\vec{\lambda}_{\tilde{h},O}$ |
|---|---|---|---|---|---|---|---|---|
| 154 | | | 24 | $(g_2, h_{102}^{(2)})$ | 1/48 | 0 | $(-\frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, -\frac{1}{4})$ | $(-\frac{1}{3}, -\frac{7}{24}, \frac{1}{6}, \frac{5}{12}, \frac{11}{24}, -\frac{5}{12}, \frac{1}{3}, \frac{1}{12}, -\frac{5}{24}, -\frac{1}{6}, -\frac{1}{12}, \frac{1}{24})$ |
| 155 | | A | 24 | $(g_2, h_{103}^{(2)})$ | 1/24 | 0 | – | – |
| 156 | | | 12 | $(g_2, h_{104}^{(2)})$ | 1/24 | 0 | $(-\frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4})$ | $(-\frac{5}{12}, -\frac{1}{3}, \frac{1}{4}, \frac{1}{6}, \frac{1}{2}, -\frac{5}{12}, -\frac{1}{3}, \frac{1}{12}, -\frac{1}{4}, -\frac{1}{6}, 0, -\frac{1}{12})$ |
| 157 | | A | 28 | $(g_2, h_{107}^{(2)})$ | 1/52 | 0 | – | – |
| 158 | | | 18 | $(g_2, h_{109}^{(2)})$ | 1/80 | 0 | $(\frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, -\frac{1}{4})$ | $(\frac{2}{5}, \frac{3}{10}, -\frac{1}{5}, \frac{1}{10}, \frac{2}{5}, -\frac{3}{10}, -\frac{1}{5}, -\frac{1}{10}, \frac{1}{2}, \frac{1}{2}, 0, 0)$ |
| 159 | | A | 36 | $(g_2, h_{111}^{(2)})$ | 1/80 | 0 | – | – |
| 160 | | | 4 | $(g_5, h_4^{(5)})$ | 1/3240 | 0 | $(\frac{1}{6}, -\frac{1}{6}, \frac{1}{6}, \frac{1}{6}, -\frac{1}{6}, -\frac{1}{6}, -\frac{1}{6}, \frac{1}{6}, -\frac{1}{6}, \frac{1}{6}, -\frac{1}{6}, \frac{1}{6})$ | $(0, 0, 0, 0, -\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, \frac{1}{3}, \frac{1}{3}, -\frac{1}{3}, \frac{1}{3}, -\frac{1}{3})$ |
| 161 | | | 48 | $(g_5, h_8^{(5)})$ | 1/45 | 0 | $(\frac{1}{6}, -\frac{1}{6}, -\frac{1}{6}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6}, -\frac{1}{6}, \frac{1}{6}, -\frac{1}{6}, -\frac{1}{6}, -\frac{1}{6}, \frac{1}{6})$ | $(\frac{7}{30}, -\frac{1}{10}, -\frac{1}{30}, -\frac{1}{30}, -\frac{1}{10}, \frac{3}{10}, -\frac{11}{30}, \frac{13}{30}, \frac{13}{30}, \frac{11}{30}, \frac{3}{10}, -\frac{7}{30})$ |
| 162 | | | 6 | $(g_5, h_{13}^{(5)})$ | 1/29393280 | 6 | $(\frac{1}{6}, -\frac{1}{6}, \frac{1}{6}, -\frac{1}{6}, -\frac{1}{6}, -\frac{1}{6}, \frac{1}{6}, -\frac{1}{6}, \frac{1}{6}, -\frac{1}{6}, -\frac{1}{6}, -\frac{1}{6})$ | $(\frac{1}{3}, -\frac{1}{3}, \frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, 0, 0, 0, 0, 0, 0)$ |
| 163 | | | 36 | $(g_5, h_{16}^{(5)})$ | 1/45 | 1 | $(\frac{1}{6}, \frac{1}{6}, -\frac{1}{6}, -\frac{1}{6}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6}, -\frac{1}{6}, \frac{1}{6}, \frac{1}{6}, -\frac{1}{6})$ | $(\frac{7}{30}, \frac{11}{30}, -\frac{13}{30}, \frac{13}{30}, -\frac{11}{30}, -\frac{7}{30}, \frac{1}{30}, -\frac{1}{30}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6})$ |
| 164 | | | 96 | $(g_5, h_{22}^{(5)})$ | 1/63 | 1 | $(\frac{1}{6}, -\frac{1}{6}, -\frac{1}{6}, \frac{1}{6}, -\frac{1}{6}, -\frac{1}{6}, \frac{1}{6}, -\frac{1}{6}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6}, -\frac{1}{6})$ | $(\frac{2}{7}, -\frac{5}{21}, -\frac{8}{21}, \frac{3}{7}, \frac{10}{21}, \frac{3}{7}, \frac{1}{7}, -\frac{2}{21}, -\frac{2}{7}, -\frac{1}{21}, -\frac{4}{21}, \frac{1}{7})$ |
| 165 | | | 18 | $(g_5, h_{34}^{(5)})$ | 1/10368 | 2 | $(\frac{1}{6}, -\frac{1}{6}, -\frac{1}{6}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6}, -\frac{1}{6}, \frac{1}{6}, -\frac{1}{6}, \frac{1}{6}, \frac{1}{6})$ | $(\frac{1}{3}, -\frac{1}{3}, 0, 0, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, -\frac{1}{6}, \frac{1}{6}, -\frac{1}{6}, \frac{5}{6})$ |
| 166 | | | 36 | $(g_5, h_{40}^{(5)})$ | 1/864 | 2 | $(\frac{1}{6}, \frac{1}{6}, \frac{1}{6}, -\frac{1}{6}, -\frac{1}{6}, -\frac{1}{6}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6}, -\frac{1}{6}, \frac{1}{6})$ | $(\frac{1}{12}, \frac{1}{12}, \frac{1}{4}, -\frac{1}{4}, \frac{1}{6}, \frac{1}{6}, -\frac{1}{4}, -\frac{1}{4}, \frac{1}{2}, \frac{1}{2}, \frac{5}{12}, -\frac{5}{12})$ |
| 167 | | | 72 | $(g_5, h_{43}^{(5)})$ | 1/72 | 0 | $(-\frac{1}{6}, \frac{1}{6}, \frac{1}{6}, -\frac{1}{6}, -\frac{1}{6}, -\frac{1}{6}, \frac{1}{6}, \frac{1}{6}, -\frac{1}{6}, -\frac{1}{6}, -\frac{1}{6}, -\frac{1}{6})$ | $(-\frac{1}{4}, \frac{7}{24}, \frac{3}{8}, -\frac{5}{12}, \frac{11}{24}, \frac{3}{8}, \frac{1}{8}, \frac{1}{24}, \frac{1}{12}, \frac{1}{8}, \frac{1}{4}, \frac{5}{24})$ |
| 168 | | | 12 | $(g_5, h_{46}^{(5)})$ | 1/104976 | 3 | $(-\frac{1}{6}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6}, -\frac{1}{6}, -\frac{1}{6}, \frac{1}{6}, -\frac{1}{6}, -\frac{1}{6}, \frac{1}{6}, -\frac{1}{6}, \frac{1}{6})$ | $(0, 0, 0, -\frac{1}{3}, \frac{1}{3}, \frac{1}{3}, \frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, \frac{1}{3}, -\frac{1}{3}, \frac{4}{3})$ |
| 169 | 170 | | 36 | $(g_5, h_{49}^{(5)})$ | 1/3888 | $\sqrt{7}$ | $(\frac{1}{6}, -\frac{1}{6}, \frac{1}{6}, \frac{1}{6}, -\frac{1}{6}, \frac{1}{6}, -\frac{1}{6}, -\frac{1}{6}, -\frac{1}{6}, \frac{1}{6}, \frac{1}{6}, -\frac{1}{6})$ | $(-\frac{1}{3}, -\frac{1}{6}, \frac{1}{6}, \frac{1}{3}, -\frac{1}{3}, \frac{1}{3}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6}, \frac{1}{2}, \frac{1}{2}, \frac{3}{2})$ |
| 170 | 169 | | 36 | $(g_5, h_{52}^{(5)})$ | 1/3888 | $\sqrt{7}$ | $(-\frac{1}{6}, -\frac{1}{6}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6}, -\frac{1}{6}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6})$ | $(\frac{1}{6}, -\frac{1}{3}, \frac{1}{6}, \frac{1}{6}, -\frac{1}{3}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6}, -\frac{1}{3}, -\frac{1}{3}, 0, 0, 0)$ |
| 171 | 172 | | 36 | $(g_5, h_{55}^{(5)})$ | 1/162 | $\sqrt{3}$ | $(\frac{1}{6}, \frac{1}{6}, -\frac{1}{6}, \frac{1}{6}, -\frac{1}{6}, \frac{1}{6}, \frac{1}{6}, -\frac{1}{6}, \frac{1}{6}, \frac{1}{6}, -\frac{1}{6}, \frac{1}{6})$ | $(\frac{7}{18}, \frac{1}{18}, \frac{5}{18}, -\frac{1}{6}, -\frac{5}{18}, \frac{5}{18}, -\frac{7}{18}, \frac{7}{18}, \frac{1}{6}, \frac{1}{6}, \frac{1}{18}, -\frac{1}{18})$ |
| 172 | 171 | | 36 | $(g_5, h_{58}^{(5)})$ | 1/162 | $\sqrt{3}$ | $(-\frac{1}{6}, -\frac{1}{6}, \frac{1}{6}, \frac{1}{6}, -\frac{1}{6}, \frac{1}{6}, -\frac{1}{6}, \frac{1}{6}, -\frac{1}{6}, -\frac{1}{6}, \frac{1}{6}, \frac{1}{6})$ | $(-\frac{4}{9}, -\frac{1}{9}, 0, -\frac{2}{9}, -\frac{2}{9}, \frac{2}{9}, \frac{1}{9}, -\frac{1}{9}, \frac{1}{3}, \frac{1}{3}, -\frac{4}{9}, \frac{5}{9})$ |
| 173 | 174 | | 108 | $(g_5, h_{61}^{(5)})$ | 1/54 | 1 | $(\frac{1}{6}, \frac{1}{6}, \frac{1}{6}, -\frac{1}{6}, -\frac{1}{6}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6}, -\frac{1}{6}, \frac{1}{6}, -\frac{1}{6})$ | $(\frac{2}{9}, \frac{1}{3}, \frac{7}{18}, -\frac{4}{9}, \frac{4}{9}, \frac{1}{9}, \frac{1}{18}, -\frac{1}{9}, -\frac{1}{6}, \frac{2}{9}, -\frac{1}{3}, \frac{5}{18})$ |
| 174 | 173 | | 108 | $(g_5, h_{64}^{(5)})$ | 1/54 | 1 | $(\frac{1}{6}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6}, -\frac{1}{6}, -\frac{1}{6}, -\frac{1}{6}, -\frac{1}{6}, -\frac{1}{6}, \frac{1}{6}, -\frac{1}{6}, \frac{1}{6})$ | $(\frac{1}{9}, \frac{1}{6}, -\frac{1}{18}, -\frac{1}{9}, -\frac{2}{9}, -\frac{5}{18}, -\frac{1}{3}, \frac{2}{9}, -\frac{4}{9}, -\frac{4}{9}, \frac{7}{18}, -\frac{1}{3})$ |
| 175 | | | 36 | $(g_5, h_{82}^{(5)})$ | 1/1296 | 1 | $(-\frac{1}{6}, \frac{1}{6}, \frac{1}{6}, -\frac{1}{6}, \frac{1}{6}, -\frac{1}{6}, -\frac{1}{6}, -\frac{1}{6}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6}, -\frac{1}{6})$ | $(\frac{1}{2}, -\frac{1}{6}, \frac{1}{3}, -\frac{1}{3}, \frac{1}{6}, -\frac{1}{6}, 0, 0, -\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, \frac{4}{3})$ |
| 176 | | | 72 | $(g_5, h_{85}^{(5)})$ | 1/216 | 1 | $(-\frac{1}{6}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6}, -\frac{1}{6}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6}, -\frac{1}{6}, -\frac{1}{6}, \frac{1}{6})$ | $(-\frac{1}{4}, \frac{1}{2}, -\frac{5}{12}, -\frac{1}{4}, \frac{1}{12}, -\frac{1}{6}, -\frac{5}{12}, \frac{5}{12}, \frac{1}{6}, \frac{1}{6}, \frac{1}{12}, \frac{13}{12})$ |
| 177 | | | 72 | $(g_5, h_{91}^{(5)})$ | 1/72 | $\sqrt{3}$ | $(\frac{1}{6}, \frac{1}{6}, \frac{1}{6}, -\frac{1}{6}, -\frac{1}{6}, -\frac{1}{6}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6}, -\frac{1}{6}, -\frac{1}{6}, -\frac{1}{6})$ | $(-\frac{1}{12}, \frac{1}{4}, -\frac{1}{6}, -\frac{1}{3}, \frac{1}{4}, -\frac{5}{12}, -\frac{1}{3}, \frac{1}{2}, \frac{1}{12}, -\frac{1}{12}, \frac{5}{12}, \frac{5}{12})$ |
| 178 | | | 12 | $(g_5, h_{97}^{(5)})$ | 1/72 | 0 | $(-\frac{1}{6}, -\frac{1}{6}, -\frac{1}{6}, -\frac{1}{6}, -\frac{1}{6}, \frac{1}{6}, -\frac{1}{6}, -\frac{1}{6}, -\frac{1}{6}, -\frac{1}{6}, \frac{1}{6}, -\frac{1}{6})$ | $(-\frac{1}{4}, -\frac{1}{4}, -\frac{1}{12}, -\frac{1}{12}, \frac{1}{12}, -\frac{1}{12}, -\frac{5}{12}, -\frac{5}{12}, \frac{5}{12}, \frac{5}{12}, -\frac{1}{4}, \frac{1}{4})$ |
| 179 | | | 24 | $(g_5, h_{98}^{(5)})$ | 1/36 | 0 | $(\frac{1}{6}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6}, -\frac{1}{6}, -\frac{1}{6}, -\frac{1}{6}, -\frac{1}{6}, -\frac{1}{6}, -\frac{1}{6}, -\frac{1}{6}, -\frac{1}{6})$ | $(\frac{7}{24}, \frac{5}{24}, \frac{3}{8}, \frac{11}{24}, \frac{11}{24}, \frac{3}{8}, -\frac{1}{8}, \frac{7}{24}, -\frac{1}{24}, \frac{5}{24}, \frac{1}{24}, \frac{1}{8})$ |
| 180 | | | 36 | $(g_5, h_{117}^{(5)})$ | 1/144 | 0 | $(\frac{1}{6}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6}, -\frac{1}{6}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6}, -\frac{1}{6}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6})$ | $(\frac{1}{4}, \frac{5}{12}, -\frac{1}{4}, -\frac{1}{12}, -\frac{1}{6}, \frac{1}{6}, 0, 0, \frac{1}{2}, \frac{1}{2}, -\frac{1}{3}, -\frac{1}{3})$ |
| 181 | | | 1 | $(g_8, h_4^{(8)})$ | 1/330301440 | 8 | $(0, 0, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, 0, 0, \frac{1}{2}, 0, 0, 0, 1)$ | $(0, 0, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, 0, 0, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2})$ |
| 182 | | | 6 | $(g_8, h_5^{(8)})$ | 1/49152 | 0 | $(\frac{1}{2}, 0, 0, 0, 0, 0, 0, 0, 0, \frac{1}{2}, \frac{1}{2}, \frac{3}{2})$ | $(0, 0, 0, -\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2})$ |
| 183 | | | 3 | $(g_8, h_6^{(8)})$ | 1/1474560 | 0 | $(\frac{1}{2}, 0, 0, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, 0, 0, 0, 0, 0, 0)$ | $(0, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, 0, 0, 0, 0, 1)$ |
| 184 | | | 18 | $(g_8, h_{10}^{(8)})$ | 1/40 | 0 | $(0, \frac{1}{2}, 0, \frac{1}{2}, 0, 0, 0, \frac{1}{2}, 0, 0, 0, \frac{3}{2})$ | $(\frac{1}{10}, \frac{3}{10}, -\frac{3}{10}, \frac{1}{10}, 0, -\frac{1}{5}, -\frac{1}{5}, \frac{1}{2}, \frac{2}{5}, -\frac{2}{5}, \frac{1}{2}, 0)$ |
| 185 | | | 12 | $(g_8, h_{14}^{(8)})$ | 1/256 | 0 | $(\frac{1}{2}, 0, 0, 0, 0, 0, 0, 0, \frac{1}{2}, 0, \frac{1}{2}, \frac{1}{2})$ | $(-\frac{1}{4}, 0, \frac{1}{8}, -\frac{1}{8}, -\frac{1}{4}, \frac{1}{4}, -\frac{3}{8}, -\frac{3}{8}, 0, \frac{1}{2}, \frac{1}{2}, \frac{3}{2})$ |
| 186 | | | 12 | $(g_8, h_{15}^{(8)})$ | 1/1536 | 0 | $(\frac{1}{2}, \frac{1}{2}, 0, 0, 0, \frac{1}{2}, \frac{1}{2}, 0, 0, 0, 0, 1)$ | $(-\frac{1}{4}, 0, -\frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, 0, 0, 1)$ |
| 187 | | | 24 | $(g_8, h_{17}^{(8)})$ | 1/576 | 0 | $(\frac{1}{2}, \frac{1}{2}, 0, 0, 0, 0, 0, 0, 0, 0, \frac{1}{2}, \frac{1}{2})$ | $(-\frac{1}{3}, \frac{1}{2}, \frac{1}{2}, \frac{1}{6}, -\frac{1}{6}, \frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, 0, 0, 0, 0)$ |
| 188 | | | 48 | $(g_8, h_{19}^{(8)})$ | 1/48 | 0 | $(\frac{1}{2}, \frac{1}{2}, 0, 0, 0, 0, \frac{1}{2}, 0, 0, 0, \frac{1}{2}, 1)$ | $(\frac{1}{3}, \frac{1}{4}, \frac{1}{4}, -\frac{1}{3}, -\frac{5}{12}, \frac{1}{12}, \frac{1}{2}, 0, \frac{1}{6}, \frac{1}{6}, 0, \frac{3}{2})$ |
| 189 | | | 3 | $(g_8, h_{21}^{(8)})$ | 1/147456 | 0 | $(0, 0, 0, 0, 0, 0, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, 0, 0)$ | $(0, 0, -\frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, 0, 0, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, -\frac{1}{2})$ |
| 190 | | | 3 | $(g_8, h_{23}^{(8)})$ | 1/786432 | 0 | $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, 0, 0, 0, 0, 0, 0, 0, 0)$ | $(0, \frac{1}{2}, \frac{1}{2}, 0, 0, 0, 0, 0, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{3}{2})$ |
| 191 | | | 1 | $(g_8, h_{24}^{(8)})$ | 1/7077888 | 0 | $(0, \frac{1}{2}, \frac{1}{2}, 0, \frac{1}{2}, \frac{1}{2}, 0, 0, 0, 0, 0, 1)$ | $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, 0, 0, 0, 0, 0, 0, 0, 1)$ |
| 192 | | | 12 | $(g_8, h_{25}^{(8)})$ | 1/3456 | 0 | $(\frac{1}{2}, 0, 0, \frac{1}{2}, 0, 0, 0, 0, \frac{1}{2}, \frac{1}{2}, 0, 0)$ | $(-\frac{1}{3}, -\frac{1}{3}, 0, 0, -\frac{1}{6}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{3}{2})$ |
| 193 | | | 4 | $(g_8, h_{28}^{(8)})$ | 1/184320 | 4 | $(0, 0, 0, 0, 0, 0, 0, 0, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2})$ | $(\frac{1}{6}, -\frac{1}{6}, \frac{1}{6}, \frac{1}{6}, -\frac{1}{3}, \frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, \frac{1}{6}, -\frac{1}{6}, -\frac{1}{6}, \frac{1}{6})$ |
| 194 | | | 3 | $(g_8, h_{32}^{(8)})$ | 1/24576 | 4 | $(0, 0, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, 0, 0, 0, 0, 0, 0)$ | $(\frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, \frac{1}{2}, \frac{1}{2}, 0, 0, 0, \frac{1}{2}, \frac{1}{2}, \frac{3}{2})$ |
| 195 | | | 6 | $(g_8, h_{33}^{(8)})$ | 1/147456 | 4 | $(\frac{1}{2}, \frac{1}{2}, 0, 0, \frac{1}{2}, \frac{1}{2}, 0, 0, 0, 0, 0, 0)$ | $(-\frac{1}{4}, \frac{1}{4}, 0, 0, 0, 0, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, \frac{1}{4})$ |
| 196 | | | 12 | $(g_8, h_{35}^{(8)})$ | 1/2304 | 0 | $(0, 0, 0, 0, \frac{1}{2}, \frac{1}{2}, 0, 0, 0, 0, \frac{1}{2}, \frac{1}{2})$ | $(-\frac{1}{12}, -\frac{1}{12}, -\frac{1}{6}, \frac{1}{6}, -\frac{1}{3}, \frac{1}{3}, -\frac{5}{12}, \frac{5}{12}, -\frac{1}{3}, -\frac{1}{3}, \frac{1}{6}, -\frac{1}{6})$ |
| 197 | | | 4 | $(g_8, h_{37}^{(8)})$ | 1/27648 | 0 | $(\frac{1}{2}, \frac{1}{2}, 0, 0, \frac{1}{2}, \frac{1}{2}, 0, 0, 0, 0, 0, 0)$ | $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}, -\frac{1}{3}, \frac{1}{6}, \frac{1}{6}, -\frac{1}{6}, -\frac{1}{6}, \frac{1}{6}, -\frac{1}{6}, \frac{1}{6}, \frac{1}{6})$ |
| 198 | | | 24 | $(g_8, h_{38}^{(8)})$ | 1/1152 | 2 | $(\frac{1}{2}, \frac{1}{2}, 0, 0, \frac{1}{2}, \frac{1}{2}, 0, 0, 0, 0, 0, 0)$ | $(\frac{5}{12}, -\frac{1}{12}, -\frac{1}{6}, \frac{1}{6}, -\frac{1}{6}, -\frac{1}{6}, -\frac{1}{12}, -\frac{1}{12}, \frac{1}{12}, \frac{5}{12}, -\frac{5}{12}, -\frac{5}{12})$ |
| 199 | | | 3 | $(g_8, h_{44}^{(8)})$ | 1/196608 | 0 | $(0, 0, 0, 0, 0, 0, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, 0, 1)$ | $(0, 0, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, \frac{1}{2}, \frac{1}{2})$ |
| 200 | | | 3 | $(g_8, h_{45}^{(8)})$ | 1/2949120 | 0 | $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, 0, 0, 0, 0, 0, 0, 0, 0)$ | $(-\frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, 0, 0, 0, 0)$ |

| No. | P | com. | size | rep. | $[h]_{\subset C_g}/\lvert C_g\rvert$ | $\lvert Z_{\mathrm{R}g}^{\mathrm{R}h}\rvert$ | $\vec{\lambda}_{\tilde{g},O}$ | $\vec{\lambda}_{\tilde{h},O}$ |
|---|---|---|---|---|---|---|---|---|
| 201 | | | 12 | $(g_8, h_{46}^{(8)})$ | 1/4608 | 0 | $(0,0,\frac{1}{2},\frac{1}{2},\frac{1}{2},\frac{1}{2},0,0,0,0,0,1)$ | $(\frac{1}{12},-\frac{1}{12},-\frac{5}{12},-\frac{5}{12},-\frac{1}{12},\frac{1}{12},-\frac{5}{12},-\frac{5}{12},\frac{1}{3},\frac{1}{3},\frac{1}{3},\frac{1}{3})$ |
| 202 | | | 12 | $(g_8, h_{48}^{(8)})$ | 1/1536 | 0 | $(\frac{1}{2},\frac{1}{2},0,0,0,0,0,0,0,0,\frac{1}{2},\frac{1}{2})$ | $(-\frac{5}{12},-\frac{5}{12},-\frac{1}{12},\frac{1}{12},\frac{1}{6},-\frac{1}{6},-\frac{5}{12},-\frac{5}{12},\frac{1}{3},\frac{1}{3},-\frac{1}{12},\frac{11}{12})$ |
| 203 | | | 6 | $(g_8, h_{52}^{(8)})$ | 1/3072 | 0 | $(\frac{1}{2},\frac{1}{2},0,0,0,0,\frac{1}{2},\frac{1}{2},0,0,0,1)$ | $(-\frac{3}{8},-\frac{3}{8},\frac{1}{8},-\frac{1}{8},-\frac{3}{8},-\frac{3}{8},-\frac{1}{8},\frac{1}{8},-\frac{1}{4},\frac{1}{4},-\frac{1}{4},\frac{1}{4})$ |
| 204 | | | 6 | $(g_8, h_{53}^{(8)})$ | 1/1024 | 0 | $(\frac{1}{2},\frac{1}{2},0,0,0,0,\frac{1}{2},\frac{1}{2},0,0,0,0)$ | $(-\frac{3}{8},\frac{3}{8},-\frac{1}{8},\frac{1}{8},-\frac{3}{8},-\frac{3}{8},\frac{1}{8},-\frac{1}{8},0,0,\frac{1}{2},\frac{3}{2})$ |
| 205 | | | 6 | $(g_8, h_{54}^{(8)})$ | 1/1024 | 0 | $(\frac{1}{2},\frac{1}{2},0,0,0,0,0,0,0,0,\frac{1}{2},\frac{3}{2})$ | $(-\frac{3}{8},\frac{3}{8},0,\frac{1}{8},-\frac{1}{8},\frac{1}{2},-\frac{3}{8},\frac{3}{8},\frac{1}{4},-\frac{1}{4},-\frac{1}{8},-\frac{1}{8})$ |
| 206 | | | 6 | $(g_8, h_{56}^{(8)})$ | 1/3072 | 0 | $(0,0,\frac{1}{2},\frac{1}{2},0,0,0,0,\frac{1}{2},\frac{1}{2},0,1)$ | $(\frac{1}{8},-\frac{1}{8},\frac{3}{8},-\frac{3}{8},-\frac{1}{4},\frac{1}{4},\frac{3}{8},\frac{3}{8},\frac{1}{8},\frac{1}{8},\frac{1}{2},\frac{3}{2})$ |
| 207 | | | 24 | $(g_8, h_{58}^{(8)})$ | 1/96 | 0 | $(\frac{1}{2},\frac{1}{2},0,0,\frac{1}{2},0,0,0,0,\frac{1}{2},0,0)$ | $(\frac{11}{24},-\frac{7}{24},-\frac{11}{24},\frac{5}{12},\frac{5}{24},\frac{7}{24},\frac{5}{24},-\frac{1}{24},\frac{1}{12},-\frac{1}{24},\frac{1}{6},-\frac{1}{6})$ |
| 208 | | | 24 | $(g_8, h_{59}^{(8)})$ | 1/96 | 0 | $(\frac{1}{2},0,0,\frac{1}{2},0,\frac{1}{2},\frac{1}{2},0,0,0,0,0)$ | $(\frac{11}{24},-\frac{1}{24},\frac{5}{24},\frac{7}{24},-\frac{7}{24},-\frac{5}{24},-\frac{1}{24},-\frac{11}{24},\frac{1}{12},-\frac{1}{12},\frac{5}{12},\frac{5}{12})$ |
| 209 | | | 4 | $(g_8, h_{61}^{(8)})$ | 1/1152 | 2 | $(0,0,\frac{1}{2},\frac{1}{2},0,0,0,0,\frac{1}{2},\frac{1}{2},0,1)$ | $(\frac{1}{6},\frac{1}{6},-\frac{1}{6},\frac{1}{6},-\frac{1}{3},\frac{1}{3},0,0,\frac{1}{2},\frac{1}{2},\frac{1}{2},\frac{1}{2})$ |
| 210 | | | 12 | $(g_8, h_{65}^{(8)})$ | 1/288 | 0 | $(\frac{1}{2},0,0,\frac{1}{2},0,0,\frac{1}{2},\frac{1}{2},0,0,0,0)$ | $(\frac{5}{12},\frac{1}{12},\frac{5}{12},-\frac{1}{12},\frac{1}{4},-\frac{1}{4},\frac{1}{4},-\frac{1}{4},\frac{1}{3},\frac{1}{3},0,0)$ |
| 211 | | | 4 | $(g_8, h_{68}^{(8)})$ | 1/864 | 0 | $(\frac{1}{2},0,\frac{1}{2},\frac{1}{2},0,\frac{1}{2},0,0,0,0,0,0)$ | $(\frac{1}{3},-\frac{1}{3},\frac{1}{6},\frac{1}{2},0,0,-\frac{1}{6},\frac{1}{6},-\frac{1}{6},\frac{1}{2},\frac{1}{2},\frac{3}{2})$ |
| 212 | | | 24 | $(g_8, h_{69}^{(8)})$ | 1/288 | 0 | $(0,0,\frac{1}{2},0,\frac{1}{2},0,\frac{1}{2},0,0,0,0,\frac{3}{2})$ | $(-\frac{1}{12},-\frac{1}{6},\frac{1}{3},-\frac{1}{3},\frac{1}{6},\frac{5}{12},\frac{1}{2},0,\frac{1}{4},\frac{1}{4},\frac{1}{2},1)$ |
| 213 | | | 12 | $(g_8, h_{76}^{(8)})$ | 1/1728 | 3 | $(\frac{1}{2},0,0,\frac{1}{2},\frac{1}{2},\frac{1}{2},0,0,0,0,0,1)$ | $(-\frac{1}{6},-\frac{1}{3},-\frac{1}{3},\frac{1}{3},-\frac{1}{3},\frac{1}{2},\frac{1}{6},-\frac{1}{6},\frac{1}{6},-\frac{1}{6},\frac{1}{2},\frac{1}{2})$ |
| 214 | | | 4 | $(g_8, h_{79}^{(8)})$ | 1/864 | 0 | $(\frac{1}{2},0,\frac{1}{2},\frac{1}{2},0,\frac{1}{2},0,0,0,0,0,0)$ | $(-\frac{1}{3},-\frac{1}{3},-\frac{1}{6},\frac{1}{2},0,0,\frac{1}{6},-\frac{1}{6},-\frac{1}{6},\frac{1}{2},\frac{1}{2},\frac{1}{2})$ |
| 215 | | | 3 | $(g_8, h_{80}^{(8)})$ | 1/49152 | 4 | $(0,0,0,0,0,0,\frac{1}{2},\frac{1}{2},\frac{1}{2},\frac{1}{2},0,0)$ | $(0,0,\frac{1}{4},-\frac{1}{4},\frac{1}{4},\frac{1}{4},-\frac{1}{4},\frac{1}{4},-\frac{1}{4},-\frac{1}{4},\frac{1}{2},\frac{1}{2})$ |
| 216 | | | 12 | $(g_8, h_{82}^{(8)})$ | 1/96 | 2 | $(\frac{1}{2},0,0,\frac{1}{2},0,0,0,0,0,0,\frac{1}{2},\frac{1}{2},0)$ | $(-\frac{5}{12},\frac{1}{12},-\frac{1}{6},-\frac{1}{12},-\frac{5}{12},-\frac{1}{3},0,-\frac{1}{4},-\frac{1}{4},-\frac{1}{4},\frac{1}{4},\frac{1}{2})$ |
| 217 | | | 3 | $(g_8, h_{85}^{(8)})$ | 1/8192 | 0 | $(\frac{1}{2},\frac{1}{2},0,0,0,0,0,0,0,0,\frac{1}{2},\frac{1}{2},0,1)$ | $(\frac{1}{2},\frac{1}{2},-\frac{1}{4},\frac{1}{4},-\frac{1}{4},-\frac{1}{4},0,0,0,0,\frac{1}{2},-\frac{1}{2})$ |
| 218 | | | 3 | $(g_8, h_{87}^{(8)})$ | 1/49152 | 0 | $(\frac{1}{2},\frac{1}{2},\frac{1}{2},\frac{1}{2},0,0,0,0,0,0,0,0)$ | $(0,\frac{1}{2},\frac{1}{2},0,0,0,0,0,0,\frac{1}{2},\frac{1}{2},\frac{1}{2},-\frac{1}{2})$ |
| 219 | | | 12 | $(g_8, h_{88}^{(8)})$ | 1/96 | 0 | $(\frac{1}{2},\frac{1}{2},\frac{1}{2},0,0,0,0,0,\frac{1}{2},0,0,0,1)$ | $(-\frac{1}{3},-\frac{1}{6},\frac{1}{2},-\frac{1}{6},-\frac{1}{6},\frac{1}{3},-\frac{1}{3},0,0,0,\frac{1}{2},-\frac{1}{2})$ |
| 220 | | | 24 | $(g_8, h_{89}^{(8)})$ | 1/96 | 0 | $(\frac{1}{2},0,0,\frac{1}{2},\frac{1}{2},0,0,0,0,0,0,-\frac{1}{2})$ | $(-\frac{1}{3},-\frac{5}{12},\frac{1}{12},-\frac{1}{6},\frac{1}{2},0,-\frac{1}{4},\frac{1}{4},-\frac{1}{6},\frac{1}{6},\frac{1}{2},1)$ |
| 221 | | | 12 | $(g_8, h_{91}^{(8)})$ | 1/384 | 0 | $(\frac{1}{2},\frac{1}{2},\frac{1}{2},0,0,0,0,0,\frac{1}{2},0,0,0,1)$ | $(\frac{1}{3},-\frac{1}{6},\frac{1}{2},\frac{1}{6},-\frac{1}{6},\frac{1}{3},-\frac{1}{3},0,0,0,\frac{1}{2},-\frac{1}{2})$ |
| 222 | | | 48 | $(g_8, h_{97}^{(8)})$ | 1/288 | 1 | $(\frac{1}{2},0,\frac{1}{2},0,0,0,0,0,0,0,\frac{1}{2},\frac{1}{2})$ | $(-\frac{5}{12},-\frac{1}{3},-\frac{1}{12},0,-\frac{1}{12},-\frac{1}{12},\frac{5}{12},\frac{5}{12},-\frac{1}{4},-\frac{1}{4},-\frac{1}{6},-\frac{1}{6})$ |
| 223 | | | 24 | $(g_8, h_{100}^{(8)})$ | 1/288 | 2 | $(\frac{1}{2},0,0,0,\frac{1}{2},\frac{1}{2},0,0,0,0,0,-\frac{1}{2})$ | $(\frac{1}{3},-\frac{1}{3},-\frac{5}{12},-\frac{5}{12},-\frac{1}{4},-\frac{1}{4},-\frac{1}{4},-\frac{1}{4},-\frac{1}{12},\frac{1}{12},0,0)$ |
| 224 | | | 12 | $(g_8, h_{102}^{(8)})$ | 1/48 | 1 | $(\frac{1}{2},\frac{1}{2},0,0,\frac{1}{2},0,0,0,0,0,0,\frac{1}{2})$ | $(-\frac{5}{12},-\frac{1}{3},\frac{5}{12},-\frac{1}{12},-\frac{1}{12},0,\frac{1}{6},\frac{1}{6},\frac{1}{3},-\frac{1}{3},\frac{1}{2},1)$ |
| 225 | | | 6 | $(g_8, h_{110}^{(8)})$ | 1/960 | 2 | $(\frac{1}{2},\frac{1}{2},0,0,0,0,0,0,0,0,\frac{1}{2},\frac{1}{2})$ | $(\frac{2}{5},\frac{2}{5},-\frac{1}{10},-\frac{1}{10},-\frac{1}{5},-\frac{1}{5},\frac{2}{5},\frac{2}{5},\frac{3}{10},\frac{3}{10},-\frac{1}{5},-\frac{1}{5})$ |
| 226 | | | 6 | $(g_8, h_{116}^{(8)})$ | 1/128 | 0 | $(\frac{1}{2},0,0,0,0,0,\frac{1}{2},\frac{1}{2},0,0,\frac{1}{2},0)$ | $(\frac{1}{2},\frac{1}{8},\frac{1}{8},0,\frac{1}{4},\frac{1}{4},-\frac{1}{4},\frac{1}{4},-\frac{3}{8},-\frac{3}{8},0,\frac{1}{2})$ |
| 227 | | | 6 | $(g_8, h_{117}^{(8)})$ | 1/256 | 0 | $(\frac{1}{2},\frac{1}{2},\frac{1}{2},\frac{1}{2},0,0,0,0,0,0,0,0)$ | $(\frac{1}{2},-\frac{1}{4},-\frac{1}{4},0,0,0,-\frac{1}{4},\frac{1}{4},-\frac{1}{4},\frac{1}{4},\frac{1}{2},\frac{1}{2})$ |
| 228 | | | 18 | $(g_8, h_{118}^{(8)})$ | 1/80 | 0 | $(\frac{1}{2},\frac{1}{2},0,0,0,0,\frac{1}{2},\frac{1}{2},0,0,0,1)$ | $(\frac{9}{20},\frac{7}{20},-\frac{1}{20},-\frac{3}{20},\frac{9}{20},\frac{7}{20},-\frac{1}{20},\frac{3}{20},\frac{1}{10},-\frac{1}{10},-\frac{3}{10},-\frac{3}{10})$ |
| 229 | | | 6 | $(g_8, h_{123}^{(8)})$ | 1/256 | 2 | $(0,0,0,\frac{1}{2},\frac{1}{2},0,0,0,0,0,\frac{1}{2},\frac{1}{2})$ | $(0,\frac{1}{8},\frac{1}{8},-\frac{3}{8},-\frac{3}{8},\frac{1}{2},-\frac{3}{8},\frac{3}{8},\frac{1}{4},-\frac{1}{4},-\frac{1}{8},\frac{1}{8})$ |
| 230 | | | 6 | $(g_8, h_{127}^{(8)})$ | 1/1024 | 0 | $(\frac{1}{2},\frac{1}{2},\frac{1}{2},\frac{1}{2},0,0,0,0,0,0,0,1)$ | $(\frac{1}{2},-\frac{1}{4},\frac{1}{4},0,0,0,-\frac{1}{4},-\frac{1}{4},\frac{1}{4},\frac{1}{4},\frac{1}{2},\frac{1}{2})$ |
| 231 | | | 3 | $(g_8, h_{128}^{(8)})$ | 1/4096 | 0 | $(\frac{1}{2},\frac{1}{2},\frac{1}{2},0,0,\frac{1}{2},0,0,0,0,0,0)$ | $(\frac{1}{2},\frac{1}{4},-\frac{1}{4},-\frac{1}{4},-\frac{1}{4},0,\frac{1}{2},\frac{1}{2},0,0,0,\frac{3}{2})$ |
| 232 | | | 3 | $(g_8, h_{129}^{(8)})$ | 1/8192 | 0 | $(\frac{1}{2},\frac{1}{2},\frac{1}{2},0,0,\frac{1}{2},0,0,0,0,0,0)$ | $(\frac{1}{2},\frac{1}{4},\frac{1}{4},-\frac{1}{4},-\frac{1}{4},0,0,0,\frac{1}{2},\frac{1}{2},\frac{1}{2},\frac{1}{2})$ |
| 233 | | | 3 | $(g_8, h_{130}^{(8)})$ | 1/8192 | 0 | $(\frac{1}{2},\frac{1}{2},\frac{1}{2},0,0,\frac{1}{2},0,0,0,0,0,0)$ | $(\frac{1}{2},\frac{1}{4},-\frac{1}{4},\frac{1}{2},0,0,-\frac{1}{4},\frac{1}{4},-\frac{1}{4},\frac{1}{4},-\frac{1}{4},-\frac{1}{4})$ |
| 234 | | | 6 | $(g_8, h_{131}^{(8)})$ | 1/128 | 2 | $(0,\frac{1}{2},\frac{1}{2},\frac{1}{2},0,0,0,\frac{1}{2},0,0,0,0)$ | $(-\frac{1}{8},\frac{3}{8},\frac{1}{8},-\frac{1}{4},\frac{3}{8},-\frac{1}{4},-\frac{1}{4},\frac{1}{2},0,0,\frac{1}{2},\frac{3}{2})$ |
| 235 | | | 12 | $(g_8, h_{133}^{(8)})$ | 1/256 | 2 | $(\frac{1}{2},\frac{1}{2},0,\frac{1}{2},0,\frac{1}{2},0,0,0,0,0,0)$ | $(-\frac{3}{8},\frac{1}{4},\frac{1}{4},\frac{1}{8},0,0,-\frac{1}{8},-\frac{1}{8},\frac{1}{8},-\frac{3}{8},\frac{3}{8},-\frac{3}{8})$ |
| 236 | | | 16 | $(g_8, h_{141}^{(8)})$ | 1/28 | 1 | $(0,0,0,\frac{1}{2},\frac{1}{2},0,\frac{1}{2},0,0,0,\frac{1}{2},1)$ | $(-\frac{1}{14},\frac{1}{7},-\frac{3}{14},\frac{5}{14},\frac{1}{14},-\frac{2}{7},\frac{3}{14},\frac{5}{14},\frac{3}{7},0,\frac{1}{2},\frac{3}{2})$ |
| 237 | | | 6 | $(g_8, h_{148}^{(8)})$ | 1/128 | 0 | $(0,\frac{1}{2},\frac{1}{2},0,0,\frac{1}{2},\frac{1}{2},0,0,0,0,1)$ | $(-\frac{1}{8},-\frac{3}{8},-\frac{1}{8},-\frac{3}{8},0,\frac{1}{2},0,\frac{1}{4},-\frac{1}{4},-\frac{1}{4},\frac{1}{2},\frac{3}{2})$ |
| 238 | | | 6 | $(g_8, h_{149}^{(8)})$ | 1/128 | 0 | $(\frac{1}{2},\frac{1}{2},\frac{1}{2},0,\frac{1}{2},0,0,0,0,0,0,0)$ | $(-\frac{3}{8},-\frac{1}{8},\frac{1}{2},0,0,\frac{1}{2},\frac{1}{8},\frac{1}{8},\frac{1}{8},\frac{3}{8},\frac{3}{8},\frac{11}{8})$ |
| 239 | | | 48 | $(g_8, h_{153}^{(8)})$ | 1/60 | 1 | $(\frac{1}{2},0,0,0,\frac{1}{2},0,0,0,0,0,\frac{1}{2},\frac{1}{2})$ | $(\frac{7}{15},-\frac{1}{30},\frac{1}{15},-\frac{2}{15},\frac{4}{15},-\frac{7}{15},\frac{13}{30},-\frac{11}{30},-\frac{4}{15},-\frac{7}{30},-\frac{2}{15},\frac{16}{15})$ |
| 240 | | | 4 | $(g_9, h_{16}^{(9)})$ | 1/52488 | 3 | $(-\frac{1}{3},\frac{1}{3},\frac{1}{3},\frac{1}{3},-\frac{1}{3},-\frac{1}{3},0,0,0,0,0,0)$ | $(-\frac{1}{3},\frac{1}{3},\frac{1}{3},-\frac{1}{3},\frac{1}{3},\frac{1}{3},-\frac{1}{3},\frac{1}{3},-\frac{1}{3},0,0,1)$ |
| 241 | | | 12 | $(g_9, h_{22}^{(9)})$ | 1/648 | 1 | $(0,\frac{1}{3},\frac{1}{3},-\frac{1}{3},-\frac{1}{3},0,0,-\frac{1}{3},-\frac{1}{3},0,0,0)$ | $(-\frac{1}{6},\frac{1}{6},-\frac{1}{6},-\frac{1}{3},-\frac{1}{3},\frac{1}{3},-\frac{1}{3},\frac{1}{3},\frac{1}{3},\frac{1}{2},0,0)$ |
| 242 | | | 6 | $(g_9, h_{76}^{(9)})$ | 1/5832 | 3 | $(\frac{1}{3},\frac{1}{3},0,0,0,0,\frac{1}{3},-\frac{1}{3},-\frac{1}{3},\frac{1}{3},0,0)$ | $(\frac{1}{6},-\frac{1}{6},\frac{1}{6},-\frac{1}{6},\frac{1}{6},\frac{1}{6},\frac{1}{2},\frac{1}{2},\frac{1}{2},\frac{1}{2},\frac{1}{2},\frac{1}{2})$ |
| 243 | | | 36 | $(g_9, h_{79}^{(9)})$ | 1/324 | 1 | $(0,\frac{1}{3},-\frac{1}{3},0,\frac{1}{3},-\frac{1}{3},\frac{1}{3},\frac{1}{3},0,0,0,0)$ | $(\frac{1}{6},\frac{1}{3},\frac{1}{3},0,0,0,\frac{1}{2},\frac{1}{2},\frac{1}{2},-\frac{1}{3},-\frac{1}{3},\frac{4}{3})$ |
| 244 | | | 18 | $(g_9, h_{82}^{(9)})$ | 1/648 | 1 | $(\frac{1}{3},-\frac{1}{3},\frac{1}{3},-\frac{1}{3},0,0,0,0,\frac{1}{3},-\frac{1}{3},0,0)$ | $(\frac{1}{3},\frac{1}{3},\frac{1}{2},\frac{1}{2},\frac{1}{3},\frac{1}{3},\frac{1}{6},\frac{1}{6},0,0,0,0)$ |
| 245 | | | 36 | $(g_9, h_{85}^{(9)})$ | 1/36 | 1 | $(-\frac{1}{3},-\frac{1}{3},-\frac{1}{3},0,0,-\frac{1}{3},0,0,-\frac{1}{3},-\frac{1}{3},0,0)$ | $(-\frac{1}{3},\frac{1}{2},\frac{5}{12},\frac{5}{12},\frac{1}{3},-\frac{1}{6},\frac{1}{12},\frac{1}{6},-\frac{1}{12},\frac{1}{6},0,\frac{3}{2})$ |
| 246 | | | 4 | $(g_9, h_{97}^{(9)})$ | 1/162 | 0 | $(\frac{1}{3},-\frac{1}{3},-\frac{1}{3},-\frac{1}{3},\frac{1}{3},-\frac{1}{3},0,0,0,0,0,0)$ | $(\frac{1}{6},-\frac{1}{6},\frac{1}{6},\frac{1}{6},\frac{1}{6},\frac{1}{2},\frac{1}{2},\frac{1}{6},-\frac{1}{6},-\frac{1}{6},\frac{1}{6},\frac{1}{2},-\frac{1}{2})$ |
| 247 | | | 12 | $(g_9, h_{98}^{(9)})$ | 1/18 | 0 | $(-\frac{1}{3},-\frac{1}{3},-\frac{1}{3},\frac{1}{3},-\frac{1}{3},-\frac{1}{3},0,0,0,0,0,0)$ | $(-\frac{1}{3},-\frac{1}{6},\frac{1}{2},0,\frac{1}{3},\frac{1}{6},0,-\frac{1}{6},\frac{1}{6},\frac{1}{3},-\frac{1}{3},\frac{3}{2})$ |

| No. | P | com. | size | rep. | $[h]_{\subset C_g}/\lvert C_g\rvert$ | $\lvert Z_{\mathrm{R}g}^{\mathrm{R}h}\rvert$ | $\vec{\lambda}_{\tilde{g},O}$ | $\vec{\lambda}_{\tilde{h},O}$ |
|---|---|---|---|---|---|---|---|---|
| 248 | | | 4 | $(g_9, h_{99}^{(9)})$ | 1/2916 | 0 | $(-\frac{1}{3}, -\frac{1}{3}, \frac{1}{3}, -\frac{1}{3}, 0, -\frac{1}{3}, -\frac{1}{3}, 0, 0, 0, 0, 0)$ | $(-\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, \frac{1}{3}, 0, 0, 0, \frac{1}{3}, -\frac{1}{3}, \frac{1}{3}, -\frac{1}{3}, \frac{4}{3})$ |
| 249 | | | 1 | $(g_9, h_{100}^{(9)})$ | 1/5832 | 0 | $(-\frac{1}{3}, \frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, 0, 0, \frac{1}{3}, \frac{1}{3}, 0, 0, 0, 0)$ | $(-\frac{1}{3}, \frac{1}{3}, \frac{1}{3}, \frac{1}{3}, -\frac{1}{3}, \frac{1}{3}, 0, 0, 0, 0, 0, 0)$ |
| 250 | | | 3 | $(g_9, h_{101}^{(9)})$ | 1/72 | 0 | $(-\frac{1}{3}, -\frac{1}{3}, \frac{1}{3}, -\frac{1}{3}, 0, 0, -\frac{1}{3}, -\frac{1}{3}, 0, 0, 0, 0)$ | $(\frac{1}{2}, -\frac{1}{3}, -\frac{1}{3}, -\frac{1}{6}, \frac{1}{3}, -\frac{1}{6}, \frac{1}{6}, 0, 0, 0, \frac{1}{2}, \frac{3}{2})$ |
| 251 | | | 18 | $(g_9, h_{102}^{(9)})$ | 1/648 | 2 | $(\frac{1}{3}, 0, 0, \frac{1}{3}, \frac{1}{3}, \frac{1}{3}, -\frac{1}{3}, \frac{1}{3}, 0, 0, 0, 0)$ | $(\frac{1}{3}, \frac{1}{3}, -\frac{1}{6}, -\frac{1}{6}, -\frac{1}{3}, -\frac{1}{3}, 0, 0, 0, 0, \frac{1}{2}, \frac{3}{2})$ |
| 252 | | | 4 | $(g_9, h_{105}^{(9)})$ | 1/81 | 0 | $(-\frac{1}{3}, 0, \frac{1}{3}, 0, \frac{1}{3}, -\frac{1}{3}, \frac{1}{3}, 0, -\frac{1}{3}, 0, 0, 0)$ | $(-\frac{4}{9}, \frac{1}{9}, \frac{2}{9}, \frac{2}{9}, \frac{1}{9}, \frac{1}{9}, -\frac{4}{9}, -\frac{4}{9}, \frac{2}{9}, \frac{1}{3}, \frac{1}{3}, -\frac{1}{3})$ |
| 253 | | | 4 | $(g_9, h_{106}^{(9)})$ | 1/81 | 0 | $(-\frac{1}{3}, \frac{1}{3}, \frac{1}{3}, 0, 0, 0, -\frac{1}{3}, 0, \frac{1}{3}, -\frac{1}{3}, 0, 0)$ | $(-\frac{4}{9}, -\frac{4}{9}, \frac{2}{9}, -\frac{4}{9}, \frac{1}{9}, -\frac{1}{3}, -\frac{1}{9}, -\frac{2}{9}, -\frac{2}{9}, \frac{1}{9}, 0, 0)$ |
| 254 | | | 4 | $(g_9, h_{107}^{(9)})$ | 1/81 | 0 | $(\frac{1}{3}, \frac{1}{3}, -\frac{1}{3}, 0, 0, 0, \frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, 0, 0, 0)$ | $(\frac{4}{9}, -\frac{4}{9}, -\frac{2}{9}, \frac{4}{9}, -\frac{1}{9}, -\frac{2}{9}, \frac{1}{9}, \frac{1}{9}, \frac{2}{9}, 0, \frac{1}{3}, \frac{2}{3})$ |
| 255 | | | 24 | $(g_9, h_{111}^{(9)})$ | 1/108 | 1 | $(-\frac{1}{3}, -\frac{1}{3}, \frac{1}{3}, 0, 0, -\frac{1}{3}, 0, \frac{1}{3}, \frac{1}{3}, 0, 0, 0)$ | $(-\frac{5}{12}, -\frac{1}{4}, -\frac{1}{3}, -\frac{5}{12}, \frac{1}{3}, \frac{1}{4}, \frac{1}{12}, 0, -\frac{1}{12}, 0, \frac{1}{4}, \frac{1}{4})$ |
| 256 | | | 36 | $(g_9, h_{117}^{(9)})$ | 1/324 | 2 | $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}, \frac{1}{3}, \frac{1}{3}, \frac{1}{3}, 0, 0, 0, 0, 0, 0)$ | $(\frac{1}{3}, \frac{1}{2}, 0, -\frac{1}{6}, \frac{1}{6}, \frac{1}{6}, \frac{1}{3}, -\frac{1}{3}, \frac{1}{2}, \frac{1}{6}, -\frac{1}{6}, \frac{1}{6})$ |
| 257 | | | 3 | $(g_{10}, h_{15}^{(10)})$ | 1/8192 | 0 | $(\frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, -\frac{1}{2})$ | $(\frac{1}{2}, \frac{1}{2}, 0, 0, \frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, \frac{1}{4})$ |
| 258 | | | 3 | $(g_{10}, h_{19}^{(10)})$ | 1/49152 | 4 | $(-\frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, -\frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, \frac{3}{4})$ | $(-\frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, 0, 0, 0, 0)$ |
| 259 | | | 12 | $(g_{10}, h_{27}^{(10)})$ | 1/3072 | $2\sqrt{2}$ | $(\frac{1}{2}, \frac{1}{2}, -\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, \frac{1}{2}, \frac{1}{2})$ | $(-\frac{3}{8}, \frac{3}{8}, \frac{3}{8}, -\frac{3}{8}, \frac{1}{8}, -\frac{1}{8}, 0, 0, \frac{1}{4}, \frac{1}{4}, \frac{1}{8}, -\frac{1}{8})$ |
| 260 | | | 12 | $(g_{10}, h_{30}^{(10)})$ | 1/1024 | 0 | $(-\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{2}, \frac{1}{2}, -\frac{1}{4}, -\frac{1}{4}, \frac{1}{2}, \frac{1}{2}, \frac{1}{4}, -\frac{1}{4})$ | $(-\frac{1}{4}, \frac{1}{2}, 0, -\frac{1}{4}, -\frac{3}{8}, \frac{3}{8}, -\frac{3}{8}, -\frac{3}{8}, \frac{1}{8}, -\frac{1}{8}, -\frac{1}{8}, \frac{1}{8})$ |
| 261 | | | 12 | $(g_{10}, h_{32}^{(10)})$ | 1/384 | 2 | $(-\frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, \frac{1}{2}, \frac{1}{2}, \frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{2}, \frac{1}{2})$ | $(-\frac{1}{3}, \frac{1}{6}, \frac{1}{3}, \frac{1}{6}, \frac{5}{12}, \frac{5}{12}, \frac{5}{12}, -\frac{5}{12}, \frac{1}{12}, \frac{1}{12}, \frac{1}{12}, \frac{13}{12})$ |
| 262 | | | 48 | $(g_{10}, h_{37}^{(10)})$ | 1/96 | $\sqrt{2}$ | $(\frac{1}{2}, \frac{1}{4}, -\frac{1}{4}, \frac{1}{2}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{2}, -\frac{1}{4}, -\frac{1}{4}, \frac{3}{2})$ | $(-\frac{11}{24}, \frac{5}{12}, \frac{11}{24}, \frac{7}{24}, \frac{5}{24}, \frac{1}{6}, \frac{1}{12}, \frac{1}{24}, -\frac{5}{24}, \frac{7}{24}, \frac{1}{6}, -\frac{1}{24})$ |
| 263 | | | 6 | $(g_{10}, h_{40}^{(10)})$ | 1/384 | 0 | $(-\frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, \frac{1}{2}, \frac{1}{2}, -\frac{1}{4}, -\frac{1}{4}, \frac{1}{2}, \frac{1}{2}, \frac{1}{4}, -\frac{1}{4})$ | $(-\frac{1}{8}, \frac{1}{8}, -\frac{3}{8}, -\frac{3}{8}, \frac{3}{8}, \frac{3}{8}, \frac{3}{8}, \frac{3}{8}, -\frac{1}{8}, -\frac{1}{8}, -\frac{1}{8}, \frac{1}{8})$ |
| 264 | | | 24 | $(g_{10}, h_{45}^{(10)})$ | 1/48 | 0 | $(\frac{1}{4}, -\frac{1}{4}, \frac{1}{2}, -\frac{1}{4}, \frac{1}{2}, \frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, \frac{1}{2}, \frac{1}{4}, \frac{1}{2}, \frac{1}{4})$ | $(\frac{5}{24}, -\frac{7}{24}, -\frac{11}{24}, -\frac{11}{24}, -\frac{7}{24}, -\frac{11}{24}, -\frac{1}{24}, \frac{1}{24}, -\frac{1}{24}, -\frac{5}{24}, -\frac{5}{24}, \frac{17}{24})$ |
| 265 | | | 3 | $(g_{10}, h_{50}^{(10)})$ | 1/49152 | 0 | $(-\frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2})$ | $(-\frac{1}{4}, -\frac{1}{4}, 0, 0, \frac{1}{2}, \frac{1}{2}, -\frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, -\frac{1}{4})$ |
| 266 | | | 2 | $(g_{10}, h_{51}^{(10)})$ | 1/1474560 | $4\sqrt{2}$ | $(\frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{3}{2})$ | $(\frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, -\frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, -\frac{1}{4})$ |
| 267 | | | 8 | $(g_{10}, h_{53}^{(10)})$ | 1/2304 | $2\sqrt{2}$ | $(\frac{1}{4}, -\frac{1}{4}, \frac{1}{2}, \frac{1}{2}, -\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, \frac{1}{2}, \frac{1}{2})$ | $(\frac{1}{6}, -\frac{1}{6}, \frac{5}{12}, \frac{5}{12}, \frac{5}{12}, -\frac{5}{12}, -\frac{1}{12}, -\frac{1}{12}, \frac{1}{6}, \frac{1}{6}, -\frac{1}{12}, -\frac{1}{12})$ |
| 268 | | | 12 | $(g_{10}, h_{57}^{(10)})$ | 1/384 | 0 | $(-\frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2})$ | $(-\frac{1}{3}, \frac{1}{6}, -\frac{5}{12}, \frac{1}{12}, -\frac{5}{12}, \frac{1}{3}, \frac{1}{12}, -\frac{1}{6}, -\frac{5}{12}, \frac{5}{12}, \frac{1}{12}, \frac{11}{12})$ |
| 269 | | | 3 | $(g_{10}, h_{58}^{(10)})$ | 1/6144 | 0 | $(-\frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{3}{2})$ | $(-\frac{1}{4}, -\frac{1}{4}, 0, 0, -\frac{1}{4}, \frac{1}{4}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, 0, 0)$ |
| 270 | | | 6 | $(g_{10}, h_{59}^{(10)})$ | 1/49152 | 0 | $(\frac{1}{2}, \frac{1}{2}, -\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, \frac{1}{2}, \frac{3}{2})$ | $(\frac{1}{2}, \frac{1}{2}, -\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, 0, 1)$ |
| 271 | | | 6 | $(g_{10}, h_{63}^{(10)})$ | 1/256 | 0 | $(-\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{2}, \frac{1}{4}, \frac{1}{4}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, -\frac{1}{4}, \frac{1}{4})$ | $(0, 0, \frac{1}{4}, \frac{1}{4}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, 0, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{3}{4})$ |
| 272 | | | 6 | $(g_{10}, h_{64}^{(10)})$ | 1/512 | 0 | $(\frac{1}{4}, -\frac{1}{4}, \frac{1}{2}, \frac{1}{4}, -\frac{1}{4}, \frac{1}{2}, \frac{1}{2}, -\frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, -\frac{1}{2})$ | $(\frac{1}{8}, -\frac{1}{8}, \frac{1}{2}, \frac{3}{8}, -\frac{3}{8}, -\frac{1}{4}, \frac{1}{4}, \frac{3}{8}, \frac{3}{8}, -\frac{1}{8}, \frac{1}{8}, 1)$ |
| 273 | | | 12 | $(g_{10}, h_{65}^{(10)})$ | 1/1024 | 2 | $(-\frac{1}{4}, -\frac{1}{4}, \frac{1}{2}, \frac{1}{2}, \frac{1}{4}, -\frac{1}{4}, \frac{1}{2}, \frac{1}{2}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{5}{4})$ | $(-\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, 0, 0, 0, 0, 1)$ |
| 274 | | | 3 | $(g_{10}, h_{67}^{(10)})$ | 1/1024 | 0 | $(\frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, -\frac{1}{2})$ | $(\frac{1}{4}, \frac{1}{4}, 0, 0, \frac{1}{4}, \frac{1}{4}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, 0, 0)$ |
| 275 | | | 6 | $(g_{10}, h_{71}^{(10)})$ | 1/6144 | $2\sqrt{2}$ | $(-\frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, -\frac{1}{2})$ | $(-\frac{1}{4}, 0, \frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, 0, 0, \frac{1}{2}, \frac{1}{2})$ |
| 276 | | | 24 | $(g_{10}, h_{79}^{(10)})$ | 1/96 | 0 | $(-\frac{1}{4}, -\frac{1}{4}, \frac{1}{2}, \frac{1}{4}, -\frac{1}{4}, \frac{1}{2}, \frac{1}{2}, \frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, \frac{3}{2})$ | $(-\frac{1}{12}, -\frac{5}{12}, \frac{1}{3}, -\frac{5}{12}, \frac{1}{12}, -\frac{1}{6}, \frac{1}{2}, \frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, 0)$ |
| 277 | | | 24 | $(g_{10}, h_{87}^{(10)})$ | 1/48 | $\sqrt{2}$ | $(\frac{1}{4}, \frac{1}{4}, \frac{1}{2}, \frac{1}{4}, \frac{1}{4}, \frac{1}{2}, \frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, \frac{1}{2}, \frac{3}{2})$ | $(\frac{1}{4}, \frac{1}{6}, \frac{1}{3}, \frac{1}{4}, -\frac{5}{12}, -\frac{1}{6}, -\frac{1}{4}, -\frac{1}{4}, 0, -\frac{1}{12}, \frac{1}{2}, 1)$ |
| 278 | | | 6 | $(g_{10}, h_{91}^{(10)})$ | 1/128 | 0 | $(-\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{2}, \frac{1}{2}, \frac{1}{4}, \frac{1}{4}, \frac{1}{2}, \frac{1}{2}, -\frac{1}{4}, \frac{1}{4})$ | $(-\frac{1}{8}, \frac{1}{8}, \frac{3}{8}, \frac{3}{8}, \frac{3}{8}, \frac{3}{8}, -\frac{3}{8}, -\frac{3}{8}, \frac{1}{8}, \frac{1}{8}, \frac{1}{8}, -\frac{1}{8})$ |
| 279 | | | 12 | $(g_{10}, h_{92}^{(10)})$ | 1/256 | 2 | $(\frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, \frac{1}{2}, \frac{1}{2}, \frac{1}{4}, -\frac{1}{4}, \frac{1}{2}, \frac{3}{2})$ | $(\frac{3}{8}, \frac{1}{4}, \frac{3}{8}, \frac{1}{8}, -\frac{1}{4}, \frac{1}{8}, -\frac{3}{8}, \frac{3}{8}, 0, 0, -\frac{1}{8}, -\frac{1}{8})$ |
| 280 | | | 12 | $(g_{10}, h_{95}^{(10)})$ | 1/256 | 0 | $(\frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, -\frac{1}{4}, \frac{5}{4})$ | $(\frac{1}{8}, \frac{3}{8}, \frac{1}{8}, \frac{3}{8}, 0, 0, \frac{1}{8}, -\frac{1}{8}, \frac{3}{8}, \frac{3}{8}, \frac{1}{2}, \frac{1}{2})$ |
| 281 | | | 6 | $(g_{10}, h_{97}^{(10)})$ | 1/512 | 0 | $(\frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{2}, \frac{1}{2}, -\frac{1}{4}, -\frac{1}{4}, \frac{1}{2}, \frac{1}{2})$ | $(\frac{1}{8}, \frac{1}{8}, \frac{1}{8}, \frac{1}{8}, \frac{3}{8}, \frac{3}{8}, -\frac{1}{4}, -\frac{1}{4}, \frac{3}{8}, \frac{3}{8}, \frac{1}{2}, \frac{1}{2})$ |
| 282 | | | 12 | $(g_{10}, h_{98}^{(10)})$ | 1/6144 | 2 | $(\frac{1}{4}, -\frac{1}{4}, \frac{1}{2}, \frac{1}{2}, -\frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, \frac{1}{2}, -\frac{1}{2})$ | $(0, 0, -\frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{2}, -\frac{1}{2})$ |
| 283 | | | 12 | $(g_{10}, h_{102}^{(10)})$ | 1/64 | $\sqrt{2}$ | $(\frac{1}{4}, -\frac{1}{4}, \frac{1}{2}, \frac{1}{4}, -\frac{1}{4}, \frac{1}{2}, \frac{1}{2}, -\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, \frac{1}{2})$ | $(\frac{1}{4}, -\frac{1}{8}, \frac{3}{8}, \frac{1}{2}, \frac{3}{8}, -\frac{1}{8}, \frac{1}{2}, \frac{1}{4}, -\frac{1}{4}, 0, 0, 0)$ |
| 284 | | | 12 | $(g_{10}, h_{104}^{(10)})$ | 1/32 | 0 | $(\frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, \frac{1}{2}, \frac{1}{4}, -\frac{1}{4}, \frac{1}{2}, \frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, \frac{1}{2}, \frac{1}{2})$ | $(\frac{3}{16}, \frac{5}{16}, -\frac{1}{16}, -\frac{3}{8}, \frac{7}{16}, \frac{1}{16}, -\frac{1}{8}, -\frac{3}{16}, \frac{5}{16}, \frac{7}{16}, \frac{1}{2}, 0)$ |
| 285 | | | 12 | $(g_{10}, h_{119}^{(10)})$ | 1/192 | 0 | $(-\frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2})$ | $(-\frac{1}{6}, -\frac{1}{12}, \frac{1}{12}, \frac{1}{6}, \frac{1}{3}, -\frac{5}{12}, \frac{1}{3}, -\frac{5}{12}, \frac{1}{6}, -\frac{1}{6}, \frac{1}{3}, -\frac{1}{3})$ |
| 286 | | | 8 | $(g_{10}, h_{124}^{(10)})$ | 1/144 | $\sqrt{2}$ | $(\frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, \frac{1}{2}, \frac{1}{2}, -\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{2}, \frac{1}{2}, -\frac{1}{4}, -\frac{1}{4})$ | $(\frac{1}{6}, -\frac{1}{12}, \frac{1}{6}, -\frac{1}{12}, -\frac{5}{12}, \frac{5}{12}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{2}, \frac{3}{2})$ |
| 287 | | | 48 | $(g_{10}, h_{126}^{(10)})$ | 1/48 | 1 | $(\frac{1}{4}, \frac{1}{2}, \frac{1}{4}, -\frac{1}{4}, \frac{1}{2}, -\frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, \frac{1}{2}, \frac{1}{2})$ | $(\frac{1}{3}, \frac{5}{12}, \frac{1}{2}, -\frac{5}{12}, -\frac{1}{3}, -\frac{1}{6}, -\frac{1}{3}, -\frac{1}{12}, 0, -\frac{1}{6}, -\frac{1}{12}, 1)$ |
| 288 | | | 12 | $(g_{10}, h_{130}^{(10)})$ | 1/32 | 0 | $(-\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{2}, -\frac{1}{4}, -\frac{1}{4}, \frac{1}{2}, \frac{1}{4}, \frac{1}{2}, -\frac{1}{4}, -\frac{1}{4}, -\frac{1}{2})$ | $(-\frac{1}{16}, \frac{3}{16}, -\frac{1}{16}, -\frac{1}{8}, \frac{3}{16}, -\frac{5}{16}, \frac{1}{4}, \frac{7}{16}, \frac{3}{8}, \frac{5}{16}, \frac{7}{16}, \frac{1}{2})$ |
| 289 | | | 48 | $(g_{10}, h_{131}^{(10)})$ | 1/128 | $\sqrt{2}$ | $(-\frac{1}{4}, -\frac{1}{4}, \frac{1}{2}, \frac{1}{2}, \frac{1}{4}, \frac{1}{2}, -\frac{1}{4}, \frac{1}{2}, \frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, \frac{1}{4})$ | $(-\frac{1}{8}, 0, \frac{3}{8}, -\frac{1}{4}, -\frac{3}{8}, -\frac{1}{8}, \frac{1}{4}, \frac{1}{2}, \frac{3}{8}, \frac{3}{8}, -\frac{1}{8}, -\frac{1}{8})$ |
| 290 | | | 24 | $(g_{10}, h_{135}^{(10)})$ | 1/80 | $\sqrt{2}$ | $(\frac{1}{2}, \frac{1}{2}, \frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, \frac{1}{2}, \frac{1}{2}, \frac{1}{4}, \frac{5}{4})$ | $(-\frac{9}{20}, -\frac{7}{20}, \frac{7}{20}, \frac{3}{10}, \frac{9}{20}, -\frac{3}{20}, \frac{1}{10}, \frac{3}{10}, \frac{3}{20}, -\frac{1}{20}, -\frac{1}{10}, -\frac{1}{20})$ |
| 291 | | | 6 | $(g_{11}, h_{25}^{(11)})$ | 1/576 | 2 | $(-\frac{1}{6}, \frac{1}{6}, 0, 0, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{3}, \frac{1}{3}, -\frac{1}{6}, \frac{7}{6})$ | $(0, 0, \frac{1}{6}, -\frac{1}{6}, -\frac{1}{6}, -\frac{1}{6}, \frac{1}{3}, -\frac{1}{3}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, -\frac{1}{2})$ |
| 292 | | | 18 | $(g_{11}, h_{76}^{(11)})$ | 1/72 | 0 | $(\frac{1}{2}, -\frac{1}{6}, \frac{1}{3}, -\frac{1}{6}, \frac{1}{2}, \frac{1}{6}, -\frac{1}{6}, -\frac{1}{3}, \frac{1}{2}, 0, \frac{1}{2}, 0)$ | $(-\frac{1}{3}, -\frac{1}{6}, \frac{1}{6}, \frac{1}{2}, \frac{1}{6}, 0, \frac{1}{6}, \frac{1}{6}, \frac{1}{2}, 0, 0, \frac{1}{2})$ |
| 293 | | | 36 | $(g_{11}, h_{97}^{(11)})$ | 1/108 | $\sqrt{3}$ | $(0, -\frac{1}{6}, \frac{1}{3}, -\frac{1}{6}, 0, \frac{1}{6}, -\frac{1}{3}, -\frac{1}{6}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{3}{2})$ | $(-\frac{1}{6}, 0, 0, -\frac{1}{3}, \frac{1}{3}, -\frac{1}{3}, \frac{1}{2}, \frac{1}{2}, 0, \frac{1}{3}, \frac{1}{3}, \frac{3}{2})$ |
| 294 | | | 36 | $(g_{11}, h_{106}^{(11)})$ | 1/144 | 0 | $(\frac{1}{2}, -\frac{1}{3}, \frac{1}{2}, -\frac{1}{6}, \frac{1}{6}, 0, \frac{1}{6}, 0, \frac{1}{3}, \frac{1}{2}, \frac{1}{2}, -\frac{1}{6})$ | $(-\frac{5}{12}, \frac{1}{2}, -\frac{1}{3}, -\frac{1}{4}, \frac{1}{2}, -\frac{1}{6}, 0, -\frac{1}{3}, 0, -\frac{1}{12}, -\frac{1}{6}, \frac{5}{4})$ |

| No. | P | com. | size | rep. | $[h]_{\subset C_g}/\lvert C_g\rvert$ | $\lvert Z_{\mathrm{R}g}^{\mathrm{R}h}\rvert$ | $\vec{\lambda}_{\tilde{g},O}$ | $\vec{\lambda}_{\tilde{h},O}$ |
|---|---|---|---|---|---|---|---|---|
| 295 | | | 12 | $(g_{11}, h_{109}^{(11)})$ | 1/432 | 0 | $(0, -\frac{1}{6}, \frac{1}{3}, \frac{1}{2}, 0, \frac{1}{3}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{6}, -\frac{1}{6}, \frac{5}{6})$ | $(-\frac{1}{6}, 0, 0, -\frac{1}{3}, \frac{1}{3}, \frac{1}{2}, \frac{1}{6}, \frac{1}{6}, -\frac{1}{6}, \frac{1}{2}, \frac{1}{2}, -\frac{1}{2})$ |
| 296 | | | 24 | $(g_{11}, h_{136}^{(11)})$ | 1/108 | 0 | $(\frac{1}{2}, -\frac{1}{6}, -\frac{1}{6}, 0, -\frac{1}{3}, \frac{1}{6}, 0, -\frac{1}{6}, \frac{1}{3}, \frac{1}{2}, \frac{1}{2}, \frac{3}{2})$ | $(\frac{1}{3}, 0, -\frac{1}{6}, -\frac{1}{6}, \frac{1}{3}, \frac{1}{2}, -\frac{1}{3}, \frac{1}{6}, -\frac{1}{6}, \frac{1}{2}, -\frac{1}{6}, \frac{7}{6})$ |
| 297 | | | 24 | $(g_{11}, h_{142}^{(11)})$ | 1/216 | 0 | $(\frac{1}{3}, \frac{1}{6}, \frac{1}{3}, \frac{1}{6}, -\frac{1}{6}, -\frac{1}{6}, \frac{1}{2}, 0, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, 1)$ | $(\frac{1}{3}, \frac{1}{6}, \frac{1}{6}, \frac{1}{2}, 0, \frac{1}{6}, \frac{1}{2}, 0, -\frac{1}{6}, -\frac{1}{6}, 0, \frac{3}{2})$ |
| 298 | | | 72 | $(g_{11}, h_{151}^{(11)})$ | 1/432 | $\sqrt{3}$ | $(0, \frac{1}{3}, \frac{1}{3}, 0, \frac{1}{2}, \frac{1}{2}, -\frac{1}{6}, \frac{1}{6}, \frac{1}{2}, \frac{1}{2}, -\frac{1}{6}, \frac{1}{6})$ | $(-\frac{1}{6}, -\frac{1}{6}, -\frac{1}{3}, -\frac{1}{3}, \frac{1}{3}, \frac{1}{3}, -\frac{1}{3}, \frac{1}{3}, \frac{1}{6}, \frac{1}{6}, \frac{1}{3}, -\frac{1}{3})$ |
| 299 | | | 18 | $(g_{11}, h_{184}^{(11)})$ | 1/144 | 0 | $(0, -\frac{1}{6}, \frac{1}{3}, \frac{1}{2}, \frac{1}{2}, \frac{1}{6}, \frac{1}{3}, 0, \frac{1}{2}, \frac{1}{2}, \frac{1}{6}, \frac{1}{6})$ | $(-\frac{1}{12}, -\frac{1}{4}, \frac{1}{4}, \frac{5}{12}, \frac{1}{12}, -\frac{1}{4}, -\frac{1}{4}, -\frac{5}{12}, -\frac{1}{6}, \frac{1}{6}, \frac{1}{2}, \frac{1}{2})$ |
| 300 | | | 6 | $(g_{12}, h_{9}^{(12)})$ | 1/256 | 0 | $(\frac{1}{8}, \frac{1}{8}, 0, \frac{3}{8}, -\frac{3}{8}, \frac{1}{2}, -\frac{1}{8}, -\frac{1}{8}, 0, \frac{1}{2}, -\frac{3}{8}, \frac{11}{8})$ | $(\frac{1}{8}, \frac{1}{8}, 0, \frac{3}{8}, -\frac{3}{8}, \frac{1}{2}, \frac{3}{8}, \frac{3}{8}, \frac{1}{2}, 0, \frac{1}{8}, \frac{7}{8})$ |
| 301 | | | 3 | $(g_{12}, h_{17}^{(12)})$ | 1/64 | 0 | $(-\frac{3}{8}, -\frac{1}{8}, -\frac{1}{8}, -\frac{3}{8}, -\frac{3}{8}, \frac{1}{8}, \frac{1}{8}, \frac{3}{8}, \frac{1}{2}, 0, 0, \frac{1}{2})$ | $(-\frac{3}{8}, -\frac{1}{8}, -\frac{3}{8}, \frac{3}{8}, -\frac{1}{8}, -\frac{1}{8}, -\frac{3}{8}, -\frac{1}{8}, \frac{1}{2}, 0, \frac{1}{2}, 0)$ |
| 302 | | A | 96 | $(g_{12}, h_{35}^{(12)})$ | 1/48 | 0 | – | – |
| 303 | | A | 12 | $(g_{12}, h_{43}^{(12)})$ | 1/768 | 0 | – | – |
| 304 | | A | 6 | $(g_{12}, h_{47}^{(12)})$ | 1/128 | 0 | – | – |
| 305 | | A | 3 | $(g_{12}, h_{53}^{(12)})$ | 1/128 | 0 | – | – |
| 306 | | A | 3 | $(g_{12}, h_{55}^{(12)})$ | 1/256 | 0 | – | – |
| 307 | | A | 6 | $(g_{12}, h_{57}^{(12)})$ | 1/512 | 0 | – | – |
| 308 | | A | 6 | $(g_{12}, h_{61}^{(12)})$ | 1/32 | 0 | – | – |
| 309 | | A | 6 | $(g_{12}, h_{63}^{(12)})$ | 1/128 | 0 | – | – |
| 310 | | A | 12 | $(g_{12}, h_{71}^{(12)})$ | 1/64 | 0 | – | – |
| 311 | | A | 12 | $(g_{12}, h_{75}^{(12)})$ | 1/32 | 0 | – | – |
| 312 | | | 12 | $(g_{12}, h_{77}^{(12)})$ | 1/64 | 0 | $(-\frac{1}{8}, 0, -\frac{1}{8}, \frac{3}{8}, \frac{1}{2}, \frac{3}{8}, \frac{1}{2}, -\frac{3}{8}, -\frac{3}{8}, 0, -\frac{1}{8}, \frac{9}{8})$ | $(-\frac{1}{8}, -\frac{1}{8}, -\frac{1}{4}, -\frac{1}{8}, -\frac{1}{8}, \frac{3}{8}, \frac{3}{8}, \frac{1}{2}, \frac{1}{4}, -\frac{3}{8}, \frac{3}{8}, \frac{1}{2})$ |
| 313 | | | 12 | $(g_{12}, h_{81}^{(12)})$ | 1/32 | 0 | $(\frac{3}{8}, \frac{1}{2}, -\frac{3}{8}, -\frac{1}{8}, -\frac{1}{8}, 0, -\frac{1}{8}, 0, -\frac{1}{8}, -\frac{3}{8}, \frac{1}{2}, -\frac{3}{8})$ | $(\frac{3}{8}, \frac{3}{8}, \frac{3}{8}, -\frac{3}{8}, \frac{3}{8}, \frac{3}{8}, -\frac{1}{8}, \frac{1}{8}, \frac{1}{8}, -\frac{1}{8}, -\frac{1}{8}, \frac{1}{8})$ |
| 314 | | | 6 | $(g_{12}, h_{83}^{(12)})$ | 1/32 | 0 | $(\frac{1}{8}, \frac{3}{8}, -\frac{3}{8}, \frac{1}{8}, -\frac{3}{8}, 0, \frac{1}{8}, \frac{1}{2}, -\frac{3}{8}, -\frac{1}{8}, \frac{1}{2}, 0)$ | $(0, \frac{1}{2}, 0, \frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, \frac{1}{2}, 0, \frac{3}{2})$ |
| 315 | | | 6 | $(g_{12}, h_{89}^{(12)})$ | 1/128 | 0 | $(-\frac{1}{8}, -\frac{3}{8}, -\frac{1}{8}, -\frac{3}{8}, -\frac{1}{8}, \frac{3}{8}, \frac{3}{8}, -\frac{1}{8}, 0, \frac{1}{2}, \frac{1}{2}, 0)$ | $(-\frac{1}{8}, -\frac{1}{8}, \frac{1}{8}, -\frac{3}{8}, -\frac{3}{8}, -\frac{1}{8}, -\frac{3}{8}, \frac{3}{8}, 0, \frac{1}{2}, 0, -\frac{1}{2})$ |
| 316 | | | 144 | $(g_{13}, h_{73}^{(13)})$ | 1/72 | 1 | $(-\frac{1}{12}, \frac{1}{4}, 0, \frac{1}{3}, \frac{1}{3}, -\frac{1}{4}, -\frac{5}{12}, -\frac{1}{4}, -\frac{1}{12}, 0, \frac{1}{4}, -\frac{5}{12})$ | $(0, \frac{1}{6}, \frac{1}{6}, \frac{1}{2}, \frac{1}{3}, -\frac{1}{3}, \frac{1}{3}, \frac{1}{3}, -\frac{1}{3}, \frac{1}{3}, -\frac{1}{6}, 0)$ |
| 317 | | | 24 | $(g_{13}, h_{115}^{(13)})$ | 1/144 | $\sqrt{2}$ | $(0, -\frac{1}{4}, \frac{1}{4}, -\frac{1}{12}, -\frac{1}{4}, -\frac{1}{3}, -\frac{5}{12}, -\frac{5}{12}, -\frac{1}{4}, -\frac{1}{3}, \frac{1}{12}, 0)$ | $(-\frac{1}{12}, -\frac{1}{12}, \frac{1}{6}, -\frac{1}{4}, \frac{1}{6}, -\frac{1}{4}, -\frac{1}{4}, \frac{1}{2}, -\frac{5}{12}, \frac{1}{4}, \frac{1}{2}, \frac{5}{12})$ |
| 318 | | | 1 | $(g_{15}, h_{3}^{(15)})$ | 1/491520 | 0 | $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, 0, 0, 0, 0, 0, 0, \frac{1}{2}, \frac{1}{2}, -\frac{1}{2})$ | $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, 0, 0, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, 0, 0, 0, 0, 0)$ |
| 319 | | | 3 | $(g_{15}, h_{7}^{(15)})$ | 1/2048 | 0 | $(0, \frac{1}{2}, \frac{1}{2}, 0, 0, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, 0, 0, 1)$ | $(0, \frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, 0, 0, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2})$ |
| 320 | | | 3 | $(g_{15}, h_{8}^{(15)})$ | 1/2048 | 0 | $(0, 0, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, 0, 0, \frac{1}{2}, \frac{1}{2}, 0, 1)$ | $(-\frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{2}, \frac{1}{2}, 0, 0, 0, 0, \frac{1}{2}, \frac{1}{2})$ |
| 321 | | | 3 | $(g_{15}, h_{9}^{(15)})$ | 1/8192 | 0 | $(\frac{1}{2}, 0, \frac{1}{2}, 0, 0, 0, 0, 0, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2})$ | $(\frac{1}{2}, 0, 0, \frac{1}{2}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4})$ |
| 322 | | | 1 | $(g_{15}, h_{13}^{(15)})$ | 1/8192 | 0 | $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, 0, 0, 0, 0, 0, 0, \frac{1}{2}, \frac{1}{2}, \frac{1}{2})$ | $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, 0, 0, 0, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, 0, 0, 1)$ |
| 323 | | A | 6 | $(g_{15}, h_{24}^{(15)})$ | 1/128 | 0 | – | – |
| 324 | | A | 3 | $(g_{15}, h_{26}^{(15)})$ | 1/128 | 0 | – | – |
| 325 | | A | 6 | $(g_{15}, h_{27}^{(15)})$ | 1/768 | 0 | – | – |
| 326 | | A | 48 | $(g_{15}, h_{35}^{(15)})$ | 1/48 | 0 | – | – |
| 327 | | | 6 | $(g_{15}, h_{43}^{(15)})$ | 1/64 | 0 | $(\frac{1}{2}, 0, 0, 0, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, 0, \frac{1}{2}, 0, 0)$ | $(-\frac{3}{8}, \frac{1}{8}, \frac{1}{4}, -\frac{3}{8}, -\frac{1}{8}, \frac{1}{2}, \frac{1}{4}, -\frac{1}{4}, 0, 0, \frac{1}{2}, \frac{1}{2})$ |
| 328 | | | 3 | $(g_{15}, h_{47}^{(15)})$ | 1/512 | 0 | $(0, 0, \frac{1}{2}, \frac{1}{2}, 0, 0, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, 0, 1)$ | $(-\frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, 0, 0, \frac{1}{2}, \frac{1}{2}, 0, 0, \frac{1}{2}, \frac{1}{2})$ |
| 329 | | | 3 | $(g_{15}, h_{48}^{(15)})$ | 1/512 | 0 | $(0, 0, 0, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, 0, 0, 0)$ | $(0, \frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, \frac{1}{2}, \frac{1}{2}, 0, 0, \frac{1}{2}, \frac{1}{2}, \frac{1}{2})$ |
| 330 | | | 6 | $(g_{15}, h_{50}^{(15)})$ | 1/32 | 0 | $(0, \frac{1}{2}, 0, \frac{1}{2}, 0, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, 0, \frac{1}{2}, 0, 0)$ | $(-\frac{1}{8}, -\frac{3}{8}, -\frac{1}{4}, -\frac{1}{8}, -\frac{3}{8}, \frac{1}{2}, -\frac{1}{4}, -\frac{1}{4}, \frac{1}{2}, 0, 0, 1)$ |
| 331 | | | 6 | $(g_{15}, h_{51}^{(15)})$ | 1/32 | 0 | $(\frac{1}{2}, 0, 0, \frac{1}{2}, 0, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, 0, 0, 0, \frac{1}{2})$ | $(-\frac{3}{8}, \frac{1}{8}, \frac{3}{8}, \frac{1}{8}, 0, \frac{1}{2}, \frac{1}{4}, \frac{1}{4}, \frac{1}{2}, -\frac{1}{4}, -\frac{1}{4}, 0)$ |
| 332 | | | 3 | $(g_{15}, h_{54}^{(15)})$ | 1/2048 | 0 | $(\frac{1}{2}, \frac{1}{2}, 0, 0, \frac{1}{2}, \frac{1}{2}, 0, 0, 0, 0, \frac{1}{2}, \frac{1}{2})$ | $(-\frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{2}, \frac{1}{2}, 0, 0, \frac{1}{2}, \frac{1}{2}, 0, 0)$ |
| 333 | | | 3 | $(g_{15}, h_{55}^{(15)})$ | 1/2048 | 0 | $(0, \frac{1}{2}, \frac{1}{2}, 0, 0, 0, 0, 0, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2})$ | $(0, \frac{1}{2}, 0, \frac{1}{2}, -\frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, \frac{1}{4})$ |
| 334 | | | 1 | $(g_{15}, h_{59}^{(15)})$ | 1/30720 | 0 | $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, 0, 0, 0, 0, 0, 0, \frac{1}{2}, \frac{1}{2}, \frac{3}{2})$ | $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, 0, 0, 0, 0, 0, 0)$ |
| 335 | | A | 6 | $(g_{15}, h_{61}^{(15)})$ | 1/64 | 0 | – | – |
| 336 | | A | 6 | $(g_{15}, h_{62}^{(15)})$ | 1/64 | 0 | – | – |
| 337 | | A | 3 | $(g_{15}, h_{63}^{(15)})$ | 1/192 | 0 | – | – |
| 338 | | A | 3 | $(g_{15}, h_{66}^{(15)})$ | 1/512 | 0 | – | – |
| 339 | | A | 6 | $(g_{15}, h_{67}^{(15)})$ | 1/1024 | 0 | – | – |
| 340 | | A | 3 | $(g_{15}, h_{71}^{(15)})$ | 1/7680 | 0 | – | – |
| 341 | | | 12 | $(g_{15}, h_{72}^{(15)})$ | 1/160 | 0 | $(0, 0, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, 0, 0, 0, \frac{1}{2}, \frac{1}{2}, 0)$ | $(-\frac{1}{10}, -\frac{1}{5}, \frac{2}{5}, \frac{3}{10}, \frac{1}{10}, -\frac{1}{5}, \frac{3}{10}, \frac{2}{5}, 0, 0, \frac{1}{2}, -\frac{1}{2})$ |

| No. | P | com. | size | rep. | $[h]_{\subset C_g}/\lvert C_g\rvert$ | $\lvert Z_{\mathrm{R}g}^{\mathrm{R}h}\rvert$ | $\vec{\lambda}_{\tilde{g},O}$ | $\vec{\lambda}_{\tilde{h},O}$ |
|---|---|---|---|---|---|---|---|---|
| 342 | | A | 36 | $(g_{15}, h_{78}^{(15)})$ | 1/40 | 0 | – | – |
| 343 | | | 6 | $(g_{15}, h_{84}^{(15)})$ | 1/80 | 0 | $(\frac{1}{2}, \frac{1}{2}, 0, 0, 0, \frac{1}{2}, 0, \frac{1}{2}, \frac{1}{2}, 0, 0, \frac{1}{2})$ | $(-\frac{2}{5}, \frac{3}{10}, -\frac{1}{10}, -\frac{1}{5}, \frac{2}{5}, \frac{1}{5}, \frac{3}{10}, \frac{1}{10}, \frac{1}{2}, 0, \frac{1}{2}, 0)$ |
| 344 | | | 4 | $(g_{15}, h_{86}^{(15)})$ | 1/96 | 0 | $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, 0, 0, 0, \frac{1}{2}, \frac{1}{2}, 0, 0, \frac{1}{2}, 1)$ | $(\frac{1}{2}, \frac{1}{3}, -\frac{1}{3}, 0, -\frac{1}{6}, \frac{1}{6}, -\frac{1}{6}, -\frac{1}{6}, \frac{1}{3}, \frac{1}{3}, 0, \frac{3}{2})$ |
| 345 | | | 4 | $(g_{15}, h_{91}^{(15)})$ | 1/96 | 0 | $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, 0, 0, 0, \frac{1}{2}, \frac{1}{2}, 0, 0, 0, \frac{1}{2})$ | $(\frac{1}{2}, -\frac{1}{3}, -\frac{1}{3}, 0, \frac{1}{6}, -\frac{1}{6}, -\frac{1}{6}, -\frac{1}{6}, \frac{1}{3}, -\frac{1}{3}, \frac{1}{2}, 0)$ |
| 346 | | A | 12 | $(g_{15}, h_{92}^{(15)})$ | 1/24 | 0 | – | – |
| 347 | | A | 12 | $(g_{15}, h_{94}^{(15)})$ | 1/24 | 0 | – | – |
| 348 | | | 15 | $(g_{16}, h_{11}^{(16)})$ | 1/250 | $\frac{1}{2}(3-\sqrt{5})$ | $(\frac{2}{5}, \frac{1}{5}, \frac{1}{5}, -\frac{2}{5}, -\frac{2}{5}, -\frac{2}{5}, 0, 0, 0, 0, -\frac{1}{5}, \frac{1}{5})$ | $(-\frac{3}{10}, -\frac{3}{10}, \frac{1}{10}, \frac{1}{10}, -\frac{3}{10}, -\frac{3}{10}, -\frac{3}{10}, -\frac{3}{10}, -\frac{1}{10}, \frac{1}{10}, \frac{1}{10}, -\frac{1}{10})$ |
| 349 | | | 15 | $(g_{16}, h_{16}^{(16)})$ | 1/250 | $\frac{1}{2}(3+\sqrt{5})$ | $(-\frac{1}{5}, -\frac{1}{5}, \frac{2}{5}, \frac{2}{5}, 0, 0, 0, 0, -\frac{2}{5}, \frac{2}{5}, \frac{1}{5}, -\frac{1}{5})$ | $(-\frac{1}{10}, \frac{1}{10}, \frac{3}{10}, -\frac{3}{10}, \frac{1}{10}, \frac{1}{10}, \frac{3}{10}, -\frac{3}{10}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, -\frac{1}{2})$ |
| 350 | | | 10 | $(g_{16}, h_{31}^{(16)})$ | 1/1500 | $\sqrt{5}-1$ | $(-\frac{2}{5}, -\frac{2}{5}, \frac{1}{5}, \frac{1}{5}, 0, 0, \frac{1}{5}, -\frac{1}{5}, 0, 0, -\frac{2}{5}, \frac{2}{5})$ | $(\frac{3}{10}, \frac{3}{10}, \frac{3}{10}, \frac{3}{10}, \frac{1}{10}, \frac{1}{10}, \frac{1}{10}, -\frac{1}{10}, -\frac{3}{10}, -\frac{3}{10}, -\frac{1}{10}, \frac{1}{10})$ |
| 351 | | | 10 | $(g_{16}, h_{36}^{(16)})$ | 1/1500 | $1+\sqrt{5}$ | $(-\frac{1}{5}, -\frac{1}{5}, 0, 0, -\frac{1}{5}, \frac{1}{5}, \frac{2}{5}, \frac{2}{5}, -\frac{2}{5}, -\frac{2}{5}, 0, 0)$ | $(-\frac{1}{5}, -\frac{1}{5}, \frac{1}{5}, -\frac{1}{5}, \frac{1}{5}, -\frac{1}{5}, \frac{2}{5}, \frac{2}{5}, \frac{2}{5}, \frac{2}{5}, -\frac{2}{5}, -\frac{2}{5})$ |
| 352 | | | 30 | $(g_{16}, h_{46}^{(16)})$ | 1/100 | 0 | $(0, -\frac{1}{5}, 0, \frac{1}{5}, \frac{2}{5}, -\frac{2}{5}, \frac{2}{5}, 0, \frac{1}{5}, \frac{2}{5}, 0, \frac{1}{5})$ | $(-\frac{1}{20}, -\frac{3}{20}, \frac{3}{20}, -\frac{1}{20}, \frac{3}{20}, -\frac{1}{20}, -\frac{9}{20}, \frac{7}{20}, \frac{9}{20}, -\frac{7}{20}, -\frac{9}{20}, \frac{13}{20})$ |
| 353 | | | 30 | $(g_{16}, h_{51}^{(16)})$ | 1/100 | 0 | $(0, 0, \frac{1}{5}, -\frac{1}{5}, \frac{2}{5}, \frac{2}{5}, \frac{2}{5}, \frac{1}{5}, 0, -\frac{1}{5}, 0, -\frac{2}{5})$ | $(-\frac{1}{20}, -\frac{3}{20}, \frac{3}{20}, -\frac{1}{4}, \frac{1}{20}, -\frac{9}{20}, \frac{1}{4}, -\frac{1}{4}, -\frac{9}{20}, \frac{7}{20}, \frac{7}{20}, \frac{1}{4})$ |
| 354 | | | 6 | $(g_{16}, h_{61}^{(16)})$ | 1/125 | 1 | $(-\frac{2}{5}, \frac{2}{5}, \frac{1}{5}, \frac{1}{5}, 0, 0, -\frac{2}{5}, \frac{1}{5}, \frac{1}{5}, -\frac{2}{5}, 0, 0)$ | $(-\frac{2}{5}, -\frac{2}{5}, \frac{2}{5}, \frac{1}{5}, \frac{2}{5}, \frac{1}{5}, \frac{1}{5}, -\frac{1}{5}, 0, 0, 0, 0)$ |
| 355 | | | 40 | $(g_{16}, h_{66}^{(16)})$ | 1/75 | $\frac{1}{2}(\sqrt{5}-1)$ | $(\frac{2}{5}, -\frac{1}{5}, \frac{2}{5}, -\frac{1}{5}, 0, 0, -\frac{1}{5}, 0, 0, \frac{2}{5}, -\frac{1}{5}, \frac{2}{5})$ | $(-\frac{7}{15}, -\frac{7}{15}, \frac{4}{15}, -\frac{4}{15}, \frac{7}{15}, -\frac{4}{15}, -\frac{2}{15}, -\frac{2}{15}, -\frac{1}{15}, -\frac{2}{15}, \frac{1}{15}, -\frac{1}{15})$ |
| 356 | | | 40 | $(g_{16}, h_{71}^{(16)})$ | 1/75 | $\frac{1}{2}(1+\sqrt{5})$ | $(-\frac{2}{5}, -\frac{2}{5}, \frac{1}{5}, 0, -\frac{1}{5}, 0, \frac{1}{5}, -\frac{2}{5}, 0, 0, -\frac{1}{5}, \frac{2}{5})$ | $(-\frac{7}{15}, -\frac{4}{15}, \frac{7}{15}, \frac{7}{15}, \frac{4}{15}, -\frac{4}{15}, \frac{2}{15}, -\frac{2}{15}, \frac{1}{15}, -\frac{2}{15}, -\frac{1}{15}, -\frac{1}{15})$ |
| 357 | | | 45 | $(g_{16}, h_{81}^{(16)})$ | 1/50 | $\frac{1}{2}(\sqrt{5}-1)$ | $(\frac{2}{5}, \frac{1}{5}, 0, \frac{1}{5}, -\frac{1}{5}, \frac{2}{5}, 0, 0, \frac{1}{5}, 0, \frac{2}{5}, -\frac{2}{5})$ | $(\frac{2}{5}, \frac{3}{10}, \frac{1}{10}, \frac{1}{10}, -\frac{1}{5}, -\frac{3}{10}, \frac{2}{5}, \frac{3}{10}, -\frac{3}{10}, -\frac{1}{5}, \frac{1}{10}, \frac{1}{10})$ |
| 358 | | | 45 | $(g_{16}, h_{86}^{(16)})$ | 1/50 | $\frac{1}{2}(1+\sqrt{5})$ | $(0, \frac{1}{5}, \frac{1}{5}, 0, -\frac{2}{5}, \frac{1}{5}, 0, \frac{2}{5}, 0, \frac{2}{5}, -\frac{2}{5}, -\frac{1}{5})$ | $(-\frac{1}{10}, 0, \frac{1}{5}, -\frac{1}{5}, 0, -\frac{1}{5}, \frac{3}{10}, \frac{2}{5}, \frac{2}{5}, \frac{1}{2}, \frac{2}{5}, \frac{1}{2})$ |
| 359 | 360 | | 1 | $(g_{21}, h_{27}^{(21)})$ | 1/25 | $\frac{1}{2}(3+\sqrt{5})$ | $(\frac{1}{10}, -\frac{1}{10}, -\frac{3}{10}, \frac{3}{10}, \frac{1}{10}, \frac{1}{2}, -\frac{3}{10}, \frac{1}{2}, -\frac{3}{10}, \frac{1}{10}, -\frac{3}{10}, \frac{1}{10})$ | $(0, -\frac{1}{5}, -\frac{1}{5}, 0, -\frac{1}{5}, \frac{1}{5}, -\frac{2}{5}, \frac{2}{5}, \frac{1}{5}, \frac{2}{5}, \frac{2}{5}, \frac{3}{5})$ |
| 360 | 359 | | 1 | $(g_{21}, h_{28}^{(21)})$ | 1/25 | $\frac{1}{2}(3+\sqrt{5})$ | $(-\frac{3}{10}, -\frac{3}{10}, \frac{1}{2}, \frac{1}{10}, \frac{1}{10}, -\frac{3}{10}, -\frac{1}{10}, -\frac{1}{10}, -\frac{3}{10}, \frac{1}{10}, \frac{1}{2}, -\frac{3}{10})$ | $(\frac{1}{2}, -\frac{3}{10}, -\frac{3}{10}, \frac{3}{10}, \frac{1}{2}, \frac{3}{10}, \frac{3}{10}, -\frac{1}{10}, -\frac{1}{10}, -\frac{1}{10}, -\frac{1}{10}, \frac{11}{10})$ |
| 361 | 362 | | 1 | $(g_{21}, h_{39}^{(21)})$ | 1/25 | $\frac{1}{2}(3-\sqrt{5})$ | $(-\frac{1}{10}, \frac{1}{10}, -\frac{3}{10}, \frac{3}{10}, -\frac{1}{10}, \frac{3}{10}, \frac{1}{2}, \frac{1}{2}, \frac{3}{10}, \frac{1}{10}, \frac{1}{10}, \frac{3}{10})$ | $(0, \frac{1}{5}, 0, \frac{1}{5}, \frac{1}{5}, \frac{2}{5}, -\frac{2}{5}, \frac{1}{5}, -\frac{2}{5}, \frac{2}{5}, -\frac{2}{5}, \frac{4}{5})$ |
| 362 | 361 | | 1 | $(g_{21}, h_{40}^{(21)})$ | 1/25 | $\frac{1}{2}(3-\sqrt{5})$ | $(-\frac{1}{10}, \frac{1}{2}, -\frac{3}{10}, -\frac{1}{10}, \frac{3}{10}, \frac{3}{10}, \frac{1}{10}, \frac{3}{10}, \frac{1}{10}, \frac{1}{2}, -\frac{1}{10}, -\frac{3}{10})$ | $(0, -\frac{2}{5}, -\frac{2}{5}, -\frac{1}{5}, \frac{1}{5}, 0, -\frac{1}{5}, -\frac{2}{5}, \frac{2}{5}, \frac{1}{5}, \frac{2}{5}, \frac{6}{5})$ |
| 363 | | | 4 | $(g_{23}, h_{19}^{(23)})$ | 1/243 | 0 | $(-\frac{1}{3}, \frac{1}{3}, 0, -\frac{1}{3}, -\frac{1}{3}, \frac{1}{3}, 0, 0, 0, \frac{1}{3}, -\frac{1}{3}, \frac{1}{3})$ | $(-\frac{1}{6}, \frac{1}{6}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, -\frac{1}{6}, \frac{1}{6}, -\frac{1}{6}, -\frac{1}{6}, \frac{1}{6}, \frac{5}{6})$ |
| 364 | | | 12 | $(g_{23}, h_{28}^{(23)})$ | 1/27 | 0 | $(0, \frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, 0, -\frac{1}{3}, \frac{1}{3}, -\frac{1}{3}, 0, -\frac{1}{3}, \frac{1}{3}, 0)$ | $(-\frac{1}{9}, \frac{2}{9}, -\frac{1}{3}, -\frac{1}{9}, \frac{2}{9}, -\frac{4}{9}, -\frac{4}{9}, \frac{1}{3}, -\frac{4}{9}, \frac{2}{9}, -\frac{1}{9}, 1)$ |
| 365 | | | 12 | $(g_{23}, h_{31}^{(23)})$ | 1/27 | 0 | $(\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, 0, -\frac{1}{3}, 0, 0, 0, -\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3})$ | $(\frac{7}{18}, \frac{7}{18}, -\frac{5}{18}, \frac{7}{18}, \frac{5}{18}, -\frac{5}{18}, -\frac{1}{18}, \frac{1}{6}, -\frac{1}{18}, \frac{1}{18}, -\frac{1}{6}, -\frac{1}{6})$ |
| 366 | | A | 4 | $(g_{24}, h_{43}^{(24)})$ | 1/36 | 0 | – | – |
| 367 | | A | 4 | $(g_{24}, h_{52}^{(24)})$ | 1/36 | 0 | – | – |
| 368 | | A | 20 | $(g_{25}, h_{41}^{(25)})$ | 1/100 | 0 | – | – |
| 369 | | A | 20 | $(g_{25}, h_{71}^{(25)})$ | 1/100 | 0 | – | – |
| 370 | 371 | | 1 | $(g_{28}, h_{37}^{(28)})$ | 1/4374 | 3 | $(-\frac{1}{3}, -\frac{1}{3}, \frac{1}{3}, \frac{1}{3}, \frac{1}{3}, -\frac{1}{3}, 0, 0, 0, -\frac{1}{3}, \frac{1}{3}, \frac{4}{3})$ | $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{6}, \frac{1}{6}, -\frac{1}{6}, -\frac{1}{6}, -\frac{1}{6}, -\frac{1}{6}, \frac{1}{6}, -\frac{1}{6}, \frac{5}{6})$ |
| 371 | 370 | | 1 | $(g_{28}, h_{49}^{(28)})$ | 1/4374 | 3 | $(\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, \frac{1}{3}, -\frac{1}{3}, \frac{1}{3}, 0, 0, 0, \frac{1}{3}, \frac{1}{3}, \frac{2}{3})$ | $(0, 0, 0, \frac{1}{3}, -\frac{1}{3}, \frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, \frac{4}{3})$ |
| 372 | 373 | | 4 | $(g_{28}, h_{59}^{(28)})$ | 1/27 | $\sqrt{3}$ | $(\frac{1}{3}, \frac{1}{3}, 0, 0, -\frac{1}{3}, -\frac{1}{3}, \frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, \frac{1}{3}, \frac{1}{3}, 1)$ | $(\frac{1}{3}, \frac{2}{9}, \frac{1}{9}, \frac{2}{9}, -\frac{1}{9}, 0, -\frac{1}{9}, \frac{2}{9}, -\frac{4}{9}, -\frac{4}{9}, -\frac{1}{3}, \frac{13}{9})$ |
| 373 | 372 | | 4 | $(g_{28}, h_{63}^{(28)})$ | 1/27 | $\sqrt{3}$ | $(0, -\frac{1}{3}, -\frac{1}{3}, 0, \frac{1}{3}, \frac{1}{3}, \frac{1}{3}, \frac{1}{3}, \frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, 1)$ | $(-\frac{1}{9}, -\frac{2}{9}, -\frac{1}{9}, \frac{2}{9}, 0, -\frac{1}{9}, \frac{1}{3}, -\frac{2}{9}, \frac{4}{9}, \frac{4}{9}, \frac{1}{3}, \frac{13}{9})$ |
| 374 | 375 | | 3 | $(g_{28}, h_{71}^{(28)})$ | 1/162 | $\sqrt{7}$ | $(0, -\frac{1}{3}, -\frac{1}{3}, \frac{1}{3}, \frac{1}{3}, -\frac{1}{3}, \frac{1}{3}, -\frac{1}{3}, 0, 0, -\frac{1}{3}, \frac{4}{3})$ | $(-\frac{1}{6}, -\frac{1}{6}, \frac{1}{2}, -\frac{1}{6}, 0, 0, \frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, \frac{1}{3}, -\frac{1}{3})$ |
| 375 | 374 | | 3 | $(g_{28}, h_{79}^{(28)})$ | 1/162 | $\sqrt{7}$ | $(0, \frac{1}{3}, -\frac{1}{3}, \frac{1}{3}, \frac{1}{3}, \frac{1}{3}, -\frac{1}{3}, 0, 0, -\frac{1}{3}, \frac{4}{3})$ | $(\frac{1}{6}, \frac{1}{2}, -\frac{1}{6}, -\frac{1}{6}, 0, 0, \frac{1}{3}, -\frac{1}{3}, \frac{1}{3}, \frac{1}{3}, -\frac{1}{3})$ |
| 376 | | | 6 | $(g_{33}, h_{17}^{(33)})$ | 1/256 | 0 | $(\frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, 0, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, 0, 0)$ | $(\frac{1}{4}, 0, -\frac{1}{4}, \frac{1}{2}, 0, -\frac{1}{4}, -\frac{1}{4}, \frac{1}{2}, 0, 0, \frac{1}{2}, \frac{3}{2})$ |
| 377 | | | 1 | $(g_{33}, h_{21}^{(33)})$ | 1/512 | $2\sqrt{2}$ | $(-\frac{1}{4}, -\frac{1}{4}, \frac{1}{2}, 0, -\frac{1}{4}, \frac{1}{4}, \frac{1}{2}, \frac{1}{2}, 0, 0, \frac{1}{2}, \frac{1}{2})$ | $(-\frac{1}{4}, \frac{1}{2}, -\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, 0, \frac{1}{2}, \frac{1}{2}, 0, 0, 0, 0)$ |
| 378 | | | 3 | $(g_{33}, h_{26}^{(33)})$ | 1/256 | 0 | $(\frac{1}{4}, \frac{1}{2}, 0, -\frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, 0, 0, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{3}{2})$ | $(\frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, \frac{1}{2}, \frac{1}{4}, 0, 0, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, 0, 0)$ |
| 379 | | | 1 | $(g_{33}, h_{27}^{(33)})$ | 1/512 | 0 | $(-\frac{1}{4}, \frac{1}{2}, 0, -\frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, 0, \frac{1}{2}, 0, \frac{1}{2}, \frac{1}{2}, -\frac{1}{2})$ | $(-\frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, 0, -\frac{1}{4}, \frac{1}{2}, 0, \frac{1}{2}, \frac{1}{2}, 0, 0, 1)$ |
| 380 | | | 3 | $(g_{33}, h_{28}^{(33)})$ | 1/1024 | 0 | $(0, \frac{1}{4}, 0, \frac{1}{2}, \frac{1}{2}, 0, -\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{2}, \frac{1}{2}, -\frac{1}{2})$ | $(-\frac{1}{4}, -\frac{1}{4}, 0, \frac{1}{2}, 0, \frac{1}{2}, -\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, -\frac{1}{4})$ |
| 381 | | | 6 | $(g_{33}, h_{33}^{(33)})$ | 1/1024 | 0 | $(-\frac{1}{4}, \frac{1}{4}, \frac{1}{2}, -\frac{1}{4}, \frac{1}{4}, \frac{1}{2}, \frac{1}{2}, 0, 0, 0, \frac{1}{2}, -\frac{1}{2})$ | $(0, \frac{1}{2}, \frac{1}{2}, -\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, \frac{1}{2}, 0, 0, 0, 0)$ |
| 382 | | | 3 | $(g_{33}, h_{42}^{(33)})$ | 1/512 | 2 | $(\frac{1}{2}, \frac{1}{4}, -\frac{1}{4}, \frac{1}{2}, \frac{1}{2}, 0, 0, 0, \frac{1}{4}, \frac{1}{4}, \frac{1}{2}, -\frac{1}{2})$ | $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, 0, 0, 0, 0, 1)$ |
| 383 | | | 6 | $(g_{33}, h_{47}^{(33)})$ | 1/256 | 0 | $(0, 0, \frac{1}{2}, \frac{1}{4}, \frac{1}{4}, 0, \frac{1}{2}, \frac{1}{4}, -\frac{1}{4}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2})$ | $(\frac{1}{4}, 0, \frac{1}{2}, \frac{1}{4}, \frac{1}{4}, \frac{1}{2}, 0, -\frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, -\frac{1}{4})$ |
| 384 | | | 3 | $(g_{33}, h_{51}^{(33)})$ | 1/256 | 0 | $(-\frac{1}{4}, \frac{1}{2}, 0, -\frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, 0, 0, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, -\frac{1}{2})$ | $(-\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, 0, \frac{1}{4}, \frac{1}{2}, 0, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, 0, 0)$ |
| 385 | | | 1 | $(g_{33}, h_{52}^{(33)})$ | 1/512 | 0 | $(-\frac{1}{4}, \frac{1}{2}, 0, -\frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, 0, \frac{1}{2}, 0, \frac{1}{2}, \frac{1}{2}, \frac{1}{2})$ | $(-\frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, 0, \frac{1}{4}, \frac{1}{2}, 0, 0, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, -\frac{1}{2})$ |
| 386 | | | 3 | $(g_{33}, h_{57}^{(33)})$ | 1/512 | 2 | $(-\frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, \frac{1}{2}, \frac{1}{2}, 0, 0, 0, \frac{1}{2}, \frac{1}{2}, \frac{1}{2})$ | $(0, -\frac{1}{4}, \frac{1}{2}, \frac{1}{4}, \frac{1}{2}, 0, -\frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, \frac{5}{4})$ |
| 387 | | | 6 | $(g_{33}, h_{84}^{(33)})$ | 1/32 | $\sqrt{2}$ | $(\frac{1}{2}, \frac{1}{4}, \frac{1}{4}, \frac{1}{2}, 0, 0, -\frac{1}{4}, \frac{1}{2}, -\frac{1}{4}, \frac{1}{2}, 0, \frac{1}{2})$ | $(-\frac{3}{8}, \frac{1}{4}, \frac{1}{2}, \frac{1}{4}, \frac{3}{8}, -\frac{1}{8}, \frac{1}{4}, \frac{1}{8}, 0, \frac{1}{2}, 0, 1)$ |
| 388 | | | 3 | $(g_{33}, h_{89}^{(33)})$ | 1/32 | 0 | $(\frac{1}{4}, \frac{1}{4}, 0, -\frac{1}{4}, -\frac{1}{4}, \frac{1}{2}, 0, \frac{1}{2}, 0, \frac{1}{2}, \frac{1}{2}, \frac{3}{2})$ | $(\frac{1}{4}, \frac{1}{2}, \frac{1}{4}, \frac{1}{4}, 0, \frac{1}{2}, 0, 0, \frac{1}{2}, -\frac{1}{4}, -\frac{1}{4}, \frac{5}{4})$ |

| No. | P | com. | size | rep. | $[h]_{\subset C_g}/\lvert C_g\rvert$ | $\lvert Z_{\mathrm{R}g}^{\mathrm{R}h}\rvert$ | $\vec{\lambda}_{\tilde{g},O}$ | $\vec{\lambda}_{\tilde{h},O}$ |
|---|---|---|---|---|---|---|---|---|
| 389 | | | 12 | $(g_{33}, h_{90}^{(33)})$ | 1/64 | 0 | $(\frac{1}{4}, \frac{1}{2}, \frac{1}{2}, 0, -\frac{1}{4}, \frac{1}{2}, \frac{1}{4}, \frac{1}{4}, 0, \frac{1}{2}, 0, \frac{1}{2})$ | $(\frac{1}{8}, \frac{3}{8}, -\frac{1}{4}, \frac{1}{4}, 0, -\frac{1}{8}, -\frac{1}{4}, -\frac{3}{8}, 0, \frac{1}{2}, \frac{1}{2}, 1)$ |
| 390 | | | 3 | $(g_{33}, h_{95}^{(33)})$ | 1/32 | $\sqrt{2}$ | $(0, \frac{1}{4}, \frac{1}{2}, \frac{1}{4}, 0, \frac{1}{2}, \frac{1}{4}, 0, \frac{1}{2}, \frac{1}{4}, \frac{1}{2}, \frac{1}{2})$ | $(-\frac{1}{8}, \frac{1}{8}, \frac{3}{8}, \frac{3}{8}, \frac{1}{4}, \frac{1}{4}, -\frac{3}{8}, \frac{3}{8}, -\frac{1}{8}, -\frac{1}{8}, \frac{1}{2}, 0)$ |
| 391 | 392 | | 14 | $(g_{37}, h_{29}^{(37)})$ | 1/49 | $\sqrt{2}$ | $(\frac{1}{14}, \frac{3}{14}, \frac{1}{14}, \frac{5}{14}, \frac{1}{2}, \frac{1}{2}, \frac{5}{14}, \frac{1}{2}, \frac{5}{14}, \frac{3}{14}, -\frac{3}{14}, -\frac{1}{14})$ | $(\frac{1}{14}, \frac{3}{14}, \frac{3}{14}, \frac{1}{14}, -\frac{1}{14}, \frac{3}{14}, \frac{5}{14}, -\frac{5}{14}, \frac{1}{2}, -\frac{5}{14}, \frac{1}{2}, \frac{3}{2})$ |
| 392 | 391 | | 14 | $(g_{37}, h_{36}^{(37)})$ | 1/49 | $\sqrt{2}$ | $(-\frac{1}{14}, -\frac{1}{14}, \frac{3}{14}, -\frac{1}{14}, \frac{1}{2}, -\frac{5}{14}, \frac{1}{2}, \frac{1}{2}, \frac{5}{14}, \frac{3}{14}, -\frac{5}{14}, -\frac{3}{14})$ | $(-\frac{1}{7}, \frac{1}{7}, -\frac{1}{7}, -\frac{2}{7}, -\frac{1}{7}, -\frac{2}{7}, -\frac{2}{7}, \frac{3}{7}, \frac{3}{7}, \frac{3}{7}, \frac{2}{7}, \frac{3}{7})$ |
| 393 | | | 3 | $(g_{48}, h_{15}^{(48)})$ | 1/1024 | 0 | $(0, 0, \frac{1}{2}, \frac{1}{2}, 0, -\frac{1}{4}, -\frac{1}{4}, \frac{1}{2}, \frac{1}{2}, \frac{1}{4}, -\frac{1}{4}, 0)$ | $(\frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, 0, 0, 0, 0, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2})$ |
| 394 | | | 3 | $(g_{48}, h_{19}^{(48)})$ | 1/2048 | 0 | $(\frac{1}{2}, 0, \frac{1}{4}, \frac{1}{4}, 0, \frac{1}{2}, \frac{1}{2}, 0, 0, -\frac{1}{4}, -\frac{1}{4}, \frac{3}{2})$ | $(\frac{1}{2}, 0, \frac{1}{4}, \frac{1}{4}, \frac{1}{2}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, 0)$ |
| 395 | | | 6 | $(g_{48}, h_{59}^{(48)})$ | 1/128 | 0 | $(\frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, 0, 0, 0, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, 1)$ | $(\frac{1}{4}, 0, \frac{1}{4}, \frac{1}{2}, 0, \frac{1}{4}, \frac{1}{4}, 0, \frac{1}{4}, -\frac{1}{4}, \frac{1}{2}, \frac{1}{2})$ |
| 396 | | | 3 | $(g_{48}, h_{69}^{(48)})$ | 1/1024 | 0 | $(0, 0, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{4}, \frac{1}{4}, 0, 0, \frac{1}{4}, -\frac{1}{4}, \frac{1}{2})$ | $(-\frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, 0, 0, 0, 1)$ |
| 397 | | | 6 | $(g_{48}, h_{73}^{(48)})$ | 1/128 | 0 | $(0, -\frac{1}{4}, \frac{1}{2}, \frac{1}{4}, 0, \frac{1}{2}, \frac{1}{4}, 0, \frac{1}{4}, \frac{1}{2}, \frac{1}{2}, 1)$ | $(-\frac{1}{8}, -\frac{1}{8}, \frac{3}{8}, \frac{3}{8}, \frac{1}{4}, \frac{1}{4}, -\frac{3}{8}, -\frac{3}{8}, -\frac{1}{8}, \frac{1}{8}, 0, \frac{3}{2})$ |
| 398 | | | 3 | $(g_{48}, h_{81}^{(48)})$ | 1/64 | 0 | $(-\frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, \frac{1}{2}, 0, \frac{1}{2}, \frac{1}{2}, 0, 0, 0, -\frac{1}{2})$ | $(-\frac{1}{4}, \frac{1}{2}, -\frac{1}{4}, 0, \frac{1}{2}, \frac{1}{2}, -\frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, 0, 1)$ |
| 399 | | | 3 | $(g_{48}, h_{97}^{(48)})$ | 1/64 | 0 | $(\frac{1}{2}, -\frac{1}{4}, 0, \frac{1}{4}, \frac{1}{4}, 0, -\frac{1}{4}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, 0, 0)$ | $(-\frac{3}{8}, -\frac{3}{8}, \frac{1}{8}, \frac{1}{8}, -\frac{3}{8}, \frac{3}{8}, \frac{1}{8}, -\frac{1}{8}, \frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, \frac{1}{4})$ |
| 400 | | A | 6 | $(g_{48}, h_{104}^{(48)})$ | 1/256 | 0 | – | – |
| 401 | | A | 3 | $(g_{48}, h_{106}^{(48)})$ | 1/256 | 0 | – | – |
| 402 | | A | 3 | $(g_{48}, h_{108}^{(48)})$ | 1/256 | 0 | – | – |
| 403 | | A | 6 | $(g_{48}, h_{110}^{(48)})$ | 1/128 | 0 | – | – |
| 404 | | A | 6 | $(g_{48}, h_{117}^{(48)})$ | 1/128 | 0 | – | – |
| 405 | | A | 1 | $(g_{48}, h_{120}^{(48)})$ | 1/256 | 0 | – | – |
| 406 | | A | 24 | $(g_{48}, h_{122}^{(48)})$ | 1/192 | 0 | – | – |
| 407 | | A | 4 | $(g_{48}, h_{123}^{(48)})$ | 1/192 | 0 | – | – |
| 408 | | A | 16 | $(g_{48}, h_{127}^{(48)})$ | 1/2304 | 0 | – | – |
| 409 | | A | 16 | $(g_{48}, h_{131}^{(48)})$ | 1/144 | 0 | – | – |
| 410 | | A | 1 | $(g_{48}, h_{135}^{(48)})$ | 1/36864 | 0 | – | – |
| 411 | | A | 3 | $(g_{48}, h_{136}^{(48)})$ | 1/4096 | 0 | – | – |
| 412 | | A | 3 | $(g_{48}, h_{141}^{(48)})$ | 1/1024 | 0 | – | – |
| 413 | | A | 2 | $(g_{48}, h_{144}^{(48)})$ | 1/2048 | 0 | – | – |
| 414 | | A | 1 | $(g_{48}, h_{148}^{(48)})$ | 1/6144 | 0 | – | – |
| 415 | | A | 1 | $(g_{48}, h_{149}^{(48)})$ | 1/2048 | 0 | – | – |
| 416 | | A | 3 | $(g_{48}, h_{150}^{(48)})$ | 1/3072 | 0 | – | – |
| 417 | | A | 3 | $(g_{48}, h_{151}^{(48)})$ | 1/32 | 0 | – | – |
| 418 | | A | 12 | $(g_{48}, h_{152}^{(48)})$ | 1/64 | 0 | – | – |
| 419 | | A | 1 | $(g_{48}, h_{161}^{(48)})$ | 1/128 | 0 | – | – |
| 420 | | A | 6 | $(g_{48}, h_{162}^{(48)})$ | 1/192 | 0 | – | – |
| 421 | | A | 48 | $(g_{48}, h_{163}^{(48)})$ | 1/48 | 0 | – | – |
| 422 | | A | 24 | $(g_{51}, h_{82}^{(51)})$ | 1/144 | 0 | – | – |
| 423 | | A | 24 | $(g_{51}, h_{94}^{(51)})$ | 1/144 | 0 | – | – |
| 424 | | A | 3 | $(g_{57}, h_{33}^{(57)})$ | 1/256 | 0 | – | – |
| 425 | | A | 6 | $(g_{57}, h_{35}^{(57)})$ | 1/1536 | 0 | – | – |
| 426 | | A | 48 | $(g_{57}, h_{87}^{(57)})$ | 1/96 | 0 | – | – |
| 427 | | | 3 | $(g_{62}, h_{13}^{(62)})$ | 1/8192 | 0 | $(\frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, \frac{1}{2}, \frac{1}{2}, 0, 0, \frac{1}{4}, -\frac{1}{4})$ | $(\frac{1}{2}, \frac{1}{2}, 0, 0, -\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, \frac{1}{4})$ |
| 428 | | | 1 | $(g_{62}, h_{47}^{(62)})$ | 1/8192 | 4 | $(\frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, \frac{1}{2}, \frac{1}{2}, 0, 0, -\frac{1}{4}, \frac{1}{4})$ | $(0, 0, \frac{1}{2}, \frac{1}{2}, \frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{3}{4})$ |
| 429 | | | 1 | $(g_{62}, h_{49}^{(62)})$ | 1/49152 | 4 | $(-\frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, 0, 0, \frac{1}{2}, \frac{1}{2}, -\frac{1}{4}, -\frac{1}{4})$ | $(0, 0, \frac{1}{2}, \frac{1}{2}, -\frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{5}{4})$ |
| 430 | | | 8 | $(g_{62}, h_{53}^{(62)})$ | 1/768 | 2 | $(-\frac{1}{4}, 0, \frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, \frac{1}{2}, -\frac{1}{4}, \frac{1}{2}, \frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, 0)$ | $(-\frac{1}{6}, -\frac{1}{12}, \frac{1}{12}, \frac{1}{3}, \frac{1}{12}, -\frac{1}{12}, \frac{1}{6}, -\frac{5}{12}, \frac{5}{12}, \frac{5}{12}, -\frac{1}{3}, \frac{7}{12})$ |
| 431 | | | 1 | $(g_{62}, h_{59}^{(62)})$ | 1/512 | 2 | $(\frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, \frac{1}{2}, \frac{1}{2}, 0, 0, \frac{1}{4}, \frac{3}{4})$ | $(\frac{1}{2}, \frac{1}{2}, 0, 0, -\frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, \frac{1}{4})$ |
| 432 | | A | 1 | $(g_{62}, h_{73}^{(62)})$ | 1/256 | 0 | – | – |
| 433 | | A | 6 | $(g_{62}, h_{75}^{(62)})$ | 1/128 | 0 | – | – |
| 434 | | | 1 | $(g_{62}, h_{84}^{(62)})$ | 1/3072 | 4 | $(-\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, \frac{1}{2}, \frac{1}{2}, -\frac{1}{4}, -\frac{1}{4}, 0, 1)$ | $(\frac{1}{2}, \frac{1}{2}, 0, 0, -\frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, \frac{5}{4})$ |
| 435 | | | 8 | $(g_{62}, h_{87}^{(62)})$ | 1/192 | 2 | $(-\frac{1}{4}, 0, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{2}, -\frac{1}{4}, \frac{1}{2}, \frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, 0)$ | $(-\frac{1}{6}, -\frac{1}{12}, \frac{1}{12}, \frac{1}{3}, -\frac{1}{12}, \frac{1}{12}, \frac{1}{6}, \frac{5}{12}, \frac{5}{12}, \frac{5}{12}, -\frac{1}{3}, \frac{17}{12})$ |
| 436 | | A | 3 | $(g_{62}, h_{125}^{(62)})$ | 1/128 | 0 | – | – |

| No. | P | com. | size | rep. | $[h]_{\subset C_g}/\lvert C_g\rvert$ | $\lvert Z_{\mathrm{R}g}^{\mathrm{R}h}\rvert$ | $\vec{\lambda}_{\tilde{g},O}$ | $\vec{\lambda}_{\tilde{h},O}$ |
|---|---|---|---|---|---|---|---|---|
| 437 | | | 3 | $(g_{62}, h_{137}^{(62)})$ | 1/128 | 2 | $(-\frac{1}{4}, 0, \frac{1}{4}, -\frac{1}{4}, \frac{1}{2}, \frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, \frac{1}{2}, 0, \frac{5}{4})$ | $(-\frac{1}{4}, -\frac{1}{8}, \frac{1}{8}, -\frac{3}{8}, \frac{3}{8}, 0, \frac{1}{2}, \frac{3}{8}, \frac{1}{8}, -\frac{1}{8}, -\frac{3}{8}, \frac{3}{4})$ |
| 438 | | | 3 | $(g_{62}, h_{150}^{(62)})$ | 1/256 | 0 | $(0, -\frac{1}{4}, \frac{1}{4}, \frac{1}{2}, -\frac{1}{4}, \frac{1}{2}, 0, \frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, \frac{5}{4})$ | $(\frac{1}{8}, -\frac{1}{8}, -\frac{1}{8}, \frac{1}{8}, -\frac{3}{8}, \frac{3}{8}, -\frac{3}{8}, -\frac{3}{8}, \frac{1}{4}, -\frac{1}{4}, \frac{1}{2}, \frac{1}{2})$ |
| 439 | | | 3 | $(g_{62}, h_{151}^{(62)})$ | 1/256 | 0 | $(0, -\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{2}, -\frac{1}{4}, \frac{1}{2}, \frac{1}{4}, \frac{1}{4}, 0, \frac{5}{4})$ | $(-\frac{1}{8}, -\frac{1}{8}, 0, \frac{1}{4}, -\frac{1}{8}, -\frac{1}{8}, -\frac{3}{8}, -\frac{3}{8}, \frac{1}{2}, -\frac{1}{4}, -\frac{3}{8}, \frac{5}{8})$ |
| 440 | | A | 3 | $(g_{62}, h_{181}^{(62)})$ | 1/64 | 0 | – | – |
| 441 | | | 3 | $(g_{62}, h_{183}^{(62)})$ | 1/32 | $\sqrt{2}$ | $(\frac{1}{4}, \frac{1}{4}, 0, \frac{1}{4}, \frac{1}{4}, \frac{1}{2}, -\frac{1}{4}, \frac{1}{2}, \frac{1}{4}, -\frac{1}{4}, 0, \frac{3}{4})$ | $(\frac{1}{4}, \frac{1}{8}, -\frac{1}{8}, 0, \frac{3}{8}, \frac{3}{8}, \frac{1}{8}, -\frac{1}{8}, \frac{1}{2}, \frac{3}{8}, \frac{3}{8}, \frac{5}{4})$ |
| 442 | | | 4 | $(g_{72}, h_{87}^{(72)})$ | 1/128 | $\sqrt{4-2\sqrt{2}}$ | $(-\frac{1}{8}, -\frac{1}{4}, -\frac{1}{8}, \frac{1}{2}, -\frac{3}{8}, \frac{3}{8}, \frac{1}{2}, -\frac{3}{8}, \frac{3}{8}, -\frac{1}{8}, \frac{1}{8}, \frac{3}{4})$ | $(-\frac{1}{4}, -\frac{1}{8}, \frac{1}{8}, -\frac{1}{8}, \frac{1}{8}, -\frac{1}{4}, -\frac{3}{8}, \frac{1}{2}, -\frac{3}{8}, \frac{3}{8}, \frac{1}{2}, \frac{3}{8})$ |
| 443 | | | 4 | $(g_{72}, h_{91}^{(72)})$ | 1/128 | $\sqrt{2(2+\sqrt{2})}$ | $(\frac{1}{2}, \frac{3}{8}, \frac{3}{8}, -\frac{1}{4}, \frac{1}{8}, \frac{1}{4}, -\frac{1}{8}, -\frac{1}{8}, \frac{1}{2}, -\frac{1}{8}, -\frac{3}{8}, -\frac{3}{8})$ | $(\frac{3}{8}, \frac{1}{4}, -\frac{3}{8}, \frac{3}{8}, -\frac{3}{8}, \frac{1}{8}, 0, \frac{1}{4}, -\frac{1}{8}, \frac{1}{8}, 0, \frac{1}{8})$ |
| 444 | | A | 4 | $(g_{75}, h_{81}^{(75)})$ | 1/128 | 0 | – | – |
| 445 | | A | 4 | $(g_{75}, h_{82}^{(75)})$ | 1/128 | 0 | – | – |

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